



10. If the direction of the initial velocity of the charged particle is neither along nor perpendicular to that of the magnetic field, then the orbit will be [MP PET 1993]
 (a) A straight line (b) An ellipse
 (c) A circle (d) A helix
11. Particles having positive charges occasionally come with high velocity from the sky towards the earth. On account of the magnetic field of earth, they would be deflected towards the
 (a) North (b) South
 (c) East (d) West
12. A 2 MeV proton is moving perpendicular to a uniform magnetic field of 2.5 tesla. The force on the proton is [CPMT 1989]
 (a) $2.5 \times 10^{-10} \text{ N}$ (b) $7.6 \times 10^{-11} \text{ N}$
 (c) $2.5 \times 10^{-11} \text{ N}$ (d) $7.6 \times 10^{-12} \text{ N}$
13. A charged particle moves with velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$. The magnetic force experienced by the particle is
 (a) Always zero
 (b) Never zero
 (c) Zero, if $\vec{B}$ and $\vec{v}$ are perpendicular
 (d) Zero, if $\vec{B}$ and $\vec{v}$ are parallel
14. A proton is moving along Z-axis in a magnetic field. The magnetic field is along X-axis. The proton will experience a force along
 (a) X-axis (b) Y-axis
 (c) Z-axis (d) Negative Z-axis
15. A proton of mass m and charge $+e$ is moving in a circular orbit in a magnetic field with energy 1 MeV. What should be the energy of α -particle (mass = $4m$ and charge = $+2e$), so that it can revolve in the path of same radius [BHU 1997]
 (a) 1 MeV (b) 4 MeV
 (c) 2 MeV (d) 0.5 MeV
16. An electron is moving with a speed of 10^8 m/sec perpendicular to a uniform magnetic field of intensity B . Suddenly intensity of the magnetic field is reduced to $B/2$. The radius of the path becomes from the original value of r [MP PET 1993]
 (a) No change (b) Reduces to $r/2$
 (c) Increases to $2r$ (d) Stops moving
17. A proton and an α -particle enter a uniform magnetic field perpendicularly with the same speed. If proton takes $25 \mu \text{ sec}$ to make 5 revolutions, then the periodic time for the α -particle would be [MP PET 1993]
 (a) $50 \mu \text{ sec}$ (b) $25 \mu \text{ sec}$
 (c) $10 \mu \text{ sec}$ (d) $5 \mu \text{ sec}$
18. A proton (mass = $1.67 \times 10^{-27} \text{ kg}$ and charge = $1.6 \times 10^{-19} \text{ C}$) enters perpendicular to a magnetic field of intensity 2 weber / m^2 with a velocity $3.4 \times 10^7 \text{ m/sec}$. The acceleration of the proton should be [DPMT 1999]
 (a) $6.5 \times 10^{15} \text{ m/sec}^2$ (b) $6.5 \times 10^{13} \text{ m/sec}^2$
 (c) $6.5 \times 10^{11} \text{ m/sec}^2$ (d) $6.5 \times 10^9 \text{ m/sec}^2$
19. An α -particle travels in a circular path of radius 0.45 m in a magnetic field $B = 1.2 \text{ Wb/m}^2$ with a speed of $2.6 \times 10^7 \text{ m/sec}$. The period of revolution of the α -particle is
 (a) $1.1 \times 10^{-5} \text{ sec}$ (b) $1.1 \times 10^{-6} \text{ sec}$
 (c) $1.1 \times 10^{-7} \text{ sec}$ (d) $1.1 \times 10^{-8} \text{ sec}$
20. A uniform magnetic field B is acting from south to north and is of magnitude 1.5 Wb/m^2 . If a proton having mass = $1.7 \times 10^{-27} \text{ kg}$ and charge = $1.6 \times 10^{-19} \text{ C}$ moves in this field vertically downwards with energy 5 MeV, then the force acting on it will be [NCERT 1977]
 (a) $7.4 \times 10^{12} \text{ N}$ (b) $7.4 \times 10^{-12} \text{ N}$
 (c) $7.4 \times 10^{19} \text{ N}$ (d) $7.4 \times 10^{-19} \text{ N}$
21. A strong magnetic field is applied on a stationary electron, then [BIT 1989; MP PET 1993]
 (a) The electron moves in the direction of the field
 (b) The electron moves in an opposite direction
 (c) The electron remains stationary
 (d) The electron starts spinning
22. A uniform magnetic field acts at right angles to the direction of motion of electrons. As a result, the electron moves in a circular path of radius 2 cm. If the speed of the electrons is doubled, then the radius of the circular path will be [CBSE PMT 1991]
 (a) 2.0 cm (b) 0.5 cm
 (c) 4.0 cm (d) 1.0 cm
23. A deuteron of kinetic energy 50 keV is describing a circular orbit of radius 0.5 metre in a plane perpendicular to magnetic field $\vec{B}$. The kinetic energy of the proton that describes a circular orbit of radius 0.5 metre in the same plane with the same $\vec{B}$ is
 (a) 25 keV (b) 50 keV
 (c) 200 keV (d) 100 keV
24. If a proton is projected in a direction perpendicular to a uniform magnetic field with velocity v and an electron is projected along the lines of force, what will happen to proton and electron
 (a) The electron will travel along a circle with constant speed and the proton will move along a straight line
 (b) Proton will move in a circle with constant speed and there will be no effect on the motion of electron
 (c) There will not be any effect on the motion of electron and proton
 (d) The electron and proton both will follow the path of a parabola
25. An electron is travelling horizontally towards east. A magnetic field in vertically downward direction exerts a force on the electron along
 (a) East (b) West
 (c) North (d) South
26. Lorentz force can be calculated by using the formula [MP PET 1994, 2002, 03; CBSE PMT 2002]
 (a) $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$ (b) $\vec{F} = q(\vec{E} - \vec{v} \times \vec{B})$
 (c) $\vec{F} = q(\vec{E} + \vec{v} \cdot \vec{B})$ (d) $\vec{F} = q(\vec{E} \times \vec{B} + \vec{v})$
27. A magnetic field [MP PET 1994; Pb PMT 2003]
 (a) Always exerts a force on a charged particle
 (b) Never exerts a force on a charged particle
 (c) Exerts a force, if the charged particle is moving across the magnetic field lines
 (d) Exerts a force, if the charged particle is moving along the magnetic field lines

28. A proton enters a magnetic field of flux density $1.5 \text{ weber} / \text{m}^2$ with a velocity of $2 \times 10^7 \text{ m/sec}$ at an angle of 30° with the field. The force on the proton will be
[MP PET 1994 ; Pb. PMT 2004]
- (a) $2.4 \times 10^{-12} \text{ N}$ (b) $0.24 \times 10^{-12} \text{ N}$
(c) $24 \times 10^{-12} \text{ N}$ (d) $0.024 \times 10^{-12} \text{ N}$
29. If a particle of charge 10^{-12} coulomb moving along the $\hat{x}$ - direction with a velocity 10^5 m/s experiences a force of 10^{-10} newton in $\hat{y}$ - direction due to magnetic field, then the minimum magnetic field is
[MP PMT 1994]
- (a) $6.25 \times 10^3 \text{ tesla}$ in $\hat{z}$ - direction
(b) 10^{-15} tesla in $\hat{z}$ - direction
(c) $6.25 \times 10^{-3} \text{ tesla}$ in $\hat{z}$ - direction
(d) 10^{-3} tesla in $\hat{z}$ - direction
30. If a proton, deuteron and α - particle on being accelerated by the same potential difference enters perpendicular to the magnetic field, then the ratio of their kinetic energies is
[MP PMT 2003; J & K CET 2005]
- (a) $1 : 2 : 2$ (b) $2 : 2 : 1$
(c) $1 : 2 : 1$ (d) $1 : 1 : 2$
31. Which of the following statement is true
[Manipal MEE 1995]
- (a) The presence of a large magnetic flux through a coil maintains a current in the coil if the circuit is continuous
(b) A coil of a metal wire kept stationary in a non-uniform magnetic field has an e.m.f. induced in it
(c) A charged particle enters a region of uniform magnetic field at an angle of 85° to the magnetic lines of force; the path of the particle is a circle
(d) There is no change in the energy of a charged particle moving in a magnetic field although a magnetic force is acting on it
32. An electron and a proton enter region of uniform magnetic field in a direction at right angles to the field with the same kinetic energy. They describe circular paths of radius r_e and r_p respectively. Then
- (a) $r_e = r_p$
(b) $r_e < r_p$
(c) $r_e > r_p$
(d) r_e may be less than or greater than r_p depending on the direction of the magnetic field
33. A proton of mass $1.67 \times 10^{-27} \text{ kg}$ and charge $1.6 \times 10^{-19} \text{ C}$ is projected with a speed of $2 \times 10^6 \text{ m/s}$ at an angle of 60° to the X - axis. If a uniform magnetic field of 0.104 Tesla is applied along Y - axis, the path of proton is [IIT-JEE 1995]
- (a) A circle of radius = 0.2 m and time period $\pi \times 10^{-7} \text{ s}$
(b) A circle of radius = 0.1 m and time period $2\pi \times 10^{-7} \text{ s}$
(c) A helix of radius = 0.1 m and time period $2\pi \times 10^{-7} \text{ s}$
(d) A helix of radius = 0.2 m and time period $4\pi \times 10^{-7} \text{ s}$
34. A proton and a deuteron both having the same kinetic energy, enter perpendicularly into a uniform magnetic field B . For motion of proton and deuteron on circular path of radius R_p and R_d respectively, the correct statement is
[MP PET 1995]
- (a) $R_d = \sqrt{2} R_p$ (b) $R_d = R_p / \sqrt{2}$
(c) $R_d = R_p$ (d) $R_d = 2R_p$
35. A proton (or charged particle) moving with velocity v is acted upon by electric field E and magnetic field B . The proton will move undeflected if
[MP PMT 1995, 2003; UPSEAT 2002; DPMT 2003]
- (a) E is perpendicular to B
(b) E is parallel to v and perpendicular to B
(c) E , B and v are mutually perpendicular and $v = \frac{E}{B}$
(d) E and B both are parallel to v
36. A proton and an electron both moving with the same velocity v enter into a region of magnetic field directed perpendicular to the velocity of the particles. They will now move in circular orbits such that
[MP PMT 1995]
- (a) Their time periods will be same
(b) The time period for proton will be higher
(c) The time period for electron will be higher
(d) Their orbital radii will be same
37. A charge $+Q$ is moving upwards vertically. It enters a magnetic field directed to the north. The force on the charge will be towards
[MP PMT 1995; AIIMS 1995]
- (a) North (b) South
(c) East (d) West
38. An electron is moving on a circular path of radius r with speed v in a transverse magnetic field B . e/m for it will be
[MP PMT 2003]
- (a) $\frac{v}{Br}$ (b) $\frac{B}{rv}$
(c) Bvr (d) $\frac{vr}{B}$
39. A beam of well collimated cathode rays travelling with a speed of $5 \times 10^6 \text{ ms}^{-1}$ enter a region of mutually perpendicular electric and magnetic fields and emerge undeviated from this region. If $|B| = 0.02 \text{ T}$, the magnitude of the electric field is
- (a) 10^5 Vm^{-1} (b) $2.5 \times 10^8 \text{ Vm}^{-1}$
(c) $1.25 \times 10^{10} \text{ Vm}^{-1}$ (d) $2 \times 10^3 \text{ Vm}^{-1}$
40. An electron having charge $1.6 \times 10^{-19} \text{ C}$ and mass $9 \times 10^{-31} \text{ kg}$ is moving with $4 \times 10^6 \text{ ms}^{-1}$ speed in a magnetic field $2 \times 10^{-1} \text{ tesla}$ in a circular orbit. The force acting on electron and the radius of the circular orbit will be
[MP PET 1996; JIPMER 2000; BVP 2003]
- (a) $12.8 \times 10^{-13} \text{ N}$, $1.1 \times 10^{-4} \text{ m}$

(b) $1.28 \times 10^{-14} \text{ N}, 1.1 \times 10^{-3} \text{ m}$

(c) $1.28 \times 10^{-13} \text{ N}, 1.1 \times 10^{-3} \text{ m}$

(d) $1.28 \times 10^{-13} \text{ N}, 1.1 \times 10^{-4} \text{ m}$

41. An electron enters a magnetic field whose direction is perpendicular to the velocity of the electron. Then

[MP PMT 1996; CBSE PMT 2003]

- (a) The speed of the electron will increase
 (b) The speed of the electron will decrease
 (c) The speed of the electron will remain the same
 (d) The velocity of the electron will remain the same

42. An electron is moving in the north direction. It experiences a force in vertically upward direction. The magnetic field at the position of the electron is in the direction of

[MP PET 2003]

- (a) East (b) West
 (c) North (d) South

43. A current carrying long solenoid is placed on the ground with its axis vertical. A proton is falling along the axis of the solenoid with a velocity v . When the proton enters into the solenoid, it will

- (a) Be deflected from its path
 (b) Be accelerated along the same path
 (c) Be decelerated along the same path
 (d) Move along the same path with no change in velocity

44. A charged particle of mass m and charge q describes circular motion of radius r in a uniform magnetic field of strength B . The frequency of revolution is

[MP PET 1997; RPET 2001]

- (a) $\frac{Bq}{2\pi m}$ (b) $\frac{Bq}{2\pi m}$
 (c) $\frac{2\pi m}{Bq}$ (d) $\frac{Bm}{2\pi q}$

45. An electron is accelerated by a potential difference of 12000 volts. It then enters a uniform magnetic field of 10^{-3} T applied perpendicular to the path of electron. Find the radius of path. Given mass of electron $= 9 \times 10^{-31} \text{ kg}$ and charge on electron $= 1.6 \times 10^{-19} \text{ C}$

[MP PET 1997]

- (a) 36.7 m (b) 36.7 cm
 (c) 3.67 m (d) 3.67 cm

46. The charge on a particle Y is double the charge on particle X . These two particles X and Y after being accelerated through the same potential difference enter a region of uniform magnetic field and describe circular paths of radii R_1 and R_2 respectively. The ratio of the mass of X to that of Y is

[MP PET 1997]

- (a) $\left(\frac{2R_1}{R_2}\right)^2$ (b) $\left(\frac{R_1}{2R_2}\right)^2$
 (c) $\frac{R_1^2}{2R_2^2}$ (d) $\frac{2R_1}{R_2}$

47. A particle with 10^{-11} coulomb of charge and 10^{-7} kg mass is moving with a velocity of 10^8 m/s along the y -axis. A uniform static magnetic field $B = 0.5 \text{ Tesla}$ is acting along the x -direction. The force on the particle is

[MP PMT 1997]

- (a) $5 \times 10^{-11} \text{ N}$ along $\hat{i}$ (b) $5 \times 10^3 \text{ N}$ along $\hat{k}$
 (c) $5 \times 10^{-11} \text{ N}$ along $-\hat{j}$ (d) $5 \times 10^{-4} \text{ N}$ along $-\hat{k}$

48. A particle of charge q and mass m moving with a velocity v along the x -axis enters the region $x > 0$ with uniform magnetic field B along the $\hat{k}$ direction. The particle will penetrate in this region in the x -direction upto a distance d equal to

- (a) Zero (b) $\frac{mv}{qB}$
 (c) $\frac{2mv}{qB}$ (d) Infinity

49. A charged particle is moving with velocity v in a magnetic field of induction B . The force on the particle will be maximum when

- (a) v and B are in the same direction
 (b) v and B are in opposite directions
 (c) v and B are perpendicular
 (d) v and B are at an angle of 45°

50. A charged particle enters a magnetic field H with its initial velocity making an angle of 45° with H . The path of the particle will be

[MP PET 1999]

- (a) A straight line (b) A circle
 (c) An ellipse (d) A helix

51. An electron and a proton enter a magnetic field perpendicularly. Both have same kinetic energy. Which of the following is true

- (a) Trajectory of electron is less curved
 (b) Trajectory of proton is less curved
 (c) Both trajectories are equally curved
 (d) Both move on straight-line path

52. A charged particle moves in a uniform magnetic field. The velocity of the particle at some instant makes an acute angle with the magnetic field. The path of the particle will be

[MP PMT 1999]

- (a) A straight line
 (b) A circle
 (c) A helix with uniform pitch
 (d) A helix with non-uniform pitch

53. An electron is moving along positive x -axis. To get it moving on an anticlockwise circular path in x - y plane, a magnetic field is applied

- (a) Along positive y -axis (b) Along positive z -axis
 (c) Along negative y -axis (d) Along negative z -axis

54. A moving charge will gain energy due to the application of

[CPMT 1999]

- (a) Electric field (b) Magnetic field
 (c) Both of these (d) None of these

55. A proton, a deuteron and an α -particle having the same kinetic energy are moving in circular trajectories in a constant magnetic field. If r_p, r_d and r_α denote respectively the radii of the trajectories of these particles, then

[IIT 1997 Re-Exam]

- (a) $r_\alpha = r_p < r_d$ (b) $r_\alpha > r_d > r_p$
(c) $r_\alpha = r_d > r_p$ (d) $r_p = r_d = r_\alpha$

56. When a magnetic field is applied in a direction perpendicular to the direction of cathode rays, then their

[EAMCET 1994; BHU 2005]

- (a) Energy decreases
(b) Energy increases
(c) Momentum increases
(d) Momentum and energy remain unchanged

57. A charge moves in a circle perpendicular to a magnetic field. The time period of revolution is independent of

[RPET 1997; AIEEE 2002]

- (a) Magnetic field (b) Charge
(c) Mass of the particle (d) Velocity of the particle

58. A proton of energy 200 MeV enters the magnetic field of 5 T. If direction of field is from south to north and motion is upward, the force acting on it will be [RPET 1997]

- (a) Zero (b) $1.6 \times 10^{-10} \text{ N}$
(c) $3.2 \times 10^{-8} \text{ N}$ (d) $1.6 \times 10^{-6} \text{ N}$

59. An electron enters a region where magnetic (B) and electric (E) fields are mutually perpendicular to one another, then

[CBSE PMT1993]

- (a) It will always move in the direction of B
(b) It will always move in the direction of E
(c) It always possess circular motion
(d) It can go undeflected also

60. A charge moving with velocity v in X -direction is subjected to a field of magnetic induction in the negative X -direction. As a result, the charge will [CBSE PMT1993]

- (a) Remain unaffected
(b) Start moving in a circular path Y - Z plane
(c) Retard along X -axis
(d) Move along a helical path around X -axis

61. An electron and a proton with equal momentum enter perpendicularly into a uniform magnetic field, then

[BHU 1997; AIEEE 2002; MH CET (Med.) 2000]

- (a) The path of proton shall be more curved than that of electron
(b) The path of proton shall be less curved than that of electron
(c) Both are equally curved
(d) Path of both will be straight line

62. A positively charged particle moving due east enters a region of uniform magnetic field directed vertically upwards. The particle will

- (a) Get deflected vertically upwards
(b) Move in a circular orbit with its speed increased
(c) Move in a circular orbit with its speed unchanged
(d) Continue to move due east

63. A particle moving in a magnetic field increases its velocity, then its radius of the circle [BHU 1998]

- (a) Decreases (b) Increases
(c) Remains the same (d) Becomes half

64. A particle is moving in a uniform magnetic field, then

[BHU 1998]

- (a) Its momentum changes but total energy remains the same
(b) Both momentum and total energy remain the same
(c) Both will change
(d) Total energy changes but momentum remains the same

65. If an electron is going in the direction of magnetic field $\vec{B}$ with the velocity of $\vec{v}$ then the force on electron is

[RPMT 1999]

- (a) Zero (b) $e(\vec{v} \cdot \vec{B})$
(c) $e(\vec{v} \times \vec{B})$ (d) None of these

66. One proton beam enters a magnetic field of 10^{-4} T normally, Specific charge = 10^{11} C/kg , velocity = 10^7 m/s . What is the radius of the circle described by it [DCE 1999]

- (a) 0.1 m (b) 1 m
(c) 10 m (d) None of these

67. In a cyclotron, the angular frequency of a charged particle is independent of [CPMT 1999]

- (a) Mass (b) Speed
(c) Charge (d) Magnetic field

68. A charged particle is moving in a uniform magnetic field in a circular path. Radius of circular path is R . When energy of particle is doubled, then new radius will be

[CPMT 1999; Pb. PET 2002]

- (a) $R\sqrt{2}$ (b) $R\sqrt{3}$
(c) $2R$ (d) $3R$

69. The radius of curvature of the path of a charged particle moving in a static uniform magnetic field is [Roorkee 1999]

- (a) Directly proportional to the magnitude of the charge on the particle
(b) Directly proportional to the magnitude of the linear momentum of the particle
(c) Directly proportional to the kinetic energy of the particle
(d) Inversely proportional to the magnitude of the magnetic field

70. A proton moving with a velocity, $2.5 \times 10^7 \text{ m/s}$, enters a magnetic field of intensity 2.5 T making an angle 30° with the magnetic field. The force on the proton is

[AFMC 2000; CBSE PMT 2000]

- (a) $3 \times 10^{-12} \text{ N}$ (b) $5 \times 10^{-12} \text{ N}$
(c) $6 \times 10^{-12} \text{ N}$ (d) $9 \times 10^{-12} \text{ N}$

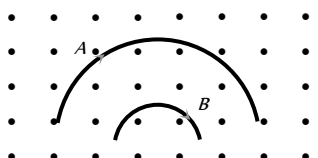
71. Maximum kinetic energy of the positive ion in the cyclotron is

- (a) $\frac{q^2 B r_0}{2m}$ (b) $\frac{q B^2 r_0}{2m}$
(c) $\frac{q^2 B^2 r_0^2}{2m}$ (d) $\frac{q B r_0}{2m^2}$

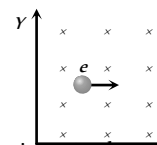
72. A charge q is moving in a magnetic field then the magnetic force does not depend upon [RPET 2000]

- (a) Charge (b) Mass
(c) Velocity (d) Magnetic field

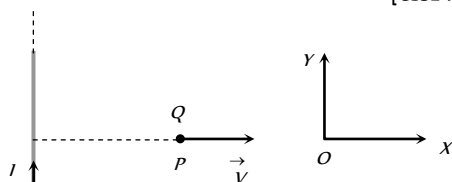
73. An electron is travelling in east direction and a magnetic field is applied in upward direction then electron will deflect in

- (a) South (b) North
(c) West (d) East
74. A charge of $1C$ is moving in a magnetic field of 0.5 Tesla with a velocity of 10 m/sec Perpendicular to the field. Force experienced is
(a) 5 N (b) 10 N
(c) 0.5 N (d) 0 N
75. An electron of mass m and charge q is travelling with a speed v along a circular path of radius r at right angles to a uniform magnetic field B . If speed of the electron is doubled and the magnetic field is halved, then resulting path would have a radius of
[Kerala PMT 2004; KCET 2000, 05]
(a) $\frac{r}{4}$ (b) $\frac{r}{2}$
(c) $2r$ (d) $4r$
76. If an electron enters a magnetic field with its velocity pointing in the same direction as the magnetic field, then
[MP PMT 2000]
(a) The electron will turn to its right
(b) The electron will turn to its left
(c) The velocity of the electron will increase
(d) The velocity of the electron will remain unchanged
77. A particle of mass m and charge q enters a magnetic field B perpendicularly with a velocity v . The radius of the circular path described by it will be
[MP PMT 2000]
(a) Bq/mv (b) mq/Bv
(c) mB/qv (d) mv/Bq
78. An electron moving towards the east enters a magnetic field directed towards the north. The force on the electron will be directed
(a) Vertically upward (b) Vertically downward
(c) Towards the west (d) Towards the south
79. An electron (mass = $9.0 \times 10^{-31}\text{ kg}$ and charge = $1.6 \times 10^{-19}\text{ coulomb}$) is moving in a circular orbit in a magnetic field of $1.0 \times 10^{-4}\text{ weber/m}^2$. Its period of revolution is
(a) $3.5 \times 10^{-7}\text{ sec}$ (b) $7.0 \times 10^{-7}\text{ sec}$
(c) $1.05 \times 10^{-6}\text{ sec}$ (d) $2.1 \times 10^{-6}\text{ sec}$
80. An electron (charge q coulomb) enters a magnetic field of H weber/m^2 with a velocity of $v\text{ m/s}$ in the same direction as that of the field the force on the electron is [MP PET 2000]
(a) Hqv Newton's in the direction of the magnetic field
(b) Hqv dynes in the direction of the magnetic field
(c) Hqv Newton's at right angles to the direction of the magnetic field
(d) Zero
81. A homogeneous electric field E and a uniform magnetic field $\vec{B}$ are pointing in the same direction. A proton is projected with its velocity parallel to $\vec{E}$. It will [Roorkee 2000]
(a) Go on moving in the same direction with increasing velocity
(b) Go on moving in the same direction with constant velocity
(c) Turn to its right
(d) Turn to its left
82. The radius of circular path of an electron when subjected to a perpendicular magnetic field is
[Pb. PMT 1999; DCE 2000; MH CET (Med.) 2000]
(a) $\frac{mv}{Be}$ (b) $\frac{me}{Be}$
(c) $\frac{mE}{Be}$ [RPMT 2000] (d) $\frac{Be}{mv}$
83. Cyclotron is used to accelerate [AIIMS 2001; BCECE 2004]
(a) Electrons (b) Neutrons
(c) Positive ions (d) Negative ions
84. Two particles A and B of masses m_A and m_B respectively and having the same charge are moving in a plane. A uniform magnetic field exists perpendicular to this plane. The speeds of the particles are v_A and v_B respectively, and the trajectories are as shown in the figure. Then [IIT-JEE (Screening) 2001]
(a) $m_A v_A < m_B v_B$
(b) $m_A v_A > m_B v_B$
(c) $m_A < m_B$ and $v_A < v_B$
(d) $m_A = m_B$ and $v_A = v_B$
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85. A proton and an alpha particle are separately projected in a region where a uniform magnetic field exists. Their initial velocities are perpendicular to direction of magnetic field. If both the particles move around magnetic field in circles of equal radii, the ratio of momentum of proton to alpha particle $\left(\frac{P_p}{P_\alpha}\right)$ is
(a) 1 (b) $\frac{1}{2}$
(c) 2 [MP PET 2000] (d) $\frac{1}{4}$
86. A particle of mass 0.6 g and having charge of 25 nC is moving horizontally with a uniform velocity $1.2 \times 10^4\text{ ms}^{-1}$ in a uniform magnetic field, then the value of the magnetic induction is ($g = 10\text{ ms}^{-2}$) [EAMCET 2001]
[MP PET 2000; Pb PET 2003]
(a) Zero (b) 10 T
(c) 20 T (d) 200 T
87. An α particle and a proton travel with same velocity in a magnetic field perpendicular to the direction of their velocities, find the ratio of the radii of their circular path
[AIIMS 2004; DCE 2001, 03; Kerala PMT 2005]
(a) 4 : 1 (b) 1 : 4
(c) 2 : 1 (d) 1 : 2
88. Motion of a moving electron is not affected by [AMU (Engg.) 2001]
(a) An electric field applied in the direction of motion
(b) Magnetic field applied in the direction of motion
(c) Electric field applied perpendicular to the direction of motion
(d) Magnetic field applied perpendicular to the direction of motion
89. When a charged particle enters a uniform magnetic field its kinetic energy [MP PMT 2001; MP PET 2002]
(a) Remains constant (b) Increases
(c) Decreases (d) Becomes zero
90. If cathode rays are projected at right angles to a magnetic field, their trajectory is [JIPMER 2002]
(a) Ellipse (b) Circle
(c) Parabola (d) None of these

91. At a specific instant emission of radioactive compound is deflected in a magnetic field. The compound can emit
(i) Electrons (ii) Protons
(iii) He^{2+} (iv) Neutrons
The emission at the instant can be [AIEEE 2002]
(a) i, ii, iii (b) i, ii, iii, iv
(c) iv (d) ii, iii
92. Which particles will have minimum frequency of revolution when projected with the same velocity perpendicular to a magnetic field
(a) Li (b) Electron
(c) Proton (d) He^+
93. Mixed He^+ and O^{2+} ions (mass of $He^+ = 4 amu$ and that of $O^{2+} = 16 amu$) beam passes a region of constant perpendicular magnetic field. If kinetic energy of all the ions is same then
(a) He^+ ions will be deflected more than those of O^{2+}
(b) He^+ ions will be deflected less than those of O^{2+}
(c) All the ions will be deflected equally
(d) No ions will be deflected
94. An electron (mass = $9 \times 10^{-31} kg$. Charge = $1.6 \times 10^{-19} C$) whose kinetic energy is $7.2 \times 10^{-18} joule$ is moving in a circular orbit in a magnetic field of $9 \times 10^{-4} weber/m$. The radius of the orbit is
(a) 1.25 cm (b) 2.5 cm
(c) 12.5 cm (d) 25.0 cm
95. An electron enters a region where electrostatic field is $20 N/C$ and magnetic field is $5 T$. If electron passes undeflected through the region, then velocity of electron will be [DPMPT 2002]
(a) $0.25 ms^{-1}$ (b) $2 ms^{-1}$
(c) $4 ms^{-1}$ (d) $8 ms^{-1}$
96. A charged particle is released from rest in a region of steady uniform electric and magnetic fields which are parallel to each other the particle will move in a [IIT-JEE 1999; DPMPT 2000; UPSEAT 2003]
(a) Straight line (b) Circle
(c) Helix (d) Cycloid
97. A particle of mass M and charge Q moving with velocity $\vec{v}$ describes a circular path of radius R when subjected to a uniform transverse magnetic field of induction B . The work done by the field when the particle completes one full circle is
(a) $BQv2\pi R$ (b) $\left(\frac{Mv^2}{R}\right)2\pi R$
(c) Zero (d) $BQ2\pi R$
98. A particle of charge $-16 \times 10^{-18} coulomb$ moving with velocity $10 ms^{-1}$ along the x -axis enters a region where a magnetic field of induction B is along the y -axis, and an electric field of magnitude $10^4 V/m$ is along the negative z -axis. If the charged particle continues moving along the x -axis, the magnitude of B is
(a) $10^{-3} Wb/m^2$ (b) $10^3 Wb/m^2$
(c) $10^5 Wb/m^2$ (d) $10^{16} Wb/m^2$
99. Two ions having masses in the ratio 1 : 1 and charges 1 : 2 are projected into uniform magnetic field perpendicular to the field with speeds in the ratio 2 : 3. The ratio of the radii of circular paths along which the two particles move is [EAMCET 2003]
(a) 4 : 3 (b) 2 : 3
(c) 3 : 1 (d) 1 : 4
100. An electron is travelling along the x -direction. It encounters a magnetic field in the y -direction. Its subsequent motion will be [Orissa JEE 2002]
(a) Straight line along the x -direction
(b) A circle in the xz -plane
(c) A circle in the yz -plane
(d) A circle in the xy -plane
101. An electron and a proton have equal kinetic energies. They enter in a magnetic field perpendicularly, Then [UPSEAT 2003]
(a) Both will follow a circular path with same radius
(b) Both will follow a helical path
(c) Both will follow a parabolic path
(d) All the statements are false
102. Electrons move at right angles to a magnetic field of $1.5 \times 10^{-2} Tesla$ with a speed of $6 \times 10^7 m/s$. If the specific charge of the electron is $1.7 \times 10^{11} C/kg$. The radius of the circular path will be [MP PMT 2002] [BHU 2003]
(a) 2.9 cm (b) 3.9 cm
(c) 2.35 cm (d) 3 cm
103. The cyclotron frequency of an electron grating in a magnetic field of 1 T is approximately [AIIMS 2004]
(a) 28 MHz (b) 280 MHz
(c) 2.8 GHz (d) 28 GHz
104. In the given figure, the electron enters into the magnetic field. It deflects in direction [Orissa PMT 2004]
(a) $+ve X$ direction
(b) $-ve X$ direction
(c) $+ve Y$ direction
(d) $-ve Y$ direction
105. A proton of energy 8 eV is moving in a circular path in a uniform magnetic field. The energy of an alpha particle moving in the same magnetic field and along the same path will be
(a) 4 eV (b) 2 eV
(c) 8 eV (d) 6 eV [AIEEE 2003]
106. An electron, a proton, a deuteron and an alpha particle, each having the same speed are in a region of constant magnetic field perpendicular to the direction of the velocities of the particles. The radius of the circular orbits of these particles are respectively R_e , R_p , R_d and R_α . It follows that
(a) $R_e = R_p$ (b) $R_p = R_d$
(c) $R_d = R_\alpha$ (d) $R_p = R_\alpha$
107. An electron moving with a uniform velocity along the positive x -direction enters a magnetic field directed along the positive y -direction. The force on the electron is directed along [AIEEE 2003]
(a) Positive y -direction (b) Negative y -direction



- (c) Positive z -direction (d) Negative z -direction
108. An electron is projected along the axis of a circular conductor carrying some current. Electron will experience force
(a) Along the axis
(b) Perpendicular to the axis
(c) At an angle of 45° with axis
(d) No force experienced
109. A very high magnetic field is applied to a stationary charge. Then the charge experiences [DCE 2004]
(a) A force in the direction of magnetic field
(b) A force perpendicular to the magnetic field
(c) A force in an arbitrary direction
(d) No force
110. A electron ($q = 1.6 \times 10^{-19} \text{ C}$) is moving at right angle to the uniform magnetic field $3.534 \times 10^{-3} \text{ T}$. The time taken by the electron to complete a circular orbit is [MH CET 2004]
(a) $2 \mu\text{s}$ (b) $4 \mu\text{s}$
(c) $3 \mu\text{s}$ (d) $1 \mu\text{s}$
111. In case Hall effect for a strip having charge Q and area of cross-section A , the Lorentz force is [DCE 2004]
(a) Directly proportional to Q
(b) Inversely proportional to Q
(c) Inversely proportional to A
(d) Directly proportional to A
112. A charged particle of mass m and charge q travels on a circular path of radius r that is perpendicular to a magnetic field B . The time taken by the particle to complete one revolution is
(a) $\frac{2\pi q B}{m}$ (b) $\frac{2\pi m}{q B}$
(c) $\frac{2\pi m q}{B}$ (d) $\frac{2\pi q^2 B}{m}$
113. A very long straight wire carries a current I . At the instant when a charge $+Q$ at point P has velocity $\vec{V}$, as shown, the force on the charge is [CBSE PMT 2005]

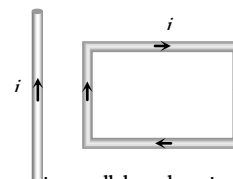


- (a) Opposite to OX (b) Along OX
(c) Opposite to OY (d) Along OY
114. The electron in the beam of a television tube move horizontally from south to north. The vertical component of the earth's magnetic field points down. The electron is deflected towards
(a) West (b) No deflection
(c) East (d) North to south
115. An electron moves in a circular orbit with a uniform speed v . It produces a magnetic field B at the centre of the circle. The radius of the circle is proportional to [CBSE PMT 2005]
(a) $\frac{B}{v}$ (b) $\frac{v}{B}$
(c) $\sqrt{\frac{v}{B}}$ (d) $\sqrt{\frac{B}{v}}$

116. An electric field of 1500 V/m and a magnetic field of 0.40 weber/meter act on a moving electron. The minimum uniform speed along a straight line the electron could have is [DCE 2002]
(a) $1.6 \times 10^{-6} \text{ m/s}$ (b) $6 \times 10^{-6} \text{ m/s}$
(c) $3.75 \times 10^{-6} \text{ m/s}$ (d) $3.75 \times 10^{-5} \text{ m/s}$
117. An electron (mass = $9.1 \times 10^{-31} \text{ kg}$; charge = $1.6 \times 10^{-19} \text{ C}$) experiences no deflection if subjected to an electric field of $3.2 \times 10^5 \text{ V/m}$, and a magnetic fields of $2.0 \times 10^{-3} \text{ Wb/m}$. Both the fields are normal to the path of electron and to each other. If the electric field is removed, then the electron will revolve in an orbit of radius [BCECE 2005]
(a) 45 m (b) 4.5 m
(c) 0.45 m (d) 0.045 m
118. An electron, moving in a uniform magnetic field of induction of intensity $\vec{B}$, has its radius directly proportional to [DPMT 2005]
(a) Its charge (b) Magnetic field
(c) Speed (d) None of these

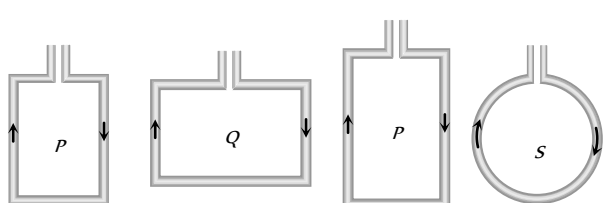
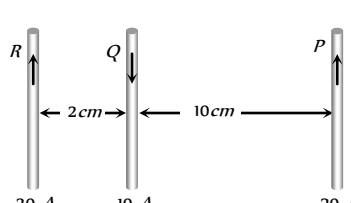
Force and Torque on a Current Carrying Conductor

1. Two free parallel wires carrying currents in opposite direction [CPMT 1977; MP PMT 1993; AFMC 2002; CPMT 2003]
(a) Attract each other
(b) Repel each other
(c) Neither attract nor repel
(d) Get rotated to be perpendicular to each other
2. A rectangular loop carrying a current i is situated near a long straight wire such that the wire is parallel to the one of the sides of the loop and is in the plane of the loop. If a steady current I is established in wire as shown in figure, the loop will [IIT 1985; MP PET 1995; MP PMT 1995, 99; AIIMS 2003]



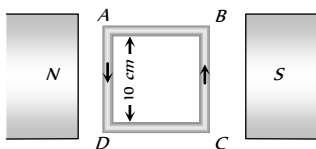
- (a) Rotate about an axis parallel to the wire
(b) Move away from the wire or towards right
(c) Move towards the wire
(d) Remain stationary
3. A circular coil of radius 4 cm and of 20 turns carries a current of 3 amperes . It is placed in a magnetic field of intensity of 0.5 weber/m^2 . The magnetic dipole moment of the coil is [IIT 1983; MP PMT 1999; AFMC 2002]
(a) 0.15 ampere-m^2 (b) 0.3 ampere-m^2
(c) 0.45 ampere-m^2 (d) 0.6 ampere-m^2
4. A conducting circular loop of radius r carries a constant current i . It is placed in a uniform magnetic field $\vec{B}$, such that $\vec{B}$ is perpendicular to the plane of the loop. The magnetic force acting on the loop is [BIT 1992; MP PET 1994; IIT 1983; MP PMT 1999; AMU (Engg.) 2000]
(a) $i r \vec{B}$ (b) $2\pi i \vec{B}$

- (c) Zero (d) $\pi i \vec{B}$
5. Two thin long parallel wires separated by a distance b are carrying a current i amp each. The magnitude of the force per unit length exerted by one wire on the other is
[CPMT 1991; IIT 1986; Bihar MEE 1995; RPMT 1997; MP PET 1996; MP PMT 1994, 96, 99; UPSEAT 2001, 03]
- (a) $\frac{\mu_0 i^2}{b^2}$ (b) $\frac{\mu_0 i^2}{2\pi b}$
(c) $\frac{\mu_0 i}{2\pi b}$ (d) $\frac{\mu_0 i}{2\pi b^2}$
6. Through two parallel wires A and B , 10 and 2 ampere of currents are passed respectively in opposite direction. If the wire A is infinitely long and the length of the wire B is 2 m, the force on the conductor B , which is situated at 10 cm distance from A will be [CPMT 1988; MP PMT 1994]
- (a) $8 \times 10^{-5} N$ (b) $4 \times 10^{-7} N$
(c) $4 \times 10^{-5} N$ (d) $4\pi \times 10^{-7} N$
7. If two streams of protons move parallel to each other in the same direction, then they [MP PET 1999; AIIMS 2004]
- (a) Do not exert any force on each other
(b) Repel each other
(c) Attract each other
(d) Get rotated to be perpendicular to each other
8. A straight wire carrying a current i_1 amp runs along the axis of a circular current i_2 amp. Then the force of interaction between the two current carrying conductors is
- (a) ∞ (b) Zero
(c) $\frac{\mu_0}{4\pi} \frac{2i_1 i_2}{r} N/m$ (d) $\frac{2i_1 i_2}{r} N/m$
9. Two parallel wires are carrying electric currents of equal magnitude and in the same direction. They exert
[CPMT 1990; MP PET/PMT 1988; Orissa JEE 2003; AFMC 2003]
- (a) An attractive force on each other
(b) A repulsive force on each other
(c) No force on each other
(d) A rotational torque on each other
10. Two long and parallel wires are at a distance of 0.1 m and a current of 5 A is flowing in each of these wires. The force per unit length due to these wires will be [CPMT 1977]
- (a) $5 \times 10^{-5} N/m$ (b) $5 \times 10^{-3} N/m$
(c) $2.5 \times 10^{-5} N/m$ (d) $2.5 \times 10^{-4} N/m$
11. Two straight parallel wires, both carrying 10 ampere in the same direction attract each other with a force of $1 \times 10^{-3} N$. If both currents are doubled, the force of attraction will be [MP PET 1994]
- (a) $1 \times 10^{-3} N$ (b) $2 \times 10^{-3} N$
(c) $4 \times 10^{-3} N$ (d) $0.25 \times 10^{-3} N$
12. A circular coil of radius 4 cm has 50 turns. In this coil a current of 2 A is flowing. It is placed in a magnetic field of 0.1 weber / m². The amount of work done in rotating it through 180° from its equilibrium position will be [CPMT 1977]
- (a) 0.1 J (b) 0.2 J
(c) 0.4 J (d) 0.8 J
13. 3 A of current is flowing in a linear conductor having a length of 40 cm. The conductor is placed in a magnetic field of strength 500 gauss and makes an angle of 30° with the direction of the field. It experiences a force of magnitude [MP PET 1993]
- (a) 3×10^4 newton (b) 3×10^2 newton
(c) 3×10^{-2} newton (d) 3×10^{-4} newton
14. The radius of a circular loop is r and a current i is flowing in it. The equivalent magnetic moment will be [CPMT 1990]
- (a) ir (b) $2\pi ir$
(c) $i\pi r^2$ (d) $\frac{1}{r^2}$
15. A current carrying loop is placed in a uniform magnetic field. The torque acting on it does not depend upon [CPMT 1985; RPMT 1997; Kerala PMT 2002]
- (a) Shape of the loop (b) Area of the loop
(c) Value of the current (d) Magnetic field
16. To make the field radial in a moving coil galvanometer [MP PET 1993]
- (a) The number of turns in the coil is increased
(b) Magnet is taken in the form of horse-shoe
(c) Poles are cylindrically cut
(d) Coil is wound on aluminium frame
17. The deflection in a moving coil galvanometer is [MP PMT 1993]
- (a) Directly proportional to the torsional constant
(b) Directly proportional to the number of turns in the coil
(c) Inversely proportional to the area of the coil
(d) Inversely proportional to the current flowing
18. A moving coil sensitive galvanometer gives at once much more deflection. To control its speed of deflection [MP PET 1985]
- (a) A high resistance is to be connected across its terminals
(b) A magnet should be placed near the coil
(c) A small copper wire should be connected across its terminals
(d) The body of galvanometer should be earthed
19. In a moving coil galvanometer, the deflection of the coil θ is related to the electrical current i by the relation [MP PMT 1996, 2000, 03; RPMT 1997; CPMT 1975; MP PET 1999]
- (a) $i \propto \tan \theta$ (b) $i \propto \theta$
(c) $i \propto \theta^2$ (d) $i \propto \sqrt{\theta}$
20. The unit of electric current "ampere" is the current which when flowing through each of two parallel wires spaced 1 m apart in vacuum and of infinite length will give rise to a force between them equal to [BIT 1987; CBSE PMT 1998; MP PET 1999; MP PMT 2002]
- (a) 1 N / m (b) $2 \times 10^{-7} N/m$

- (c) $1 \times 10^{-2} \text{ N/m}$ (d) $4\pi \times 10^{-7} \text{ N/m}$
21. A moving coil galvanometer has N number of turns in a coil of effective area A , it carries a current I . The magnetic field B is radial. The torque acting on the coil is [MP PMT 1994]
- (a) NA^2B^2I (b) $NAB I^2$
(c) N^2ABI (d) $NABI$
22. A small coil of N turns has an effective area A and carries a current I . It is suspended in a horizontal magnetic field $\vec{B}$ such that its plane is perpendicular to $\vec{B}$. The work done in rotating it by 180° about the vertical axis is [MP PMT 1994]
- (a) $NAIB$ (b) $2NAIB$
(c) $2\pi NAIB$ (d) $4\pi NAIB$
23. A small coil of N turns has area A and a current I flows through it. The magnetic dipole moment of this coil will be [MP PMT 1994]
- (a) NI/A (b) NI^2A
(c) N^2AI (d) NAI
24. A current of 10 ampere is flowing in a wire of length 1.5 m. A force of 15 N acts on it when it is placed in a uniform magnetic field of 2 tesla. The angle between the magnetic field and the direction of the current is [MP PMT 1994]
- (a) 30° (b) 45°
(c) 60° (d) 90°
25. A rectangular loop carrying a current i is placed in a uniform magnetic field B . The area enclosed by the loop is A . If there are n turns in the loop, the torque acting on the loop is given by
- (a) $ni\vec{A} \times \vec{B}$ (b) $ni\vec{A} \cdot \vec{B}$
(c) $\frac{1}{n}(i\vec{A} \times \vec{B})$ (d) $\frac{1}{n}(i\vec{A} \cdot \vec{B})$
26. An electron moves with a constant speed v along a circle of radius r . Its magnetic moment will be (e is the electron's charge)
- (a) evr (b) $\frac{1}{2}evr$
(c) $\pi^2 ev$ (d) $2\pi ev$
27. Four wires each of length 2.0 metres are bent into four loops P , Q , R and S and then suspended into uniform magnetic field. Same current is passed in each loop. Which statement is correct
- 
- (a) Couple on loop P will be the highest
(b) Couple on loop Q will be the highest
(c) Couple on loop R will be the highest
(d) Couple on loop S will be the highest
28. A current carrying rectangular coil is placed in a uniform magnetic field. In which orientation, the coil will not tend to rotate
- (a) The magnetic field is parallel to the plane of the coil
(b) The magnetic field is perpendicular to the plane of the coil
(c) The magnetic field is at 45° with the plane of the coil
(d) Always in any orientation
29. A current carrying circular loop is freely suspended by a long thread. The plane of the loop will point in the direction [MP PMT 1995]
- (a) Wherever left free
(b) North-south
(c) East-west
(d) At 45° with the east-west direction
30. A current carrying loop is free to turn in a uniform magnetic field. The loop will then come into equilibrium when its plane is inclined at [CBSE PMT 1992; Haryana CEE 1996]
- (a) 0° to the direction of the field
(b) 45° to the direction of the field
(c) 90° to the direction of the field
(d) 135° to the direction of the field
31. The expression for the torque acting on a coil having area of cross-section A , number of turns n , placed in a magnetic field of strength B , making an angle θ with the normal to the plane of the coil, when a current i is flowing in it, will be [MP PET 1996]
- (a) $niAB \tan \theta$ (b) $niAB \cos \theta$
(c) $niAB \sin \theta$ (d) $niAB$
32. The pole pieces of the magnet used in a pivoted coil galvanometer are [MP PET 1996]
- (a) Plane surfaces of a bar magnet
(b) Plane surfaces of a horse-shoe magnet [MP PMT 1994]
(c) Cylindrical surfaces of a bar magnet
(d) Cylindrical surfaces of a horse-shoe magnet
33. The sensitiveness of a moving coil galvanometer can be increased by decreasing [MP PMT 1996]
- (a) The number of turns in the coil [MP PMT 1994]
(b) The area of the coil
(c) The magnetic field
(d) The couple per unit twist of the suspension
34. A metallic loop is placed in a magnetic field. If a current is passed through it, then [UPSEAT 2003]
- [MP PET 1995; DPMT 1999]
- (a) The ring will feel a force of attraction
(b) The ring will feel a force of repulsion
(c) It will move to and fro about its centre of gravity
(d) None of these
35. Two parallel conductors A and B of equal lengths carry currents I and $10 I$, respectively, in the same direction. Then [MP PET 2003]
- (a) A and B will repel each other with same force
(b) A and B will attract each other with same force
(c) A will attract B , but B will repel A
(d) A and B will attract each other with different forces
36. Three long, straight and parallel wires carrying currents are arranged as shown in figure. The force experienced by 10 cm length of wire Q is [MP PET 1997]
- 

- (a) $1.4 \times 10^{-4} \text{ N}$ towards the right
(b) $1.4 \times 10^{-4} \text{ N}$ towards the left
(c) $2.6 \times 10^{-4} \text{ N}$ to the right
(d) $2.6 \times 10^{-4} \text{ N}$ to the left

37. A 100 turns coil shown in figure carries a current of 2 amp in a magnetic field $B = 0.2 \text{ Wb/m}^2$. The torque acting on the coil is



- (a) 0.32 Nm tending to rotate the side AD out of the page
(b) 0.32 Nm tending to rotate the side AD into the page
(c) 0.0032 Nm tending to rotate the side AD out of the page
(d) 0.0032 Nm tending to rotate the side AD into the page
38. A current of 5 ampere is flowing in a wire of length 1.5 metres. A force of 7.5 N acts on it when it is placed in a uniform magnetic field of 2 Tesla. The angle between the magnetic field and the direction of the current is

[MP PET 1997; Pb. PET 2003]

- (a) 30° (b) 45°
(c) 60° (d) 90°
39. A conductor in the form of a right angle ABC with $AB = 3 \text{ cm}$ and $BC = 4 \text{ cm}$ carries a current of 10 A. There is a uniform magnetic field of 5 T perpendicular to the plane of the conductor. The force on the conductor will be

[MP PMT 1997]

- (a) 1.5 N (b) 2.0 N
(c) 2.5 N (d) 3.5 N
40. The coil of a galvanometer consists of 100 turns and effective area of 1 square cm. The restoring couple is $10^{-8} \text{ N-m/radian}$. The magnetic field between the pole pieces is 5 T. The current sensitivity of this galvanometer will be
- (a) $5 \times 10^4 \text{ rad/}\mu\text{amp}$ (b) $5 \times 10^{-6} \text{ per amp}$
(c) $2 \times 10^{-7} \text{ per amp}$ (d) $5 \text{ rad/}\mu\text{amp}$
41. A rectangular coil $20 \text{ cm} \times 20 \text{ cm}$ has 100 turns and carries a current of 1 A. It is placed in a uniform magnetic field $B = 0.5 \text{ T}$ with the direction of magnetic field parallel to the plane of the coil. The magnitude of the torque required to hold this coil in this position is

[MP PMT 1997]

- (a) Zero (b) 200 N-m
(c) 2 N-m (d) 10 N-m
42. If a current is passed in a spring, it

[MP PMT/PET 1998; AIEEE 2002]

- (a) Gets compressed (b) Gets expanded
(c) Oscillates (d) Remains unchanged

43. A current carrying small loop behaves like a small magnet. If A be its area and M its magnetic moment, the current in the loop will be

[MP PMT/PET 1998; RPET 2001; MP PMT 2003]

- (a) M/A (b) A/M
(c) MA (d) A^2M

44. In hydrogen atom, the electron is making $6.6 \times 10^{15} \text{ rev/sec}$ around the nucleus in an orbit of radius $0.528 \text{ \AA}$. The magnetic moment ($A - m^2$) will be

[MP PET 1999]

- (a) 1×10^{-23} (b) 1×10^{-10}
(c) 1×10^{-23} (d) 1×10^{-27}

45. A triangular loop of side l carries a current I. It is placed in a magnetic field B such that the plane of the loop is in the direction of B. The torque on the loop is

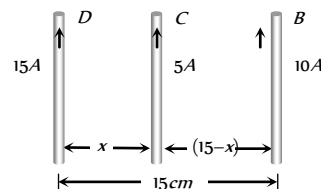
[MP PET 2003]

- (a) Zero (b) IBl
(c) $\frac{\sqrt{3}}{2} I^2 B^2$ (d) $\frac{\sqrt{3}}{4} IBl^2$

46. Three long, straight and parallel wires carrying currents are arranged as shown in the figure. The wire C which carries a current of 5.0 amp is so placed that it experiences no force. The distance of wire C from wire D is then

[AMU 1995]

- (a) 9 cm
(b) 7 cm
(c) 5 cm
(d) 3 cm



47. A vertical wire carrying a current in the upward direction is placed in horizontal magnetic field directed towards north. The wire will experience a force directed towards

[SCRA 1994]

- (a) North (b) South
(c) East (d) West

48. A coil carrying electric current is placed in uniform magnetic field, then

[CBSE PMT 1993]

- (a) Torque is formed
(b) E.M.F. is induced
(c) Both (a) and (b) are correct
(d) None of these

49. A circular loop carrying a current is replaced by an equivalent magnetic dipole. A point on the axis of the loop is

- (a) An end-on position (b) A broad side-on position
(c) Both (a) and (b) (d) Neither (a) nor (b)

50. A power line lies along the east-west direction and carries a current of 10 ampere. The force per metre due to the earth's magnetic field of 10^{-4} tesla is

[Roorkee 1992]

- (a) 10^{-5} N (b) 10^{-4} N
(c) 10^{-3} N (d) 10^{-2} N

51. A straight wire of length 0.5 metre and carrying a current of 1.2 ampere placed in a uniform magnetic field of induction 2 Tesla. The

magnetic field is perpendicular to the length of the wire. The force on the wire is

[CBSE PMT 1992; BHU 1998; DPMT 2001; RPET 2003]

- (a) 2.4 N (b) 1.2 N
(c) 3.0 N (d) 2.0 N

52. Two parallel wires in free space are 10 cm apart and each carries a current of 10 A in the same direction. The force one wire exerts on the other per metre of length is

[CBSE PMT 1997; AFMC 1999]

- (a) $2 \times 10^{-4} \text{ N}$, attractive (b) $2 \times 10^{-4} \text{ N}$, repulsive
(c) $2 \times 10^{-7} \text{ N}$, attractive (d) $2 \times 10^{-7} \text{ N}$, repulsive

53. The current sensitivity of a moving coil galvanometer can be increased by [Roorkee 1999]

- (a) Increasing the magnetic field of the permanent magnet
(b) Increasing the area of the deflecting coil
(c) Increasing the number of turns in the coil
(d) Increasing the restoring couple of the coil

54. A circular coil of diameter 7 cm has 24 turns of wire carrying current of 0.75 A . The magnetic moment of the coil is

[AMU (Med.) 1999]

- (a) $6.9 \times 10^{-2} \text{ amp-m}^2$ (b) $2.3 \times 10^{-2} \text{ amp-m}^2$
(c) 10^{-2} amp-m^2 (d) 10^{-3} amp-m^2

55. Two long parallel wires carrying equal current separated by 1 m , exert a force of $2 \times 10^{-7} \text{ N/m}$ on one another. The current flowing through them is [AMU (Engg.) 1999]

- (a) 2.0 A (b) $2.0 \times 10^{-7} \text{ A}$
(c) 1.0 A (d) $1.0 \times 10^{-7} \text{ A}$

56. Two parallel beams of electrons moving in the same direction produce a mutual force [MP PET 1996; DCE 1999]

- (a) Of attraction in plane of paper
(b) Of repulsion in plane of paper
(c) Upwards perpendicular to plane of paper
(d) Downwards perpendicular to plane of paper

57. A circular loop of area 0.01 m^2 carrying a current of 10 A , is held perpendicular to a magnetic field of intensity 0.1 T . The torque acting on the loop is [Pb. PMT 2000]

- (a) Zero (b) 0.01 N-m
(c) 0.001 N-m (d) 0.8 N-m

58. Magnetic dipole moment of a rectangular loop is

[RPET 2000]

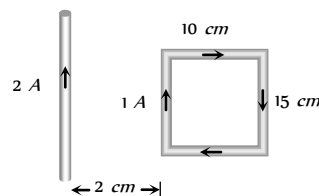
- (a) Inversely proportional to current in loop
(b) Inversely proportional to area of loop
(c) Parallel to plane of loop and proportional to area of loop
(d) Perpendicular to plane of loop and proportional to area of loop

59. If m is magnetic moment and B is the magnetic field, then the torque is given by [DCE 2000]

- (a) $\vec{m} \cdot \vec{B}$ (b) $\frac{|\vec{m}|}{|\vec{B}|}$
(c) $\vec{m} \times \vec{B}$ (d) $|\vec{m}| \cdot |\vec{B}|$

60. What is the net force on the square coil

[DCE 2000; RPMT 2000]



- (a) $25 \times 10^{-7} \text{ N}$ moving towards wire
(b) $25 \times 10^{-7} \text{ N}$ moving away from wire
(c) $35 \times 10^{-7} \text{ N}$ moving towards wire
(d) $35 \times 10^{-7} \text{ N}$ moving away from wire

61. Two long parallel copper wires carry currents of 5 A each in opposite directions. If the wires are separated by a distance of 0.5 m , then the force between the two wires is

[EAMCET (Engg.) 2000]

- (a) 10^{-5} N , attractive (b) 10^{-5} N , repulsive
(c) $2 \times 10^{-5} \text{ N}$, attractive (d) $2 \times 10^{-5} \text{ N}$, repulsive

62. In order to increase the sensitivity of a moving coil galvanometer, one should decrease [MP PMT 2000]

- (a) The strength of its magnet
(b) The torsional constant of its suspension
(c) The number of turns in its coil
(d) The area of its coil

63. A circular loop has a radius of 5 cm and it is carrying a current of 0.1 amp . Its magnetic moment is [MP PMT 2000]

- (a) $1.32 \times 10^{-4} \text{ amp-m}^2$
(b) $2.62 \times 10^{-4} \text{ amp-m}^2$
(c) $5.25 \times 10^{-4} \text{ amp-m}^2$
(d) $7.85 \times 10^{-4} \text{ amp-m}^2$

64. Due to the flow of current in a circular loop of radius R , the magnetic induction produced at the centre of the loop is B . The magnetic moment of the loop is

(μ_0 = permeability constant)

[MP PET 2000]

- (a) $BR^3 / 2\pi\mu_0$ (b) $2\pi BR^3 / \mu_0$
(c) $BR^2 / 2\pi\mu_0$ (d) $2\pi BR^2 / \mu_0$

65. The magnetic moment of a circular coil carrying current is

[MP PET 2000]

- (a) Directly proportional to the length of the wire in the coil
(b) Inversely proportional to the length of the wire in the coil
(c) Directly proportional to the square of the length of the wire in the coil
(d) Inversely proportional to the square of the length of the wire in the coil

66. A long wire A carries a current of 10 amp . Another long wire B , Which is parallel to A and separated by 0.1 m from A , carries a current of 5 amp , in the opposite direction to that in A . what is the magnitude and nature of the force experienced per unit length of B

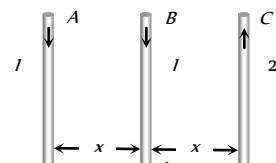
($\mu_0 = 4\pi \times 10^{-7} \text{ weber / amp-m}$)

[MP PET 2000]

- (a) Repulsive force of 10^{-4} N/m
 (b) Attractive force of 10^{-4} N/m
 (c) Repulsive force of $2\pi \times 10^{-5} \text{ N/m}$
 (d) Attractive force of $2\pi \times 10^{-5} \text{ N/m}$
67. A stream of electrons is projected horizontally to the right. A straight conductor carrying a current is supported parallel to electron stream and above it. If the current in the conductor is from left to right then what will be the effect on electron stream
 (a) The electron stream will be pulled upward
 (b) The electron stream will be pulled downward
 (c) The electron stream will be retarded
 (d) The electron beam will be speeded up towards the right
68. The relation between voltage sensitivity (σ_V) and current sensitivity (σ_i) of a moving coil galvanometer is (Resistance of Galvanometer = G) [CPMT 2001]
 (a) $\frac{\sigma_i}{G} = \sigma_V$ (b) $\frac{\sigma_V}{G} = \sigma_i$
 (c) $\frac{G}{\sigma_V} = \sigma_i$ (d) $\frac{G}{\sigma_i} = \sigma_V$
69. What is shape of magnet in moving coil galvanometer to make the radial magnetic field [RPET 2001]
 (a) Concave (b) Horse shoe magnet
 (c) Convex (d) None of these
70. If a wire of length 1 meter placed in uniform magnetic field 1.5 Tesla at angle 30° with magnetic field. The current in a wire 10 amp. Then force on a wire will be [RPET 2001]
 (a) 7.5 N (b) 1.5 N
 (c) 0.5 N (d) 2.5 N
71. A current i flows in a circular coil of radius r . If the coil is placed in a uniform magnetic field B with its plane parallel to the field, magnitude of the torque that acts on the coil is [MP PET 2001]
 (a) Zero (b) $2\pi r i B$
 (c) $\pi r^2 i B$ (d) $2\pi r^2 i B$
72. An arbitrary shaped closed coil is made of a wire of length L and a current I ampere is flowing in it. If the plane of the coil is perpendicular to magnetic field $\vec{B}$, the force on the coil is [MP PMT 2001]
 (a) Zero (b) IBL
 (c) $2IBL$ (d) $\frac{1}{2}IBL$
73. A circular coil having N turns is made from a wire of length L meter. If a current I ampere is passed through it and is placed in a magnetic field of B Tesla, the maximum torque on it is
 (a) Directly proportional to N
 (b) Inversely proportional to N
 (c) Inversely proportional to N^2
 (d) Independent of N
74. A small cylindrical soft iron piece is kept in a galvanometer so that
 (a) A radial uniform magnetic field is produced
 (b) A uniform magnetic field is produced
 (c) There is a steady deflection of the coil
 (d) All of these
75. A , B and C are parallel conductors of equal length carrying currents I , I and $2I$ respectively. Distance between A and B is x . Distance

between B and C is also x . F_1 is the force exerted by B on A and F_2 is the force exerted by B on C choose the correct answer

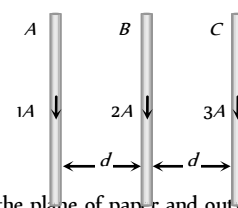
- (a) $F_1 = 2F_2$
 (b) $F_2 = 2F_1$
 (c) $F_1 = F_2$
 (d) $F_1 = -F_2$



76. A straight conductor carries a current of 5A. An electron travelling with a speed of $9 \times 10^6 \text{ ms}^{-1}$ parallel to the wire at a distance of 0.1m from the conductor, experiences a force of [Jee 2000]
 (a) $8 \times 10^{-20} \text{ N}$ (b) $3.2 \times 10^{-19} \text{ N}$
 (c) $8 \times 10^{-18} \text{ N}$ (d) $1.6 \times 10^{-19} \text{ N}$
77. Two galvanometers A and B require 3mA and 5mA respectively to produce the same deflection of 10 divisions. Then
 (a) A is more sensitive than B
 (b) B is more sensitive than A
 (c) A and B are equally sensitive
 (d) Sensitiveness of B is $5/3$ times that of A
78. Two long straight parallel conductors separated by a distance of 0.5m carry currents of 5A and 8A in the same direction. The force per unit length experienced by each other is
 (a) $1.6 \times 10^{-5} \text{ N}$ (attractive) (b) $1.6 \times 10^{-5} \text{ N}$ (repulsive)
 (c) $16 \times 10^{-5} \text{ N}$ (attractive) (d) $16 \times 10^{-5} \text{ N}$ (repulsive)
79. If the current is doubled, the deflection is also doubled in [Orissa JEE 2002]

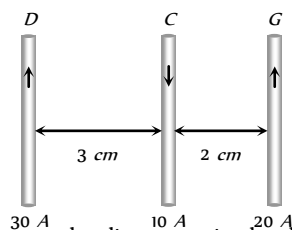
- (a) A tangent galvanometer
 (b) A moving coil galvanometer
 (c) Both (a) and (b)
 (d) None of these

80. Which is a vector quantity [AFMC 2003]
 (a) Density (b) Magnetic flux
 (c) Intensity of magnetic field (d) Magnetic potential
81. Three long straight wires A , B and C are carrying current as shown figure. Then the resultant force on B is directed [KCET 2004]



- (a) Towards A
 (b) Towards C
 (c) Perpendicular to the plane of paper and outward
 (d) Perpendicular to the plane of paper and inward
82. Two long conductors, separated by a distance d carry current I and I in the same direction. They exert a force F on each other. Now the current in one of them is increased to two times and its directions is reversed. The distance is also increased to $3d$. The new value of the force between them is [AIEEE 2004]
 (a) $-2F$ (b) $F/3$
 (c) $2F$ (d) $-F/3$
83. The resultant magnetic moment of neon atom will be [J & K CET 2004]
 (a) Infinity (b) μ
 (c) Zero (d) $\mu/2$

84. A one metre long wire is lying at right angles to the magnetic field. A force of 1 kg wt. is acting on it in a magnetic field of 0.98 Tesla . The current flowing in it will be [J & K CET 2004]
- (a) 100 A (b) 10 A
(c) 1 A (d) Zero
85. A beam of electrons and protons move parallel to each other in the same direction, then they [DCE 2004]
- (a) Attract each other (b) Repel each other
(c) No relation (d) Neither attract nor repel
86. Two parallel wires of length 9 m each are separated by a distance 0.15 m . If they carry equal currents in the same direction and exerts a total force of $30 \times 10^{-7} \text{ N}$ on each other, then the value of current must be [MH CET 2003]
- (a) 2.5 amp (b) 3.5 amp
(c) 1.5 amp (d) 0.5 amp
87. Current i is carried in a wire of length L . If the wire is turned into a circular coil, the maximum magnitude of torque in a given magnetic field B will be [Pb. PET 2004]
- (a) $\frac{LiB^2}{2}$ (b) $\frac{Li^2B}{2}$
(c) $\frac{L^2iB}{4\pi}$ (d) $\frac{Li^2B}{4\pi}$
88. In ballistic galvanometer, the frame on which the coil is wound is non-metallic to [MH CET 2004]
- (a) Avoid the production of induced e.m.f.
(b) Avoid the production of eddy currents
(c) Increase the production of eddy currents
(d) Increase the production of induced e.m.f.
89. Two thin, long, parallel wires, separated by a distance ' d ' carry a current of ' i ' A in the same direction. They will [AIEEE 2005]
- (a) Attract each other with a force of $\mu_0 i^2 / (2\pi d^2)$
(b) Repel each other with a force of $\mu_0 i^2 / (2\pi d^2)$
(c) Attract each other with a force of $\mu_0 i^2 / (2\pi d)$
(d) Repel each other with a force of $\mu_0 i^2 / (2\pi d)$
90. Three long, straight parallel wires carrying current, are arranged as shown in figure. The force experienced by a 25 cm length of wire C is [KCET 2005]



- (a) 10^{-7} N
(b) $2.5 \times 10^{-7} \text{ N}$
(c) Zero
(d) $1.5 \times 10^{-7} \text{ N}$
91. A circular coil of 20 turns and radius 10 cm is placed in uniform magnetic field of 0.10 T normal to the plane of the coil. If the current in coil is 5 A , then the torque acting on the coil will be [J & K CET 2005]
- (a) 31.4 Nm (b) 3.14 Nm
(c) 0.314 Nm (d) Zero

Critical Thinking

Objective Questions

1. A circular current carrying coil has a radius R . The distance from the centre of the coil on the axis where the magnetic induction will be $\frac{1}{8}$ th to its value at the centre of the coil, is

[MP PMT 1997]

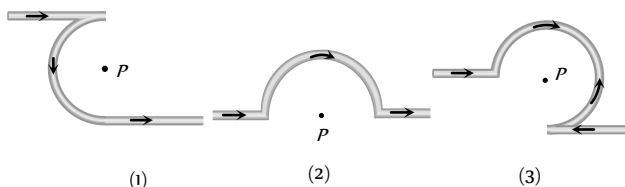
- (a) $\frac{R}{\sqrt{3}}$ (b) $R\sqrt{3}$
(c) $2\sqrt{3}R$ (d) $\frac{2}{\sqrt{3}}R$

2. The field normal to the plane of a wire of n turns and radius r which carries a current i is measured on the axis of the coil at a small distance h from the centre of the coil. This is smaller than the field at the centre by the fraction

- (a) $\frac{3}{2} \frac{h^2}{r^2}$ (b) $\frac{2}{3} \frac{h^2}{r^2}$
(c) $\frac{3}{2} \frac{r^2}{h^2}$ (d) $\frac{2}{3} \frac{r^2}{h^2}$

3. The magnetic field at the centre of a circular coil of radius r is π times that due to a long straight wire at a distance r from it, for equal currents. Figure here shows three cases : in all cases the circular part has radius r and straight ones are infinitely long. For same current the B field at the centre P in cases 1, 2, 3 have the ratio

[CPMT 1989]



- (a) $\left(-\frac{\pi}{2}\right) : \left(\frac{\pi}{2}\right) : \left(\frac{3\pi}{4} - \frac{1}{2}\right)$
(b) $\left(-\frac{\pi}{2} + 1\right) : \left(\frac{\pi}{2} + 1\right) : \left(\frac{3\pi}{4} + \frac{1}{2}\right)$
(c) $-\frac{\pi}{2} : \frac{\pi}{2} : 3\frac{\pi}{4}$
(d) $\left(-\frac{\pi}{2} - 1\right) : \left(\frac{\pi}{2} - \frac{1}{4}\right) : \left(\frac{3\pi}{4} + \frac{1}{2}\right)$

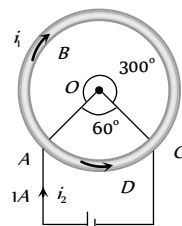
4. Two straight long conductors AOB and COD are perpendicular to each other and carry currents i_1 and i_2 . The magnitude of the magnetic induction at a point P at a distance a from the point O in a direction perpendicular to the plane $ACBD$ is

- (a) $\frac{\mu_0}{2\pi a}(i_1 + i_2)$ (b) $\frac{\mu_0}{2\pi a}(i_1 - i_2)$
(c) $\frac{\mu_0}{2\pi a}(i_1^2 + i_2^2)^{1/2}$ (d) $\frac{\mu_0}{2\pi a} \frac{i_1 i_2}{(i_1 + i_2)}$

5. A cell is connected between the points A and C of a circular conductor $ABCD$ of centre O with angle $AOC = 60^\circ$ If B_1 and

B_2 are the magnitudes of the magnetic fields at O due to the currents in ABC and ADC respectively, the ratio $\frac{B_1}{B_2}$ is

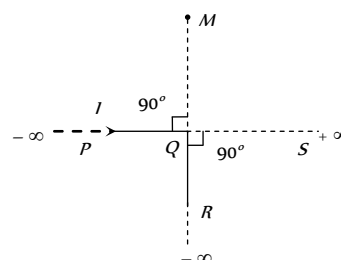
- (a) 0.2
(b) 6
(c) 1
(d) 5



6. An infinitely long conductor PQR is bent to form a right angle as shown. A current I flows through PQR . The magnetic field due to this current at the point M is H_1 . Now another infinitely long straight conductor QS is connected at Q so that the current is $I/2$ in QR as well as in QS . The current in PQ remaining unchanged. The magnetic field at M is now H_2 . The ratio H_1/H_2 is given by

[IIT-JEE (Screening) 2000]

- (a) $\frac{1}{2}$
(b) 1
(c) $\frac{2}{3}$
(d) 2



7. Two coaxial solenoids 1 and 2 of the same length are set so that one is inside the other. The number of turns per unit length are n_1 and n_2 . The currents i_1 and i_2 are flowing in opposite directions. The magnetic field inside the inner coil is zero. This is possible when

- (a) $i_1 \neq i_2$ and $n_1 = n_2$
(b) $i_1 = i_2$ and $n_1 \neq n_2$
(c) $i_1 = i_2$ and $n_1 = n_2$
(d) $i_1 n_1 = i_2 n_2$

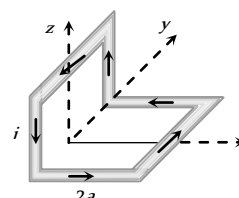
8. A coil having N turns is wound tightly in the form of a spiral with inner and outer radii a and b respectively. When a current I passes through the coil, the magnetic field at the centre is

- (a) $\frac{\mu_0 NI}{b}$ (b) $\frac{2\mu_0 NI}{a}$
(c) $\frac{\mu_0 NI}{2(b-a)} \ln \frac{b}{a}$ (d) $\frac{\mu_0 I^N}{2(b-a)} \ln \frac{b}{a}$

9. A non-uniformly conducting wire carrying a current I is placed as shown in the figure. Each of the straight sections of the loop is of length $2a$. The magnetic field due to this loop at the point $P(a, 0, a)$ points in the direction

[IIT-JEE (Screening) 2001]

- (a) $\frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$
(b) $\frac{1}{\sqrt{3}}(-\hat{j} + \hat{k} + \hat{i})$



(c) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$

(d) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{k})$

10. A long straight wire along the z -axis carries a current I in the negative z direction. The magnetic vector field $\vec{B}$ at a point having coordinates (x, y) in the $z = 0$ plane is

[IIT-JEE (Screening) 2002]

(a) $\frac{\mu_0 I(y\hat{i} - x\hat{j})}{2\pi(x^2 + y^2)}$

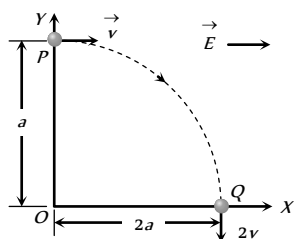
(b) $\frac{\mu_0 I(x\hat{i} + y\hat{j})}{2\pi(x^2 + y^2)}$

(c) $\frac{\mu_0 I(x\hat{j} - y\hat{i})}{2\pi(x^2 + y^2)}$

(d) $\frac{\mu_0 I(x\hat{i} - y\hat{j})}{2\pi(x^2 + y^2)}$

11. A particle of charge $+q$ and mass m moving under the influence of a uniform electric field $E\hat{i}$ and a uniform magnetic field $B\hat{k}$ follows trajectory from P to Q as shown in figure. The velocities at P and Q are $v\hat{i}$ and $-2v\hat{j}$ respectively. Which of the following statement(s) is/are correct

[IIT 1991; BVP 2003]



(a) $E = \frac{3}{4} \frac{mv^2}{qa}$

(b) Rate of work done by electric field at P is $\frac{3}{4} \frac{mv^3}{a}$

(c) Rate of work done by electric field at P is zero

(d) Rate of work done by both the fields at Q is zero

12. H^+ , He^+ and O^{++} ions having same kinetic energy pass through a region of space filled with uniform magnetic field B directed perpendicular to the velocity of ions. The masses of the ions H^+ , He^+ and O^{++} are respectively in the ratio $1:4:16$. As a result

(a) H^+ ions will be deflected most

(b) O^{++} ions will be deflected least

(c) He^+ and O^{++} ions will suffer same deflection

(d) All ions will suffer the same deflection

13. An ionized gas contains both positive and negative ions. If it is subjected simultaneously to an electric field along the $+x$ direction and a magnetic field along the $+z$ direction, then

[IIT-JEE (Screening) 2000]

(a) Positive ions deflect towards $+y$ direction and negative ions towards $-y$ direction

(b) All ions deflect towards $+y$ direction

(c) All ions deflect towards $-y$ direction

(d) Positive ions deflect towards $-y$ direction and negative ions towards $+y$ direction

14. An electron moves with speed 2×10^5 m/s along the positive x -direction in the presence of a magnetic induction

$B = \hat{i} + 4\hat{j} - 3\hat{k}$ (in Tesla.) The magnitude of the force experienced by the electron in Newton's is (charge on the electron $= 1.6 \times 10^{-19}$ C)

[EAMCET 2001]

(a) 1.18×10^{-13}

(b) 1.28×10^{-13}

(c) 1.6×10^{-13}

(d) 1.72×10^{-13}

15. A particle of mass m and charge q moves with a constant velocity v along the positive x direction. It enters a region containing a uniform magnetic field B directed along the negative z direction, extending from $x = a$ to $x = b$. The minimum value of v required so that the particle can just enter the region $x > b$ is

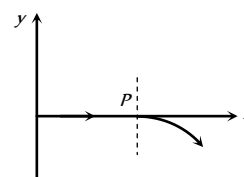
(a) qbB/m

(b) $q(b-a)B/m$

(c) qaB/m

(d) $q(b+a)B/2m$

16. For a positively charged particle moving in a x - y plane initially along the x -axis, there is a sudden change in its path due to the presence of electric and/or magnetic fields beyond P . The curved path is shown in the x - y plane and is found to be non-circular. Which one of the following combinations is possible



(a) $\vec{E} = 0; \vec{B} = b\hat{i} + c\hat{k}$

(b) $\vec{E} = a\hat{i}; \vec{B} = c\hat{k} + a\hat{i}$

(c) $\vec{E} = 0; \vec{B} = c\hat{j} + b\hat{k}$

(d) $\vec{E} = a\hat{i}; \vec{B} = c\hat{k} + b\hat{j}$

17. A horizontal rod of mass 10 gm and length 10 cm is placed on a smooth plane inclined at an angle of 60° with the horizontal, with the length of the rod parallel to the edge of the inclined plane. A uniform magnetic field of induction B is applied vertically downwards. If the current through the rod is 1.73 ampere, then the value of B for which the rod remains stationary on the inclined plane is

(a) 1.73 Tesla

(b) $\frac{1}{1.73}$ Tesla

(c) 1 Tesla

(d) None of the above

18. Two long wires are hanging freely. They are joined first in parallel and then in series and then are connected with a battery. In both cases, which type of force acts between the two wires

(a) Attraction force when in parallel and repulsion force when in series

(b) Repulsion force when in parallel and attraction force when in series

(c) Repulsion force in both cases

(d) Attraction force in both cases

19. A wire of length L metre carrying a current of I ampere is bent in the form of a circle. Its magnitude of magnetic moment will be [MP PET 1995; M

(a) $\frac{IL}{4\pi}$

(b) $\frac{IL^2}{4\pi}$

(c) $\frac{I^2 L^2}{4\pi}$

(d) $\frac{I^2 L}{4\pi}$

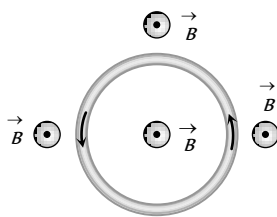
20. A thin circular wire carrying a current I has a magnetic moment M . The shape of the wire is changed to a square and it carries the same current. It will have a magnetic moment [MP PET 2003; MP PMT 2004]

- (a) M (b) $\frac{4}{\pi^2} M$
(c) $\frac{4}{\pi} M$ (d) $\frac{\pi}{4} M$

21. A particle of charge q and mass m moves in a circular orbit of radius r with angular speed ω . The ratio of the magnitude of its magnetic moment to that of its angular momentum depends on

- (a) ω and q (b) ω , q and m
(c) q and m (d) ω and m

22. An elastic circular wire of length l carries a current I . It is placed in a uniform magnetic field $\vec{B}$ (Out of paper) such that its plane is perpendicular to the direction of $\vec{B}$. The wire will experience

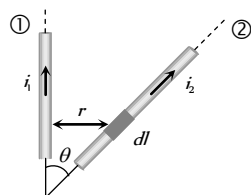


- (a) No force (b) A stretching force
(c) A compressive force (d) A torque

23. A and B are two conductors carrying a current i in the same direction. x and y are two electron beams moving in the same direction [Karnataka CET (Engg./Med.) 2002]



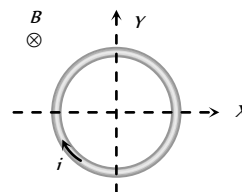
- (a) There will be repulsion between A and B attraction between x and y
(b) There will be attraction between A and B , repulsion between x and y
(c) There will be repulsion between A and B and also x and y
(d) There will be attraction between A and B and also x and y
24. Wires 1 and 2 carrying currents i_1 and i_2 respectively are inclined at an angle θ to each other. What is the force on a small element dl of wire 2 at a distance of r from wire 1 (as shown in figure) due to the magnetic field of wire 1 [AIEEE 2002]



- (a) $\frac{\mu_0}{2\pi} i_1 i_2 dl \tan \theta$
(b) $\frac{\mu_0}{2\pi} i_1 i_2 dl \sin \theta$
(c) $\frac{\mu_0}{2\pi} i_1 i_2 dl \cos \theta$
(d) $\frac{\mu_0}{4\pi} i_1 i_2 dl \sin \theta$

25. A conducting loop carrying a current I is placed in a uniform magnetic field pointing into the plane of the paper as shown. The loop will have a tendency to

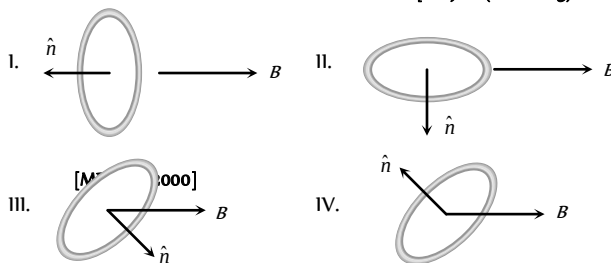
[IIT-JEE (Screening) 2003]



- (a) Contract
(b) Expand
(c) Move towards $+ve$ x -axis
(d) Move towards $-ve$ x -axis

26. A current carrying loop is placed in a uniform magnetic field in four different orientations, I, II, III & IV arrange them in the decreasing order of potential Energy [IIT-JEE (Screening) 2000]

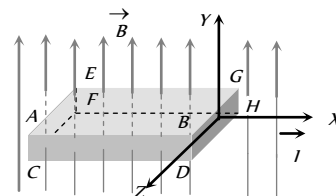
[IIT-JEE (Screening) 2003]



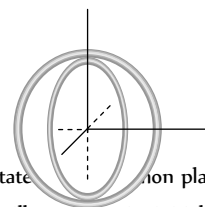
- (a) $I > III > II > IV$ (b) $I > II > III > IV$
(c) $I > IV > II > III$ (d) $III > IV > I > II$

27. A metallic block carrying current I is subjected to a uniform magnetic induction $\vec{B}$ as shown in the figure. The moving charges experience a force $\vec{F}$ given by which results in the lowering of the potential of the face Assume the speed of the carriers to be v [IIT 1996]

- (a) $eVB\hat{k}$, $ABCD$
(b) $eVB\hat{k}$, $EFGH$
(c) $-eVB\hat{k}$, $ABCD$
(d) $-eVB\hat{k}$, $EFGH$



28. Two insulated rings, one of slightly smaller diameter than the other are suspended along their common diameter as shown. Initially the planes of the rings are mutually perpendicular. When a steady current is set up in each of them [IIT 1995]



- (a) The two rings rotate about their common plane
(b) The inner ring oscillates about its initial position
(c) The inner ring stays stationary while the outer one moves into the plane of the inner ring
(d) The outer ring stays stationary while the inner one moves into the plane of the outer ring

29. Two particles each of mass m and charge q are attached to the two ends of a light rigid rod of length $2R$. The rod is rotated at constant angular speed about a perpendicular axis passing through its centre. The ratio of the magnitudes of the magnetic moment of the system and its angular momentum about the centre of the rod is [IIT 1998]

- (a) $\frac{q}{2m}$ (b) $\frac{q}{m}$
(c) $\frac{2q}{m}$ (d) $\frac{q}{\pi m}$

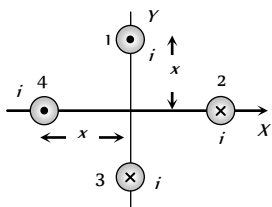
30. Two very long, straight and parallel wires carry steady currents I and I respectively. The distance between the wires is d . At a certain instant of time, a point charge q is at a point equidistant from the two wires in the plane of the wires. Its instantaneous velocity v is perpendicular to this plane. The magnitude of the force due to the magnetic field acting on the charge at this instant is [IIT 1998]

- (a) $\frac{\mu_0 I q v}{2\pi d}$ (b) $\frac{\mu_0 I q v}{\pi d}$
(c) $\frac{2\mu_0 I q v}{\pi d}$ (d) 0

31. A ring of radius R , made of an insulating material carries a charge Q uniformly distributed on it. If the ring rotates about the axis passing through its centre and normal to plane of the ring with constant angular speed ω , then the magnitude of the magnetic moment of the ring is [MP PET 2001]

- (a) $Q\omega R^2$ (b) $\frac{1}{2}Q\omega R^2$
(c) $Q\omega^2 R$ (d) $\frac{1}{2}Q\omega^2 R$

32. What will be the resultant magnetic field at origin due to four infinite length wires. If each wire produces magnetic field ' B ' at origin



- (a) $4B$ (b) $\sqrt{2}B$
(c) $2\sqrt{2}B$ (d) Zero

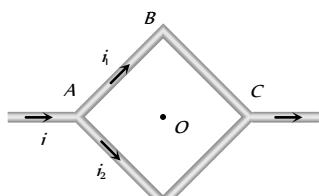
33. The ratio of the magnetic field at the centre of a current carrying circular wire and the magnetic field at the centre of a square coil made from the same length of wire will be

- (a) $\frac{\pi^2}{4\sqrt{2}}$ (b) $\frac{\pi^2}{8\sqrt{2}}$
(c) $\frac{\pi}{2\sqrt{2}}$ (d) $\frac{\pi}{4\sqrt{2}}$

34. Two infinite length wires carries currents $8A$ and $6A$ respectively and placed along X and Y -axis. Magnetic field at a point $P(0,0,d)m$ will be

- (a) $\frac{7\mu_0}{\pi d}$ (b) $\frac{10\mu_0}{\pi d}$
(c) $\frac{14\mu_0}{\pi d}$ (d) $\frac{5\mu_0}{\pi d}$

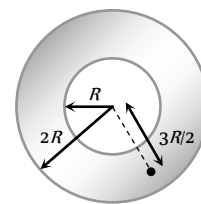
35. Figure shows a square loop $ABCD$ with edge length a . The resistance of the wire ABC is r and that of ADC is $2r$. The value of magnetic field at the centre of the loop assuming uniform wire is



- (a) $\frac{\sqrt{2}\mu_0 i}{3\pi a} \odot$ (b) $\frac{\sqrt{2}\mu_0 i}{3\pi a} \otimes$
(c) $\frac{\sqrt{2}\mu_0 i}{\pi a} \odot$ (d) $\frac{\sqrt{2}\mu_0 i}{\pi a} \otimes$

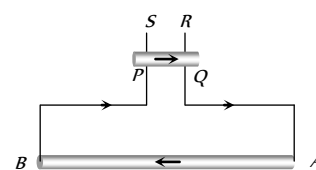
36. Figure shows the cross-sectional view of the hollow cylindrical conductor with inner radius ' R ' and outer radius ' $2R$ ', cylinder carrying uniformly distributed current along its axis. The magnetic induction at point ' P ' at a distance $\frac{3R}{2}$ from the axis of the cylinder will be

- (a) Zero
(b) $\frac{5\mu_0 i}{72\pi R}$
(c) $\frac{7\mu_0 i}{18\pi R}$
(d) $\frac{5\mu_0 i}{36\pi R}$

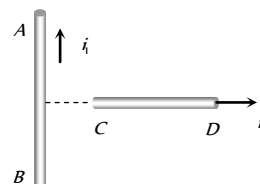


37. A long wire AB is placed on a table. Another wire PQ of mass $1.0g$ and length $50cm$ is set to slide on two rails PS and QR . A current of $50A$ is passed through the wires. At what distance above AB , will the wire PQ be in equilibrium

- (a) $25mm$
(b) $50mm$
(c) $75mm$
(d) $100mm$



38. An infinitely long, straight conductor AB is fixed and a current is passed through it. Another movable straight wire CD of finite length and carrying current is held perpendicular to it and released. Neglect weight of the wire



- (a) The rod CD will move upwards parallel to itself
(b) The rod CD will move downward parallel to itself
(c) The rod CD will move upward and turn clockwise at the same time
(d) The rod CD will move upward and turn anti-clockwise at the same time

39. A steady current i flows in a small square loop of wire of side L in a horizontal plane. The loop is now folded about its middle such that half of it lies in a vertical plane. Let $\vec{\mu}_1$ and $\vec{\mu}_2$ respectively denote the magnetic moments due to the current loop before and after folding. Then

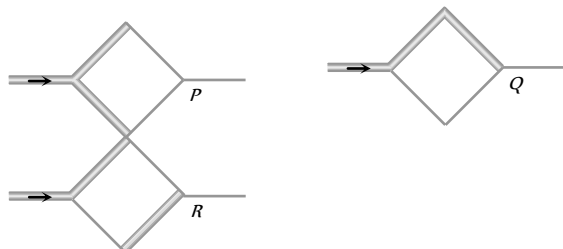
[IIT-JEE 1993]

- (a) $\vec{\mu}_2 = 0$
 (b) $\vec{\mu}_1$ and $\vec{\mu}_2$ are in the same direction
 (c) $\frac{|\vec{\mu}_1|}{|\vec{\mu}_2|} = \sqrt{2}$
 (d) $\frac{|\vec{\mu}_1|}{|\vec{\mu}_2|} = \left(\frac{1}{\sqrt{2}}\right)$

40. A current i is flowing in a straight conductor of length L . The magnetic induction at a point distant $\frac{L}{4}$ from its centre will be

- (a) $\frac{4\mu_0 i}{\sqrt{5}\pi L}$ (b) $\frac{\mu_0 i}{2\pi L}$
 (c) $\frac{\mu_0 i}{\sqrt{2}L}$ (d) Zero

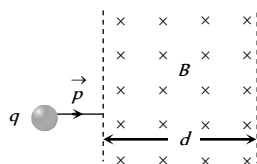
41. Two thick wires and two thin wires, all of the same materials and same length form a square in the three different ways P , Q and R as shown in fig with current connection shown, the magnetic field at the centre of the square is zero in cases



- (a) In P only (b) In P and Q only
 (c) In Q and R only (d) P and R only

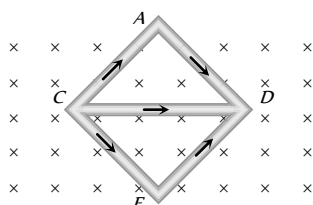
42. A particle with charge q , moving with a momentum p , enters a uniform magnetic field normally. The magnetic field has magnitude B and is confined to a region of width d , where $d < \frac{p}{Bq}$. The particle is deflected by an angle θ in crossing the field

- (a) $\sin\theta = \frac{Bqd}{p}$
 (b) $\sin\theta = \frac{p}{Bqd}$
 (c) $\sin\theta = \frac{Bp}{qd}$
 (d) $\sin\theta = \frac{pd}{Bq}$



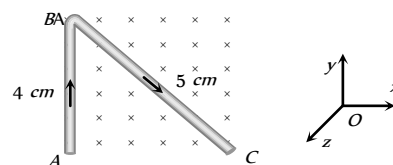
43. Same current $i = 2A$ is flowing in a wire frame as shown in figure. The frame is a combination of two equilateral triangles ACD and CDE of side $1m$. It is placed in uniform magnetic field $B = 4T$ acting perpendicular to the plane of frame. The magnitude of magnetic force acting on the frame is

- (a) $24 N$
 (b) Zero
 (c) $16 N$



- (d) $8 N$

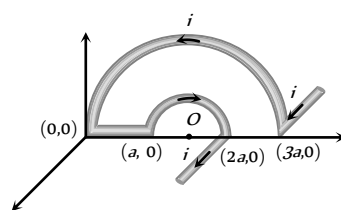
44. A uniform conducting wire ABC has a mass of $10g$. A current of $2A$ flows through it. The wire is kept in a uniform magnetic field $B = 2T$. The acceleration of the wire will be



- (a) Zero
 (b) $12 ms^{-2}$ along y -axis
 (c) $1.2 \times 10^{-3} ms^{-2}$ along y -axis
 (d) $0.6 \times 10^{-3} ms^{-2}$ along y -axis

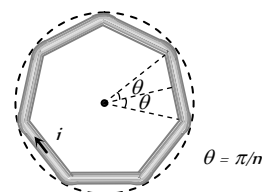
45. In the given figure net magnetic field at O will be

- (a) $\frac{2\mu_0 i}{3\pi a} \sqrt{4 - \pi^2}$
 (b) $\frac{\mu_0 i}{3\pi a} \sqrt{4 + \pi^2}$
 (c) $\frac{2\mu_0 i}{3\pi a^2} \sqrt{4 + \pi^2}$
 (d) $\frac{2\mu_0 i}{3\pi a} \sqrt{(4 - \pi^2)}$



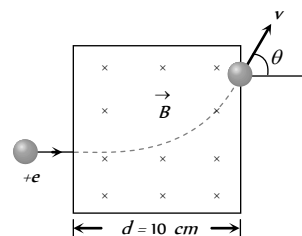
46. In the following figure a wire bent in the form of a regular polygon of n sides is inscribed in a circle of radius a . Net magnetic field at centre will be

- (a) $\frac{\mu_0 i}{2\pi a} \tan \frac{\pi}{n}$
 (b) $\frac{\mu_0 ni}{2\pi a} \tan \frac{\pi}{n}$
 (c) $\frac{2}{\pi} \frac{ni}{a} \mu_0 \tan \frac{\pi}{n}$
 (d) $\frac{ni}{2a} \mu_0 \tan \frac{\pi}{n}$



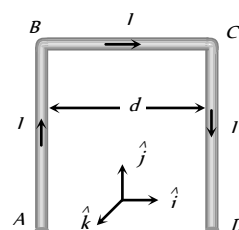
47. A proton accelerated by a potential difference $500 KV$ moves through a transverse magnetic field of $0.51 T$ as shown in figure. The angle θ through which the proton deviates from the initial direction of its motion is

- (a) 15°
 (b) 30°
 (c) 45°
 (d) 60°



48. AB and CD are long straight conductor, distance d apart, carrying a current I . The magnetic field at the midpoint of BC is

- (a) $\frac{-\mu_0 I}{2\pi d} \hat{k}$
 (b) $\frac{-\mu_0 I}{\pi d} \hat{k}$
 (c) $\frac{-\mu_0 I}{4\pi d} \hat{k}$
 (d) $\frac{-\mu_0 I}{8\pi d} \hat{k}$

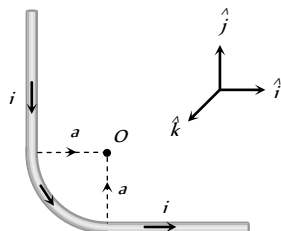


49. An electron is moving along the positive X -axis. You want to apply a magnetic field for a short time so that the electron may reverse its direction and move parallel to the negative X -axis. This can be done by applying the magnetic field along

(a) Y -axis (b) X -axis
(c) Y -axis only (d) None of these

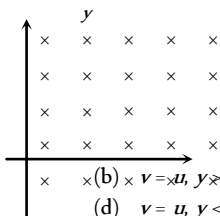
50. The unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ are as shown below. What will be the magnetic field at O in the following figure

- (a) $\frac{\mu_0}{4\pi} \frac{i}{a} \left(2 - \frac{\pi}{2} \right) \hat{j}$
(b) $\frac{\mu_0}{4\pi} \frac{i}{a} \left(2 + \frac{\pi}{2} \right) \hat{j}$
(c) $\frac{\mu_0}{4\pi} \frac{i}{a} \left(2 + \frac{\pi}{2} \right) \hat{i}$
(d) $\frac{\mu_0}{4\pi} \frac{i}{a} \left(2 + \frac{\pi}{2} \right) \hat{k}$



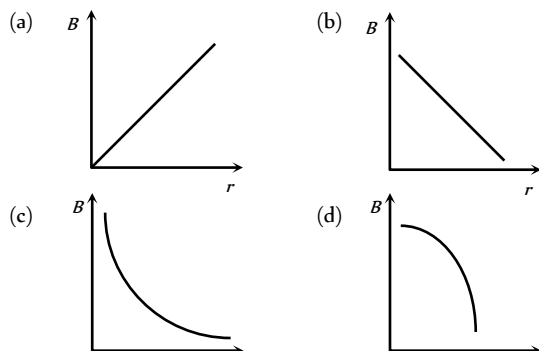
51. An electron moving with a speed u along the positive x -axis at $y = 0$ enters a region of uniform magnetic field $\vec{B} = -B_0 \hat{k}$ which exists to the right of y -axis. The electron exits from the region after some time with the speed v at co-ordinate y , then [IIT-JEE (Screening 2004)]

- (a) $v > u, y < 0$ (b) $v = u, y > 0$
(c) $v > u, y > 0$ (d) $v = u, y < 0$



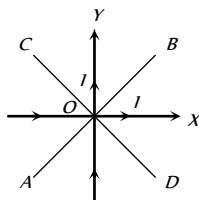
Graphical Questions

1. Which of the following graphs shows the variation of magnetic induction B with distance r from a long wire carrying current [NCERT 1984; MNR 1998; MP PMT 1999]



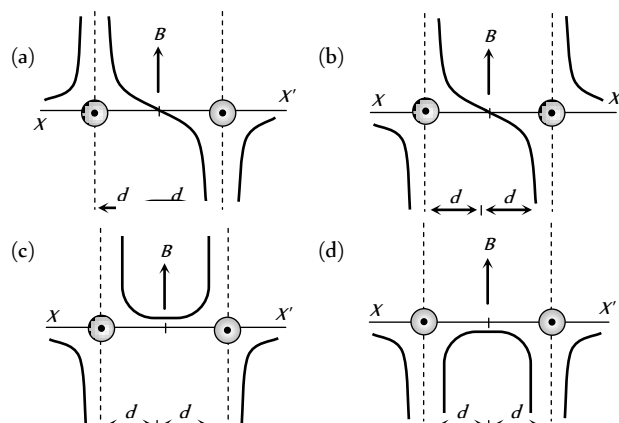
2. Two very thin metallic wires placed along X and Y -axis carry equal currents as shown here. AB and CD are lines at 45° with the axes with origin of axes at O . The magnetic field will be zero on the line [MP PMT 1995; CBSE PMT 1996]

- (a) AB
(b) CD
(c) Segment OB only of line AB

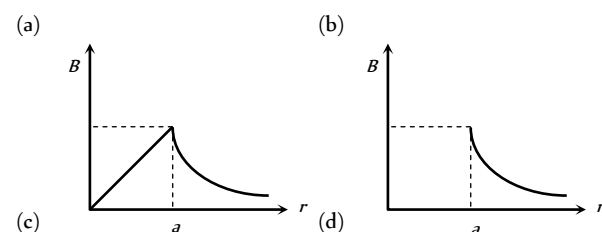


- (d) Segment OC only of line CD

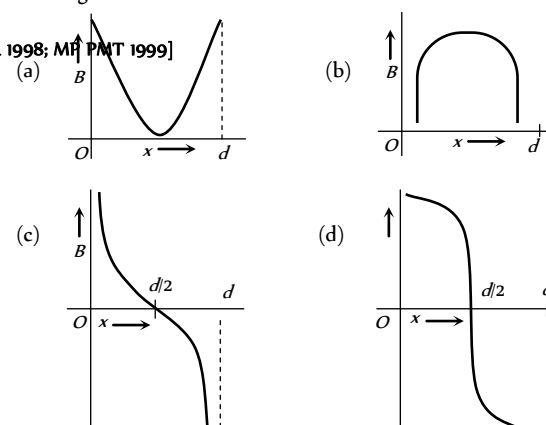
3. Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper, as shown. The variation of the magnetic field B along the line XX' is given by



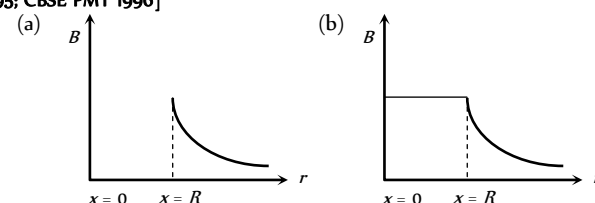
4. The magnetic field due to a straight conductor of radius a and carrying a steady current is represented by



5. Two parallel beams of protons and electrons, carrying equal currents are fixed at a separation d . The protons and electrons move in opposite directions. P is a point on a line joining the beams, at a distance x from any one beam. The magnetic field at P is B . If B is plotted against x , which of the following best represents the resulting curve



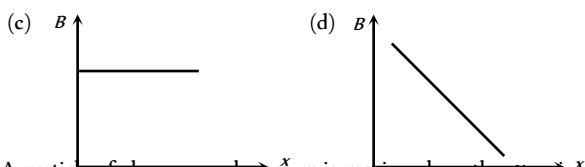
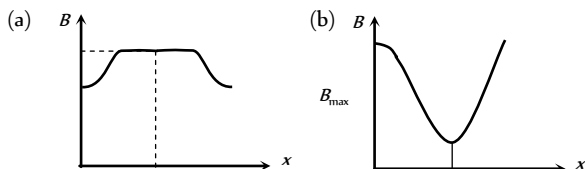
6. A long thin hollow metallic cylinder of radius ' R ' has a current i ampere. The magnetic induction ' B ' away from the axis at a distance r from the axis varies as shown in



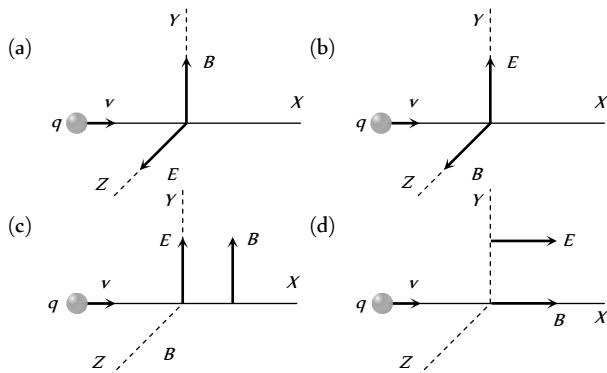
(c)

(d)

7. The correct curve between the magnetic induction (B) along the axis of a long solenoid due to current flow i in it and distance x from one end is

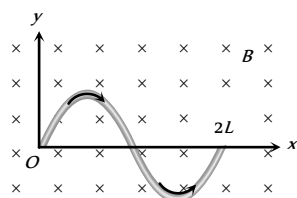


8. A particle of charge q and mass m is moving along the x -axis with a velocity v and enters a region of electric field E and magnetic field B as shown in figure below for which figure the net force on the charge may be zero

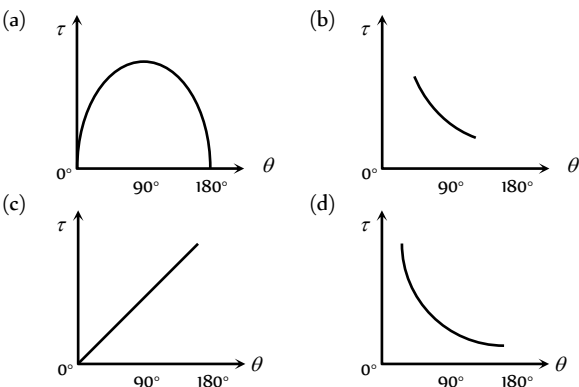


9. A wire carrying a current i is placed in a uniform magnetic field in the form of the curve $y = a \sin\left(\frac{\pi x}{L}\right)$ $0 \leq x \leq 2L$. The force acting on the wire is

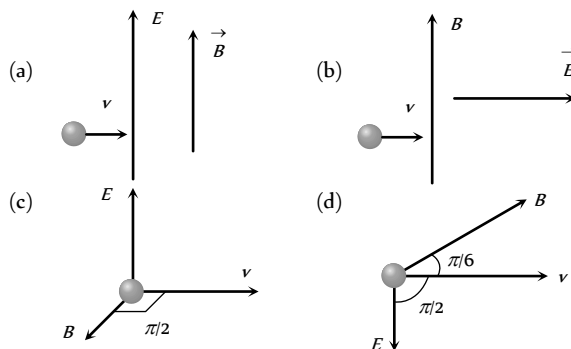
- (a) $\frac{iBL}{\pi}$
(b) $iBL\pi$
(c) $2iBL$
(d) Zero



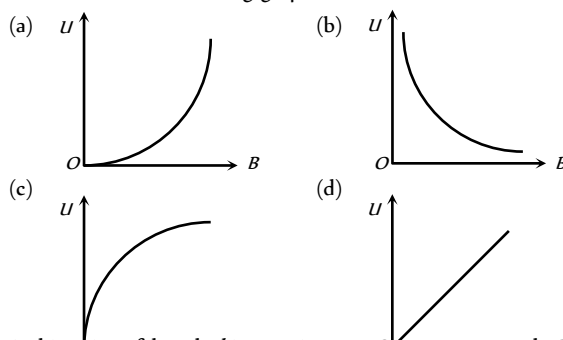
10. The $(\tau - \theta)$ graph for a coil is



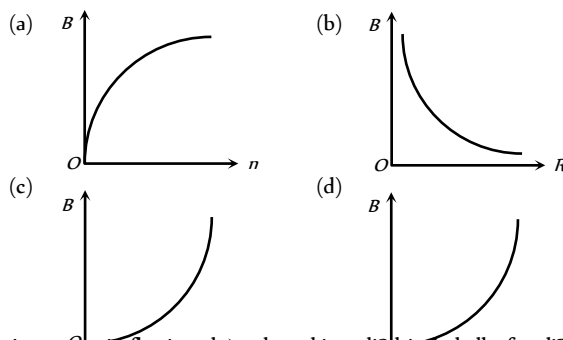
11. A uniform magnetic field B and a uniform electric field E act in a common region. An electron is entering this region of space. The correct arrangement for it to escape undeviated is



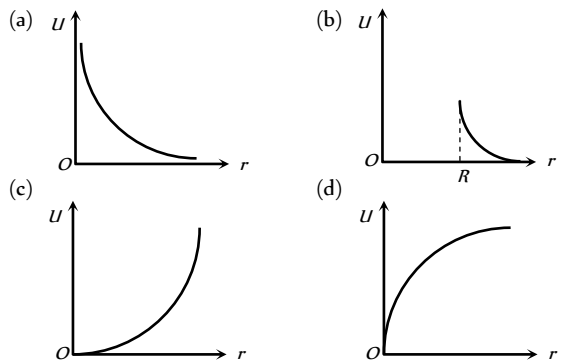
12. If induction of magnetic field at a point is B and energy density is U then which of the following graphs is correct



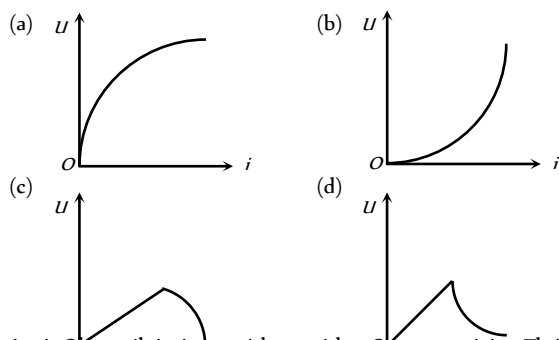
13. A thin wire of length L is carrying a constant current. The wire is bent to form a circular coil. If radius of the coil, thus formed, is equal to R and number of turns in it is equal to n , then which of the following graphs represent (s) variation of magnetic field induction (B) at centre of the coil



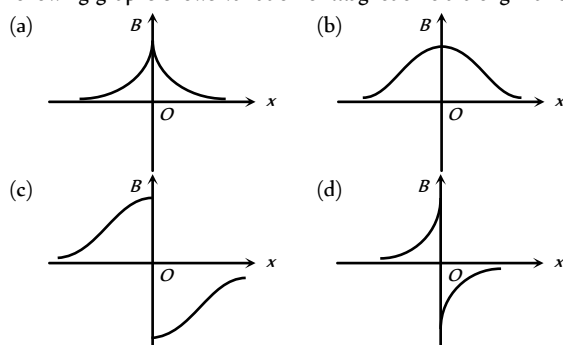
14. A current is flowing through a thin cylindrical shell of radius R . If energy density in the medium, due to magnetic field, at a distance r from axis of the shell is equal to U then which of the following graphs is correct



15. If current flowing through shell of previous objective is equal to i , then energy density at a point distance $2R$ from axis of the shell varies according to the graph



16. A circular coil is in yz plane with centre at origin. The coil is carrying a constant current. Assuming direction of magnetic field at $x = -25 \text{ cm}$ to be positive direction of magnetic field, which of the following graphs shows variation of magnetic field along x -axis



Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

- Assertion : Cyclotron does not accelerate electron.
Reason : Mass of the electron is very small.
[AIIMS 2000]
- Assertion : Cyclotron is a device which is used to accelerate the positive ion.
Reason : Cyclotron frequency depends upon the velocity.[AIIMS 1997]
- Assertion : Magnetic field interacts with a moving charge and not with a stationary charge.
Reason : A moving charge produces a magnetic field.
- Assertion : If an electron is not deflected while passing through a certain region of space, then only possibility is that there is no magnetic region.
Reason : Force is directly proportional to the magnetic field applied.
- Assertion : Free electron always keep on moving in a conductor even then no magnetic force act on them in magnetic field unless a current is passed through it.

- Reason : The average velocity of free electron is zero.
- Assertion : The ion cannot move with a speed beyond a certain limit in a cyclotron.
Reason : As velocity increases time taken by ion increases.
- Assertion : The coil is bound over the metallic frame in moving coil galvanometer.
Reason : The metallic frame help in making steady deflection without any oscillation.
- Assertion : A circular loop carrying current lies in XY plane with its center at origin having a magnetic flux in negative Z -axis.
Reason : Magnetic flux direction is independent of the direction of current in the conductor.
- Assertion : The energy of charged particle moving in a uniform magnetic field does not change.
Reason : Work done by magnetic field on the charge is zero.
- Assertion : If an electron, while coming vertically from outerspace, enter the earth's magnetic field, it is deflected towards west.
Reason : Electron has negative charge.
- Assertion : A direct current flows through a metallic rod, produced magnetic field only outside the rod.
Reason : There is no flow of charge carriers inside the rod.
- Assertion : An electron and proton enters a magnetic field with equal velocities, then, the force experienced by the proton will be more than electron.
Reason : The mass of proton is 1837 times more than electron.
- Assertion : Torque on the coil is the maximum, when coil is suspended in a radial magnetic field.
Reason : The torque tends to rotate the coil on its own axis.
- Assertion : A loosely round helix made of stiff wire is suspended vertically with the lower end just touching a dish of mercury. When a current is passed through the wire, the helical wire executes oscillatory motion with the lower end jumping out of and inside of mercury.
Reason : When electric current is passed through helix, a magnetic field is produced both inside and outside the helix.
- Assertion : The magnetic field at the ends of a very long current carrying solenoid is half of that at the center.
Reason : If the solenoid is sufficiently long, the field within it is uniform.
- Assertion : If a charged particle is moving on a circular path in a perpendicular magnetic field, the momentum of the particle is not changing.
Reason : Velocity of the particle is not changing in the magnetic field.
- Assertion : If a proton and an α -particle enter a uniform magnetic field perpendicularly, with the same speed, then the time period of revolution of the α -particle is double than that of proton.
Reason : In a magnetic field, the time period of revolution of a charged particle is directly proportional to mass.
- Assertion : If two long wires, hanging freely are connected to a battery in series, they come closer to each other.



1216 Magnetic Effect of Current

Reason : Force of attraction acts between the two wires carrying current.

19. Assertion : A current I flows along the length of an infinitely long straight and thin walled pipe. Then the magnetic field at any point inside the pipe is zero.

Reason : $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$

Answers

Biot-Savart's Law and Amperes Law

1	c	2	b	3	c	4	b	5	d
6	c	7	b	8	d	9	b	10	d
11	b	12	a	13	a	14	c	15	c
16	a	17	b	18	a	19	b	20	d
21	c	22	c	23	a	24	d	25	b
26	d	27	c	28	a	29	b	30	d
31	b	32	a	33	a	34	d	35	d
36	d	37	b	38	c	39	a	40	c
41	c	42	b	43	d	44	b	45	d
46	b	47	a	48	a	49	b	50	d
51	c	52	b	53	c	54	b	55	d
56	a	57	b	58	a	59	d	60	d
61	c	62	c	63	d	64	b	65	a
66	c	67	a	68	c	69	a	70	b
71	c	72	c	73	b	74	b	75	d
76	b	77	b	78	b	79	c	80	b
81	d	82	c	83	c	84	d	85	c
86	b	87	d	88	a	89	a	90	b
91	c	92	d	93	d	94	a	95	b
96	b	97	a	98	d	99	c	100	d
101	b	102	c	103	b	104	d	105	a
106	c	107	b	108	a	109	a	110	c
111	d	112	a	113	b	114	a	115	b
116	d	117	c	118	b	119	a	120	a
121	a	122	a	123	c	124	d	125	c
126	b								

Motion of Charged Particle In Magnetic Field

1	abd	2	d	3	c	4	a	5	b
6	b	7	c	8	c	9	c	10	d
11	c	12	d	13	d	14	b	15	a
16	c	17	c	18	a	19	c	20	b
21	c	22	c	23	d	24	b	25	d
26	a	27	c	28	a	29	d	30	d
31	d	32	b	33	c	34	a	35	c
36	b	37	d	38	a	39	a	40	d
41	c	42	a	43	d	44	a	45	b
46	c	47	d	48	b	49	c	50	d
51	b	52	c	53	b	54	a	55	a
56	d	57	d	58	b	59	d	60	a
61	c	62	c	63	b	64	a	65	a

66	b	67	b	68	a	69	bd	70	b
71	c	72	b	73	b	74	a	75	d
76	d	77	d	78	b	79	a	80	d
81	a	82	a	83	c	84	b	85	b
86	c	87	c	88	b	89	a	90	b
91	a	92	a	93	c	94	d	95	c
96	a	97	c	98	b	99	a	100	b
101	d	102	c	103	d	104	d	105	c
106	c	107	d	108	d	109	d	110	d
111	a	112	b	113	d	114	c	115	c
116	c	117	c	118	c				

Force and Torque on a Current Carrying Conductor

1	b	2	c	3	b	4	c	5	b
6	a	7	b	8	b	9	a	10	a
11	c	12	a	13	c	14	c	15	a
16	c	17	b	18	b	19	b	20	b
21	d	22	b	23	d	24	a	25	a
26	b	27	d	28	b	29	c	30	c
31	c	32	d	33	d	34	d	35	b
36	a	37	a	38	a	39	c	40	d
41	c	42	a	43	a	44	c	45	d
46	a	47	d	48	a	49	a	50	c
51	b	52	a	53	abc	54	a	55	c
56	b	57	a	58	d	59	c	60	a
61	b	62	b	63	d	64	b	65	c
66	a	67	b	68	a	69	a	70	a
71	c	72	a	73	a	74	d	75	d
76	c	77	a	78	a	79	b	80	c
81	b	82	c	83	c	84	b	85	a
86	d	87	c	88	b	89	c	90	c
91	d								

Critical Thinking Questions

1	b	2	a	3	a	4	c	5	c
6	c	7	cd	8	c	9	d	10	a
11	abd	12	ac	13	c	14	c	15	b
16	b	17	c	18	a	19	b	20	d
21	c	22	b	23	b	24	c	25	b
26	c	27	a	28	a	29	a	30	d
31	b	32	c	33	b	34	d	35	b
36	d	37	a	38	c	39	c	40	a
41	d	42	a	43	a	44	b	45	b
46	b	47	b	48	b	49	a	50	d
51	d								

Graphical Questions

1	c	2	a	3	b	4	a	5	c
6	a	7	a	8	b	9	c	10	a
11	c	12	a	13	bc	14	b	15	b
16	b								

Assertion and Reason

1	a	2	c	3	a	4	e	5	a
6	c	7	a	8	c	9	a	10	b
11	d	12	e	13	b	14	b	15	b
16	d	17	b	18	d	19	a		

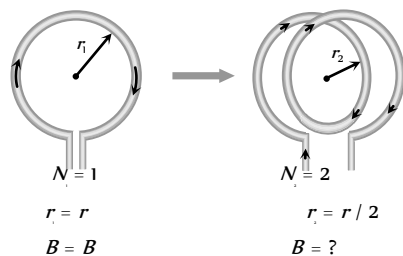
AS Answers and Solutions

Biot-Savart's Law and Amperes Law

1. (c) Magnetic field at the centre of current carrying coil is given by

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi Ni}{r} \Rightarrow B \propto \frac{N}{r} \Rightarrow \frac{B_1}{B_2} = \frac{N_1}{N_2} \times \frac{r_2}{r_1}$$

The following figure shows that single turn coil changes to double turn coil.

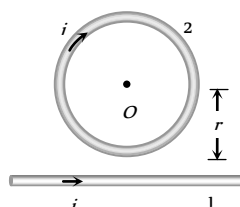


$$\Rightarrow \frac{B}{B_2} = \frac{1}{2} \times \frac{r/2}{r} = \frac{1}{4} \Rightarrow B_2 = 4B$$

Short trick : For such type of problems remember $B_2 = n^2 B_1$

2. (b) If distance is same field will be same ($\because B = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r}$)
3. (c) Magnetic field lies inside as well as outside the solid current carrying conductor.
4. (b) Because for inside the pipe $i = 0$
 $\therefore B = \frac{\mu_0 i}{2\pi r} = 0$
5. (d) $dB = \frac{\mu_0}{4\pi} \cdot \frac{idl \sin \theta}{r^2} \Rightarrow d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{i(d\vec{l} \times \vec{r})}{r^3}$
6. (c) The magnetic field at the centre of the circle
 $= \frac{\mu_0}{4\pi} \times \frac{2\pi i}{r} = 10^{-7} \times \frac{2\pi(nq)}{r} = \frac{2\pi nq}{r} \times 10^{-7} \text{ N/A-m}$
7. (b) The given shape is equivalent to the following diagram

The field at O due to straight part of conductor



is $B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r} \odot$. The field at O due to circular coil is

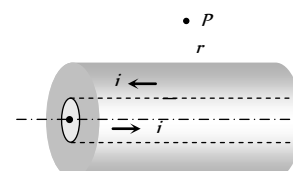
$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} \otimes$. Both fields will act in the opposite direction, hence the total field at O .

$$i.e. B = B_2 - B_1 = \left(\frac{\mu_0}{4\pi} \right) \times (\pi - 1) \frac{2i}{r} = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r} (\pi - 1)$$

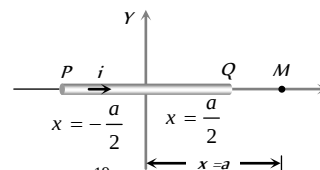
$$8. (d) B = \frac{\mu_0}{4\pi} \cdot \frac{(2\pi - \theta)i}{R} = \frac{\mu_0}{4\pi} \cdot \frac{\left(2\pi - \frac{\pi}{2} \right) \times i}{R} = \frac{3\mu_0 i}{8R}$$

9. (b) The respective figure is shown below

Magnetic field at P due to inner and outer conductors are equal and opposite. Hence net magnetic field at P will be zero.



10. (d) Magnetic field at a point on the axis of a current carrying wire is always zero.



$$11. (b) i = \frac{q}{T} = \frac{2 \times 1.6 \times 10^{-19}}{2} = 1.6 \times 10^{-19} \text{ A}$$

$$\therefore B = \frac{\mu_0 i}{2r} = \frac{\mu_0 \times 1.6 \times 10^{-19}}{2 \times 0.8} = \mu_0 \times 10^{-19}$$

$$12. (a) B = \mu_0 ni = 4\pi \times 10^{-7} \times 5 \times 1000 = 2\pi \times 10^{-3} \text{ Tesla}$$

$$13. (a) B = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r} \Rightarrow \frac{B_1}{B_2} = \frac{r_2}{r_1} \Rightarrow \frac{10^{-8}}{B_2} = \frac{12}{4}$$

$$\Rightarrow B_2 = 3.33 \times 10^{-9} \text{ Tesla}$$

$$14. (c) B \propto \frac{1}{r} \Rightarrow \frac{B_1}{B_2} = \frac{r_2}{r_1} \Rightarrow \frac{B}{B_2} = \frac{r/2}{r} \Rightarrow B_2 = 2B$$

$$15. (c) \text{Field at the centre of a circular coil of radius } r \text{ is } B = \frac{\mu_0 I}{2r}$$

$$16. (a) B = \frac{\mu_0 Ni}{2r} = \frac{4\pi \times 10^{-7} \times 100 \times 0.1}{2 \times 5 \times 10^{-2}} = 4\pi \times 10^{-5} \text{ Tesla}$$

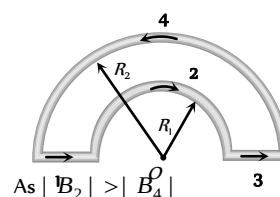
$$17. (b) \text{Magnetic field inside the solenoid } B_{in} = \mu_0 ni$$

18. (a) In the following figure, magnetic fields at O due to sections 1, 2, 3 and 4 are considered as B_1, B_2, B_3 and B_4 respectively.

$$B_1 = B_3 = 0$$

$$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{R_1} \otimes$$

$$B_4 = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{R_2} \odot$$



As $|B_2| > |B_4|$

$$\text{So } B_{\text{net}} = B_2 - B_4 \Rightarrow B_{\text{net}} = \frac{\mu_0 i}{4} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \otimes$$

19. (b) $B = \mu_0 ni$
20. (d) The magnetic induction at O due to the current in portion AB will be zero because O lies on AB when extended.
21. (c) The induction due to AB and CD will be zero. Hence the whole induction will be due to the semicircular part BC .

$$B = \frac{\mu_0 i}{4r}$$

22. (c) The magnetic induction due to both semicircular parts will be in the same direction perpendicular to the paper inwards.

$$\therefore B = B_1 + B_2 = \frac{\mu_0 i}{4r_1} + \frac{\mu_0 i}{4r_2} = \frac{\mu_0 i}{4} \left(\frac{r_1 + r_2}{r_1 r_2} \right) \otimes$$

23. (a) Field at a point x from the centre of a current carrying loop on the axis is

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2M}{x^3} = \frac{10^{-7} \times 2 \times 2.1 \times 10^{-25}}{(10^{-10})^3}$$

$$= 4.2 \times 10^{-32} \times 10^{30} = 4.2 \times 10^{-2} \text{ W/m}^2$$

24. (d) At these points, the resultant field = 0

25. (b) $i = \frac{q}{t} = 100 \times e$

$$B_{\text{centre}} = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} = \frac{\mu_0}{4\pi} \cdot \frac{2\pi \times 100e}{r}$$

$$= \frac{\mu_0 \times 200 \times 1.6 \times 10^{-19}}{4 \times 0.8} = 10^{-17} \mu_0$$

26. (d) $B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi Ni R^2}{r^3} \Rightarrow B \propto \frac{1}{r^3}$

27. (c) $B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi(qv)}{r}$
- $$= 10^{-7} \times \frac{2 \times 3.14 \times (1.6 \times 10^{-19} \times 6.6 \times 10^{15})}{0.53 \times 10^{-10}} = 12.5 \text{ Wb/m}^2$$

28. (a)

29. (b)

30. (d) Two coils carrying current in opposite direction, hence net magnetic field at centre will be difference of the two fields.

$$\text{i.e. } B_{\text{net}} = \frac{\mu_0}{4\pi} \cdot 2\pi N \left[\frac{Ni_1}{r_1} - \frac{i_2}{r_2} \right] = \frac{10\mu_0}{2} \left[\frac{0.2}{0.2} - \frac{0.3}{0.4} \right] = \frac{5}{4} \mu_0$$

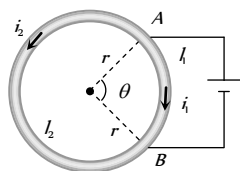
31. (b) Because $B = \mu_0 ni \Rightarrow B \propto ni$.

32. (a) See solution 34.

33. (a) $B = \mu_0 ni \Rightarrow i = \frac{B}{\mu_0 n} = \frac{20 \times 10^{-3}}{4\pi \times 10^{-7} \times 20 \times 100}$
- $$= 7.9 \text{ amp} = 8 \text{ amp}$$

34. (d) Directions of currents in two parts are different, so directions of magnetic fields due to these currents are opposite. Also applying Ohm's law across AB

$$i_1 R_1 = i_2 R_2 \Rightarrow i_1 l_2 = i_2 l_1$$



$$\left(\because R = \rho \frac{l}{A} \right)$$

$$\text{Also } B_1 = \frac{\mu_0}{4\pi} \times \frac{i_1 l_1}{r^2} \text{ and } B_2 = \frac{\mu_0}{4\pi} \times \frac{i_2 l_2}{r^2} \quad (\because l = r\theta)$$

$$\therefore \frac{B_2}{B_1} = \frac{i_1 l_1}{i_2 l_2} = 1$$

Hence, two field induction's are equal but of opposite direction. So, resultant magnetic induction at the centre is zero and is independent of θ .

35. (d) The magnetic field at any point on the axis of wire be zero.
36. (d) Magnetic field inside the hollow conductor (tube) is zero.
37. (b) If a wire of length l is bent in the form of a circle of radius r then $2\pi r = l \Rightarrow$

$$r = \frac{l}{2\pi}$$

$$l = \pi^2 m$$

$$r = \frac{l}{2\pi} = \frac{\pi^2}{2\pi} = \frac{\pi}{2}$$

$$i = 2A$$

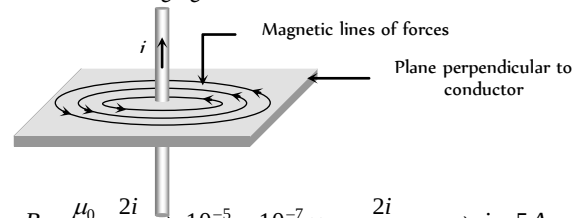
Magnetic field due to

$$\text{straight wire } B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r} = \frac{\mu_0}{4\pi} \times \frac{2 \times 2}{1 \times 10^{-2}} \text{ also magnetic}$$

$$\text{field due to circular loop } B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} = \frac{\mu_0}{4\pi} \cdot \frac{2\pi \times 2}{\pi/2} \Rightarrow$$

$$\frac{B_2}{B_1} = \frac{1}{50}$$

38. (c) See the following figure



39. (a) $B = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r} \Rightarrow 10^{-5} = 10^{-7} \times \frac{2i}{(10 \times 10^{-2})} \Rightarrow i = 5A$

40. (c) $B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi Ni}{r}$

$$\Rightarrow 3.14 \times 10^{-3} = \frac{10^{-7} \times 2 \times 3.14 \times N \times 10}{(10 \times 10^{-2})} \Rightarrow N = 50$$

41. (c) The magnetic field in the solenoid along its axis (i) At an internal point = $\mu_0 ni$

$$= 4\pi \times 10^{-7} \times 5000 \times 4 = 25.1 \times 10^{-3} \text{ Wb/m}^2$$

(Here $n = 50 \text{ turns/cm} = 5000 \text{ turns/m}$)

(ii) At one end

$$B_{\text{end}} = \frac{1}{2} B_{\text{in}} = \frac{\mu_0 ni}{2} = \frac{25.1 \times 10^{-3}}{2} = 12.6 \times 10^{-3} \text{ Wb/m}^2$$

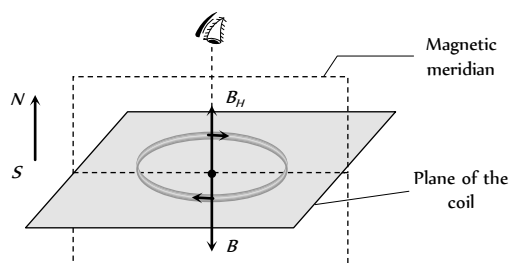
42. (b) Magnetic field at the centre of solenoid (B) = $\mu_0 ni$

Where n = Number of turns / meter

$$\therefore B = 4\pi \times 10^{-7} \times 4250 \times 5 = 2.7 \times 10^{-2} \text{ Wb/m}^2$$

43. (d) Use Right hand palm rule, or Maxwell's Cork screw rule or any other.

44. (b) $B' = n^2 B \Rightarrow B' = (4)^2 B = B' = 16B$
45. (d) $B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} \Rightarrow 12.56 = 10^{-7} \times \frac{2\pi \times i}{5.2 \times 10^{-11}}$
 $\Rightarrow i = 1.04 \times 10^{-3} \text{ A}$
46. (b) $B = \frac{\mu_0}{4\pi} \cdot \frac{\theta}{r} = \frac{\mu_0}{4\pi} \times \frac{\pi}{2} \times \frac{i}{R} = \frac{\mu_0 i}{8R}$
47. (a) $B \propto \frac{1}{r} \Rightarrow \frac{B_1}{B_2} = \frac{r_2}{r_1} \Rightarrow \frac{0.04}{B_2} = \frac{40}{10} \Rightarrow B_2 = 0.01 \text{ T}$
48. (a) See solution 34.
49. (b) $B = \frac{\mu_0 i}{2\pi r}$ or $B \propto \frac{1}{r}$
50. (d) $B = \mu_0 n i = \mu_0 \frac{N}{L} i$
51. (c) Here $B = \mu_0 n i$
 where n is number of turns per unit length $= \frac{N}{l}$
52. (b) $\frac{\mu_0}{4\pi} \times \frac{2\pi i}{r} = H \Rightarrow \frac{(10^{-7}) \times 2 \times 3.142 \times i}{0.05} = 7 \times 10^{-5}$
 $\therefore i = \frac{7 \times 0.05 \times 10^{-5}}{2 \times 3.142 \times 10^{-7}} = \frac{35}{2 \times 3.142} = 5.6 \text{ amp}$
53. (c) $B = \frac{\mu_0 N i}{2r} = \frac{4\pi \times 10^{-7} \times 1000 \times 0.1}{2 \times 0.1} = 6.28 \times 10^{-4} \text{ T}$
54. (b) $B = \frac{\mu_0 N i}{2r} = \frac{4\pi \times 10^{-7} \times 50 \times 2}{2 \times 0.5} = 1.25 \times 10^{-4} \text{ T}$
55. (d) $B \propto \frac{i}{r}$
56. (a) $B = \frac{\mu_0 i}{2\pi r}$ i.e. $B \propto \frac{1}{r}$ i.e. when r is doubled, B is halved.
57. (b) Applying Ampere's law $\oint B \cdot dl = \mu_0 i$ to any closed path inside the pipe we find no current is enclosed. Hence $B = 0$.
58. (a) Magnetic field at the centre of current carrying coil is
 $B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} = \frac{\mu_0 n i}{2r}$
59. (d) The magnetic field is given by $B = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r}$.
 It is independent of the radius of the wire.
60. (d) Magnetic meridian is a vertical N - S plane, the earth's magnetic field (B_H) lies in it. (For more details see magnetism).
 To obtain neutral point at the centre of coil, magnetic field due to current (B) and B_H must cancel each other. Hence plane of the coil and magnetic meridian must be perpendicular to each other as shown



61. (c) $1 \text{ Tesla} = 10^4 \text{ Gauss}$
62. (c)
63. (d)
64. (b) $B = \frac{\mu_0}{4\pi} \times \frac{2i}{r} = 10^{-7} \times \frac{2 \times 1}{10^{-2}} = 2 \times 10^{-5} \text{ Tesla}$
65. (a) Magnetic field due to one side of the square at centre O
 $B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2i \sin 45^\circ}{a/2} \Rightarrow B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2\sqrt{2} i}{a}$
 Hence magnetic field at centre due to all side
 $B = 4B_1 = \frac{\mu_0 (2\sqrt{2} i)}{\pi a}$
 Magnetic field due to n turns
 $B_{\text{net}} = nB = \frac{\mu_0 2\sqrt{2} n i}{\pi a} = \frac{\mu_0 2\sqrt{2} n i}{\pi (2l)} = \frac{\sqrt{2} \mu_0 n i}{\pi l} \quad (\because a = 2l)$
66. (c)
67. (a)
68. (c) Magnetic field on the axis of circular current
 $B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i r^2}{(x^2 + r^2)^{3/2}} \Rightarrow B \propto \frac{nr^2}{(x^2 + r^2)^{3/2}}$
69. (a) $r_1 : r_2 = 1 : 2$ and $B_1 : B_2 = 1 : 3$ We know that
 $B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} \Rightarrow \frac{i_1}{i_2} = \frac{B_1 r_1}{B_2 r_2} = \frac{1 \times 1}{3 \times 2} = \frac{1}{6}$
70. (b) $B = 10^{-7} \cdot \frac{2i}{r} = 10^{-7} \times \frac{2 \times 2}{5} = 8 \times 10^{-8} \text{ T}$
71. (c) $B = \frac{\mu_0}{4\pi} \times \frac{\pi i}{r} \Rightarrow B = 10^{-7} \times \frac{\pi \times 10}{5 \times 10^{-2}} = 6.28 \times 10^{-5} \text{ T}$
72. (c) Magnetic field due to solenoid is independent of diameter (Because $B = \mu_0 n i$).
73. (b) $B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} = 10^{-7} \times \frac{2\pi \times 2}{0.0157} = 8 \times 10^{-5} \text{ Wb/m}^2$
74. (b) $B = \mu_0 n i = 4\pi \times 10^{-7} \times \frac{200}{10^{-2}} \times 2.5 = 6.28 \times 10^{-2} \text{ Wb/m}^2$
75. (d) Magnetic field at centre due to smaller loop
 $B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i_1}{r_1} \dots (i)$

Due to Bigger loop $B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i_2}{r_2}$ So net magnetic field at centre

$$B = B_1 - B_2 = \frac{\mu_0}{4\pi} \times 2\pi \left(\frac{i_1}{r_1} - \frac{i_2}{r_2} \right)$$

According to question $B = \frac{1}{2} \times B_1$

$$\Rightarrow \frac{\mu_0}{4\pi} \cdot 2\pi \left(\frac{i_1}{r_1} - \frac{i_2}{r_2} \right) = \frac{1}{2} \times \frac{\mu_0}{4\pi} \cdot \frac{2\pi i_1}{r_1}$$

$$\frac{i_1}{r_1} - \frac{i_2}{r_2} = \frac{i_1}{2r_1} \Rightarrow \frac{i_1}{2r_1} = \frac{i_2}{r_2} \Rightarrow \frac{i_1}{i_2} = 1 \quad \{r_2 = 2r_1\}$$

76. (b)

$$77. (b) \quad B = \frac{\mu_0}{4\pi} \times \frac{2\pi NiR^2}{(R^2 + x^2)^{3/2}} \Rightarrow B \propto \frac{1}{(r^2 + x^2)^{3/2}}$$

$$\Rightarrow \frac{8}{1} = \frac{(R^2 + x_2^2)^{3/2}}{(R^2 + x_1^2)^{3/2}} \Rightarrow \left(\frac{8}{1} \right)^{2/3} = \frac{R^2 + 0.04}{R^2 + 0.0025}$$

$$\Rightarrow \frac{4}{1} = \frac{R^2 + 0.04}{R^2 + 0.0025} \text{ . On solving } R = 0.1m$$

$$78. (b) \quad B = 10^{-7} \frac{2i}{r} \Rightarrow \frac{B}{B'} = \frac{20}{5} \Rightarrow B' = B/4$$

$$79. (c) \quad B = 10^{-7} \frac{2\pi i}{r} = 10^{-7} \times \frac{2 \times \pi \times 25 \times 4}{5 \times 10^{-2}} = 1.257 \times 10^{-3} T$$

$$80. (b) \quad F = Bil \Rightarrow [B] = \frac{[F]}{[i][l]} = \frac{MLT^{-2}}{AL} = MT^{-2}A^{-1}$$

81. (d) Magnetic field on the axis of conductor is zero.

$$82. (c) \quad B \propto \frac{1}{r} \Rightarrow \frac{B_1}{B_2} = \frac{r_2}{r_1} = \frac{2r}{r} = 2$$

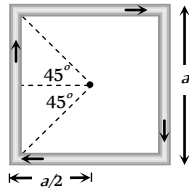
$$83. (c) \quad B = 10^{-7} \times \frac{2i}{r} = 10^{-7} \times \frac{2 \times 1}{1} = 2 \times 10^{-7} T$$

84. (d) At midpoint, magnetic fields due to both the wires are equal and opposite. So $B_{\text{net}} = 0$.

$$85. (c) \quad B_0 = 4 \times \frac{\mu_0}{4\pi} \times \frac{i}{(a/2)} (\sin 45^\circ + \sin 45^\circ)$$

$$= 4 \times \frac{\mu_0}{4\pi} \times \frac{2i}{a} \times \frac{2}{\sqrt{2}}$$

$$= \frac{\mu_0 i 2\sqrt{2}}{\pi a}$$



$$86. (b) \quad B = 10^{-7} \frac{2\pi i}{r}; \text{ according to question } B_{\text{net}} = B$$

$$\Rightarrow 5 \times 10^{-5} = 10^{-7} \times \frac{2 \times 3.14 \times i}{5 \times 10^{-2}} \Rightarrow i = 4A$$

$$87. (d) \quad B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi Ni}{r} = \frac{10^{-7} \times 2\pi \times 100 \times 0.1}{5 \times 10^{-2}} = 4\pi \times 10^{-5} T$$

88. (a) Corresponding current $i = en$

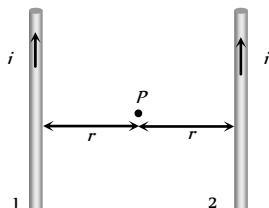
$$\text{So } B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi(en)}{r} = \frac{\mu_0 ne}{2r}$$

$$89. (a) \quad B \text{ at ends of solenoid is } \frac{\mu_0}{2} ni$$

90. (b) Use Right hand palm rule or Maxwell's Cork screw rule.

91. (c) At P

$$\Rightarrow B_{\text{net}} = B_1 - B_2$$



$$\text{Since } |B_1| = |B_2|$$

$$\text{So, } B_{\text{net}} = 0$$

92. (d) A moving charge and changing electric field both produces magnetic field.

93. (d)

$$94. (a) \quad B = \frac{\mu_0}{2\pi} \frac{i}{r} \Rightarrow 5 \times 10^{-5} = \frac{\mu_0}{2\pi} \times \frac{\pi}{r} \Rightarrow r = 10^4 \mu_0 \text{ metre}$$

$$95. (b) \quad B' = n^2 B = (3)^2 B = 9B$$

96. (b) B represents the magnetic field.

97. (a)

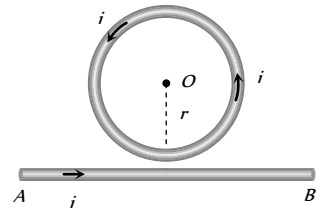
$$98. (d) \quad \frac{B_c}{B_a} = \left(1 + \frac{x^2}{a^2} \right)^{3/2} = \left(1 + \frac{a^2}{a^2} \right)^{3/2} = (1+1)^{3/2} = 2\sqrt{2}$$

99. (c) The given circuit can be considered as follows

$$B_{\text{loop}} = \frac{\mu_0 i}{2r} \odot$$

$$B_{\text{conductor}} = \frac{\mu_0 i}{2\pi r} \odot$$

$$B_{\text{net}} = \frac{\mu_0 i}{2\pi r} (\pi + 1) \odot$$



$$100. (d) \quad B = \frac{\mu_0 \mu_r Ni}{2\pi r} \Rightarrow 1 = \frac{4\pi \times 10^{-7} \times \mu_r \times 400 \times 2}{0.4} \Rightarrow \mu_r = 400$$

101. (b)

102. (c)

$$103. (b) \quad B = 10^{-7} \times \frac{\pi \times i}{r} = 10^{-7} \times \frac{\pi \times 10}{20 \times 10^{-2}} = B = 5\pi \mu T$$

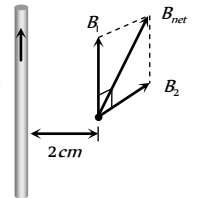
$$104. (d) \quad B' = n^2 B = (2)^2 B = 4B$$

$$105. (a) \quad B = \mu_0 ni \Rightarrow \frac{B}{B'} = \frac{n}{n'} \times \frac{i}{i'} = \frac{1}{(1/2)} \times \frac{1}{2} = 1 \Rightarrow B' = B$$

$$106. (c) \quad B_1 = 4 \times 10^{-4} T$$

$$B_2 = 10^{-7} \times \frac{2 \times 30}{2 \times 10^{-2}} = 3 \times 10^{-4} T$$

$$\therefore B_{\text{net}} = \sqrt{B_1^2 + B_2^2} = 5 \times 10^{-4} T$$



107. (b) Magnetic field at the centre of circular loop

$$B = \frac{\mu_0}{4\pi} \frac{2\pi i}{r} \Rightarrow 0.5 \times 10^{-5} = \frac{10^{-7} \times 2 \times 3.14 \times i}{5 \times 10^{-2}}$$

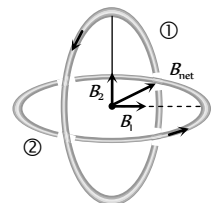
$$i = 0.4 A$$

108. (a)

$$109. (a) \quad B_1 = B_2 = B = \frac{\mu_0}{4\pi} \times \frac{2\pi i}{r}$$

$$B_{\text{net}} = \sqrt{2} B$$

$$\Rightarrow \frac{B}{B_{\text{net}}} = \frac{1}{\sqrt{2}}$$



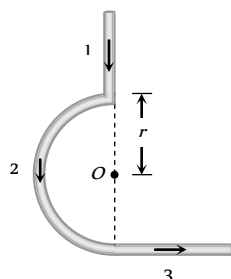
110. (c) Magnetic field due to different parts are

$$B = 0$$

$$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{r} \odot$$

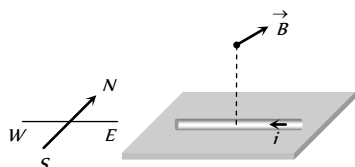
$$B_3 = \frac{\mu_0}{4\pi} \cdot \frac{i}{r} \odot$$

$$\therefore B_{\text{net}} = B_2 + B_3 = \frac{\mu_0 i}{4r} + \frac{\mu_0 i}{4\pi r}$$



111. (d) $M = niA = n(\pi r^2) \Rightarrow M \propto r^2$

112. (a)



113. (b) Magnetic field at the center of single turn loop

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r}, \text{ magnetic field at the center of } n\text{-turn loop}$$

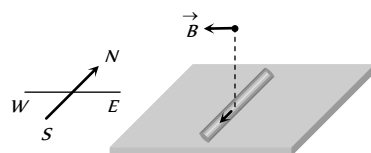
$$B_n = \left(\frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r/n} \right) \times n \Rightarrow B_n = n^2 B$$

114. (a) $\frac{B_{\text{center}}}{B_{\text{axis}}} = \left(1 + \frac{x^2}{r^2} \right)^{3/2} \Rightarrow \frac{B_{\text{center}}}{54} = \left(1 + \left(\frac{4}{3} \right)^2 \right)^{3/2} = \frac{125}{27}$

$$B_{\text{center}} = 250 \mu T$$

115. (b) $B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} \Rightarrow B \propto \frac{1}{r}$

116. (d)



117. (c) Suppose length of each wire is l . $A_{\text{square}} = \left(\frac{l}{4} \right)^2 = \frac{l^2}{16}$

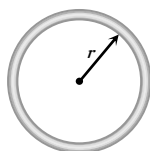
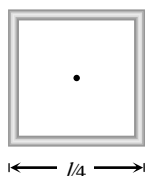
$$A_{\text{circle}} = \pi r^2 = \pi \left(\frac{l}{2\pi} \right)^2 = \frac{l^2}{4\pi}$$

$\therefore$ Magnetic moment

$$M = iA$$

$$\Rightarrow \frac{M_{\text{square}}}{M_{\text{circle}}} = \frac{A_{\text{square}}}{A_{\text{circle}}}$$

$$= \frac{l^2 / 16}{l^2 / 4\pi} = \frac{\pi}{4}$$



118. (b) $B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} \Rightarrow B \propto ni$

119. (a) Magnetic field due to revolution of electron

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} = \frac{\mu_0}{4\pi} \cdot \frac{2\pi \cdot \left(\frac{e\omega}{2\pi} \right)}{r} = 10^{-7} \times \frac{e\omega}{r}$$

$$\Rightarrow 16 = 10^{-7} \times \frac{1.6 \times 10^{-19} \omega}{1 \times 10^{-10}} \Rightarrow \omega = 10^{17} \text{ rad/sec.}$$

120. (a) $B = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r} = 10^{-7} \times \frac{2 \times 20}{10 \times 10^{-2}} = 4 \times 10^{-5} \text{ Wb/m}^2$

121. (a) $B = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r} \Rightarrow B \propto i$

122. (a) $B_{\text{net}} = \sqrt{B_1^2 + B_2^2} = \frac{\mu_0}{4\pi} \cdot \frac{2\pi}{r} \sqrt{i_1^2 + i_2^2}$
 $= 10^{-7} \times \frac{2\pi}{2\pi \times 10^{-2}} \sqrt{(3)^2 + (4)^2} = 5 \times 10^{-5} \text{ wb/m}^2$

123. (c) When two parallel conductors carrying current I and $2I$ in same direction, then magnetic field at the midpoint is

$$B = \frac{\mu_0 2I}{2\pi r} - \frac{\mu_0 I}{2\pi r} = \frac{\mu_0 I}{2\pi r}$$

When current $2I$ is switched off then magnetic field due to

$$\text{conductor carrying current } I \text{ is } B = \frac{\mu_0 I}{2\pi r}.$$

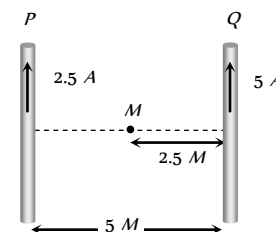


124. (d) In the following figure magnetic field at mid point M is given by

$$B_{\text{net}} = B_Q - B_P$$

$$= \frac{\mu_0}{4\pi} \cdot \frac{2}{r} (i_Q - i_P)$$

$$= \frac{\mu_0}{4\pi} \times \frac{2}{2.5} (5 - 2.5) = \frac{\mu_0}{2\pi}$$



125. (c)

126. (b) The magnetic field due to small element of conductor of length

$$\text{is given by } dB = \frac{\mu_0}{4\pi} \frac{Idl \sin \theta}{r^2}$$

This value will be maximum when

$$\sin \theta = 1 = \sin 90^\circ \text{ or, } \theta = 90^\circ$$

Motion of Charged Particle in Magnetic Field

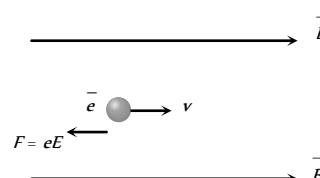
1. (a, b, d)

Here the proton has no acceleration so $E = B = 0$.

When $E = 0$ but $B \neq 0$, but parallel to the motion of proton, there will be no force acting.

When $E \neq 0$ and $B \neq 0$ and E, B and motion of proton (v) are mutually perpendicular, there may be no net force. Forces due to E and B cancel each other.

2. (d) Since electron is moving parallel to the magnetic field, hence magnetic force on it $F_m = 0$.



The only force acting on the electron is electric force which reduces its speed.

$$3. \quad (c) \quad r = \frac{\sqrt{2mk}}{qB} = \frac{1}{B} \sqrt{\frac{2mV}{q}} \Rightarrow r \propto \sqrt{m} \Rightarrow \frac{m_1}{m_2} = \left(\frac{R_1}{R_2} \right)^2$$

$$4. \quad (a) \quad r = \frac{mv}{Bq} = \frac{v}{(q/m)B} = \frac{2 \times 10^5}{5 \times 10^7 \times 4 \times 10^{-2}} = 0.1m$$

$$5. \quad (b) \quad r = \frac{p}{qB} \Rightarrow r \propto p$$

$$6. \quad (b) \quad B = \frac{mv}{qr} = \frac{9 \times 10^{-31} \times 10^6}{1.6 \times 10^{-19} \times 0.1} = 5.6 \times 10^{-5} T$$

$$7. \quad (c) \quad r = \frac{\sqrt{2mK}}{qB} \text{ i.e. } r \propto \frac{\sqrt{m}}{q}$$

Here kinetic energy K and B are same.

$$\therefore \frac{r_p}{r_\alpha} = \frac{\sqrt{m_p}}{\sqrt{m_\alpha}} \cdot \frac{q_\alpha}{q_p} = \frac{\sqrt{m_p}}{\sqrt{4m_p}} \cdot \frac{2q_p}{q_p} = 1$$

$$8. \quad (c) \quad \vec{F} = q\vec{v} \times \vec{B}$$

$$9. \quad (c)$$

$$10. \quad (d)$$

$$11. \quad (c) \quad \text{East, (By } \vec{F} = q(\vec{v} \times \vec{B})) \text{ or by applying Fleming's left hand rule.}$$

$$12. \quad (d) \quad F = qvB = 1.6 \times 10^{-19} \times \left[\sqrt{\frac{2E}{m}} \right] \times 2.5$$

$$= 4 \times 10^{-19} \sqrt{\frac{2 \times 2 \times 1.6 \times 10^{-19} \times 10^6}{1.66 \times 10^{-27}}} = 7.6 \times 10^{-12} N$$

$$13. \quad (d) \quad \vec{F} = q(\vec{v} \times \vec{B}); \text{ if } \vec{v} \parallel \vec{B} \text{ then } \vec{F} = 0$$

$$14. \quad (b) \quad \text{This is according to the cross product } \vec{F} = q(\vec{v} \times \vec{B}) \text{ otherwise can be evaluated by the left-hand rule of Fleming.}$$

$$15. \quad (a) \quad r = \frac{\sqrt{2mK}}{qB} \Rightarrow K \propto \frac{q^2}{m} \Rightarrow \frac{K_p}{K_\alpha} = \left(\frac{q_p}{q_\alpha} \right)^2 \times \frac{m_\alpha}{m_p}$$

$$\Rightarrow \frac{1}{K_\alpha} = \left(\frac{q_p}{2q_p} \right)^2 \times \frac{4m_p}{m_p} = 1 \Rightarrow K_\alpha = 1 \text{ MeV.}$$

$$16. \quad (c) \quad r \propto \frac{1}{B} \text{ i.e. } \frac{r_1}{r_2} = \frac{B_2}{B_1} \Rightarrow r_2 = \frac{B_1}{B_1/2} \times r = 2r$$

$$17. \quad (c) \quad \text{Time period of proton } T_p = \frac{2\pi m}{5} = 5 \mu \text{ sec}$$

$$\text{By using } T = \frac{2\pi m}{qB} \Rightarrow \frac{T_\alpha}{T_p} = \frac{m_\alpha}{m_p} \times \frac{q_p}{q_\alpha} = \frac{4m_p}{m_p} \times \frac{q_p}{2q_p}$$

$$\Rightarrow T_\alpha = 2T_p = 10 \mu \text{ sec.}$$

$$18. \quad (a) \quad F = ma = qvB \Rightarrow a = \frac{qvB}{m} = \frac{1.6 \times 10^{-19} \times 2 \times 3.4 \times 10^7}{1.67 \times 10^{-27}}$$

$$= 6.5 \times 10^8 \text{ m/sec}$$

$$19. \quad (c) \quad T = \frac{2\pi m}{qB} = \frac{2\pi}{v} = \frac{2 \times 3.14 \times 0.45}{2.6 \times 10^7} = 1.08 \times 10^{-7} \text{ sec}$$

$$20. \quad (b) \quad F = qvB \text{ and } K = \frac{1}{2}mv^2 \Rightarrow F = qB\sqrt{\frac{2K}{m}}$$

$$= 1.6 \times 10^{-19} \times 1.5 \sqrt{\frac{2 \times 5 \times 10^6 \times 1.6 \times 10^{-19}}{1.7 \times 10^{-27}}}$$

$$= 7.344 \times 10^{-12} N$$

$$21. \quad (c) \quad \text{Magnetic force acts on a moving charge.}$$

$$22. \quad (c) \quad r = \frac{mv}{qB} \Rightarrow r \propto v, \Rightarrow r_2 = 2r_1 = 2 \times 2 = 4 \text{ cm}$$

$$23. \quad (d) \quad r = \frac{\sqrt{2mK}}{qB} \Rightarrow K \propto \frac{q^2}{m}$$

$$\Rightarrow \frac{K_p}{K_d} = \left(\frac{q_p}{q_d} \right)^2 \times \frac{m_d}{m_p} = \left(\frac{1}{1} \right)^2 \times \frac{2}{1} = \frac{2}{1}$$

$$\Rightarrow K_p = 2 \times 50 = 100 \text{ keV.}$$

$$24. \quad (b) \quad \text{Maximum force will act on proton so it will move on a circular path. Force on electron will be zero because it is moving parallel to the field.}$$

$$25. \quad (d) \quad \text{Fleming's left hand rule is used to determine the direction of force.}$$

$$26. \quad (a) \quad \text{Lorentz force is given by}$$

$$\vec{F} = \vec{F}_e + \vec{F}_m = q\vec{E} + q(\vec{v} \times \vec{B}) = q[\vec{E} + (\vec{v} \times \vec{B})]$$

$$27. \quad (c) \quad \vec{F} = q\vec{v} \times \vec{B}$$

$$28. \quad (a) \quad F = qvB \sin \theta$$

$$= 1.6 \times 10^{-19} \times 2 \times 10^7 \times 1.5 \sin 30^\circ$$

$$= 1.6 \times 10^{-19} \times 2 \times 10^7 \times 1.5 \times \frac{1}{2} = 2.4 \times 10^{-12} N$$

$$29. \quad (d) \quad F = qvB \sin \theta \Rightarrow B = \frac{F}{qv \sin \theta}$$

$$B_{\min} = \frac{F}{qv} \quad (\text{when } \theta = 90^\circ)$$

$$\therefore B_{\min} = \frac{F}{qv} = \frac{10^{-10}}{10^{-12} \times 10^5} = 10^{-3} \text{ Tesla in } \hat{z} \text{-direction.}$$

$$30. \quad (d) \quad \text{Kinetic energy in magnetic field remains constant and it is}$$

$$K = qV \Rightarrow K \propto q \quad (V = \text{constant})$$

$$\therefore K_p : K_d : K_\alpha = q_p : q_d : q_\alpha = 1 : 1 : 2$$

$$31. \quad (d) \quad \text{When charged particle enters perpendicularly in a magnetic field, it moves on a circular path with a constant speed. Hence its kinetic energy also remains constant.}$$

$$32. \quad (b) \quad r = \frac{\sqrt{2mK}}{qB} \text{ i.e. } r \propto \frac{\sqrt{m}}{q}$$

$$\therefore \frac{r_e}{r_p} = \sqrt{\frac{m_e}{m_p}} \times \frac{q_p}{q_e} \Rightarrow \frac{r_e}{r_p} = \sqrt{\frac{m_e}{m_p}} \quad (\because q_e = q_p)$$

Here kinetic energy K and B are same.

Since $m_e < m_p$, therefore $r_e < r_p$.

33. (c) Path of the proton will be a helix of radius $r = \frac{mv \sin \theta}{qB}$

(where θ = Angle between $\vec{B}$ and $\vec{v}$)

$$\Rightarrow r = \frac{1.67 \times 10^{-27} \times 2 \times 10^6 \times \sin 30^\circ}{1.6 \times 10^{-19} \times 0.104}$$

$$= 0.1 \text{ m}$$

$$\text{Time period } T = \frac{2\pi m}{qB} = \frac{2\pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.104}$$

$$= 2\pi \times 10^{-7} \text{ sec}$$

34. (a) $\frac{mv^2}{R} = qvB$. For proton $R_p = \frac{mv}{qB} = \frac{\sqrt{2m_p E}}{qB}$

$$\text{and for deuteron } R_d = \frac{\sqrt{2m_d E}}{qB}$$

$$\Rightarrow \frac{R_d}{R_p} = \sqrt{\frac{m_d}{m_p}} = \sqrt{2} \Rightarrow R_d = \sqrt{2} R_p$$

35. (c) In this case $|\vec{F}_e| = |\vec{F}_m|$ and both forces are opposite to each other.

36. (b) We know that time period $T = \frac{2\pi m}{qB}$ i.e. $T \propto m$

(Since q and B are same)

$\therefore$ Mass of proton > Mass of electron

$\therefore$ Time period of proton > Time period of electron

37. (d) According to Fleming's right hand rule.

38. (a) $r = \frac{mv}{eB} \Rightarrow \frac{e}{m} = \frac{v}{rB}$

39. (a) Using $eE = evB \Rightarrow E = vB = 5 \times 10^6 \times 0.02 = 10^5 \text{ Vm}^{-1}$

40. (d) $F = evB = 1.6 \times 10^{-19} \times 4 \times 10^6 \times 2 \times 10^{-1} = 1.28 \times 10^{-13} \text{ N}$

$$\text{Also } \frac{mv^2}{r} = evB \Rightarrow r = \frac{mv}{eB}$$

$$\Rightarrow r = \frac{9 \times 10^{-31} \times 4 \times 10^6}{1.6 \times 10^{-19} \times 2 \times 10^{-1}} = 1.1 \times 10^{-4} \text{ m}$$

41. (c) Force acts perpendicular to the velocity in a magnetic field, so speed of electron will remain same.

42. (a) By Fleming left hand rule.

43. (d) Direction of motion of proton is same as that of direction of magnetic field.

44. (a) Time period is given by $T = \frac{2\pi m}{qB}$

$$\Rightarrow \text{Frequency } \nu = \frac{1}{T} = \frac{qB}{2\pi m}$$

45. (b) $r = \frac{\sqrt{2mK}}{qB} = \frac{1}{B} \sqrt{\frac{2mV}{q}}$

$$= \frac{1}{10^{-3}} \sqrt{\frac{2 \times 9 \times 10^{-31} \times 12000}{1.6 \times 10^{-19}}} = 0.367 \text{ m} = 36.7 \text{ cm}$$

46. (c) $r = \frac{1}{B} \sqrt{\frac{2mV}{q}} \Rightarrow r \propto \sqrt{\frac{m}{q}} \Rightarrow \frac{r_x}{r_y} = \sqrt{\frac{m_x}{m_y} \times \frac{q_y}{q_x}}$

$$\Rightarrow \frac{R_1}{R_2} = \sqrt{\frac{m_x}{m_y} \times \frac{2}{1}} \Rightarrow \frac{m_x}{m_y} = \frac{R_1^2}{2R_2^2}$$

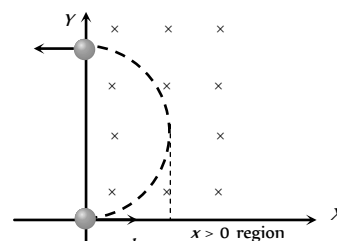
47. (d) $\vec{F} = q(\vec{v} \times \vec{B}) = 10^{-11} (10^8 \hat{j} \times 0.5 \hat{i})$

$$= 5 \times 10^{-4} (\hat{j} \times \hat{i}) = 5 \times 10^{-4} \text{ N}(-\hat{k})$$

48. (b) It is easy to understand the given problem, along with the following figure.

d = radius of path

$$= \frac{mv}{qB}$$



49. (c) Lorentz force $F = q(\vec{v} \times \vec{B})$ or $|F| = qvB \sin \theta$

F will be maximum, when $\theta = 90^\circ$

50. (d) The component of velocity perpendicular to H will make the motion circular while that parallel to H will make it move along a straight line. The two together will make the motion helical.

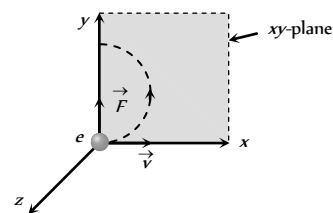
51. (b) We have $qvB = \frac{mv^2}{r}$ or $r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$

For same kinetic energy K , we have $r \propto \sqrt{m}$

Hence path of proton will have larger r and is therefore less curved.

52. (c) When particle enters at angle other than 0° or 90° or 180° , path followed is helix.

53. (b) To move the electron in xy plane, force on it must be acting in the y -direction initially. The direction of $\vec{F}$ is known, and the direction of $\vec{v}$ is known, hence by applying Fleming's left hand rule, the direction of magnetic field is also determined.



54. (a) A moving charge gains energy in electric field only because in magnetic field energy remains constant.

55. (a) Given that $K_p = K_d = K_\alpha = K$ (say)

We know that $m_e = m$, $m_p = 2m$ and $m_\alpha = 4m$ and $q_e = e$, $q_p = e$ and $q_\alpha = 2e$

$$\text{Further } r = \frac{\sqrt{2mK}}{qB} \Rightarrow r_p = \frac{\sqrt{2mK}}{eB}, r_d = \frac{\sqrt{2(2m)K}}{eB} = \sqrt{2}r_p$$

$$\text{and } r_\alpha = \frac{\sqrt{2(4m)K}}{(2e)B} = r_p. \text{ Hence } r_\alpha = r_p < r_d$$

56. (d) Since force is perpendicular to direction of motion, energy and magnitude of momentum remains constant.

57. (d) $T = \frac{2\pi m}{qB} \Rightarrow T \propto v^0$

58. (b) $F = qvB$ also Kinetic energy $K = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2K}{m}}$

$$\therefore F = q\sqrt{\frac{2K}{m}}B$$

$$= 1.6 \times 10^{-19} \sqrt{\frac{2 \times 200 \times 10^6 \times 1.6 \times 10^{-19}}{1.67 \times 10^{-27}}} \times 5$$

$$= 1.6 \times 10^{-10} \text{ N}$$

59. (d) The deflection produced by the electric field may be nullified by that produced by magnetic field.

60. (a) $\vec{F}_m = q(\vec{v} \times \vec{B})$

When the angle between $\vec{v}$ and $\vec{B}$ is 180° , $F_m = 0$

61. (c) $r = mv / qB$

Since both have same momentum, therefore the circular path of both will have the same radius.

62. (c) When particle enters perpendicularly in a magnetic field, it moves along a circular path with constant speed.

63. (b) For motion of a charged particle in a magnetic field, we have $r = mv/qB$ i.e. $r \propto v$

64. (a) The charged particle moving in a magnetic field does not gain energy. However, the direction of its velocity changes continuously. Hence momentum changes.

65. (a) $F = qvB \sin \theta = qvB \sin 0 = 0$

66. (b) $r = \frac{mv}{qB} = \frac{10^7}{10^{11} \times 10^{-4}} = 1 \text{ m } (\because q/m = 10^{11} \text{ C/kg})$

67. (b) $\omega = \frac{2\pi}{T} = \frac{qB}{m} \Rightarrow \omega \propto v^0 \left(\because T = \frac{2\pi m}{qB} \right)$

68. (a) $r = \frac{\sqrt{2mK}}{qB} \Rightarrow r \propto \sqrt{K} \Rightarrow \frac{R}{R_2} = \sqrt{\frac{K}{2K}} \Rightarrow R_2 = R\sqrt{2}$

69. (b, d) $r = \frac{mv}{qB} = \frac{P}{qB}$

70. (b) $F = qvB \sin \theta = 1.6 \times 10^{-19} \times 2.5 \times 2.5 \times 10^7 \sin 30^\circ$

$$F = 1.6 \times 10^{-19} \times 6.25 \times 10^7 \times \frac{1}{2} = 5 \times 10^{-12} \text{ N}$$

71. (c) $K_{\max} = \frac{1}{2}mv^2$ and $r_0 = \frac{mv}{qB} \Rightarrow v = \frac{qBr_0}{m}$

$$\Rightarrow K_{\max} = \frac{1}{2}m \left(\frac{qBr_0}{m} \right)^2 = \frac{q^2 B^2 r_0^2}{2m}$$

72. (b) $F = qvB \sin \theta$; Independent of mass

73. (b) By Fleming left hand rule.

74. (a) $F = qBv = 1 \times 0.5 \times 10 = 5 \text{ N}$

75. (d) $r = \frac{mv}{qB} \Rightarrow \frac{r_1}{r_2} = \frac{v_1}{v_2} \times \frac{B_2}{B_1} \Rightarrow \frac{r_1}{r_2} = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

$$r_2 = 4r_1$$

76. (d) Magnetic force on charge will be zero.

77. (d)

78. (b) Apply Fleming's left hand rule.

79. (a) $T = \frac{2\pi m}{qB} = \frac{2 \times 3.14 \times 9 \times 10^{-31}}{1.6 \times 10^{-19} \times 1 \times 10^{-4}} = 3.5 \times 10^{-7} \text{ sec}$

80. (d) $\vec{F} = q(\vec{v} \times \vec{B}) = 0$ as $\vec{v}$ and $\vec{B}$ are parallel.

81. (a) Here magnetic force is zero, but the velocity increases due to electric force.

82. (a)

83. (c)

84. (b) $r = \frac{mv}{qB} \Rightarrow r \propto mv$ (q and B are constant)

$$\therefore r_A > r_B \Rightarrow m_A v_A > m_B v_B$$

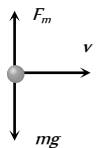
85. (b) $r = \frac{p}{qB} \Rightarrow p \propto q$ ($\because r$ and B are constant)

$$\frac{p_p}{p_\alpha} = \frac{q_p}{q_\alpha} = \frac{q_p}{(2q_p)} = \frac{1}{2}$$

86. (c) Particle will move with uniform velocity when its acceleration is zero.

$$\text{i.e. } |F_m| = mg \Rightarrow mg = qvB$$

$$\Rightarrow B = \frac{mg}{qv} = \frac{0.6 \times 10^{-3} \times 10}{25 \times 10^{-9} \times 1.2 \times 10^4} = 20 \text{ T}$$



87. (c) $r = \frac{mv}{qB} \Rightarrow \frac{r_\alpha}{r_p} = \frac{m_\alpha}{m_p} \times \frac{q_p}{q_\alpha} = \frac{4}{1} \times \frac{1}{2} = \frac{2}{1}$

88. (b) When field is parallel to the direction of motion of charge, magnetic force on it is zero.

89. (a) Since $\vec{F}$ and $\vec{v}$ are perpendicular to each other work done by force is zero. Hence K.E. is constant.

90. (b)

91. (a) Charged particles deflects in magnetic field.

92. (a) $v = \frac{qB}{2\pi m} \Rightarrow v \propto \frac{q}{m}$

$$\left(\frac{q}{m} \right)_{Li^+} \text{ is minimum so } v_{Li^+} \text{ is minimum.}$$

93. (c) $r = \frac{\sqrt{2mK}}{qB} \Rightarrow r \propto \frac{\sqrt{m}}{q} \Rightarrow \frac{r_{He^+}}{r_{O^{++}}} = \sqrt{\frac{m_{He^+}}{m_{O^{++}}}} \times \frac{q_{O^{++}}}{q_{He^+}}$

$$= \sqrt{\frac{4}{16}} \times \frac{2}{1} = \frac{1}{1}. \text{ Then will deflect equally.}$$

$$94. (d) r = \frac{\sqrt{2mE}}{qB} = \frac{\sqrt{2 \times 9 \times 10^{-31} \times 7.2 \times 10^{-18}}}{1.6 \times 10^{-19} \times 9 \times 10^{-5}} \\ = 0.25 \text{ m} = 25 \text{ cm}$$

$$95. (c) v = \frac{E}{B} = \frac{20}{5} = 4 \text{ m/sec}$$

96. (a) Because magnetic force on charge will be zero.

$$97. (c) W = F.d \cos 90^\circ = 0$$

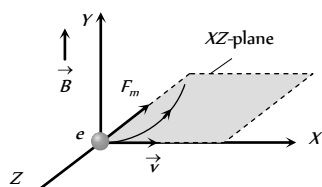
98. (b) Since particle is moving undeflected.

$$\text{So } qE = qvB \Rightarrow B = E/v = \frac{10^4}{10} = 10^3 \text{ Wb/m}^2$$

$$99. (a) r = \frac{mv}{qB} \Rightarrow \frac{r_1}{r_2} = \frac{m_1 v_1}{m_2 v_2} \times \frac{q_2}{q_1} = \frac{1 \times 2}{1 \times 3} \times \frac{2}{1} = \frac{4}{3}$$

$$100. (b) \vec{F} = -e(\vec{v} \times \vec{B}) \Rightarrow \vec{F} = -e[\hat{v} \times \hat{B}] = evB[-\hat{k}]$$

i.e. Force on electron is acting towards negative z -axis. Hence particle will move on a circle in xz -plane.



101. (d) Particles entering perpendicularly, hence they will describe circular path. Since their masses are different so they will describe path of different radii.

$$102. (c) r = \frac{mv}{qB} = \frac{6 \times 10^7}{1.7 \times 10^{11} \times 1.5 \times 10^{-2}} = 2.35 \text{ cm}$$

$$103. (d) \text{ Cyclotron frequency } \nu = \frac{Bq}{2\pi m}$$

$$\Rightarrow \nu = \frac{1 \times 1.6 \times 10^{-19}}{2 \times 3.14 \times 9.1 \times 10^{-31}} = 2.79 \times 10^{10} \text{ Hz}$$

$$= 27.9 \times 10^9 \text{ Hz} \approx 28 \text{ GHz}$$

104. (d) By Fleming's left hand rule.

$$105. (c) r = \frac{\sqrt{2mK}}{qB} \Rightarrow q \propto \sqrt{mK} \Rightarrow K \propto \frac{q^2}{m}$$

$$\Rightarrow \frac{K_\alpha}{K_p} = \left(\frac{q_\alpha}{q_p} \right)^2 \times \frac{m_p}{m_\alpha} \Rightarrow \frac{K_\alpha}{8} = \left(\frac{2q_p}{q_p} \right)^2 \times \frac{m_p}{4m_p} = 1$$

$$\Rightarrow K_\alpha = 8 \text{ eV}$$

$$106. (c) \text{ By using } r = \frac{mv}{qB} = \frac{v}{\left(\frac{q}{m}\right)B} \Rightarrow r \propto \frac{1}{\left(\frac{q}{m}\right)}$$

$$\therefore \left(\frac{q}{m} \right)_{e^-} > \left(\frac{q}{m} \right)_{p^+} > \left\{ \left(\frac{q}{m} \right)_d = \left(\frac{q}{m} \right)_\alpha \right\}$$

$$\therefore R_d = R_\alpha$$

107. (d) By using Fleming's left hand rule.

108. (d) Along the axis of coil, $\vec{v}$ and $\vec{B}$ are parallel, so $F = 0$

109. (d) $F_m = qvB \sin \theta$, if $v = 0 \Rightarrow F_m = 0$

$$110. (d) T = \frac{2\pi m}{qB} = \frac{2 \times 3.14 \times 9.1 \times 10^{-31}}{1.6 \times 10^{-19} \times 3.534 \times 10^{-5}} \\ = 1 \times 10^{-6} \text{ sec} = 1 \mu\text{sec.}$$

111. (a)

112. (b)

113. (d) Magnetic field produced by wire at the location of charge is perpendicular to the paper inwards. Hence by applying Fleming's left hand rule, force is directed along OY .

114. (c) From Fleming's left hand rule the force on electron is towards the east means it is deflected towards east.

115. (c) Electric current corresponds to the revolution of electron is

$$i = \frac{ev}{2\pi r}$$

Magnetic field due to circular current at the centre

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} = \frac{\mu_0}{4\pi} \cdot \frac{ev}{r^2} \Rightarrow r = \sqrt{\frac{\mu_0}{4\pi} \cdot \frac{ev}{B}} \Rightarrow r \propto \sqrt{\frac{v}{B}}$$

116. (c) When electron moves in both electric and magnetic field then $qE = qvB$.

$$\therefore v = \frac{E}{B} = \frac{1500}{0.40} = 3750 \text{ m/s} = 3.75 \times 10^3 \text{ m/s}$$

117. (c) For no deflection in mutually perpendicular electric and

$$\text{magnetic field } v = \frac{E}{B} = \frac{3.2 \times 10^5}{2 \times 10^{-3}} = 1.6 \times 10^8 \text{ m/s}$$

If electric field is removed then due to only magnetic field radius of the path described by electron

$$r = \frac{mv}{qB} = \frac{9.1 \times 10^{-31} \times 1.6 \times 10^8}{1.6 \times 10^{-19} \times 2 \times 10^{-3}} = 0.45 \text{ m}$$

$$118. (c) r = \frac{mv}{qB} \Rightarrow r \propto v$$

Force and Torque on Current Carrying Conductor

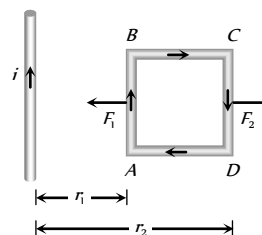
1. (b) Two wires, if carries current in opposite direction, they repel each other.

2. (c) $\therefore r_1 < r_2$

$$\text{So } F_1 > F_2$$

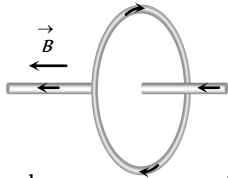
$$\Rightarrow F_{\text{net}} = (F_1 - F_2)$$

towards the wire.



$$3. (b) M = NiA = 20 \times \frac{22}{7} (4 \times 10^{-2})^2 3 = 0.3 \text{ A-m}^2$$

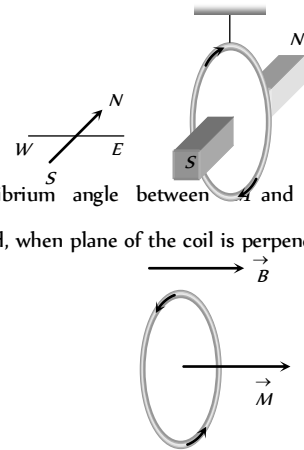
4. (c) Net force on a current carrying closed loop is always zero, if it is placed in a uniform magnetic field.
5. (b) Force per unit length $= \frac{\mu_0}{4\pi} \cdot \frac{2i_1 i_2}{r} = \frac{\mu_0}{2\pi} \cdot \frac{i^2}{b}$
6. (a) $F = \frac{\mu_0}{4\pi} \cdot \frac{2i_1 i_2}{a} \times l \Rightarrow F = 10^{-7} \times \frac{2 \times 10 \times 2}{(10 \times 10^{-2})} \times 2 = 8 \times 10^{-5} \text{ N}$
7. (b) For charge particles, if they are moving freely in space, electrostatic force is dominant over magnetic force between them. Hence due to electric force they repel each other.
8. (b) As shown in the following figure straight wire is placed parallel to the magnetic field produced by circular current. Hence force on wire $F = 0$



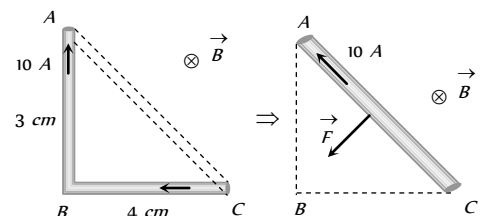
9. (a) Two straight conductors carry current in same direction, then attractive force acts between them.
10. (a) $F = \frac{\mu_0}{4\pi} \cdot \frac{2 \times i_1 i_2}{a} = \frac{10^{-7} \times 2 \times 5 \times 5}{0.1} = 5 \times 10^{-5} \text{ N/m}$
11. (c) $F = \frac{\mu_0}{4\pi} \cdot \frac{2i_1 i_2}{a} = 10^{-3} \text{ N}$
When current in both the wires is doubled, then
 $F' = \frac{\mu_0}{4\pi} \cdot \frac{2(2i_1 \times 2i_2)}{a} = 4 \times 10^{-3} \text{ N}$
12. (a) The magnetic moment of current carrying loop
 $M = niA = ni(\pi r^2)$
Hence the work done in rotating it through 180°
 $W = MB(1 - \cos \theta) = 2MB = 2(ni\pi r^2)B$
 $= 2 \times (50 \times 2 \times 3.14 \times 16 \times 10^{-4}) \times 0.1 = 0.1 \text{ J}$
13. (c) $F = Bil \sin \theta$
 $= 500 \times 10^{-4} \times 3 \times (40 \times 10^{-2}) \times \frac{1}{2} = 3 \times 10^{-2} \text{ N}$
14. (c) $M = i\pi r^2$
15. (a) Because $\tau = NiAB \cos \theta$
16. (c)
17. (b) $\theta = \frac{NiAB}{C} \Rightarrow \theta \propto N$ (Number of turns)
18. (b) Magnet provides damping.
19. (b) $i = \frac{C\theta}{NAB} \Rightarrow i \propto \theta$
20. (b) Force per unit length on two parallel current carrying conductor is given by $\frac{F}{l} = 10^{-7} \times 2 \frac{i_1 i_2}{a}$
 $\Rightarrow \frac{F}{l} = 10^{-7} \times 2 \times \frac{1 \times 1}{1} = 2 \times 10^{-7} \text{ N/m}$
21. (d) $\tau = MB \sin \theta \Rightarrow \tau_{\max} = NiAB$, ($\theta = 90^\circ$)
22. (b) $W = MB(\cos \theta_1 - \cos \theta_2)$

$$= (NiA) B(\cos 0^\circ - \cos 180^\circ) = 2NiAB$$

23. (d) Magnetic dipole moment of coil $= NiA$
24. (a) $F = Bil \sin \theta \Rightarrow \sin \theta = \frac{F}{Bil} = \frac{15}{2 \times 10 \times 1.5} = \frac{1}{2} \Rightarrow \theta = 30^\circ$
25. (a)
26. (b) $M = i(\pi r^2) = \frac{ev}{2\pi r} \times \pi r^2 \Rightarrow M = \frac{1}{2} e v r$
27. (d) Couple of force on loop S will be maximum because for same perimeter the area of loop will be maximum and magnetic moment of loop $= i \times A$. So, it will also be maximum for loop S .
28. (b) According to the definition.
29. (c) Current carrying loop, behaves as a bar magnet. A freely suspended bar magnet stays in the $N-S$ direction.



30. (c) In equilibrium angle between $\vec{M}$ and $\vec{B}$ is zero. It is happened, when plane of the coil is perpendicular to $\vec{B}$
31. (c)
32. (d)
33. (d) Sensitivity $S = \frac{\theta}{i} = \frac{nBA}{C}$
34. (d)
35. (b) By Fleming left hand rule.
36. (a) Force on wire Q due to wire P is
 $F_P = 10^{-7} \times \frac{2 \times 30 \times 10}{0.1} \times 0.1 = 6 \times 10^{-5} \text{ N}$ (Towards left)
Force on wire Q due to wire R is
 $F_R = 10^{-7} \times \frac{2 \times 20 \times 10}{0.02} \times 0.1 = 20 \times 10^{-5} \text{ N}$ (Towards right)
Hence $F_{\text{net}} = F_R - F_P = 14 \times 10^{-5} \text{ N} = 1.4 \times 10^{-4} \text{ N}$
(Towards right)
37. (a) $\tau = NBiA = 100 \times 0.2 \times 2 \times (0.08 \times 0.1) = 0.32 \text{ N}\cdot\text{m}$
Direction can be found by Fleming's left hand rule.
38. (a) $F = Bil \sin \theta \Rightarrow 7.5 = 2 \times 5 \times 1.5 \sin \theta \Rightarrow \theta = 30^\circ$
39. (c) According to the question figure can be drawn as shown below.



Force on the conductor ABC = Force on the conductor AC

$$= 5 \times 10 \times (5 \times 10^{-7}) = 2.5 \text{ N}$$

40. (d) Current sensitivity $\frac{\theta}{i} = \frac{NBA}{C}$

$$\Rightarrow \frac{\theta}{i} = \frac{100 \times 5 \times 10^{-4}}{10^{-8}} = 5 \text{ rad} / \mu \text{ Amp}$$

41. (c) $\tau = NBiA = 100 \times 0.5 \times 1 \times 400 \times 10^{-4} = 2 \text{ N-m}$

42. (a) When current is passed through a spring, it gets compressed.

43. (a) $M = iA \Rightarrow i = M / A$

44. (c) $i = 6.6 \times 10^{15} \times 1.6 \times 10^{-19} = 10.5 \times 10^{-4} \text{ amp}$

$$A = \pi R^2 = 3.142 \times (0.528)^2 \times 10^{-20} \text{ m}^2$$

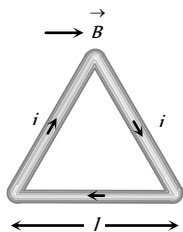
$$\Rightarrow M = iA = 10.5 \times 10^{-4} \times 3.142 \times (0.528)^2 \times 10^{-20}$$

$$= 10 \times 10^{-24} \text{ units} = 1 \times 10^{-23} \text{ units}$$

45. (d) Since $\theta = 90^\circ$

Hence $\tau = NIAB = 1 \times I \times \left(\frac{\sqrt{3}}{4} l^2 \right) B$

$$= \frac{\sqrt{3}}{4} l^2 B$$



46. (a) For no force on wire C , force on wire C due to wire D = force on wire C due to wire B

$$\Rightarrow \frac{\mu_0}{4\pi} \times \frac{2 \times 15 \times 5}{x} \times l = \frac{\mu_0}{4\pi} \times \frac{2 \times 5 \times 10}{(15-x)} \times l \Rightarrow x = 9 \text{ cm.}$$

47. (d) By Fleming's left hand rule.

48. (a)

49. (a)

50. (c) Force on the wire = $Bi l$

$$\text{Force per unit length} = Bi = 10^{-4} \times 10 = 10^{-3} \text{ N}$$

51. (b) $F = Bi l = 2 \times 1.2 \times 0.5 = 1.2 \text{ N}$

52. (a) $F = \frac{\mu_0}{4\pi} \frac{2i_1 i_2}{a} = 10^{-7} \times \frac{2 \times 10 \times 10}{0.1} = 2 \times 10^{-4} \text{ N}$

Direction of current is same, so force is attractive.

53. (a,b,c) Sensitivity $\frac{\theta}{i} = \frac{NAB}{C}$

54. (a) $M = NiA = 24 \times 0.75 \times 3.14 \times (3.5 \times 10^{-2})^2$
 $= 6.9 \times 10^{-2} \text{ A-m}^2$

55. (c) $\frac{F}{l} = \frac{\mu_0}{4\pi} \times \frac{2i_1 i_2}{a} = \frac{\mu_0}{4\pi} \frac{2i^2}{a} \quad (\because i_1 = i_2 = i)$

$$\Rightarrow 2 \times 10^{-7} = 10^{-7} \times \frac{2i^2}{1} \Rightarrow i = 1 \text{ A}$$

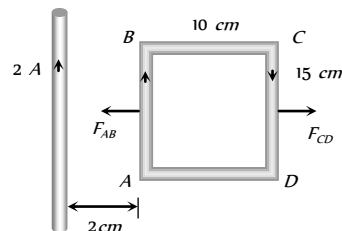
56. (b)

57. (a) $\tau = NiAB \sin \theta = 0 \quad (\because \theta = 0^\circ)$

58. (d) $M = NiA$

59. (c)

60. (a) Force on side BC and AD are equal but opposite so their net will be zero.



$$\text{But } F_{AB} = 10^{-7} \times \frac{2 \times 2 \times 1}{2 \times 10^{-2}} \times 15 \times 10^{-2} = 3 \times 10^{-6} \text{ N}$$

$$\text{and } F_{CD} = 10^{-7} \times \frac{2 \times 2 \times 1}{(12 \times 10^{-2})} \times 15 \times 10^{-2} = 0.5 \times 10^{-6} \text{ N}$$

$$\Rightarrow F_{\text{net}} = F_{AB} - F_{CD} = 2.5 \times 10^{-6} \text{ N}$$

$= 25 \times 10^{-7} \text{ N}$, towards the wire.

61. (b) $F = 10^{-7} \frac{2i_1 i_2}{a} = 10^{-7} \times \frac{2 \times 5 \times 5}{0.5} = 10^{-5} \text{ N}$ (repulsive)

62. (b) Sensitivity $= \frac{NAB}{C}$

63. (d) $M = iA = 0.1 \times \pi \times (0.05)^2$
 $= (0.1) \times 3.14 \times 25 \times 10^{-4} = 7.85 \times 10^{-4} \text{ amp-m}^2$

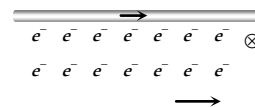
64. (b) $B = \frac{\mu_0 i}{2R} \Rightarrow i = \frac{B \times 2R}{\mu_0}$

$$\text{Now, } M = i \times A = i \pi R^2 = \frac{B \times 2R}{\mu_0} \times \pi R^2 = \frac{2\pi B R^3}{\mu_0}$$

65. (c) $M = NiA \Rightarrow M \propto A \Rightarrow M \propto r^2 \quad (\text{As } l = 2\pi r \Rightarrow l \propto r)$
 $\Rightarrow M \propto l^2$

66. (a) $F = \frac{\mu_0}{4\pi} \frac{2i_1 i_2}{a} = 10^{-7} \times \frac{2 \times 10 \times 5}{0.1} = 10^{-4} \text{ N}$ (Repulsive)

67. (b) According to Fleming's left hand rule, magnetic force on electrons will be downward.

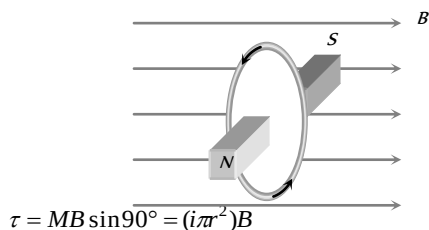


68. (a) $\sigma_i = \frac{\theta}{i} = \frac{\theta}{iG} \cdot G = \sigma_v G \Rightarrow \frac{\sigma_i}{G} = \sigma_v$

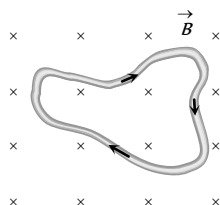
69. (a)

70. (a) $F = Bi l \sin 30^\circ = 1.5 \times 10 \times 1 \times \frac{1}{2} = 7.5 \text{ N}$

71. (c) As shown in the following figure, the given situation is similar to a bar magnet placed in a uniform magnetic field perpendicularly. Hence torque on it



72. (a) As shown in figure, since $\vec{L} = 0$



Hence according to $\vec{F} = i(\vec{L} \times \vec{B}) \Rightarrow \vec{F} = 0$

73. (a) Because $\tau_{\max} = BiNA \Rightarrow \tau \propto N$.

74. (d)

75. (d) $F = \frac{\mu_0}{4\pi} \frac{2i_1 i_2}{a}$

$$F_1 = \frac{\mu_0}{4\pi} \frac{2i^2}{x} \quad (\text{Attraction})$$

$$F_2 = \frac{\mu_0}{4\pi} \frac{2i \times 2i}{2x} = \frac{\mu_0}{4\pi} \frac{2i^2}{x} \quad (\text{Repulsion})$$

Thus $F_1 = -F_2$

76. (c) Magnetic field produced by wire is perpendicular to the motion of electron and it is given by

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2i}{a} = 10^{-7} \times \frac{2 \times 5}{0.1} = 10^{-5} \text{ Wb/m}^2$$

Hence force on electron

$$F = qvB = (1.6 \times 10^{-19}) \times 5 \times 10^6 \times 10^{-5} = 8 \times 10^{-18} \text{ N}$$

77. (a) Sensitivity $(S) = \frac{\theta}{i} \Rightarrow \frac{S_A}{S_B} = \frac{i_B}{i_A} = \frac{5}{3} \Rightarrow S_A > S_B$

78. (a) $F = 10^{-7} \frac{2i_1 i_2}{a} = 10^{-7} \times \frac{2 \times 5 \times 8}{0.5} = 1.6 \times 10^{-5}$ (Attractive)

79. (b) In moving coil galvanometer $i \propto \theta$.

80. (c)

81. (b) $F \propto i_1 i_2$, so force on B due to C will be greater than that due to A. Hence net force on B acts towards C.

82. (c) $F \propto \frac{i_1 i_2}{a}$; Since one of the current increase two times and distance increases three times, so force become $\frac{2}{3}$ times. Also

due to the reversal of direction of current force becomes negative.

83. (c) Neon molecule is diatomic, so its net magnetic moment is zero.

84. (b) $F = Bil \Rightarrow 1 \times 9.8 = 0.98 \times i \times 1 \Rightarrow i = 10 \text{ A}$

85. (a)

86. (d) $F = 10^{-7} \times \frac{2i^2}{a} \times l \Rightarrow 30 \times 10^{-7} = 10^{-7} \times \frac{2i^2}{0.15} \times 9$
 $\Rightarrow i = 0.5 \text{ A}$

87. (c) $\tau_{\max} = NiAB = 1 \times i \times (\pi r^2) \times B$ $\left(2\pi r = L, \Rightarrow r = \frac{L}{2\pi}\right)$

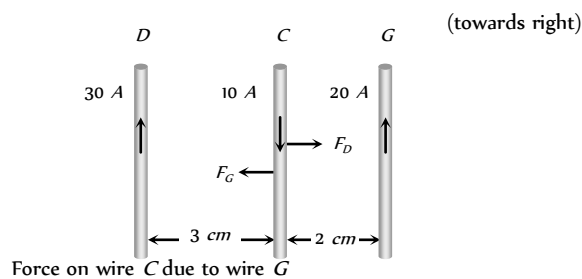
$$\tau_{\max} = \pi \left(\frac{L}{2\pi}\right)^2 B = \frac{L^2 i B}{4\pi}$$

88. (b)

89. (c) $\left(\frac{F}{l}\right) = \frac{\mu_0}{4\pi} \cdot \frac{2i_1 i_2}{a} \Rightarrow \left(\frac{F}{l}\right) = \frac{\mu_0}{4\pi} \cdot \frac{2i^2}{d} = \frac{\mu_0 i^2}{2\pi d}$ (Attractive)

90. (c) Force on wire C due to wire D

$$F_D = 10^{-7} \times \frac{2 \times 30 \times 10}{2 \times 10^{-2}} \times 25 \times 10^{-2} = 5 \times 10^{-4} \text{ N}$$



$$F_G = 10^{-7} \times \frac{2 \times 20 \times 10}{2 \times 10^{-2}} \times 25 \times 10^{-2} = 5 \times 10^{-4} \text{ N}$$

(towards left)

$\Rightarrow$ Net force on wire C is $F_{\text{net}} = F_D - F_G = 0$

91. (d) Since $\theta = 0^\circ$ so $\tau = 0$ ($\because \tau = NiAB \sin \theta$)

Critical Thinking Questions

1. (b) $\frac{B_{\text{centre}}}{B_{\text{axis}}} = \left(1 + \frac{x^2}{R^2}\right)^{3/2}$, also $B_{\text{axis}} = \frac{1}{8} B_{\text{centre}}$
 $\Rightarrow \frac{8}{1} = \left(1 + \frac{x^2}{R^2}\right)^{3/2} \Rightarrow 2 = \left(1 + \frac{x^2}{R^2}\right)^{1/2}$
 $\Rightarrow 4 = 1 + \frac{x^2}{R^2} \Rightarrow 3 = \frac{x^2}{R^2} \Rightarrow x^2 = 3R^2 \Rightarrow x = \sqrt{3}R$

2. (a) Field at the centre $B_1 = \frac{\mu_0}{4\pi} \times \frac{2\pi n i}{r} = \frac{\mu_0}{2} \cdot \frac{n i}{r}$

Field at a distance h from the centre

$$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2\pi n i r^2}{(r^2 + h^2)^{3/2}} = \frac{\mu_0}{2} \cdot \frac{n i r^2}{r^3 \left(1 + \frac{h^2}{r^2}\right)^{3/2}}$$

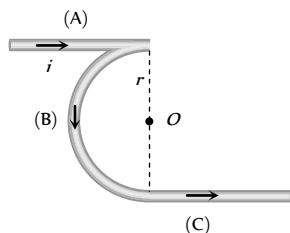
$$= B_1 \left(1 + \frac{h^2}{r^2} \right)^{-3/2} = B_1 \left(1 - \frac{3}{2} \cdot \frac{h^2}{r^2} \right) \text{ (By binomial theorem)}$$

Hence B_1 is less than B by a fraction $= \frac{3}{2} \frac{h^2}{r^2}$

3. (a) **Case 1:** $B_A = \frac{\mu_0}{4\pi} \cdot \frac{i}{r} \otimes$

$$B_B = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{r} \odot$$

$$B_C = \frac{\mu_0}{4\pi} \cdot \frac{i}{r} \odot$$

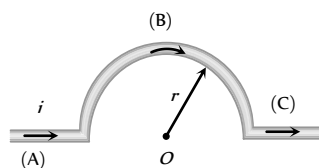


So net magnetic field at the centre of case 1

$$B_1 = B_B - B_C - B_A \Rightarrow B_1 = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{r} \odot \dots (i)$$

Case 2: As we discussed before magnetic field at the centre O in this case

$$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{r} \otimes \dots (ii)$$

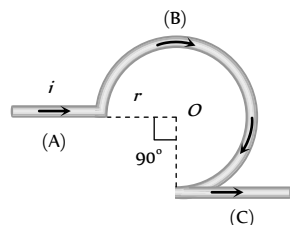


Case 3: $B_A = 0$

$$B_B = \frac{\mu_0}{4\pi} \cdot \frac{(2\pi - \pi/2)i}{r} \otimes$$

$$B_C = \frac{\mu_0}{4\pi} \cdot \frac{i}{r} \odot$$

$$= \frac{\mu_0}{4\pi} \cdot \frac{3\pi i}{2r} \otimes$$



So net magnetic field at the centre of case 3

$$B_3 = \frac{\mu_0}{4\pi} \cdot \frac{i}{r} \left(\frac{3\pi}{2} - 1 \right) \otimes \dots (iii)$$

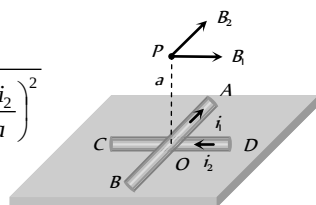
From equation (i), (ii) and (iii)

$$B_1 : B_2 : B_3 = \pi \odot : \pi \otimes : \left(\frac{3\pi}{2} - 1 \right) \otimes = -\frac{\pi}{2} : \frac{\pi}{2} : \left(\frac{3\pi}{2} - 1 \right)$$

4. (c) At P : $B_{net} = \sqrt{B_1^2 + B_2^2}$

$$= \sqrt{\left(\frac{\mu_0}{4\pi} \cdot \frac{2i_1}{a} \right)^2 + \left(\frac{\mu_0}{4\pi} \cdot \frac{2i_2}{a} \right)^2}$$

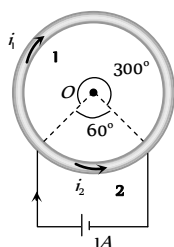
$$= \frac{\mu_0}{2\pi a} (i_1^2 + i_2^2)^{1/2}$$



5. (c) $B = \frac{\mu_0}{4\pi} \frac{\theta i}{r} \Rightarrow B \propto \theta i$ (but $\frac{i_1}{i_2} = \frac{l_2}{l_1} = \frac{\theta_2}{\theta_1}$)

$$\Rightarrow \frac{B_1}{B_2} = \frac{\theta_1}{\theta_2} \cdot \frac{i_1}{i_2}$$

$$\text{So, } \frac{B_1}{B_2} = \frac{\theta_1}{\theta_2} \times \frac{\theta_2}{\theta_1}$$



$$\Rightarrow B_1 = B_2$$

6. (c) Magnetic field at any point lying on the current carrying straight conductor is zero.

Here H_1 = Magnetic field at M due to current in PQ .

H_2 = Magnetic field at M due to QR

+ magnetic field at M due to QS

+ magnetic field at M due to PQ

$$= 0 + \frac{H_1}{2} + H_1 = \frac{3}{2} H_1 \Rightarrow \frac{H_1}{H_2} = \frac{2}{3}$$

7. (c, d) $B_{net} = B_1 - B_2 \Rightarrow B_1 - B_2 = 0 \Rightarrow B_1 = B_2$

$$\Rightarrow B \propto ni. \text{ So } n_1 i_1 = n_2 i_2 \text{ or } n_1 = n_2 \text{ and } i_1 = i_2$$

8. (c) Number of turns per unit width $= \frac{N}{b-a}$

Consider an elemental ring of radius x and with thickness dx

$$\text{Number of turns in the ring} = dN = \frac{Nd x}{b-a}$$

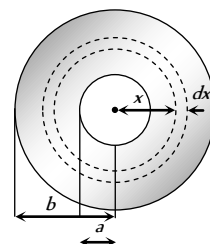
Magnetic field at the centre due to the ring element

$$dB = \frac{\mu_0 (dN) i}{2x} = \frac{\mu_0 i}{2} \cdot \frac{Nd x}{(b-a)} \cdot \frac{1}{x}$$

$\therefore$ Field at the centre

$$= \int dB = \frac{\mu_0 N i}{2(b-a)} \int_a^b \frac{dx}{x}$$

$$= \frac{\mu_0 N i}{2(b-a)} \ln \frac{b}{a}$$

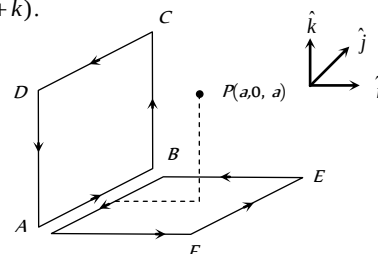


9. (d) The magnetic field at $P(a, 0, a)$ due to the loop is equal to the vector sum of the magnetic fields produced by loops $ABCD$ and $AFEBA$ as shown in the figure.

Magnetic field due to loop $ABCD$ will be along $\hat{i}$ and due to loop $AFEBA$, along $\hat{k}$. Magnitude of magnetic field due to both the loops will be equal.

Therefore, direction of resultant magnetic field at P will be

$$\frac{1}{\sqrt{2}} (\hat{i} + \hat{k}).$$

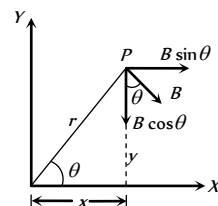


10. (a) Magnetic field at P is $\vec{B}$, perpendicular to OP in the direction shown in figure.

$$\text{So, } \vec{B} = B \sin \theta \hat{i} - B \cos \theta \hat{j}$$

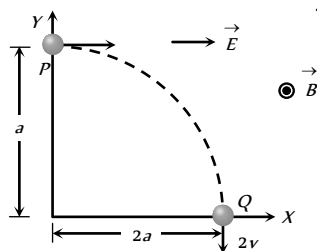
$$\text{Here } B = \frac{\mu_0}{2\pi} \frac{I}{r}$$

$$\sin \theta = \frac{y}{r} \text{ and } \cos \theta = \frac{x}{r}$$



$$\therefore \vec{B} = \frac{\mu_0 I}{2\pi} \cdot \frac{1}{r^2} (y\hat{i} - x\hat{j}) = \frac{\mu_0 I(y\hat{i} - x\hat{j})}{2\pi(x^2 + y^2)} \text{ (as } r^2 = x^2 + y^2 \text{)}$$

11. (a, b, d) Kinetic energy of the particle at point $P = \frac{1}{2}mv^2$



$$\text{K.E. of the particle at point } Q = \frac{1}{2}m(2v)^2$$

$$\text{Increase in K.E.} = \frac{3}{2}mv^2$$

It comes from the work done by the electric force qE on the particle as it covers a distance $2a$ along the x -axis. Thus

$$\frac{3}{2}mv^2 = qE \times 2a \Rightarrow E = \frac{3}{4} \frac{mv^2}{qa}$$

$$\text{the electric field at } P = F \times v = qE \times v = 3 \frac{mv^3}{4a}$$

At Q , $\vec{F}_e = q\vec{E}$ is along x -axis while velocity is along negative y -axis. Hence rate of work done by electric field $= \vec{F}_e \cdot \vec{v} = 0$ ($\because \theta = 90^\circ$). Similarly, according to equation $\vec{F}_m = q(\vec{v} \times \vec{B})$

Force $\vec{F}_m$ is also perpendicular to velocity vector $\vec{v}$.

Hence the rate of work done by the magnetic field = 0

$$12. (a, c) r \propto \frac{\sqrt{m}}{q} \Rightarrow r_H : r_{He} : r_o = \frac{\sqrt{1}}{1} : \frac{\sqrt{4}}{1} : \frac{\sqrt{16}}{2} = 1 : 2 : 2$$

Radius is smallest for H^+ , so it is deflected most.

13. (c) As the electric field is switched on, positive ion will start to move along positive x -direction and negative ion along negative x -direction. Current associated with motion of both types of ions is along positive x -direction. According to Fleming's left hand rule force on both types of ions will be along negative y -direction.

$$14. (c) \vec{v} = 2 \times 10^5 \hat{i} \text{ and } \vec{B} = (\hat{i} + 4\hat{j} - 3\hat{k})$$

$$\vec{F} = q(\vec{v} \times \vec{B}) = -1.6 \times 10^{-19} [2 \times 10^5 \hat{i} \times (\hat{i} + 4\hat{j} - 3\hat{k})]$$

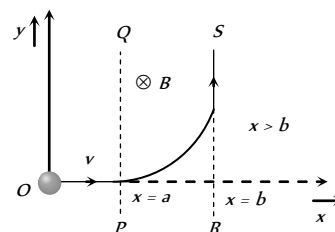
$$= -1.6 \times 10^{-19} \times 2 \times 10^5 [\hat{i} \times \hat{i} + 4(\hat{i} \times \hat{j}) - 3(\hat{i} \times \hat{k})]$$

$$= -3.2 \times 10^{-14} [0 + 4\hat{k} + 3\hat{j}] = 3.2 \times 10^{-14} (-4\hat{k} - 3\hat{j})$$

$$\Rightarrow |\vec{F}| = 3.2 \times 10^{-14} \times 5 = 1.6 \times 10^{-13} \text{ N.}$$

15. (b) In the figure, the z -axis points out of the paper, and the magnetic field is directed into the paper, existing in the region between PQ and RS . The particle moves in a circular path of

radius r in the magnetic field. It can just enter the region $x > b$ for $r \geq (b - a)$



$$\text{Now, } r = \frac{mv}{qB} \geq (b - a)$$

$$\text{or } v \geq \frac{q(b - a)B}{m} \Rightarrow v_{\min} = \frac{q(b - a)B}{m}$$

16. (b) Electric field can deviate the path of the particle in the shown direction only when it is along negative y -direction. In the given options $\vec{E}$ is either zero or along x -direction. Hence it is the magnetic field which is really responsible for its curved path. Options (a) and (c) can't be accepted as the path will be helix in that case (when the velocity vector makes an angle other than 0° , 180° or 90° with the magnetic field, path is a helix) option (d) is wrong because in that case component of net force on the particle also comes in k direction which is not acceptable as the particle is moving in x - y plane. Only in option (b) the particle can move in x - y plane.

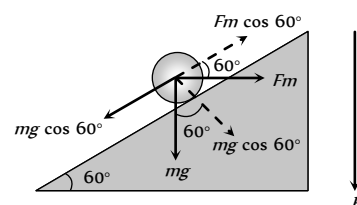
$$\text{In option (d) : } \vec{F}_{\text{net}} = q\vec{E} + q(\vec{v} \times \vec{B})$$

Initial velocity is along x -direction. So let $\vec{v} = v\hat{i}$

$$\therefore \vec{F}_{\text{net}} = q\hat{i} + q[(v\hat{i}) \times (c\hat{k} + b\hat{j})] = q\hat{i} - qvc\hat{j} + qvb\hat{k}$$

$$\text{In option (b) } \vec{F}_{\text{net}} = q(\hat{i}) + q[(v\hat{i}) \times (c\hat{k} + a\hat{i})] = q\hat{i} - qvc\hat{j}$$

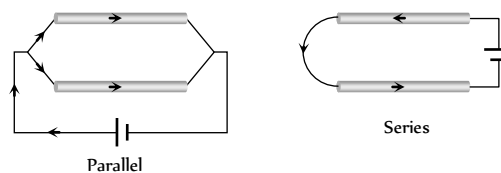
17. (c) The given situation can be drawn as follows



$$F = ilB \Rightarrow mg \sin 60^\circ = ilB \cos 60^\circ$$

$$\Rightarrow B = \frac{0.01 \times 10 \times \sqrt{3}}{0.1 \times 1.73} = 1 \text{ T}$$

18. (a) When connected in parallel the current will be in the same direction and when connected in series the current will be in the opposite direction.



19. (b) If the radius of circle is r , then $2\pi r = L \Rightarrow r = \frac{L}{2\pi}$

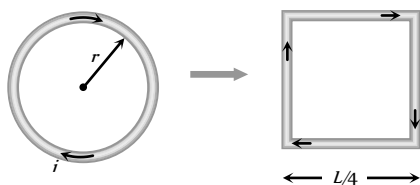
$$\text{Area} = \pi r^2 = \frac{\pi L^2}{4\pi^2} = \frac{L^2}{4\pi}$$

$$\text{Magnetic moment} = IA = \frac{IL^2}{4\pi}$$

20. (d) Initially for circular coil $L = 2\pi r$ and $M = i \times \pi r^2$

$$= i \times \pi \left(\frac{L}{2\pi} \right)^2 = \frac{iL^2}{4\pi} \dots (i)$$

$$\text{Finally for square coil } M' = i \times \left(\frac{L}{4} \right)^2 = \frac{iL^2}{16} \dots (ii)$$



$$\text{Solving equation (i) and (ii) } M' = \frac{\pi M}{4}$$

21. (c) The effective current $i = \frac{q\omega}{2\pi}$ and $A = \pi r^2$.

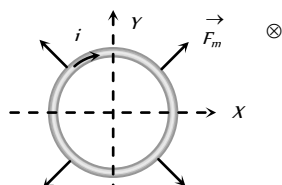
$$\text{Magnetic moment } M = iA = \frac{1}{2} q\omega r^2$$

$$\text{Angular momentum } L = I\omega = mr^2\omega \Rightarrow \frac{M}{L} = \frac{q}{2m}$$

22. (b) On applying Fleming's left hand rule.
23. (b) Current carrying conductors will attract each other, while electron beams will repel each other.
24. (c) Length of the component dl which is parallel to wire (1) is $dl \cos \theta$, so force on it.

$$F = \frac{\mu_0}{4\pi} \cdot \frac{2i_1 i_2}{r} (dl \cos \theta) = \frac{\mu_0 i_1 i_2}{2\pi} \frac{dl \cos \theta}{r}$$

25. (b) Net force on a current carrying loop in uniform magnetic field is zero. Hence the loop can't translate. So, options (c) and (d) are wrong.



From Fleming's left hand rule we can see that if magnetic field is perpendicular to paper inwards and current in the loop is clockwise (as shown) the magnetic force $\vec{F}_m$ on each element of the loop is radially outwards, or the loops will have a tendency to expand.

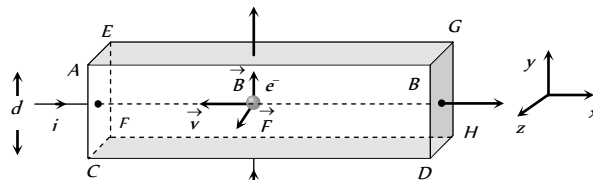
26. (c) $U = -MBC \cos \theta$; where θ = Angle between normal to the plane of the coil and direction of magnetic field.

27. (a) As the block is of metal, the charge carriers are electrons, so for current along positive x -axis, the electrons are moving along negative x -axis, i.e. $\vec{v} = -v\hat{i}$

and as the magnetic field is along the y -axis, i.e. $\vec{B} = B\hat{j}$

so $\vec{F} = q(\vec{v} \times \vec{B})$ for this case yield $\vec{F} = (-e)[-v\hat{i} \times B\hat{j}]$

i.e., $\vec{F} = evB\hat{k}$ [As $\hat{i} \times \hat{j} = \hat{k}$]



As force on electrons is towards the face ABCD, the electrons will accumulate on it and hence it will acquire lower potential.

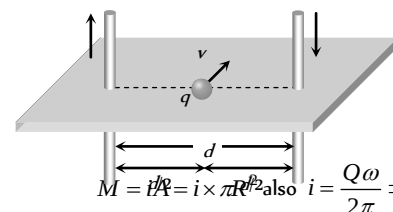
28. (a)

$$29. (a) i = \frac{2q\omega}{2\pi} = \frac{q\omega}{\pi}; \therefore M = iA = \frac{q\omega}{\pi} \pi R^2 = q\omega R^2$$

$$L = 2R.mv = 2R.mR\omega = 2mR^2\omega \quad (\because v = R\omega)$$

$$\Rightarrow \frac{M}{L} = \frac{q}{2m}$$

30. (d) According to gives information following figure can be drawn, which shows that direction of magnetic field is along the direction of motion of charge so net force on it is zero.



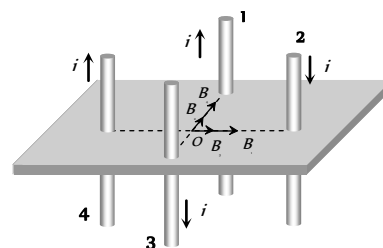
31. (b)

$$M = iA = i \times \pi R^2 \text{ also } i = \frac{Q\omega}{2\pi} \Rightarrow M = \frac{1}{2} Q\omega R^2$$

32. (c) Direction of magnetic field (B_1, B_2, B_3 and B_4) at origin due to wires 1, 2, 3 and 4 are shown in the following figure.

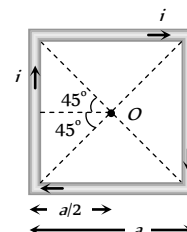
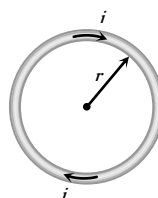
$$B_1 = B_2 = B_3 = B_4 = \frac{\mu_0}{4\pi} \cdot \frac{2i}{x} = B. \text{ So net magnetic field at origin } O$$

$$B_{\text{net}} = \sqrt{(B_1 + B_3)^2 + (B_2 + B_4)^2} = \sqrt{(2B)^2 + (2B)^2} = 2\sqrt{2}B$$



33. (b) Circular coil

Square coil



Length $L = 2\pi r$ Length $L = 4a$

Magnetic field at the centre of circular coil

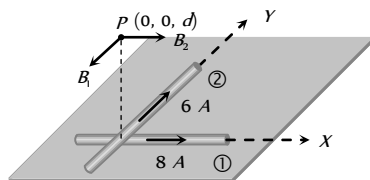
$$B_{\text{circular}} = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} = \frac{\mu_0}{4\pi} \cdot \frac{4\pi^2 i}{L}$$

Magnetic field at the centre of square coil

$$B_{\text{square}} = \frac{\mu_0}{4\pi} \cdot \frac{8\sqrt{2} i}{a} = \frac{\mu_0}{4\pi} \cdot \frac{32\sqrt{2} i}{L}$$

$$\text{Hence } \frac{B_{\text{circular}}}{B_{\text{square}}} = \frac{\pi^2}{8\sqrt{2}}$$

34. (d) Magnetic field at P due to wire 1, $B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2(8)}{d}$

and due to wire 2, $B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2(6)}{d}$

$$\Rightarrow B_{\text{net}} = \sqrt{B_1^2 + B_2^2} = \sqrt{\left(\frac{\mu_0}{4\pi} \cdot \frac{16}{d}\right)^2 + \left(\frac{\mu_0}{4\pi} \cdot \frac{12}{d}\right)^2}$$

$$= \frac{\mu_0}{4\pi} \times \frac{2}{d} \times 10 = \frac{5\mu_0}{\pi d}$$

35. (b) According to question resistance of wire ADC is twice that of wire ABC . Hence current flows through ADC is half that of ABC i.e. $\frac{i_2}{i_1} = \frac{1}{2}$. Also $i_1 + i_2 = i \Rightarrow i_1 = \frac{2i}{3}$ and $i_2 = \frac{i}{3}$

Magnetic field at centre O due to wire AB and BC (part 1 and

$$2) B_1 = B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2i_1 \sin 45^\circ}{a/2} \otimes = \frac{\mu_0}{4\pi} \cdot \frac{2\sqrt{2} i_1}{a} \otimes$$

and magnetic field at centre O due to wires AD and DC (i.e.

$$\text{part 3 and 4) } B_3 = B_4 = \frac{\mu_0}{4\pi} \cdot \frac{2\sqrt{2} i_2}{a} \odot$$

Also $i_1 = 2i_2$. So $(B_1 = B_2) > (B_3 = B_4)$ Hence net magnetic field at centre O

$$B_{\text{net}} = (B_1 + B_2) - (B_3 + B_4)$$

$$= 2 \times \frac{\mu_0}{4\pi} \cdot \frac{2\sqrt{2} \times \left(\frac{2}{3}i\right)}{a} - \frac{\mu_0}{4\pi} \cdot \frac{2\sqrt{2} \left(\frac{i}{3}\right) \times 2}{a}$$

$$= \frac{\mu_0}{4\pi} \cdot \frac{4\sqrt{2} i}{3a} (2-1) \otimes = \frac{\sqrt{2} \mu_0 i}{3\pi a} \otimes$$

36. (d) By using $B = \frac{\mu_0 i}{2\pi r} \left(\frac{r^2 - a^2}{b^2 - a^2} \right)$ here $r = \frac{3R}{2}$, $a = R$, $b = 2R$

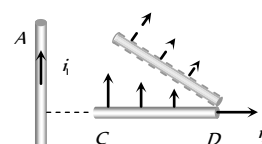
$$\Rightarrow B = \frac{\mu_0 i}{2\pi \left(\frac{3R}{2} \right)} \times \left\{ \frac{\left(\frac{3R}{2} \right)^2 - R^2}{(2R)^2 - R^2} \right\} = \frac{5\mu_0 i}{36\pi R}$$

37. (a) Suppose in equilibrium wire PQ lies at a distance r above the wire AB

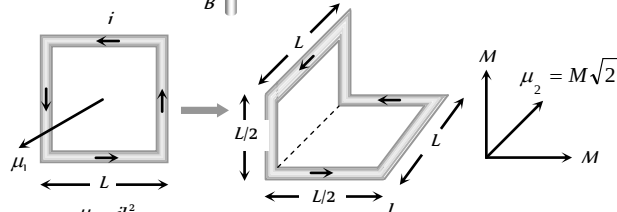
$$\text{Hence in equilibrium } mg = Bil \Rightarrow mg = \frac{\mu_0}{4\pi} \left(\frac{2i}{r} \right) \times il$$

$$\Rightarrow 10^{-3} \times 10 = 10^{-7} \times \frac{2 \times (50)^2}{r} \times 0.5 \Rightarrow r = 25 \text{ mm}$$

38. (c) Since the force on the rod CD is non-uniform it will experience force and torque. From the left hand side it can be seen that the force will be upward and torque is clockwise.



39. (c) Initial magnetic moment $\mu_1 = iL$

After folding the loop, $M =$ magnetic moment due to each part

$$= i \left(\frac{L}{2} \right) \times L = \frac{iL^2}{2} = \frac{\mu_1}{2}$$

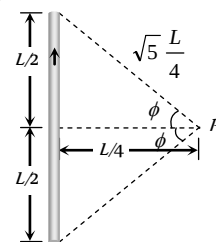
$$\Rightarrow \mu_2 = M\sqrt{2} = \frac{\mu_1}{2} \times \sqrt{2} = \frac{\mu_1}{\sqrt{2}}$$

40. (a) By using $B = \frac{\mu_0}{4\pi} \cdot \frac{i}{a} (\sin \phi_1 + \sin \phi_2)$

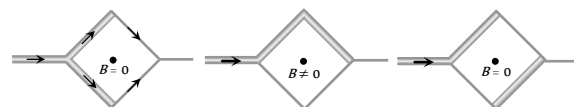
$$\Rightarrow B = \frac{\mu_0}{4\pi} \cdot \frac{i}{(L/4)} (2 \sin \phi)$$

$$\text{Also } \sin \phi = \frac{L/2}{\sqrt{5}L/4} = \frac{2}{\sqrt{5}}$$

$$\Rightarrow B = \frac{4\mu_0 i}{\sqrt{5}\pi L}$$



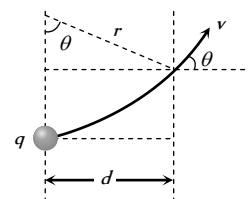
41. (d) In P and R loops, currents are divided in same proportion because the branches have equal resistance. Hence magnetic field produced at centre due to each segment is of equal magnitude but of opposite direction, so net field is zero.



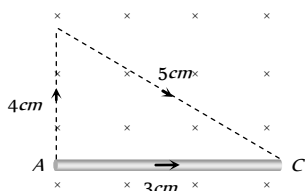
42. (a) From figure it is clear that

$$\sin \theta = \frac{d}{r} \text{ also } r = \frac{p}{qB}$$

$$\therefore \sin \theta = \frac{Bqd}{p}$$



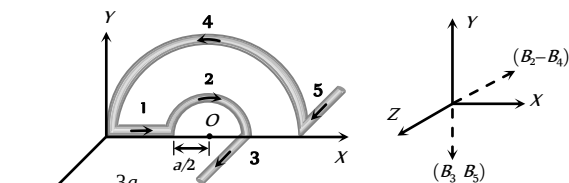
43. (a) $\vec{F}_{CAD} = \vec{F}_{CD} = \vec{F}_{CED}$
 $\therefore$ Net force on frame $= 3\vec{F}_{CD} = (3)(2)(1)(4) \quad (F = i\ell B)$
 $= 24 \text{ N}$
44. (b) The given curved wire can be treated as a straight wire as shown



Force acting on the wire AC, $F = Bi\ell = 2 \times 2 \times 3 \times 10^{-2}$
 $= 12 \times 10^{-2} \text{ N}$ along y-axis.

So acceleration of wire $= \frac{F}{m} = \frac{12 \times 10^{-2}}{10 \times 10^{-3}} = 12 \text{ m/s}^2$

45. (b)



Magnetic field at O due to

Part (1): $B_1 = 0$

Part (2): $B_2 = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{(a/2)} \otimes$ (along -Z-axis)

Part (3): $B_3 = \frac{\mu_0}{4\pi} \cdot \frac{i}{(a/2)} \downarrow$ (along -Y-axis)

Part (4): $B_4 = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{(3a/2)} \odot$ (along +Z-axis)

Part (5): $B_5 = \frac{\mu_0}{4\pi} \cdot \frac{i}{(3a/2)} \downarrow$ (along -Y-axis)

$B_2 - B_4 = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{a} \left(2 - \frac{2}{3} \right) = \frac{\mu_0 i}{3a} \otimes$ (along -Z-axis)

$B_3 + B_5 = \frac{\mu_0}{4\pi} \cdot \frac{1}{a} \left(2 + \frac{2}{3} \right) = \frac{8\mu_0 i}{12\pi a} \downarrow$ (along -Y-axis)

Hence net magnetic field

$$B_{\text{net}} = \sqrt{(B_2 - B_4)^2 + (B_3 + B_5)^2} = \frac{\mu_0 i}{3\pi a} \sqrt{\pi^2 + 4}$$

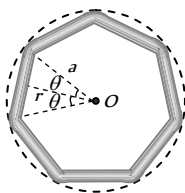
46. (b) Magnetic field at the centre due to one side

$$B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2i \sin \theta}{r} \text{ where } r = a \cos \theta$$

$$\text{So } B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2i \sin \theta}{a \cos \theta} = \frac{\mu_0 i}{2\pi a} \tan \theta$$

Hence net magnetic field

$$B_{\text{net}} = n \times \frac{\mu_0 i}{2\pi a} \tan \frac{\pi}{n}$$



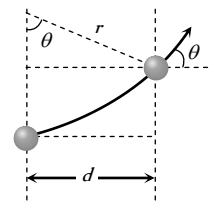
47. (b) According to following figure $\sin \theta = \frac{d}{r}$

$$\text{also } r = \frac{\sqrt{2mk}}{qB} = \frac{1}{B} \sqrt{\frac{2mV}{q}}$$

$$\therefore \sin \theta = Bd \sqrt{\frac{q}{2mV}}$$

$$= 0.51 \times 0.1 \sqrt{\frac{1.6 \times 10^{-19}}{2 \times 1.67 \times 10^{-27} \times 500 \times 10^3}}$$

$$= \frac{1}{2} \Rightarrow \theta = 30^\circ$$



48. (b) The field at the midpoint of BC due to AB is $\left(-\frac{\mu_0}{4\pi} \cdot \frac{i}{d/2} \hat{k} \right)$ and the same is due to CD. Therefore the total field is $\left[-\left(\frac{\mu_0 i}{\pi d} \right) \hat{k} \right]$

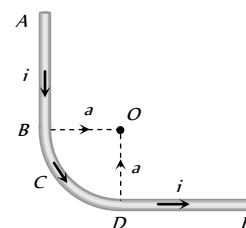
49. (a) The electron reverses its direction. It can be done by covering semi-circular path in x-z or x-y plane.

50. (d) The field at O due to AB is $\frac{\mu_0}{4\pi} \cdot \frac{i}{a} \hat{k}$ and that due to DE is

$$\text{also } \frac{\mu_0}{4\pi} \cdot \frac{i}{a} \hat{k}$$

However the field due to BCD is $\frac{\mu_0}{4\pi} \cdot \frac{i}{a} \left(\frac{\pi}{2} \right) \hat{k}$.

Thus the total field at O is $\frac{\mu_0}{4\pi} \cdot \frac{i}{a} \left(2 + \frac{\pi}{2} \right) \hat{k}$



51. (d) The energy of a charged particle moving in magnetic field remains constant because the magnetic field does not do any work. Therefore kinetic energy is constant i.e. $u = v$.

The force on electron will act along negative y-axis initially. The electron will undergo circular motion in clockwise direction and emerge out the field. So $y < 0$.

Graphical Questions

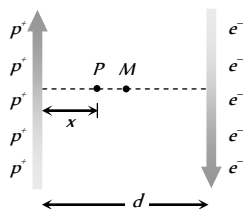
- (c) $|\vec{B}| = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r} \Rightarrow |\vec{B}| \propto \frac{1}{r}$
- (a) Every point on line AB will be equidistant from X and Y-axis. So magnetic field at every point on line AB due to wire 1 along X-axis is equal in magnitude but opposite in direction to the magnetic field due to wire along Y-axis. Hence B_{net} on AB = 0
- (b) If the current flows out of the paper, the magnetic field at points to the right of the wire will be upwards and to the left

will be downward. Now magnetic field at C , is zero. The field in the region BX' will be upwards ($+ve$) because all points lying in this region are to the right of both the wires. Similarly, magnetic field in the region AX will be downwards ($-ve$). The field in the region AC will be upwards ($+ve$) because points are closer to A compared to B . Similarly magnetic field in region BC will be downward ($-ve$). Graph (b) satisfies all these conditions.

4. (a) Magnetic field inside the conductor $B_{in} \propto r$ and magnetic field outside the conductor $B_{out} \propto \frac{1}{r}$
(where r is the distance of observation point from axis)

5. (c) The magnetic field at points to the right of the proton beam acts perpendicular to the paper inwards ($\times$). The magnetic field at points to the left of the electron beam acts perpendicular to the paper outwards ($\cdot$).

Magnetic field at mid point M is zero.



Magnetic field at the points closer to proton beam acts perpendicular to the paper inwards (*i.e.* ($\times$)) and at the points closer to electron beam it acts outwards *i.e.* ($\cdot$). In the given options graph (c) satisfies all the conditions.

6. (a) Magnetic field inside the hollow metallic cylinder $B_{in} = 0$, and magnetic field outside it $B_{out} \propto \frac{1}{r}$
7. (a) Magnetic field in the middle of the solenoid's is maximum, magnetic field at the end $B_{end} = \frac{1}{2} B_{\infty}$.
8. (b) The charge will not experience any force if $|\vec{F}_e| = |\vec{F}_m|$. This condition is satisfied in option (b) only.
9. (c) The given portion of the curved wire may be treated as a straight wire of length $2L$ which experiences a magnetic force $F_m = Bi(2L)$
10. (a) $\tau = NBiA \sin \theta$ so the graph between τ and θ is a sinusoidal graph.
11. (c) For undeviated motion $|\vec{F}_e| = |\vec{F}_m|$, which happened when $\vec{v}$, $\vec{E}$ and $\vec{B}$ are mutually perpendicular to each other.

12. (a) If at a place, magnetic induction is B , then energy density will be equal to $U = \frac{B^2}{2\mu_0}$. It means, graph between U and B will be a parabola passing through origin and symmetric about U -axis.

13. (b, c) Since length of the wire is equal to l , therefore, $2\pi Rn = l$ or $n = \frac{l}{2\pi R}$.

Magnetic induction at centre of a circular coil is given by

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi ni}{R} = \frac{\mu_0 l i}{4\pi R^2} \Rightarrow B \propto \frac{1}{R^2}$$

It means, when $R \rightarrow 0$, $B \rightarrow \infty$ and $R \rightarrow \infty$, $B \rightarrow 0$,

Hence (b) is correct and (d) is wrong.

$$\text{Substituting } R = \frac{l}{2\pi n} \text{ in } B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi ni}{R}$$

$B \propto n^2$. It means graph between B and n will be a parabola having increasing slope and passing through origin. Hence (c) is correct and (a) is wrong.

14. (b) When a current flows through cylindrical shell, then according to Ampere circuital law, magnetic induction inside it will be equal to zero. Hence energy density at $r < R$ is equal to zero.

Therefore, (a), (c) and (d) are wrong.

$$\text{When } r > R, B = \frac{\mu_0 i}{2\pi r}$$

$$\text{Since } U = \frac{B^2}{2\mu_0}, \text{ therefore, outside the shell, } U = \frac{\mu_0 i^2}{8\pi^2 r^2}. \text{ It}$$

$$\text{means, just outside the shell, } U = \frac{\mu_0 i^2}{8\pi^2 R^2} \text{ and when}$$

$$r \rightarrow \infty, U \rightarrow 0.$$

Hence (b) is correct.

15. (b) Energy density in previous objective, at $r = 2R$, will be equal to $U = \frac{\mu_0 i^2}{32\pi^2 R^2}$ or $U \propto i^2$. It means, graph-between U and i will be a parabola, passing through origin, symmetric about U -axis and having increasing slope. Hence (b) is correct.

16. (b) Direction of magnetic field at every point on axis of a current carrying coil remains same though magnitude varies. Hence magnetic induction for whole the x -axis will remain positive.

Therefore, (c) and (d) are wrong.

Magnitude of magnetic field will vary with x according to law,

$$B = \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}}$$

Hence, at $x = 0$, $B = \frac{\mu_0 NI}{2R}$ and when $x \rightarrow \infty$, $B \rightarrow 0$.

Slope of the graph will be $\frac{dB}{dx} = -\frac{3\mu_0 NI R^2 \cdot x}{2(R^2 + x^2)^{5/2}}$.

It means, at $x = 0$, slope is equal to zero or tangent to the graph at $x = 0$, must be parallel to x -axis.

Hence (b) is correct and (a) is wrong.

Assertion and Reason

- (a) Cyclotron is suitable for accelerating heavy particles like protons, α -particles etc, and not for electrons because of low mass. Because electrons acquire very high velocities very near to velocity of light and appreciable variation in their mass, occurs.
- (c) Cyclotron is utilised to accelerate the positive ion. And cyclotron frequency is given by $\nu = \frac{Be}{2\pi m}$. It means cyclotron frequency doesn't depends upon velocity. Therefore, assertion is true and reason false.
- (a) A moving charge experiences a force in magnetic field. It is because of interaction of two magnetic fields, one which is produced due to the motion of charge and other in which charge is moving.
- (e) In this case we can not be sure about the absence of the magnetic field because if the electron moving parallel to the direction of magnetic field, the angle between velocity and applied magnetic field is zero ($F = 0$). Then also electron passes without deflection. Also $F = evB \sin\theta \Rightarrow F \propto B$.
- (a) In the absence of the electric current, the free electrons in a conductor are in a state of random motion, like molecules in a gas. Their average velocity is zero. i.e. they do not have any net velocity in a direction. As a result, there is no net magnetic force on the free electrons in the magnetic field. On passing the current, the free electrons acquire drift velocity in a definite direction, hence magnetic force acts on them, unless the field has no perpendicular component.
- (c) Time taken is independent of velocity and radius of path. However, maximum velocity will be given by $v_{\max} = \frac{qBR}{m}$ where R is radius of Dee's.
- (a) Due to metallic frame the deflection is only due to current in a coil and magnetic field, not due to vibration in the strings. If string start oscillating, presence of metallic frame in the field make these oscillations damped.
- (c) The direction of magnetic field due to current carrying conductor can be found by applying right hand thumb rule or right hand palm rule. When electric current is passed through a circular conductor, the magnetic field lines near the center of the conductor are almost straight lines. Magnetic flux direction is determined only by the direction of current.
- (a) The force on a charged particle moving in a uniform magnetic field always acts in direction perpendicular to the direction of motion of the charge. As work done by magnetic field on the

charge is zero, [$W = FS \cos\theta$], so the energy of the charged particle does not change.

- (b) We know that the direction of the earth's magnetic field is toward north and the velocity of electron is vertically downward. Applying Fleming's left hand rule, the direction of force is towards west. Therefore, an electron coming from outer space will be deflected toward west.
- (d) In the case of metallic rod, the charge carriers flow through whole of the cross section. Therefore, the magnetic field exists both inside as well as outside. However magnetic field inside the rod will go on decreasing as we go towards the axis.
- (e) The force experienced by a charge particle in a magnetic field is given by, $\vec{F} = q(\vec{v} \times \vec{B})$ which is independent of mass. As q , v and B are same for both the electron and proton, hence both will experience same force.
- (b) The torque on the coil in a magnetic field is given by $\tau = nIBA \cos\theta$
For radial field, the coil is set with its plane parallel to the direction of the magnetic field B , then $\theta = 0$ and $\cos\theta = 1 \Rightarrow$ Torque = $nIBA$ (i) = $nIBA$ (maximum).
- (b) The winding of helix carry currents in the same direction therefore they experience an attractive force pulling the lower end out of mercury. As a result of this, the circuit breaks, current becomes zero and hence the force of attraction vanishes. Therefore helix comes back to its final position, completing the circuit again. In this way, the process is repeated and helix executes oscillatory motion.
- (b) For a solenoid $B_{\text{end}} = \frac{1}{2}(B_{\text{in}})$. Also for a long solenoid, magnetic field is uniform within it but this reason is not explaining the assertion.
- (d) When a charged particle is moving on a circular path in a magnetic field, the magnitude of velocity does not change but direction of velocity is changing every moment. Hence velocity is changing, so momentum ($m\vec{v}$) is also changing.
- (b) Time period, $T = \frac{2\pi m}{Bq}$ as $\left(\frac{m}{q}\right)_\alpha = 2\left(\frac{m}{q}\right)_p \Rightarrow T_\alpha = 2T_p$
Also $T \propto m$, but then $T_\alpha = 4T_p$ which is not the case.
- (d) When two long parallel wires, are connected to a battery in series. They carry currents in opposite directions, hence they repel each other.
- (a) Here, both Assertion and Reason are correct and reason is the correct explanation of assertion.

Magnetic Effect of Current

Self Evaluation Test -21

1. In the hydrogen atom, the electron is making 6.6×10^{13} r.p.s. If the radius of the orbit is 0.53×10^{-10} metre, then magnetic field produced at the centre of the orbit is

(a) 140 Tesla (b) 12.5 Tesla
(c) 1.4 Tesla (d) 0.14 Tesla

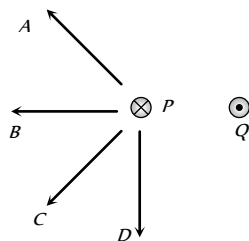
2. A coil carrying a heavy current and having large number of turns mounted in a *N-S* vertical plane and a current flows in clockwise direction. A small magnetic needle at its centre will have its north pole in

(a) East-north direction (b) West-north direction
(c) East-south direction (d) West-south direction

3. A charged particle is projected in a plane perpendicular to a uniform magnetic field. The area bounded by the path described by the particle is proportional to

(a) The velocity (b) The momentum
(c) The kinetic energy (d) None of these

4. In figure shows three long straight wires *P*, *Q* and *R* carrying currents normal to the plane of the paper. All three currents have the same magnitude. Which arrow best shows the direction of the resultant force on the wire *P*



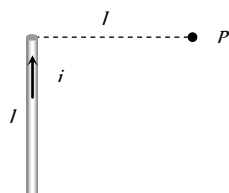
(a) A (b) B
(c) C (d) D

5. A moving coil galvanometer has 48 turns and area of coil is $4 \times 10^{-2} \text{ m}^2$. If the magnetic field is 0.2 T, then to increase the current sensitivity by 25% without changing area (*A*) and field (*B*) the number of turns should become

(a) 24 (b) 36
(c) 60 (d) 54

6. Figure shows a straight wire of length *l* current *i*. The magnitude of magnetic field produced by the current at point *P* is

(a) $\frac{\sqrt{2}\mu_0 i}{\pi l}$
(b) $\frac{\mu_0 i}{4\pi l}$
(c) $\frac{\sqrt{2}\mu_0 i}{8\pi l}$
(d) $\frac{\mu_0 i}{2\sqrt{2}\pi l}$

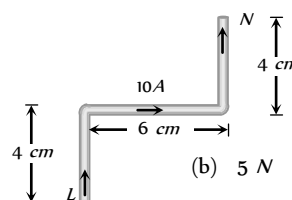


7. A winding wire which is used to frame a solenoid can bear a maximum 10 A current. If length of solenoid is 80 cm and it's cross sectional radius is 3 cm then required length of winding wire is (*B* = 0.2 T)

(a) $1.2 \times 10^2 \text{ m}$ (b) $4.8 \times 10^2 \text{ m}$

(c) $2.4 \times 10^3 \text{ m}$ (d) $6 \times 10^3 \text{ m}$

8. A current carrying wire *LN* is bent in the form shown below. If wire carries a current of 10 A and it is placed in a magnetic field of 5 T which acts perpendicular to the paper outwards then it will experience a force



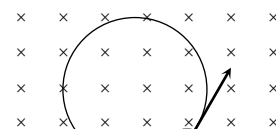
(a) Zero (b) 5 N
(c) 30 N (d) 20 N

9. A wire of length *L* is bent in the form of a circular coil and current *i* is passed through it. If this coil is placed in a magnetic field then the torque acting on the coil will be maximum when the number of turns is

(a) As large as possible (b) Any number
(c) 2 (d) 1

10. A particle having a charge of $10.0 \mu\text{C}$ and mass $1 \mu\text{g}$ moves in a circle of radius 10 cm under the influence of a magnetic field of induction 0.1 T. When the particle is at a point *P*, a uniform electric field is switched on so that the particle starts moving along the tangent with a uniform velocity. The electric field is

(a) 0.1 V/m
(b) 1.0 V/m
(c) 10.0 V/m
(d) 100 V/m



11. Two parallel long wires carry currents i_1 and i_2 with $i_1 > i_2$. When the currents are in the same direction, the magnetic field midway between the wires is $10 \mu\text{T}$. When the direction of i_1 is reversed, it becomes $40 \mu\text{T}$. the ratio i_1 / i_2 is

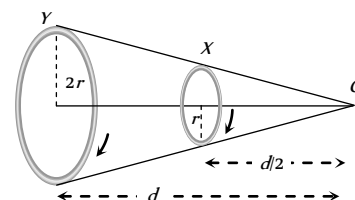
(a) 3 : 4 (b) 11 : 7
(c) 7 : 11 (d) 5 : 3

12. Two circular coils *X* and *Y*, having equal number of turns, carry equal currents in the same sense and subtend same solid angle at point *O*. If the smaller coil, *X* is midway between *O* and *Y*, then if we represent the magnetic induction due to bigger coil *Y* at *O* as *B_y* and that due to smaller coil *X* at *O* as *B_x*, then

(a) $\frac{B_y}{B_x} = 1$

(b) $\frac{B_y}{B_x} = 2$

(c) $\frac{B_y}{B_x} = \frac{1}{2}$

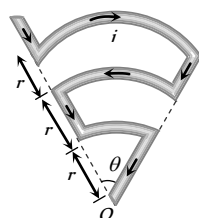


(d) $\frac{B_y}{B_x} = \frac{1}{4}$

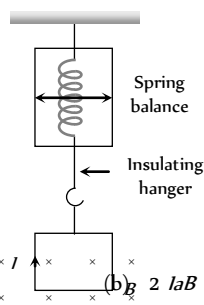
13. A fixed horizontal wire carries a current of 200 A. Another wire having a mass per unit length 10^{-2} kg/m is placed below the first wire at a distance of 2 cm and parallel to it. How much current must be passed through the second wire if it floats in air without any support? What should be the direction of current in it
- (a) 25 A (direction of current is same to first wire)
 (b) 25 A (direction of current is opposite to first wire)
 (c) 49 A (direction of current is same to first wire)
 (d) 49 A (direction of current is opposite to first wire)

14. Find magnetic field at O

- (a) $\frac{5\mu_0 i \theta}{24\pi r}$
 (b) $\frac{\mu_0 i \theta}{24\pi r}$
 (c) $\frac{11\mu_0 i \theta}{24\pi r}$
 (d) Zero



15. A square loop of side a hangs from an insulating hanger of spring balance. The magnetic field of strength B occurs only at the lower edge. It carries a current I. Find the change in the reading of the spring balance if the direction of current is reversed



- (a) IaB
 (b) $2IaB$
 (c) $\frac{IaB}{2}$
 (d) $\frac{3}{2}IaB$

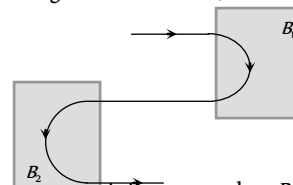
16. A charge of $2.0 \mu\text{C}$ moves with a speed of $3.0 \times 10^6 \text{ ms}^{-1}$ along +ve X-axis. A magnetic field of strength $\vec{B} = -0.2 \hat{k} \text{ Tesla}$ exists in space. What is the magnetic force ($\vec{F}_m$) on the charge
- (a) $F_m = 1.2 \text{ N}$ along +ve x - direction
 (b) $F_m = 1.2 \text{ N}$ along -ve x - direction

(c) $F_m = 1.2 \text{ N}$ along +ve y - direction

(d) $F_m = 1.2 \text{ N}$ along -ve y - direction

17. Five very long, straight wires are bound together to form a small cable. Currents carried by the wires are $I_1 = 20 \text{ A}$, $I_2 = -6 \text{ A}$, $I_3 = 12 \text{ A}$, $I_4 = -7 \text{ A}$, $I_5 = 18 \text{ A}$. The magnetic induction at a distance of 10 cm from the cable is
- (a) $34 \mu\text{T}$ (b) 74 mT
 (c) 34 mT (d) $74 \mu\text{T}$

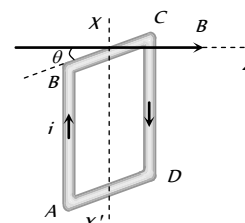
18. Following figure shows the path of an electron that passes through two regions containing uniform magnetic fields of magnitudes B and B_2 . Its path in each region is a half circle, choose the correct option



- (a) B is into the page and it is stronger than B_2
 (b) B is into the page and it is weaker than B_2
 (c) B is out of the page and it is weaker than B_2
 (d) B is out of the page and it is stronger than B_2

19. The square loop ABCD, carrying a current i, is placed in uniform magnetic field B, as shown. The loop can rotate about the axis XX'. The plane of the loop makes an angle θ ($\theta < 90^\circ$) with the direction of B. Through what angle will the loop rotate by itself before the torque on it becomes zero

- (a) θ
 (b) $90^\circ - \theta$
 (c) $90^\circ + \theta$
 (d) $180^\circ - \theta$



20. A cylindrical conductor of radius 'R' carries a current 'i'. The value of magnetic field at a point which is R/4 distance inside from the surface is 10 T. Find the value of magnetic field at point which is 4R distance outside from the surface

- (a) $\frac{4}{3} \text{ T}$ (b) $\frac{8}{3} \text{ T}$
 (c) $\frac{40}{3} \text{ T}$ (d) $\frac{80}{3} \text{ T}$

21. Three long straight wires are connected parallel to each other across a battery of negligible internal resistance. The ratio of their resistances are 3 : 4 : 5. What is the ratio of distances of middle wire from the others if the net force experienced by it is zero
- (a) 4 : 3 (b) 3 : 1
 (c) 5 : 3 (d) 2 : 3

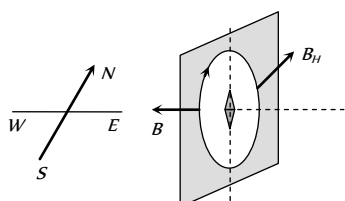
AS Answers and Solutions

(SET -21)

1. (b) $i = qv$

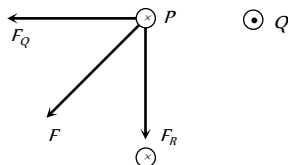
$$B = \frac{\mu_0 i}{2r} = \frac{\mu_0 qv}{2r} = \frac{4\pi \times 10^{-7} \times 1.6 \times 10^{-19} \times 6.6 \times 10^{15}}{2 \times 0.53 \times 10^{-10}}$$

$$= \frac{2\pi \times 1.6 \times 6.6}{5.3} = 12.518 \text{ Tesla}$$



2. (b)

3. (c) $r = \frac{\sqrt{2mK}}{qB}$ and $A = \pi r^2 \Rightarrow A = \frac{\pi(2mK)}{q^2 B^2} \Rightarrow A \propto K$.
4. (c) The forces F_Q and F_R are the forces applied by wires Q and R respectively on the wire P as shown in figure. Their resultant force F is best shown by C .



5. (c) As we know
Current sensitivity $S_i = \frac{NBA}{C}$
 $\Rightarrow S_i \propto N \Rightarrow \frac{(S_i)_1}{(S_i)_2} = \frac{N_1}{N_2} \Rightarrow \frac{100}{125} = \frac{48}{N_2} \Rightarrow N_2 = 60$.

6. (c) The given situation can be redrawn as follow.

As we know the general formula for finding the magnetic field due to a finite length wire

$$B = \frac{\mu_0}{4\pi} \cdot \frac{i}{r} (\sin\phi_1 + \sin\phi_2)$$

Here $\phi_1 = 0^\circ$, $\phi_2 = 45^\circ$

$$\therefore B = \frac{\mu_0}{4\pi} \cdot \frac{i}{r} (\sin 0^\circ + \sin 45^\circ) = \frac{\mu_0}{4\pi} \cdot \frac{i}{\sqrt{2}l} \Rightarrow B = \frac{\sqrt{2}\mu_0 i}{8\pi l}$$

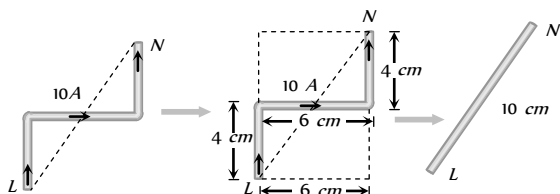
7. (c) $B = \frac{\mu_0 Ni}{l}$ where N = Total number of turns, l = length of the solenoid

$$\Rightarrow 0.2 = \frac{4\pi \times 10^{-7} \times N \times 10}{0.8} \Rightarrow N = \frac{4 \times 10^4}{\pi}$$

Since N turns are made from the winding wire so length of the wire (L) = $2\pi \times N$ [2π = length of each turns]

$$\Rightarrow L = 2\pi \times 3 \times 10^{-2} \times \frac{4 \times 10^4}{\pi} = 2.4 \times 10^3 \text{ m}$$

8. (b) The given wire can be replaced by a straight wire as shown below



Hence force experienced by the wire

$$F = BIl = 5 \times 10 \times 0.1 = 5 \text{ N}$$

9. (d) $\tau_{\max} = MB$ or $\tau_{\max} = ni\pi r^2 B$. Let number of turns in length l is n so $l = n(2\pi r)$ or $\alpha = \frac{l}{2\pi r}$

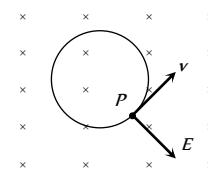
$$\Rightarrow \tau_{\max} = \frac{ni\pi r^2 B}{4\pi^2 n^2} = \frac{l^2 i B}{4\pi n_{\min}} \Rightarrow \tau_{\max} \propto \frac{1}{n_{\min}} \Rightarrow n_{\min} = 1$$

10. (c) When the particle moves along a circle in the magnetic field B , the magnetic force is radially inward. If an electric field of proper magnitude is switched on which is directed radially outwards, the particle may experience no force. It will then move along a straight line with uniform velocity. This will be the case when $qE = qvB \Rightarrow E = vB$

$$\text{also } r = \frac{mv}{qB} \Rightarrow v = \frac{qBr}{m}$$

$$\text{So } E = \frac{qB^2 r}{m}$$

$$= \frac{(10 \times 10^{-6}) \times (0.1)^2 \times 10 \times 10^{-2}}{1 \times 10^{-3} \times 10^{-6}} = 10 \text{ V/m}$$

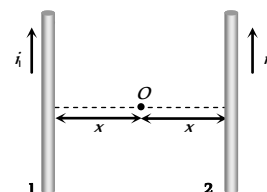


11. (d) Initially when wires carry currents in the same direction as shown.

Magnetic field at mid point O due to wires 1 and 2 are respectively

$$B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2i_1}{x} \otimes$$

$$\text{and } B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2i_2}{x} \odot$$



$$\text{Hence net magnetic field at } O \quad B_{\text{net}} = \frac{\mu_0}{4\pi} \times \frac{2}{x} \times (i_1 - i_2)$$

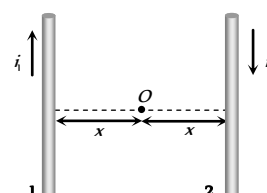
$$\Rightarrow 10 \times 10^{-6} = \frac{\mu_0}{4\pi} \cdot \frac{2}{x} (i_1 - i_2) \quad \dots (i)$$

If the direction of i_2 is reversed then

$$B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2i_1}{x} \otimes$$

$$\text{and } B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2i_2}{x} \otimes$$

$$\text{So } B_{\text{net}} = \frac{\mu_0}{4\pi} \cdot \frac{2}{x} (i_1 + i_2)$$



$$\Rightarrow 40 \times 10^{-6} = \frac{\mu_0}{4\pi} \cdot \frac{2}{x} (i_1 + i_2) \quad \dots (ii)$$

$$\text{Dividing equation (ii) by (i)} \quad \frac{i_1 + i_2}{i_1 - i_2} = \frac{4}{1} \Rightarrow \frac{i_1}{i_2} = \frac{5}{3}$$

12. (c) Magnetic field at O due to bigger coil Y , is

$$B_Y = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i(2r)^2}{\{d^2 + (2r)^2\}^{3/2}} = \frac{\mu_0}{4\pi} \cdot \frac{8\pi i r^2}{(d^2 + 4r^2)^{3/2}}$$

Magnetic field at O due to smaller coil X is

$$B_X = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i r^2}{\left\{\left(\frac{d}{2}\right)^2 + r^2\right\}^{3/2}} = \frac{\mu_0}{4\pi} \cdot \frac{16\pi i r^2}{(d^2 + 4r^2)^{3/2}}$$

$$\Rightarrow \frac{B_y}{B_x} = \frac{1}{2}$$

13. (c) For floating the second wire

$$\left| \begin{array}{l} \text{Down ward weight} \\ \text{of second wire} \end{array} \right| = \left| \begin{array}{l} \text{Magnetic force} \\ \text{on it} \end{array} \right|$$

$$\Rightarrow mg = \frac{\mu_0}{4\pi} \cdot \frac{2i_1 i_2}{a} \times l$$

$$\Rightarrow \left(\frac{m}{l} \right) g = \frac{\mu_0}{4\pi} \cdot \frac{2i_1 i_2}{a}$$

$$\Rightarrow 10^{-2} \times 9.8 = 10^{-7} \times \frac{2 \times 200 \times i}{2 \times 10^{-2}} \Rightarrow i = 49 \text{ A}$$

(Direction of current is same to first wire)

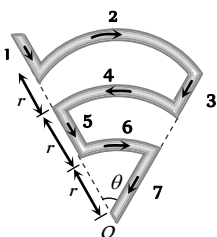
14. (a) $B_1 = B_3 = B_5 = 0$

$$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{\theta i}{3r} \otimes, B_4 = \frac{\mu_0}{4\pi} \cdot \frac{\theta i}{2r} \odot$$

$$\text{and } B_6 = \frac{\mu_0}{4\pi} \cdot \frac{\theta i}{r} \otimes$$

$\therefore$ Net magnetic field at O,

$$B_{\text{net}} = B_2 - B_4 + B_6 = \frac{\mu_0}{4\pi} \cdot \frac{\theta i}{r} \left(\frac{1}{3} - \frac{1}{2} + 1 \right) = \frac{5\mu_0 \theta i}{24\pi r}$$



15. (b) Initially $F_1 = mg + IaB$ (down wards)

when the direction of current is reversed

$$F_2 = mg - IaB \text{ (down wards)} \Rightarrow \Delta F = 2IaB$$

16. (c) By using $\vec{F}_m = q(\vec{v} \times \vec{B})$

$$\Rightarrow \vec{F}_m = 2 \times 10^{-6} \{ 3 \times 10^6 \hat{i} \times (-0.2) \hat{k} \} = -1.2 (\hat{i} \times \hat{k}) = +1.2 \hat{j}$$

i.e., 1.2 N in positive y direction.

17. (d) $i_{\text{net}} = 20 - 6 + 12 - 7 + 18 = 37 \text{ A}$ so $B = \frac{\mu_0}{4\pi} \cdot \frac{2i}{a}$

$$= 10^{-7} \times \frac{2 \times 37}{10 \times 10^{-2}} = 74 \times 10^{-6} \text{ T} = 74 \mu\text{T}$$

18. (a) Direction of field can be find using Fleming left hand rule and

$$r \propto \frac{1}{B}$$

19. (c) In the position shown, AB is outside and CD is inside the plane of the paper. The Ampere force on AB acts into the paper. The torque on the loop will be clockwise, as seen from above. The loop must rotate through an angle $(90^\circ + \theta)$ before the plane of the loop becomes normal to the direction of the direction of B and the torque becomes zero.

20. (b) Magnetic field inside the cylindrical conductor

$$B_{\text{in}} = \frac{\mu_0}{4\pi} \cdot \frac{2ir}{R^2} \quad (R = \text{Radius of cylinder, } r = \text{distance of observation point from axis of cylinder})$$

Magnetic field out side the cylinder at a distance r' from it's

$$\text{axis } B_{\text{out}} = \frac{\mu_0}{4\pi} \cdot \frac{2i}{r'}$$

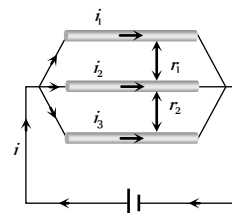
$$\Rightarrow \frac{B_{\text{in}}}{B_{\text{out}}} = \frac{r'}{R^2} \Rightarrow \frac{10}{B_{\text{out}}} = \frac{\left(R - \frac{R}{4} \right) (R + 4R)}{R^2} \Rightarrow B_{\text{out}} = \frac{8}{3} \text{ T}$$

21. (c) The wires are in parallel and ratio of their resistances are 3 : 4 : 5, Hence currents in wires are in the ratio $\frac{1}{3} : \frac{1}{4} : \frac{1}{5}$

$$i_1 = \frac{k}{3}$$

$$i_2 = \frac{k}{4}$$

$$i_3 = \frac{k}{5}$$



$$\text{Force between top and middle wire } F_1 = \frac{\mu_0}{4\pi} \cdot \frac{2i_1 i_2}{r_1}$$

$$= \frac{\mu_0}{4\pi} \times 2 \left(\frac{1}{3} \right) \left(\frac{1}{4} \right) k^2$$

$$\text{Force between bottom and middle wire } F_2 = \frac{\mu_0}{4\pi} \cdot \frac{2i_2 i_3}{r_2}$$

$$\text{As the forces are equal and opposite so } F_1 = F_2 \Rightarrow \frac{r_1}{r_2} = \frac{5}{3}$$



Chapter 22 Magnetism

The molecular theory of magnetism was given by Weber and modified later by Ewing. According to this theory.

Every molecule of a substance is a complete magnet in itself. However, in an **magnetic** substance the molecular magnets are randomly oriented to give net zero magnetic moment. On magnetising, the molecular magnets are realigned in a specific direction leading to a net magnetic moment.

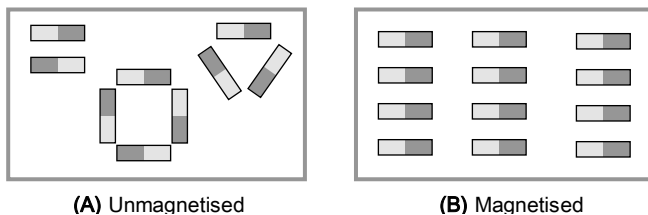


Fig. 22.1

Bar Magnet

A bar magnet consists of two equal and opposite magnetic poles separated by a small distance. Poles are not exactly at the ends. The shortest distance between two poles is called effective length (L_e) and is less than its geometric length (L_g). For a bar magnet $L_e = (5/6) L_g$.

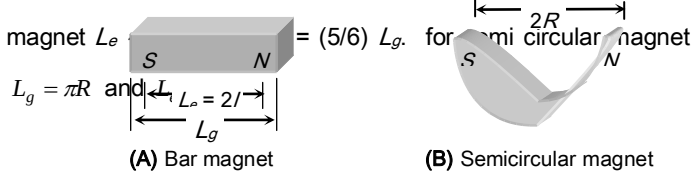


Fig. 22.2

(1) **Directive properties** : When a magnet is suspended freely it stays in the earth's $N-S$ direction (in magnetic meridian).

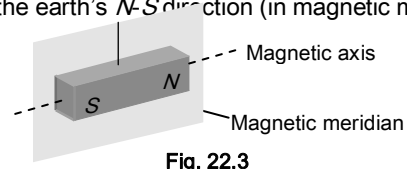


Fig. 22.3

(2) **Monopole concept** : If a magnet is broken into number of pieces, each piece becomes a magnet. This in turn implies that monopoles do not exist. (*i.e.*, ultimate individual unit of magnetism in any magnet is called dipole).

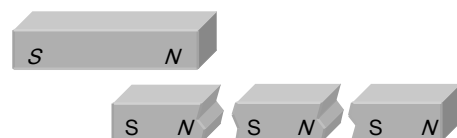


Fig. 22.4

(3) For two rods as shown, if both the rods attract in figure (A) and don't attract in figure (B) then, Q is a magnetic and P is simple iron rod. Repulsion is sure test of magnetism.

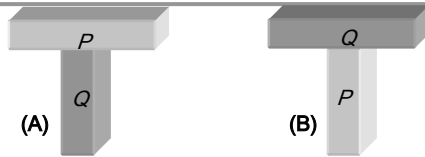


Fig. 22.5

length as well as perpendicular to the length simultaneously as shown in the figure then

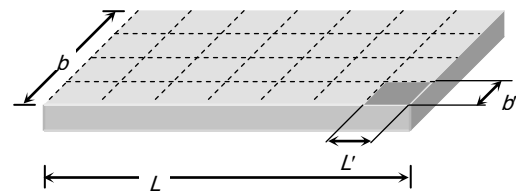


Fig. 22.8

Length of each part $L' = \frac{L}{n}$, breadth of each part $b' = \frac{b}{n}$,
 Mass of each part $w' = \frac{w}{n}$, pole strength of each part $m' = \frac{m}{n}$,
 Magnetic moment of each part $M' = m' L' = \frac{m}{n} \times \frac{L}{n} = \frac{M}{n^2}$

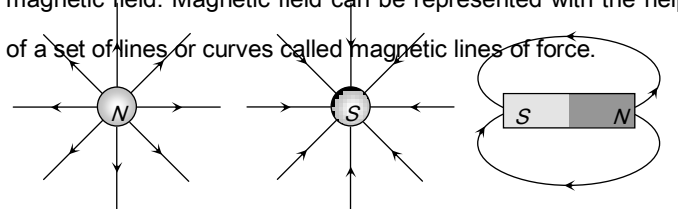
If initially moment of inertia of bar magnet about the axes passing from centre and perpendicular to its length is

$$I = w \left(\frac{L^2 + b^2}{12} \right) \text{ then moment of inertia of each part } I' = \frac{I}{n^2}$$

(7) **Cutting of a thin bar magnet** : For thin magnet $b = 0$ so
 $L' = \frac{L}{n}$, $w' = \frac{w}{n}$, $m' = \frac{m}{n}$, $I' = \frac{I}{n^3}$

Various Terms Related to Magnetism

(1) **Magnetic field and magnetic lines of force** : Space around a magnetic pole or magnet or current carrying wire within which its effect can be experienced is defined as magnetic field. Magnetic field can be represented with the help of a set of lines or curves called magnetic lines of force.



(A) Isolated north pole (B) Isolated south pole (C) Magnetic dipole

Fig. 22.9

(4) **Pole strength (m)** : The strength of a magnetic pole to attract magnetic materials towards itself is known as pole strength.

- (i) It is a scalar quantity.
- (ii) Pole strength of N and S pole of a magnet is conventionally represented by $+m$ and $-m$ respectively.
- (iii) Its SI unit is $\text{amp} \times \text{m}$ or N Tesla and dimensions are $[LA]$.
- (iv) Pole strength of the magnet depends on the nature of material of magnet and area of cross section. It doesn't depend upon

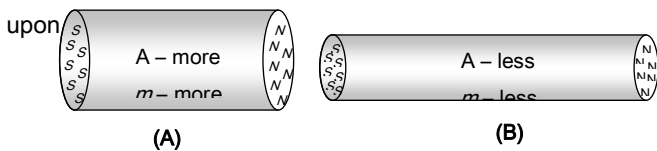


Fig. 22.6

(5) **Magnetic moment or magnetic dipole moment ($\vec{M}$)** : It represents the strength of magnet. Mathematically it is defined as the product of the strength of either pole and effective length. i.e.

$$\vec{M} = m(2\vec{l})$$

Fig. 22.7

- (i) It is a vector quantity directed from south to north.
- (ii) Its S.I. unit $\text{amp} \times \text{m}^2$ or N-m / Tesla and dimensions $[AL^2]$

(6) **Cutting of a rectangular bar magnet** : Suppose we have a rectangular bar magnet having length, breadth and mass are L , b and w respectively if it is cut in n equal parts along the

(2) **Magnetic flux (ϕ) and flux density (B)**

(i) The number of magnetic lines of force passing normally through a surface is defined as magnetic flux (ϕ). Its S.I. unit is *weber (wb)* and CGS unit is *Maxwell*.

Remember $1 \text{ wb} = 10^8 \text{ Maxwell}$.

(ii) When a piece of a magnetic substance is placed in an external magnetic field the substance becomes magnetised. The number of magnetic lines of induction inside a magnetised substance crossing unit area normal to their direction is called magnetic induction or magnetic flux density ($\vec{B}$). It is a vector quantity.

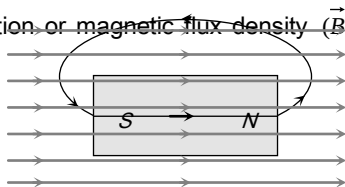


Fig. 22.10

Its SI unit is *Tesla* which is equal to $\frac{\text{wb}}{\text{m}^2} = \frac{\text{N}}{\text{amp} \times \text{m}} = \frac{\text{J}}{\text{amp} \times \text{m}^2} = \frac{\text{volt} \times \text{sec}}{\text{m}^2}$

and CGS unit is *Gauss*. Remember $1 \text{ Tesla} = 10^4 \text{ Gauss}$.

(3) **Magnetic permeability** : It is the degree or extent to which magnetic lines of force can enter a substance and is denoted by μ . Or characteristic of a medium which allows magnetic flux to pass through it is called its permeability. e.g. permeability of soft iron is 1000 times greater than that of air.

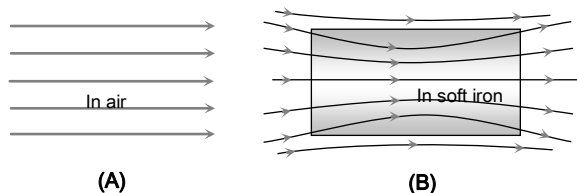


Fig. 22.11

Also $\mu = \mu_0 \mu_r$; where μ_0 = absolute permeability of air or free space = $4\pi \times 10^{-7} \text{ tesla} \times \text{m} / \text{amp}$.

and μ_r = Relative permeability of the medium = $\frac{B}{B_0} = \frac{\text{flux density in material}}{\text{flux density in vacuum}}$.

(4) **Intensity of magnetising field ($\vec{H}$) (magnetising field)** :

It is the degree or extent to which a magnetic field can magnetise a substance. Also $H = \frac{B}{\mu}$.

Its SI unit is

$$\text{A/m} = \frac{\text{N}}{\text{m}^2 \times \text{Tesla}} = \frac{\text{N}}{\text{wb}} = \frac{\text{J}}{\text{m}^3 \times \text{Tesla}} = \frac{\text{J}}{\text{m} \times \text{wb}}$$

Its CGS unit is *Oersted*. Also $1 \text{ Oersted} = 80 \text{ A/m}$

(5) **Intensity of magnetisation (I)** : It is the degree to which a substance is magnetised when placed in a magnetic field.

It can also be defined as the pole strength per unit cross sectional area of the substance or the induced dipole moment per unit volume.

Hence $I = \frac{m}{A} = \frac{M}{V}$. It is a vector quantity, its S.I. unit is *Ampl/m*.

(6) **Magnetic susceptibility (χ_m)** : It is the property of the substance which shows how easily a substance can be magnetised. It can also be defined as the ratio of intensity of magnetisation (I) in a substance to the magnetic intensity (H) applied to the substance, i.e. $\chi_m = \frac{I}{H}$. It is a scalar quantity with no units and dimensions.

(7) **Relation between permeability and susceptibility** : Total magnetic flux density B in a material is the sum of magnetic flux density in vacuum B_0 produced by magnetising force and magnetic flux density due to magnetisation of material B_m . i.e. $B = B_0 + B_m \Rightarrow B = \mu_0 H + \mu_0 I = \mu_0 (H + I) = \mu_0 H (1 + \chi_m)$. Also $\mu_r = (1 + \chi_m)$

Force and Field

(1) **Coulombs law in magnetism** : The force between two magnetic poles of strength m_1 and m_2 lying at a distance r is

given by $F = k \cdot \frac{m_1 m_2}{r^2}$. In S.I. units $k = \frac{\mu_0}{4\pi} = 10^{-7} \text{ wb / Amp} \times \text{m}$,

In CGS units $k = 1$

(2) Magnetic field

(i) Magnetic field due to an imaginary magnetic pole (Pole strength m): Is given by $B = \frac{F}{m_0}$ also $B = \frac{\mu_0}{4\pi} \cdot \frac{m}{d^2}$

(ii) Magnetic field due to a bar magnet: At a distance r from the centre of magnet

(a) On axial position

$$B_a = \frac{\mu_0}{4\pi} \frac{2Mr}{(r^2 - l^2)^2}; \text{ If } l \ll r \text{ then } B_a = \frac{\mu_0}{4\pi} \frac{2M}{r^3}$$

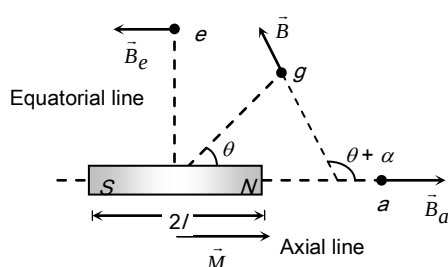


Fig. 22.12

(b) On equatorial position: $B_e = \frac{\mu_0}{4\pi} \frac{M}{(r^2 + l^2)^{3/2}}$; If $l \ll r$;

$$\text{then } B_e = \frac{\mu_0}{4\pi} \frac{M}{r^3}$$

(c) General position: In general position for a short bar magnet $B_g = \frac{\mu_0}{4\pi} \frac{M}{r^3} \sqrt{3 \cos^2 \theta + 1}$

(3) **Bar magnet in magnetic field**: When a bar magnet is left free in an uniform magnetic field, it aligns itself in the directional field.

(i) Torque: $\tau = MB \sin \theta \Rightarrow \vec{\tau} = \vec{M} \times \vec{B}$

(ii) Work: $W = MB(1 - \cos \theta)$

(iii) Potential energy: $U = -MB \cos \theta = -\vec{M} \cdot \vec{B}$; (θ = Angle made by the dipole with the field)

(4) **Gauss's law in magnetism**: Net magnetic flux through

any closed surface is always zero i.e. $\oint \vec{B} \cdot d\vec{s} = 0$

Earth's Magnetic Field (Terrestrial Magnetism)

As per the most established theory it is due to the rotation of the earth where by the various charged ions present in the molten state in the core of the earth rotate and constitute a current.

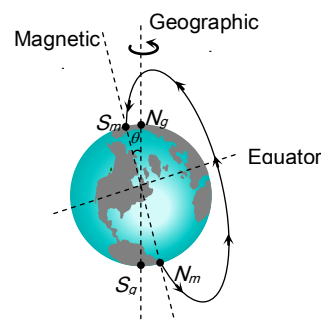


Fig. 22.13

(1) The magnetic field of earth is similar to one which would be obtained if a huge magnet is assumed to be buried deep inside the earth at its centre.

(2) The axis of rotation of earth is called geographic axis and the points where it cuts the surface of earth are called geographical poles (N_g, S_g). The circle on the earth's surface perpendicular to the geographical axis is called equator.

(3) A vertical plane passing through the geographical axis is called geographical meridian.

(4) The axis of the huge magnet assumed to be lying inside the earth is called magnetic axis of the earth. The points where the magnetic axis cuts the surface of earth are called magnetic poles. The circle on the earth's surface perpendicular to the magnetic axis is called magnetic equator.

(5) Magnetic axis and Geographical axis don't coincide but they make an angle of 17.5° with each other.

(6) Magnetic equator divides the earth into two hemispheres. The hemisphere containing south polarity of

earth's magnetism is called northern hemisphere while the other, the southern hemisphere.

(7) The magnetic field of earth is not constant but changes irregularly from place to place on the surface of the earth and even at a given place it varies with time too.

(8) Direction of earth's magnetic field is from *S* (geographical south) to *N* (geographical north).

Elements of Earth's Magnetic Field

The magnitude and direction of the magnetic field of the earth at a place are completely given by certain quantities known as magnetic elements.

(1) **Magnetic Declination (θ)** : It is the angle between geographic and the magnetic meridian planes.

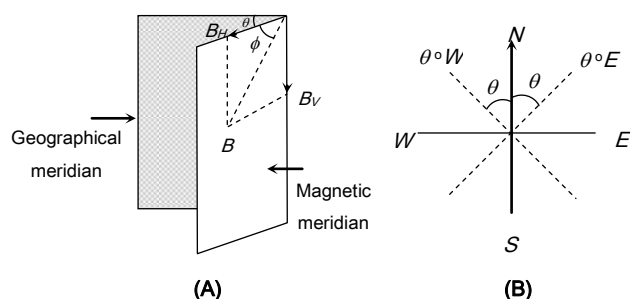


Fig. 22.14

Declination at a place is expressed at $\theta^\circ E$ or $\theta^\circ W$ depending upon whether the north pole of the compass needle lies to the east or to the west of the geographical axis.

(2) **Angle of inclination or Dip (ϕ)** : It is the angle between the direction of intensity of total magnetic field of earth and a horizontal line in the magnetic meridian.

(3) **Horizontal component of earth's magnetic field (B_H)** : Earth's magnetic field is horizontal only at the magnetic equator. At any other place, the total intensity can be resolved into horizontal component (B_H) and vertical component (B_V).

$$\text{Also } B_H = B \cos \phi \dots\dots (i) \text{ and } B_V = B \sin \phi \dots\dots (ii)$$

By squaring and adding equation (i) and (ii)

$$B = \sqrt{B_H^2 + B_V^2}$$

$$\text{Dividing equation (ii) by equation (i) } \tan \phi = \frac{B_V}{B_H}$$

Magnetic Maps and Neutral Points

(1) **Magnetic maps** : Magnetic maps (*i.e.* Declination, dip and horizontal component) over the earth vary in magnitude from place to place. It is found that many places have the same value of magnetic elements. The lines are drawn joining all place on the earth having same value of a magnetic element. These lines form magnetic map.

(i) **Isogonic lines** : These are the lines on the magnetic map joining the places of equal declination.

(ii) **Agonic line** : The line which passes through places having zero declination is called agonic line.

(iii) **Isoclinic lines** : These are the lines joining the points of equal dip or inclination.

(iv) **Aclinic line** : The line joining places of zero dip is called acclinic line (or magnetic equator)

(v) **Isodynamic lines** : The lines joining the points or places having the same value of horizontal component of earth's magnetic field are called isodynamic lines.

(2) **Neutral points** : A neutral point is a point at which the resultant magnetic field is zero. In general the neutral point is obtained when horizontal component of earth's field is balanced by the field produced by the magnet.

Tangent Law

When a small magnet is suspended in two uniform magnetic fields B and B_H which are at right angles to each other, the magnet comes to rest at an angle θ with respect to B_H .

In equilibrium

$$MB_H \sin \theta = MB \sin(90^\circ - \theta)$$

$$\Rightarrow B = B_H \tan \theta. \text{ This is called tangent law.}$$

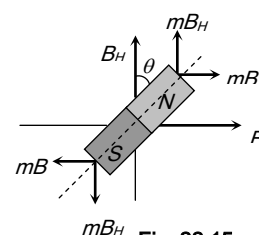


Fig. 22.15

Tangent Galvanometer

It consists of three circular coils of insulated copper wire wound on a vertical circular frame made of nonmagnetic material as ebonite or wood. A small magnetic compass needle is pivoted at the centre of the vertical circular frame. When the coil of the tangent galvanometer is kept in magnetic meridian and current passes through any of the coil then the needle at the centre gets deflected and comes to an equilibrium position under the action of two perpendicular field : one due to horizontal component of earth and the other due to field (B) set up by the coil due to current.

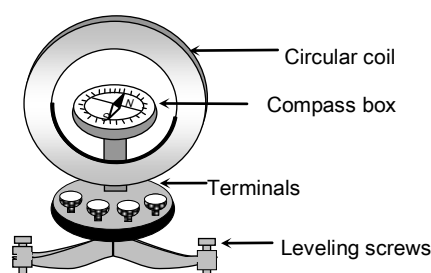


Fig. 22.16

In equilibrium $B = B_H \tan \theta$ where $B = \frac{\mu_0 n i}{2r}$; n = number of turns, r = radius of coil, i = the current to be measured, θ = angle made by needle from the direction of B_H in equilibrium.

$$\text{Hence } \frac{\mu_0 N i}{2r} = B_H \tan \theta \Rightarrow i = k \tan \theta \text{ where } k = \frac{2r B_H}{\mu_0 N} \text{ is}$$

called reduction factor.

Deflection Magnetometer

It's working is based on the principle of tangent law. It consists of a small compass needle, pivoted at the centre of a circular box. The box is kept in a wooden frame having two meter scale fitted on it's two arms. Reading of a scale at any point directly gives the distance of that point from the centre of compass needle.



Fig. 22.17

(1) **Tan A position** : In this position the magnetometer is set perpendicular to magnetic meridian. So that, magnetic field due to magnet, is in axial position and perpendicular to earth's field.

$$\text{Hence } B_H \tan \theta = \frac{\mu_0}{4\pi} \cdot \frac{2Mr}{(r^2 - l^2)^2} \text{ or } B_H \tan \theta = \frac{\mu_0}{4\pi} \cdot \frac{2M}{r^3}$$

(2) **Tan B position** : The arms of magnetometer are set in magnetic meridian, so that the magnetic field due to magnet is at it's equatorial position. Hence $B_H \tan \theta = \frac{\mu_0}{4\pi} \cdot \frac{M}{(r^2 + l^2)^{3/2}}$ or

$$B_H \tan \theta = \frac{\mu_0}{4\pi} \cdot \frac{M}{r^3}$$

(3) **Comparison of magnetic moments** : According to deflection method $\frac{M_1}{M_2} = \frac{\tan \theta_1}{\tan \theta_2}$

$$\text{According to null deflection method } \frac{M_1}{M_2} = \left(\frac{d_1}{d_2} \right)^3$$

Vibration Magnetometer

Vibration magnetometer is used for comparison of magnetic moments and magnetic fields. This device works on the principle, that whenever a freely suspended magnet in a uniform magnetic field, is disturbed from it's equilibrium position, it starts vibrating about the mean position.

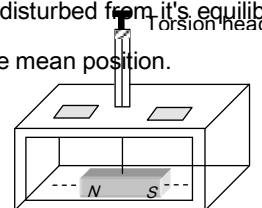


Fig. 22.18

Time period of oscillation of experimental bar magnet (magnetic moment M) in earth's magnetic field (B_H) is given by

the formula. $T = 2\pi \sqrt{\frac{I}{MB_H}}$; where, I = moment of inertia of

short bar magnet = $\frac{wL^2}{12}$ (w = mass of bar magnet)

(1) **Determination of magnetic moment of a magnet** : The experimental (given) magnet is put into vibration magnetometer and it's time period T is determined. Now

$$T = 2\pi \sqrt{\frac{I}{MB_H}} \Rightarrow M = \frac{4\pi^2 I}{B_H \cdot T^2}$$

(2) Comparison of horizontal components of earth's magnetic field at two places

$$T = 2\pi \sqrt{\frac{I}{MB_H}}; \text{ since } I \text{ and } M \text{ of the magnet are constant,}$$

$$\text{So } T^2 \propto \frac{1}{B_H} \Rightarrow \frac{(B_H)_1}{(B_H)_2} = \frac{T_2^2}{T_1^2}$$

(3) Comparison of magnetic moment of two magnets of same size and mass

$$T = 2\pi \sqrt{\frac{I}{M \cdot B_H}}; \text{ Here } I \text{ and } B_H \text{ are constants.}$$

$$\text{So } M \propto \frac{1}{T^2} \Rightarrow \frac{M_1}{M_2} = \frac{T_2^2}{T_1^2}$$

(4) Comparison of magnetic moments by sum and difference method

Sum position

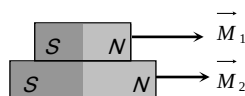


Fig. 22.19

Net magnetic moment $M_s = M_1 + M_2$

Net moment of inertia $I_s = I_1 + I_2$

Time period of oscillation of this pair in earth's magnetic field (B_H)

$$T_s = 2\pi \sqrt{\frac{I_s}{M_s B_H}} = 2\pi \sqrt{\frac{I_1 + I_2}{(M_1 + M_2) B_H}} \quad \dots(i)$$

$$\text{Frequency } \nu_s = \frac{1}{2\pi} \sqrt{\frac{(M_1 + M_2) B_H}{I_s}}$$

Difference position

Net magnetic moment

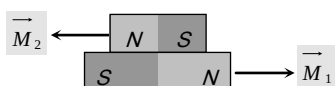


Fig. 22.20

$M_d = M_1 + M_2$

Net moment of inertia $I_d = I_1 + I_2$

$$\text{and } T_d = 2\pi \sqrt{\frac{I_d}{M_d B_H}} = 2\pi \sqrt{\frac{I_1 + I_2}{(M_1 - M_2) B_H}} \quad \dots(ii)$$

$$\text{and } \nu_d = \frac{1}{2\pi} \sqrt{\frac{(M_1 + M_2) B_H}{(I_1 + I_2)}}. \text{ From equation (i) and (ii) we}$$

get

$$\frac{T_s}{T_d} = \sqrt{\frac{M_1 - M_2}{M_1 + M_2}} \Rightarrow \frac{M_1}{M_2} = \frac{T_d^2 + T_s^2}{T_d^2 - T_s^2} = \frac{\nu_s^2 + \nu_d^2}{\nu_s^2 - \nu_d^2}$$

(5) To find the ratio of magnetic field : Suppose it is required to find the ratio $\frac{B}{B_H}$ where B is the field created by magnet and B_H is the horizontal component of earth's magnetic field.

To determine $\frac{B}{B_H}$ a primary (main) magnet is made to first oscillate in earth's magnetic field (B_H) alone and its time period of oscillation (T) is noted.

$$T = 2\pi \sqrt{\frac{I}{M B_H}}$$

$$\text{and frequency } \nu = \frac{1}{2\pi} \sqrt{\frac{M B_H}{I}}$$

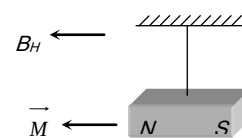


Fig. 22.21

Now a secondary magnet placed near the primary magnet so primary magnet oscillate in a new field with is the resultant of B and B_H and now time period, is noted again.

$$T' = 2\pi \sqrt{\frac{I}{M(B + B_H)}}$$

$$\text{or } \nu' = \frac{1}{2\pi} \sqrt{\frac{M(B + B_H)}{I}}$$

$$\Rightarrow \frac{B}{B_H} = \left(\frac{\nu'}{\nu} \right)^2 - 1$$

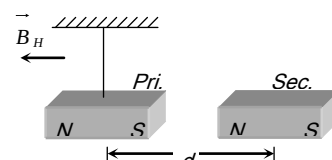


Fig. 22.22

Magnetic Materials

On the basis of mutual interactions or behaviour of various materials in an external magnetic field, the materials are divided in three main categories.

(1) **Diamagnetic materials** : Diamagnetism is the intrinsic property of every material and it is generated due to mutual interaction between the applied magnetic field and orbital motion of electrons.

(2) **Paramagnetic materials** : In these substances the inner orbits of atoms are incomplete. The electron spins are uncoupled, consequently on applying a magnetic field the magnetic moment generated due to spin motion align in the direction of magnetic field and induces magnetic moment in its direction due to which the material gets feebly magnetised. In these materials the electron number is odd.

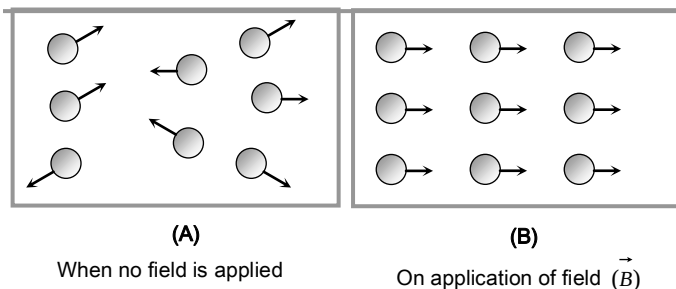


Fig. 22.23

(3) **Ferromagnetic materials** : In some materials, the permanent atomic magnetic moments have strong tendency to align themselves even without any external field.

These materials are called ferromagnetic materials.

In every unmagnetised ferromagnetic material, the atoms form domains inside the material. Different domains, however, have different directions of magnetic moment and hence the materials remain unmagnetised. On applying an external magnetic field, these domains rotate and align in the direction of magnetic field.

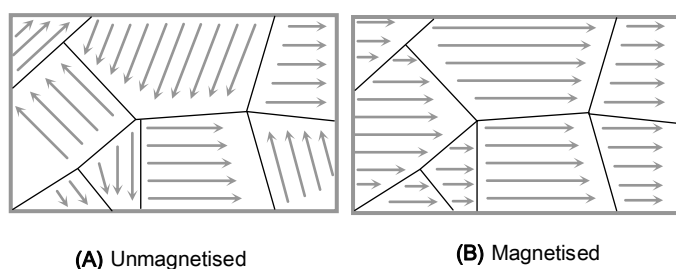


Fig. 22.24

(4) **Curie Law** : The magnetic susceptibility of paramagnetic substances is inversely proportional to its absolute temperature *i.e.* $\chi \propto \frac{1}{T} \Rightarrow \chi \propto \frac{C}{T}$; where C = Curie constant, T = absolute temperature.

On increasing temperature, the magnetic susceptibility of paramagnetic materials decreases and vice versa.

The magnetic susceptibility of ferromagnetic substances does not change according to Curie law.

(5) **Curie temperature (T_c)** : The temperature above which a ferromagnetic material behaves like a paramagnetic material is defined as Curie temperature (T_c).

or

The minimum temperature at which a ferromagnetic substance is converted into paramagnetic substance is defined as Curie temperature. For various ferromagnetic materials its values are different, *e.g.* for Ni , $T_{c_{Ni}} = 358^\circ C$ for Fe , $T_{c_{Fe}} = 770^\circ C$

for Co , $T_{c_{Co}} = 1120^\circ C$

At this temperature the ferromagnetism of the substances suddenly vanishes.

(6) **Curie-weiss law** : At temperatures above Curie temperature the magnetic susceptibility of ferromagnetic materials is inversely proportional to $(T - T_c)$

$$i.e. \chi \propto \frac{1}{T - T_c}$$

$$\Rightarrow \chi = \frac{C}{(T - T_c)}$$

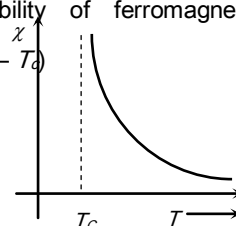


Fig. 22.25

Here T_c = Curie temperature

χ - T curve is shown (for Curie-Weiss Law)

Hysteresis Curve

For ferromagnetic materials, by removing external magnetic field *i.e.* $H = 0$. The magnetic moment of some domains remain aligned in the applied direction of previous magnetising field which results into a residual magnetism.

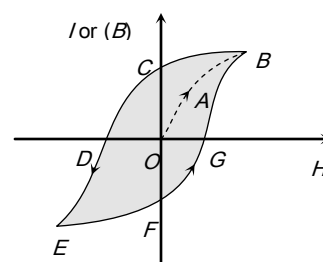


Fig. 22.26

The lack of retracibility as shown in figure is called hysteresis and the curve is known as hysteresis loop.

(1) **Retentivity** : When H is reduced, I reduces but is not zero when $H = 0$. The remainder value OC of magnetisation when $H = 0$ is called the residual magnetism or retentivity.

1250 Magnetism

The property by virtue of which the magnetism (I) remains in a material even on the removal of magnetising field is called Retentivity or Residual magnetism.

(2) **Corecivity or corecive force** : When magnetic field H is reversed, the magnetisation decreases and for a particular value of H , denoted by H_c , it becomes zero *i.e.*, $H_c = OD$ when $I = 0$. This value of H is called the corecivity.

Magnetic hard substance (steel) $\rightarrow$ High corecivity

Magnetic soft substance (soft iron) $\rightarrow$ Low corecivity

(3) When field H is further increased in reverse direction, the intensity of magnetisation attains saturation value in reverse direction (*i.e.* point E)

(4) When H is decreased to zero and changed direction in steps, we get the part $EFGB$.

Thus complete cycle of magnetisation and demagnetisation is represented by $BCDEFG$. This curve is known as hysteresis curve

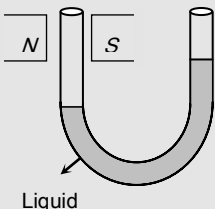
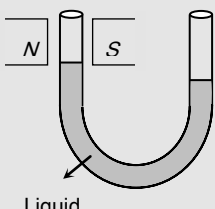
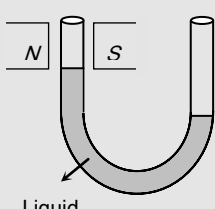
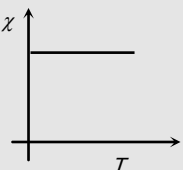
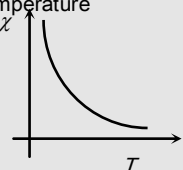
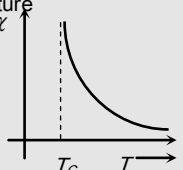
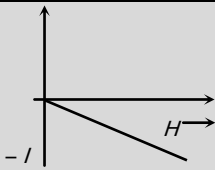
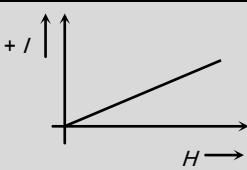
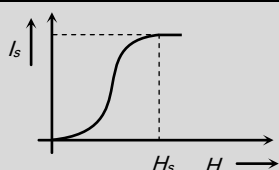
The area of hysteresis loop is less (low energy loss)	The area of hysteresis loop is large (high energy loss)
Less retentivity and corecive force	More retentivity and corecive force
Magnetic permeability is high	Magnetic permeability is less
I and χ both are high	I and χ both are low
It magnetised and demagnetised easily	Magnetisation and demagnetisation is not easy
Used in dynamo, transformer, electromagnet tape recorder and tapes <i>etc.</i>	Used for making permanent magnet.

Table 22.1 : Comparison between soft iron and steel

Soft iron	Steel
-----------	-------

Table 22.2 : Comparative study of magnetic materials

Property	Diamagnetic substances	Paramagnetic substances	Ferromagnetic substances
Cause of magnetism	Orbital motion of electrons	Spin motion of electrons	Formation of domains
Explanation of magnetism	On the basis of orbital motion of electrons	On the basis of spin and orbital motion of electrons	On the basis of domains formed
Behaviour In a non-uniform magnetic field	These are repelled in an external magnetic field <i>i.e.</i> have a tendency to move from high to low field region.	These are feebly attracted in an external magnetic field <i>i.e.</i>, have a tendency to move from low to high field region.	These are strongly attracted in an external magnetic field <i>i.e.</i> they easily move from low to high field region.

State of magnetisation	These are weakly magnetised in a direction opposite to that of applied magnetic field	These get weakly magnetised in the direction of applied magnetic field	These get strongly magnetised in the direction of applied magnetic field
When the material in the form of liquid is filled in the U-tube and placed between pole pieces.	Liquid level in that limb gets depressed 	Liquid level in that limb rises up 	Liquid level in that limb rises up very much 
On placing the gaseous materials between pole pieces	The gas expands at right angles to the magnetic field.	The gas expands in the direction of magnetic field.	The gas rapidly expands in the direction of magnetic field
The value of magnetic induction B	$B < B_0$ (where B_0 is the magnetic induction in vacuum)	$B > B_0$	$B \gg B_0$
Magnetic susceptibility χ	Low and negative $ \chi \approx 1$	Low but positive $\chi \approx 1$	Positive and high $\chi \approx 10^2$
Dependence of χ on temperature	Does not depend on temperature (except B_i at low temperature) 	On cooling, these get converted to ferromagnetic materials at Curie temperature 	These get converted into paramagnetic materials at Curie temperature 
Relative permeability (μ_r)	$\mu_r < 1$	$\mu_r > 1$	$\mu_r \gg 1$ $\mu_r = 10^2$
Intensity of magnetisation (I)	I is in a direction opposite to that of H and its value is very low	I is in the direction of H but value is low	I is in the direction of H and value is very high.
I - H curves			

Magnetic moment (M)	Very low (≈ 0)	Very low	Very high
Examples	$Cu, Ag, Au, Zn, Bi, Sb, NaCl, H_2O$ air and diamond etc.	$Al, Mn, Pt, Na, CuCl_2, O_2$ and crown glass	Fe, Co, Ni, Cd, Fe_3O_4 etc.

Tips & Tricks

✍ Bohr magneton $\mu_B = \frac{eh}{4\pi m} = 9.27 \times 10^{-24} \text{ A/m}^2$. It serves as natural unit of magnetic moment. Bohr magneton can be defined as the orbital magnetic moment of an electron circulating in inner most orbit.

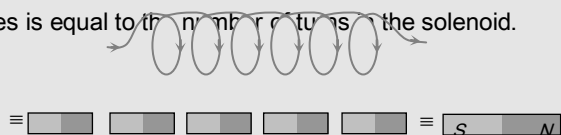
✍ Magnetic moment of straight current carrying wire is zero.

✍ Magnetic moment of toroid is zero

✍ Atoms which have paired electron have the magnetic moment zero.

✍ Magnetostriction : The length of an iron bar changes when it is magnetised, when an iron bar magnetised its length increases due to alignment of spins parallel to the field. This increase is in the direction of magnetisation. This effect is known as magnetostriction.

✍ A current carrying solenoid can be treated as the arrangement of small magnetic dipoles placed in line with each other as shown. The number of such small magnetic dipoles is equal to the number of turns in the solenoid.

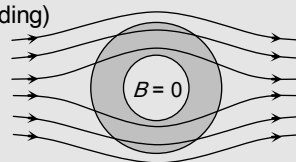


✍ When a magnetic dipole of moment M moves from unstable equilibrium to stable equilibrium position in a

magnetic field B , the kinetic energy will decrease by $2 MB$.

✍ Intensity of magnetisation (I) is produced in materials due to spin motion of electrons.

✍ For protecting a sensitive equipment from the external magnetic field it should be placed inside an iron can. (magnetic shielding)

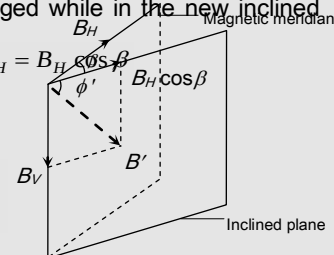


✍ **Apparent dip** : In a vertical plane inclined at an angle β to the magnetic meridian, vertical component of earth's magnetic field remains unchanged while in the new inclined plane horizontal component $B'_H = B_H \cos \beta$

$\phi' =$ apparent angle of dip

$$\text{and } \tan \phi' = \frac{B_V}{B'_H} = \frac{B_V}{B_H \cos \beta}$$

$$\Rightarrow \tan \phi' = \frac{\tan \phi}{\cos \beta}$$



✍ If at any place the angle of dip is θ and magnetic latitude is λ then $\tan \theta = 2 \tan \lambda$

✍ At the poles and equator of earth the values of total intensity are 0.66 and 0.33 Oersted respectively.

✍ Remember time period of oscillation in difference position is greater than that in sum position $T_d > T_s$.

✍ If a rectangular bar magnet is cut in n equal parts then time period of each part will be $\frac{1}{\sqrt{n}}$ times that of complete magnet (i.e. $T' = \frac{T}{\sqrt{n}}$) while for short magnet $T' = \frac{T}{n}$. If

nothing is said then bar magnet is treated as short magnet.

✍ Suppose a magnetic needle is vibrating in earth's magnetic field. With temperature rise M decreases hence time period (T) increases but at 770°C (Curie temperature) it stops vibrating.

✍ An iron cored coil and a bulb are connected in series with an ac generator. If an iron rod is introduced inside a coil, then the intensity of bulb will decrease, because some energy lost in magnetising the rod.

✍ Hysteresis energy loss = Area bound by the hysteresis loop = $VAnf \text{ Joule}$; Where, V = Volume of ferromagnetic sample, A = Area of $B-H$ loop, n = Frequency of alternating magnetic field and t = Time



Ordinary Thinking

Objective Questions

Magnet and its Properties

1. An iron rod of length L and magnetic moment M is bent in the form of a semicircle. Now its magnetic moment will be

[CPMT 1984; MP Board 1986; NCERT 1975; MP PET/PMT 1988; EAMCET (Med.) 1995;

Manipal MEE 1995; RPMT 1996; BHU 1995; MP PET 2002]

- (a) M (b) $\frac{2M}{\pi}$
(c) $\frac{M}{\pi}$ (d) $M\pi$

2. Unit of magnetic flux density (or magnetic induction) is

[DPMT 1988; CPMT 1984, 78, 90; MP PMT 1992; MH CET 2004]

- (a) *Tesla* (b) *Weber/metre²*
(c) *Newton/ampere-metre* (d) All of the above

3. Magnetic intensity for an axial point due to a short bar magnet of magnetic moment M is given by

[MP PET 1984; CPMT 1974; Pb. PMT 1999]

- (a) $\frac{\mu_0}{4\pi} \times \frac{M}{d^3}$ (b) $\frac{\mu_0}{4\pi} \times \frac{M}{d^2}$

- (c) $\frac{\mu_0}{2\pi} \times \frac{M}{d^3}$ (d) $\frac{\mu_0}{2\pi} \times \frac{M}{d^2}$

4. A magnet is placed in iron powder and then taken out, then maximum iron powder is at

- (a) Some away from north pole
(b) Some away from south pole
(c) The middle of the magnet
(d) The end of the magnet

5. A magnet of magnetic moment M and pole strength m is divided in two equal parts, then magnetic moment of each part will be

[MP Board 1985; MP PET 1984, 2000;

NCERT 1974; AFMC 1996; MP PMT 2002;

MH CET (Med.) 2001; CPMT 1983, 84; KCET 1994, 2001]

- (a) M (b) $M/2$
(c) $M/4$ (d) $2M$

6. Points A and B are situated along the extended axis of 2 cm long bar magnet at a distance x and $2x \text{ cm}$ respectively. From the pole nearer to the points, the ratio of the magnetic field at A and B will be

[EAMCET 1984; CPMT 1986]

- (a) 4 : 1 exactly (b) 4 : 1 approx.
(c) 8 : 1 exactly (d) 8 : 1 approx.

7. If a magnet of pole strength m is divided into four parts such that the length and width of each part is half that of initial one, then the pole strength of each part will be

- (a) $m/4$ (b) $m/2$
(c) $m/8$ (d) $4m$

8. The distance of two points on the axis of a magnet from its centre is 10 cm and 20 cm respectively. The ratio of magnetic intensity at these points is 12.5 : 1. The length of the magnet will be

- (a) 5 cm (b) 25 cm
(c) 10 cm (d) 20 cm

9. Ratio of magnetic intensities for an axial point and a point on broad side-on position at equal distance d from the centre of magnet will be or The magnetic field at a



1254 Magnetism

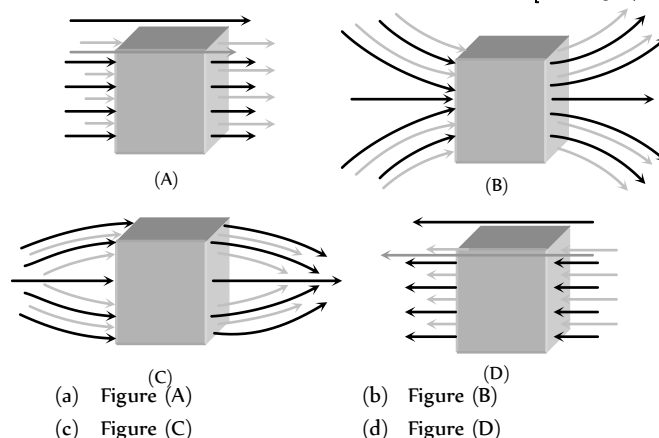
distance d from a short bar magnet in longitudinal and transverse positions are in the ratio [CPMT 1978, 82; KCET 1998]

- | | |
|-----------|-----------|
| (a) 1 : 1 | (b) 2 : 3 |
| (c) 2 : 1 | (d) 3 : 2 |

10. The magnetism of magnet is due to [JIPMER 1997]
 (a) The spin motion of electron
 (b) Earth
 (c) Pressure of big magnet inside the earth
 (d) Cosmic rays
11. The pole strength of a bar magnet is 48 ampere-metre and the distance between its poles is 25 cm. The moment of the couple by which it can be placed at an angle of 30° with the uniform magnetic intensity of flux density 0.15 Newton / ampere-metre will be
 (a) 12 Newton $\times$ metre (b) 18 Newton $\times$ metre
 (c) 0.9 Newton $\times$ metre (d) None of the above
12. The magnetic field at a point x on the axis of a small bar magnet is equal to the field at a point y on the equator of the same magnet. The ratio of the distances of x and y from the centre of the magnet is [MP PMT 1990]
 (a) 2^{-3} (b) $2^{-1/3}$
 (c) 2^3 (d) $2^{1/3}$
13. A magnet of magnetic moment 20 C.G.S. units is freely suspended in a uniform magnetic field of intensity 0.3 C.G.S. units. The amount of work done in deflecting it by an angle of 30° in C.G.S. units is
 (a) 6 (b) $3\sqrt{3}$
 (c) $3(2 - \sqrt{3})$ (d) 3
14. A bar magnet having centre O has a length of 4 cm. Point P is in the broad side-on and P is in the end side-on position with $OP = 10$ metres. The ratio of magnetic intensities H at P and P is
 (a) $H_1 : H_2 = 16 : 100$ (b) $H_1 : H_2 = 1 : 2$
 (c) $H_1 : H_2 = 2 : 1$ (d) $H_1 : H_2 = 100 : 16$
15. The magnetic field due to a short magnet at a point on its axis at distance X cm from the middle point of the magnet is 200 Gauss. The magnetic field at a point on the neutral axis at a distance X cm from the middle of the magnet is [CPMT 1971, 88; MP PET 1985]
 (a) 100 Gauss (b) 400 Gauss
 (c) 50 Gauss (d) 200 Gauss
16. Which of the following, the most suitable material for making permanent magnet is
 (a) Steel (b) Soft iron
 (c) Copper (d) Nickel
17. In the case of bar magnet, lines of magnetic induction [CPMT 1975; CBSE PMT 1990]
 (a) Start from the north pole and end at the south pole
 (b) Run continuously through the bar and outside
 (c) Emerge in circular paths from the middle of the bar
 (d) Are produced only at the north pole like rays of light from a bulb
18. A sensitive magnetic instrument can be shielded very effectively from outside magnetic fields by placing it inside a box of
 (a) Teak wood
 (b) Plastic material
 (c) Soft iron of high permeability
 (d) A metal of high conductivity
19. The field due to a magnet at a distance R from the centre of the magnet is proportional to [MP PET 1996]

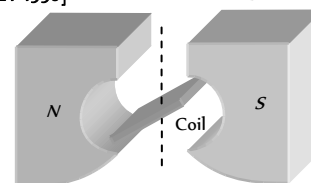
- (a) R^2 (b) R^3
 (c) $1/R^2$ (d) $1/R^3$

20. A uniform magnetic field, parallel to the plane of the paper existed in space initially directed from left to right. When a bar of soft iron is placed in the field parallel to it, the lines of force passing through it will be represented by [CPMT 1986, 88]



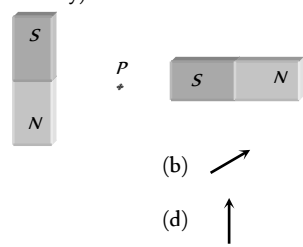
21. The figure below shows the north and south poles of a permanent magnet in which n turn coil of area of cross-section A is resting, such that for a current i passed through the coil, the plane of the coil makes an angle θ with respect to the direction of magnetic field B. If the plane of the magnetic field and the coil are horizontal and vertical respectively, the torque on the coil will be [MP PET 1990]

[MP PET 1990] [CPMT 1986, 88; DPMT 2002]



- (a) $\tau = niAB \cos \theta$
 (b) $\tau = niAB \sin \theta$
 (c) $\tau = niAB$
 (d) None of the above, since the magnetic field is radial
22. Points A and B are situated perpendicular to the axis of a 2 cm long bar magnet at large distances X and $3X$ from its centre on opposite sides. The ratio of the magnetic fields at A and B will be approximately equal to [CPMT 1988]
 (a) 1 : 9 (b) 2 : 9
 (c) 27 : 1 (d) 9 : 1
23. Two short magnets with their axes horizontal and perpendicular to the magnetic meridian are placed with their centres 40 cm east and 50 cm west of magnetic needle. If the needle remains undeflected, the ratio of their magnetic moments $M_1 : M_2$ is [MP PET 1990]
 (a) 4 : 5 (b) 16 : 25
 (c) 64 : 125 (d) $2 : \sqrt{5}$
24. If a bar magnet of magnetic moment M is freely suspended in a uniform magnetic field of strength B , the work done in rotating the magnet through an angle θ is [AFMC 1997; MNR 1998; RPET 1999; MP PMT 1989, 96, 99; MP PET 1984, 89, 2000; UPSEAT 1999, 2000, 05]
 (a) $MB(1 - \sin \theta)$ (b) $MB \sin \theta$
 (c) $MB \cos \theta$ (d) $MB(1 - \cos \theta)$

25. Two small bar magnets are placed in a line with like poles facing each other at a certain distance d apart. If the length of each magnet is negligible as compared to d , the force between them will be inversely proportional to
[CPMT 1971; NCERT 1971; MP PMT 1992]
- (a) d (b) d^2
(c) $\frac{1}{d^2}$ (d) d^4
26. A magnet of magnetic moment M is situated with its axis along the direction of a magnetic field of strength B . The work done in rotating it by an angle of 180° will be
[MP PMT 1985; MP PET 1997]
- (a) $-MB$ (b) $+MB$
(c) 0 (d) $+2MB$
27. A long magnet is cut in two parts in such a way that the ratio of their lengths is 2 : 1. The ratio of pole strengths of both the section is
[CPMT 1986]
- (a) Equal (b) In the ratio of 2 : 1
(c) In the ratio of 1 : 2 (d) In the ratio of 4 : 1
28. A bar magnet of length 10 cm and having the pole strength equal to 10^{-7} weber is kept in a magnetic field having magnetic induction (B) equal to $4\pi \times 10^{-3}$ Tesla. It makes an angle of 30° with the direction of magnetic induction. The value of the torque acting on the magnet is
[MP PMT 1993]
- (a) $2\pi \times 10^{-7} N \times m$ (b) $2\pi \times 10^{-5} N \times m$
(c) $0.5 N \times m$ (d) $0.5 \times 10^2 N \times m$
($\mu_0 = 4\pi \times 10^{-7}$ weber / amp $\times m$)
29. Magnetic field intensity is defined as [MP PET 1993]
- (a) Magnetic moment per unit volume
(b) Magnetic induction force acting on a unit magnetic pole
(c) Number of lines of force crossing per unit area
(d) Number of lines of force crossing per unit volume
30. If the magnetic flux is expressed in weber, then magnetic induction can be expressed in
[CPMT 1974, 77, 83, 86, 87; MP PET 1989]
- (a) Weber/m (b) Weber/m
(c) Weber-m (d) Weber-m
31. A magnetic needle is kept in a non-uniform magnetic field. It experiences
[MP PMT 1987; IIT 1982; Kerala PET 2002; AMU 1999; AIEEE 2005]
- (a) A force and a torque
(b) A force but not a torque
(c) A torque but not a force
(d) Neither a torque nor a force
32. The magnetic induction in air at a distance d from an isolated point pole of strength m unit will be [MNR 1987; CPMT 1991; MP PET 1995; AMU 1999; J & K CET 2005]
- (a) $\frac{m}{d}$ (b) $\frac{m}{d^2}$
(c) md (d) md^2
33. A magnetic needle lying parallel to a magnetic field requires W units of work to turn it through 60° . The torque required to maintain the needle in this position will be
[AIEEE 2003; UPSEAT 2000; BHU 2004; Pb PET 2004]
- (a) $\sqrt{3} W$ (b) W
(c) $\frac{\sqrt{3}}{2} W$ (d) $2W$
34. A long magnetic needle of length $2L$, magnetic moment M and pole strength m units is broken into two pieces at the middle. The magnetic moment and pole strength of each piece will be
- (a) $\frac{M}{2}, \frac{m}{2}$ (b) $M, \frac{m}{2}$
(c) $\frac{M}{2}, m$ (d) M, m
35. Two identical thin bar magnets each of length l and pole strength m are placed at right angle to each other with north pole of one touching south pole of the other. Magnetic moment of the system is
- (a) ml (b) $2ml$
(c) $\sqrt{2}ml$ (d) $\frac{1}{2}ml$
36. Magnetic induction is a [AFMC 1986]
- (a) Scalar quantity (b) Vector quantity
(c) Both (a) and (b) (d) None of the above
37. What happens to the force between magnetic poles when their pole strength and the distance between them are both doubled [CPMT 1978, 80, 84, 85]
- (a) Force increases to two times the previous value
(b) No change
(c) Force decreases to half the previous value
(d) Force increases to four times the previous value
38. Force between two unit pole strength placed at a distance of one metre is [CPMT 1987]
- (a) 1 N (b) $\frac{10^{-7}}{4\pi} N$
(c) $10^{-7} N$ (d) $4\pi \times 10^{-7} N$
39. A small bar magnet of moment M is placed in a uniform field H . If magnet makes an angle of 30° with field, the torque acting on the magnet is [CPMT 1989]
- (a) MH (b) $\frac{MH}{2}$
(c) $\frac{MH}{3}$ (d) $\frac{MH}{4}$
40. If a hole is made at the centre of a bar magnet, then its magnetic moment will
- (a) Increase (b) Decrease
(c) Not change (d) None of these
41. The small magnets each of magnetic moment $10 A-m$ are placed end-on position 0.1 m apart from their centres. The force acting between them is [MNR 1994]
- (a) $0.6 \times 10^7 N$ (b) $0.06 \times 10^7 N$
(c) $0.6 N$ (d) $0.06 N$
42. Magnetic lines of force [MP PET 1994]
- (a) Always intersect
(b) Are always closed
(c) Tend to crowd far away from the poles of magnet

- (d) Do not pass through vacuum
43. Rate of change of torque τ with deflection θ is maximum for a magnet suspended freely in a uniform magnetic field of induction B , when [MP PET 1994]
 (a) $\theta = 0^\circ$ (b) $\theta = 45^\circ$
 (c) $\theta = 60^\circ$ (d) $\theta = 90^\circ$
44. A magnet of magnetic moment M is rotated through 360° in a magnetic field H , the work done will be [KCET 1998; MP PMT 1994; Roorkee 2000]
 (a) MH (b) $2MH$
 (c) $2\pi MH$ (d) Zero
45. The direction of line of magnetic field of bar magnet is [AFMC 1995]
 (a) From south pole to north pole
 (b) From north pole to south pole
 (c) Across the bar magnet
 (d) From south pole to north pole inside the magnet and from north pole to south pole outside the magnet
46. The work done in turning a magnet of magnetic moment ' M ' by an angle of 90° from the meridian is ' n ' times the corresponding work done to turn it through an angle of 60° , where ' n ' is given by [CBSE PMT 1995; MP PET 2003]
 (a) $1/2$ (b) 2
 (c) $1/4$ (d) 1
47. Force between two identical bar magnets whose centres are r metre apart is 4.8 N , when their axes are in the same line. If separation is increased to $2r$, the force between them is reduced to
 (a) 2.4 N (b) 1.2 N
 (c) 0.6 N (d) 0.3 N
48. A bar magnet of magnetic moment 10 J/T is free to rotate in a horizontal plane. The work done in rotating the magnet slowly from a direction parallel to a horizontal magnetic field of $4 \times 10^{-2}\text{ T}$ to a direction 60° from the field will be [MP PET 1995]
 (a) 0.2 J (b) 2.0 J
 (c) 4.18 J (d) $2 \times 10\text{ J}$
49. Magnetic lines of force due to a bar magnet do not intersect because
 (a) A point always has a single net magnetic field
 (b) The lines have similar charges and so repel each other
 (c) The lines always diverge from a single point
 (d) The lines need magnetic lenses to be made to intersect
50. The unit of magnetic moment is [MP PET 1996; AMU 2000; MP PMT 1995, 2002]
 (a) Wb/m (b) Wb.m^2
 (c) A.m (d) A.m^2
51. The dipole moment of a short bar magnet is 1.25 A.m . The magnetic field on its axis at a distance of 0.5 metre from the centre of the magnet is
 (a) $1.0 \times 10^{-4}\text{ Newton / amp - metre}$
 (b) $4 \times 10^{-2}\text{ Newton / amp - metre}$
 (c) $2 \times 10^{-6}\text{ Newton / amp - metre}$
 (d) $6.64 \times 10^{-8}\text{ Newton / amp - metre}$
52. A permanent magnet [MP PET 1996]
 (a) Attracts all substances
 (b) Attracts only magnetic substances
 (c) Attracts magnetic substances and repels all non-magnetic substances
 (d) Attracts non-magnetic substances and repels magnetic substances
53. Two equal bar magnets are kept as shown in the figure. The direction of resultant magnetic field, indicated by arrow head at the point P is (approximately)

54. The S.I. unit of magnetic permeability is [MP PET 1997]
 (a) Am^{-1}
 (b) Am
 (c) Henry m^{-1}
 (d) No unit, it is a dimensionless number
55. A short bar magnet placed with its axis at 30° with a uniform external magnetic field of 0.16 Tesla experiences a torque of magnitude 0.032 Joule . The magnetic moment of the bar magnet will be [MP PMT 1997; UPSEAT 2004]
 (a) 0.23 Joule/Tesla (b) 0.40 Joule/Tesla
 (c) 0.80 Joule/Tesla (d) Zero
56. The magnetic field to a small magnetic dipole of magnetic moment M , at distance r from the centre on the equatorial line is given by (in M.K.S. system) [MP PMT/PET 1998]
 (a) $\frac{\mu_0}{4\pi} \times \frac{M}{r^2}$ (b) $\frac{\mu_0}{4\pi} \times \frac{M}{r^3}$
 (c) $\frac{\mu_0}{4\pi} \times \frac{2M}{r^2}$ (d) $\frac{\mu_0}{4\pi} \times \frac{2M}{r^3}$
57. The incorrect statement regarding the lines of force of the magnetic field B is [MP PET 1999]
 (a) Magnetic intensity is a measure of lines of force passing through unit area held normal to it
 (b) Magnetic lines of force form a close curve
 (c) Inside a magnet, its magnetic lines of force move from north pole of a magnet towards its south pole
 (d) Due to a magnet magnetic lines of force never cut each other
58. A straight wire carrying current i is turned into a circular loop. If the magnitude of magnetic moment associated with it in M.K.S. unit is M , the length of wire will be [MP PET 1999]

- (a) $4\pi iM$ (b) $\sqrt{\frac{4\pi M}{i}}$
- (c) $\sqrt{\frac{4\pi i}{M}}$ (d) $\frac{M\pi}{4i}$
59. A bar magnet of magnetic moment $\vec{M}$ is placed in a magnetic field of induction $\vec{B}$. The torque exerted on it is
[EAMCET (Engg.) 1995; CBSE PMT 1999; BHU 2003; CPMT 2004; MP PMT 2001, 05]
- (a) $\vec{M} \cdot \vec{B}$ (b) $-\vec{M} \cdot \vec{B}$
- (c) $\vec{M} \times \vec{B}$ (d) $\vec{B} \times \vec{M}$
60. For protecting a sensitive equipment from the external magnetic field, it should be
[KCET 1993; CBSE PMT 1998]
- (a) Placed inside an aluminium cane
- (b) Placed inside an iron cane
- (c) Wrapped with insulation around it when passing current through it
- (d) Surrounded with fine copper sheet
61. If a piece of metal was thought to be magnet, which one of the following observations would offer conclusive evidence
[KCET 1994]
- (a) It attracts a known magnet
- (b) It repels a known magnet
- (c) Neither (a) nor (b)
- (d) It attracts a steel screw driver
62. The magnet can be completely demagnetized by
[KCET 1994]
- (a) Breaking the magnet into small pieces
- (b) Heating it slightly
- (c) Dropping it into ice cold water
- (d) A reverse field of appropriate strength
63. A current loop placed in a magnetic field behaves like a
[AFMC 1994]
- (a) Magnetic dipole (b) Magnetic substance
- (c) Magnetic pole (d) All are true
64. A magnet when placed perpendicular to a uniform field of strength 10^{-4} Wb/m^2 experiences a maximum couple of moment $4 \times 10^{-5} \text{ N/m}$. What is its magnetic moment
[Bihar MEE 1995]
- (a) $0.4 \text{ A} \times \text{m}^2$ (b) $0.2 \text{ A} \times \text{m}^2$
- (c) $0.16 \text{ A} \times \text{m}^2$ (d) $0.04 \text{ A} \times \text{m}^2$
- (e) $0.06 \text{ A} \times \text{m}^2$
65. Weber/m is equal to
[CPMT 1985; AFMC 1997]
- (a) Volt (b) Henry
- (c) Tesla (d) All of these
66. Two magnets, each of magnetic moment ' M ' are placed so as to form a cross at right angles to each other. The magnetic moment of the system will be
[AFMC 1999; Pb PET 2001]
- (a) $2M$ (b) $\sqrt{2}M$
- (c) $0.5M$ (d) M
67. Two like magnetic poles of strength 10 and 40 SI units are separated by a distance 30 cm. The intensity of magnetic field is zero on the line joining them
[JIPMER 1999]
- (a) At a point 10 cm from the stronger pole
- (b) At a point 20 cm from the stronger pole
- (c) At the mid-point
- (d) At infinity
68. If a magnet of length 10 cm and pole strength 40 A-m is placed at an angle of 45° in a uniform induction field of intensity $2 \times 10^{-7} \text{ T}$, the couple acting on it is
[Pb. PMT 1999; MH CET (Med.) 1999]
- (a) $0.5656 \times 10^{-7} \text{ N-m}$ (b) $0.5656 \times 10^{-8} \text{ N-m}$
- (c) $0.656 \times 10^{-7} \text{ N-m}$ (d) $0.656 \times 10^{-8} \text{ N-m}$
69. The intensity of magnetic field is H and moment of magnet is M . The maximum potential energy is
[Pb. PMT 1999; MH CET (Med.) 1999]
- (a) MH (b) $2MH$
- (c) $3MH$ (d) $4MH$
70. A bar magnet of magnetic moment 200 A-m is suspended in a magnetic field of intensity 0.25 N/A-m . The couple required to deflect it through 30° is
[AFMC 1999; Pb. PET 2000]
- (a) 50 N-m (b) 25 N-m
- (c) 20 N-m (d) 15 N-m
71. Two similar bar magnets P and Q , each of magnetic moment M , are taken, If P is cut along its axial line and Q is cut along its equatorial line, all the four pieces obtained have
[EAMCET (Engg.) 2000]
- (a) Equal pole strength (b) Magnetic moment $\frac{M}{4}$
- (c) Magnetic moment $\frac{M}{2}$ (d) Magnetic moment M
72. A magnet of magnetic moment $50 \hat{i} \text{ A-m}^2$ is placed along the x -axis in a magnetic field $\vec{B} = (0.5 \hat{i} + 3.0 \hat{j}) \text{ T}$. The torque acting on the magnet is
[MP PMT 2000]
- (a) $175 \hat{k} \text{ N-m}$ (b) $150 \hat{k} \text{ N-m}$
- (c) $75 \hat{k} \text{ N-m}$ (d) $25\sqrt{37} \hat{k} \text{ N-m}$
73. A bar magnet is held perpendicular to a uniform magnetic field. If the couple acting on the magnet is to be halved by rotating it, then the angle by which it is to be rotated is
[CBSE PMT 2000]
- (a) 30° (b) 45°
- (c) 60° (d) 90°
74. There is no couple acting when two bar magnets are placed coaxially separated by a distance because
[EAMCET (Engg.) 2000]
- (a) There are no forces on the poles
- (b) The forces are parallel and their lines of action do not coincide
- (c) The forces are perpendicular to each other
- (d) The forces act along the same line
75. A bar magnet of magnetic moment 3.0 A-m is placed in a uniform magnetic induction field of $2 \times 10^{-7} \text{ T}$. If each pole of the

magnet experiences a force of $6 \times 10^{-4} \text{ N}$, the length of the magnet is [EAMCET (Med.) 2000]

- (a) 0.5 m (b) 0.3 m
(c) 0.2 m (d) 0.1 m

76. A bar magnet when placed at an angle of 30° to the direction of magnetic field induction of $5 \times 10^{-4} \text{ T}$, experiences a moment of couple $25 \times 10^{-4} \text{ N-m}$. If the length of the magnet is 5 cm its pole strength is [EAMCET (Med.) 2000]

- (a) $2 \times 10^{-4} \text{ A-m}$ (b) $5 \times 10^{-4} \text{ A-m}$
(c) 2 A-m (d) 5 A-m

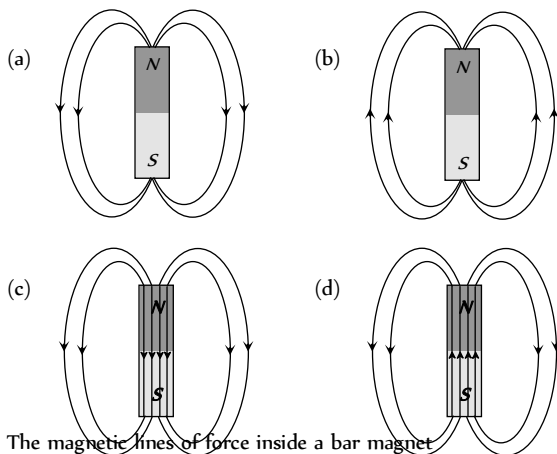
77. Two lines of force due to a bar magnet [MP PMT 2002]

- (a) Intersect at the neutral point
(b) Intersect near the poles of the magnet
(c) Intersect on the equatorial axis of the magnet
(d) Do not intersect at all

78. The ultimate individual unit of magnetism in any magnet is called [MP PET 2002; J & K CET 2004]

- (a) North pole (b) South pole
(c) Dipole (d) Quadrupole

79. The magnetic field lines due to a bar magnet are correctly shown in



80. The magnetic lines of force inside a bar magnet [AIEEE 2003]

- (a) Are from south-pole to north-pole of the magnet
(b) Are from north-pole to south-pole of the magnet
(c) Do not exist
(d) Depend upon the area of cross-section of the bar magnet

81. If a magnet is hanged with its magnetic axis then it stops in [AFMC 2003]

- (a) Magnetic meridian (b) Geometric meridian
(c) Angle of dip (d) None of these

82. The work done in rotating a magnet of magnetic moment 2 A-m in a magnetic field of $5 \times 10^{-4} \text{ T}$ from the direction along the magnetic field to opposite direction to the magnetic field, is

- (a) Zero (b) $2 \times 10^{-4} \text{ J}$
(c) 10^{-4} J (d) 10 J

83. The torque on a bar magnet due to the earth's magnetic field is maximum when the axis of the magnet is [MP PMT 2004]

- (a) Perpendicular to the field of the earth

- (b) Parallel to the vertical component of the earth's field

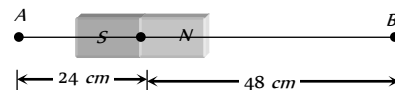
- (c) At an angle of 33° with respect to the $N-S$ direction

- (d) Along the North-South ($N-S$) direction

84. Magnetic dipole moment is a [AFMC 2004]

- (a) Scalar quantity (b) Vector quantity
(c) Constant quantity (d) None of these

85. A bar magnet of length 3 cm has points A and B along its axis at distances of 24 cm and 48 cm on the opposite sides. Ratio of magnetic fields at these points will be [DPMT 2004]



- (a) 8 (b) $1/2 \sqrt{2}$
(c) $\frac{3}{4}$ (d) 4

86. A magnet of magnetic moment 2 J T^{-1} is aligned in the direction of magnetic field of 0.1 T . What is the net work done to bring the [IIT-JEE (Screening) 2002]

[DCE 2004]

- (a) 0.1 J (b) 0.2 J
(c) 1 J (d) 2 J

87. The magnetic moment of a magnet of length 10 cm and pole strength 4.0 Am will be [DPMT 2003]

- (a) 0.4 Am^2 (b) 1.6 Am^2
(c) 20 Am^2 (d) 8.0 Am^2

88. The effective length of a magnet is 31.4 cm and its pole strength is 0.5 Am . The magnetic moment, if it is bent in the form of a semicircle will be [DPMT 2003]

- (a) 0.1 Am^2 (b) 0.01 Am^2
(c) 0.2 Am^2 (d) 1.2 Am^2

89. The magnetic potential at a point on the axial line of a bar magnet of dipole moment M is V . What is the magnetic potential due to a bar magnet of dipole moment $\frac{M}{4}$ at the same point

- (a) $4 V$ (b) $2 V$
(c) $\frac{V}{2}$ (d) $\frac{V}{4}$

90. A small bar magnet has a magnetic moment 1.2 A-m . The magnetic field at a distance 0.1 m on its axis will be : ($\mu_0 = 4\pi \times 10^{-7} \text{ T-m/A}$) [MP PET 2003]

- (a) $1.2 \times 10^{-4} \text{ T}$ (b) $2.4 \times 10^{-4} \text{ T}$
(c) $2.4 \times 10^{-4} \text{ T}$ (d) $1.2 \times 10^{-4} \text{ T}$

91. Two identical short bar magnets, each having magnetic moment of 10 Am , are arranged such that their axial lines are perpendicular to each other and their centres be along the same straight line in a horizontal plane. If the distance between their centres is 0.2 m , the resultant magnetic induction at a point midway between them is

$$(\mu_0 = 4\pi \times 10^{-7} \text{ Hm}^{-1})$$

[EAMCET 2005]

- (a) $\sqrt{2} \times 10^{-7} \text{ Tesla}$ (b) $\sqrt{5} \times 10^{-7} \text{ Tesla}$
(c) $\sqrt{2} \times 10^{-3} \text{ Tesla}$ (d) $\sqrt{5} \times 10^{-3} \text{ Tesla}$

92. A magnet of length 0.1 m and pole strength 10^{-4} A.m. is kept in a magnetic field of 30 Wb/m^2 at an angle 30° . The couple acting on it is $\times 10^{-4} \text{ Nm}$. [MP PET 2005]
(a) 7.5 (b) 3.0
(c) 1.5 (d) 6.0

Earth Magnetism

1. A very small magnet is placed in the magnetic meridian with its south pole pointing north. The null point is obtained 20 cm away from the centre of the magnet. If the earth's magnetic field (horizontal component) at this point be 0.3 gauss, the magnetic moment of the magnet is

[CPMT 1987; MNR 1978]

- (a) $8.0 \times 10^2 \text{ e.m.u.}$ (b) $1.2 \times 10^3 \text{ e.m.u.}$
(c) $2.4 \times 10^3 \text{ e.m.u.}$ (d) $3.6 \times 10^3 \text{ e.m.u.}$

2. Intensity of magnetic field due to earth at a point inside a hollow steel box is [MP PET 1995]

- (a) Less than outside (b) More than outside
(c) Same (d) Zero

3. Earth's magnetic field always has a horizontal component except at or Horizontal component of earth's magnetic field remains zero at

- (a) Equator (b) Magnetic poles
(c) A latitude of 60° (d) An altitude of 60°

4. A dip needle in a plane perpendicular to magnetic meridian will remain [NCERT 1975; MP PMT 1984; MP PET 1995]

- (a) Vertical
(b) Horizontal
(c) In any direction
(d) At an angle of dip to the horizontal

5. At magnetic poles of earth, angle of dip is

[CPMT 1977, 91; NCERT 1981; MP PET 1997; Pb PET 2002]

- (a) Zero (b) 45°
(c) 90° (d) 180°

6. The correct relation is

[CPMT 1986; MP PET 1981; AFMC 1996]

- (a) $B = \frac{B_V}{B_H}$ (b) $B = B_V \times B_H$
(c) $|B| = \sqrt{B_H^2 + B_V^2}$ (d) $B = B_H + B_V$

(Where B_H = Horizontal component of earth's magnetic field;
 B_V = Vertical component of earth's magnetic field and B = Total intensity of earth's magnetic field)

7. At a certain place, the horizontal component of earth's magnetic field is $\sqrt{3}$ times the vertical component. The angle of dip at that place is [MP PMT 1984, 85; AFMC 2000]

- (a) 60° (b) 45°
(c) 90° (d) 30°

8. The vertical component of earth's magnetic field is zero at or The earth's magnetic field always has a vertical component except at the

[NCERT 1980, 88; CPMT 1983; MP PMT 1996]

- (a) Magnetic poles (b) Geographical poles
(c) Every place (d) Magnetic equator

9. The angle between the magnetic meridian and geographical meridian is called

[MNR 1990; UPSEAT 1999, 2000; MP PMT 2000]

- (a) Angle of dip (b) Angle of declination
(c) Magnetic moment (d) Power of magnetic field

10. The lines of forces due to earth's horizontal component of magnetic field are

[CPMT 1985; MP PMT 1980; AIIMS 1998]

- (a) Parallel straight lines (b) Concentric circles
(c) Elliptical (d) Parabolic

11. At a place, if the earth's horizontal and vertical components of magnetic fields are equal, then the angle of dip will be

[SCRA 1994; DCE 2001; MP PMT 2002]

- (a) 30° (b) 90°
(c) 45° (d) 0°

12. If the angles of dip at two places are 30° and 45° respectively, then the ratio of horizontal components of earth's magnetic field at the two places will be [MP PET 1989]

- (a) $\sqrt{3} : \sqrt{2}$ (b) $1 : \sqrt{2}$
(c) $1 : \sqrt{3}$ (d) $1 : 2$

13. At a place the earth's horizontal component of magnetic field is $0.36 \times 10^{-4} \text{ weber/m}^2$. If the angle of dip at that place is 60° , then the vertical component of earth's field at that place in weber/m^2 will be approximately [MP PMT 1985]

- (a) 0.12×10^{-4} (b) 0.24×10^{-4}
(c) 0.40×10^{-4} (d) 0.62×10^{-4}

14. The angle of dip at a place is 40.6° and the intensity of the vertical component of the earth's magnetic field $V = 6 \times 10^{-5} \text{ Tesla}$. The total intensity of the earth's magnetic field (I) at this place is

- (a) $7 \times 10^{-5} \text{ tesla}$ (b) $6 \times 10^{-5} \text{ tesla}$
(c) $5 \times 10^{-5} \text{ tesla}$ (d) $9.2 \times 10^{-5} \text{ tesla}$

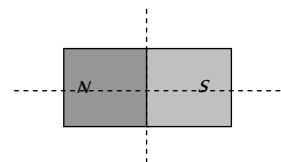
15. The angle of dip is the angle [CPMT 1978]

- (a) Between the vertical component of earth's magnetic field and magnetic meridian
(b) Between the vertical component of earth's magnetic field and geographical meridian
(c) Between the earth's magnetic field direction and horizontal direction
(d) Between the magnetic meridian and the geographical meridian

16. At a certain place the angle of dip is 30° and the horizontal component of earth's magnetic field is 0.50 Oersted . The earth's total magnetic field is [CPMT 1990]
- (a) $\sqrt{3}$ (b) 1
(c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{2}$
17. The angle of dip at the magnetic equator is [MP PET 1984; MP PMT 1987; CBSE PMT 1989, 90; MP Board 1980; CPMT 1977, 87, 90; Manipal MEE 1995]
- (a) 0° (b) 45°
(c) 30° (d) 90°
18. The line on the earth's surface joining the points where the field is horizontal is [MNR 1985; UPSEAT 1999; Pb PET 2004]
- (a) Magnetic meridian (b) Magnetic axis
(c) Magnetic line (d) Magnetic equator
(e) Isogonic line
19. The angle between the earth's magnetic and the earth's geographical axes is [MNR 1979]
- (a) Zero (b) 17°
(c) 23° (d) None of these
20. The lines joining the places of the same horizontal intensity are known as [MNR 1984]
- (a) Isogonic lines (b) Aclinic lines
(c) Isoclinic lines (d) Agonic lines
(e) Isodynamic lines
21. Ratio between total intensity of magnetic field at equator to poles is
- (a) 1 : 1 (b) 1 : 2
(c) 2 : 1 (d) 1 : 4
22. A line passing through places having zero value of magnetic dip is called [CPMT 1987]
- (a) Isoclinic line (b) Agonic line
(c) Isogonic line (d) Aclinic line
23. At a place, the horizontal and vertical intensities of earth's magnetic field is 0.30 Gauss and 0.173 Gauss respectively. The angle of dip at this place is [MP PMT 1986]
- (a) 30° (b) 90°
(c) 60° (d) 45°
24. The angle of dip at a place is 60° . At this place the total intensity of earth's magnetic field is 0.64 units . The horizontal intensity of earth's magnetic field at this place is [MP PET 1984]
- (a) 1.28 units (b) 0.64 units
(c) 0.16 units (d) 0.32 units
25. The magnetic compass is not useful for navigation near the magnetic poles because [BIT Ranchi 1982]
- (a) The magnetic field near the poles is zero
(b) The magnetic field near the poles is almost vertical
(c) At low temperature, the compass needle loses its magnetic properties
(d) Neither of the above
26. The angle of dip at a place on the earth gives
- (a) The horizontal component of the earth's magnetic field
(b) The location of the geographic meridian
(c) The vertical component of the earth's field
(d) The direction of the earth's magnetic field
27. At the magnetic north pole of the earth, the value of horizontal component of earth's magnetic field and angle of dip are, respectively [MP PMT 1994]
- (a) Zero, maximum (b) Maximum, minimum
(c) Maximum, maximum (d) Minimum, minimum
28. At a place, the magnitudes of the horizontal component and total intensity of the magnetic field of the earth are 0.3 and 0.6 Oersted respectively. The value of the angle of dip at this place will be
- (a) 60° (b) 45°
(c) 30° (d) 0°
29. A dip circle is at right angle to the magnetic meridian. What will be the apparent dip [AFMC 1995]
- (a) 0° (b) 30°
(c) 60° (d) 90°
30. A bar magnet is placed north-south with its north pole due north. The points of zero magnetic field will be in which direction from the centre of the magnet [MNR 1995; MP PMT 1995; UPSEAT 2000]
- (a) North and south
(b) East and west
(c) North-east and south-west
(d) North-west and south-east
31. In two separate experiments the neutral points due to two small magnets are at a distance of r and $2r$ in broad side-on position. The ratio of their magnetic moments will be [MP PET 1981; CPMT 1981]
- (a) 4 : 1 (b) 1 : 2
(c) 2 : 1 (d) 1 : 8
32. The magnetic field due to the earth is closely equivalent to that due to [BIT Ranchi 1982]
- (a) A large magnet of length equal to the diameter of the earth
(b) A magnetic dipole placed at the centre of the earth
(c) A large coil carrying current
(d) Neither of the above
33. The earth's magnetic field at a certain place has a horizontal component 0.3 Gauss and the total strength 0.5 Gauss . The angle of dip is [MP PMT 1995]
- (a) $\tan^{-1} \frac{3}{4}$ (b) $\sin^{-1} \frac{3}{4}$
(c) $\tan^{-1} \frac{4}{3}$ (d) $\sin^{-1} \frac{3}{5}$
34. The value of the horizontal component of the earth's magnetic field and angle of dip are $1.8 \times 10^{-5} \text{ Weber/m}^2$ and 30° respectively at some place. The total intensity of earth's magnetic field at that place will be [MP PET 1996]
- (a) $2.08 \times 10^{-5} \text{ Weber/m}^2$ (b) $3.67 \times 10^{-5} \text{ Weber/m}^2$

- (c) $3.18 \times 10^{-5} \text{ Weber/m}^2$ (d) $5.0 \times 10^{-5} \text{ Weber/m}^2$
35. When the N -pole of a bar magnet points towards the south and S -pole towards the north, the null points are at the
[MP PMT 1996]
- (a) Magnetic axis
(b) Magnetic centre
(c) Perpendicular divider of magnetic axis
(d) N and S poles
36. Lines which represent places of constant angle of dip are called
(a) Isobaric lines (b) Isogonic lines
(c) Isoclinic lines (d) Isodynamic lines
37. The vertical component of the earth's magnetic field is zero at a place where the angle of dip is [MP PMT/PET 1998]
(a) 0° (b) 45°
(c) 60° (d) 90°
38. At a certain place, the horizontal component B_0 and the vertical component V_0 of the earth's magnetic field are equal in magnitude. The total intensity at the place will be
[MP PMT 1999, 2003]
- (a) B_0 (b) B_0^2
(c) $2B_0$ (d) $\sqrt{2}B_0$
39. A compass needle will show which one of the following directions at the earth's magnetic pole [KCET 1993, 94]
(a) Vertical (b) No particular direction
(c) Bent at 45° to the vertical (d) Horizontal
40. The north pole of the earth's magnet is near the geographical
(a) South (b) East
(c) West (d) North
41. The magnetic field of earth is due to [JIPMER 1997]
(a) Motion and distribution of some material in and outside the earth
(b) Interaction of cosmic rays with the current of earth
(c) A magnetic dipole buried at the centre of the earth
(d) Induction effect of the sun
42. A short magnet of moment 6.75 Am produces a neutral point on its axis. If horizontal component of earth's magnetic field is $5 \times 10^{-5} \text{ Wb/m}^2$, then the distance of the neutral point should be
(a) 10 cm (b) 20 cm
(c) 30 cm (d) 40 cm
43. Due to the earth's magnetic field, charged cosmic ray particles
(a) Require greater kinetic energy to reach the equator than the poles
(b) Require less kinetic energy to reach the equator than the poles
(c) Can never reach the equator
(d) Can never reach the poles
44. Two bar magnets with magnetic moments $2M$ and M are fastened together at right angles to each other at their centres to form a crossed system, which can rotate freely about a vertical axis through the centre. The crossed system sets in earth's magnetic field with magnet having magnetic moment $2M$ making an angle θ with the magnetic meridian such that
[AFMC 1999]
- (a) $\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (b) $\theta = \tan^{-1}(\sqrt{3})$
(c) $\theta = \tan^{-1}\left(\frac{1}{2}\right)$ (d) $\theta = \tan^{-1}\left(\frac{3}{4}\right)$
45. Angle of dip is 90° at [AIIMS 1999]
(a) Poles (b) Equator
(c) Both (a) and (b) (d) None of these
46. At a certain place the horizontal component of the earth's magnetic field is B and the angle of dip is 45° . The total intensity of the field at that place will be [MP PET 2000; Pb PET 2003]
(a) B (b) $\sqrt{2}B_0$
(c) $2B$ (d) B_0^2
47. The value of angle of dip is zero at the magnetic equator because on it [MP PET 2001]
(a) V and H are equal
(b) The value of V and H is zero
(c) The value of V is zero
(d) The value of H is zero
48. Which of the following relation is correct in magnetism [KCET (Engg./Med) 2001]
(a) $I^2 = V^2 + H^2$ (b) $I = V + H$
(c) $V = I^2 + H^2$ (d) $V^2 = I + H$
49. The direction of the null points is on the equatorial line of a bar magnet, when the north pole of the magnet is pointing [AFMC 1999; Pb. PMT 2000; CPMT 2001; MH CET 2003]
(a) North (b) South
(c) East (d) West
50. Magnetic meridian is a [Orissa JEE 2002]
(a) Point (b) Horizontal plane
(c) Vertical plane (d) Line along $N-S$
51. The angle of dip at a certain place is 30° . If the horizontal component of the earth's magnetic field is H , the intensity of the total magnetic field is [SCRA 1994]
[UPSEAT 1993, 2000; MP PMT 2002]
- (a) $\frac{H}{2}$ (b) $\frac{2H}{\sqrt{3}}$
(c) $H\sqrt{2}$ (d) $H\sqrt{3}$
52. The horizontal component of the earth's magnetic field is 0.22 Gauss and total magnetic field is 0.4 Gauss . The angle of dip is
(a) $\tan^{-1}(1)$ (b) $\tan^{-1}(\infty)$
(c) $\tan^{-1}(1.518)$ (d) $\tan^{-1}(\pi)$
53. A bar magnet is situated on a table along east-west direction in the magnetic field of earth. The number of neutral points, where the magnetic field is zero, are [MP PMT 2004]
(a) 2 (b) 0
(c) 1 (d) 4

54. At which place, earth's magnetism become horizontal [CPMT 1973, 76, 87; MP PET 1994, 96]
[AFMC 2004]
(a) Magnetic pole (b) Geographical pole
(c) Magnetic meridian (d) Magnetic equator
55. Isogonic lines on magnetic map will have [AFMC 2004]
(a) Zero angle of dip
(b) Zero angle of declination
(c) Same angle of declination
(d) Same angle of dip
56. A current carrying coil is placed with its axis perpendicular to N - S direction. Let horizontal component of earth's magnetic field be H and magnetic field inside the loop is H . If a magnet is suspended inside the loop, it makes angle θ with H . Then $\theta =$
(a) $\tan^{-1}\left(\frac{H_0}{H}\right)$ (b) $\tan^{-1}\left(\frac{H}{H_0}\right)$
(c) $\operatorname{cosec}^{-1}\left(\frac{H}{H_0}\right)$ (d) $\cot^{-1}\left(\frac{H_0}{H}\right)$
57. Let V and H be the vertical and horizontal components of earth's magnetic field at any point on earth. Near the north pole
(a) $V \gg H$ (b) $V \ll H$
(c) $V = H$ (d) $V = H = 0$
58. At the magnetic poles of the earth, a compass needle will be [DCE 2003]
(a) Vertical
(b) Bent slightly
(c) Horizontal
(d) Inclined at 45° to the horizontal
59. If magnetic lines of force are drawn by keeping magnet vertical, then number of neutral points will be [MP PMT 1985; CPMT 1985]
(a) One (b) Two
(c) Four (d) Five
4. In sum and difference method in vibration magnetometer, the time period is more if [MP PMT 1989; MP PET/PMT 1988]
(a) Similar poles of both magnets are on same sides
(b) Opposite poles of both magnets are on same sides
(c) Both magnets are perpendicular to each other
(d) Nothing can be said [Onissa PMT 2004]
5. At a certain place a magnet makes 30 oscillations per minute. At another place where the magnetic field is double, its time period will be [MP PMT 1989; MP PET/PMT 1988]
(a) 4 sec (b) 2 sec
(c) $\frac{1}{2}$ sec (d) $\sqrt{2}$ sec
6. Vibration magnetometer is used for comparing [IIPSEAT 2004]
(a) Magnetic fields (b) Earth's field
(c) Magnetic moments (d) All of the above [MP PET/PMT 1988]
7. Two magnets of same size and mass make respectively 10 and 15 oscillations per minute at certain place. The ratio of their magnetic moments is [Bihar PET 1984; MP PET/PMT 1988; MP PET 1992]
(a) 4 : 9 (b) 9 : 4
(c) 2 : 3 (d) 3 : 2
8. Time period for a magnet is T . If it is divided in four equal parts along its axis and perpendicular to its axis as shown then time period for each part will be



- (a) $4T$ (b) $T/4$
(c) $T/2$ (d) T

9. Keeping dissimilar poles of two magnets of equal pole strength and length same side, their time period will be [DPMT 2001]
(a) Zero (b) One second
(c) Infinity (d) Any value
10. Time period in vibration magnetometer will be infinity at
(a) Magnetic equator (b) Magnetic poles
(c) Equator (d) At all places
11. Twists of suspension fibre should be removed in vibration magnetometer so that
(a) Time period be less
(b) Time period be more
(c) Magnet may vibrate freely

Magnetic Equipments

1. Time period of a freely suspended magnet does not depend upon [NCERT 1980; CPMT 1989; MP PET 1997]
(a) Length of the magnet
(b) Pole strength of the magnet
(c) Horizontal component of earth's magnetic field
(d) Length of the suspension thread
2. Magnetic moments of two bar magnets may be compared with the help of [MP PET/PMT 1988]
(a) Deflection magnetometer
(b) Vibration magnetometer
(c) Both of the above
(d) None of the above
3. The time period of oscillation of a freely suspended bar magnet with usual notations is given by

- (d) Cannot be said with certainty
12. The period of oscillation of a magnet in vibration magnetometer is 2 sec. The period of oscillation of a magnet whose magnetic moment is four times that of the first magnet is [CPMT 1975, 77, 79, 89, 90; MP PMT 1986]
- (a) 1 sec (b) 4 sec
(c) 8 sec (d) 0.5 sec
13. Moment of inertia of a magnetic needle is 40 gm-cm has time period 3 seconds in earth's horizontal field = 3.6×10^{-5} weber/m. Its magnetic moment will be
- (a) $0.5 A \times m^2$ (b) $5 A \times m^2$
(c) $0.250 A \times m^2$ (d) $5 \times 10^2 A \times m^2$
14. Vibration magnetometer before use, should be set
- (a) In magnetic meridian
(b) In geographical meridian
(c) Perpendicular to magnetic meridian
(d) In any position
15. If a brass bar is placed on a vibrating magnet, then its time period
- (a) Decreases
(b) Increases
(c) Remains unchanged
(d) First increases then decreases
16. A magnetic needle is made to vibrate in uniform field H , then its time period is T . If it vibrates in the field of intensity $4H$, its time period will be
- [MP Board 1988; MP PMT 1992; MH CET (Med.) 1999]
- (a) $2T$ (b) $T/2$
(c) $2/T$ (d) T
17. Two bar magnets of the same mass, length and breadth but magnetic moments M and $2M$ respectively, when placed in same position, time period is 3 sec. What will be the time period when they are placed in different position
- [NCERT 1977; DPMT 1999]
- (a) $\sqrt{3}$ sec (b) $3\sqrt{3}$ sec
(c) 3 sec (d) 6 sec
18. To compare magnetic moments of two magnets by vibration magnetometer, 'sum and difference method' is better because
- (a) Determination of moment of inertia is not needed which minimises the errors
(b) Less observations are required
(c) Comparatively less calculations
(d) All the above
19. A magnet is suspended in such a way that it oscillates in the horizontal plane. It makes 20 oscillations per minute at a place where dip angle is 30° and 15 oscillations per minute at a place where dip angle is 60° . The ratio of total earth's magnetic field at the two places is
- [MP PMT 1991; BHU 1997]
- (a) $3\sqrt{3} : 8$ (b) $16 : 9\sqrt{3}$
(c) $4 : 9$ (d) $2\sqrt{3} : 9$
20. The time period of oscillation of a magnet in a vibration magnetometer is 1.5 seconds. The time period of oscillation of another magnet similar in size, shape and mass but having one-fourth magnetic moment than that of first magnet, oscillating at same place will be
- [MP PMT 1991; MP PMT 2002]
- (a) 0.75 sec (b) 1.5 sec
(c) 3 sec (d) 6 sec
21. A bar magnet A of magnetic moment M_1 is found to oscillate at a frequency twice that of magnet B of magnetic moment M_2 when placed in a vibrating *magneto-meter*. We may say that
- (a) $M_A = 2M_B$ (b) $M_A = 8M_B$
(c) $M_A = 4M_B$ (d) $M_B = 8M_A$
22. Two magnets A and B are identical in mass, length and breadth but have different magnetic moments. In a vibration magnetometer, if the time period of B is twice the time period of A . The ratio of the magnetic moments M_A / M_B of the magnets will be [MP PET 1990; MP PMT 1991]
- (a) $1/2$ (b) 2
(c) 4 (d) $1/4$
23. A magnet of magnetic moment M oscillating freely in earth's horizontal magnetic field makes n oscillations per minute. If the magnetic moment is quadrupled and the earth's field is doubled, the number of oscillations made per minute would be
- (a) $\frac{n}{2\sqrt{2}}$ (b) $\frac{n}{\sqrt{2}}$
(c) $2\sqrt{2}n$ (d) $\sqrt{2}n$
24. A magnetic needle suspended horizontally by an unspun silk fibre, oscillates in the horizontal plane because of the restoring force originating mainly from [CPMT 1980, 89]
- (a) The torsion of the silk fibre
(b) The force of gravity
(c) The horizontal component of earth's magnetic field
(d) All the above factors
25. At two places A and B using vibration magnetometer, a magnet vibrates in a horizontal plane and its respective periodic time are 2 sec and 3 sec and at these places the earth's horizontal components are H_1 and H_2 respectively. Then the ratio between H_1 and H_2 will be [MP PMT 1985, 89]
- (a) 9 : 4 (b) 3 : 2
(c) 4 : 9 (d) 2 : 3
26. The time period of a bar magnet suspended horizontally in the earth's magnetic field and allowed to oscillate
- [MP PET 1992]
- (a) Is directly proportional to the square root of its mass
(b) Is directly proportional to its pole strength
(c) Is inversely proportional to its magnetic moment
(d) Decreases if the length increases but pole strength remains same
27. Magnets A and B are geometrically similar but the magnetic moment of A is twice that of B . If T_1 and T_2 be the time periods of the oscillation when their like poles and unlike poles are kept together respectively, then $\frac{T_1}{T_2}$ will be
- [SCRA 1998]
- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$
(c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{3}$
28. A small bar magnet A oscillates in a horizontal plane with a period T at a place where the angle of dip is 60° . When the same needle is made to oscillate in a vertical plane coinciding with the magnetic meridian, its period will be
- [MP PMT 1992]

- (a) $\frac{T}{\sqrt{2}}$ (b) T
(c) $\sqrt{2}T$ (d) $2T$
29. Vibration magnetometer works on the principle of [MP PET 1993]
(a) Torque acting on the bar magnet
(b) Force acting on the bar magnet
(c) Both the force and the torque acting on the bar magnet
(d) None of these
30. Tangent galvanometer is used to measure [MP PET 1993]
(a) Steady currents
(b) Current impulses
(c) Magnetic moments of bar magnets
(d) Earth's magnetic field
31. A tangent galvanometer has a coil with 50 turns and radius equal to 4 cm. A current of 0.1 A is passing through it. The plane of the coil is set parallel to the earth's magnetic meridian. If the value of the earth's horizontal component of the magnetic field is 7×10^{-5} tesla and $\mu_0 = 4\pi \times 10^{-7}$ weber / amp $\times m$, then the deflection in the galvanometer needle will be [MP PMT 1993]
(a) 45° (b) 48.2°
(c) 50.7° (d) 52.7°
32. A bar magnet has a magnetic moment equal to 5×10^{-5} weber $\times m$. It is suspended in a magnetic field which has a magnetic induction (B) equal to $8\pi \times 10^{-4}$ tesla. The magnet vibrates with a period of vibration equal to 15 sec. The moment of inertia of the magnet is [MP PMT 1993; CBSE PMT 2001]
(a) $22.5 kg \times m^2$ (b) $11.25 \times kg \times m^2$
(c) $5.62 \times kg \times m^2$ (d) $7.16 \times 10^{-7} kg \times m^2$
33. The time period of a freely suspended magnet is 4 seconds. If it is broken in length into two equal parts and one part is suspended in the same way, then its time period will be [NCERT 1984; CPMT 1991; MP PMT 1994; MH CET 2004]
(a) 4 sec (b) 2 sec
(c) 0.5 sec (d) 0.25 sec
34. Which of the following statement is true about magnetic moments of atoms of different elements [CPMT 1977]
(a) All have a magnetic moment
(b) None has a magnetic moment
(c) All acquire a magnetic moment under external magnetic field and in same direction as the field
(d) None of the above statements are accurate
35. The number of turns and radius of cross-section of the coil of a tangent galvanometer are doubled. The reduction factor K will be [NCERT 1983; MP PMT 2002]
(a) K (b) $2K$
(c) $4K$ (d) $K/4$
36. A magnetic needle suspended by a silk thread is vibrating in the earth's magnetic field. If the temperature of the needle is increased by $500^\circ C$, then [MNR 1994]
(a) The time period decreases
(b) The time period remains unchanged
(c) The time period increases
(d) The needle stops vibrating
37. The sensitivity of a tangent galvanometer is increased if [AFMC 1995]
(a) Number of turn decreases (b) Number of turn increases
(c) Field increases (d) None of the above
38. Two tangent galvanometers having coils of the same radius are connected in series. A current flowing in them produces deflections of 60° and 45° respectively. The ratio of the number of turns in the coils is [MP PET 1995; MP PMT 1999]
(a) 4/3 (b) $(\sqrt{3} + 1)/1$
(c) $(\sqrt{3} + 1)/(\sqrt{3} - 1)$ (d) $\sqrt{3}/1$
39. Using a bar magnet P , a vibration magnetometer has time period 2 seconds. When a bar Q (identical to P in mass and size) is placed on top of P , the time period is unchanged. Which of the following statements is true [MP PMT 1995]
(a) Q is of non-magnetic material
(b) Q is a bar magnet identical to P , and its north pole placed on top of P 's north pole
(c) Q is of unmagnetized ferromagnetic material
(d) Nothing can be said about Q 's properties
40. The strength of the magnetic field in which the magnet of a vibration magnetometer is oscillating is increased 4 times its original value. The frequency of oscillation would then become
(a) Twice its original value
(b) Four times its original value
(c) Half its original value
(d) One-fourth its original value
41. A certain amount of current when flowing in a properly set tangent galvanometer, produces a deflection of 45° . If the current be reduced by a factor of $\sqrt{3}$, the deflection would [MP PMT 1996; DPMT 2005]
(a) Decrease by 30° (b) Decrease by 15°
(c) Increase by 15° (d) Increase by 30°
42. Two normal uniform magnetic field contain a magnetic needle making an angle 60° with F . Then the ratio of $\frac{F}{H}$ is [CPMT 1987; DPMT 2001]
(a) 1 : 2 (b) 2 : 1
(c) $\sqrt{3} : 1$ (d) $1 : \sqrt{3}$
43. A short magnetic needle is pivoted in a uniform magnetic field of strength 1 T. When another magnetic field of strength $\sqrt{3}$ T is applied to the needle in a perpendicular direction, the needle deflects through an angle θ , where θ is [KCET 1999]
(a) 30° (b) 45°
(c) 90° (d) 60°
44. Two magnets are held together in a vibration magnetometer and are allowed to oscillate in the earth's magnetic field with like poles

together, 12 oscillations per minute are made but for unlike poles together only 4 oscillations per minute are executed. The ratio of their magnetic moments is

[MP PMT 1996; CPMT 2002]

- (a) 3 : 1 (b) 1 : 3
(c) 3 : 5 (d) 5 : 4

45. To measure which of the following, is a tangent galvanometer used [MP PET 1997; CPMT 2001]

- (a) Charge (b) Angle
(c) Current (d) Magnetic intensity

46. When $\sqrt{3}$ ampere current is passed in a tangent galvanometer, there is a deflection of 30° in it. The deflection obtained when 3 amperes current is passed, is

[MP PMT 1997]

- (a) 30° (b) 45°
(c) 60° (d) 75°

47. The period of oscillations of a magnetic needle in a magnetic field is 1.0 sec. If the length of the needle is halved by cutting it, the time period will be

[MP PMT/PET 1998]

- (a) 1.0 sec (b) 0.5 sec
(c) 0.25 sec (d) 2.0 sec

48. The time period of a freely suspended magnet is 2 sec. If it is broken in length into two equal parts and one part is suspended in the same way, then its time period will be

[MP PMT 1999]

- (a) 4 sec (b) 2 sec
(c) $\sqrt{2}$ sec (d) 1 sec

49. The bob of a simple pendulum is replaced by a magnet. The oscillations are set along the length of the magnet. A copper coil is added so that one pole of the magnet passes in and out of the coil. The coil is short-circuited. Then which one of the following happens

[KCET 1994]

- (a) Period decreases
(b) Period does not change
(c) Oscillations are damped
(d) Amplitude increases

50. The period of oscillation of a vibration magnetometer depends on which of the following factors

[KCET 1994]

- (a) I and M only (b) M and H only
(c) I and H only (d) I , M and H only

where I is the moment of inertia of the magnet about the axis of suspension, M is the magnetic moment of the magnet and H is the external magnetic field

51. The time period of oscillation of a bar magnet suspended horizontally along the magnetic meridian is T . If this magnet is replaced by another magnet of the same size and pole strength but with double the mass, the new time period will be

- (a) $\frac{T_0}{2}$ (b) $\frac{T_0}{\sqrt{2}}$
(c) $\sqrt{2}T_0$ (d) $2T_0$

52. Two short magnets having magnetic moments in the ratio 27 : 8, when placed on opposite sides of a deflection magnetometer, produce no deflection. If the distance of the weaker magnet is 0.12 m from the centre of deflection magnetometer, the distance of the stronger magnet from the centre is

- (a) 0.06 m (b) 0.08 m

- (c) 0.12 m (d) 0.18 m

53. The magnet of a vibration magnetometer is heated so as to reduce its magnetic moment by 19%. By doing this the periodic time of the magnetometer will

[MP PMT 2000, 01]

- (a) Increase by 19% (b) Decrease by 19%
(c) Increase by 11% (d) Decrease by 21%

54. A magnet makes 40 oscillations per minute at a place having magnetic field intensity of 0.1×10^{-7} T. At another place, it takes 2.5 sec to complete one vibration. The value of earth's horizontal field at that place is

[AIIMS 2000; CPMT 2000; Pb PET 2002]

- (a) 0.25×10^{-7} T (b) 0.36×10^{-7} T
(c) 0.66×10^{-7} T (d) 1.2×10^{-7} T

55. A tangent galvanometer has a coil of 25 turns and radius of 15 cm. The horizontal component of the earth's magnetic field is 3×10^{-7} T. The current required to produce a deflection of 45° in it, is

[MP PMT 2000]

- (a) 0.29 A (b) 1.2 A
(c) 3.6×10^{-7} A (d) 0.14 A

56. The time period of a vibration magnetometer is T . Its magnet is replaced by another magnet whose moment of inertia is 3 times and magnetic moment is $1/3$ of the initial magnet. The time period now will be

[MP PMT 2000]

- (a) $3T$ (b) T
(c) $T_0 / \sqrt{3}$ (d) $T/3$

57. The error in measuring the current with a tangent galvanometer is minimum when the deflection is about

[MP PET 2001]

- (a) 0° (b) 30°
(c) 45° (d) 60°

58. Before using the tangent galvanometer, its coil is set in

[MP PMT 2001; CPMT 2005]

- (a) Magnetic meridian (or vertically north south)
(b) Perpendicular to magnetic meridian
(c) At angle of 45° to magnetic meridian
(d) It does not require any setting

59. The time period of a thin bar magnet in earth's magnetic field is T . If the magnet is cut into two equal parts perpendicular to its length, the time period of each part in the same field will be

- (a) $\frac{T}{2}$ (b) T
(c) $\sqrt{2}T$ (d) $2T$

60. A magnet freely suspended in a vibration magnetometer makes 10 oscillations per minute at a place A and 20 oscillations per minute at a place B. If the horizontal component of earth's magnetic field at A is 36×10^{-6} T, then its value at B is

[SCRA 1994; JPMER 2001, 02]

- (a) 36×10^{-6} T (b) 72×10^{-6} T
(c) 144×10^{-6} T (d) 288×10^{-6} T

61. When 2 amperes current is passed through a tangent galvanometer, it gives a deflection of 30° . For 60° deflection, the current must be

[EAMCET (Med) 2000]

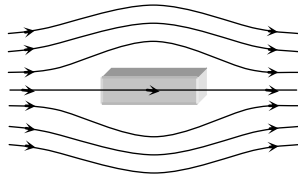
- (a) 1 amp (b) $2\sqrt{3}$ amp
(c) 4 amp (d) 6 amp

62. Which of the following statement is not the true
[KCET (Engg./Med.) 2001]
- While taking reading of tangent galvanometer, the plane of the coil must be set at right angles to the earth's magnetic meridian
 - A short magnet is used in a tangent galvanometer since a long magnet would be heavy and may not easily move
 - Measurements with the tangent galvanometer will be more accurate when the deflection is around 45°
 - A tangent galvanometer can not be used in the polar region
63. The period of oscillations of a magnet is 2 sec. When it is remagnetised so that the pole strength is 4 times its period will be
(a) 4 sec (b) 2 sec
(c) 1 sec (d) $1/2$ sec
64. When two magnetic moments are compared using equal distance method the deflections produced are 45° and 30° . If the length of magnets are in the ratio 1 : 2, the ratio of their pole strengths is
(a) 3 : 1 (b) 3 : 2
(c) $\sqrt{3} : 1$ (d) $2\sqrt{3} : 1$
65. The magnetic needle of a tangent galvanometer is deflected at an angle 30° due to a magnet. The horizontal component of earth's magnetic field $0.34 \times 10^{-4} T$ is along the plane of the coil. The magnetic intensity is
[AIIMS 2000, 2002; BHU 2000; AFMC 2000; KCET (Engg./Med.) 1999]
- $1.96 \times 10^{-4} T$
 - $1.96 \times 10^{-4} T$
 - $1.96 \times 10^{-4} T$
 - $1.96 \times 10^{-4} T$
66. In a tangent galvanometer a current of 0.1 A produces a deflection of 30° . The current required to produce a deflection of 60° is
(a) 0.2 A (b) 0.3 A
(c) 0.4 A (d) 0.5 A
67. A bar magnet is oscillating in the Earth's magnetic field with a period T . What happens to its period and motion if its mass is quadrupled
[CBSE PMT 2003]
- Motion remains S.H.M. with time period = $2T$
 - Motion remains S.H.M. with time period = $4T$
 - Motion remains S.H.M. and period remains nearly constant
 - Motion remains S.H.M. with time period = $\frac{T}{2}$
68. A thin rectangular magnet suspended freely has a period of oscillation equal to T . Now it is broken into two equal halves (each having half of the original length) and one piece is made to oscillate freely in the same field. If its period of oscillation is T' , then ratio $\frac{T'}{T}$ is
[AIEEE 2003]
- $\frac{1}{4}$
 - $\frac{1}{2\sqrt{2}}$
 - $\frac{1}{2}$
 - 2
69. A bar magnet is oscillating in the earth's magnetic field with time period T . If its mass is increased four times then its time period will be
[J & K CET 2004]
- 4 T
 - 2 T
 - T
 - $7/2 T$
70. The length of a magnet is large compared to its width and breadth. The time period of its oscillation in a vibration magnetometer is 2 s. The magnet is cut along its length into three equal parts and three parts are then placed on each other with their like poles together. The time period of this combination will be
(a) 2 s (b) $2/3$ s
(c) $2\sqrt{3}$ s (d) $2/\sqrt{3}$ s
71. A magnet oscillating in a horizontal plane has a time period of 2 second at a place where the angle of dip is 30° and 3 seconds at another place where the angle of dip is 60° . The ratio of resultant magnetic fields at the two places is
[Pb. PET 2001]
- $\frac{4\sqrt{3}}{9\sqrt{3}}$
 - $\frac{4}{9\sqrt{3}}$
 - $\frac{9}{4\sqrt{3}}$
 - $\frac{9}{\sqrt{3}}$
72. Two identical bar magnets are placed on above the other such that they are mutually perpendicular and bisect each other. The time period of this combination in a horizontal magnetic field is T . The time period of each magnet in the same field is
(a) $\sqrt{2} T$ (b) $2^{\frac{1}{4}} T$
(c) $2^{-\frac{1}{4}} T$ (d) $2^{-\frac{1}{2}} T$
73. The radius of the coil of a Tangent galvanometer, which has 10 turns is 0.1m. The current required to produce a deflection of 60° ($B_H = 4 \times 10^{-5} T$) is
[MP PET 2005]
- 3 A
 - 1.1 A
 - 2.1 A
 - 1.5 A

Magnetic Materials

- Magnets cannot be made from which of the following substances
(a) Iron (b) Nickel
(c) Copper (d) All of the above
- The magnetic moment of atomic neon is
[NCERT 1984]
(a) Zero (b) $\mu_B / 2$
(c) μ_B (d) $3\mu_B / 2$
- Which of the following is most suitable for the core of electromagnets
[AIIMS 1980; NCERT 1980; AFMC 1988; CBSE PMT 1990]
(a) Soft iron (b) Steel
(c) Copper-nickel alloy (d) Air
- Demagnetisation of magnets can be done by
[DPMT 1984; CBSE PMT 1988]
(a) Rough handling
(b) Heating
(c) Magnetising in the opposite direction
(d) All the above

5. A ferromagnetic material is heated above its curie temperature. Which one is a correct statement
[MP PET 1995]
- Ferromagnetic domains are perfectly arranged
 - Ferromagnetic domains becomes random
 - Ferromagnetic domains are not influenced
 - Ferromagnetic material changes itself into diamagnetic material
6. If a diamagnetic substance is brought near north or south pole of a bar magnet, it is
[EAMCET (Engg.) 1995; CBSE PMT 1999; AFMC 2003]
- Attracted by the poles
 - Repelled by the poles
 - Repelled by the north pole and attracted by the south pole
 - Attracted by the north pole and repelled by the south pole
7. The material of permanent magnet has
[KCET 1994, 2003; AFMC 2004]
- High retentivity, low coercivity
 - Low retentivity, high coercivity
 - Low retentivity, low coercivity
 - High retentivity, high coercivity
8. The permanent magnet is made from which one of the following substances
[Bihar MEE 1995]
- Diamagnetic
 - Paramagnetic
 - Ferromagnetic
 - Electromagnetic
9. Temperature above which a ferromagnetic substance becomes paramagnetic is called
[SCRA 1994; J & K CET 2004]
- Critical temperature
 - Boyle's temperature
 - Debye's temperature
 - Curie temperature
10. When a magnetic substance is heated, then it
[AIIMS 1999]
- Becomes a strong magnet
 - Losses its magnetism
 - Does not effect the magnetism
 - Either (a) or (c)
11. The only property possessed by ferromagnetic substance is
[KCET 1999]
- Hysteresis
 - Susceptibility
 - Directional property
 - Attracting magnetic substances
12. Substances in which the magnetic moment of a single atom is not zero, is known as
[AFMC 1999]
- Diamagnetism
 - Ferromagnetism
 - Paramagnetism
 - Ferrimagnetism
13. Diamagnetic substances are
[AFMC 1999]
- Feebly attracted by magnets
 - Strongly attracted by magnets
 - Feebly repelled by magnets
 - Strongly repelled by magnets
14. The magnetic susceptibility is
[RPMT 1999]
- $\chi = \frac{I}{H}$
 - $\chi = \frac{B}{H}$
 - $\chi = \frac{M}{V}$
 - $\chi = \frac{M}{H}$
15. Which of the following statements are true about the magnetic susceptibility χ_m of paramagnetic substance
[Roorkee 1999]
- Value of χ_m is inversely proportional to the absolute temperature of the sample
 - χ is positive at all temperature
 - χ_m is negative at all temperature
 - χ_m does not depend on the temperature of the sample
16. Relative permeability of iron is 5500, then its magnetic susceptibility will be
[KCET 2000; Kerala PMT 2004]
- 5500×10
 - 5500×10^3
 - 5501
 - 5499
17. An example of a diamagnetic substance is
[KCET 2000]
- Aluminium
 - Copper
 - Iron
 - Nickel
18. The use of study of hysteresis curve for a given material is to estimate the
[KCET (Engg./Med.) 2000]
- Voltage loss
 - Hysteresis loss
 - Current loss
 - All of these
19. Magnetic permeability is maximum for
[AIIMS 2000; MH CET 2003; DPMT 2003]
- Diamagnetic substance
 - Paramagnetic substance
 - Ferromagnetic substance
 - All of these
20. If a diamagnetic solution is poured into a U-tube and one arm of this U-tube placed between the poles of a strong magnet with the meniscus in a line with the field, then the level of the solution will
[AIIMS 2000]
- Rise
 - Fall
 - Oscillate slowly
 - Remain as such
21. The relative permeability is represented by μ and the susceptibility is denoted by χ for a magnetic substance. Then for a paramagnetic substance
[KCET (Engg./Med.) 2001]
- $\mu < 1, \chi < 0$
 - $\mu < 1, \chi > 0$
 - $\mu > 1, \chi < 0$
 - $\mu > 1, \chi > 0$
22. Which of the following is true
[BHU 2001]
- Diamagnetism is temperature dependent
 - Paramagnetism is temperature dependent
 - Paramagnetism is temperature independent
 - None of these
23. The magnetic susceptibility does not depend upon the temperature in
[CBSE PMT 2001]
- Ferrite substances
 - Ferromagnetic substances
 - Diamagnetic substances
 - Paramagnetic substances
24. Identify the paramagnetic substance
[KCET 2001]
- Iron
 - Aluminium
 - Nickel
 - Hydrogen
25. If a magnetic substance is kept in a magnetic field, then which of the following is thrown out
[DCE 1999, 2001]

- (a) Paramagnetic (b) Ferromagnetic
(c) Diamagnetic (d) Antiferromagnetic [AIIMS 2003]
26. If the angular momentum of an electron is $\vec{J}$ then the magnitude of the magnetic moment will be [MP PET 2002]
(a) $\frac{eJ}{m}$ (b) $\frac{eJ}{2m}$
(c) $e/2m$ (d) $\frac{2m}{eJ}$
27. The magnetic susceptibility is negative for [AIEEE 2002]
(a) Paramagnetic materials
(b) Diamagnetic materials
(c) Ferromagnetic materials
(d) Paramagnetic and ferromagnetic materials
28. The universal property of all substances is [CPMT 2002]
(a) Diamagnetism (b) Ferromagnetism
(c) Paramagnetism (d) All of these
29. Which of the following statements is incorrect about hysteresis
(a) This effect is common to all ferromagnetic substances
(b) The hysteresis loop area is proportional to the thermal energy developed per unit volume of the material
(c) The hysteresis loop area is independent of the thermal energy developed per unit volume of the material
(d) The shape of the hysteresis loop is characteristic of the material
30. Curies law can be written as [MH CET 2002; CBSE PMT 2003]
(a) $\chi \propto (T - T_c)$ (b) $\chi \propto \frac{1}{T - T_c}$
(c) $\chi \propto \frac{1}{T}$ (d) $\chi \propto T$
31. A superconductor exhibits perfect [KCET 2002]
(a) Ferrimagnetism (b) Ferromagnetism
(c) Paramagnetism (d) Diamagnetism
32. A small rod of bismuth is suspended freely between the poles of a strong electromagnet. It is found to arrange itself at right angles to the magnetic field. This observation establishes that bismuth is
(a) Diamagnetic (b) Paramagnetic
(c) Ferri-magnetic (d) Antiferro-magnetic
33. A diamagnetic material in a magnetic field moves [Pb. PMT 1999; AIIMS 2000; MH CET 2000; CBSE PMT 2003]
(a) From weaker to the stronger parts of the field
(b) Perpendicular to the field
(c) From stronger to the weaker parts of the field
(d) In none of the above directions
34. Curie temperature is the temperature above which [DCE 2002; AIEEE 2003]
(a) A paramagnetic material becomes ferromagnetic
(b) A ferromagnetic material becomes paramagnetic
(c) A paramagnetic material becomes diamagnetic
(d) A ferromagnetic material becomes diamagnetic
35. A frog can be deviated in a magnetic field produced by a current in a vertical solenoid placed below the frog. This is possible because the body of the frog behaves as
(a) Paramagnetic (b) Diamagnetic
(c) Ferromagnetic (d) Antiferromagnetic [MP PET 2004]
36. Which one of the following is a non-magnetic substance [MP PET 2004]
(a) Iron (b) Nickel
(c) Cobalt (d) Brass
37. Liquid oxygen remains suspended between two pole faces of a magnet because it is [AIIMS 2004]
(a) Diamagnetic (b) Paramagnetic
(c) Ferromagnetic (d) Antiferromagnetic
38. Curie-Weiss law is obeyed by iron at a temperature [KCET 2004]
(a) Below Curie temperature (b) Above Curie temperature
(c) At Curie temperature only (d) At all temperatures
39. The materials suitable for making electromagnets should have [UPSEAT 2002]
(a) High retentivity and high coercivity
(b) Low retentivity and low coercivity
(c) High retentivity and low coercivity
(d) Low retentivity and high coercivity
40. The given figure represents a material which is [Orissa PMT 2004]

(a) Paramagnetic (b) Diamagnetic
(c) Ferromagnetic (d) None of these
41. For an isotropic medium B , μ , H and M are related as (where B , μ_0 , H and M have their usual meaning in the context of magnetic material) [Kerala 2002]
(a) $(B - M) = \mu_0 H$ (b) $M = \mu_0 (H + M)$
(c) $H = \mu_0 (H + M)$ (d) $B = \mu_0 (H + M)$
42. The magnetic susceptibility of any paramagnetic material changes with absolute temperature T as [UPSEAT 2004; DCE 2005]
(a) Directly proportional to T
(b) Remains constant
(c) Inversely proportional to T
(d) Exponentially decaying with T
43. When a piece of a ferromagnetic substance is put in a uniform magnetic field, the flux density inside it is four times the flux density away from the piece. The magnetic permeability of the material is
(a) 1 (b) 2
(c) 3 (d) 4
44. Which of the following is diamagnetism [DCE 2002]
(a) Aluminium (b) Quartz

- (c) Nickel (d) Bismuth
45. If a ferromagnetic material is inserted in a current carrying solenoid, the magnetic field of solenoid [DCE 2004]
(a) Largely increases (b) Slightly increases
(c) Largely decreases (d) Slightly decreases
46. In the hysteresis cycle, the value of H needed to make the intensity of magnetisation zero is called [DCE 2004]
(a) Retentivity (b) Coercive force
(c) Lorentz force (d) None of the above
47. If the magnetic dipole moment of an atom of diamagnetic material, paramagnetic material and ferromagnetic material denoted by μ_d, μ_p, μ_f respectively then [CBSE PMT 2005]
(a) $\mu_d \neq 0$ and $\mu_f \neq 0$ (b) $\mu_p = 0$ and $\mu_f \neq 0$
(c) $\mu_d = 0$ and $\mu_p \neq 0$ (d) $\mu_d \neq 0$ and $\mu_p = 0$
48. Among the following properties describing diamagnetism identify the property that is wrongly stated [KCET 2005]
(a) Diamagnetic material do not have permanent magnetic moment
(b) Diamagnetism is explained in terms of electromagnetic induction
(c) Diamagnetic materials have a small positive susceptibility
(d) The magnetic moment of individual electrons neutralize each other
49. Susceptibility of ferromagnetic substance is [Orissa JEE 2005]
(a) > 1 (b) < 1
(c) 0 (d) 1
50. When a ferromagnetic material is heated to temperature above its Curie temperature, the material [UPSEAT 2005]
(a) Is permanently magnetized
(b) Remains ferromagnetic
(c) Behaves like a diamagnetic material
(d) Behaves like a paramagnetic material
- (b) Cube of the distance
(c) Distance
(d) Fourth power of the distance
3. Two identical short bar magnets, each having magnetic moment M , are placed a distance of $2d$ apart with axes perpendicular to each other in a horizontal plane. The magnetic induction at a point midway between them is [IIT-JEE (Screening) 2000]
(a) $\frac{\mu_0}{4\pi}(\sqrt{2})\frac{M}{d^3}$ (b) $\frac{\mu_0}{4\pi}(\sqrt{3})\frac{M}{d^3}$
(c) $\left(\frac{2\mu_0}{\pi}\right)\frac{M}{d^3}$ (d) $\frac{\mu_0}{4\pi}(\sqrt{5})\frac{M}{d^3}$
4. If a magnet is suspended at an angle 30° to the magnetic meridian, it makes an angle of 45° with the horizontal. The real dip is
(a) $\tan^{-1}(\sqrt{3}/2)$ (b) $\tan^{-1}(\sqrt{3})$
(c) $\tan^{-1}(\sqrt{3}/2)$ (d) $\tan^{-1}(2/\sqrt{3})$
5. A short bar magnet with its north pole facing north forms a neutral point at P in the horizontal plane. If the magnet is rotated by 90° in the horizontal plane, the net magnetic induction at P is (Horizontal component of earth's magnetic field = B_H)
(a) 0 (b) $2 B_H$
(c) $\frac{\sqrt{5}}{2} B_H$ (d) $\sqrt{5} B_H$
6. The true value of angle of dip at a place is 60° , the apparent dip in a plane inclined at an angle of 30° with magnetic meridian is
(a) $\tan^{-1} \frac{1}{2}$ (b) $\tan^{-1}(2)$
(c) $\tan^{-1}\left(\frac{2}{3}\right)$ (d) None of these
7. A vibration magnetometer consists of two identical bar magnets placed one over the other such that they are perpendicular and bisect each other. The time period of oscillation in a horizontal magnetic field is $2^{5/4}$ seconds. One of the magnets is removed and if the other magnet oscillates in the same field, then the time period in seconds is
(a) $2^{1/4}$ (b) $2^{1/2}$
(c) 2 (d) $2^{3/4}$
8. In a vibration magnetometer, the time period of a bar magnet oscillating in horizontal component of earth's magnetic field is 2 sec. When a magnet is brought near and parallel to it, the time period reduces to 1 sec. The ratio H/F of the horizontal component H and the field F due to magnet will be [MP PMT 1990; Pb PET 2000]
(a) 3 (b) $1/3$
(c) $\sqrt{3}$ (d) $1/\sqrt{3}$
9. A cylindrical rod magnet has a length of 5 cm and a diameter of 1 cm. It has a uniform magnetisation of $5.30 \times 10^4 \text{ Amp/m}$. What its magnetic dipole moment
(a) $1 \times 10^{-2} \text{ J/T}$ (b) $2.08 \times 10^{-2} \text{ J/T}$

Critical Thinking

Objective Questions

1. Two identical magnetic dipoles of magnetic moments i.o. $A\text{-}m$ each, placed at a separation of $2m$ with their axis perpendicular to each other. The resultant magnetic field at a point midway between the dipoles is [Roorkee 1995]
(a) $5 \times 10^{-7} \text{ T}$ (b) $\sqrt{5} \times 10^{-7} \text{ T}$
(c) 10^{-7} T (d) None of these
2. Two short magnets placed along the same axis with their like poles facing each other repel each other with a force which varies inversely as
(a) Square of the distance

- (c) $3.08 \times 10^{-2} \text{ J/T}$ (d) $1.52 \times 10^{-2} \text{ J/T}$

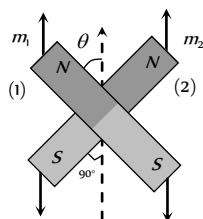
10. Two magnets of equal mass are joined at right angles to each other as shown the magnet 1 has a magnetic moment 3 times that of magnet 2. This arrangement is pivoted so that it is free to rotate in the horizontal plane. In equilibrium what angle will the magnet 1 subtend with the magnetic meridian

(a) $\tan^{-1}\left(\frac{1}{2}\right)$

(b) $\tan^{-1}\left(\frac{1}{3}\right)$

(c) $\tan^{-1}(1)$

(d) 0°



11. The dipole moment of each molecule of a paramagnetic gas is $1.5 \times 10^{-23} \text{ amp} \times \text{m}$. The temperature of gas is 27°C and the number of molecules per unit volume in it is $2 \times 10^{23} \text{ m}^{-3}$. The maximum possible intensity of magnetisation in the gas will be

(a) $3 \times 10^2 \text{ amp/m}$

(b) $4 \times 10^2 \text{ amp/m}$

(c) $5 \times 10^2 \text{ amp/m}$

(d) $6 \times 10^2 \text{ amp/m}$

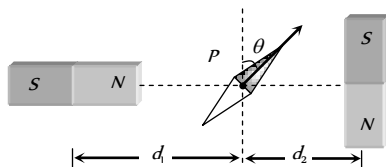
12. Two magnets A and B are identical and these are arranged as shown in the figure. Their length is negligible in comparison to the separation between them. A magnetic needle is placed between the magnets at point P which gets deflected through an angle θ under the influence of magnets. The ratio of distance d_1 and d_2 will be

(a) $(2 \tan \theta)^{1/3}$

(b) $(2 \tan \theta)^{-1/3}$

(c) $(2 \cot \theta)^{1/3}$

(d) $(2 \cot \theta)^{-1/3}$



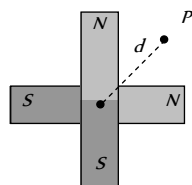
13. Two short magnets of equal dipole moments M are fastened perpendicularly at their centre (figure). The magnitude of the magnetic field at a distance d from the centre on the bisector of the right angle is

(a) $\frac{\mu_0}{4\pi} \frac{M}{d^3}$

(b) $\frac{\mu_0}{4\pi} \frac{M\sqrt{2}}{d^3}$

(c) $\frac{\mu_0}{4\pi} \frac{2\sqrt{2}M}{d^3}$

(d) $\frac{\mu_0}{4\pi} \frac{2M}{d^3}$



14. A small coil C with $N = 200$ turns is mounted on one end of a balance beam and introduced between the poles of an electromagnet as shown in figure. The cross sectional area of coil is $A = 1.0 \text{ cm}^2$, length of arm OA of the balance beam is $l = 30 \text{ cm}$. When there is no current in the coil the balance is in equilibrium. On passing a current $I = 22 \text{ mA}$ through the coil the equilibrium is restored by putting the additional counter weight of mass $\Delta m = 60 \text{ mg}$ on the

balance pan. Find the magnetic induction at the spot where coil is located.

(a) 0.4 T

(b) 0.3 T

(c) 0.2 T

(d) 0.1 T

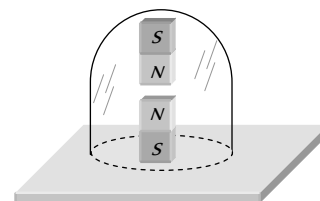
15. Two identical bar magnets with a length 10 cm and weight 50 gm-weight are arranged freely with their like poles facing in a inverted vertical glass tube. The upper magnet hangs in the air above the lower one so that the distance between the nearest pole of the magnet is 3 mm . Pole strength of the each magnet will be

(a) $6.64 \text{ amp} \times \text{m}$

(b) $2 \text{ amp} \times \text{m}$

(c) $10.25 \text{ amp} \times \text{m}$

(d) None of these



16. If ϕ_1 and ϕ_2 be the angles of dip observed in two vertical planes at right angles to each other and ϕ be the true angle of dip, then

(a) $\cos^2 \phi = \cos^2 \phi_1 + \cos^2 \phi_2$

(b) $\sec^2 \phi = \sec^2 \phi_1 + \sec^2 \phi_2$

(c) $\tan^2 \phi = \tan^2 \phi_1 + \tan^2 \phi_2$

(d) $\cot^2 \phi = \cot^2 \phi_1 + \cot^2 \phi_2$

17. Each atom of an iron bar ($5 \text{ cm} \times 1 \text{ cm} \times 1 \text{ cm}$) has a magnetic moment $1.8 \times 10^{-23} \text{ Am}^2$. Knowing that the density of iron is $7.78 \times 10^3 \text{ kg m}^{-3}$, atomic weight is 56 and Avogadro's number is 6.02×10^{23} the magnetic moment of bar in the state of magnetic saturation will be

(a) 4.75 Am^2

(b) 5.74 Am^2

(c) 7.54 Am^2

(d) 75.4 Am^2

18. An iron rod of volume 10^{-4} m^3 and relative permeability 1000 is placed inside a long solenoid wound with 5 turns/cm. If a current of 0.5 A is passed through the solenoid, then the magnetic moment of the rod is

(a) 10 Am^2

(b) 15 Am^2

(c) 20 Am^2

(d) 25 Am^2

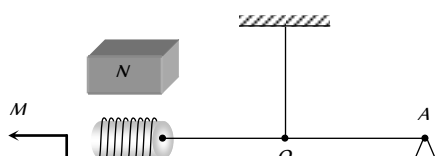
19. A bar magnet has coercivity $4 \times 10^3 \text{ Am}^{-1}$. It is desired to demagnetise it by inserting it inside a solenoid 12 cm long and having 60 turns. The current that should be sent through the solenoid is

(a) 2 A

(b) 4 A

(c) 6 A

(d) 8 A

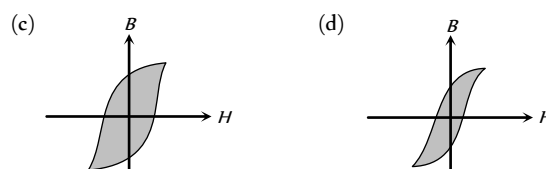
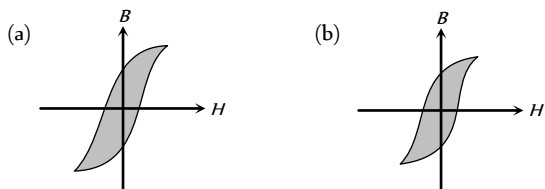


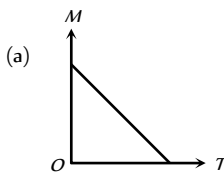
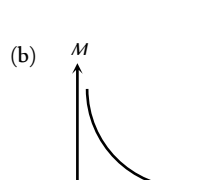
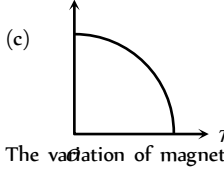
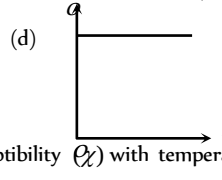
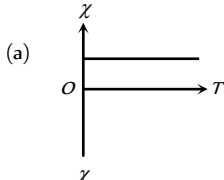
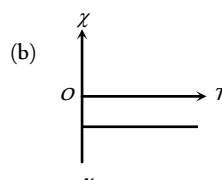
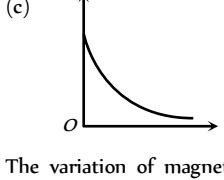
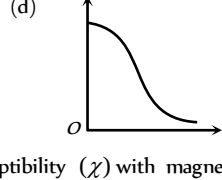
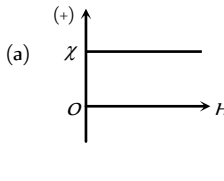
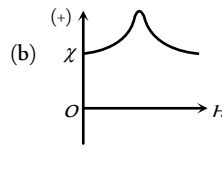
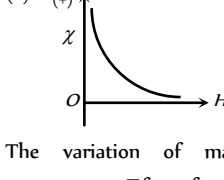
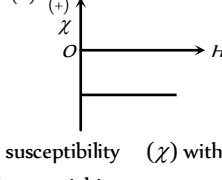
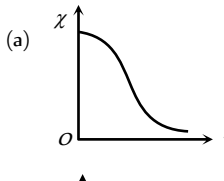
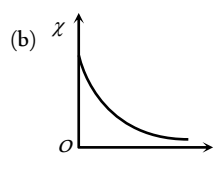
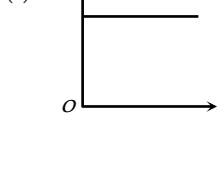
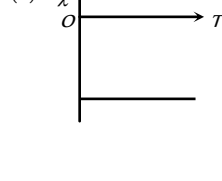
20. A magnet is suspended in the magnetic meridian with an untwisted wire. The upper end of wire is rotated through 180° to deflect the magnet by 30° from magnetic meridian. When this magnet is replaced by another magnet, the upper end of wire is rotated through 270° to deflect the magnet 30° from magnetic meridian. The ratio of magnetic moments of magnets is
- (a) 1 : 5 (b) 1 : 8
(c) 5 : 8 (d) 8 : 5
21. A dip needle vibrates in the vertical plane perpendicular to the magnetic meridian. The time period of vibration is found to be 2 seconds. The same needle is then allowed to vibrate in the horizontal plane and the time period is again found to be 2 seconds. Then the angle of dip is
- (a) 0° (b) 30°
(c) 45° (d) 90°
22. The unit for molar susceptibility is
- (a) m (b) $kg\text{-}m$
(c) $kg\text{-}m$ (d) No units
23. A short magnet oscillates with a time period 0.1 s at a place where horizontal magnetic field is $24\mu T$. A downward current of 18 A is established in a vertical wire 20 cm east of the magnet. The new time period of oscillator
- (a) 0.1 s (b) 0.089 s
(c) 0.076 s (d) 0.057 s
24. A dip needle lies initially in the magnetic meridian when it shows an angle of dip θ at a place. The dip circle is rotated through an angle x in the horizontal plane and then it shows an angle of dip θ' . Then $\frac{\tan \theta'}{\tan \theta}$ is
- (a) $\frac{1}{\cos x}$ (b) $\frac{1}{\sin x}$
(c) $\frac{1}{\tan x}$ (d) $\cos x$
25. A dip circle is adjusted so that its needle moves freely in the magnetic meridian. In this position, the angle of dip is 40° . Now the dip circle is rotated so that the plane in which the needle moves makes an angle of 30° with the magnetic meridian. In this position the needle will dip by an angle

[DCE 2005]

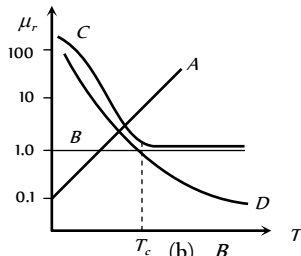
Graphical Questions

1. For substances hysteresis ($B-H$) curves are given as shown in figure. For making temporary magnet which of the following is best.

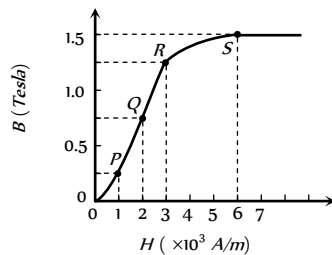


2. A curve between magnetic moment and temperature of magnet is
- (a)  (b) 
(c)  (d) 
3. The variation of magnetic susceptibility (χ) with temperature for a diamagnetic substance is best represented by
- (a)  (b) 
(c)  (d) 
4. The variation of magnetic susceptibility (χ) with magnetising field for a paramagnetic substance is
- (a)  (b) 
(c)  (d) 
5. The variation of magnetic susceptibility (χ) with absolute temperature T for a ferromagnetic material is
- (a)  (b) 
(c)  (d) 

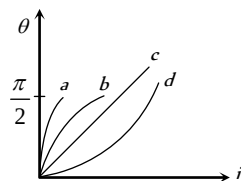
6. The relative permeability (μ_r) of a ferromagnetic substance varies with temperature (T) according to the curve



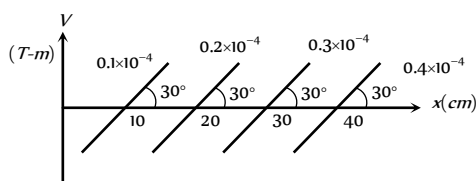
- (a) A (b) B (c) C (d) D
7. The basic magnetization curve for a ferromagnetic material is shown in figure. Then, the value of relative permeability is highest for the point



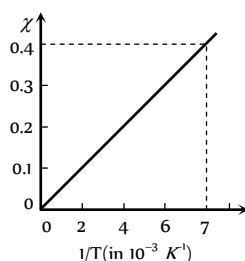
- (a) P (b) Q (c) R (d) S
8. Which curve may best represent the current deflection in a tangent galvanometer



- (a) A (b) B (c) C (d) D
9. Some equipotential surfaces of the magnetic scalar potential are shown in the figure. Magnetic field at a point in the region is



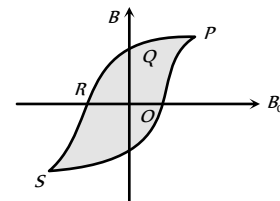
- (a) $10^{-4} T$ (b) $2 \times 10^{-4} T$ (c) $0.5 \times 10^{-4} T$ (d) None of these
10. The $\chi - 1/T$ graph for an alloy of paramagnetic nature is shown in Fig. The Curie constant is, then



- (a) 57 K (b) $2.8 \times 10^{-3} K$ (c) 570 K (d) $17.5 \times 10^{-3} K$

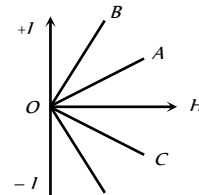
11. The figure illustrates how B , the flux density inside a sample of unmagnetised ferromagnetic material varies with B_0 , the magnetic flux density in which the sample is kept. For the sample to be suitable for making a permanent magnet

[AMU 2001]

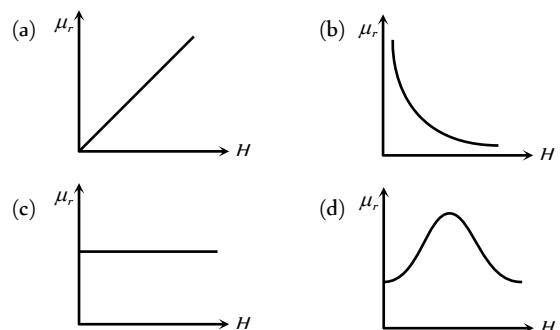


- (a) OQ should be large, OR should be small (b) OQ and OR should both be large (c) OQ should be small and OR should be large (d) OQ and OR should both be small
12. The variation of the intensity of magnetisation (I) with respect to the magnetising field (H) in a diamagnetic substance is described by the graph

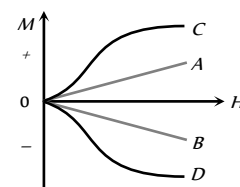
[KCET 2002]



- (a) OD (b) OC (c) OB (d) OA
13. For ferromagnetic material, the relative permeability (μ_r), versus magnetic intensity (H) has the following shape



14. The most appropriate magnetization M versus magnetising field H curve for a paramagnetic substance is



- (a) A (b) B
(c) C (d) D

Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(c) If assertion is true but reason is false.
(d) If the assertion and reason both are false.
(e) If assertion is false but reason is true.

- Assertion : We cannot think of magnetic field configuration with three poles.
Reason : A bar magnet does exert a torque on itself due to its own field. [AIIMS 2002]
- Assertion : The poles of magnet cannot be separated by breaking into two pieces.
Reason : The magnetic moment will be reduced to half when a magnet is broken into two equal pieces. [SCRA 1994]
- Assertion : Basic difference between an electric line and magnetic line of force is that former is discontinuous and the latter is continuous or endless.
Reason : No electric lines of forces exist inside a charged body but magnetic lines do exist inside a magnet.
- Assertion : Magnetic moment of an atom is due to both, the orbital motion and spin motion of every electron.
Reason : A charged particle produces a magnetic field.
- Assertion : When radius of circular loop carrying current is doubled, its magnetic moment becomes four times.
Reason : Magnetic moment depends on area of the loop.
- Assertion : The earth's magnetic field is due to iron present in its core.
Reason : At a high temperature magnet losses its magnetic property or magnetism.
- Assertion : A compass needle when placed on the magnetic north pole of the earth rotates in vertical direction.
Reason : The earth has only horizontal component of its magnetic field at the north poles.
- Assertion : The tangent galvanometer can be made more sensitive by increasing the number of turns of its coil.
Reason : Current through galvanometer is proportional to the number of turns of coil.
- Assertion : The ferromagnetic substance do not obey Curie's law.
Reason : At Curie point a ferromagnetic substance start behaving as a paramagnetic substance.
- Assertion : The properties of paramagnetic and ferromagnetic substance are not effected by heating.

- Reason : As temperature rises, the alignment of molecular magnets gradually decreases.
- Assertion : Soft iron is used as transformer core.
Reason : Soft iron has narrow hysteresis loop.
- Assertion : Magnetism is relativistic.
Reason : When we move along with the charge so that there is no motion relative to us, we find no magnetic field associated with the charge.
- Assertion : The earth's magnetic field does not affect the working of a moving coil galvanometer.
Reason : Earth's magnetic field is very weak.
- Assertion : A paramagnetic sample display greater magnetisation (for the same magnetising field) when cooled.
Reason : The magnetisation does not depend on temperature.
- Assertion : Electromagnets are made of soft iron.
Reason : Coercivity of soft iron is small.
- Assertion : To protect any instrument from external magnetic field, it is put inside an iron body.
Reason : Iron is a magnetic substance.
- Assertion : When a magnet is brought near iron nails, only translatory force act on it.
Reason : The field due to a magnet is generally uniform.
- Assertion : When a magnetic dipole is placed in a non uniform magnetic field, only a torque acts on the dipole.
Reason : Force would also acts on dipole if magnetic field were uniform.
- Assertion : Reduction factor (K) of a tangent galvanometer helps in reducing deflection to current.
Reason : Reduction factor increases with increase of current.
- Assertion : The susceptibility of diamagnetic materials does not depend upon temperature.
Reason : Every atom of a diamagnetic material is not a complete magnet in itself.
- Assertion : The permeability of a ferromagnetic material is independent of the magnetic field.
Reason : Permeability of a material is a constant quantity.
- Assertion : For a perfectly diamagnetic substance permeability is always one.
Reason : The ability of a material of permit the passage of magnetic lines of force through it is called magnetic permeability.
- Assertion : Gauss theorem is not applicable in magnetism.
Reason : Mono magnetic pole does not exist.
- Assertion : Magnetic moment of helium atom is zero.
Reason : All the electron are paired in helium atom orbitals.
- Assertion : For making permanent magnets, steel is preferred over soft iron.
Reason : As retentivity of steel is smaller.

Answers

Magnet and It's Properties

1	b	2	d	3	c	4	d	5	b
6	d	7	b	8	c	9	c	10	a
11	c	12	d	13	c	14	b	15	a
16	a	17	b	18	c	19	d	20	b
21	a	22	c	23	c	24	d	25	d
26	d	27	a	28	a	29	b	30	a
31	a	32	b	33	a	34	c	35	c
36	b	37	b	38	c	39	b	40	c
41	c	42	b	43	a	44	d	45	d
46	b	47	d	48	a	49	a	50	d
51	c	52	b	53	b	54	c	55	b
56	b	57	c	58	b	59	c	60	b
61	b	62	d	63	a	64	a	65	c
66	b	67	b	68	b	69	a	70	b
71	c	72	b	73	c	74	d	75	d
76	a	77	d	78	c	79	d	80	a
81	a	82	b	83	a	84	b	85	a
86	b	87	a	88	a	89	d	90	b
91	d	92	c						

26	a	27	c	28	a	29	a	30	a
31	b	32	d	33	b	34	d	35	a
36	c	37	b	38	d	39	b	40	a
41	b	42	d	43	d	44	d	45	c
46	b	47	b	48	d	49	c	50	d
51	c	52	d	53	c	54	b	55	a
56	a	57	c	58	a	59	a	60	c
61	d	62	a	63	c	64	d	65	b
66	b	67	a	68	c	69	b	70	b
71	c	72	c	73	b				

Magnetic Materials

1	c	2	a	3	a	4	d	5	b
6	b	7	d	8	c	9	d	10	b
11	a	12	c	13	c	14	a	15	ab
16	d	17	b	18	b	19	c	20	b
21	d	22	b	23	c	24	b	25	c
26	b	27	b	28	a	29	c	30	c
31	d	32	a	33	c	34	b	35	b
36	d	37	b	38	b	39	c	40	b
41	d	42	c	43	d	44	d	45	a
46	b	47	c	48	c	49	a	50	d

Earth Magnetism

1	b	2	d	3	b	4	a	5	c
6	c	7	d	8	d	9	b	10	a
11	c	12	a	13	d	14	d	15	c
16	c	17	a	18	d	19	b	20	e
21	b	22	d	23	a	24	d	25	b
26	d	27	a	28	a	29	d	30	b
31	d	32	a	33	c	34	a	35	a
36	c	37	a	38	d	39	a	40	a
41	a	42	c	43	c	44	c	45	a
46	b	47	c	48	a	49	a	50	c
51	b	52	c	53	b	54	d	55	c
56	a	57	a	58	b	59	a		

Magnetic Equipments

1	d	2	c	3	a	4	b	5	d
6	d	7	a	8	c	9	c	10	b
11	c	12	a	13	a	14	a	15	b
16	b	17	b	18	d	19	b	20	c
21	c	22	c	23	c	24	c	25	a

Critical Thinking Questions

1	b	2	d	3	d	4	a	5	d
6	b	7	c	8	b	9	b	10	b
11	a	12	c	13	c	14	a	15	a
16	d	17	c	18	d	19	d	20	c
21	c	22	a	23	c	24	a	25	c

Graphical Questions

1	d	2	c	3	b	4	a	5	a
6	c	7	b	8	b	9	b	10	a
11	b	12	b	13	d	14	a		

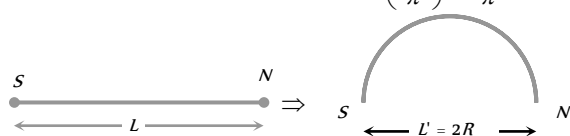
Assertion and Reason

1	d	2	b	3	a	4	c	5	b
6	e	7	d	8	b	9	b	10	e
11	a	12	a	13	a	14	c	15	a
16	a	17	d	18	d	19	c	20	c
21	d	22	e	23	a	24	a	25	b

Magnet and it's Properties

1. (b) On bending a rod it's pole strength remains unchanged where as its magnetic moment changes.

$$\text{New magnetic moment } M' = m(2R) = m\left(\frac{2L}{\pi}\right) = \frac{2M}{\pi}$$



2. (d)

$$3. (c) B_a = \frac{\mu_0}{4\pi} \frac{2M}{d^3} = \frac{\mu_0}{2\pi} \frac{M}{d^3}$$

4. (d)

5. (b) If cut along the axis of magnet of length l , then new pole strength $m' = \frac{m}{2}$ and new length $l' = l$

$$\therefore \text{New magnetic moment } M' = \frac{m}{2} \times l = \frac{ml}{2} = \frac{M}{2}$$



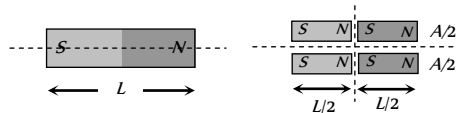
If cut perpendicular to the axis of magnet, then new pole strength $m' = m$ and new length, $l' = l/2$

$$\therefore \text{New magnetic moment } M' = m \times \frac{l}{2} = \frac{ml}{2} = \frac{M}{2}$$

6. (d) For a magnet $B = \frac{\mu_0}{4\pi} \cdot \frac{2M}{x^3}$ (Nearly)

$$\Rightarrow \frac{B_1}{B_2} = \left(\frac{x_1}{x_2}\right)^3 = \left(\frac{x}{2x}\right)^3 = \frac{1}{8} \quad (\text{Approx.})$$

7. (b) For each part $m' = \frac{m}{2}$



8. (c) $\frac{B_1}{B_2} = \frac{d_1}{d_2} \left(\frac{d_2^2 - l^2}{d_1^2 - l^2} \right)^2 \Rightarrow \frac{12.5}{1} = \frac{10}{20} \left(\frac{400 - l^2}{100 - l^2} \right)^2$

$$\Rightarrow l = 5 \text{ cm}$$

Hence length of magnet $= 2l = 10 \text{ cm}$

9. (c) $B_1 = \frac{2M}{d^3}, B_2 = \frac{M}{d^3}; \therefore \frac{B_1}{B_2} = 2:1$

10. (a)

11. (c) $\tau = MB \sin \theta = 48 \times 25 \times 10^{-2} \times 0.15 \times \frac{1}{2} = 0.9 \text{ N} \times \text{m}$

12. (d) $B_1 = \frac{2M}{x^3}$ and $B_2 = \frac{M}{y^3}$

As $B_1 = B_2$

Hence $\frac{2M}{x^3} = \frac{M}{y^3}$ or $\frac{x^3}{y^3} = 2$ or $\frac{x}{y} = 2^{1/3}$

13. (c) Work done $W = MB_H (1 - \cos \theta)$

$$= 20 \times 0.3 (1 - \cos 30^\circ) = 6 \left(1 - \frac{\sqrt{3}}{2} \right) = 3(2 - \sqrt{3})$$

14. (b) Magnetic intensity on end side-on position is twice than broad side on position.

15. (a) Along the axis of magnet $B_a = \frac{2M}{X^3} = 200 \text{ gauss}$

$$\Rightarrow B_a = \frac{M}{X^3} = 100 \text{ gauss}$$

16. (a)

17. (b)

18. (c)

19. (d) Provided length of magnet is $\ll$ the distance.

20. (b) Permeability of soft iron is maximum, so maximum lines of force tries to pass through the soft iron.

21. (a) Plane of coil is having angle θ with the magnetic field.
 $\therefore \tau = MB \sin(90 - \theta)$ or $\tau = niAB \cos \theta$ [As $M = niA$]

22. (c) $B \propto \frac{1}{x^3} \Rightarrow \frac{B_1}{B_2} = \left(\frac{x_2}{x_1}\right)^3 = \left(\frac{3x}{x}\right)^3 = \frac{27}{1}$

23. (c) For null deflection $\frac{M_1}{M_2} = \left(\frac{d_1}{d_2}\right)^3 = \left(\frac{40}{50}\right)^3 = \frac{64}{125}$

24. (d)

25. (d) $F = \frac{\mu_0}{4\pi} \left(\frac{6MM'}{d^4} \right)$ in end-on position.

26. (d) Work done $MB(\cos \theta_1 - \cos \theta_2)$

$$\theta_1 = 0^\circ \text{ and } \theta_2 = 180^\circ$$

$$\Rightarrow W = MB(\cos 0 - \cos 180) = 2MB$$

27. (a) Pole strength doesn't depend upon the length.

28. (a) Torque $\tau = MB_H \sin \theta$

$$= 0.1 \times 10^{-3} \times 4\pi \times 10^{-3} \times \sin 30^\circ = 10^{-7} \times 4\pi \times \frac{1}{2}$$

$$= 2\pi \times 10^{-7} \text{ N} \times \text{m}$$

29. (b) Number of lines of force passing through per unit area normally is intensity of magnetic field, hence option (c) is incorrect. The correct option is (b).

30. (a) Flux $= B \times A; \therefore B = \frac{\text{Flux}}{A} = \text{Weber} / \text{m}^2$

31. (a)

32. (b) $B = \frac{m}{d^2}$ in C.G.S. system.

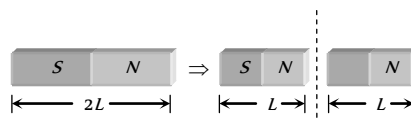
33. (a) $W = MB(\cos \theta_1 - \cos \theta_2) = MB(\cos 0^\circ - \cos 60^\circ)$

$$= MB \left(1 - \frac{1}{2} \right) = \frac{MB}{2}$$

$$\text{and } \tau = MB \sin \theta = MB \sin 60^\circ = MB \frac{\sqrt{3}}{2}$$

$$\therefore \tau = \left(\frac{MB}{2} \right) \sqrt{3} \Rightarrow \tau = \sqrt{3} W$$

34. (c)

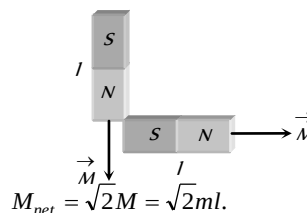


Pole strength of each part $= m$

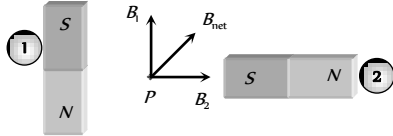
Magnetic moment of each part

$$= M' = m'L' = mL = \frac{M}{2}$$

35. (c)



36. (b)

37. (b) $F \propto \frac{m_1 m_2}{r^2}$
38. (c) $F = 10^{-7} \times \frac{m^2}{r^2} = \frac{10^{-7} (1)^2}{(1)^2} = 10^{-7} \text{ N}$
39. (b) $\tau = MH \sin \theta = MH \sin 30^\circ = \frac{MH}{2}$
40. (c)
41. (c) $F = \frac{\mu_0}{4\pi} \left(\frac{6MM'}{d^4} \right)$ in end-on position between two small magnets.
 $\therefore F = 10^{-7} \left(\frac{6 \times 10 \times 10}{(0.1)^4} \right) = 0.6 \text{ N}$
42. (b)
43. (a) $\tau = MB_H \sin \theta$ or $\frac{d\tau}{d\theta} = MB_H \cos \theta$
 This will be maximum, when $\theta = 0^\circ$.
44. (d) $W = MB(\cos \theta_1 - \cos \theta_2)$; $\theta_1 = 0^\circ$ and $\theta_2 = 360^\circ \Rightarrow W = 0$
45. (d)
46. (b) $W_1 = MB(\cos 0^\circ - \cos 90^\circ) = MB(1 - 0) = MB$
 $W_2 = MB(\cos 0^\circ - \cos 60^\circ) = MB\left(1 - \frac{1}{2}\right) = \frac{MB}{2}$
 $\therefore W_1 = 2W_2 \Rightarrow n = 2$
47. (d) In magnetic dipole, force $\propto \frac{1}{r^4}$
 Hence new force $= \frac{4.8}{2^4} = \frac{4.8}{16} = 0.3 \text{ N}$
48. (a) Magnetic moment of bar $M = 10^4 \text{ J/T}$
 $B = 4 \times 10^{-5} \text{ T}$
 Hence work done $W = \vec{M} \cdot \vec{B}$
 $= 10^4 \times 4 \times 10^{-5} \times \cos 60^\circ = 0.2 \text{ J}$
49. (a)
50. (d)
51. (c) $B = \frac{\mu_0}{4\pi} \frac{2M}{d^3} = 10^{-7} \times \frac{2 \times 1.25}{(0.5)^3} = 2 \times 10^{-6} \text{ N/A-m}$
52. (b)
53. (b)
- 
54. (c)
55. (b) $\tau = MB_H \sin \theta \Rightarrow 0.032 = M \times 0.16 \times \sin 30^\circ$
 $\Rightarrow M = 0.4 \text{ J/tesla}$
56. (b) $B_{\text{equatorial}} = \frac{\mu_0}{4\pi} \frac{M}{r^3}$

57. (c) Inside a magnet, magnetic lines of force move from south pole to north pole.

58. (b) Magnetic moment of circular loop carrying current

$$M = IA = I(\pi R^2) = I\pi \left(\frac{L}{2\pi} \right)^2 = \frac{IL^2}{4\pi} \Rightarrow L = \sqrt{\frac{4\pi M}{I}}$$

59. (c)

60. (b) Concept of magnetic screening.

61. (b) Repulsion is the sure test of magnetism.

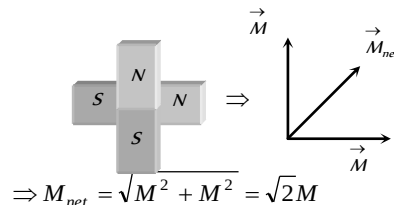
62. (d)

63. (a)

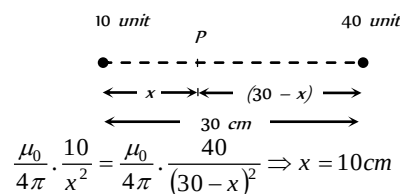
64. (a) $C_{\text{max}} = MB \Rightarrow 4 \times 10^{-5} = M \times 10^{-4} \Rightarrow M = 0.4 \text{ A} \times \text{m}^2$

65. (c) Magnetic flux $\phi = BA \Rightarrow B = \frac{\phi}{A} = \frac{\text{Weber}}{\text{m}^2} = \text{Tesla}$

66. (b)



67. (b) Suppose magnetic field is zero at point P. Which lies at a distance x from 10 unit pole. Hence at P



So from stronger pole distance is 20 cm.

68. (b) $\tau = MB \sin \theta = (mL)B \sin \theta$
 $= (40 \times 10 \times 10^{-2}) \times 2 \times 10^{-4} \times \sin 45^\circ$
 $= 0.565 \times 10^{-3} \text{ N-m}$

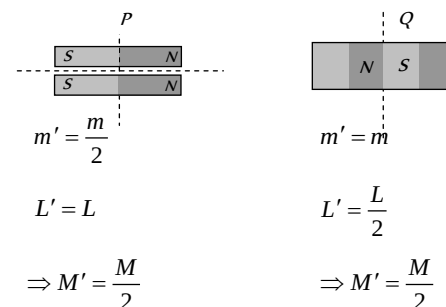
69. (a) Potential energy $U = -MB \cos \theta$

$$\Rightarrow U_{\text{max}} = MH (\text{at } \theta = 180^\circ)$$

70. (b) $\tau = MB \sin \theta$

$$\tau = 200 \times 0.25 \times \sin 30^\circ = 25 \text{ N} \times \text{m}.$$

71. (c) If pole strength, magnetic moment and length of each part are m', M' and L' respectively then



72. (b) $\vec{\tau} = \vec{M} \times \vec{B} \Rightarrow \vec{\tau} = 50\hat{i} \times (0.5\hat{i} + 3\hat{j})$
 $= 150(\hat{i} \times \hat{j}) = 150\hat{k} \text{ N} \times \text{m}.$

73. (c) $\tau = MB \sin \theta \Rightarrow \tau \propto \sin \theta$

$$\Rightarrow \frac{\tau_1}{\tau_2} = \frac{\sin \theta_1}{\sin \theta_2} \Rightarrow \frac{\tau}{\tau/2} = \frac{\sin 90}{\sin \theta_2}$$

$$\Rightarrow \sin \theta_2 = \frac{1}{2} \Rightarrow \theta_2 = 30^\circ$$

$$\Rightarrow \text{angle of rotation} = 0^\circ - 30^\circ = 60^\circ$$

74. (d)

$$75. (d) F = mB \Rightarrow F = \frac{M}{L} \times B$$

$$\Rightarrow 6 \times 10^{-4} = \frac{3}{L} \times 2 \times 10^{-5} \Rightarrow L = 0.1 \text{ m.}$$

$$76. (a) \tau = MB \sin \theta \Rightarrow \tau = (mL)B \sin \theta$$

$$\Rightarrow 25 \times 10^{-6} = (m \times 5 \times 10^{-2}) \times 5 \times 10^{-2} \times \sin 30$$

$$\Rightarrow m = 2 \times 10^{-2} \text{ A-m.}$$

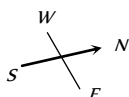
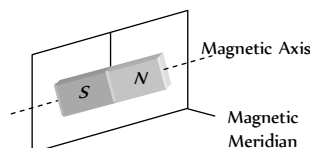
77. (d)

78. (c) Monopole do not exists.

79. (d)

80. (a)

81. (a)



$$82. (b) W = MB(1 - \cos \theta); \text{ where } \theta = 180^\circ$$

$$\Rightarrow W = 2MB \Rightarrow W = 2 \times 2 \times 5 \times 10^{-3} = 2 \times 10^{-2} \text{ J}$$

83. (a) Torque on a bar magnet in earth's magnetic field (B) is $\tau = MB_H \sin \theta$. τ will be maximum if $\sin \theta = \text{maximum}$ i.e. $\theta = 90^\circ$. Hence axis of the magnet is perpendicular to the field of earth.

84. (b)

85. (a) Both points A and B lying on the axis of the magnet and on axial position

$$B \propto \frac{1}{d^3} \Rightarrow \frac{B_A}{B_B} = \left(\frac{d_B}{d_A} \right)^3 = \left(\frac{48}{24} \right)^3 = \frac{8}{1}$$

$$86. (b) W = MB(1 - \cos \theta) = 2 \times 0.1 \times (1 - \cos 90^\circ) = 0.2 \text{ J}$$

$$87. (a) M = mL = 4 \times 10 \times 10^{-2} = 0.4 \text{ A-m}^2$$

88. (a) Similar to solution (1)

New magnetic moment

$$M' = \frac{2M}{\pi} = \frac{2mL}{\pi} = \frac{2 \times 0.5 \times 31.4 \times 10^{-2}}{3.14} = 0.1 \text{ amp-m}^2$$

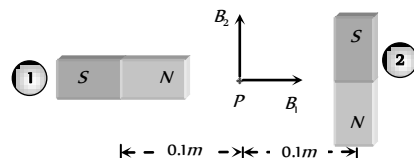
89. (d) Magnetic potential at a distance d from the bar magnet on its axial line is given by

$$V = \frac{\mu_0}{4\pi} \cdot \frac{M}{d^2} \Rightarrow V \propto M \Rightarrow \frac{V_1}{V_2} = \frac{M_1}{M_2}$$

$$\Rightarrow \frac{V}{V_2} = \frac{M}{M/4} \Rightarrow V_2 = \frac{V}{4}$$

$$90. (b) B = \frac{\mu_0}{4\pi} \cdot \frac{2M}{d^3} \Rightarrow B = 10^{-7} \times \frac{2 \times 1.2}{(0.1)^3} = 2.4 \times 10^{-4} \text{ T}$$

91. (d)



$$\text{From figure } B_{\text{net}} = \sqrt{B_a^2 + B_e^2}$$

$$= \sqrt{\left(\frac{\mu_0}{4\pi} \cdot \frac{2M}{d^3} \right)^2 + \left(\frac{\mu_0}{4\pi} \cdot \frac{M}{d^3} \right)^2}$$

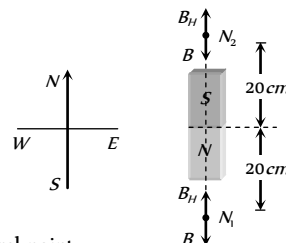
$$= \sqrt{5} \cdot \frac{\mu_0}{4\pi} \cdot \frac{M}{d^3} = \sqrt{5} \times 10^{-7} \times \frac{10}{(0.1)^3} = \sqrt{5} \times 10^{-3} \text{ Tesla.}$$

$$92. (c) \tau = MB \sin \theta = m \times (2l) \times B \sin \theta$$

$$= 10^{-4} \times 0.1 \times 30 \sin 30^\circ = 1.5 \times 10^{-4} \text{ Nm}$$

Earth Magnetism

1. (b)



At neutral point

$$|B| = |B_H| \Rightarrow \frac{2M}{(20)^3} = 0.3 \Rightarrow M = 1.2 \times 10^3 \text{ emu.}$$

2. (d)

No magnetic lines of force passes through the steel box.

3. (b)

At magnetic poles, the angle of dip is 90° . Hence the horizontal component $B_H = B \cos \theta = 0$.

4. (a)

5. (c)

6. (c)

$$7. (d) B_H = \sqrt{3} B_V, \text{ also } \tan \theta = \frac{B_V}{B_H} = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$$

8. (d)

At magnetic equator, the angle of dip is 0° . Hence the vertical component $V = I \sin \phi = 0$.

9. (b)

10. (a)

$$11. (c) B_V = H_H \tan \phi; \text{ If } B_V = B_H, \text{ then } \tan \phi = 1 \text{ or } \phi = 45^\circ$$

$$12. (a) \text{ The horizontal components are } (B_H)_1 = B \cos \phi_1 \text{ and } (B_H)_2 = B \cos \phi_2$$

$$\therefore \frac{(B_H)_1}{(B_H)_2} = \frac{\cos \phi_1}{\cos \phi_2} = \frac{\cos 30^\circ}{\cos 45^\circ} = \frac{\sqrt{3}}{2} \times \sqrt{2} = \frac{\sqrt{3}}{2}$$

$$13. (d) \text{ From the relation } B_H = B \cos \phi \text{ and } B_V = B \sin \phi$$

$$\frac{B_V}{B_H} = \tan \phi \text{ or } B_V = B_H \tan \phi$$

$$= 0.36 \times 10^{-4} \times \tan 60^\circ = 0.623 \times 10^{-4} \text{ Wb/m}^2$$

14. (d) From the relation $B_V = I \sin \phi$

$$I = \frac{V}{\sin \phi} = \frac{6 \times 10^{-5}}{\sin 40.6^\circ} = \frac{6 \times 10^{-5}}{0.65} = 9.2 \times 10^{-5} \text{ tesla}$$

15. (c)

16. (c) $B_H = B \cos \phi; \therefore B = \frac{B_H}{\cos \phi} = \frac{0.5}{\cos 30^\circ} = \frac{0.5}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$

17. (a)

18. (d)

19. (b)

20. (e)

21. (b)

22. (d)

23. (a) $\tan \phi = \frac{B_V}{B_H} = \frac{0.173}{0.30} = \frac{1.73}{3.0} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}} \Rightarrow \phi = 30^\circ$

24. (d) $B_H = B \cos \phi = 0.64 \times \cos 60^\circ = 0.64 \times \frac{1}{2} = 0.32 \text{ units}$

25. (b)

26. (d)

27. (a)

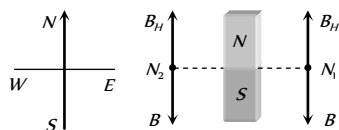
28. (a) $B_H = 0.3 \text{ Oersted}, I = 0.6 \text{ Oersted}$

$$\text{We have } B_H = I \cos \phi \Rightarrow \cos \phi = \frac{B_H}{I} = \frac{0.3}{0.6} = \frac{1}{2}$$

$$\therefore \phi = 60^\circ$$

29. (d)

30. (b)



31. (d) At broad side-on position $B = \frac{M}{d^3}$

$$\therefore \frac{M_1}{d_1^3} = \frac{M_2}{d_2^3} \text{ or } \frac{M_1}{r^3} = \frac{M_2}{8r^3} \text{ or } \frac{M_1}{M_2} = \frac{r^3}{8r^3} = \frac{1}{8}$$

32. (a)

33. (c) $B^2 = B_V^2 + B_H^2 \Rightarrow B_V = \sqrt{B^2 - B_H^2} = \sqrt{(0.5)^2 - (0.3)^2} = 0.4$

$$\text{Now } \tan \phi = \frac{B_V}{B_H} = \frac{0.4}{0.3} = \frac{4}{3} \Rightarrow \phi = \tan^{-1}\left(\frac{4}{3}\right)$$

34. (a) Horizontal component $B_H = B \cos \phi$

$$\text{Total intensity of earth magnetic field } B = \frac{B_H}{\cos \phi} \\ = \frac{1.8 \times 10^{-5}}{\cos 30^\circ} = \frac{1.8 \times 10^{-5}}{\sqrt{3}/2} = 2.08 \times 10^{-5} \text{ Wb/m}^2$$

35. (a)

36. (c)

37. (a) The vertical component of earth's magnetic field is zero at equator where angle of dip is also zero.

38. (d) $B_0 = V_0$ also total intensity $B = \sqrt{B_0^2 + V_0^2} \Rightarrow B = \sqrt{2} B_0$

39. (a) At poles magnetic field is perpendicular to the surface of earth.

40. (a)

41. (a)

42. (c) At neutral point

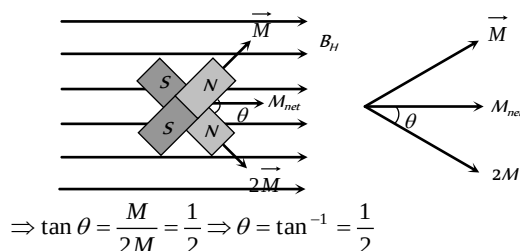
$$\left| \begin{array}{c} \text{Magnetic field due} \\ \text{to magnet} \end{array} \right| = \left| \begin{array}{c} \text{Magnetic field due} \\ \text{to earth} \end{array} \right|$$

$$\frac{\mu_0}{4\pi} \cdot \frac{2M}{d^3} = 5 \times 10^{-5} \Rightarrow 10^{-7} \times \frac{2 \times 6.75}{d^3} = 5 \times 10^{-5}$$

$$\Rightarrow d = 0.3 \text{ m} = 30 \text{ cm}$$

43. (c) As they enter the magnetic field of the earth, they are deflected away from the equator.

44. (c)



$$\Rightarrow \tan \theta = \frac{M}{2M} = \frac{1}{2} \Rightarrow \theta = \tan^{-1} \frac{1}{2}$$

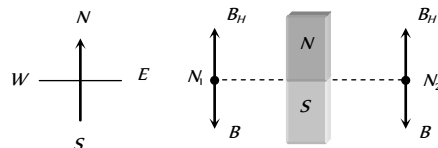
45. (a)

46. (b) $B_H = B \sin \phi \Rightarrow B = \frac{B_H}{\sin \phi} \Rightarrow B = \frac{B_0}{\sin 45^\circ} = \sqrt{2} B_0$

47. (c)

48. (a)

49. (a)



N_1 and N_2 are two null points. And

B_H = Horizontal component of earth's magnetic field

B = Magnetic field due to bar magnet.

50. (c)

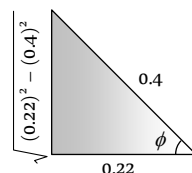
51. (b) $B_H = B \cos \phi \Rightarrow B = \frac{B_H}{\cos \phi} \Rightarrow B = \frac{B_H}{\cos 30^\circ} = \frac{2B_H}{\sqrt{3}}$

52. (c) By using $B_H = B \cos \phi$

$$\Rightarrow \cos \phi = \frac{B_H}{B} = \frac{0.22}{0.4}$$

$$\Rightarrow \tan \phi = \frac{\sqrt{(0.4)^2 - (0.22)^2}}{0.22}$$

$$\Rightarrow \phi = \tan^{-1}(1.518)$$



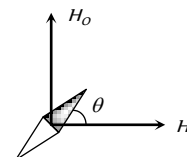
53. (b)

54. (d) At equator angle of dip is zero.

55. (c)

56. (a) In given case H and H_0 are perpendicular to each other.

$$\text{From figure } \tan \theta = \frac{H_0}{H}$$



$$\Rightarrow \theta = \tan^{-1} \left(\frac{H_0}{H} \right)$$

57. (a)

58. (b)

59. (a)

Magnetic Equipments

1. (d)

2. (c)

3. (a)

4. (b) In sum position : $T_s = 2\pi \sqrt{\frac{I_s}{(M_1 + M_2)B_H}}$

In difference position : $T_d = 2\pi \sqrt{\frac{I_d}{(M_1 - M_2)B_H}}$

It is clear that $T_d > T_s$

5. (d) $T = 2\pi \sqrt{\frac{I}{MB_H}}; \therefore \frac{T_1}{T_2} = \sqrt{\frac{(B_H)_2}{(B_H)_1}} \Rightarrow T_2 = T_1 \sqrt{\frac{(B_H)_1}{(B_H)_2}}$

Here $n=30$ oscillation/min $= \frac{1}{2}$ oscillation/sec

$$\therefore T_1 = \frac{1}{n_1} = 2 \text{ sec}$$

$$\therefore T_2 = 2 \sqrt{\frac{B_H}{2B_H}} = 2 \times \frac{1}{\sqrt{2}} = \sqrt{2} \text{ sec}$$

6. (d)

7. (a) $T = 2\pi \sqrt{\frac{1}{MB_H}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{M_2}{M_1}}$

$$\Rightarrow \frac{M_1}{M_2} = \frac{T_2^2}{T_1^2} = \frac{(60/15)^2}{(60/10)^2} = \frac{4}{9}$$

8. (c) When magnet of length l is cut into four equal parts. then

$$m' = \frac{m}{2} \text{ and } l' = \frac{l}{2}; \therefore M' = \frac{m}{2} \times \frac{l}{2} = \frac{ml}{4} = \frac{M}{4}$$

$$\text{New moment of inertia } I' = \frac{wl^2}{12} = \frac{\frac{w}{4} \cdot \left(\frac{l}{2}\right)^2}{12} = \frac{1}{16} \cdot \frac{wl^2}{12}$$

Here w is the mass of magnet.

$$\therefore I' = \frac{1}{16} I; \text{ Time period of each part } T' = 2\pi \sqrt{\frac{I'}{M'B_H}}$$

$$= 2\pi \sqrt{\frac{I/16}{(M/4)B_H}} = 2\pi \sqrt{\frac{I}{4MB_H}} = \frac{T}{2}$$

9. (c) $T = 2\pi \sqrt{\frac{I_1 + I_2}{(M_1 - M_2)B_H}}$

Here $M_1 = M_2 = M, \therefore T = \infty$

10. (b) Time period in vibration magnetometer

$$T = 2\pi \sqrt{\frac{I}{MB_H}}, \text{ At poles } B_H = 0 \text{ so } T = \infty$$

11. (c)

12. (a) $\frac{T_1}{T_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{4M}{M}} = 2 \Rightarrow \frac{2}{T_2} = 2 \Rightarrow T_2 = 1 \text{ sec}$

13. (a) $T = 2\pi \sqrt{\frac{I}{MB_H}}$

$$I = 40 \text{ gm-cm}^2 = 400 \times 10^{-8} \text{ kg-m}^2$$

$$\therefore 3 = 2\pi \sqrt{\frac{400 \times 10^{-8}}{36 \times 10^{-6} \times M}}$$

$$\Rightarrow \frac{1}{M} = \frac{9}{4\pi^2} \times \frac{36}{4} \Rightarrow M = 0.5 \text{ A} \times \text{m}^2$$

14. (a)

15. (b) Because moment of inertia increases i.e. $T \propto \sqrt{I}$

16. (b) $T = 2\pi \sqrt{\frac{I}{MB_H}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{(B_H)_2}{(B_H)_1}}$

$$\Rightarrow T_2 = T \sqrt{\frac{(B_H)_1}{(B_H)_2}} = \frac{T}{2} \quad (\because (B_H)_2 = 4(B_H)_1)$$

17. (b) In sum position $T \propto \frac{1}{\sqrt{M_1 + M_2}}$ and in difference position

$$T \propto \frac{1}{\sqrt{M_1 - M_2}}$$

$$\Rightarrow \frac{3^2}{T^2} = \frac{2M - M}{2M + M} \Rightarrow T^2 = 9 \times 3 \text{ sec}^2$$

$$\therefore T = 3\sqrt{3} \text{ sec}$$

18. (d)

19. (b) Given $v_1 = \frac{20}{60} = \frac{1}{3} \text{ sec}^{-1}$ and $v_2 = \frac{15}{60} = \frac{1}{4} \text{ sec}^{-1}$

$$\text{Now } v = \frac{1}{2\pi} \sqrt{\frac{MB_H}{I}} = \frac{1}{2\pi} \sqrt{\frac{MB \cos \phi}{I}} \quad (\because B_H = B \cos \phi)$$

$$\therefore \frac{v_1}{v_2} = \sqrt{\frac{B_1 \cos \phi_1}{B_2 \cos \phi_2}} \Rightarrow \frac{B_1}{B_2} = \left(\frac{v_1}{v_2} \right)^2 \left(\frac{\cos \phi_2}{\cos \phi_1} \right)^2$$

$$\Rightarrow \frac{B_1}{B_2} = \left(\frac{1/3}{1/4} \right)^2 \frac{\cos 60^\circ}{\cos 30^\circ} = \frac{16}{9} \times \frac{1/2}{\sqrt{3}/2} = \frac{16}{9\sqrt{3}}$$

20. (c) $T \propto \frac{1}{\sqrt{M}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{M_2}{M_1}} \Rightarrow \frac{1.5}{T_2} = \sqrt{\frac{M_1/4}{M_1}} = \frac{1}{2}$

$$\Rightarrow T_2 = 3 \text{ sec}$$

21. (c) $v = \frac{1}{2\pi} \sqrt{\frac{MB_H}{I}} \Rightarrow v \propto \sqrt{M}$

$$\Rightarrow \frac{v_A}{v_B} = \sqrt{\frac{M_A}{M_B}} \Rightarrow \frac{2}{1} = \sqrt{\frac{M_A}{M_B}} \Rightarrow M_A = 4M_B$$

22. (c) $T = 2\pi\sqrt{\frac{I}{MB_H}} \Rightarrow T \propto \frac{1}{\sqrt{M}} \Rightarrow \frac{M_A}{M_B} = \left(\frac{T_B}{T_A}\right)^2 = \frac{4}{1}$

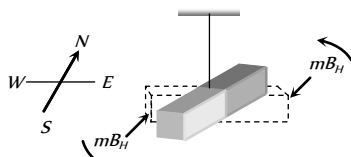
23. (c) No. of oscillation per minute $= \frac{1}{2\pi} \sqrt{\frac{MB_H}{I}}$

$$\Rightarrow n \propto \sqrt{MB_H}; M \rightarrow 4 \text{ times}$$

$$B_H \rightarrow 2 \text{ times}$$

$$\text{So } v \rightarrow \sqrt{8} \text{ times i.e. } v' = \sqrt{8}v = 2\sqrt{2}n$$

24. (c)



25. (a) $T = 2\pi\sqrt{\frac{I}{MH}} \Rightarrow T \propto \frac{1}{\sqrt{H}} \Rightarrow \frac{T_A}{T_B} = \sqrt{\frac{H_B}{H_A}}$

$$\Rightarrow \frac{H_A}{H_B} = \left(\frac{T_B}{T_A}\right)^2 = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$$

26. (a) $T = 2\pi\sqrt{\frac{I}{MB_H}}$ and $I = \frac{w(l^2 + b^2)}{12}$; $\therefore T \propto \sqrt{w}$

(w = Mass of the magnet)

27. (c) $T_{\text{Sum}} = 2\pi\sqrt{\frac{(I_1 + I_2)}{(M_1 + M_2)B_H}}$

$$T_{\text{diff}} = 2\pi\sqrt{\frac{I_1 + I_2}{(M_1 - M_2)B_H}}$$

$$\Rightarrow \frac{T_s}{T_d} = \frac{T_1}{T_2} = \sqrt{\frac{M_1 - M_2}{M_1 + M_2}} = \sqrt{\frac{2M - M}{2M + M}} = \frac{1}{\sqrt{3}}$$

28. (a) $T = 2\pi\sqrt{\frac{I}{MB}} \Rightarrow \frac{T}{T'} = \sqrt{\frac{B'}{B}} = \sqrt{\frac{B}{B_H}}$

$$\Rightarrow \frac{T}{T'} = \sqrt{\frac{1}{\cos \phi}} = \sqrt{\frac{1}{\cos 60^\circ}} = \sqrt{2} \Rightarrow T' = \frac{T}{\sqrt{2}}$$

29. (a)

30. (a)

31. (b) For tangent galvanometer $I = \frac{2rB}{\mu_0 n} \tan \theta$

$$\therefore \tan \theta = \frac{I\mu_0 n}{2rB} = \frac{0.1 \times 4\pi \times 10^{-7} \times 50}{0.04 \times 7 \times 10^{-5} \times 2} = 1.12$$

$$\text{or } \theta = \tan^{-1}(1.12) = 48.2^\circ$$

32. (d) Time period of a magnet $T = 2\pi\sqrt{\frac{I}{MB}}$

$$\text{or } I = \frac{T^2 MB}{4\pi^2} = \frac{225 \times 5 \times 10^{-5} \times 8\pi \times 10^{-4}}{4\pi^2}$$

$$\therefore I = 7.16 \times 10^{-7} \text{ kg-m}^2$$

33. (b) $T = 2\pi\sqrt{\frac{I}{MB_H}} = 4 \text{ sec}$

When magnet is cut into two equal halves, then New magnetic

$$\text{moment } M' = \frac{M}{2}$$

$$\text{New moment of inertia } I' = \frac{(w/2)(l/2)^2}{12} = \frac{1}{8} \cdot \frac{wl^2}{12}$$

Where w is the initial mass of the magnet

$$\text{But } I = \frac{wl^2}{12}; \therefore I' = \frac{I}{8}$$

$$\therefore \text{New time period } T' = 2\pi\sqrt{\frac{I'}{M'B_H}}$$

$$= 2\pi\sqrt{\frac{I/8}{(M/2)B_H}} = \frac{1}{2} 2\pi\sqrt{\frac{I}{MB_H}} = \frac{1}{2} \times T = \frac{1}{2} \times 4 = 2 \text{ sec}$$

34. (d)

35. (a) $K = \frac{2RB_H}{\mu_0 N}$ (R = radius, N = number of turns)

36. (c) $T \propto \frac{1}{\sqrt{M}}$. Since magnetic moment decreases with increase in temperature hence time period T increases.

37. (b) Sensitivity $S = \frac{\theta}{i} = \frac{\theta}{K \tan \theta}$ where $K = \frac{2RB_H}{\mu_0 N}$

For increasing sensitivity K should be decreased and hence number of turns should be increased.

38. (d) In the first galvanometer

$$i_1 = K_1 \tan \theta_1 = K_1 \tan 60^\circ = K_1 \sqrt{3}$$

In the second galvanometer

$$i_2 = K_2 \tan \theta_2 = K_2 \tan 45^\circ = K_2$$

$$\text{In series } i = i_1 \Rightarrow K_1 \sqrt{3} = K_2 \Rightarrow \frac{K_1}{K_2} = \frac{1}{\sqrt{3}}$$

$$\text{But } K \propto \frac{1}{n} \Rightarrow \frac{K_1}{K_2} = \frac{n_2}{n_1} \therefore \frac{n_1}{n_2} = \frac{\sqrt{3}}{1}$$

39. (b) $T = 2\pi\sqrt{\frac{I}{MB_H}}$. If Q is an identical bar magnet then time

$$\text{period of system will be } T' = 2\pi\sqrt{\frac{2I}{(2M)B_H}} = T$$

40. (a) Frequency $\nu \propto \sqrt{B_H}$

41. (b) In tangent galvanometer, $I \propto \tan \theta$

$$\therefore \frac{I_1}{I_2} = \frac{\tan \theta_1}{\tan \theta_2} \Rightarrow \frac{I_1}{I_1 / \sqrt{3}} = \frac{\tan 45^\circ}{\tan \theta_2}$$

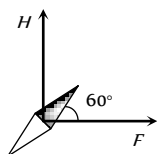
$$\Rightarrow \sqrt{3} \tan \theta_2 = 1 \Rightarrow \tan \theta_2 = \frac{1}{\sqrt{3}} \Rightarrow \theta_2 = 30^\circ$$

So deflection will decrease by $45^\circ - 30^\circ = 15^\circ$.

42. (d) From figure at equilibrium

$$\tan 60^\circ = \frac{H}{F}$$

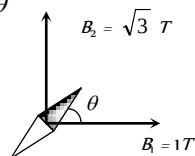
$$\Rightarrow \sqrt{3} = \frac{H}{F} \Rightarrow \frac{F}{H} = \frac{1}{\sqrt{3}}$$



43. (d) In balance condition $B_2 = B_1 \tan \theta$

$$\Rightarrow \tan \theta = \frac{\sqrt{3}}{1}$$

$$\Rightarrow \theta = 60^\circ$$



44. (d) In the sum and difference method of vibration magnetometer

$$\frac{M_1}{M_2} = \frac{T_2^2 + T_1^2}{T_2^2 - T_1^2}$$

$$\text{Here } T_1 = \frac{1}{n_1} = \frac{60}{12} = 5 \text{ sec. } T_2 = \frac{1}{n_2} = \frac{60}{4} = 15 \text{ sec}$$

$$\therefore \frac{M_1}{M_2} = \frac{15^2 + 5^2}{15^2 - 5^2} = \frac{225 + 25}{225 - 25} = \frac{5}{4}$$

45. (c)

46. (b) $i \propto \tan \theta \Rightarrow \frac{i_1}{i_2} = \frac{\tan \theta_1}{\tan \theta_2} \Rightarrow \frac{\sqrt{3}}{3} = \frac{\tan 30^\circ}{\tan \theta_2} \Rightarrow \theta = 45^\circ$

47. (b) $T = 2\pi \sqrt{\frac{I}{MB}} = 2\pi \sqrt{\frac{wl^2 / 12}{\text{Pole strength} \times 2l \times B}}$

$$\therefore T \propto \sqrt{wl}$$

$$\therefore \frac{T_2}{T_1} = \sqrt{\frac{w_2 \times l_2}{w_1 \times l_1}} = \sqrt{\frac{w_1 / 2 \times l_1 / 2}{w_1 \times l_1}} = \frac{1}{2}$$

$$\Rightarrow T_2 = \frac{T_1}{2} = 0.5 \text{ sec}$$

48. (d) $T' = \frac{T}{n} \Rightarrow T' = \frac{2}{2} = 1 \text{ sec}$

49. (c) It is due to the magnetic field produced by coil.

50. (d)

51. (c) $T = 2\pi \sqrt{\frac{I}{MB_H}} \Rightarrow T \propto \sqrt{I} \propto \sqrt{w} \Rightarrow T' = \sqrt{2} T_0$

52. (d) $\frac{M_1}{M_2} = \left(\frac{d_1}{d_2}\right)^3 \Rightarrow \frac{27}{8} = \left(\frac{d_1}{0.12}\right)^3$

$$\Rightarrow \frac{3}{2} = \frac{d_1}{0.12} \Rightarrow 0.18 \text{ m}$$

53. (c) $T = 2\pi \sqrt{\frac{I}{MB_H}} \Rightarrow T \propto \frac{1}{\sqrt{M}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{M_2}{M_1}}$

If $M_1 = 100$ than $M_2 (100 - 19) = 81$

$$\text{So } \frac{T_1}{T_2} = \sqrt{\frac{81}{100}} = \frac{9}{10} \Rightarrow T_2 = \frac{10}{9} T_1 = 1.11 T_1$$

$\Rightarrow$ Time period increases by 11%

54. (b) $T = 2\pi \sqrt{\frac{I}{M \times B_H}} \Rightarrow T \propto \frac{1}{\sqrt{B_H}}$

$$\Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{(B_H)_2}{(B_H)_1}} \Rightarrow \frac{60/40}{2.5} = \sqrt{\frac{(B_H)_2}{0.1 \times 10^{-5}}}$$

$$\Rightarrow (B_H)_2 = 0.36 \times 10^{-6} \text{ T}$$

55. (a) $i = \frac{2rB_H}{\mu_0 N} \tan \theta$

$$\Rightarrow i = \frac{2 \times 15 \times 10^{-2} \times 3 \times 10^{-5}}{4\pi \times 10^{-7} \times 25} \times \tan 45^\circ \Rightarrow i = 0.29 \text{ A}$$

56. (a) $T = 2\pi \sqrt{\frac{I}{MB_H}}; I \rightarrow 3 \text{ times and } M \rightarrow \frac{1}{3} \text{ times}$

So $T \rightarrow 3 \text{ times i.e. } T' = 3T_0$

57. (c) In case of tangent galvanometer as

$$i = k \tan \phi$$

Differentiating both side w.r.t. ϕ

$$\frac{di}{d\phi} = k \sec^2 \phi \Rightarrow di = k \sec^2 \phi d\phi$$

$$\Rightarrow \frac{di}{i} = \frac{d\phi}{\sin \phi \cos \phi} = \frac{2d\phi}{\sin 2\phi}$$

Hence the error in the measurement will be least when

$$\sin 2\phi = \max = 1 \Rightarrow 2\phi = 90^\circ \Rightarrow \phi = 45^\circ$$

58. (a)

59. (a) $T' = \frac{T}{n}$

60. (c) $\frac{T_A}{T_B} = \sqrt{\frac{(B_H)_B}{(B_H)_A}} \Rightarrow \frac{60/10}{60/20} = \sqrt{\frac{(B_H)_B}{36 \times 10^{-6}}}$

$$\Rightarrow (B_H)_B = 144 \times 10^{-6} \text{ T}$$

61. (d) $i \propto \tan \phi \Rightarrow \frac{i_1}{i_2} = \frac{\tan \phi_1}{\tan \phi_2}$

$$\Rightarrow \frac{2}{i_2} = \frac{\tan 30^\circ}{\tan 60^\circ} \Rightarrow i_2 = 6 \text{ amp}$$

62. (a) In tangent galvanometer experiment. The plane of the coil firstly set in the magnetic meridian.

63. (c) $T \propto \frac{1}{\sqrt{M}} \Rightarrow T \propto \frac{1}{\sqrt{m}}$; If $m \rightarrow 4$ times.

$$T \rightarrow \frac{1}{2} \text{ times i.e. } T' = \frac{T}{2} = \frac{2}{2} = 1 \text{ sec}$$

64. (d) $\frac{M_1}{M_2} = \frac{\tan \theta_1}{\tan \theta_2} \Rightarrow \frac{m_1 L_1}{m_2 L_2} = \frac{\tan \theta_1}{\tan \theta_2}$

$$\Rightarrow \frac{m_1}{m_2} = \frac{2}{1} \times \frac{\tan 45^\circ}{\tan 30^\circ} = \frac{2\sqrt{3}}{1}$$

65. (b) $B = B_H \tan \theta = 0.34 \times 10^{-4} \tan 30^\circ = 1.96 \times 10^{-5} T$

66. (b) $i \propto \tan \phi \Rightarrow \frac{i_1}{i_2} = \frac{\tan \phi_1}{\tan \phi_2}$

$$\Rightarrow \frac{0.1}{i_2} = \frac{\tan 30^\circ}{\tan 60^\circ} = \frac{1}{3} \Rightarrow i_2 = 0.3 A$$

67. (a) As $T \propto \sqrt{I}$; where I = moment of inertia

$$= \frac{wL^2}{12} \Rightarrow T \propto \sqrt{w} \quad (w = \text{Mass of magnet. If } w \rightarrow \text{quadrupled, then } T \rightarrow \text{doubled i.e. } T' = 2T)$$

68. (c) Oscillation of n part of magnet $T' = \frac{T}{n}$

$$\Rightarrow \frac{T'}{T} = \frac{1}{n}; \text{ here } n = 2 \text{ so } \frac{T'}{T} = \frac{1}{2}.$$

69. (b) $T = 2\pi \sqrt{\frac{I}{M B_H}}$; where $I = \frac{w(L^2 + b^2)}{12}$

(w = Mass of magnet)

$$\Rightarrow T \propto \sqrt{w}, \text{ If } w \rightarrow \text{four times then } T \rightarrow \text{Two times}$$

70. (b) Initially, the time period of the magnet

$$T = 2 = 2\pi \sqrt{\frac{I}{MB}} \quad \dots (i)$$

For each part, it's moment of inertia $= \frac{I}{27}$ and magnetic

$$\text{moment} = \frac{M}{3}$$

$$\therefore \text{Moment of inertia of system } I_s = \frac{I}{27} \times 3 = \frac{I}{9}$$

$$\text{Magnetic moment of system } M_s = \frac{M}{3} \times 3 = M$$

Time period of system

$$T_s = 2\pi \sqrt{\frac{I_s}{M_s B}} = \frac{1}{3} \times 2\pi \sqrt{\frac{I}{MB}} = \frac{T}{3} = \frac{2}{3} \text{ sec}$$

71. (c) $T \propto \frac{1}{\sqrt{B_H}} = \frac{1}{\sqrt{B \cos \phi}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{B_2 \cos \phi_2}{B_1 \cos \phi_1}}$

$$\Rightarrow \frac{B_1}{B_2} = \frac{T_2^2}{T_1^2} \times \frac{\cos \phi_2}{\cos \phi_1} = \left(\frac{3}{2}\right)^2 \times \frac{\cos 60^\circ}{\cos 30^\circ} \Rightarrow \frac{B_1}{B_2} = \frac{9}{4\sqrt{3}}$$

72. (c) Time period of combination

$$T = 2\pi \sqrt{\frac{2I}{\sqrt{2} M.H}} \quad \dots (i)$$

and time period of each magnet

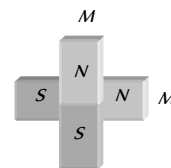
$$T' = 2\pi \sqrt{\frac{I}{MH}} \quad \dots (ii)$$

from (i) and (ii) we get

$$T' = \frac{T}{2^{1/4}} = 2^{-1/4} T$$

73. (b) $B = B_H \tan \theta \Rightarrow \frac{\mu_0 n i}{2r} = B_H \tan \theta$

$$\Rightarrow i = \frac{2r \cdot B_H \tan \theta}{\mu_0 n} = \frac{2 \times 0.1 \times 4 \times 10^{-5}}{10 \times 4\pi \times 10^{-7}} = 1.1 A$$



Magnetic Materials

- (c)
- (a) Neon atom is diamagnetic, hence it's net magnetic moment is zero.
- (a) Soft iron is highly ferromagnetic.
- (d)
- (b) On heating, different domains have net magnetisation in them which are randomly distributes. Thus the net magnetisation of the substance due to various domains decreases to minimum.
- (b) Repelled due to induction of similar poles.
- (d) From the characteristic of B - H curve.
- (c)
- (d)
- (b)
- (a)
- (c) The property of paramagnetism is found in these substances whose atoms have an excess of electrons spinning in the same direction. Hence atoms of paramagnetic substances have a net non-zero magnetic moment of their own.
- (c)
- (a)
- (a, b)
- (d) $\chi_m = (\mu_r - 1) \Rightarrow \chi_m = (5500 - 1) = 5499$
- (b)
- (b)
- (c)
- (b) Because, diamagnetic substance, moves from stronger magnetic field to weaker field.
- (d)

22. (b) With rise in temperature their magnetic susceptibility decreases

$$\text{i.e. } \chi_m \propto \frac{1}{T}$$

23. (c)

24. (b)

25. (c) Diamagnetic substances are repelled by magnetic field.

26. (b) As we know for circulating electron magnetic moment

$$M = \frac{1}{2} e v r \quad \dots\dots (i)$$

and angular momentum $J = m v r \quad \dots\dots (ii)$

$$\text{From equation (i) and (ii) } M = \frac{eJ}{2m}$$

27. (b)

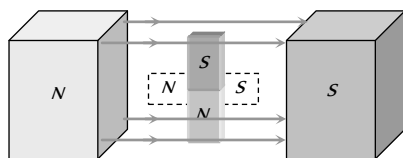
28. (a)

29. (c) The energy lost per unit volume of a substance in a complete cycle of magnetisation is equal to the area of the hysteresis loop.

30. (c)

31. (d)

32. (a) A diamagnetic rod set itself perpendicular to the field if free to rotate between the poles of a magnet as in this situation the field is strongest near the poles.



33. (c)

34. (b)

35. (b) Diamagnetic substances are repelled by the magnetic field.

36. (d)

37. (b)

38. (b)

39. (c)

40. (b)

41. (d) Net magnetic induction $B = B_0 + B_m = \mu_0 H + \mu_0 M$

42. (c)

43. (d) $\mu_r = \frac{B}{B_0} = 4$

44. (d)

45. (a)

46. (b)

47. (c)

48. (c) Susceptibility of diamagnetic substance is negative and it does not change with temperature.

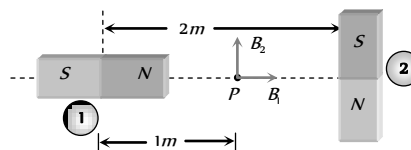
49. (a)

50. (d) When a ferromagnetic material is heated above its Curie temperature then it behaves like paramagnetic material.

Critical Thinking Questions

1. (b) With respect to 1st magnet, P lies in end side-on position

$$\therefore B_1 = \frac{\mu_0}{4\pi} \left(\frac{2M}{d^3} \right) \quad (\text{RHS})$$



With respect to 2nd magnet, P lies in broad side on position.

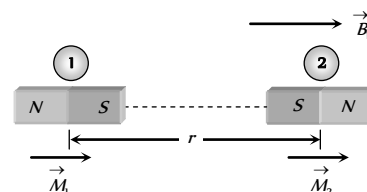
$$\therefore B_2 = \frac{\mu_0}{4\pi} \left(\frac{M}{d^3} \right) \quad (\text{Upward})$$

$$B_1 = 10^{-7} \times \frac{2 \times 1}{1} = 2 \times 10^{-7} \text{ T}, \quad B_2 = \frac{B_1}{2} = 10^{-7} \text{ T}$$

As B_1 and B_2 are mutually perpendicular, hence the resultant magnetic field

$$B_R = \sqrt{B_1^2 + B_2^2} = \sqrt{(2 \times 10^{-7})^2 + (10^{-7})^2} = \sqrt{5} \times 10^{-7} \text{ T}$$

2. (d)



Both the magnets are placed in the field of one another, hence potential energy of dipole (2) is

$$U_2 = -M_2 B_1 \cos 0 = -M_2 B_1 = M_2 \times \frac{\mu_0}{4\pi} \cdot \frac{2M_1}{r^3}$$

By using $F = -\frac{dU}{dr}$, Force on magnet (2) is

$$F_2 = -\frac{dU_2}{dr} = -\frac{d}{dr} \left(\frac{\mu_0}{4\pi} \cdot \frac{2M_1 M_2}{r^3} \right) = -\frac{\mu_0}{4\pi} \cdot 6 \frac{M_1 M_2}{r^4}$$

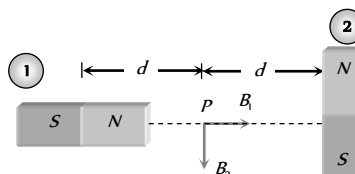
$$\text{It can be proved } |F_1| = |F_2| = F = \frac{\mu_0}{4\pi} \cdot \frac{6M_1 M_2}{r^4}$$

$$\Rightarrow F \propto \frac{1}{r^4}$$

3. (d) At point P net magnetic field $B_{\text{net}} = \sqrt{B_1^2 + B_2^2}$

$$\text{where } B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2M}{d^3} \text{ and } B_2 = \frac{\mu_0}{4\pi} \cdot \frac{M}{d^3}$$

$$\Rightarrow B_{\text{net}} = \frac{\mu_0}{4\pi} \cdot \frac{\sqrt{5}M}{d^3}$$



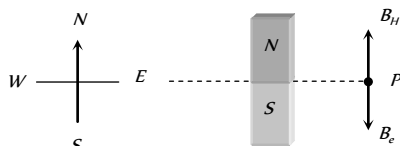
4. (a) Let the real dip be ϕ then $\tan \phi = \frac{B_V}{B_H}$

For apparent dip,

$$\tan \phi' = \frac{B_V}{B_H \cos \beta} = \frac{B_V}{B_H \cos 30^\circ} = \frac{2B_V}{\sqrt{3}B_H}$$

$$\text{or } \tan 45^\circ = \frac{2}{\sqrt{3}} \cdot \tan \phi \text{ or } \phi = \tan^{-1} \left(\frac{\sqrt{3}}{2} \right)$$

5. (d) Initially

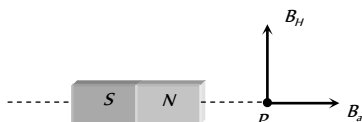


Neutral point obtained on equatorial line and at neutral point

$$|B_H| = |B_e|$$

where B_e = Horizontal component of earth's magnetic field, B_e = Magnetic field due to bar magnet on it's equatorial line

Finally



Point P comes on axial line of the magnet and at P, net magnetic field $B = \sqrt{B_a^2 + B_H^2}$

$$= \sqrt{(2B_e)^2 + (B_H)^2} = \sqrt{(2B_H)^2 + B_H^2} = \sqrt{5} B_H$$

6. (b) $\tan \phi' = \frac{\tan \phi}{\cos \beta}$; where ϕ' = Apparent angle of dip,

ϕ = True angle of dip, β = Angle made by vertical plane with magnetic meridian.

$$\Rightarrow \tan \phi' = \frac{\tan 60^\circ}{\cos 30^\circ} = 2 \Rightarrow \phi' = \tan^{-1}(2)$$

7. (c) Initially magnetic moment of system

$$M_1 = \sqrt{M^2 + M^2} = 2M \text{ and moment of inertia}$$

$$I_1 = I + I = 2I.$$

Finally when one of the magnet is removed then

$$M_2 = M \text{ and } I_2 = I$$

$$\text{So } T = 2\pi \sqrt{\frac{I}{M B_H}}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{I_1 \times M_2}{I_2 \times M_1}} = \sqrt{\frac{2I \times M}{I \times 2M}} \Rightarrow T_2 = \frac{2^{5/4}}{2^{1/4}} = 2 \text{ sec.}$$

8. (b) $T \propto \frac{1}{\sqrt{H}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{H_2}{H_1}} \Rightarrow \frac{2}{1} = \sqrt{\frac{H+F}{H}} \Rightarrow F = 3H$

$$\text{or } \frac{H}{F} = \frac{1}{3}$$

9. (b) Relation for dipole moment is, $M = I \times V$. Volume of the cylinder $V = \pi r^2 l$, Where r is the radius and l is the length of the cylinder, then dipole moment,

$$M = I \pi r^2 l = (5.30 \times 10^3) \times \frac{22}{7} \times (0.5 \times 10^{-2})^2 (5 \times 10^{-2})$$

$$= 2.08 \times 10^{-2} \text{ J/T}$$

10. (b) For equilibrium of the system torques on M and M due to B_e must counter balance each other i.e. $M_1 \times B_H = M_2 \times B_H$. If θ

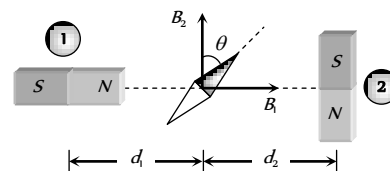
is the angle between M and B_e will be $(90 - \theta)$; so

$$M_1 B_H \sin \theta = M_2 B_H \sin(90 - \theta)$$

$$\Rightarrow \tan \theta = \frac{M_2}{M_1} = \frac{M}{3M} = \frac{1}{3} \Rightarrow \theta = \tan^{-1} \left(\frac{1}{3} \right)$$

11. (a) $I = \frac{M}{V} = \frac{\mu N}{V} = \frac{1.5 \times 10^{-23} \times 2 \times 10^{26}}{1} = 3 \times 10^3 \text{ Amp/m}$

12. (c) In equilibrium $B_1 = B_2 \tan \theta$

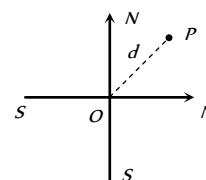


$$\Rightarrow \frac{\mu_0}{4\pi} \cdot \frac{2M}{d_1^3} = \frac{\mu_0}{4\pi} \cdot \frac{M}{d_2^3} \tan \theta$$

$$\Rightarrow \frac{d_1}{d_2} = (2 \cot \theta)^{1/3}$$

13. (c) Resultant magnetic moment of the two magnets is

$$M_{net} = \sqrt{M^2 + M^2} = \sqrt{2}M$$



Imagine a short magnet lying along OP with magnetic moment equal to $M\sqrt{2}$. Thus point P lies on the axial line of the magnet.

$\therefore$ Magnitude of magnetic field at P is given by

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2\sqrt{2}M}{d^3}$$

14. (a) On passing current through the coil, it acts as a magnetic dipole. Torque acting on magnetic dipole is counter balanced by the moment of additional weight about position O . Torque acting on a magnetic dipole

$$\tau = MB \sin \theta = (NiA)B \sin 90^\circ = NiAB.$$

Again $\tau = \text{Force} \times \text{Lever arm} = \Delta mg \times l$

$$\Rightarrow NiAB = \Delta mg l$$

$$\Rightarrow B = \frac{\Delta mg l}{NiA} = \frac{60 \times 10^{-3} \times 9.8 \times 30 \times 10^{-2}}{200 \times 22 \times 10^{-3} \times 1 \times 10^{-4}} = 0.4 \text{ T}$$

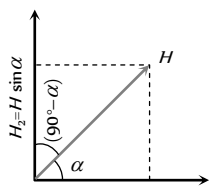
15. (a) The weight of upper magnet should be balanced by the repulsion between the two magnet

$$\therefore \frac{\mu}{4\pi} \cdot \frac{m^2}{r^2} = 50 \text{ gm} - wt$$

$$\Rightarrow 10^{-7} \times \frac{m^2}{(9 \times 10^{-6})} = 50 \times 10^{-3} \times 9.8$$

$$\Rightarrow m = 6.64 \text{ amp} \times m$$

16. (d) Let α be the angle which one of the planes make with the magnetic meridian the other plane makes an angle $(90^\circ - \alpha)$ with it. The components of H in these planes will be $H \cos \alpha$ and $H \sin \alpha$ respectively. If ϕ_1 and ϕ_2 are the apparent dips in these two planes, then



$$\tan \phi_1 = \frac{V}{H \cos \alpha} \quad \text{i.e.} \quad \cos \alpha = \frac{V}{H \tan \phi_1} \quad \dots (i)$$

$$\tan \phi_2 = \frac{V}{H \sin \alpha} \quad \text{i.e.} \quad \sin \alpha = \frac{V}{H \tan \phi_2} \quad \dots (ii)$$

Squaring and adding (i) and (ii), we get

$$\cos^2 \alpha + \sin^2 \alpha = \left(\frac{V}{H} \right)^2 \left(\frac{1}{\tan^2 \phi_1} + \frac{1}{\tan^2 \phi_2} \right)$$

$$\text{i.e. } 1 = \frac{V^2}{H^2} (\cot^2 \phi_1 + \cot^2 \phi_2)$$

$$\text{or } \frac{H^2}{V^2} = \cot^2 \phi_1 + \cot^2 \phi_2 \quad \text{i.e.} \quad \cot^2 \phi = \cot^2 \phi_1 + \cot^2 \phi_2$$

This is the required result.

17. (c) The number of atoms per unit volume in a specimen,

$$n = \frac{\rho N_A}{A}$$

For iron, $\rho = 7.8 \times 10^{-3} \text{ kg m}^{-3}$,

$$N_A = 6.02 \times 10^{26} / \text{kg mol}, \quad A = 56$$

$$\Rightarrow n = \frac{7.8 \times 10^3 \times 6.02 \times 10^{26}}{56} = 8.38 \times 10^{28} \text{ m}^{-3}$$

Total number of atoms in the bar is

$$N_0 = nV = 8.38 \times 10^{28} \times (5 \times 10^{-2} \times 1 \times 10^{-2} \times 1 \times 10^{-2})$$

$$N_0 = 4.19 \times 10^{23}$$

The saturated magnetic moment of bar

$$= 4.19 \times 10^{23} \times 1.8 \times 10^{-23} = 7.54 \text{ Am}^2$$

18. (d) We have, $B = \mu_0 H + \mu_0 I$

$$\text{or } I = \frac{B - \mu_0 H}{\mu_0} \quad \text{or } I = \frac{\mu H - \mu_0 H}{\mu_0} = \left(\frac{\mu}{\mu_0} - 1 \right) H$$

$$I = (\mu_r - 1) H$$

For a solenoid of n -turns per unit length and current i

$$H = ni$$

$$\therefore I = (\mu_r - 1) ni = (1000 - 1) \times 500 \times 0.5$$

$$I = 2.5 \times 10^5 \text{ Am}^{-1}$$

$$\therefore \text{Magnetic moment } M = IV$$

$$M = 2.5 \times 10^5 \times 10^{-4} = 25 \text{ Am}^2$$

19. (d) The bar magnet coercivity $4 \times 10^3 \text{ Am}^{-1}$ i.e., it requires a magnetic intensity $H = 4 \times 10^3 \text{ Am}^{-1}$ to get demagnetised. Let i be the current carried by solenoid having n number of turns per metre length, then by definition $H = ni$. Here

$$n = \frac{N}{l} = \frac{60}{0.12} = 500 \text{ turn metre}$$

$$\Rightarrow i = \frac{H}{n} = \frac{4 \times 10^3}{500} = 8.0 \text{ A}$$

20. (c) Let M and M_l be the magnetic moments of magnets and H the horizontal component of earth's field.

We have $\tau = MH \sin \theta$. If ϕ is the twist of wire, then $\tau = C\phi$, C being restoring couple per unit twist of wire

$$\Rightarrow C\phi = MH \sin \theta$$

$$\text{Here } \phi_1 = (180^\circ - 30^\circ) = 150^\circ = 150 \times \frac{\pi}{180} \text{ rad}$$

$$\phi_2 = (270^\circ - 30^\circ) = 240^\circ = 240 \times \frac{\pi}{180} \text{ rad}$$

So, $C\phi_1 = M_1 H \sin \theta$ (For deflection $\theta = 30^\circ$ of I magnet)

$$C\phi_2 = M_2 H \sin \theta \quad (\text{For deflection } \theta = 30^\circ \text{ of II magnet})$$

$$\text{Dividing } \frac{\phi_1}{\phi_2} = \frac{M_1}{M_2}$$

$$\Rightarrow \frac{M_1}{M_2} = \frac{\phi_1}{\phi_2} = \frac{150 \times \left(\frac{\pi}{180} \right)}{240 \times \left(\frac{\pi}{180} \right)} = \frac{15}{24} = \frac{5}{8}$$

$$\Rightarrow M_1 : M_2 = 5 : 8$$

21. (c) In vertical plane perpendicular to magnetic meridian.

$$T = 2\pi \sqrt{\frac{I}{MB_V}} \quad \dots (i)$$

In horizontal plane $T = 2\pi \sqrt{\frac{I}{MB_H}}$ (ii)

Equation (i) and (ii) gives $B_V = B_H$

Hence by using $\tan \phi = \frac{B_V}{B_H} \Rightarrow \tan \phi = 1 \Rightarrow \phi = 45^\circ$

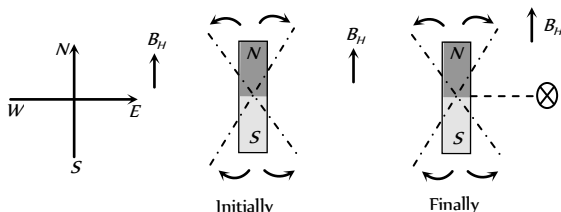
22. (a) Molar susceptibility

$$= \frac{\text{Volume susceptibility}}{\text{Density of material}} \times \text{molecular weight}$$

$$= \frac{I/H}{\rho} \times M = \frac{I/H}{M/V} \times M$$

So it's unit is m .

23. (c)



Initially $T = 2\pi \sqrt{\frac{I}{mB_H}}$, Finally $T' = 2\pi \sqrt{\frac{I}{m(B + B_H)}}$

Where B = Magnetic field due to down ward conductor

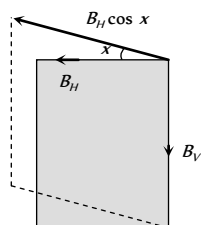
$$= \frac{\mu_0}{4\pi} \cdot \frac{2i}{a} = 18 \mu T$$

$$\therefore \frac{T'}{T} = \sqrt{\frac{B_H}{B + B_H}} \Rightarrow \frac{T'}{0.1} = \frac{24}{18 + 24} \Rightarrow T' = 0.076 \text{ s.}$$

24. (a) In first case $\tan \theta = \frac{B_V}{B_H}$ (i)

Second case $\tan \theta' = \frac{B_V}{B_H \cos x}$ (ii)

From equation (i) and (ii), $\frac{\tan \theta'}{\tan \theta} = \frac{1}{\cos x}$



25. (c) $\tan \theta = \frac{B_V}{B_H}$... (i)

If apparent dip is θ' then

$$\tan \theta' = \frac{B'_V}{B'_H} = \frac{B_V}{B_H \cos 30^\circ} = \frac{B_V}{B_H \times \frac{\sqrt{3}}{2}}$$

$$\Rightarrow \tan \theta' = \left(\frac{2}{\sqrt{3}} \right) \tan \theta \Rightarrow \tan \theta' > \tan \theta \Rightarrow \theta' > \theta$$

Graphical Questions

- (d) For a temporary magnet the hysteresis loop should be long and narrow.
- (c) Magnetism of a magnet falls with rise of temperature and becomes practically zero above curie temperature.
- (b) For a diamagnetic substance χ is small, negative and independent of temperature.
- (a) Susceptibility of a paramagnetic substance is independent of magnetising field.
- (a) Susceptibility of a ferromagnetic substance falls with rise of temperature $\left(\chi = \frac{C}{T - T_c} \right)$ and the substance becomes paramagnetic above curie temperature, so magnetic susceptibility becomes very small above curie temperature.

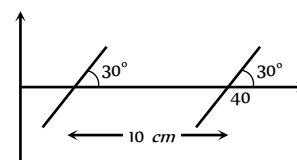
6. (c)

7. (b) $B = \mu_0 \mu_r H \Rightarrow \mu_r \propto \frac{B}{H}$ = slope of $B-H$ curve

According to the given graph, slope of the graph is highest point Q .

8. (b) $i \propto \tan \theta$

9. (b) $|B| = \frac{\Delta V}{\Delta x} = \frac{0.1 \times 10^{-4}}{0.1 \sin 30^\circ} = 2 \times 10^{-4} T$



10. (a) $X = C \times \frac{1}{T} = \frac{0.4}{7 \times 10^{-3}} = 57 K$

11. (b) In the given figure OQ refers to retentivity while OR refers to coercivity, for permanents both retentivity and coercivity should be high.

12. (b) Intensity of magnetisation of diamagnetic substance is very small and negative.

13. (d) $\mu_r = 1 + \frac{I}{H}$; as we know I dependent on H , initially value of $\frac{I}{H}$ is smaller so value of μ increases with H but slowly but

with further increases of H value of $\frac{I}{H}$ also increases i.e. μ increases speedily. When material fully magnetised I becomes constant then with the increase of H ($\frac{I}{H}$ decreases) μ decreases. This is confirm with the option (d).

14. (a) For paramagnetic substance magnetization M proportional to magnetising field H , and M is positive.

Assertion and Reason

1. (d) It is quite clear that magnetic poles always exists in pairs. Since, one can imagine magnetic field configuration with three poles. When north poles or south poles of two magnets are glued together. They provide a three pole field configuration. It is also known that a bar magnet does not exert a torque on itself due to own its field.
2. (b) As we know every atom of a magnet acts as a dipole, So poles cannot be separated. When magnet is broken into two equal pieces, magnetic moment of each part will be half of the original magnet.
3. (a) In case of the electric field of an electric dipole, the electric lines of force originate from positive charge and end at negative charge. Since isolated magnetic lines are closed continuous loops extending through out the body of the magnet.
4. (c) In an atom, electrons revolve around the nucleus and as such the circular orbits of electrons may be considered as the small current loops. In addition to orbital motion, an electron has got spin motion also. So the total magnetic moment of electron is the vector sum of its magnetic moments due to orbital and spin motion. Charge particles at rest do not produce electric field.
5. (b) Magnetic dipole moment of the current loop

$$= \text{Ampere turns} \times \text{Area of the coil}$$

Initially magnetic moment $M = i\pi r^2$, new magnetic moment $M' = i\pi(2r)^2 = 4i(\pi r^2) = 4M$.

So magnetic moment becomes four times when radius is doubled.

6. (e) The temperature inside the earth is so high that it is impossible for iron core to behave as magnet and act as a source of magnetic field. The magnetic field of earth is considered to be due to circulating electric current in the iron (In molten state) and other conducting materials inside the earth.
7. (d) The earth has only vertical component of its magnetic field at the magnetic poles. Since compass needle is only free to rotate in horizontal plane. At north pole the vertical component of earth's field will exert torque on the magnetic needle so as to align it along its direction. As the compass needle can not rotate in vertical plane, it will rest horizontally, when placed on the magnetic pole of the earth.
8. (b) In tangent galvanometer the current through the coil is given by $I = \frac{2r}{n\mu_0} \cdot B_H \tan \theta \Rightarrow \tan \theta \propto n/r$
i.e. by reducing its radius or by increasing number of turns of coil we can increase the sensitivity of tangent galvanometer.
9. (b) The susceptibility of ferromagnetic substance decreases with the rise of temperature in a complicated manner. After Curies point the susceptibility of ferromagnetic substance varies inversely with its absolute temperature. Ferromagnetic substance obey's Curies law only above its Curie point.
10. (e) The properties of substance is due to alignment of molecules in it. When these substance are heated, molecules acquire some kinetic energy. Some of molecules may get back to the closed

chain arrangement (produce zero resultant). So they lose their magnetic property or magnetism. Therefore the properties of both ferromagnetic and paramagnetic are effected by heating.

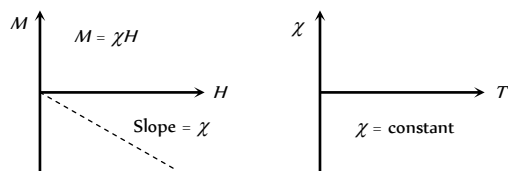
11. (a) The core of a transformer undergoes cycles of magnetisation again and again. During each cycle of magnetisation, energy numerically equal to the area of the hysteresis loop is spent per unit volume of the core. Therefore, for high efficiency of transformer, the energy loss will be lesser if the hysteresis loop is of lesser area, i.e. narrow. That's why the soft iron is used as core, which has narrow hysteresis loop (or area of $B-H$ curve is very small). Also soft iron (ferromagnetic substance) has high permeability, high retentivity, low coercivity and low hysteresis loss.
12. (a) A magnetic field is produced by the motion of electric charge. Since motion is relative, the magnetic field is also relative.
13. (a) In a moving coil galvanometer, the coil is suspended in a very strong uniform magnetic field created by two magnetic pole pieces. The earth's magnetic field is quite weak as compared to that field, therefore, it does not effect the working of magnetic field.
14. (c) A paramagnetic sample display greater magnetisation when cooled, this is because at lower temperature, the tendency to disrupt the alignment of dipoles (due to magnetising field) decreases on account of reduced random thermal motion.
15. (a) Electromagnets are magnets, which can be turned on and off by switching the current on and off. As the material in electromagnets is subjected to cyclic change (magnetisation and demagnetisation), the hysteresis loss of the material must be small. The material should attain high value of I and B with low value of magnetising field intensity H . As soft iron has small coercivity, so it is a best choice for this purpose.
16. (a) Since iron is ferromagnetic in nature, therefore, lines of force due to external magnetic field prefer to pass through iron.
17. (d) In general, the field due to a magnet is non-uniform. Therefore, it exerts both, a net force and a torque on the nails which will translate and also rotate the nails before striking to north pole of magnet with their induced south poles and vice-versa.
18. (d) In a non-uniform magnetic field, both a torque and a net force acts on the dipole. If magnetic field were uniform, net force on dipole would be zero.
19. (c) The reduction factor of tangent galvanometer is

$$K = \frac{B_H}{G} = B_H \times \frac{2r}{n\mu_0}$$

Thus reduction factor of a tangent galvanometer depends upon the geometry of its coil. It increases with increase of radius and decreases with increase in number of turn of the coil of the galvanometer.

20. (c) Diamagnetism is non-cooperative behaviour of orbiting electrons when exposed to an applied magnetic field. Diamagnetic substance are composed of atom which have no net magnetic moment (*i.e.*, all the orbital shells are filled and there are no unpaired electrons). When exposed to a field, a negative magnetization is produced and thus the susceptibility is negative.

Behaviour of diamagnetic material is that the susceptibility is temperature independent.



21. (d) The permeability of a ferromagnetic material is not independent of magnetic field, $\vec{B} = K_m \vec{B}_0$.

B_0 is applied field. The total magnetic field $\vec{B}$ inside a ferromagnet may be 10^3 or 10^4 times the applied field B_0 . The permeability K_m of a ferromagnetic material is not constant, neither the field $\vec{B}$ nor the magnetization $\vec{M}$ increases linearly with $\vec{B}$. Even at small value of B_0 . From the hysteresis curve, magnetic permeability is greater for lower field.

22. (e) For a perfectly diamagnetic substance,

$$B = \mu_0(H + I) = 0 \quad \therefore I = -H.$$

$$\text{Therefore, } \chi_m = \frac{I}{H} = -1$$

Therefore relative permeability

$$\mu_r = 1 + \chi_m = 1 - 1 = 0. \quad \therefore \mu = \mu_0 \mu_r = \text{zero.}$$

i.e. for a perfectly diamagnetic material permeability is zero.

23. (a)
24. (a) Helium atom has paired electrons so their electron spin are opposite to each other and hence it's net magnetic moment is zero.
25. (b) Steel is preferred over soft iron for making permanent magnets, because coercivity of steel is larger.

Magnetism

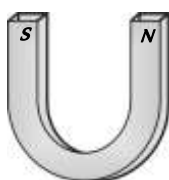
Self Evaluation Test - 22

1. A compass needle whose magnetic moment is $60 \text{ amp} \times m$ pointing geographical north at a certain place, where the horizontal component of earth's magnetic field is $40 \mu \text{ Wb/m}$, experiences a torque $1.2 \times 10^{-3} \text{ N} \times m$. What is the declination at this place

(a) 30° (b) 45°
(c) 60° (d) 25°

2. The distance between the poles of a horse shoe magnet is 0.1 m and its pole strength is 0.01 amp-m . The induction of magnetic field at a point midway between the poles will be

(a) $2 \times 10^{-5} \text{ T}$
(b) $4 \times 10^{-6} \text{ T}$
(c) $8 \times 10^{-7} \text{ T}$
(d) Zero



3. Due to a small magnet intensity at a distance x in the end on position is 9 Gauss. What will be the intensity at a distance $\frac{x}{2}$ on broad side on position

(a) 9 Gauss (b) 4 Gauss
(c) 36 Gauss (d) 4.5 Gauss

4. The magnetic moment produced in a substance of 1 gm is $6 \times 10^{-7} \text{ ampere-metre}^2$. If its density is 5 gm/cm^3 , then the intensity of magnetisation in A/m will be

(a) 8.3×10^6 (b) 3.0
(c) 1.2×10^{-7} (d) 3×10^{-6}

5. The needle of a deflection galvanometer shows a deflection of 60° due to a short bar magnet at a certain distance in $\tan A$ position. If the distance is doubled, the deflection is

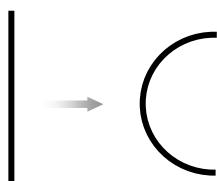
(a) $\sin^{-1}\left(\frac{\sqrt{3}}{8}\right)$ (b) $\cos^{-1}\left(\frac{\sqrt{3}}{8}\right)$
(c) $\tan^{-1}\left(\frac{\sqrt{3}}{8}\right)$ (d) $\cot^{-1}\left(\frac{\sqrt{3}}{8}\right)$

6. The area of hysteresis loop of a material is equivalent to 250 joule . When 10 kg material is magnetised by an alternating field of 50 Hz then energy lost in one hour will be if the density of material is 7.5 gm/cm^3

(a) $6 \times 10^4 \text{ J}$ (b) $6 \times 10^4 \text{ erg}$
(c) $3 \times 10^2 \text{ J}$ (d) $3 \times 10^2 \text{ erg}$

7. A magnetised wire of moment M is bent into an arc of a circle subtending an angle of 60° at the centre; then the new magnetic moment is

(a) $(2M/\pi)$



(b) (M/π)

(c) $(3\sqrt{3}M/\pi)$

(d) $(3M/\pi)$ [IIT JEE (Engg.) 1996]

8. A tangent galvanometer shows a deflection 45° when 10 mA current pass through it. If the horizontal component of the earth's field is $3.6 \times 10^{-5} \text{ T}$ and radius of the coil is 10 cm . The number of turns in the coil is

(a) 5700 turns (b) 57 turns
(c) 570 turns (d) 5.7 turns

9. A magnet is parallel to a uniform magnetic field. If it is rotated by 60° , the work done is 0.8 J . How much work is done in moving it 30° further

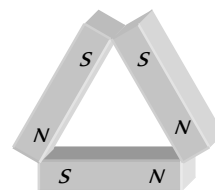
(a) $0.8 \times 10^7 \text{ ergs}$ (b) 0.4 J
(c) 8 J (d) 0.8 ergs

10. Susceptibility of Mg at 300 K is 1.2×10^{-5} . The temperature at which susceptibility will be 1.8×10^{-5} is [Roorkee 1999]

(a) 450 K (b) 200 K
(c) 375 K (d) None of these

11. Three identical bar magnets each of magnetic moment M are placed in the form of an equilateral triangle as shown. The net magnetic moment of the system is

(a) Zero
(b) $2M$
(c) $M\sqrt{3}$
(d) $\frac{3M}{2}$



12. A magnetic needle is placed on a cork floating in a still lake in the northern hemisphere. Does the needle together with the cork move towards the north of the lake

(a) Yes
(b) No
(c) May be or may not be move
(d) Nothing can be said

13. The magnet of vibration magnetometer is heated so as to reduce its magnetic moment by 36% . By doing this the periodic time of the magnetometer will

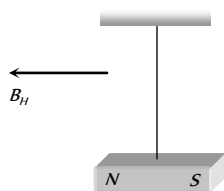
(a) Increases by 36% (b) Increases by 25%
(c) Decreases by 25% (d) Decreases by 64%

14. The ratio of magnetic moments of two bar magnet is $13 : 5$. These magnets are held together in a vibration magnetometer are allowed to oscillate in earth's magnetic field with like poles together 15 oscillation per minute are made. What will be the frequency of oscillation of system if unlike poles are together

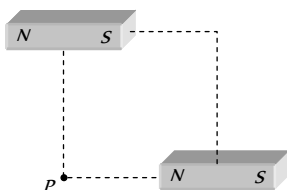
- (a) 10 oscillations/min (b) 15 oscillations/min
(c) 12 oscillations/min (d) $\frac{75}{13}$ oscillations/min

15. A magnet is suspended horizontally in the earth's magnetic field. When it is displaced and then released it oscillates in a horizontal plane with a period T . If a piece of wood of the same moment of inertia (about the axis of rotation) as the magnet is attached to the magnet what would the new period of oscillation of the system become

- (a) $\frac{T}{3}$
(b) $\frac{T}{2}$
(c) $\frac{T}{\sqrt{2}}$
(d) $T\sqrt{2}$



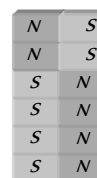
16. Two short magnets of magnetic moment 1000 Am^2 are placed as shown at the corners of a square of side 10 cm . The net magnetic induction at P is



- (a) 0.1 T (b) 0.2 T
(c) 0.3 T (d) 0.4 T

17. The length of a magnet is large compared to its width and breadth. The time period of its oscillation in a vibration magnetometer is T . The magnet is cut along its length into six parts and these parts are then placed together as shown in the figure. The time period of this combination will be

- (a) T
(b) $\frac{T}{\sqrt{3}}$
(c) $\frac{T}{2\sqrt{3}}$
(d) Zero

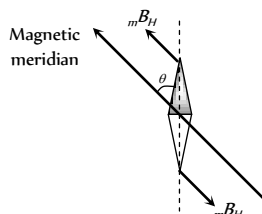


AS Answers and Solutions

(SET -22)

1. (a) As the compass needle is free to rotate in a horizontal plane and points along the magnetic meridian,

so when it is pointing along the geographic meridian, it will experience a torque due



to the horizontal component of earth's magnetic field i.e.
 $\tau = MB_H \sin \theta$

where θ = angle between geographical and magnetic meridians called angle of declination

$$\text{So, } \sin\theta = \frac{1.2 \times 10^{-3}}{60 \times 40 \times 10^{-6}} = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

2. (c) Net magnetic field at mid point P , $B = B_N + B_S$

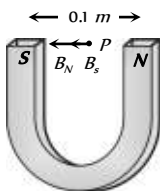
where B_N = magnetic field due to N -pole

B_S = magnetic field due to S -pole

$$B_N = B_S = \frac{\mu_0}{4\pi} \frac{m}{r^2}$$

$$= 10^{-7} \times \frac{0.01}{\left(\frac{0.1}{2}\right)^2} = 4 \times 10^{-7} \text{ T}$$

$$\therefore B_{\text{net}} = 8 \times 10^{-7} \text{ T}$$



3. (c) In C.G.S. $B_{\text{axial}} = 9 = \frac{2M}{x^3}$ (i)

$$B_{\text{equatorial}} = \frac{M}{\left(\frac{x}{2}\right)^3} = \frac{8M}{x^3} \quad \text{.....(ii)}$$

From equation (i) and (ii) $B_{\text{equatorial}} = 36 \text{ Gauss}$.

4. (b) $I = \frac{M}{V} = \frac{M}{\text{mass/density}}$,

given mass = $1 \text{ gm} = 10^{-3} \text{ kg}$,

$$\text{and density} = 5 \text{ gm/cm}^3 = \frac{5 \times 10^{-3} \text{ kg}}{(10^{-2})^3 \text{ m}^3} = 5 \times 10^3 \text{ kg/m}^3$$

$$\text{Hence } I = \frac{6 \times 10^{-7} \times 5 \times 10^3}{10^{-3}} = 3$$

5. (c) For short bar magnet in tan A -position

$$\frac{\mu_0}{4\pi} \frac{2M}{d^3} = H \tan \theta \quad \text{.....(i)}$$

When distance is doubled, then new deflection θ' is given by

$$\frac{\mu_0}{4\pi} \frac{2M}{(2d)^3} = H \tan \theta' \quad \text{.....(ii)}$$

$$\therefore \frac{\tan \theta'}{\tan \theta} = \frac{1}{8} \Rightarrow \tan \theta' = \frac{\tan \theta}{8} = \frac{\tan 60^\circ}{8} = \frac{\sqrt{3}}{8}$$

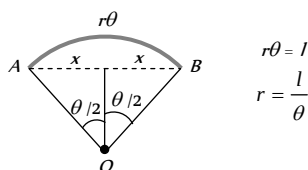
$$\Rightarrow \theta' = \tan^{-1} \left(\frac{\sqrt{3}}{8} \right)$$

6. (a) $E = nAVt = nA \frac{m}{d} t = \frac{50 \times 250 \times 10 \times 3600}{7.5 \times 10^3} = 6 \times 10^4 \text{ J}$

7. (d) From figure

$$\sin \frac{\theta}{2} = \frac{x}{r}$$

$$\Rightarrow x = r \sin \frac{\theta}{2}$$



Hence new magnetic moment $M' = m(2x) = m.2r \sin \frac{\theta}{2}$

$$= m. \frac{2l}{\theta} \sin \frac{\theta}{2} = \frac{2ml \sin \theta / 2}{\theta} = \frac{2M \sin(\pi/6)}{\pi/3} = \frac{3M}{\pi}$$

8. (c) $K = \frac{2rB_H}{\mu_0 n}$

$$\text{or } n = \frac{2rB_H}{\mu_0 K} = \frac{2 \times 0.1 \times 3.6 \times 10^{-5}}{4\pi \times 10^{-7} \times 10 \times 10^{-3}} = \frac{1.8 \times 10^3}{3.14} = 570$$

9. (a) $W = MB(\cos \theta_1 - \cos \theta_2)$

When the magnet is rotated from 0° to 60° , then work done is 0.8 J

$$0.8 = MB(\cos 0^\circ - \cos 60^\circ) = \frac{MB}{2}$$

$$\Rightarrow MB = 1.6 \text{ N-m}$$

In order to rotate the magnet through an angle of 30° , i.e., from 60° to 90° , the work done is

$$W' = MB(\cos 60^\circ - \cos 90^\circ) = MB \left(\frac{1}{2} - 0 \right)$$

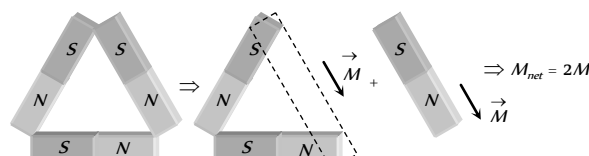
$$= \frac{MB}{2} = \frac{1.6}{2} = 0.8 \text{ J} = 0.8 \times 10^7 \text{ ergs}$$

10. (b) $\chi \propto \frac{1}{T}$

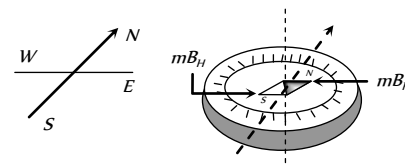
$$\therefore \chi_1 T_1 = \chi_2 T_2$$

$$\text{Hence } T_2 = \frac{1.2 \times 10^{-5} \times 300}{1.8 \times 10^{-5}} = 200 \text{ K}$$

11. (b) The resultant magnetic moment can be calculated as follows.



12. (b) Magnetic needle is a dipole which is in earth's uniform magnetic field and as a dipole in a uniform field does not experience any net force but may experience a couple as shown in figure, so the needle together with the cork will not translate i.e. move towards the north of the lake, but will rotate and set itself parallel to the field with its north pole pointing north.



13. (b) $T = 2\pi \sqrt{\frac{I}{MB_H}} \Rightarrow T \propto \frac{1}{\sqrt{M}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{M_2}{M_1}}$

$$\text{If } M_1 = 100 \text{ then } M_2 = (100 - 36) = 64$$

$$\text{So } \frac{T_1}{T_2} = \sqrt{\frac{64}{100}} = \frac{8}{10} \Rightarrow T_2 = \frac{10}{8} T_1 = 1.25 T_1$$

So % increase in time period = 25%

$$14. \quad (a) \quad \frac{M_1}{M_2} = \frac{\nu_s^2 + \nu_d^2}{\nu_s^2 - \nu_d^2} \Rightarrow \frac{13}{5} = \frac{(15)^2 + \nu_d^2}{(15)^2 - \nu_d^2}$$

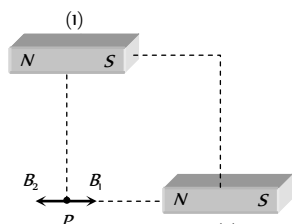
$$\Rightarrow \nu_d = 10 \text{ oscillations/min}$$

$$T' = 2\pi \sqrt{\frac{I/36}{(M/3)H}} = \frac{1}{2\sqrt{3}} 2\pi \sqrt{\frac{I}{MH}} = \frac{T}{2\sqrt{3}}$$

15. (d) Due to wood moment of inertia of the system becomes twice but there is no change in magnetic moment of the system.

$$\text{Hence by using } T = 2\pi \sqrt{\frac{I}{MB_H}} \Rightarrow T \propto \sqrt{I} \Rightarrow T' = \sqrt{2} T$$

16. (a) Point P lies on equatorial line of magnet (1) and axial line of magnet (2) as shown



$$B_1 = \frac{\mu_0}{4\pi} \cdot \frac{M}{d^3} = 10^{-7} \times \frac{1000}{(0.1)^3} = 0.1 \text{ T}$$

$$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2M}{d^3} = 10^{-7} \times \frac{2 \times 1000}{(0.1)^3} = 0.2 \text{ T}$$

$$\therefore B_{\text{net}} = B_2 - B_1 = 0.1 \text{ T}$$

$$17. \quad (c) \quad T = 2\pi \sqrt{\frac{I}{MH}}; \text{ MI of each part} = \frac{I}{6^3}$$

$$\text{and magnetic moment of each part} = \frac{M}{6}$$

$$\text{so net MI of system} = \frac{I}{6^3} \times 6 = \frac{I}{6^2}$$

$$\text{and net magnetic moment} = \frac{4M}{6} - \frac{2M}{6} = \frac{M}{3}$$

$\therefore$ time period of the system



Chapter 23

Electromagnetic Induction

Magnetic Flux

(1) The total number of magnetic lines of force passing normally through an area placed in a magnetic field is equal to the magnetic flux linked with that area.

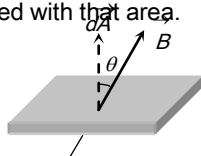


Fig. 23.1

(2) Net flux through the surface $\phi = \int \vec{B} \cdot d\vec{A} = BA \cos \theta$

(θ is the angle between area vector and magnetic field vector)

If $\theta = 0^\circ$ then $\phi = BA$, If $\theta = 90^\circ$ then $\phi = 0$

(3) **Unit and Dimension** : Magnetic flux is a scalar quantity. It's S.I. unit is *weber (wb)*, CGS unit is *Maxwell* or *Gauss* $\times cm^2$; ($1wb = 10^8 Maxwell$).

(4) Other units : *Tesla* $\times m^2$

$$= \frac{N \times m}{Amp} = \frac{Joule}{Amp} = \frac{Volt \times Coulomb}{Amp}$$

$= Volt \times sec = Ohm \times Coulomb = Henry \times Amp$. It's dimensional formula $[\phi] = [ML^2 T^{-2} A^{-1}]$

Faraday's Laws of Electromagnetic Induction

(1) **First law** : Whenever the number of magnetic lines of force (magnetic flux) passing through a circuit changes an emf

is produced in the circuit called induced emf. The induced emf persists only as long as there is change or cutting of flux.

(2) **Second law** : The induced emf is given by rate of change of magnetic flux linked with the circuit i.e. $e = -\frac{d\phi}{dt}$. For N turns $e = -\frac{N d\phi}{dt}$; Negative sign indicates that induced emf (e) opposes the change of flux.

(3) **Other formulae** : $\phi = BA \cos \theta$; Hence ϕ will change if either, B , A or θ will change

$$\begin{aligned} \text{So } e &= -N \frac{d\phi}{dt} = -\frac{N(\phi_2 - \phi_1)}{\Delta t} = -\frac{NA(B_2 - B_1) \cos \theta}{\Delta t} \\ &= -\frac{NBA(\cos \theta_2 - \cos \theta_1)}{\Delta t} \end{aligned}$$

Table 23.1 : Induced i , q and P

Induced current (i)	Induced charge (q)	Induced power (P)
$i = \frac{e}{R} = -\frac{N}{R} \cdot \frac{d\phi}{dt}$	$dq = i dt = -\frac{N}{R} \cdot d\phi$	$P = \frac{e^2}{R} = \frac{N^2}{R} \left(\frac{d\phi}{dt} \right)^2$
	Induced charge is time independent.	It depends on time and resistance

Lenz's Law

This law gives the direction of induced emf/induced current. According to this law, the direction of induced emf or current in a circuit is such as to oppose the cause that produces it. This law is based upon law of conservation of energy.

2 Electromagnetic Induction

(1) When N -pole of a bar magnet moves towards the coil, the flux associated with loop increases and an emf is induced in it. Since the circuit of loop is closed, induced current also flows in it.

(2) Cause of this induced current, is approach of north pole and therefore to oppose the cause, *i.e.*, to repel the approaching north pole, the induced current in loop is in such a direction so that the front face of loop behaves as north pole. Therefore

induced current as seen by observer O is in anticlockwise direction. (figure)

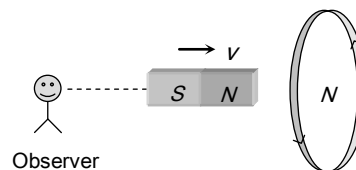


Fig. 23.2

Table 23.2 : The various positions of relative motion between the magnet and the coil

Position of magnet				
Direction of induced current	Anticlockwise direction	Clockwise direction	Clockwise direction	Anticlockwise direction
Behaviour of face of the coil	As a north pole	As a south pole	As a south pole	As a north pole
Type of magnetic force opposed	Repulsive force	Attractive force	Repulsive force	Attractive force
Magnetic field linked with the coil and it's progress as viewed from left	Cross ($\times$), Increases	Cross ($\times$), Decreases	Dots ($\cdot$) Increases	Dots ($\cdot$) Decreases

(3) If the loop is free to move the cause of induced emf in the coil can also be termed as relative motion. Therefore to oppose the cause, the relative motion between the approaching magnet and the loop should be opposed. For this, the loop will itself start moving in the direction of motion of the magnet.

(4) It is important to remember that whenever cause of induced emf is relative motion, the new motion is always in the direction of motion of the cause.

Induced Electric Field

It is non-conservative and non-electrostatic in nature. Its field lines are concentric circular closed curves.

A time varying magnetic field $\frac{dB}{dt}$ always produced induced electric field in all space surrounding it.

Induced electric field (E_{in}) is directly proportional to induced emf so $e = \oint \vec{E}_{in} \cdot d\vec{l}$ (i)

From Faraday's second laws $e = -\frac{d\phi}{dt}$ (ii)

From (i) and (ii) $e = \oint \vec{E}_{in} \cdot d\vec{l} = -\frac{d\phi}{dt}$ This is known as integral form of Faraday's laws of EMI.

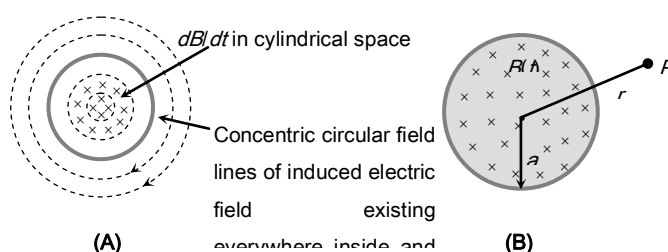


Fig. 23.3

A uniform but time varying magnetic field $B(t)$ exists in a circular region of radius ' a ' and is directed into the plane of the paper as shown, the magnitude of the induced electric field (E_{in}) at point P lies at a distance r from the centre of the circular region is calculated as follows.

$$\text{So } \oint \vec{E}_{in} d\vec{l} = e = \frac{d\phi}{dt} = A \frac{dB}{dt} \text{ i.e. } E(2\pi r) = \pi a^2 \frac{dB}{dt}$$

$$\text{where } r \geq a \text{ or } E = \frac{a^2}{2r} \frac{dB}{dt}; \quad E_{in} \propto \frac{1}{r}$$

Dynamic (Motional) EMI Due to Translatory Motion

(1) Consider a conducting rod of length l moving with a uniform velocity $\vec{v}$ perpendicular to a uniform magnetic field $\vec{B}$, directed into the plane of the paper. Let the rod be moving to the right as shown in figure. The conducting electrons also move to the right as they are trapped within the rod.

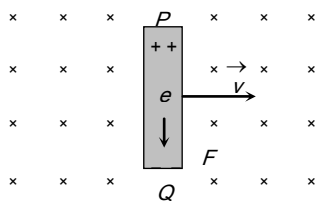


Fig. 23.4

Conducting electrons experience a magnetic force $F_m = evB$. So they move from P to Q within the rod. The end P of the rod becomes positively charged while end Q becomes negatively charged, hence an electric field is set up within the rod which opposes the further downward movement of electrons i.e. an equilibrium is reached and in equilibrium $F_e = F_m$ i.e. $eE = evB$ or $E = vB \Rightarrow$ Induced emf $e = El = Bvl$ [$E = \frac{V}{l}$]

(2) If rod is moving by making an angle θ with the direction of magnetic field or length. Induced emf $e = Bv/\sin\theta$

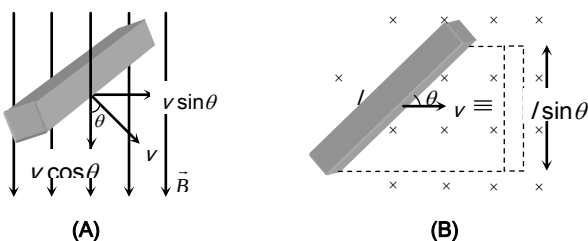


Fig. 23.5

(3) **Motion of conducting rod on an inclined plane :** When conductor starts sliding from the top of an inclined plane as shown, it moves perpendicular to its length but at an angle $(90 - \theta)$ with the direction of magnetic field.

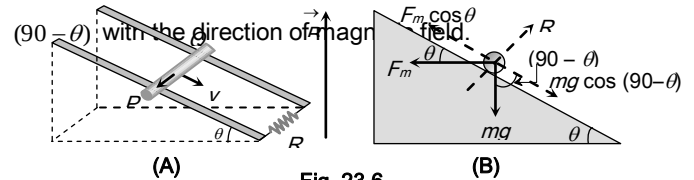


Fig. 23.6

Hence induced emf across the ends of conductor $e = Bv \sin(90 - \theta)l = Bvl \cos\theta$

$$\text{So induced current } i = \frac{Bvl \cos\theta}{R} \text{ (Directed from } Q \text{ to } P).$$

The forces acting on the bar are shown in the following figure. The rod will move down with constant velocity only if

$$F_m \cos\theta = mg \cos(90 - \theta) = mg \sin\theta \Rightarrow Bil \cos\theta = mg \sin\theta$$

$$B \left(\frac{Bv_T l \cos\theta}{R} \right) l \cos\theta = mg \sin\theta \Rightarrow v_T = \frac{mgR \sin\theta}{B^2 l^2 \cos^2 \theta}$$

Motional Emf in Loop by Generated Area

If a conducting rod moves on two parallel conducting rails as shown in the following figure, then the phenomenon of induced emf can also be understood by the concept of generated area (The area swept out by the conductor in magnetic field during its motion)

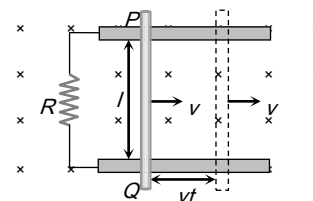


Fig. 23.7

As shown in the figure, in time t the distance travelled by the conductor is vt

Area generated $A = lvt$. Flux linked with this area $\phi = BA =$

$$Blt. \text{ Hence induced emf } |e| = \frac{d\phi}{dt} = Bvl$$

(1) **Induced current** : $i = \frac{e}{R} = \frac{Bvl}{R}$

(2) **Magnetic force** : Conductor PQ experiences a magnetic force in opposite direction of its motion and

$$F_m = Bil = B \left(\frac{Bvl}{R} \right) l = \frac{B^2 v l^2}{R}$$

(3) **Power dissipated in moving the conductor** : For uniform motion of rod PQ , the rate of doing mechanical work by external agent or mech. Power delivered by external source is given as

$$P_{mech} = P_{ext} = \frac{dW}{dt} = F_{ext} \cdot v = \frac{B^2 v l^2}{R} \times v = \frac{B^2 v^2 l^2}{R}$$

(4) **Electrical power** : Also electrical power dissipated in resistance or rate of heat dissipation across resistance is given as

$$P_{thermal} = \frac{H}{t} = i^2 R = \left(\frac{Bvl}{R} \right)^2 R ; \quad P_{thermal} = \frac{B^2 v^2 l^2}{R}$$

(It is clear that $P_{mech} = P_{thermal}$ which is consistent with the principle of conservation of energy.)

(5) **Motion of conductor rod in a vertical plane** : If conducting rod released from rest (at $t = 0$) as shown in figure then with rise in its speed (v), induces emf (e), induced current (i), magnetic force (F_m) increases but its weight remains constant.

Rod will achieve a constant maximum (terminal) velocity v_T if $F_m = mg$

$$\text{So } \frac{B^2 v_T^2 l^2}{R} = mg$$

$$\Rightarrow v_T = \frac{mgR}{B^2 l^2}$$

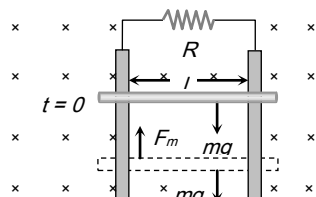


Fig. 23.8

Special cases

Motion of train and aeroplane in earth's magnetic field



(A)



(B)

Fig. 23.9

Induced emf across the axle of the wheels of the train and it is across the tips of the wing of the aeroplane is given by $e =$

$B_v l v$ where $l =$ length of the axle or distance between the tips of the wings of plane, $B_v =$ vertical component of earth's magnetic field and $v =$ speed of train or plane.

Motional EMI Due to Rotational Motion

(1) **Conducting rod** : A conducting rod of length l whose one end is fixed, is rotated about the axis passing through its fixed end and perpendicular to its length with constant angular velocity ω . Magnetic field (B) is perpendicular to the plane of the paper.

emf induces across the ends of the rod

where $\nu =$ frequency (revolution per sec) and $T =$

Time period.

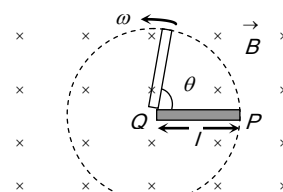


Fig. 23.10

(2) **Cycle wheel** : A conducting wheel each spoke of length l is rotating with angular velocity ω in a given magnetic field as shown below in fig.

Due to flux cutting each metal spoke becomes identical cell of emf e (say), all such identical cells connected in parallel fashion $e_{net} = e$ (emf of single cell). Let N be the number of spokes hence

$$e_{net} = \frac{1}{2} B \omega l^2 ; \quad \omega = 2\pi\nu$$

Here $e_{net} \propto N^0$ i.e. total emf does not depend on number of spokes ' N '.

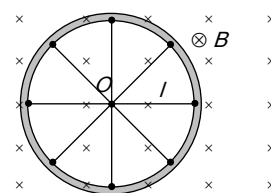


Fig. 23.11

(3) Faraday copper disc

generator : A metal disc can be assumed to be made of uncountable radial conductors when metal disc rotates in transverse magnetic field these radial conductors cut away magnetic field lines and

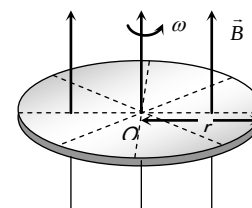


Fig. 23.12

because of this flux cutting all becomes identical cells each of emf ' e ' where $e = \frac{1}{2} B \omega r^2$,

(4) **Semicircular conducting loop** : If a semi-circular conducting loop (ACD) of radius ' r ' with centre at O , the plane of loop being in the plane of paper. The loop is now made to rotate with a constant angular velocity ω , about an axis passing through O and perpendicular to the plane of paper. The effective resistance of the loop is R .

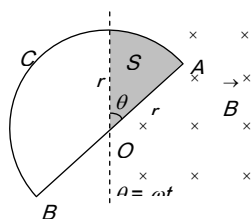


Fig. 23.13

In time t the area swept by the loop in the field *i.e.* region II

$$A = \frac{1}{2}r(r\theta) = \frac{1}{2}r^2\omega t; \quad \frac{dA}{dt} = \frac{r^2\omega}{2}$$

Flux link with the rotating loop at time t $\phi = BA$

Hence induced emf in the loop in magnitude

$$|e| = \frac{d\phi}{dt} = B \frac{dA}{dt} = \frac{B\omega r^2}{2} \text{ and induced current } i = \frac{|e|}{R} = \frac{B\omega r^2}{2R}$$

Periodic EMI

Suppose a rectangular coil having N turns placed initially in a magnetic field such that magnetic field is perpendicular to its plane as shown.

ω – Angular speed

ν – Frequency of rotation of coil

R – Resistance of coil

For uniform rotational motion

with ω , the flux linked with coil at any time t

$$\phi = NBA \cos \theta = NBA \cos \omega t$$

$$\phi = \phi_0 \cos \omega t \text{ where } \phi_0 = NBA = \text{maximum flux}$$

(1) **Induced emf in coil** : Induced emf also changes in periodic manner that's why this phenomenon called periodic EMI

$$e = -\frac{d\phi}{dt} = NBA\omega \sin \omega t \Rightarrow e = e_0 \sin \omega t \text{ where } e_0 = \text{emf}$$

amplitude or max. emf $= NBA\omega = \phi_0\omega$

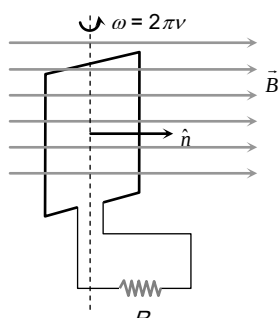


Fig. 23.14

(2) **Induced current** : At any time t

$$i = \frac{e}{R} = \frac{e_0}{R} \sin \omega t = i_0 \sin \omega t \text{ where } i_0 = \text{current amplitude or max.}$$

$$\text{current } i_0 = \frac{e_0}{R} = \frac{NBA\omega}{R} = \frac{\phi_0\omega}{R}$$

Inductance

(1) Inductance is that property of electrical circuits which opposes any change in the current in the circuit.

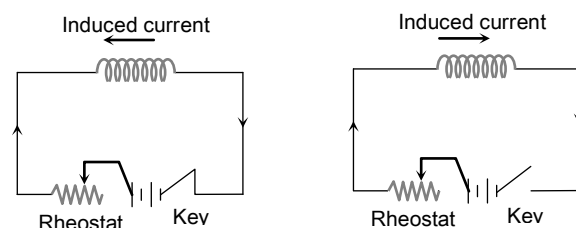
(2) Inductance is inherent property of electrical circuits. It will always be found in an electrical circuit whether we want it or not.

(3) A straight wire carrying current with no iron part in the circuit will have lesser value of inductance.

(4) Inductance is analogous to inertia in mechanics, because inductance of an electrical circuit opposes any change of current in the circuit.

Self Induction

Whenever the electric current passing through a coil or circuit changes, the magnetic flux linked with it will also change. As a result of this, in accordance with Faraday's laws of electromagnetic induction, an emf is induced in the coil or the circuit which opposes the change that causes it. This phenomenon is called 'self induction' and the emf induced is called back emf, current so produced in the coil is called induced current.



(A) Main current

(B) Main current decreasing

Fig. 23.15

(1) **Coefficient of self-induction** : Number of flux linkages with the coil is proportional to the current i . *i.e.* $N\phi \propto i$ or $N\phi = Li$ (N is the number of turns in coil and $N\phi$ – total flux linkage). Hence $L = \frac{N\phi}{i}$ = coefficient of self-induction.

1298 Electromagnetic Induction

(2) If $i = 1 \text{ amp}$, $N = 1$ then, $L = \phi \text{ i.e.}$ the coefficient of self induction of a coil is equal to the flux linked with the coil when the current in it is 1 amp .

(3) By Faraday's second law induced emf $e = -N \frac{d\phi}{dt}$.

Which gives $e = -L \frac{di}{dt}$; If $\frac{di}{dt} = 1 \text{ amp/sec}$ then $|e| = L$.

Hence coefficient of self induction is equal to the emf induced in the coil when the rate of change of current in the coil is unity.

(4) Units and dimensional formula of ' L ': It's S.I. unit

$$\frac{\text{weber}}{\text{Amp}} = \frac{\text{Tesla} \times \text{m}^2}{\text{Amp}} = \frac{N \times m}{\text{Amp}^2} = \frac{\text{Joule}}{\text{Amp}^2} = \frac{\text{Coulomb} \times \text{volt}}{\text{Amp}^2}$$

$$= \frac{\text{volt} \times \text{sec}}{\text{amp}} = \text{ohm} \times \text{sec}. \text{ But practical unit is henry (H). It's}$$

dimensional formula $[L] = [ML^2 T^{-2} A^{-2}]$

(5) Dependence of self inductance (L): ' L ' does not depend upon current flowing or change in current flowing but it depends upon number of turns (N), Area of cross section (A) and permeability of medium (μ).

' L ' does not play any role till there is a constant current flowing in the circuit. ' L ' comes in to the picture only when there is a change in current.

(6) Magnetic potential energy of inductor: In building a steady current in the circuit, the source emf has to do work against of self inductance of coil and whatever energy consumed for this work stored in magnetic field of coil this energy called as magnetic potential energy (U) of coil

$$U = \int_0^i L i di = \frac{1}{2} L i^2; \text{ Also } U = \frac{1}{2} (L i) i = \frac{N \phi i}{2}$$

(7) The various formulae for L

Condition	Figure
Circular coil $L = \frac{\mu_0 \pi N^2 r}{2}$	
Solenoid $L = \frac{\mu_0 \mu_r N^2 A}{l} = \frac{\mu N^2 A}{l} (\mu = \mu_0 \mu_r)$	

Toroid $L = \frac{\mu_0 N^2 r}{2}$	
Square coil $L = \frac{2\sqrt{2} \mu_0 N^2 a}{\pi}$	
Coaxial cylinders $L = \frac{\mu_0}{2\pi} \log_e \frac{r_2}{r_1}$ $\frac{2.303}{2\pi} \mu_0 \log_{10} \frac{r_2}{r_1}$	

Mutual Induction

Whenever the current passing through a coil or circuit changes, the magnetic flux linked with a neighbouring coil or circuit will also change. Hence an emf will be induced in the neighbouring coil or circuit. This phenomenon is called 'mutual induction'.

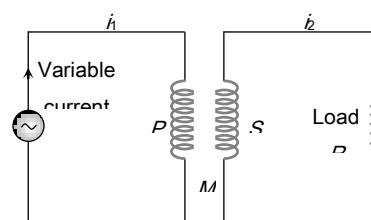


Fig. 23.16

(1) **Coefficient of mutual induction** : Total flux linked with the secondary due to current in the primary is $N_2\phi_2$ and $N_2\phi_2 \propto i_1 \Rightarrow N_2\phi_2 = Mi_1$ where N_1 - Number of turns in primary; N_2 - Number of turns in secondary; ϕ_2 - Flux linked with each turn of secondary; i_1 - Current flowing through primary; M - Coefficient of mutual induction or mutual inductance.

(2) According to Faraday's second law emf induces in secondary $e_2 = -N_2 \frac{d\phi_2}{dt}$; $e_2 = -M \frac{di_1}{dt}$

(3) If $\frac{di_1}{dt} = \frac{1 \text{ Amp}}{\text{sec}}$ then $|e_2| = M$. Hence coefficient of mutual induction is equal to the emf induced in the secondary coil when rate of change of current in primary coil is unity.

(4) **Units and dimensional formula of M** : Similar to self-inductance (L)

(5) **Dependence of mutual inductance**

(i) Number of turns (N_1 , N_2) of both coils

(ii) Coefficient of self inductances (L_1 , L_2) of both the coils

(iii) Area of cross-section of coils

(iv) Magnetic permeability of medium between the coils (μ) or nature of material on which two coils are wound

(v) Distance between two coils (As d increases so M decreases)

(vi) Orientation between primary and secondary coil (for 90° orientation no flux relation $M = 0$)

(vii) Coupling factor ' k ' between primary and secondary coil

(6) **Relation between M , L_1 and L_2** : For two magnetically coupled coils $M = k\sqrt{L_1 L_2}$; where k - coefficient of coupling or coupling factor which is defined as

$$k = \frac{\text{Magnetic flux linked in secondary}}{\text{Magnetic flux linked in primary}}; \quad 0 \leq k \leq 1$$

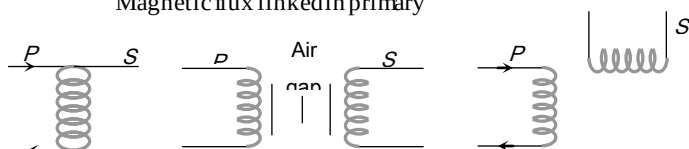


Fig. 23.17

(A) $k = 1$ (B) $0 < k < 1$ (C) $k = 0$

(7) **The various formulae for M** :

Condition	Figure
Two concentric coplanar circular coils $M = \frac{\pi\mu_0 N_1 N_2 r^2}{2R}$	
Two Solenoids $M = \frac{\mu_0 N_1 N_2 A}{l}$	
Two concentric coplanar square coils $M = \frac{\mu_0 2\sqrt{2} N_1 N_2 l^2}{\pi l}$	

Combination of Inductance

(1) **Series** : If two coils of self-inductances L_1 and L_2 having mutual inductance are in series and are far from each other, so that the mutual induction between them is negligible, then net self inductance $L_S = L_1 + L_2$

When they are situated close to each other, then net inductance $L_S = L_1 + L_2 \pm 2M$

(2) **Parallel** : If two coils of self-inductances L_1 and L_2 having mutual inductance are connected in parallel and are far from each other, then net inductance L is $\frac{1}{L_P} = \frac{1}{L_1} + \frac{1}{L_2} \Rightarrow$

$$L_P = \frac{L_1 L_2}{L_1 + L_2}$$

When they are situated close to each other, then

$$L_P = \frac{L_1 L_2 - M^2}{L_1 + L_2 \pm 2M}$$

Growth and Decay of Current In LR- Circuit

If a circuit containing a pure inductor L and a resistor R in series with a battery and a key then on closing the circuit current through the circuit rises exponentially and reaches up to a certain maximum value (steady state). If circuit is opened from its steady state condition then current through the circuit decreases exponentially.

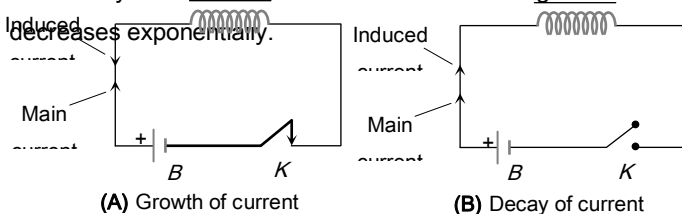


Fig. 23.18

(1) The value of current at any instant of time t after closing the circuit (*i.e.* during the rising of current) is given by

$$i = i_0 \left[1 - e^{-\frac{R}{L}t} \right]; \text{ where } i_0 = i_{\max} = \frac{E}{R} = \text{steady state current.}$$

(2) The value of current at any instant of time t after opening from the steady state condition (*i.e.* during the decaying of current) is given by

$$i = i_0 e^{-\frac{R}{L}t}$$

(3) **Time constant (τ)** : It is given as $\tau = \frac{L}{R}$; It's unit is *second*. In other words the time interval, during which the current in an inductive circuit rises to 63% of its maximum value at make, is defined as time constant or it is the time interval, during which the current after opening an inductive circuit falls to 37% of its maximum value.

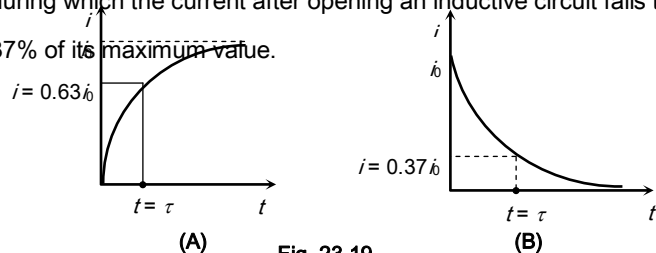


Fig. 23.19

LC- Oscillation

When a charged capacitor C having an initial charge q_0 is discharged through an inductance L , the charge and current in the circuit start oscillating simple harmonically. If the resistance of the circuit is zero, no energy is dissipated as heat. We also assume an idealized situation in which energy is not radiated away from the circuit. The total energy associated with the circuit is constant.

Frequency of oscillation is given by

$$\omega = \frac{1}{\sqrt{LC}} \text{ rad/sec}$$

$$\text{or } \nu = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$

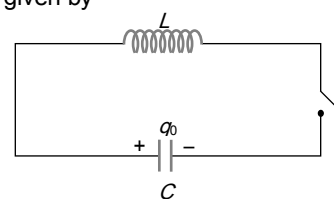


Fig. 23.20

Eddy Current

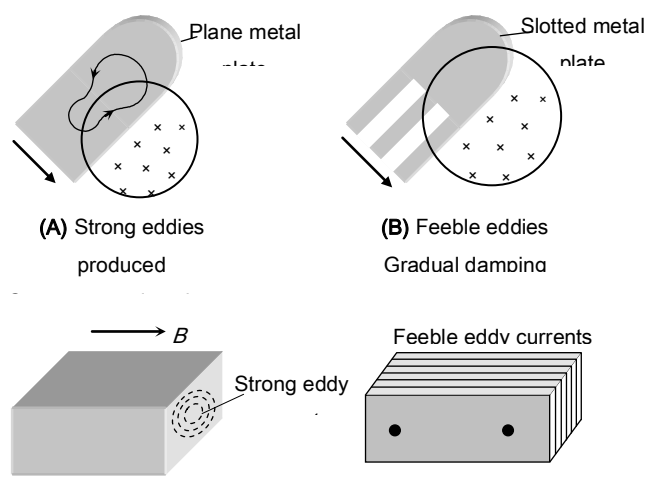
When a changing magnetic flux is applied to a bulk piece of conducting material then circulating currents called eddy currents are induced in the material. Because the resistance of the bulk conductor is usually low, eddy currents often have large magnitudes and heat up the conductor.

(1) These are circulating currents like eddies in water.

(2) Experimental concept given by Foucault hence also named as "Foucault current".

(3) The production of eddy currents in a metallic block leads to the loss of electric energy in the form of heat.

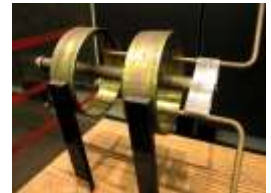
(4) By Lamination, slotting processes the resistance path for circulation of eddy current increases, resulting in to weakening them and also reducing losses caused by them



(v) **Energy meter** : In energy meters, the armature coil carries a metallic aluminium disc which rotates between the poles of a pair of permanent horse shoe magnets. As the armature rotates, the current induced in the disc tends to oppose the motion of the armature coil. Due to this braking effect, deflection is proportional to the energy consumed.

dc Motor

It is an electrical machine which converts electrical energy into mechanical energy.



(1) **Principle** : It is based on the fact that a current carrying coil

placed in the magnetic field experiences a torque. This torque rotates the coil.

(2) **Construction** : It consists of the following components

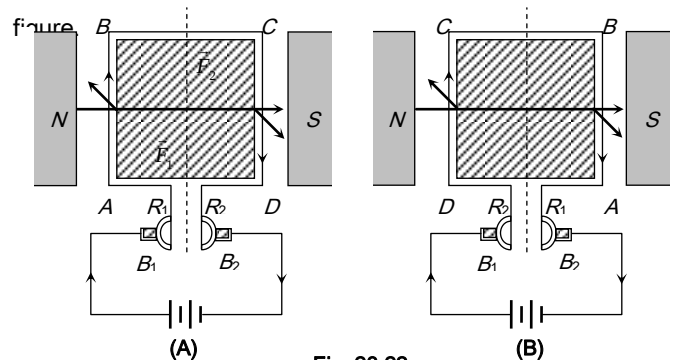


Fig. 23.22

$ABCD$ = Armature coil, S_1, S_2 = split ring commutators

B_1, B_2 = Carbon brushes, N, S = Strong magnetic poles

(3) **Working** : Force on any arm of the coil is given by $\vec{F} = i(\vec{l} \times \vec{B})$ in fig., force on AB will be perpendicular to plane of the paper and pointing inwards. Force on CD will be equal and opposite. So coil rotates in clockwise sense when viewed from top in fig. The current in AB reverses due to commutation keeping the force on AB and CD in such a direction that the coil continues to rotate in the same direction.

(5) **Application of eddy currents** : Though most of the times eddy currents are undesirable but they find some useful applications as enumerated below

(i) **Dead-beat galvanometer** : A dead beat galvanometer means one whose pointer comes to rest in the final equilibrium position immediately without any oscillation about the equilibrium position when a current is passed in its coil.

This is achieved by winding the coil on a metallic frame the large eddy currents induced in the frame provide electromagnetic damping.

(ii) **Electric-brakes** : When the train is running its wheel is moving in air and when the train is to be stopped by electric brakes the wheel is made to move in a field created by electromagnet. Eddy currents induced in the wheels due to the changing flux oppose the cause and stop the train.

(iii) **Induction furnace** : Joule's heat causes the melting of a metal piece placed in a rapidly changing magnetic field.

(iv) **Speedometer** : In the speedometer of an automobile, a magnet is geared to the main shaft of the vehicle and it rotates according to the speed of the vehicle. The magnet is mounted in an aluminium cylinder with the help of hair springs. When the magnet rotates, it produces eddy currents in the drum and drags it through an angle, which indicates the speed of the vehicle on a calibrated scale.

1302 Electromagnetic Induction

(4) **Back emf in motor** : Due to the rotation of armature coil in magnetic field a back emf is induced in the circuit. Which is given by $e = E - iR$.

Back emf directly depends upon the angular velocity ω of armature and magnetic field B . But for constant magnetic field B , value of back emf e is given by $e \propto \omega$ or $e = k\omega$ ($e = NBA\omega \sin\omega t$)

(5) **Current in the motor** : $i = \frac{E - e}{R} = \frac{E - k\omega}{R}$; When motor is just switched on *i.e.* $\omega = 0$ so $e = 0$ hence $i = \frac{E}{R} = \text{maximum}$ and at full speed, ω is maximum so back emf e is maximum and i is minimum. Thus, maximum current is drawn when the motor is just switched on which decreases when motor attains the speed.

(6) **Motor starter** : At the time of start a large current flows through the motor which may burn out it. Hence a starter is used for starting a dc motor safely. Its function is to introduce a suitable resistance in the circuit at the time of starting of the motor. This resistance decreases gradually and reduces to zero when the motor runs at full speed.

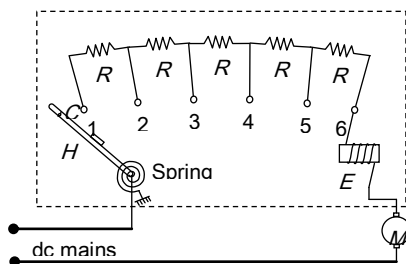


Fig. 23.23

The value of starting resistance is maximum at time $t = 0$ and its value is controlled by spring and electromagnetic system and is made to zero when the motor attains its safe speed.

(7) **Mechanical power and Efficiency of dc motor** :

$$\text{Efficiency } \eta = \frac{P_{\text{mechanical}}}{P_{\text{supplied}}} = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{e}{E} = \frac{\text{Back e.m.f.}}{\text{Supply voltage}}$$

(8) **Uses of dc motors** : They are used in electric locomotives, electric fans, rolling mills, electric cranes, electric

lifts, dc drills, fans and blowers, centrifugal pumps and air compressors, *etc.*

ac Generator/Alternator/Dynamo

An electrical machine used to convert mechanical energy into electrical energy is known as ac generator/alternator.

(1) **Principle** : It works on the principle of electromagnetic induction *i.e.*, when a coil is rotated in uniform magnetic field, an induced emf is produced in it.

(2) **Construction** : The main components of ac generator are

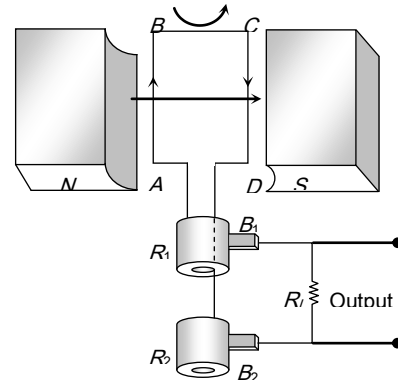


Fig. 23.24

(i) **Armature** : Armature coil ($ABCD$) consists of large number of turns of insulated copper wire wound over a soft iron core.

(ii) **Strong field magnet** : A strong permanent magnet or an electromagnet whose poles (N and S) are cylindrical in shape in a field magnet. The armature coil rotates between the pole pieces of the field magnet. The uniform magnetic field provided by the field magnet is perpendicular to the axis of rotation of the coil.

(iii) **Slip rings** : The two ends of the armature coil are connected to two brass slip rings R_1 and R_2 . These rings rotate along with the armature coil.

(iv) **Brushes** : Two carbon brushes (B_1 and B_2), are pressed against the slip rings. The brushes are fixed while slip rings rotate along with the armature. These brushes are connected to the load through which the output is obtained.

(3) **Working** : When the armature coil $ABCD$ rotates in the magnetic field provided by the strong field magnet, it cuts the magnetic lines of force. Thus the magnetic flux linked with the coil changes and hence induced emf is set up in the coil. The direction of the induced emf or the current in the coil is determined by the Fleming's right hand rule.

The current flows out through the brush B_1 in one direction of half of the revolution and through the brush B_2 in the next half revolution in the reverse direction. This process is repeated. Therefore, emf produced is of alternating nature.

$$e = -\frac{Nd\phi}{dt} = NBA\omega \sin\omega t = e_0 \sin\omega t \quad \text{where } e_0 = NBA\omega$$

$$i = \frac{e}{R} = \frac{e_0}{R} \sin\omega t = i_0 \sin\omega t \quad R \rightarrow \text{Resistance of the circuit}$$

dc Generator

If the current produced by the generator is direct current, then the generator is called dc generator.

dc generator consists of (i) Armature (coil) (ii) Magnet (iii) Commutator (iv) Brushes

In dc generator commutator is used in place of slip rings. The commutator rotates along with the coil so that in every cycle when direction of 'e' reverses, the commutator also reverses or makes contact with the other brush so that in the external load the current remains in the same direction giving dc

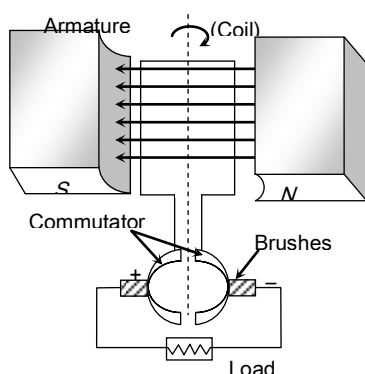


Fig. 23.25

Transformer

It is a device which raises or lowers the voltage in ac circuits through mutual induction.

It consists of two coils wound on the same core. The alternating current passing through the primary creates a continuously changing flux through the core. This changing flux induces an alternating emf in the secondary.

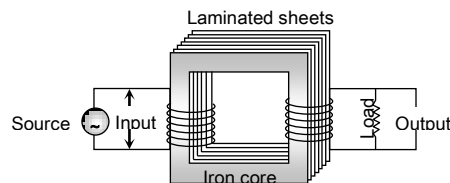


Fig. 23.26

- (1) Transformer works on ac only and never on dc.
- (2) It can increase or decrease either voltage or current but not both simultaneously.
- (3) Transformer does not change the frequency of input ac.
- (4) There is no electrical connection between the winding but they are linked magnetically.
- (5) Effective resistance between primary and secondary winding is infinite.
- (6) The flux per turn of each coil must be same i.e. $\phi_s = \phi_p$;

$$-\frac{d\phi_s}{dt} = -\frac{d\phi_p}{dt}$$
- (7) If N_p = number of turns in primary, N_s = number of turns in secondary, V_p = applied (input) voltage to primary, V_s = Voltage across secondary (load voltage or output), e_p = induced emf in primary ; e_s = induced emf in secondary, ϕ = flux linked with primary as well as secondary, i_p = current in primary; i_s = current in secondary (or load current)

As in an ideal transformer there is no loss of power i.e. $P_{out} = P_{in}$ so $V_s i_s = V_p i_p$ and $V_p \approx e_p$, $V_s \approx e_s$. Hence

$$\frac{e_s}{e_p} = \frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{i_p}{i_s} = k ; k = \text{Transformation ratio (or turn ratio)}$$

Table 23.3 : Types of transformer

Step up transformer	Step down transformer
It increases voltage and decreases current	It decreases voltage and increases current

$$V_S > V_P$$

$$N_S > N_P$$

$$E_S > E_P$$

$$i_S < i_P$$

$$R_S > R_P$$

$$t_S > t_P$$

$$k > 1$$

$$V_S < V_P$$

$$N_S < N_P$$

$$E_S < E_P$$

$$i_S > i_P$$

$$R_S < R_P$$

$$t_S > t_P$$

$$k < 1$$

(8) **Efficiency of transformer (η)** : Efficiency is defined as the ratio of output power and input power

$$i.e. \quad \eta\% = \frac{P_{out}}{P_{in}} \times 100 = \frac{V_S i_S}{V_P i_P} \times 100$$

For an ideal transformer $P_{out} = P_{in}$ so $\eta = 100\%$ (But efficiency of practical transformer lies between 70% – 90%)

For practical transformer $P_{in} = P_{out} + P_{losses}$

$$so \quad \eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{P_{out}}{(P_{out} + P_L)} \times 100 = \frac{(P_{in} - P_L)}{P_{in}} \times 100$$

(9) **Losses in transformer** : In transformers some power is always lost due to, heating effect, flux leakage eddy currents, hysteresis and humming.

(i) **Cu loss ($I^2 R$)** : When current flows through the transformer windings some power is wasted in the form of heat ($H = i^2 R t$). To minimize this loss windings are made of thick Cu wires (To reduce resistance)

(ii) **Eddy current loss** : Some electrical power is wasted in the form of heat due to eddy currents, induced in core, to minimize this loss transformers core are laminated and silicon is added to the core material as it increases the resistivity. The material of the core is then called silicon-iron (steel).

(iii) **Hysteresis loss** : The alternating current flowing through the coils magnetises and demagnetises the iron core again and again. Therefore, during each cycle of magnetisation, some

energy is lost due to hysteresis. However, the loss of energy can be minimised by selecting the material of core, which has a narrow hysteresis loop. Therefore core of transformer is made of soft iron. Now a days it is made of "Permalloy" (Fe-22%, Ni-78%).

(iv) **Magnetic flux leakage** : Magnetic flux produced in the primary winding is not completely linked with secondary because few magnetic lines of force complete their path in air only. To minimize this loss secondary winding is kept inside the primary winding.

(v) **Humming losses** : Due to the passage of alternating current, the core of the transformer starts vibrating and produces humming sound. Thus, some part (may be very small) of the electrical energy is wasted in the form of humming sounds produced by the vibrating core of the transformer.

(10) **Uses of transformer** : A transformer is used in almost all ac operations *e.g.*

(i) In voltage regulators for TV, refrigerator, computer, air conditioner etc.

(ii) In the induction furnaces.

(iii) Step down transformer is used for welding purposes.

(iv) In the transmission of ac over long distance.

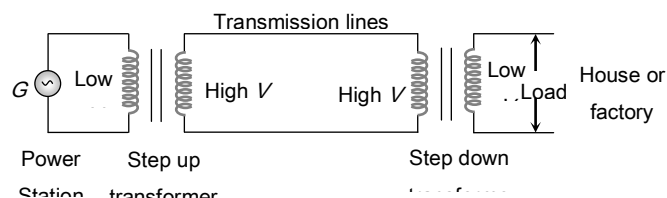


Fig. 23.27

(v) Step down and step up transformers are used in electrical power distribution.

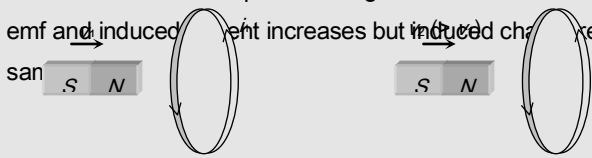
(vi) Audio frequency transformers are used in radiography, television, radio, telephone *etc.*

(vii) Radio frequency transformers are used in radio communication.

(viii) Transformers are also used in impedance matching.

Tips & Tricks

✍ If a bar magnet moves towards a fixed conducting coil, then due to the flux changes an emf, current and charge induces in the coil. If speed of magnet increases then induced emf and induced current increases but induced charge remains same.



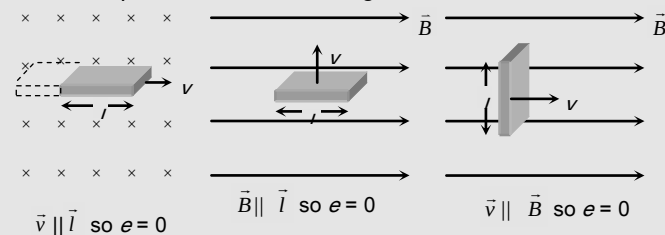
Induced parameter: e_1, i_1, q_1 $e_2 (> e_1), i_2 (> i_1), q_2 (= q_1)$

✍ Can ever electric lines of force be closed curve? Yes, when produced by a changing magnetic field.

✍ No flux cutting $\longrightarrow$ No EMI

✍ Vector form of motional emf: $e = (\vec{v} \times \vec{B}) \cdot \vec{l}$

✍ In motional emf $\vec{B}, \vec{v}$ and $\vec{l}$ are three vectors. If any two vector are parallel – No flux cutting.



✍ A piece of metal and a piece of non-metal are dropped from the same height near the surface of the earth. The non-metallic piece will reach the ground first because there will be no induced current in it.

✍ If an aeroplane is landing down or taking off and its wings are in the east-west direction, then the potential difference or emf will be induced across the wings. If an aeroplane is landing down or taking off and its wings are in

the north-south direction, then no potential difference or emf will be induced.

✍ When a conducting rod moving horizontally on equator of earth no emf induces because there is no vertical component of earth's magnetic field. But at poles B_v is maximum so maximum flux cutting hence emf induces.

✍ When a conducting rod falling freely in earth's magnetic field such that its length lies along East - West direction then induced emf continuously increases w.r.t. time and induced current flows from West - East.

✍ 1 henry = 10^9 emu of inductance or 10^9 ab-henry.

✍ Inductance at the ends of a solenoid is half of its the inductance at the centre. $\left(L_{\text{end}} = \frac{1}{2} L_{\text{centre}} \right)$.

✍ A thin long wire made up of material of high resistivity behaves predominantly as a resistance. But it has some amount of inductance as well as capacitance in it. It is thus difficult to obtain pure resistor. Similarly it is difficult to obtain pure capacitor as well as pure inductor.

✍ Due to inherent presence of self inductance in all electrical circuits, a resistive circuit with no capacitive or inductive element in it, also has some inductance associated with it.



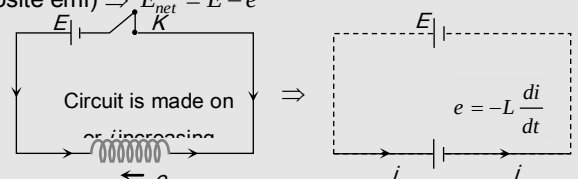
The effect of self-inductance can be eliminated

as in the coils of a resistance box by doubling

back the coil on itself.

✍ It is not possible to have mutual inductance without self inductance but it may or may not be possible self inductance without mutual inductance.

✍ If main current through a coil increases ($i \uparrow$) so $\frac{di}{dt}$ will be positive (+ve), hence induced emf e will be negative (i.e. opposite emf) $\Rightarrow E_{\text{net}} = E - e$



✍ Sometimes at sudden opening of key, because of high

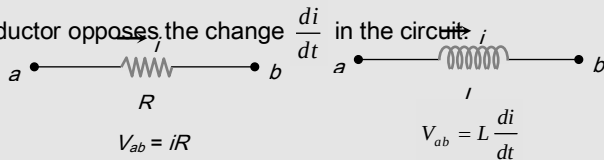
1306 Electromagnetic Induction

inductance of circuit a high momentarily induced emf produced and a sparking occurs at key position. To avoid sparking a capacitor is connected across the key.

✍ Sometimes at sudden opening of key, because of high inductance of circuit a high momentarily induced emf produced and a sparking occurs at key position. To avoid sparking a capacitor is connected across the key.

✍ One can have resistance with or without inductance but one can't have inductance without having resistance.

✍ The circuit behaviour of an inductor is quite different from that of a resistor. while a resistor opposes the current i , an inductor opposes the change $\frac{di}{dt}$ in the circuit.



✍ In RL -circuit with dc source the time taken by the current to reach half of the maximum value is called half life time and it is given by $T = 0.693 \frac{L}{R}$.

✍ dc motor is a highly versatile energy conversion device. It can meet the demand of loads requiring high starting torque, high accelerating and decelerating torque.

✍ When a source of emf is connected across the two ends of the primary winding alone or across the two ends of secondary winding alone, ohm's law can be applied. But in the transformer as a whole, ohm's law should not be applied because primary winding and secondary winding are not connected electrically.

✍ Even when secondary circuit of the transformer is open it also draws some current called no load primary current for supplying no load Cu and iron losses.

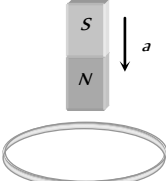
✍ Transformer has highest possible efficiency out of all the electrical machines.



Ordinary Thinking

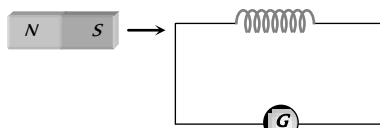
Objective Questions

Faraday's and Lenz's Law

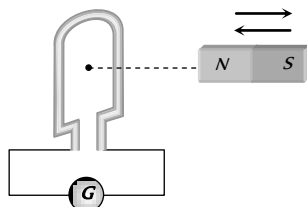
- In electromagnetic induction, the induced e.m.f. in a coil is independent of [CPMT 1984]
 - Change in the flux
 - Time
 - Resistance of the circuit
 - None of the above
- Lenz's law is consequence of the law of conservation of [JIPMER 1997; CPMT 1990; RPMT 1997; MP PET 1999; MP PMT 2000, 03; RPET 2003; AFMC 2004]
 - Charge
 - Momentum
 - Mass
 - Energy
- In electromagnetic induction, the induced charge in a coil is independent of
 - Change in the flux
 - Time
 - Resistance in the circuit
 - None of the above
- The magnetic flux through a circuit of resistance R changes by an amount $\Delta\phi$ in time Δt . Then the total quantity of electric charge Q , which passing during this time through any point of the circuit is given by [Haryana CEE 1996; CBSE PMT 2004]
 - $Q = \frac{\Delta\phi}{\Delta t}$
 - $Q = \frac{\Delta\phi}{\Delta t} \times R$
 - $Q = -\frac{\Delta\phi}{\Delta t} + R$
 - $Q = \frac{\Delta\phi}{R}$
- A cylindrical bar magnet is kept along the axis of a circular coil. If the magnet is rotated about its axis, then [CPMT 1983; BCECE 2004]
 - A current will be induced in a coil
 - No current will be induced in a coil
 - Only an e.m.f. will be induced in the coil
 - An e.m.f. and a current both will be induced in the coil
- A metallic ring is attached with the wall of a room. When the north pole of a magnet is brought near to it, the induced current in the ring will be [AFMC 1993; MP PMT/PET 1998; Pb PET 2003]
 
 - First clockwise then anticlockwise
 - In clockwise direction
 - In anticlockwise direction
 - First anticlockwise then clockwise
- A coil having an area A_0 is placed in a magnetic field which changes from B_0 to $4B_0$ in a time interval t . The e.m.f. induced in the coil will be [MP PET 1990]
 - $\frac{3A_0B_0}{t}$
 - $\frac{4A_0B_0}{t}$
 - $\frac{3B_0}{A_0t}$
 - $\frac{4B_0}{A_0t}$

- The magnetic flux linked with a coil is given by an equation ϕ (in webers) = $8t^2 + 3t + 5$. The induced e.m.f. in the coil at the fourth second will be [MP PET 1990]
 - 16 units
 - 39 units
 - 67 units
 - 145 units
- The current flowing in two coaxial coils in the same direction. On increasing the distance between the two, the electric current will
 - Increase
 - Decrease
 - Remain unchanged
 - The information is incomplete
- A copper ring is held horizontally and a bar magnet is dropped through the ring with its length along the axis of the ring. The acceleration of the falling magnet while it is passing through the ring is [CBSE PMT 1996; MP PET 1990, 99; CPMT 1991, 99; JIPMER 1997; CPMT 2003; MP PET/PMT 2001; KCET 2001; Kerala (Engg.) 2001]
 - Equal to that due to gravity
 - Less than that due to gravity
 - More than that due to gravity
 - Depends on the diameter of the ring and the length of the magnet
- A square coil $10^{-2} m^2$ area is placed perpendicular to a uniform magnetic field of intensity $10^3 Wb/m^2$. The magnetic flux through the coil is [MP PMT 1990, 2001]
 - 10 weber
 - 10^{-5} weber
 - 10^5 weber
 - 100 weber
- A magnet is brought towards a coil (i) speedily (ii) slowly then the induced e.m.f./induced charge will be respectively [RPMT 1997; MP PMT 2003]
 - More in first case / More in first case
 - More in first case/Equal in both case
 - Less in first case/More in second case
 - Less in first case/Equal in both case
- The direction of induced e.m.f. during electromagnetic induction is given by [MP PET 1994, 96]
 - Faraday's law
 - Lenz's law
 - Maxwell's law
 - Ampere's law
- In a coil of area $10 cm^2$ and 10 turns with a magnetic field directed perpendicular to the plane and is changing at the rate of $10^8 gauss/second$. The resistance of the coil is 20 ohm. The current in the coil will be [CPMT 1976]
 - 5 amp
 - 0.5 amp
 - 0.05 amp
 - 5×10^8 amp

15. As shown in the figure, a magnet is moved with a fast speed towards a coil at rest. Due to this induced electromotive force, induced current and induced charge in the coil is E , I and Q respectively. If the speed of the magnet is doubled, the incorrect statement is [MP PET 1995]



- (a) E increases (b) I increases
(c) Q remains same (d) Q increases
16. A coil having 500 square loops each of side 10 cm is placed normal to a magnetic flux which increases at the rate of 1.0 tesla/second . The induced e.m.f. in volts is [CPMT 1989, 90; DCE 2002]
- (a) 0.1 (b) 0.5
(c) 1 (d) 5
17. When a magnet is pushed in and out of a circular coil C connected to a very sensitive galvanometer G as shown in the adjoining diagram with a frequency ν , then



- (a) Constant deflection is observed in the galvanometer
(b) Visible small oscillations will be observed in the galvanometer if ν is about 50 Hz
(c) Oscillations in the deflection will be observed clearly if $\nu = 1$ or 2 Hz
(d) No variation in the deflection will be seen if $\nu = 1$ or 2 Hz
18. A coil of area 100 cm^2 has 500 turns. Magnetic field of $0.1\text{ weber / metre}^2$ is perpendicular to the coil. The field is reduced to zero in 0.1 second . The induced e.m.f. in the coil is
- (a) 1 V (b) 5 V
(c) 50 V (d) Zero
19. A 50 turns circular coil has a radius of 3 cm , it is kept in a magnetic field acting normal to the area of the coil. The magnetic field B increased from 0.10 tesla to 0.35 tesla in 2 milliseconds . The average induced e.m.f. in the coil is [MP PET 1994]
- (a) 1.77 volts (b) 17.7 volts
(c) 177 volts (d) 0.177 volts
20. A coil having an area 2 m^2 is placed in a magnetic field which changes from 1 Wb/m^2 to 4 Wb/m^2 in a interval of 2 second . The e.m.f. induced in the coil will be [DPMT 1999; MP PET 2002]
- (a) 4 V (b) 3 V

(c) 1.5 V (d) 2 V

21. A coil has 2000 turns and area of 70 cm^2 . The magnetic field perpendicular to the plane of the coil is 0.3 Wb/m^2 and takes 0.1 sec to rotate through 180° . The value of the induced e.m.f. will be [MP PET 1993; Similar to AIIMS 1997]

(a) 8.4 V (b) 84 V
(c) 42 V (d) 4.2 V

22. Two different loops are concentric and lie in the same plane. The current in the outer loop is clockwise and increasing with time. The induced current in the inner loop then, is [MP PET 1993]

(a) Clockwise
(b) Zero
(c) Counter clockwise
(d) In a direction that depends on the ratio of the loop radii

23. According to Faraday's law of electromagnetic induction [MP PET 1994]

(a) The direction of induced current is such that it opposes the cause producing it
(b) The magnitude of induced e.m.f. produced in a coil is directly proportional to the rate of change of magnetic flux
(c) The direction of induced e.m.f. is such that it opposes the cause producing it
(d) None of the above

24. The unit of magnetic flux is [MP PMT 1994; MP PET 1995; AFMC 1998]

(a) Weber / m^2 (b) Weber
(c) Henry (d) Ampere / m

25. The north pole of a long horizontal bar magnet is being brought closer to a vertical conducting plane along the perpendicular direction. The direction of the induced current in the conducting plane will be [MP PMT 1994]

(a) Horizontal (b) Vertical
(c) Clockwise (d) Anticlockwise

26. The magnetic field in a coil of 100 turns and 40 square cm area is increased from 1 Tesla to 6 Tesla in 2 second . The magnetic field is perpendicular to the coil. The e.m.f. generated in it is [MP PMT 1991; MH CET (Med.) 1999]

(a) 10^4 V (b) 1.2 V
(c) 1.0 V (d) 10^{-2} V

27. The dimensions of magnetic flux are [MP PMT 1994; CBSE PMT 1999]

(a) $MLT^{-2}A^{-2}$ (b) $ML^2T^{-2}A^{-2}$
(c) $ML^2T^{-1}A^{-2}$ (d) $ML^2T^{-2}A^{-1}$

28. Lenz's law gives [MP PMT 1994]

(a) The magnitude of the induced e.m.f.
(b) The direction of the induced current
(c) Both the magnitude and direction of the induced current
(d) The magnitude of the induced current

29. The north pole of a bar magnet is moved swiftly downward towards a closed coil and then second time it is raised upwards slowly. The

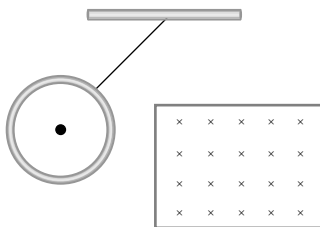
magnitude and direction of the induced currents in the two cases will be of [MP PET 1996]

First case

Second case

- | | |
|--------------------------------|----------------------------|
| (a) Low value clockwise | Higher value anticlockwise |
| (b) Low value clockwise | Equal value anticlockwise |
| (c) Higher value clockwise | Low value clockwise |
| (d) Higher value anticlockwise | Low value clockwise |

30. A metallic ring connected to a rod oscillates freely like a pendulum. If now a magnetic field is applied in horizontal direction so that the pendulum now swings through the field, the pendulum will



- (a) Keep oscillating with the old time period
(b) Keep oscillating with a smaller time period
(c) Keep oscillating with a larger time period
(d) Come to rest very soon

31. A circular coil of 500 turns of wire has an enclosed area of 0.1 m^2 per turn. It is kept perpendicular to a magnetic field of induction 0.2 T and rotated by 180° about a diameter perpendicular to the field in 0.1 sec . How much charge will pass when the coil is connected to a galvanometer with a combined resistance of 50 ohms

- | | |
|---------------------|---------------------|
| (a) 0.2 C | (b) 0.4 C |
| (c) 2 C | (d) 4 C |

32. A coil of 100 turns and area $5 \text{ square centimetre}$ is placed in a magnetic field $B = 0.2 \text{ T}$. The normal to the plane of the coil makes an angle of 60° with the direction of the magnetic field. The magnetic flux linked with the coil is

[MP PMT 1997]

- | | |
|-----------------------------------|-----------------------------------|
| (a) $5 \times 10^{-3} \text{ Wb}$ | (b) $5 \times 10^{-5} \text{ Wb}$ |
| (c) 10^{-2} Wb | (d) 10^{-4} Wb |

33. In a circuit with a coil of resistance 2 ohms , the magnetic flux changes from 2.0 Wb to 10.0 Wb in 0.2 second . The charge that flows in the coil during this time is

[MP PMT 1997]

- | | |
|---------------------------|---------------------------|
| (a) 5.0 coulomb | (b) 4.0 coulomb |
| (c) 1.0 coulomb | (d) 0.8 coulomb |

34. The direction of induced current is such that it opposes the very cause that has produced it. This is the law of

[MP PMT/PET 1998]

- | | |
|---------------|-------------|
| (a) Lenz | (b) Faraday |
| (c) Kirchhoff | (d) Fleming |

35. To induce an e.m.f. in a coil, the linking magnetic flux

[KCET 1994]

- (a) Must decrease
(b) Can either increase or decrease
(c) Must remain constant
(d) Must increase

36. A solenoid is 1.5 m long and its inner diameter is 4.0 cm . It has three layers of windings of 1000 turns each and carries a current of 2.0 amperes . The magnetic flux for a cross-section of the solenoid is nearly [AMU 1995]

- | | |
|--|---|
| (a) $2.5 \times 10^{-4} \text{ weber}$ | (b) $6.31 \times 10^{-4} \text{ weber}$ |
| (c) $5.2 \times 10^{-4} \text{ weber}$ | (d) $4.1 \times 10^{-4} \text{ weber}$ |

37. A coil of 40Ω resistance has 100 turns and radius 6 mm is connected to ammeter of resistance of 160 ohms . Coil is placed perpendicular to the magnetic field. When coil is taken out of the field, $32 \mu \text{ C}$ charge flows through it. The intensity of magnetic field will be [RPET 1997]

- | | |
|-----------------------|-----------------------|
| (a) 6.55 T | (b) 5.66 T |
| (c) 0.655 T | (d) 0.566 T |

38. Faraday's laws are consequence of conservation of

[CBSE PMT 1993; BHU 2002]

- (a) Energy
(b) Energy and magnetic field
(c) Charge
(d) Magnetic field

39. A magnetic field of $2 \times 10^{-4} \text{ T}$ acts at right angles to a coil of area 100 cm^2 with 50 turns. The average emf induced in the coil is 0.1 V , when it is removed from the field in time t . The value of t is [CBSE PMT 1992; C]

- | | |
|-----------------------|------------------------|
| (a) 0.1 sec | (b) 0.01 sec |
| (c) 1 sec | (d) 20 sec |

40. The total charge induced in a conducting loop when it is moved in magnetic field depends on [MP PET 1997]

[CBSE PMT 1992; ISM Dhanbad 1994]

- (a) The rate of change of magnetic flux
(b) Initial magnetic flux only
(c) The total change in magnetic flux
(d) Final magnetic flux only

41. A rectangular coil of 20 turns and area of cross-section 25 sq cm has a resistance of 100 ohm . If a magnetic field which is perpendicular to the plane of the coil changes at the rate of $1000 \text{ Tesla per second}$, the current in the coil is

[CBSE PMT 1992;

Very Similar to MHCET 2002; DPMT 2004]

- | | |
|--------------------------|--------------------------|
| (a) 1.0 ampere | (b) 50 ampere |
| (c) 0.5 ampere | (d) 5.0 ampere |

42. The north pole of a magnet is brought near a metallic ring. The direction of the induced current in the ring will be

[AIIMS 1999]

- | | |
|-------------------|-------------------|
| (a) Clockwise | (b) Anticlockwise |
| (c) Towards north | (d) Towards south |

43. Lenz's law applies to

[DCE 1999]

- (a) Electrostatics
(b) Lenses
(c) Electro-magnetic induction
(d) Cinema slides

44. If a coil of metal wire is kept stationary in a non-uniform magnetic field, then [BHU 2000]

- (a) An e.m.f. is induced in the coil
(b) A current is induced in the coil
(c) Neither e.m.f. nor current is induced

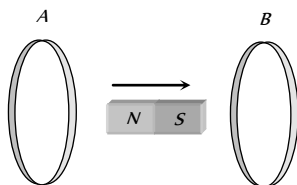
- (d) Both e.m.f. and current is induced
45. The magnetic flux linked with a coil, in *webers*, is given by the equations $\phi = 3t^2 + 4t + 9$. Then the magnitude of induced e.m.f. at $t = 2$ second will be

[KCET 2000; CPMT 2003; MP PET 2005]

- (a) 2 volt (b) 4 volt
(c) 8 volt (d) 16 volt
46. A coil has an area of 0.05 m^2 and it has 800 turns. It is placed perpendicularly in a magnetic field of strength $4 \times 10^{-5} \text{ Wb/m}^2$, it is rotated through 90° in 0.1 sec . The average e.m.f. induced in the coil is
- [CPMT 2001]
- (a) 0.056 V (b) 0.046 V
(c) 0.026 V (d) 0.016 V
47. A moving conductor coil in a magnetic field produces an induced e.m.f. This is in accordance with

[AFMC 1993; MH CET 2001, 03]

- (a) Amperes law (b) Coulomb law
(c) Lenz's law (d) Faraday's law
48. In the diagram shown if a bar magnet is moved along the common axis of two single turn coils A and B in the direction of arrow



- (a) Current is induced only in A & not in B
(b) Induced currents in A & B are in the same direction
(c) Current is induced only in B and not in A
(d) Induced currents in A & B are in opposite directions
49. Magnetic flux ϕ (in *weber*) linked with a closed circuit of resistance 10 ohm varies with time t (in seconds) as

$$\phi = 5t^2 - 4t + 1$$

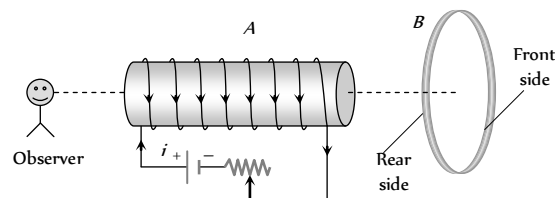
The induced electromotive force in the circuit at $t = 0.2 \text{ sec}$ is

- (a) 0.4 volts (b) -0.4 volts
(c) -2.0 volts (d) 2.0 volts
50. The formula for induced e.m.f. in a coil due to change in magnetic flux through the coil is (here A = area of the coil, B = magnetic field)

[MP PET 2002]

- (a) $e = -A \frac{dB}{dt}$ (b) $e = -B \frac{dA}{dt}$
(c) $e = -\frac{d}{dt}(A \cdot B)$ (d) $e = -\frac{d}{dt}(A \times B)$
51. Lenz's law is expressed by the following formula (here e = induced e.m.f., ϕ = magnetic flux in one turn and N = number of turns)
- (a) $e = -\phi \frac{dN}{dt}$ (b) $e = -N \frac{d\phi}{dt}$
(c) $e = -\frac{d}{dt}\left(\frac{\phi}{N}\right)$ (d) $e = N \frac{d\phi}{dt}$

52. A magnet is dropped down an infinitely long vertical copper tube
- (a) The magnet moves with continuously increasing velocity and ultimately acquires a constant terminal velocity
(b) The magnet moves with continuously decreasing velocity and ultimately comes to rest
(c) The magnet moves with continuously increasing velocity but constant acceleration
(d) The magnet moves with continuously increasing velocity and acceleration
53. An aluminium ring B faces an electromagnet A. The current I through A can be altered
- [Kerala PET 2002]



- (a) Whether I increases or decreases, B will not experience any force
(b) If I increases, B will repel B
(c) If I increases, A will attract B
(d) If I increases, A will repel B
54. The magnetic flux linked with a coil at any instant ' t ' is given by $\phi = 5t - 100t + 300$, the e.m.f. induced in the coil at $t = 2$ second is
- (a) -40 V (b) 40 V
(c) 140 V (d) 300 V
55. A coil has 1,000 turns and 500 cm^2 as its area. The plane of the coil is placed at right angles to a magnetic induction field of $2 \times 10^{-5} \text{ Wb/m}^2$. The coil is rotated through 180° in 0.2 seconds. The average e.m.f. induced in the coil, in *milli-volts*, is
- (a) 5 (b) 10
(c) 15 (d) 20
56. When a bar magnet falls through a long hollow metal cylinder fixed with its axis vertical, the final acceleration of the magnet is
- [MP PET 2001]
- (a) Equal to zero
(b) Less than g
(c) Equal to g
(d) Equal to g in to beginning and then more than g
57. The magnetic flux linked with a vector area $\vec{A}$ in a uniform magnetic field $\vec{B}$ is
- [MP PET 2003]
- (a) $\vec{B} \times \vec{A}$ (b) AB
(c) $\vec{B} \cdot \vec{A}$ (d) $\frac{B}{A}$
- [MP PET 2002]
58. The magnetic flux linked with a circuit of resistance 100 ohm increases from 10 to 60 *webers*. The amount of induced charge that flows in the circuit is (in *coulomb*)

[MP PET 2003]

- (a) 0.5 (b) 5
(c) 50 (d) 100

59. A magnet NS is suspended from a spring and while it oscillates, the magnet moves in and out of the coil C . The coil is connected to a galvanometer G . Then as the magnet oscillates,

- (a) G shows deflection to the left and right with constant amplitude
(b) G shows deflection on one side
(c) G shows no deflection.
(d) G shows deflection to the left and right but the amplitude steadily decreases.



60. A coil having n turns and resistance $R \Omega$ is connected with a galvanometer of resistance $4R \Omega$. This combination is moved in time t seconds from a magnetic field W_1 weber to W_2 weber. The induced current in the circuit is

- (a) $-\frac{W_2 - W_1}{5 Rnt}$ (b) $-\frac{n(W_2 - W_1)}{5 Rt}$
(c) $-\frac{(W_2 - W_1)}{Rnt}$ (d) $-\frac{n(W_2 - W_1)}{Rt}$



61. If a copper ring is moved quickly towards south pole of a powerful stationary bar magnet, then [Pb. PMT 2004]

- (a) Current flows through the copper ring
(b) Voltage in the magnet increase
(c) Current flows in the magnet
(d) Copper ring will get magnetised

62. The magnetic flux linked with coil, in weber is given by the equation, $\phi = 5t^2 + 3t + 16$. The induced emf in the coil in the fourth second is [Pb. PMT 2004]

- (a) 10 V (b) 30 V
(c) 45 V (d) 90 V

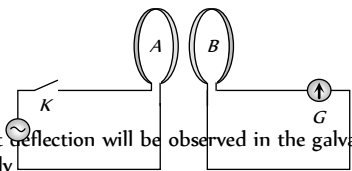
63. The coil of area 0.1 m^2 has 500 turns. After placing the coil in a magnetic field of strength $4 \times 10^{-4} \text{ Wb/m}^2$, if rotated through 90° in 0.1 s , the average emf induced in the coil is [Pb. PET 2002]

- (a) 0.012 V (b) 0.05 V
(c) 0.1 V (d) 0.2 V

64. Magnetic flux in a circuit containing a coil of resistance 2Ω changes from 2.0 Wb to 10 Wb in 0.2 sec . The charge passed through the coil in this time is [DPMT 2003]

- (a) 0.8 C (b) 1.0 C
(c) 5.0 C (d) 4.0 C

65. The diagram below shows two coils A and B placed parallel to each other at a very small distance. Coil A is connected to an ac supply. G is a very sensitive galvanometer. When the key is closed



- (a) Constant deflection will be observed in the galvanometer for 50 Hz supply
(b) Visible small variations will be observed in the galvanometer for 50 Hz input
(c) Oscillations in the galvanometer may be observed when the input ac voltage has a frequency of 1 to 2 Hz
(d) No variation will be observed in the galvanometer even when the input ac voltage is 1 or 2 Hz

66. An infinitely long cylinder is kept parallel to a uniform magnetic field B directed along positive z axis. The direction of induced current as seen from the z axis will be

- (a) Clockwise of the $+ve z$ axis

[IIT-JEE (Screening) 2005]

(b) Anticlockwise of the $+ve z$ axis

(c) Zero

(d) Along the magnetic field

67. In a magnetic field of 0.05 T , area of a coil changes from 101 cm^2 to 100 cm^2 without changing the resistance which is 2Ω . The amount of charge that flow during this period is

[Orissa PMT 2005]

- (a) $2.5 \times 10^{-6} \text{ coulomb}$ (b) $2 \times 10^{-6} \text{ coulomb}$
(c) 10^{-6} coulomb (d) $8 \times 10^{-6} \text{ coulomb}$

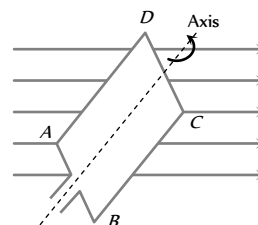
68. If a coil of 40 turns and area 4.0 cm^2 is suddenly removed from a magnetic field, it is observed that a charge of $2.0 \times 10^{-4} \text{ C}$ flows into the coil. If the resistance of the coil is 80Ω , the magnetic flux density in Wb/m^2 is [MP PET 2005]

- (a) 0.5 (b) 1.0
(c) 1.5 (d) 2.0

Motional EMI

1. A rectangular coil ABCD is rotated anticlockwise with a uniform angular velocity about the axis shown in diagram below. The axis of rotation of the coil as well as the magnetic field B are horizontal. The induced e.m.f. in the coil would be maximum when

[Haryana CEE 1996; MP PMT 1992, 94, 99]



- (a) The plane of the coil is horizontal
(b) The plane of the coil makes an angle of 45° with the magnetic field
(c) The plane of the coil is at right angles to the magnetic field
(d) The plane of the coil makes an angle of 30° with the magnetic field [CPMT 1986]

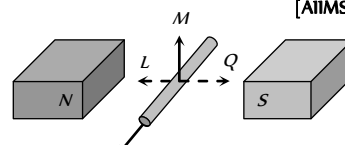
2. A 10 metre wire kept in east-west falling with velocity 5 m/sec perpendicular to the field $0.3 \times 10^{-4} \text{ Wb/m}^2$. The induced e.m.f. across the terminal will be [MP PET 2000]

- (a) 0.15 V (b) 1.5 mV
(c) 1.5 V (d) 15.0 V

3. An electric potential difference will be induced between the ends of the conductor shown in the diagram, when the conductor moves in the direction

[AIIMS 1982; DPMT 2001]

- (a) P
(b) Q
(c) L
(d) M



4. Two rails of a railway track insulated from each other and the ground are connected to a milli voltmeter. What is the reading of voltmeter, when a train travels with a speed of 180 km/hr along the

track. Given that the vertical component of earth's magnetic field is $0.2 \times 10^{-4} \text{ weber/m}^2$ and the rails are separated by 1 metre

[IIT 1981; KCET 2001]

- (a) 10^{-2} volt (b) 10^{-4} volt
(c) 10^{-3} volt (d) 1 volt

5. A conductor of 3 m in length is moving perpendicularly to magnetic field of 10^{-3} tesla with the speed of 10^2 m/s , then the e.m.f. produced across the ends of conductor will be

- (a) 0.03 volt (b) 0.3 volt
(c) $3 \times 10^{-3} \text{ volt}$ (d) 3 volt

6. When a wire loop is rotated in a magnetic field, the direction of induced e.m.f. changes once in each

[MP PMT 1991, 04]

- (a) $\frac{1}{4}$ revolution (b) $\frac{1}{2}$ revolution
(c) 1 revolution (d) 2 revolution

7. An aeroplane in which the distance between the tips of wings is 50 m is flying horizontally with a speed of 360 km/hr over a place where the vertical components of earth magnetic field is $2.0 \times 10^{-4} \text{ weber/m}^2$. The potential difference between the tips of wings would be

[CPMT 1990; MP PET 1991]

- (a) 0.1 V (b) 1.0 V
(c) 0.2 V (d) 0.01 V

8. A copper disc of radius 0.1 m is rotated about its centre with 10 revolutions per second in a uniform magnetic field of 0.1 Tesla with its plane perpendicular to the field. The e.m.f. induced across the radius of disc is

[MH CET (Med) 2001]

- (a) $\frac{\pi}{10} \text{ V}$ (b) $\frac{2\pi}{10} \text{ V}$
(c) $\pi \times 10^{-2} \text{ V}$ (d) $2\pi \times 10^{-2} \text{ V}$

9. A metal conductor of length 1m rotates vertically about one of its ends at angular velocity 5 radians per second. If the horizontal component of earth's magnetic field is $0.2 \times 10^{-4} \text{ T}$, then the e.m.f. developed between the two ends of the conductor is

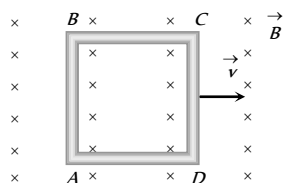
[MP PMT 1992; AIEEE 2004]

- (a) 5 mV (b) $5 \times 10^{-4} \text{ V}$
(c) 50 mV (d) 50 μV

10. A conducting square loop of side L and resistance R moves in its plane with a uniform velocity v perpendicular to one of its sides. A magnetic induction B constant in time and space, pointing perpendicular and into the plane of the loop exists everywhere. The current induced in the loop is

[IIT 1989; MP PET 1997; MP PMT 1996, 99; MP PMT 2002]

- (a) $\frac{Blv}{R}$ clockwise
(b) $\frac{Blv}{R}$ anticlockwise
(c) $\frac{2Blv}{R}$ anticlockwise
(d) Zero



11. A player with 3 m long iron rod runs towards east with a speed of 30 km/hr. Horizontal component of earth's magnetic field is $4 \times 10^{-5} \text{ Wb/m}^2$. If he is running with rod in horizontal and

vertical positions, then the potential difference induced between the two ends of the rod in two cases will be

- (a) Zero in vertical position and $1 \times 10^{-3} \text{ V}$ in horizontal position
(b) $1 \times 10^{-3} \text{ V}$ in vertical position and zero in horizontal position
(c) Zero in both cases
(d) $1 \times 10^{-3} \text{ V}$ in both cases

12. A coil of area 80 square cm and 50 turns is rotating with 2000 revolutions per minute about an axis perpendicular to a magnetic field of 0.05 Tesla. The maximum value of the e.m.f. developed in it is

[MP PMT 1994]

- (a) $200\pi \text{ volt}$ (b) $\frac{10\pi}{3} \text{ volt}$
(c) $\frac{4\pi}{3} \text{ volt}$ (d) $\frac{2}{3} \text{ volt}$

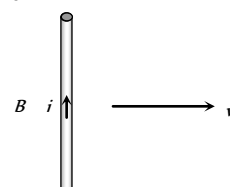
13. A conducting rod of length l is falling with a velocity v perpendicular to a uniform horizontal magnetic field B . The potential difference between its two ends will be

[MP PMT 1994]

- (a) $2Blv$ (b) Blv
(c) $\frac{1}{2}Blv$ (d) $B^2l^2v^2$

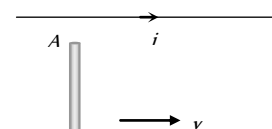
14. A conducting wire is moving towards right in a magnetic field B . The direction of induced current in the wire is shown in the figure. The direction of magnetic field will be

[MP PET 1995]



- (a) In the plane of paper pointing towards right
(b) In the plane of paper pointing towards left
(c) Perpendicular to the plane of paper and down-wards
(d) Perpendicular to the plane of paper and upwards

15. The current carrying wire and the rod AB are in the same plane. The rod moves parallel to the wire with a velocity v . Which one of the following statements is true about induced emf in the rod



- (a) End A will be at lower potential with respect to B
(b) A and B will be at the same potential
(c) There will be no induced e.m.f. in the rod
(d) Potential at A will be higher than that at B

16. A long horizontal metallic rod with length along the east-west direction is falling under gravity. The potential difference between its two ends will

[MP PMT 1997]

- (a) Be zero (b) Be constant
(c) Increase with time (d) Decrease with time

17. A two metre wire is moving with a velocity of 1 m/sec perpendicular to a magnetic field of 0.5 weber/m. The e.m.f. induced in it will be

[MP PMT/PET 1998; Pb PET 2003]

- (a) 0.5 volt (b) 0.1 volt
(c) 1 volt (d) 2 volt

18. A metal rod moves at a constant velocity in a direction perpendicular to its length. A constant uniform magnetic field exists in space in a direction perpendicular to the rod as well as its velocity. Select the correct statement(s) from the following

[CPMT 2003]

- (a) The entire rod is at the same electric potential
 (b) There is an electric field in the rod
 (c) The electric potential is highest at the centre of the rod and decreases towards its ends
 (d) The electric potential is lowest at the centre of the rod and increases towards its ends

19. A conducting wire is dropped along east-west direction, then [RPMT 1997]

- (a) No emf is induced
 (b) No induced current flows
 (c) Induced current flows from west to east
 (d) Induced current flows from east to west
20. The magnetic induction in the region between the pole faces of an electromagnet is 0.7 weber/m . The induced e.m.f. in a straight conductor 10 cm long, perpendicular to B and moving perpendicular both to magnetic induction and its own length with a velocity 2 m/sec is [AMU (Med.) 1999]
- (a) 0.08 V (b) 0.14 V
 (c) 0.35 V (d) 0.07 V

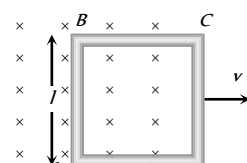
21. A straight conductor of length 0.4 m is moved with a speed of 7 m/s perpendicular to the magnetic field of intensity of 0.9 Wb/m . The induced e.m.f. across the conductor will be

[MH CET (Med.) 1999]

- (a) 7.25 V (b) 3.75 V
 (c) 1.25 V (d) 2.52 V
22. A coil of N turns and mean cross-sectional area A is rotating with uniform angular velocity ω about an axis at right angle to uniform magnetic field B . The induced e.m.f. E in the coil will be
- (a) $NBA \sin \omega t$ (b) $NB \omega \sin \omega t$
 (c) $NB/A \sin \omega t$ (d) $NBA \omega \sin \omega t$

23. A conducting square loop of side l and resistance R moves in its plane with a uniform velocity v perpendicular to one of its sides. A magnetic induction B constant in time and space, pointing perpendicular and into the plane at the loop exists everywhere with half the loop outside the field, as shown in figure. The induced e.m.f. is [AIEEE 2002]

- (a) Zero
 (b) RvB
 (c) $VB/l/R$
 (d) VB/l



24. A wheel with ten metallic spokes, each 0.5 m long is rotated with a speed of 120 rev/min in a plane normal to the earth's magnetic field at the place. If the magnitude of the field is 0.4 Gauss , the induced e.m.f. between the axle and the rim of the wheel is equal to

- (a) $1.256 \times 10^{-3} \text{ V}$ (b) $6.28 \times 10^{-4} \text{ V}$
 (c) $1.256 \times 10^{-4} \text{ V}$ (d) $6.28 \times 10^{-5} \text{ V}$

25. A metal rod of length 2 m is rotating with an angular velocity of 100 rad/sec in a plane perpendicular to a uniform magnetic field of 0.3 T . The potential difference between the ends of the rod is

- (a) 30 V (b) 40 V
 (c) 60 V (d) 600 V

26. The wing span of an aeroplane is 20 metre . It is flying in a field, where the vertical component of magnetic field of earth is $5 \times 10^{-5} \text{ tesla}$, with velocity 360 km/h . The potential difference produced between the blades will be

- (a) 0.10 V (b) 0.15 V
 (c) 0.20 V (d) 0.30 V

27. A horizontal straight conductor kept in north-south direction falls under gravity, then [MP PMT 2003]

- (a) A current will be induced from South to North
 (b) A current will be induced from North to South
 (c) No induce e.m.f. along the length of conductor
 (d) An induced e.m.f. is generated along the length of conductor

28. A rectangular coil of 300 turns has an average area of average area of $25 \text{ cm} \times 10 \text{ cm}$. The coil rotates with a speed of 50 cps in a uniform magnetic field of strength $4 \times 10^{-2} \text{ T}$ about an axis perpendicular of the field. The peak value of the induced e.m.f. is (in volt) [KCET 2004]

- (a) 3000π (b) 300π
 (c) 30π (d) 3π

29. A rod of length 20 cm is rotating with angular speed of 100 rps in a magnetic field of strength 0.5 T about its one end. What is the potential difference between two ends of the rod

[Orissa PMT 2004]

- (a) 2.28 V (b) 4.28 V
 (c) 6.28 V (d) 2.5 V

30. A circular metal plate of radius R is rotating with a uniform angular velocity ω with its plane perpendicular to a uniform magnetic field B . Then the emf developed between the centre and the rim of the plate is [UPSEAT 2004]

- (a) $\pi \omega B R^2$ (b) $\omega B R^2$
 (c) $\pi \omega B R^2 / 2$ (d) $\omega B R^2 / 2$

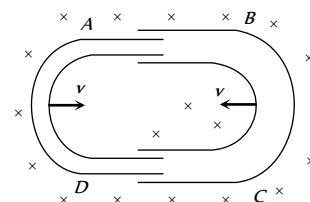
31. A circular coil of mean radius of 7 cm and having 4000 turns is rotated at the rate of 1800 revolutions per minute in the earth's magnetic field ($B = 0.5 \text{ gauss}$), the maximum e.m.f. induced in coil will be [Pb. PMT 2003]

- (a) 1.158 V (b) 0.58 V
 (c) 0.29 V (d) 5.8 V

32. One conducting U tube can slide inside another as shown in figure, maintaining electrical contacts between the tubes. The magnetic field B is perpendicular to the plane of the figure. If each tube moves towards the other at a constant speed v then the emf induced in the circuit in terms of B , l and v where l is the width of each tube, will be

[AIEEE 2005]

- (a) Zero
 (b) $2Blv$
 (c) Blv [AMU (Med.) 2002]
 (d) $-Blv$

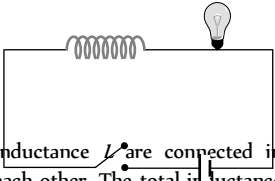


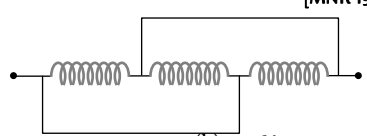
33. The magnitude of the earth's magnetic field at a place is B_0 and the angle of dip is δ . A horizontal conductor of length l lying along the magnetic north-south moves eastwards with a velocity v . The emf induced across the conductor is [MP PET 2003]

[Kerala PET 2005]

- (a) Zero (b) $B_0 l v \sin \delta$
 (c) $B_0 l v$ (d) $B_0 l v \cos \delta$

1. The back e.m.f. induced in a coil, when current changes from 1 ampere to zero in one *milli-second*, is 4 *volts*, the self inductance of the coil is [MP PET/PMT 1988]
 - (a) 1 H (b) 4 H
 - (c) 10^{-3} H (d) 4×10^{-3} H
2. An e.m.f. of 5 *volt* is produced by a self inductance, when the current changes at a steady rate from 3 A to 2 A in 1 millisecond. The value of self inductance is [CPMT 1982; MP PMT 1991; CBSE PMT 1993; AFMC 2002]
 - (a) Zero (b) 5 H
 - (c) 5000 H (d) 5 mH
3. A 50 mH coil carries a current of 2 *ampere*. The energy stored in joules is [MP PET/PMT 1988; MP PET 2005]
 - (a) 1 (b) 0.1
 - (c) 0.05 (d) 0.5
4. The current passing through a choke coil of 5 henry is decreasing at the rate of 2 *ampere/sec*. The e.m.f. developing across the coil is [CPMT 1982; MP PMT 1990; AIIMS 1997; MP PET 1999]
 - (a) 10 V (b) -10 V
 - (c) 2.5 V (d) -2.5 V
5. Average energy stored in a pure inductance L when a current i flows through it, is [MP PET/PMT 1988]
 - (a) Li^2 (b) $2Li^2$
 - (c) $\frac{Li^2}{4}$ (d) $\frac{Li^2}{2}$
6. A solenoid has 2000 turns wound over a length of 0.30 *metre*. The area of its cross-section is $1.2 \times 10^{-3} \text{ m}^2$. Around its central section, a coil of 300 turns is wound. If an initial current of 2 A in the solenoid is reversed in 0.25 *sec*, then the e.m.f. induced in the coil is [NCERT 1982; MP PMT 2003]
 - (a) 6×10^{-4} V (b) 4.8×10^{-3} V
 - (c) 6×10^{-2} V (d) 48 mV
7. A coil is wound as a transformer of rectangular cross-section. If all the linear dimensions of the transformer are increased by a factor 2 and the number of turns per unit length of the coil remain the same, the self inductance increased by a factor of
 - (a) 16 (b) 12
 - (c) 8 (d) 4
8. Two coils of self inductance L_1 and L_2 are placed closer to each other so that total flux in one coil is completely linked with other. If M is mutual inductance between them, then [DCE 2002]
 - (a) $M = L_1 L_2$ (b) $M = L_1 / L_2$
 - (c) $M = \sqrt{L_1 L_2}$ (d) $M = (L_1 L_2)^2$
9. The equivalent quantity of mass in electricity is
 - (a) Charge (b) Potential
 - (c) Inductance (d) Current
10. The momentum in mechanics is expressed as $m \times v$. The analogous expression in electricity is [MP PMT 2003]
 - (a) $I \times Q$ (b) $I \times V$
 - (c) $L \times I$ (d) $L \times Q$
11. In what form is the energy stored in an inductor or A coil of inductance L is carrying a steady current i . What is the nature of its stored energy [CBSE PMT 1990, 92; MP PMT 1996, 2000, 02; Kerala PMT 2002]
 - (a) Magnetic
 - (b) Electrical
 - (c) Both magnetic and electrical
 - (d) Heat
12. The coefficient of self inductance of a solenoid is 0.18 mH. If a core of soft iron of relative permeability 900 is inserted, then the coefficient of self inductance will become nearly
 - (a) 5.4 mH (b) 162 mH
 - (c) 0.006 mH (d) 0.0002 mH
13. In a transformer, the coefficient of mutual inductance between the primary and the secondary coil is 0.2 *henry*. When the current changes by 5 *ampere/second* in the primary, the induced e.m.f. in the secondary will be [MP PMT 1989]
 - (a) 5 V (b) 1 V
 - (c) 25 V (d) 10 V
14. When the current in a coil changes from 8 *ampere* to 2 *ampere* in 3×10^{-2} *second*, the e.m.f. induced in the coil is 2 *volt*. The self inductance of the coil (in *millihenry*) is [MNR 1991; UP SEAT 2000; Pb PET 2004]
 - (a) 1 (b) 5
 - (c) 20 (d) 10
15. The mutual inductance between two coils is 1.25 *henry*. If the current in the primary changes at the rate of 80 *ampere/second*, then the induced e.m.f. in the secondary is [MP PET 1990]
 - (a) 12.5 V (b) 64.0 V
 - (c) 0.016 V (d) 100.0 V
16. A coil of wire of a certain radius has 600 turns and a self inductance of 108 mH. The self inductance of a similar coil of 500 turns will be [MP PMT 1990]
 - (a) 74 mH (b) 75 mH
 - (c) 76 mH [AIIMS 1980] (d) 77 mH
17. When the number of turns in a coil is doubled without any change in the length of the coil, its self inductance becomes [MP PMT 1986; CBSE PMT 1992; Pb PET 2000]
 - (a) Four times (b) Doubled
 - (c) Halved (d) Unchanged
18. The average e.m.f. induced in a coil in which the current changes from 2 *ampere* to 4 *ampere* in 0.05 *second* is 8 *volt*. What is the self inductance of the coil? [NCERT 1984; CPMT 1997; MP PMT 1999, 2003; UPSEAT 2000; RPMT 2000; Pb. PMT 2002; RPET 2003; DPMT 2005]
 - (a) 0.1 H (b) 0.2 H
 - (c) 0.4 H (d) 0.8 H
19. If a current of 3.0 *amperes* flowing in the primary coil is reduced to zero in 0.001 *second*, then the induced e.m.f. in the secondary coil is 15000 *volts*. The mutual inductance between the two coils is
 - (a) 0.5 *henry* (b) 5 *henry*
 - (c) 1.5 *henry* (d) 10 *henry*
20. An e.m.f. of 12 *volts* is induced in a given coil when the current in it changes at the rate of 48 *amperes per minute*. The self inductance of the coil is [MP PMT 2000]
 - (a) 1 H (b) 10 H
 - (c) 100 H (d) 1000 H

- (a) 0.25 *henry* (b) 15 *henry*
(c) 1.5 *henry* (d) 9.6 *henry*
21. A closely wound coil of 100 turns and area of cross-section 1 cm^2 has a coefficient of self-induction 1 *mH*. The magnetic induction in the centre of the core of the coil when a current of 2 *A* flows in it, will be [MP PET 1992]
(a) 0.022 Wb m^{-2} (b) 0.4 Wb m^{-2}
(c) 0.8 Wb m^{-2} (d) 1 Wb m^{-2}
22. Two circuits have coefficient of mutual induction of 0.09 *henry*. Average e.m.f. induced in the secondary by a change of current from 0 to 20 *ampere* in 0.006 *second* in the primary will be
(a) 120 *V* (b) 80 *V*
(c) 200 *V* (d) 300 *V*
23. In the following circuit, the bulb will become suddenly bright if
(a) Contact is made or broken
(b) Contact is made
(c) Contact is broken
(d) Won't become bright at all
- 
24. Two pure inductors each of self inductance L are connected in parallel but are well separated from each other. The total inductance is [MP PET 1991; Pb. PMT 1999; BHU 1998, 05]
(a) $2L$ (b) L
(c) $\frac{L}{2}$ (d) $\frac{L}{4}$
25. A coil and a bulb are connected in series with a dc source, a soft iron core is then inserted in the coil. Then [MP PMT 1990; RPET 2001]
(a) Intensity of the bulb remains the same
(b) Intensity of the bulb decreases
(c) Intensity of the bulb increases
(d) The bulb ceases to glow
26. Self induction of a solenoid is [MP PMT 1993]
(a) Directly proportional to current flowing through the coil
(b) Directly proportional to its length
(c) Directly proportional to area of cross-section
(d) Inversely proportional to area of cross-section
27. Mutual inductance of two coils can be increased by [MP PET 1994]
(a) Decreasing the number of turns in the coils
(b) Increasing the number of turns in the coils
(c) Winding the coils on wooden core
(d) None of the above
28. The self inductance of a coil is 5 *henry*, a current of 1 *amp* change to 2 *amp* within 5 *second* through the coil. The value of induced e.m.f. will be [MP PET 1994;
Similar MP PET/PMT 1998; CBSE PMT 1990]
(a) 10 *volt* (b) 0.10 *volt*
(c) 1.0 *volt* (d) 100 *volt*
29. The unit of inductance is [MP PMT 1994, 95;
MP PET 1997; MP PMT/PET 1998; RPET 2001]
(a) *Volt/ampere* (b) *Joule/ampere*
(c) *Volt-sec/ampere* (d) *Volt-ampere/sec*
30. The current flowing in a coil of self inductance 0.4 *mH* is increased by 250 *mA* in 0.1 *sec*. The e.m.f. induced will be [MP PMT 1994]
(a) +1 *V* (b) -1 *V*
(c) +1 *mV* (d) -1 *mV*
31. 5 *cm* long solenoid having 10 *ohm* resistance and 5 *mH* inductance is joined to a 10 *volt* battery. At steady state the current through the solenoid in *ampere* will be [MP PET 1995]
(a) 5 (b) 1
(c) 2 (d) Zero
32. When current in a coil changes to 2 *ampere* from 8 *ampere* in 3×10^{-3} *second*, the e.m.f. induced in the coil is 2 *volt*. The self inductance of the coil in *millihenry* is [MP PET 1995]
[CBSE PMT 1989]
(a) 1 (b) 5
(c) 20 (d) 10
33. An ideal coil of 10 *henry* is joined in series with a resistance of 5 *ohm* and a battery of 5 *volt*. 2 *second* after joining, the current flowing in *ampere* in the circuit will be [MP PET 1995]
(a) e^{-1} (b) $(1 - e^{-1})$
(c) $(1 - e)$ (d) e
34. The number of turns of primary and secondary coils of a transformer are 5 and 10 respectively and the mutual inductance of the transformer is 25 *henry*. Now the number of turns in the primary and secondary of the transformer are made 10 and 5 respectively. The mutual inductance of the transformer in *henry* will be [MP PET 1995]
(a) 6.25 (b) 12.5
(c) 25 (d) 50
35. The inductance of a coil is 60 μH . A current in this coil increases from 1.0 *A* to 1.5 *A* in 0.1 *second*. The magnitude of the induced e.m.f. is [MP PMT 1995]
(a) $60 \times 10^{-6} \text{ V}$ (b) $300 \times 10^{-4} \text{ V}$
(c) $30 \times 10^{-4} \text{ V}$ (d) $3 \times 10^{-4} \text{ V}$
36. A circular coil of radius 5 *cm* has 500 turns of a wire. The approximate value of the coefficient of self induction of the coil will be [MP PET 1996; Pb PET 2000]
(a) 25 *millihenry* (b) 25×10^{-3} *millihenry*
(c) 50×10^{-3} *millihenry* (d) 50×10^{-3} *henry*
37. An e.m.f. of 100 *millivolts* is induced in a coil when the current in another nearby coil becomes 10 *ampere* from zero in 0.1 *second*. The coefficient of mutual induction between the two coils will be [MP PET 1996; Kerala PMT 2004]
(a) 1 *millihenry* (b) 10 *millihenry*
(c) 100 *millihenry* (d) 1000 *millihenry*
38. In a coil of self inductance 0.5 *henry*, the current varies at a constant rate from zero to 10 *amperes* in 2 *seconds*. The e.m.f. generated in the coil is [MP PMT 1996]
(a) 10 *volts* (b) 5 *volts*
(c) 2.5 *volts* (d) 1.25 *volts*
39. A coil of self inductance 50 *henry* is joined to the terminals of a battery of e.m.f. 2 *volts* through a resistance of 10 *ohm* and a steady current is flowing through the circuit. If the battery is now disconnected, the time in which the current will decay to $1/e$ of its steady value is [MP PMT 1996]

- (a) 500 seconds (b) 50 seconds
(c) 5 seconds (d) 0.5 seconds
40. The self inductance of a solenoid of length L , area of cross-section A and having N turns is [MP PET 1997; MP PET 2003]
- (a) $\frac{\mu_0 N^2 A}{L}$ (b) $\frac{\mu_0 NA}{L}$
(c) $\mu_0 N^2 LA$ (d) $\mu_0 NAL$
41. The self inductance of a coil is L . Keeping the length and area same, the number of turns in the coil is increased to four times. The self inductance of the coil will now be [MP PMT 1997]
- (a) $\frac{1}{4} L$ (b) L
(c) $4 L$ (d) $16 L$
42. The mutual inductance between a primary and secondary circuits is $0.5 H$. The resistances of the primary and the secondary circuits are 20 ohms and 5 ohms respectively. To generate a current of $0.4 A$ in the secondary, current in the primary must be changed at the rate of [MP PMT 1997]
- (a) $4.0 A/s$ (b) $16.0 A/s$
(c) $1.6 A/s$ (d) $8.0 A/s$
43. The energy stored in a 50 mH inductor carrying a current of $4 A$ will be [MP PET 1999]
- (a) $0.4 J$ (b) $4.0 J$
(c) $0.8 J$ (d) $0.04 J$
44. The average e.m.f. induced in a coil in which a current changes from 0 to $2 A$ in $0.05 s$ is $8 V$. The self inductance of the coil is
- (a) $0.1 H$ (b) $0.2 H$
(c) $0.4 H$ (d) $0.8 H$
45. If the current is halved in a coil, then the energy stored is how much times the previous value [CPMT 1999]
- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) 2 (d) 4
46. The SI unit of inductance, the henry, can be written as [IIT JEE 1998]
- (a) Weber/ampere (b) Volt-second/ampere
(c) Joule/(ampere) (d) Ohm-second
47. A varying current in a coil changes from 10 amp to zero in 0.5 sec . If average EMF is induced in the coil is 220 volts , the self inductance of coil is [EAMCET 1994; MH CET (Med.) 1999]
- (a) $5 H$ (b) $10 H$
(c) $11 H$ (d) $12 H$
48. Which of the following is wrong statement [AMU 1995]
- (a) An emf can be induced between the ends of a straight conductor by moving it through a uniform magnetic field
(b) The self induced emf produced by changing current in a coil always tends to decrease the current
(c) Inserting an iron core in a coil increases its coefficient of self induction
(d) According to Lenz's law, the direction of the induced current is such that it opposes the flux change that causes it
49. A coil has an inductance of $2.5 H$ and a resistance of 0.5Ω . If the coil is suddenly connected across a 6.0 volt battery, then the time required for the current to rise 0.63 of its final value is
- (a) 3.5 sec (b) 4.0 sec
(c) 4.5 sec (d) 5.0 sec
50. When the number of turns and the length of the solenoid are doubled keeping the area of cross-section same, the inductance [CBSE PMT 1993]
- (a) Remains the same (b) Is halved
(c) Is doubled (d) Becomes four times
51. A 100 mH coil carries a current of 1 ampere . Energy stored in its magnetic field is [CBSE PMT 1992; KCET 1998]
- (a) $0.5 J$ (b) $1 J$
(c) $0.05 J$ (d) $0.1 J$
52. The mutual inductance of an induction coil is $5 H$. In the primary coil, the current reduces from $5 A$ to zero in $10^{-3} s$. What is the induced emf in the secondary coil [RPET 1996]
- (a) $2500 V$ (b) $25000 V$
(c) $2510 V$ (d) Zero
53. The self inductance of a straight conductor is [KCET 1998]
- (a) Zero (b) Very large
(c) Infinity (d) Very small
54. What is the coefficient of mutual inductance when the magnetic flux changes by $2 \times 10^{-2} Wb$ and change in current is $0.01 A$ [BHU 1998; AIIMS 2002]
- (a) 2 henry (b) 3 henry
(c) $\frac{1}{2} \text{ henry}$ (d) Zero
55. The current in a coil changes from 4 ampere to zero in $0.1 s$. If the average e.m.f. induced is 100 volt , what is the self inductance of the coil [CPMT 1999] [MNR 1998]
- (a) $2.5 H$ (b) $25 H$
(c) $400 H$ (d) $40 H$
56. Pure inductance of $3.0 H$ is connected as shown below. The equivalent inductance of the circuit is [MNR 1998; AIEEE 2002]
- 
- (a) $1 H$ (b) $2 H$
(c) $3 H$ (d) $9 H$
57. A varying current at the rate of $3 A/s$ in a coil generates an e.m.f. of 8 mV in a nearby coil. The mutual inductance of the two coils is
- (a) 2.66 mH (b) $2.66 \times 10^{-3} \text{ mH}$
(c) $2.66 H$ (d) $0.266 H$
58. If a current of $10 A$ flows in one second through a coil, and the induced e.m.f. is $10 V$, then the self-inductance of the coil is [CPMT 2000; Pb. PMT 1999]
- (a) $\frac{2}{5} H$ (b) $\frac{4}{5} H$
(c) $\frac{5}{4} H$ (d) $1 H$
59. The inductance of a closed-packed coil of 400 turns is 8 mH . A current of 5 mA is passed through it. The magnetic flux through each turn of the coil is [Roorkee 2000]
- (a) $\frac{1}{4\pi} \mu_0 Wb$ (b) $\frac{1}{2\pi} \mu_0 Wb$
(c) $\frac{1}{3\pi} \mu_0 Wb$ (d) $0.4 \mu_0 Wb$

60. When the current through a solenoid increases at a constant rate, the induced current [UPSEAT 2000]
 (a) Is constant and is in the direction of the inducing current
 (b) Is a constant and is opposite to the direction of the inducing current
 (c) Increases with time and is in the direction of the inducing current
 (d) Increases with time and opposite to the direction of the inducing current
61. If in a coil rate of change of area is 5 m/milli second and current become 1 amp from 2 amp in $2 \times 10^{-3} \text{ sec}$. If magnitude of field is 1 tesla then self inductance of the coil is [RPET 2000]
 (a) 2 H (b) 5 H
 (c) 20 H (d) 10 H
62. The inductance of a solenoid 0.5 m long of cross-sectional area 20 cm and with 500 turns is [AMU (Med.) 2000]
 (a) 12.5 mH (b) 1.25 mH
 (c) 15.0 mH (d) 0.12 mH
63. The equivalent inductance of two inductances is 2.4 henry when connected in parallel and 10 henry when connected in series. The difference between the two inductances is [MP PMT 2000]
 (a) 2 henry (b) 3 henry
 (c) 4 henry (d) 5 henry
64. An e.m.f. of 12 volt is produced in a coil when the current in it changes at the rate of 45 amp/minute . The inductance of the coil is
 (a) 0.25 henry (b) 1.5 henry
 (c) 9.6 henry (d) 16.0 henry
65. An average induced e.m.f. of 1 V appears in a coil when the current in it is changed from 10 A in one direction to 10 A in opposite direction in 0.5 sec . Self-inductance of the coil is [CPMT 2001]
 (a) 25 mH (b) 50 mH
 (c) 75 mH (d) 100 mH
66. A coil of resistance 10Ω and an inductance 5 H is connected to a 100 volt battery. Then energy stored in the coil is [Pb. PMT 2001; CPMT 2002]
 (a) 125 erg (b) 125 J
 (c) 250 erg (d) 250 J
67. If a change in current of 0.01 A in one coil produces a change in magnetic flux of $1.2 \times 10^{-2} \text{ Wb}$ in the other coil, then the mutual inductance of the two coils in henries is [EAMCET 2001]
 (a) 0 (b) 0.5
 (c) 1.2 (d) 3
68. Energy stored in a coil of self inductance 40 mH carrying a steady current of 2 A is [Kerala (Engg.) 2001]
 (a) 0.8 J (b) 8 J
 (c) 0.08 J (d) 80 J
69. A solenoid of length $l \text{ metre}$ has self-inductance $L \text{ henry}$. If number of turns are doubled, its self inductance [MP PMT 2001]
 (a) Remains same (b) Becomes $2L \text{ henry}$
 (c) Becomes $4L \text{ henry}$ (d) Becomes $\frac{L}{\sqrt{2}} \text{ henry}$
70. Two coils A and B having turns 300 and 600 respectively are placed near each other, on passing a current of 3.0 ampere in A , the flux linked with A is $1.2 \times 10^{-4} \text{ weber}$ and with B it is $9.0 \times 10^{-5} \text{ weber}$. The mutual inductance of the system is
 (a) $2 \times 10^{-4} \text{ henry}$ (b) $3 \times 10^{-4} \text{ henry}$
 (c) $4 \times 10^{-4} \text{ henry}$ (d) $6 \times 10^{-4} \text{ henry}$
71. In a circular conducting coil, when current increases from 2 A to 18 A in 0.05 sec , the induced e.m.f. is 20 V . The self inductance of the coil is [MP PET 2001]
 (a) 62.5 mH (b) 6.25 mH
 (c) 50 mH (d) None of these
72. Find out the e.m.f. produced when the current changes from 0 to 1 A in 10 second , given $L = 10 \mu\text{H}$ [DCE 2001]
 (a) 1 V (b) $1 \mu\text{V}$
 (c) 1 mV (d) 0.1 V
73. Which of the following is not the unit of self inductance [AMU (Med.) 2001]
 (a) *Weber/Ampere* (b) *Ohm-Second*
 (c) *Joule-Ampere* (d) *Joule Ampere*
74. A coil of 100 turns carries a current of 5 mA and creates a magnetic flux of 10^{-4} weber . the inductance is [Orissa JEE 2002]
 (a) 0.2 mH (b) 2.0 mH
 (c) 0.02 mH (d) None of these
75. In circular coil, when no. of turns is doubled and resistance becomes $\frac{1}{4}$ th of initial, then inductance becomes [AIIEE 2002]
 (a) 4 times (b) 2 times
 (c) 8 times (d) No change
76. The current in a coil of inductance 5 H decreases at the rate of 2 A/s . The induced e.m.f. is [MH CET 2002]
 (a) 2 V (b) 5 V
 (c) 10 V (d) -10 V
77. The self-induced e.m.f. in a 0.1 H coil when the current in it is changing at the rate of $200 \text{ ampere/second}$ is [DPMT 2002]
 (a) $8 \times 10^{-4} \text{ V}$ (b) $8 \times 10^{-5} \text{ V}$
 (c) 20 V (d) 125 V
78. Two circuits have mutual inductance of 0.1 H . What average e.m.f. is induced in one circuit when the current in the other circuit changes from 0 to 20 A in 0.02 s [Kerala PET 2002]
 (a) 240 V (b) 230 V
 (c) 100 V (d) 300 V
79. An air core solenoid has 1000 turns and is one *metre* long. Its cross-sectional area is 10 cm . Its self inductance is [JIPMER 2002]
 (a) 0.1256 mH (b) 12.56 mH
 (c) 1.256 mH (d) 125.6 mH
80. The coefficient of mutual inductance of two coils is 6 mH . If the current flowing in one is 2 ampere , then the induced e.m.f. in the second coil will be [BVP 2003]
 (a) 3 mV (b) 2 mV
 (c) 3 V (d) Zero

81. An L - R circuit has a cell of e.m.f. E , which is switched on at time $t = 0$. The current in the circuit after a long time will be

[MP PET 2003]

- (a) Zero (b) $\frac{E}{R}$
(c) $\frac{E}{L}$ (d) $\frac{E}{\sqrt{L^2 + R^2}}$

82. Two coils are placed close to each other. The mutual inductance of the pair of coils depends upon [AIEEE 2003]

- (a) The currents in the two coils
(b) The rates at which currents are changing in the two coils
(c) Relative position and orientation of the two coils
(d) The materials of the wires of the coils

83. When the current change from $+2A$ to $-2A$ in 0.05 second, an e.m.f. of 8 V is induced in a coil. The coefficient of self-induction of the coil is [AIEEE 2003]

- (a) 0.1 H (b) 0.2 H
(c) 0.4 H (d) 0.8 H

84. A coil resistance 20Ω and inductance $5H$ is connected with a 100 V battery. Energy stored in the coil will be

[MP PMT 2003]

- (a) 41.5 J (b) 62.50 J
(c) 125 J (d) 250 J

85. Why the current does not rise immediately in a circuit containing inductance [EAMCET 1994]

- (a) Because of induced emf
(b) Because of high voltage drop
(c) Because of low power consumption
(d) Because of Joule heating

86. Two circular coils have their centres at the same point. The mutual inductance between them will be maximum when their axes

- (a) Are parallel to each other
(b) Are at 60° to each other
(c) Are at 45° to each other
(d) Are perpendicular to each other

87. The current in a coil decreases from 1 A to 0.2 A. In 10 sec. Calculate the coefficient of self inductance. If induced emf is 0.4 volt.

- (a) 5 H (b) 3 H
(c) 4 H (d) 2 H

88. The current through choke coil increases from zero to 6 A in 0.3 seconds and an induced e.m.f. of 30 V is produced. The inductance of the coil of choke is [MP PMT 2004]

- (a) 5 H (b) 2.5 H
(c) 1.5 H (d) 2 H

89. The resistance and inductance of series circuit are 5Ω and $20H$ respectively. At the instant of closing the switch, the current is increasing at the rate $4A/s$. The supply voltage is

[MP PMT 2004]

- (a) 20 V (b) 80 V
(c) 120 V (d) 100 V

90. A coil of $N = 100$ turns carries a current $I = 5$ A and creates a magnetic flux $\phi = 10^{-5} Tm^{-2}$ per turn. The value of its inductance L will be [UPSEAT 2004]

- (a) 0.05 mH (b) 0.10 mH
(c) 0.15 mH (d) 0.20 mH

91. Two identical induction coils each of inductance L joined in series are placed very close to each other such that the winding direction of one is exactly opposite to that of the other, what is the net inductance [DCE 2003]

- (a) L (b) $2L$
(c) $L/2$ (d) Zero

92. If the current 30 A flowing in the primary coil is made zero in 0.1 sec. The emf induced in the secondary coil is 1.5 volt. The mutual inductance between the coil is [Pb PMT 2003]

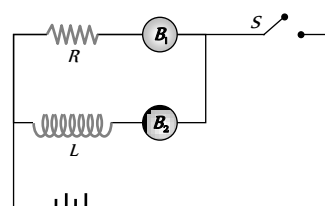
- (a) 0.05 H (b) 1.05 H
(c) 0.1 H (d) 0.2 H

93. Eddy currents are used in [AFMC 2004]

- (a) Induction furnace (b) Electromagnetic brakes
(c) Speedometers (d) All of these

94. The adjoining figure shows two bulbs B_1 and B_2 , resistor R and an inductor L . When the switch S is turned off

[CPMT 1989]



- (a) Both B and L die out promptly
(b) Both B and L die out with some delay
(c) B dies out promptly but L with some delay
(d) L dies out promptly but B with some delay

95. In L - R circuit, for the case of increasing current, the magnitude of current can be calculated by using the formula

[MP PET 1994]

- (a) $I = I_0 e^{-Rt/L}$ (b) $I = I_0 (1 - e^{-Rt/L})$
(c) $I = I_0 (1 - e^{Rt/L})$ (d) $I = I_0 e^{Rt/L}$

96. An inductor of inductance L and a resistance R are first connected to a battery. After some time the battery is disconnected but L and R remain connected in a closed circuit. Then the current reduces to 37% of its initial value in [MP PMT 1994]

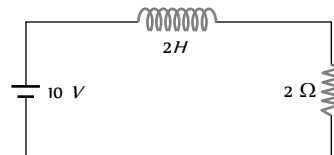
- (a) RL sec (b) $\frac{R}{L}$ sec
(c) $\frac{L}{R}$ sec (d) $\frac{1}{LR}$ sec

97. In an LR -circuit, time constant is that time in which current grows from zero to the value (where I_0 is the steady state current) [MP PMT/PET 1994]

- (a) $0.63 I_0$ (b) $0.50 I_0$
(c) $0.37 I_0$ (d) I_0

98. In the figure magnetic energy stored in the coil is

[RPET 2000]

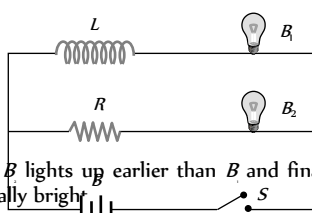


- (a) Zero (b) Infinite
(c) 25 joules (d) None of the above

99. A capacitor is fully charged with a battery. Then the battery is removed and coil is connected with the capacitor in parallel, current varies as [RPET 2000; DCE 2000]

- (a) Increases monotonically (b) Decreases monotonically
(c) Zero (d) Oscillates indefinitely

100. A coil of inductance 40 henry is connected in series with a resistance of 8 ohm and the combination is joined to the terminals of a 2 volt battery. The time constant of the circuit is
(a) 40 seconds (b) 20 seconds
(c) 8 seconds (d) 5 seconds
101. A solenoid has an inductance of 60 henrys and a resistance of 30 ohms. If it is connected to a 100 volt battery, how long will it take for the current to reach $\frac{e-1}{e} \approx 63.2\%$ of its final value
(a) 1 second (b) 2 seconds
(c) e seconds (d) $2e$ seconds
102. An inductor, L a resistance R and two identical bulbs, B_1 and B_2 are connected to a battery through a switch S as shown in the figure. The resistance R is the same as that of the coil that makes L . Which of the following statements gives the correct description of the happenings when the switch S is closed



- (a) The bulb B_1 lights up earlier than B_2 and finally both the bulbs shine equally brightly
(b) B_1 light up earlier and finally both the bulbs acquire equal brightness
(c) B_1 lights up earlier and finally B_1 shines brighter than B_2
(d) B_1 and B_2 light up together with equal brightness all the time
103. The time constant of an LR circuit represents the time in which the current in the circuit [MP PMT 2002]
(a) Reaches a value equal to about 37% of its final value
(b) Reaches a value equal to about 63% of its final value
(c) Attains a constant value
(d) Attains 50% of the constant value
104. A LC circuit is in the state of resonance. If $C = 0.1 \mu F$ and $L = 0.25$ henry. Neglecting ohmic resistance of circuit what is the frequency of oscillations [BHU 2003; MP PMT 2005]
(a) 1007 Hz (b) 100 Hz
(c) 109 Hz (d) 500 Hz
105. An oscillator circuit consists of an inductance of $0.5 mH$ and a capacitor of $20 \mu F$. The resonant frequency of the circuit is nearly
(a) 15.92 Hz (b) 159.2 Hz
(c) 1592 Hz (d) 15910 Hz
106. A coil of inductance $300 mH$ and resistance 2Ω is connected to a source of voltage $2V$. The current reaches half of its steady state value in [AIEEE 2005]
(a) 0.15 s (b) 0.3 s
(c) 0.05 s (d) 0.1 s
107. A coil having an inductance of $0.5 H$ carries a current which is uniformly varying from zero to 10 ampere in 2 second. The e.m.f. (in volts) generated in the coil is [Kerala PET 2005]
(a) 10 (b) 5
(c) 2.5 (d) 1.25
108. The square root of the product of inductance and capacitance has the dimension of [KCET 2005]
(a) Length (b) Mass

- (c) Time (d) No dimension

Application of EMI (Motor, Dynamo, Transformer...)

1. Which of the following does not depend upon the magnetic effect of some sort [MP PET 2000]
(a) Moving coil galvanometer
(b) Helium neon laser
(c) Dynamo
(d) Electric motor
2. Use of eddy currents is done in the following except
(a) Moving coil galvanometer
(b) Electric brakes
(c) Induction motor [AMU (Med.) 2002]
(d) Dynamo
3. Plane of eddy currents makes an angle with the plane of magnetic lines of force equal to
(a) 40° (b) 0°
(c) 90° (d) 180°
4. Which of the following is constructed on the principle of electromagnetic induction [MP PMT 2002]
(a) Galvanometer (b) Electric motor
(c) Generator (d) Voltmeter
5. A transformer is based on the principle of [AIIMS 1998; AFMC 2005]
(a) Mutual inductance (b) Self inductance
(c) Ampere's law (d) Lenz's law
6. Which of the following is not an application of eddy currents [CBSE PMT 1989]
(a) Induction furnace
(b) Galvanometer damping
(c) Speedometer of automobiles
(d) X-ray crystallography
7. The core of a transformer is laminated to reduce energy losses due to [Kerala PET 2002] [CBSE PMT 1990; Karnataka CET (Med.) 2001]
(a) Eddy currents (b) Hysteresis
(c) Resistance in winding (d) None of these
8. The pointer of a dead-beat galvanometer gives a steady deflection because [MP PMT 1994]
(a) Eddy currents are produced in the conducting frame over which the coil is wound
(b) Its magnet is very strong
(c) Its pointer is very light
(d) Its frame is made of abonite
9. The device that does not work on the principle of mutual induction is [KCET 1994]
(a) Induction coil (b) Motor
(c) Tesla coil (d) Transformer
10. Eddy currents are produced when [CBSE PMT 1993; AFMC 2002]
(a) A metal is kept in varying magnetic field

- (b) A metal is kept in the steady magnetic field
(c) A circular coil is placed in a magnetic field
(d) Through a circular coil, current is passed
11. If rotational velocity of a dynamo armature is doubled, then induced e.m.f. will become [MP PMT 1991; AIIMS 2000]
(a) Half (b) Two times
(c) Four times (d) Unchanged
12. Dynamo is a device for converting
(a) Electrical energy into mechanical energy
(b) Mechanical energy into electrical energy
(c) Chemical energy into mechanical energy
(d) Mechanical energy into chemical energy
13. The working of dynamo is based on principle of [CPMT 1984]
(a) Electromagnetic induction
(b) Conversion of energy into electricity
(c) Magnetic effects of current
(d) Heating effects of current
14. Choke coil works on the principle of [MP PET/PMT 1988]
(a) Transient current (b) Self induction
(c) Mutual induction (d) Wattless current
15. When the speed of a dc motor increases the armature current [CPMT 1984, 85; MP PMT 2004]
(a) Increases
(b) Decreases
(c) Does not change
(d) Increases and decreases continuously
16. The output of a dynamo using a splitting commutator is
(a) dc
(b) ac
(c) Fluctuating dc
(d) Half-wave rectified voltage
17. Which of the following statement is incorrect
(a) Both ac and dc dynamo have a field magnet
(b) Both ac and dc dynamo have an armature
(c) Both ac and dc dynamo convert mechanical energy into electrical energy
(d) Both ac and dc dynamo have slip rings
18. The coil of dynamo is rotating in a magnetic field. The developed induced e.m.f. changes and the number of magnetic lines of force also changes. Which of the following condition is correct
(a) Lines of force minimum but induced e.m.f. is zero
(b) Lines of force maximum but induced e.m.f. is zero
(c) Lines of force maximum but induced e.m.f. is not zero
(d) Lines of force maximum but induced e.m.f. is also maximum
19. Dynamo core is laminated because [MP PET 1995]
(a) Magnetic field increases
(b) Magnetic saturation level in core increases
(c) Residual magnetism in core decreases
(d) Loss of energy in core due to eddy currents decreases
20. Armature current in dc motor will be maximum when [CPMT 1986, 88; MP PET 1995]
(a) Motor has acquired maximum speed
(b) Motor has acquired intermediate speed
(c) Motor has just started moving
(d) Motor is switched off
21. The armature of dc motor has $20\ \Omega$ resistance. It draws current of 1.5 ampere when run by 220 volts dc supply. The value of back e.m.f. induced in it will be [MP PMT 1999]
(a) 150 V (b) 170 V
(c) 180 V (d) 190 V
22. In an induction coil, the secondary e.m.f. is [KCET 1994]
(a) Zero during break of the circuit
(b) Very high during make of the circuit
(c) Zero during make of the circuit
(d) Very high during break of the circuit
23. The number of turns in the coil of an ac generator is 5000 and the area of the coil is $0.25\ m^2$. The coil is rotated at the rate of 100 cycles/sec in a magnetic field of $0.2\ W/m^2$. The peak value of the emf generated is nearly [AMU 1995]
(a) 786 kV (b) 440 kV
(c) 220 kV (d) 157.1 kV
24. In a dc motor, induced e.m.f. will be maximum [RPMT 1997]
(a) When motor takes maximum speed
(b) When motor starts rotating
(c) When speed of motor increases
(d) When motor is switched off
25. Work of electric motor is [RPMT 1997]
(a) To convert ac into dc
(b) To convert dc into ac
(c) Both (a) and (b)
(d) To convert ac into mechanical work
26. In an induction coil with resistance, the induced emf will be maximum when [RPMT 1996]
(a) The switch is put on due to high resistance
(b) The switch is put off due to high resistance
(c) The switch is put on due to low resistance
(d) The switch is put off due to low resistance
27. An electric motor operating on a 60 V dc supply draws a current of 10 A. If the efficiency of the motor is 50%, the resistance of its winding is [MP PET 1993] [AMU (Engg.) 2001]
(a) $3\ \Omega$ (b) $6\ \Omega$
(c) $15\ \Omega$ (d) $30\ \Omega$
28. A device which converts electrical energy into mechanical energy is
(a) Dynamo (b) generator
(c) Electric motor (d) Induction coil
29. An electric motor operates on a 50 volt supply and a current of 12 A. If the efficiency of the motor is 30%, what is the resistance of the winding of the motor [Kerala PET 2002]
(a) $6\ \Omega$ (b) $4\ \Omega$
(c) $2.9\ \Omega$ (d) $3.1\ \Omega$

30. A motor having an armature of resistance 2Ω is designed to operate at 220 V mains. At full speed, it develops a back e.m.f. of 210 V. When the motor is running at full speed, the current in the armature is [UPSEAT 2002]
- (a) 5 A (b) 105 A
(c) 110 A (d) 215 A
31. Fan is based on [AFMC 2003]
- (a) Electric Motor (b) Electric dynamo
(c) Both (d) None of these
32. A transformer is employed to [MP PET 1985; MP PMT 1993; RPET 1999]
- (a) Obtain a suitable dc voltage
(b) Convert dc into ac
(c) Obtain a suitable ac voltage
(d) Convert ac into dc
33. What is increased in step-down transformer [MP PMT/PET 1998; CPMT 1999]
- (a) Voltage (b) Current
(c) Power (d) Current density
34. The core of a transformer is laminated so that [CPMT 1985; MP PMT 1994, 2000, 02, 03; BHU 1999]
- (a) Ratio of voltage in the primary and secondary may be increased
(b) Rusting of the core may be stopped
(c) Energy losses due to eddy currents may be reduced
(d) Change in flux is increased
35. In transformer, core is made of soft iron to reduce [AIIMS 1998; UPSEAT 2001; AFMC 2005]
- (a) Hysteresis losses
(b) Eddy current losses
(c) Force opposing electric current
(d) None of the above
36. The transformation ratio in the step-up transformer is
- (a) 1
(b) Greater than one
(c) Less than one
(d) The ratio greater or less than one depends on the other factors
37. In a transformer 220 ac voltage is increased to 2200 volts. If the number of turns in the secondary are 2000, then the number of turns in the primary will be [MP PET/PMT 1988]
- (a) 200 (b) 100
(c) 50 (d) 20
38. The ratio of secondary to the primary turns in a transformer is 3 : 2. If the power output be P , then the input power neglecting all losses must be equal to [MP PMT 1984; KCET 2003]
- (a) $5P$ (b) $1.5P$
(c) P (d) $\frac{2}{5}P$
39. The primary winding of a transformer has 100 turns and its secondary winding has 200 turns. The primary is connected to an ac supply of 120 V and the current flowing in it is 10 A. The voltage and the current in the secondary are [MP PMT 1991; DPMT 2004]
- (a) 240 V, 5 A (b) 240 V, 10 A
(c) 60 V, 20 A (d) 120 V, 20 A
40. A step-down transformer is connected to 2400 volts line and 80 amperes of current is found to flow in output load. The ratio of the turns in primary and secondary coil is 20 : 1. If transformer efficiency is 100%, then the current flowing in primary coil will be
- (a) 1600 A (b) 20 A
(c) 4 A (d) 1.5 A
41. A loss free transformer has 500 turns on its primary winding and 2500 in secondary. The meters of the secondary indicate 200 volts at 8 amperes under these conditions. The voltage and current in the primary is [MP PMT 1996]
- (a) 100 V, 16 A (b) 40 V, 40 A
(c) 160 V, 10 A (d) 80 V, 20 A
42. An ideal transformer has 100 turns in the primary and 250 turns in the secondary. The peak value of the ac is 28 V. The r.m.s. secondary voltage is nearest to [MP PMT 1992]
- (a) 50 V (b) 70 V
(c) 100 V (d) 40 V
43. A transformer is employed to reduce 220 V to 11 V. The primary draws a current of 5 A and the secondary 90 A. The efficiency of the transformer is [MP PMT 1992, 2001, 04]
- (a) 20% (b) 40%
(c) 70% (d) 90%
44. In a step-up transformer, the turn ratio is 1 : 2. A Leclanche cell (e.m.f. 1.5V) is connected across the primary. The voltage developed in the secondary would be [MP PET 1992, 99; AIIMS 2000; MP PMT 2000; RPET 2001]
- (a) 3.0 V (b) 0.75 V
(c) 1.5 V (d) Zero
45. The alternating voltage induced in the secondary coil of a transformer is mainly due to [MP PET 1992; MP PMT 1996]
- (a) A varying electric field
(b) A varying magnetic field
(c) The vibrations of the primary coil
(d) The iron core of the transformer
46. We can reduce eddy currents in the core of transformer [MP PET 1993]
- (a) By increasing the number of turns in secondary coil
(b) By taking laminated core
(c) By making step-down transformer
(d) By using a weak ac at high potential
47. A 100% efficient transformer has 100 turns in the primary and 25 turns in its secondary coil. If the current in the secondary coil is 4 amp, then the current in the primary coil is

- (a) 1 amp (b) 4 amp
(c) 8 amp (d) 16 amp
48. The efficiency of transformer is very high because [MP PET 1994]
(a) There is no moving part in a transformer
(b) It produces very high voltage
(c) It produces very low voltage
(d) None of the above
49. In a lossless transformer an alternating current of 2 amp is flowing in the primary coil. The number of turns in the primary and secondary coils are 100 and 20 respectively. The value of the current in the secondary coil is [MP PMT 1994]
(a) 0.08 A (b) 0.4 A
(c) 5 A (d) 10 A
50. A transformer connected to 220 volt line shows an output of 2 A at 11000 volt. The efficiency is 100%. The current drawn from the line is
(a) 100 A (b) 200 A
(c) 22 A (d) 11 A
51. The coils of a step down transformer have 500 and 5000 turns. In the primary coil an ac of 4 ampere at 2200 volts is sent. The value of the current and potential difference in the secondary coil will be
(a) 20 A, 220 V (b) 0.4 A, 22000 V
(c) 40 A, 220 V (d) 40 A, 22000 V
52. A power transformer is used to step up an alternating e.m.f. of 220 V to 11 kV to transmit 4.4 kW of power. If the primary coil has 1000 turns, what is the current rating of the secondary ? Assume 100% efficiency for the transformer [MP PET 1997]
(a) 4 A (b) 0.4 A
(c) 0.04 A (d) 0.2 A
53. A step up transformer connected to a 220 V AC line is to supply 22 kV for a neon sign in secondary circuit. In primary circuit a fuse wire is connected which is to blow when the current in the secondary circuit exceeds 10 mA. The turn ratio of the transformer is [MP PET 1997]
(a) 50 (b) 100
(c) 150 (d) 200
54. In a transformer the primary has 500 turns and secondary has 50 turns. 100 volts are applied to the primary coil, the voltage developed in the secondary will be [MP PMT 1997]
(a) 1 V (b) 10 V
(c) 1000 V (d) 10000 V
55. A transformer is used to [MP PET 1999]
(a) Change the alternating potential
(b) Change the alternating current
(c) To prevent the power loss in alternating current flow
(d) To increase the power of current source
56. A step-up transformer operates on a 230 V line and supplies a load of 2 ampere. The ratio of the primary and secondary windings is 1 : 25. The current in the primary is [CBSE PMT 1998]
(a) 15 A (b) 50 A
(c) 25 A (d) 12.5 A
57. The number of turns in the primary coil of a transformer is 200 and the number of turns in the secondary coil is 10. If 240 volt AC is applied to the primary, the output from the secondary will be [BHU 1997; JIPMER 1999]
(a) 48 V (b) 24 V
(c) 12 V (d) 6 V
58. The primary winding of transformer has 500 turns whereas its secondary has 5000 turns. The primary is connected to an ac supply of 20 V, 50 Hz. The secondary will have an output of [CBSE PMT 1997; AIIMS 1999]
(a) 200 V, 50 Hz (b) 2 V, 50 Hz
(c) 200 V, 500 Hz (d) 2 V, 5 Hz
59. A step-up transformer has transformation ratio of 3 : 2. What is the voltage in secondary if voltage in primary is 30 V [MP PMT 1995]
(a) 45 V (b) 15 V
(c) 90 V (d) 300 V
60. In a transformer, the number of turns in primary coil and secondary coil are 5 and 4 respectively. If 240 V is applied on the primary coil, then the ratio of current in primary and secondary coil is [AFMC 1998; CPMT 2000]
(a) 4 : 5 (b) 5 : 4
(c) 5 : 4 [MP PET 1996] (d) 9 : 5
61. A step-down transformer is connected to main supply 200 V to operate a 6 V, 30 W bulb. The current in primary is [AMU (Engg.) 1999]
(a) 3 A (b) 1.5 A
(c) 0.3 A (d) 0.15 A
62. The number of turns in primary and secondary coils of a transformer are 100 and 20 respectively. If an alternating potential of 200 volt is applied to the primary, the induced potential in secondary will be [RPET 1999]
(a) 10 V (b) 40 V
(c) 1000 V (d) 20,000 V
63. The ratio of secondary to primary turns is 9 : 4. If power input is P , what will be the ratio of power output (neglect all losses) to power input [DCE 1999]
(a) 4 : 9 (b) 9 : 4
(c) 5 : 4 (d) 1 : 1
64. Voltage in the secondary coil of a transformer does not depend upon. [BHU 2000]
(a) Voltage in the primary coil
(b) Ratio of number of turns in the two coils
(c) Frequency of the source
(d) Both (a) and (b)
65. A transformer has turn ratio 100/1. If secondary coil has 4 amp current then current in primary coil is [RPET 2000]
(a) 4 A (b) 0.04 A
(c) 0.4 A (d) 400 A
66. In a step-up transformer the turn ratio is 1:10. A resistance of 200 ohm connected across the secondary is drawing a current of 0.5 A. What is the primary voltage and current [MP PET 2000]
(a) 50 V, 1 amp (b) 10 V, 5 amp
(c) 25 V, 4 amp (d) 20 V, 2 amp

67. Large transformers, when used for some time, become hot and are cooled by circulating oil. The heating of transformer is due to
(a) Heating effect of current alone
(b) Hysteresis loss alone
(c) Both the hysteresis loss and heating effect of current
(d) None of the above
68. In a step-up transformer the voltage in the primary is 220 V and the current is 5 A. The secondary voltage is found to be 22000 V. The current in the secondary (neglect losses) is
[Kerala PMT 2002]
(a) 5 A (b) 50 A
(c) 500 A (d) 0.05 A
69. In a transformer, number of turns in the primary are 140 and that in the secondary are 280. If current in primary is 4 A then that in the secondary is
[AIEEE 2002]
(a) 4 A (b) 2 A
(c) 6 A (d) 10 A
70. A transformer has 100 turns in the primary coil and carries 8 A current. If input power is one kilowatt, the number of turns required in the secondary coil to have 500 V output will be
[MP PET 2002]
(a) 100 (b) 200
(c) 400 (d) 300
71. An ideal transformer has 500 and 5000 turn in primary and secondary windings respectively. If the primary voltage is connected to a 6V battery then the secondary voltage is
[Orissa JEE 2003]
(a) 0 (b) 60 V
(c) 0.6 V (d) 6.0 V
72. In a primary coil 5A current is flowing on 220 volts. In the secondary coil 2200 V voltage produces. Then ratio of number of turns in secondary coil and primary coil will be
[RPET 2003]
(a) 1 : 10 (b) 10 : 1
(c) 1 : 1 (d) 11 : 1
73. A step up transformer has transformation ratio 5 : 3. What is voltage in secondary if voltage in primary is 60 V
[Pb. PET 2000]
(a) 20 V (b) 60 V
(c) 100 V (d) 180 V
74. In a step up transformer, 220 V is converted into 200 V. The number of turns in primary coil is 600. What is the number of turns in the secondary coil
[DCE 2004]
(a) 60 (b) 600
(c) 6000 (d) 100
75. The output voltage of a transformer connected to 220 volt line is 1100 volt at 1 amp current. Its efficiency is 100%. The current coming from the line is
[Pb. PET 2003]
(a) 20 A (b) 10 A
(c) 11 A (d) 22 A
76. Quantity that remains unchanged in a transformer is
[MP PMT/PET 1998; AIIMS 1999; J & K CET 2005]
(a) Voltage (b) Current
(c) Frequency (d) None of the above
77. In a region of uniform magnetic induction $B = 10^{-2}$ tesla, a circular coil of radius 30 cm and resistance π ohm is rotated about an axis which is perpendicular to the direction of B and which forms a diameter of the coil. If the coil rotates at 200 rpm the amplitude of the alternating current induced in the coil is
[MP PET 2001]
(a) 4π mA (b) 30 mA
(c) 6 mA (d) 200 mA
78. In a transformer, the number of turns in primary and secondary are 500 and 2000 respectively. If current in primary is 48 A, the current in the secondary is
[Orissa PMT 2004]
(a) 12 A (b) 24 A
(c) 48 A (d) 144 A
79. In an inductor of inductance $L = 100$ mH, a current of $I = 10$ A is flowing. The energy stored in the inductor is
[Orissa PMT 2004]
(a) 5 J (b) 10 J
(c) 100 J (d) 1000 J
80. The turn ratio of a transformers is given as 2 : 3. If the current through the primary coil is 3 A, thus calculate the current through load resistance
[BHU 2005]
(a) 1 A (b) 4.5 A
(c) 2 A (d) 1.5 A
81. Core of transformer is made up of
[AFMC 2005]
(a) Soft iron (b) Steel
(c) Iron (d) Alnico
82. The induction coil works on the principle of
[KCET 2005]
(a) Self-induction (b) Mutual induction
(c) Ampere's rule (d) Fleming's right hand rule
83. A transformer with efficiency 80% works at 4 kW and 100 V. If the secondary voltage is 200 V, then the primary and secondary currents are respectively
[Kerala PMT 2005]
(a) 40 A, 16 A (b) 16 A, 40 A
(c) 20 A, 40 A (d) 40 A, 20 A
84. In a step up transformer, if ratio of turns of primary to secondary is 1 : 10 and primary voltage is 230 V. If the load current is 2 A, then the current in primary is
[Orissa PMT 2005]
(a) 20 A (b) 10 A
(c) 2 A (d) 1 A
85. If a coil made of conducting wires is rotated between poles pieces of the permanent magnet. The motion will generate a current and this device is called
[CPMT 2005]
(a) An electric motor (b) An electric generator
(c) An electromagnet (d) All of above
86. A step-down transformer is used on a 1000 V line to deliver 20 A at 120 V at the secondary coil. If the efficiency of the transformer is 80% the current drawn from the line is
[Kerala PET 2005]
(a) 3 A (b) 30 A
(c) 0.3 A (d) 2.4 A

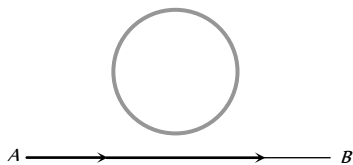


Critical Thinking

Objective Questions

1. An electron moves along the line AB , which lies in the same plane as a circular loop of conducting wires as shown in the diagram. What will be the direction of current induced if any, in the loop

[MP PET 1989; AIIMS 1982, 2001; KCET 2003; UPSEAT 2005]



- (a) No current will be induced
(b) The current will be clockwise
(c) The current will be anticlockwise
(d) The current will change direction as the electron passes by
2. A copper rod of length l is rotated about one end perpendicular to the magnetic field B with constant angular velocity ω . The induced e.m.f. between the two ends is

[MP PMT 1992; Orissa JEE 2003]

- (a) $\frac{1}{2} B \omega l^2$ (b) $\frac{3}{4} B \omega l^2$
(c) $B \omega l^2$ (d) $2 B \omega l^2$
3. Two different coils have self-inductance $L_1 = 8 \text{ mH}$, $L_2 = 2 \text{ mH}$. The current in one coil is increased at a constant rate. The current in the second coil is also increased at the same rate. At a certain instant of time, the power given to the two coils is the same. At that time the current, the induced voltage and the energy stored in the first coil are i_1 , V_1 and W_1 respectively. Corresponding values for the second coil at the same instant are i_2 , V_2 and W_2 respectively. Then

[IIT JEE 1994]

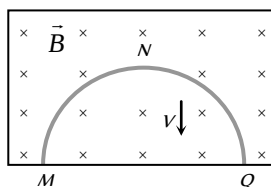
- (a) $\frac{i_1}{i_2} = \frac{1}{4}$ (b) $\frac{i_1}{i_2} = 4$
(c) $\frac{W_2}{W_1} = 4$ (d) $\frac{V_2}{V_1} = \frac{1}{4}$
4. An e.m.f. of 15 volt is applied in a circuit containing 5 henry inductance and 10 ohm resistance. The ratio of the currents at time $t = \infty$ and at $t = 1 \text{ second}$ is

[MP PMT 1994]

- (a) $\frac{e^{1/2}}{e^{1/2} - 1}$ (b) $\frac{e^2}{e^2 - 1}$
(c) $1 - e^{-1}$ (d) e^{-1}
5. Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 \gg R_2$, the mutual inductance M between them will be directly proportional to
- (a) R_1 / R_2 (b) R_2 / R_1
(c) R_1^2 / R_2 (d) R_2^2 / R_1

6. A thin semicircular conducting ring of radius R is falling with its plane vertical in a horizontal magnetic induction B . At the position MNQ , the speed of the ring is V and the potential difference developed across the ring is

[IIT JEE 1996]



- (a) Zero
(b) $B v \pi R^2 / 2$ and M is at higher potential
(c) $\pi R B V$ and Q is at higher potential
(d) $2 R B V$ and Q is at higher potential

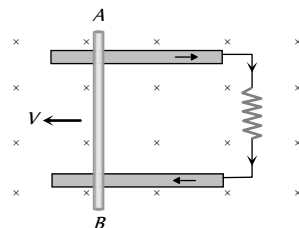
7. At a place the value of horizontal component of the earth's magnetic field H is $3 \times 10^{-5} \text{ Weber/m}^2$. A metallic rod AB of length 2 m placed in east-west direction, having the end A towards east, falls vertically downward with a constant velocity of 50 m/s. Which end of the rod becomes positively charged and what is the value of induced potential difference between the two ends

- (a) End A , $3 \times 10^{-3} \text{ mV}$ (b) End A , 3 mV

- (c) End B , $3 \times 10^{-3} \text{ mV}$ (d) End B , 3 mV

8. Consider the situation shown in the figure. The wire AB is sliding on the fixed rails with a constant velocity. If the wire AB is replaced by semicircular wire, the magnitude of the induced current will

- (a) Increase
(b) Remain the same
(c) Decrease
(d) Increase or decrease depending on whether the semicircle bulges towards the resistance or away from it



9. A circular loop of radius R carrying current I lies in $x-y$ plane with its centre at origin. The total magnetic flux through $x-y$ plane is

[IIT-JEE 1999]

- (a) Directly proportional to I
(b) Directly proportional to R
(c) Directly proportional to R^2
(d) Zero

10. Two identical circular loops of metal wire are lying on a table without touching each other. Loop-A carries a current which increases with time. In response, the loop-B

[IIT JEE 1999; UPSEAT 2003]

- (a) Remains stationary
(b) Is attracted by the loop-A
(c) Is repelled by the loop-A
(d) Rotates about its CM, with CM fixed (CM is the centre of mass)

11. Two coils have a mutual inductance 0.005 H. The current changes in the first coil according to equation $I = I_0 \sin \omega t$, where $I_0 = 10 \text{ A}$ and $\omega = 100 \pi \text{ radian/sec}$. The maximum value of e.m.f. in the second coil is

[CBSE PMT 1998; Pb. PMT 2000]

- (a) 2π (b) 5π
(c) π (d) 4π

12. A small square loop of wire of side l is placed inside a large square loop of wire of side L ($L > l$). The loop are coplanar and their centre coincide. The mutual inductance of the system is proportional to

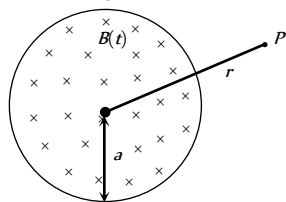
- (a) l / L (b) l^2 / L
(c) L / l (d) L^2 / l

13. A wire of length 1 m is moving at a speed of 2 ms perpendicular to its length and a homogeneous magnetic field of 0.5 T . The ends of the wire are joined to a circuit of resistance $6\ \Omega$. The rate at which work is being done to keep the wire moving at constant speed is [Roorkee 1999]

- (a) $\frac{1}{12}\text{ W}$ (b) $\frac{1}{6}\text{ W}$
(c) $\frac{1}{3}\text{ W}$ (d) 1 W

14. A uniform but time-varying magnetic field $B(t)$ exists in a circular region of radius a and is directed into the plane of the paper, as shown. The magnitude of the induced electric field at point P at a distance r from the centre of the circular region

- (a) Is zero
(b) Decreases as $\frac{1}{r}$
(c) Increases as r
(d) Decreases as $\frac{1}{r^2}$



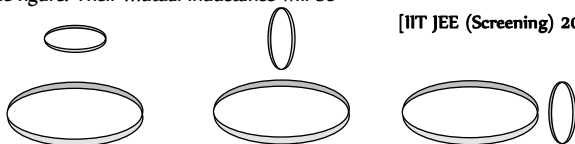
15. A coil of wire having finite inductance and resistance has a conducting ring placed coaxially within it. The coil is connected to a battery at time $t = 0$, so that a time-dependent current $I_1(t)$ starts flowing through the coil. If $I_2(t)$ is the current induced in the ring, and $B(t)$ is the magnetic field at the axis of the coil due to $I_1(t)$, then as a function of time ($t > 0$), the product $I(t) B(t)$

[IIT-JEE (Screening) 2000]

- (a) Increases with time (b) Decreases with time
(c) Does not vary with time (d) Passes through a maximum

16. Two circular coils can be arranged in any of the three situations shown in the figure. Their mutual inductance will be

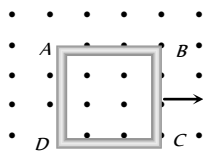
[IIT JEE (Screening) 2001]



- (a) Maximum in situation (A) (b) Maximum in situation (B)
(c) Maximum in situation (C) (d) The same in all situations

17. A metallic square loop $ABCD$ is moving in its own plane with velocity v in a uniform magnetic field perpendicular to its plane as shown in the figure. An electric field is induced

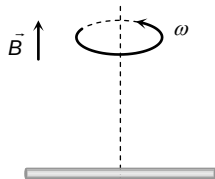
- (a) In AD , but not in BC
(b) In BC , but not in AD
(c) Neither in AD nor in BC
(d) In both AD and BC



18. A conducting rod of length $2l$ is rotating with constant angular speed ω about its perpendicular bisector. A uniform magnetic field $\vec{B}$ exists parallel to the axis of rotation. The e.m.f. induced between two ends of the rod is

[MP PET 2001]

- (a) $B\omega l$
(b) $\frac{1}{2} B\omega l^2$
(c) $\frac{1}{8} B\omega l^2$
(d) Zero

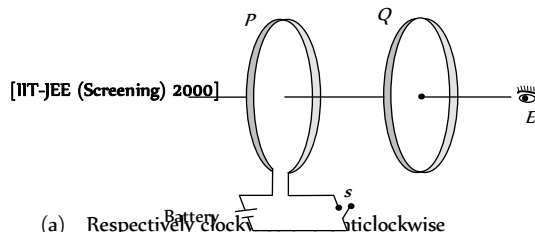


19. An inductor of 2 henry and a resistance of 10 ohms are connected in series with a battery of 5 volts . The initial rate of change of current is

[MP PMT 2001]

- (a) 0.5 amp/sec (b) 2.0 amp/sec
(c) 2.5 amp/sec (d) 0.25 amp/sec

20. As shown in the figure, P and Q are two coaxial conducting loops separated by some distance. When the switch S is closed, a clockwise current I_P flows in P (as seen by E) and an induced current I_{Q_1} flows in Q . The switch remains closed for a long time. When S is opened, a current I_{Q_2} flows in Q . Then the directions of I_{Q_1} and I_{Q_2} (as seen by E) are [IIT JEE (Screening) 2002]



- (a) Respectively clockwise and anticlockwise
(b) Both clockwise
(c) Both anticlockwise
(d) Respectively anticlockwise and clockwise

21. A short-circuited coil is placed in a time-varying magnetic field. Electrical power is dissipated due to the current induced in the coil. If the number of turns were to be quadrupled and the wire radius halved, the electrical power dissipated would be

- (a) Halved (b) The same
(c) Doubled (d) Quadrupled

22. A physicist works in a laboratory where the magnetic field is 2 T . She wears a necklace enclosing area 0.01 m^2 in such a way that the plane of the necklace is normal to the field and is having a resistance $R = 0.01\ \Omega$. Because of power failure, the field decays to 1 T in time 10^{-3} seconds . Then what is the total heat produced in her necklace? ($T = \text{Tesla}$)

[Orissa JEE 2002]

- (a) 10 J (b) 20 J
(c) 30 J (d) 40 J

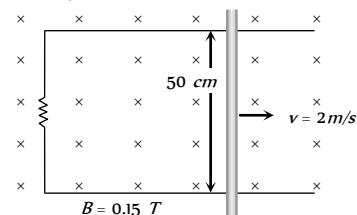
23. A coil of inductance 8.4 mH and resistance $6\ \Omega$ is connected to a 12 V battery. The current in the coil is 1.0 A at approximately the time

[IIT-JEE (Screening) 1999; UPSEAT 2003]

- (a) 500 sec (b) 20 sec
(c) 35 milli sec (d) 1 milli sec

24. As shown in the figure a metal rod makes contact and complete the circuit. The circuit is perpendicular to the magnetic field with $B = 0.15\text{ tesla}$. If the resistance is $3\ \Omega$, force needed to move the rod as indicated with a constant speed of 2 m/sec is

- (a) $3.75 \times 10^{-3}\text{ N}$
(b) $3.75 \times 10^{-2}\text{ N}$
(c) $3.75 \times 10^2\text{ N}$
(d) $3.75 \times 10^{-4}\text{ N}$

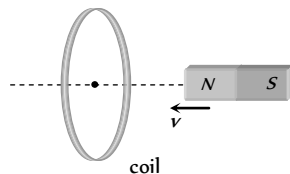


25. Two identical coaxial circular loops carry current i each circulating in the clockwise direction. If the loops are approaching each other, then [MP PMT 1995, 96]

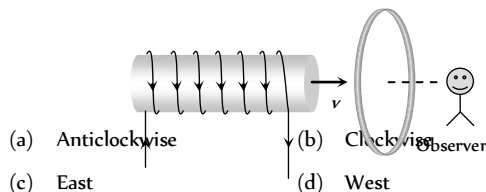
- (a) Current in each loop increases
(b) Current in each loop remains the same
(c) Current in each loop decreases
(d) Current in one-loop increases and in the other it decreases

26. In the following figure, the magnet is moved towards the coil with a speed v and induced emf is e . If magnet and coil recede away from one another each moving with speed v , the induced emf in the coil will be

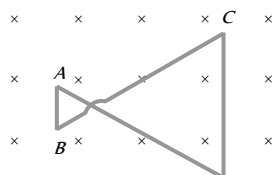
(a) e
(b) $2e$
(c) $e/2$
(d) $4e$



27. A current carrying solenoid is approaching a conducting loop as shown in the figure. The direction of induced current as observed by an observer on the other side of the loop will be

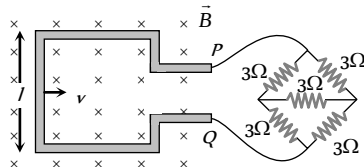


28. A conducting wire frame is placed in a magnetic field which is directed into the paper. The magnetic field is increasing at a constant rate. The directions of induced current in wires AB and CD are



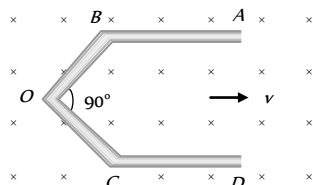
(a) B to A and D to C (b) A to B and C to D
(c) A to B and D to C (d) B to A and C to D

29. A square metallic wire loop of side 0.1 m and resistance of 1Ω is moved with a constant velocity in a magnetic field of 2 wb/m as shown in figure. The magnetic field is perpendicular to the plane of the loop, loop is connected to a network of resistances. What should be the velocity of loop so as to have a steady current of 1 mA in loop



(a) 1 cm/sec (b) 2 cm/sec
(c) 3 cm/sec (d) 4 cm/sec

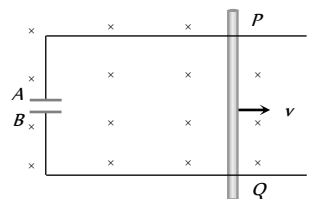
30. A conductor $ABOCD$ moves along its bisector with a velocity of 1 m/s through a perpendicular magnetic field of 1 wb/m , as shown in fig. If all the four sides are of 1 m length each, then the induced emf between points A and D is



(a) 0 (b) 1.41 volt
(c) 0.71 volt (d) None of the above

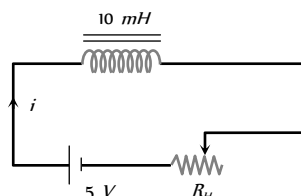
31. A conducting rod PQ of length $L = 1.0\text{ m}$ is moving with a uniform speed $v = 2\text{ m/s}$ in a uniform magnetic field $B = 4.0\text{ T}$ directed

into the paper. A capacitor of capacity $C = 10\text{ }\mu\text{F}$ is connected as shown in figure. Then



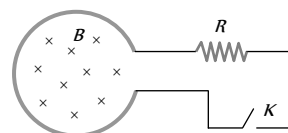
- (a) $q_+ = +80\text{ }\mu\text{C}$ and $q_- = -80\text{ }\mu\text{C}$
(b) $q_+ = -80\text{ }\mu\text{C}$ and $q_- = +80\text{ }\mu\text{C}$
(c) $q_+ = 0 = q_-$
(d) Charge stored in the capacitor increases exponentially with time

32. The resistance in the following circuit is increased at a particular instant. At this instant the value of resistance is 10Ω . The current in the circuit will be now



(a) $i = 0.5\text{ A}$ (b) $i > 0.5\text{ A}$
(c) $i < 0.5\text{ A}$ (d) $i = 0$

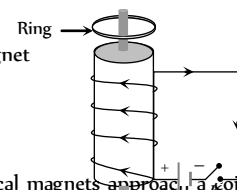
33. Shown in the figure is a circular loop of radius r and resistance R . A variable magnetic field of induction $B = B_0 e^{-t}$ is established inside the coil. If the key (K) is closed, the electrical power developed right after closing the switch is equal to



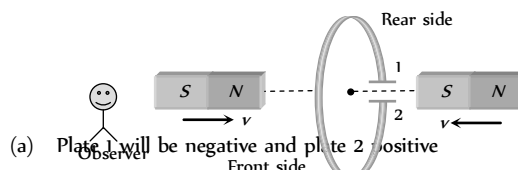
(a) $\frac{B_0^2 \pi^2}{R}$ (b) $\frac{B_0 10 r^3}{R}$
(c) $\frac{B_0^2 \pi^2 r^4}{5}$ (d) $\frac{B_0^2 \pi^2 r^4}{R}$

34. A conducting ring is placed around the core of an electromagnet as shown in fig. When key K is pressed, the ring

(a) Remain stationary
(b) Is attracted towards the electromagnet
(c) Jumps out of the core
(d) None of the above



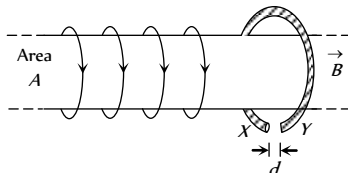
35. The north and south poles of two identical magnets approach a coil, containing a condenser, with equal speeds from opposite sides. Then



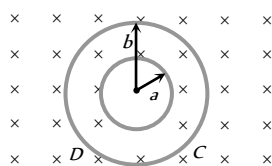
(a) Plate 1 will be negative and plate 2 positive
(b) Plate 1 will be positive and plate 2 negative
(c) Both the plates will be positive
(d) Both the plates will be negative

36. A highly conducting ring of radius R is perpendicular to and concentric with the axis of a long solenoid as shown in fig. The ring has a narrow gap of width d in its circumference. The solenoid has cross sectional area A and a uniform internal field of magnitude B . Now beginning at $t = 0$, the solenoid current is steadily increased to so that the field magnitude at any time t is given by $B(t) = B_0 + \alpha t$ where $\alpha > 0$. Assuming that no charge can flow across the gap, the end of ring which has excess of positive charge and the magnitude of induced e.m.f. in the ring are respectively

- (a) $X, A\alpha$
 (b) $X, \pi R\alpha$
 (c) $Y, \pi A\alpha$
 (d) $Y, \pi R\alpha$



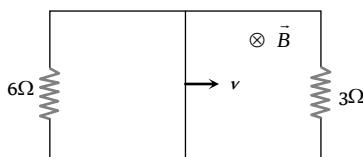
37. Plane figures made of thin wires of resistance $R = 50 \text{ milli ohm/metre}$ are located in a uniform magnetic field perpendicular into the plane of the figures and which decrease at the rate $dB/dt = 0.1 \text{ m T/s}$. Then currents in the inner and outer boundary are. (The inner radius $a = 10 \text{ cm}$ and outer radius $b = 20 \text{ cm}$)



- (a) $10 \cdot A$ (Clockwise), $2 \times 10 \cdot A$ (Clockwise)
 (b) $10 \cdot A$ (Anticlockwise), $2 \times 10 \cdot A$ (Clockwise)
 (c) $2 \times 10 \cdot A$ (clockwise), $10 \cdot A$ (Anticlockwise)
 (d) $2 \times 10 \cdot A$ (Anticlockwise), $10 \cdot A$ (Anticlockwise)

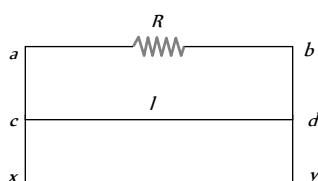
38. A rectangular loop with a sliding connector of length $l = 1.0 \text{ m}$ is situated in a uniform magnetic field $B = 2 \text{ T}$ perpendicular to the plane of loop. Resistance of connector is $r = 2\Omega$. Two resistance of 6Ω and 3Ω are connected as shown in figure. The external force required to keep the connector moving with a constant velocity $v = 2 \text{ m/s}$ is

- (a) 6 N
 (b) 4 N
 (c) 2 N
 (d) 1 N



39. A wire cd of length l and mass m is sliding without friction on conducting rails ax and by as shown. The vertical rails are connected to each other with a resistance R between a and b . A uniform magnetic field B is applied perpendicular to the plane $abcd$ such that cd moves with a constant velocity of

- (a) $\frac{mgR}{Bl}$
 (b) $\frac{mgR}{B^2 l^2}$
 (c) $\frac{mgR}{B^3 l^3}$
 (d) $\frac{mgR}{B^2 l}$



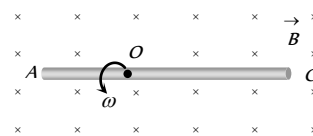
40. A conducting rod AC of length $4l$ is rotated about a point O in a uniform magnetic field $\vec{B}$ directed into the paper. $AO = l$ and $OC = 3l$. Then

(a) $V_A - V_O = \frac{B\omega l^2}{2}$

(b) $V_O - V_C = \frac{7}{2} B\omega l^2$

(c) $V_A - V_C = 4B\omega l^2$

(d) $V_C - V_O = \frac{9}{2} B\omega l^2$



41. How much length of a very thin wire is required to obtain a solenoid of length l_0 and inductance L

(a) $\sqrt{\frac{2\pi L l_0}{\mu_0}}$

(b) $\sqrt{\frac{4\pi L l_0}{\mu_0^2}}$

(c) $\sqrt{\frac{4\pi L l_0}{\mu_0}}$

(d) $\sqrt{\frac{8\pi L l_0}{\mu_0}}$

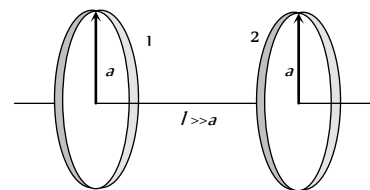
42. What is the mutual inductance of a two-loop system as shown with centre separation l

(a) $\frac{\mu_0 \pi a^4}{8l^3}$

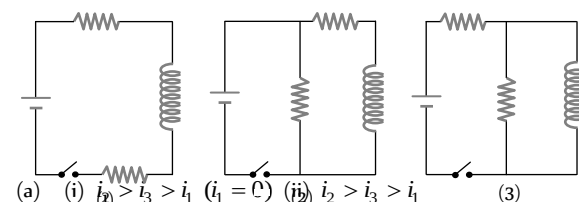
(b) $\frac{\mu_0 \pi a^4}{4l^3}$

(c) $\frac{\mu_0 \pi a^4}{6l^3}$

(d) $\frac{\mu_0 \pi a^4}{2l^3}$



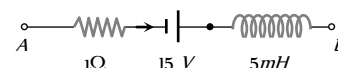
43. The figure shows three circuits with identical batteries, inductors, and resistors. Rank the circuits according to the current through the battery (i) just after the switch is closed and (ii) a long time later, greatest first



- (a) (i) $i_2 > i_3 > i_1$ ($i_1 = 0$) (ii) $i_2 > i_3 > i_1$
 (b) (i) $i_2 < i_3 < i_1$ ($i_1 \neq 0$) (ii) $i_2 > i_3 > i_1$
 (c) (i) $i_2 = i_3 = i_1$ ($i_1 = 0$) (ii) $i_2 < i_3 < i_1$
 (d) (i) $i_2 = i_3 > i_1$ ($i_1 \neq 0$) (ii) $i_2 > i_3 > i_1$

44. The network shown in the figure is a part of a complete circuit. If at a certain instant the current i is 5 A and is decreasing at the rate of 10^3 A/s then $V_A - V_B$ is

- (a) 5 V
 (b) 10 V
 (c) 15 V
 (d) 20 V



45. A 50 volt potential difference is suddenly applied to a coil with $L = 5 \times 10^{-3} \text{ henry}$ and $R = 180 \text{ ohm}$. The rate of increase of current after 0.001 second is [MP PET 1994]
 (a) 27.3 amp/sec
 (b) 27.8 amp/sec
 (c) 2.73 amp/sec
 (d) None of the above

46. The current in a LR circuit builds up to $\frac{3}{4}$ th of its steady state value in 4 s . The time constant of this circuit is

[Roorkee 2000]

- (a) $\frac{1}{\ln 2} s$ (b) $\frac{2}{\ln 2} s$
(c) $\frac{3}{\ln 2} s$ (d) $\frac{4}{\ln 2} s$

47. A conducting ring of radius 1 meter is placed in an uniform magnetic field B of 0.01 Tesla oscillating with frequency 100 Hz with its plane at right angles to B . What will be the induced electric field

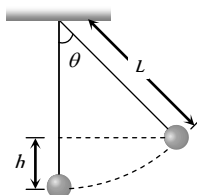
[AIIMS 2005]

- (a) $\pi \text{ volt/m}$ (b) 2 volt/m
(c) 10 volt/m (d) 62 volt/m

48. A simple pendulum with bob of mass m and conducting wire of length L swings under gravity through an angle 2θ . The earth's magnetic field component in the direction perpendicular to swing is B . Maximum potential difference induced across the pendulum is

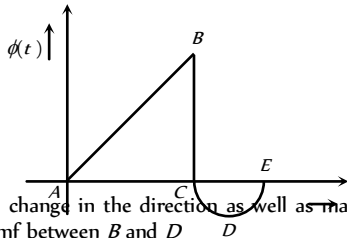
[MP PET 2005]

- (a) $2BL \sin\left(\frac{\theta}{2}\right)(gL)^{1/2}$
(b) $BL \sin\left(\frac{\theta}{2}\right)(gL)$
(c) $BL \sin\left(\frac{\theta}{2}\right)(gL)^{3/2}$
(d) $BL \sin\left(\frac{\theta}{2}\right)(gL)^2$

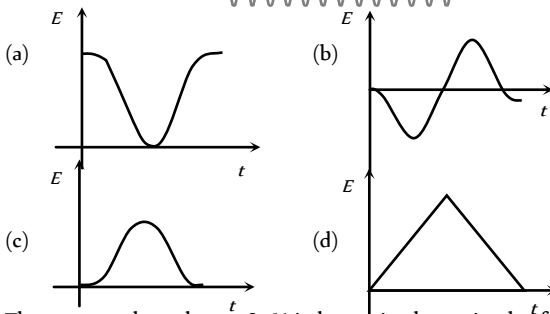
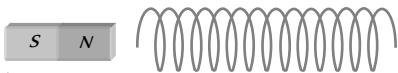


Graphical Questions

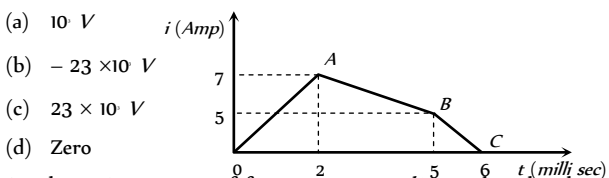
1. The graph Shows the variation in magnetic flux $\phi(t)$ with time through a coil. Which of the statements given below is not correct



- (a) There is a change in the direction as well as magnitude of the induced emf between B and D
(b) The magnitude of the induced emf is maximum between B and C
(b) There is a change in the direction as well as magnitude of induced emf between A and C
(d) The induced emf is zero at B
2. The variation of induced emf (E) with time (t) in a coil if a short bar magnet is moved along its axis with a constant velocity is best represented as [IIT-JEE (Screening) 2004]

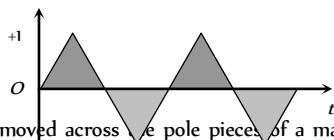


3. The current through a 4.6 H inductor is shown in the following graph. The induced emf during the time interval $t = 5 \text{ milli-sec}$ to 6 milli-sec will be

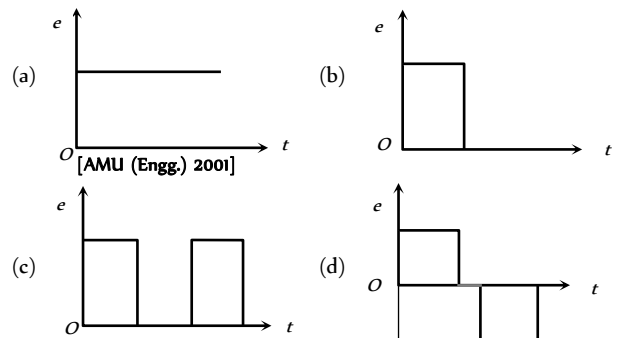
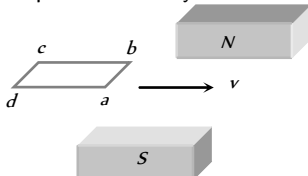


4. An alternating current of frequency 200 rad/sec and peak value 1 A as shown in the figure, is applied to the primary of a transformer. If the coefficient of mutual induction between the primary and the secondary is 1.5 H , the voltage induced in the secondary will be

- (a) 300 V
(b) 191 V
(c) 220 V
(d) 471 V

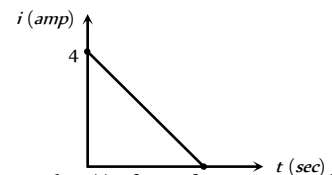


5. A horizontal loop $abcd$ is moved across the pole pieces of a magnet as shown in fig. with a constant speed v . When the edge ab of the loop enters the pole pieces at time $t = 0 \text{ sec}$. Which one of the following graphs represents correctly the induced emf in the coil



6. Some magnetic flux is changed from a coil of resistance 10 ohm . As a result an induced current is developed in it, which varies with time as shown in figure. The magnitude of change in flux through the coil in webers is

- (a) 2
(b) 4
(c) 6
(d) None of these

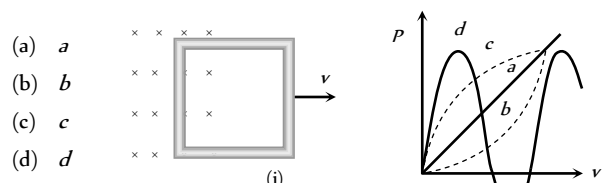


7. The graph gives the magnitude $B(t)$ of a uniform magnetic field that exists throughout a conducting loop, perpendicular to the plane of the loop. Rank the five regions of the graph according to the magnitude of the emf induced in the loop, greatest first

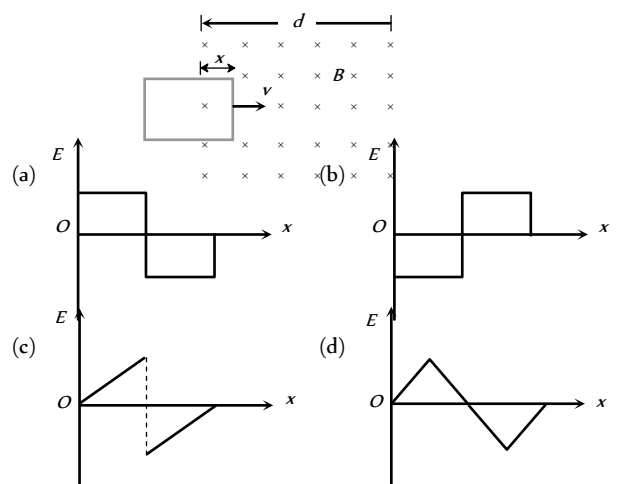
- (a) $b > (d = e) < (a = c)$
(b) $b > (d = e) > (a = c)$
(c) $b < d < e < c < a$
(d) $b > (a = c) > (d = e)$



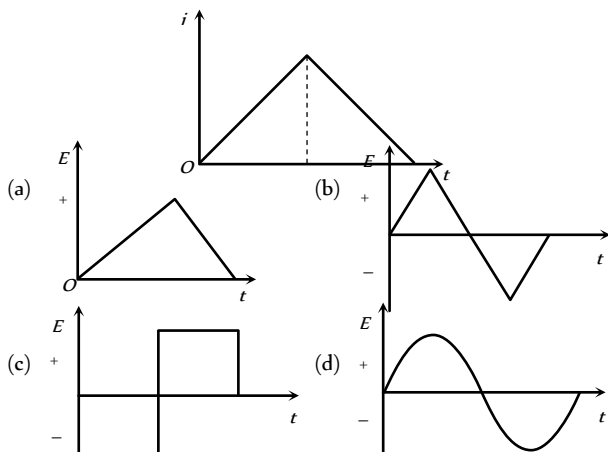
8. Figure (i) shows a conducting loop being pulled out of a magnetic field with a speed v . Which of the four plots shown in figure (ii) may represent the power delivered by the pulling agent as a function of the speed v



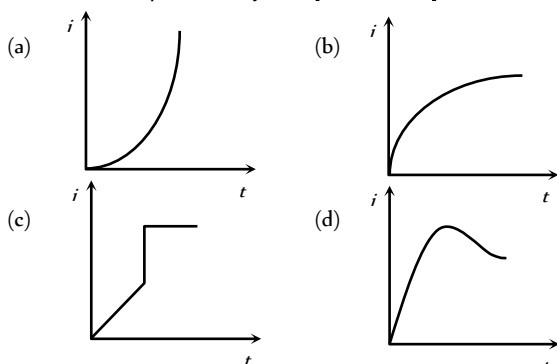
9. A rectangular loop is being pulled at a constant speed v , through a region of certain thickness d , in which a uniform magnetic field B is set up. The graph between position x of the right hand edge of the loop and the induced emf E will be



10. The current i in an inductance coil varies with time, t according to the graph shown in fig. Which one of the following plots shows the variation of voltage in the coil with time

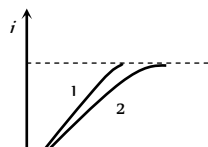


11. When a battery is connected across a series combination of self inductance L and resistance R , the variation in the current i with time t is best represented by [MP PET 2004]

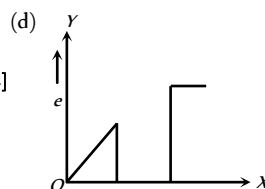
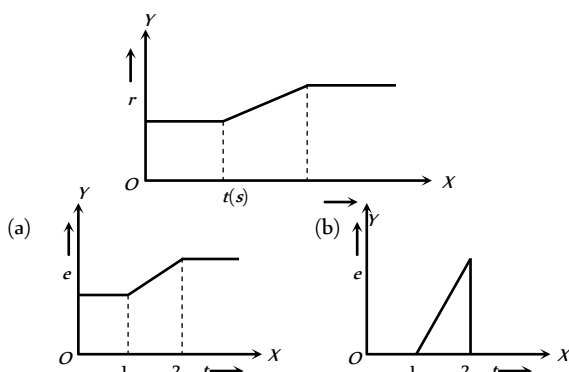


12. When a certain circuit consisting of a constant e.m.f. E an inductance L and a resistance R is closed, the current in, it increases with time according to curve 1. After one parameter (E , L or R) is changed, the increase in current follows curve 2 when the circuit is closed second time. Which parameter was changed and in what direction

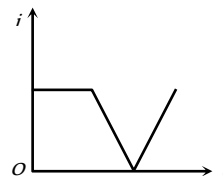
- (a) L is increased
(b) L is decreased
(c) R is increased
(d) R is decreased



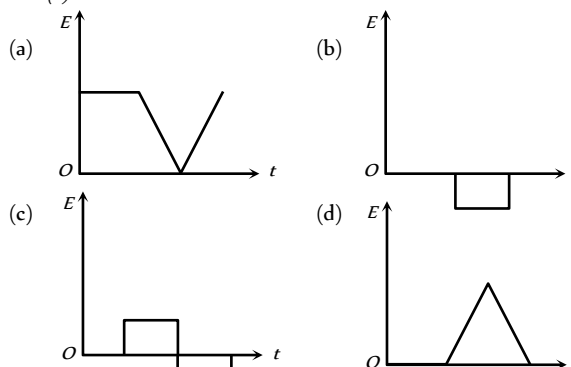
13. A flexible wire bent in the form of a circle is placed in a uniform magnetic field perpendicular to the plane of the coil. The radius of the coil changes as shown in figure. The graph of induced emf in the coil is represented by



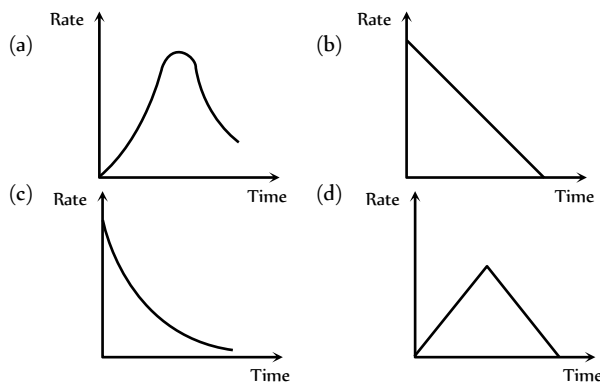
14. The current i in an induction coil varies with time t according to the graph shown



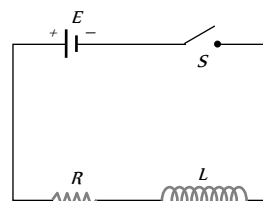
in figure. Which of the following graphs shows the induced emf (e) in the coil with time



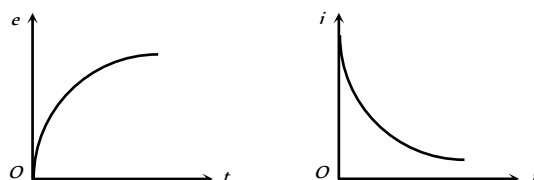
15. In an L - R circuit connected to a battery the rate at which energy is stored in the inductor is plotted against time during the growth of the current in the circuit. Which of the following best represents the resulting curve



16. Switch S of the circuit shown in figure. is closed at $t = 0$. If e denotes the induced



emf in L and i , the current flowing through the circuit at time t , which of the following graphs is correct

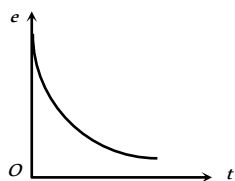


(a)

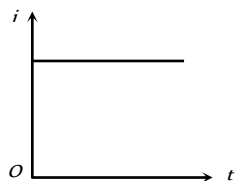
(b)

IV.

(c)

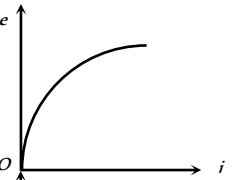


(d)

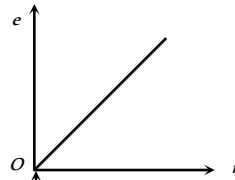


17. For previous objective, which of the following graphs is correct

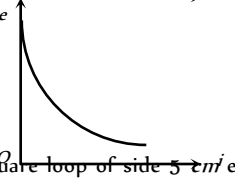
(a)



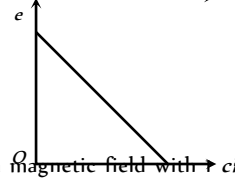
(b)



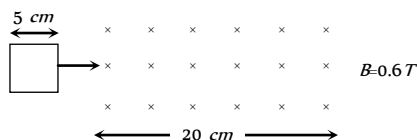
(c)



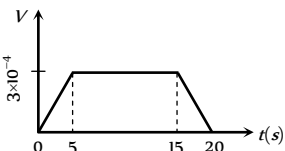
(d)



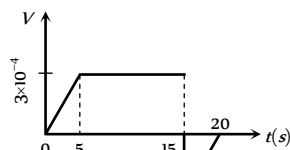
18. A square loop of side 5 cm enters a magnetic field with $\vec{B}$ as shown. The front edge enters the magnetic field at $t = 0$ then which graph best depicts emf



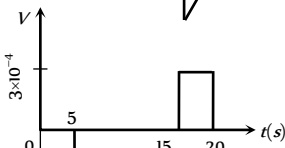
(a)



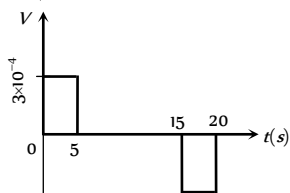
(b)



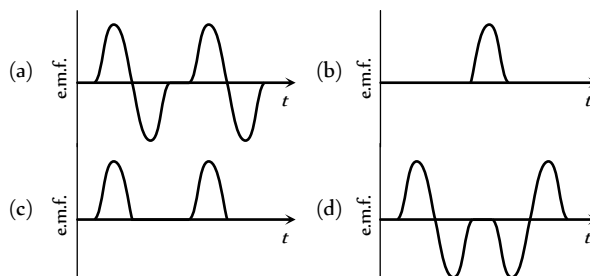
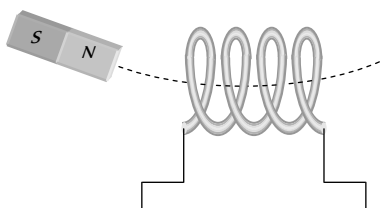
(c)



(d)



19. A magnet is made to oscillate with a particular frequency, passing through a coil as shown in figure. The time variation of the magnitude of e.m.f. generated across the coil during one cycle is



Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- If both assertion and reason are true and the reason is the correct explanation of the assertion.
- If both assertion and reason are true but reason is not the correct explanation of the assertion.
- If assertion is true but reason is false.
- If the assertion and reason both are false.
- If assertion is false but reason is true.

- Assertion : Eddy currents are produced in any metallic conductor when magnetic flux is changed around it.
Reason : Electric potential determines the flow of charge. [AIIMS 1995]
- Assertion : The quantity L/R possesses dimensions of time.
Reason : To reduce the rate of increases of current through a solenoid should increase the time constant (L/R). [AIIMS 2002]
- Assertion : Faraday's laws are consequences of conservation of energy.
Reason : In a purely resistive ac circuit, the current lags behind the e.m.f. in phase. [AIIMS 2002]
- Assertion : Only a change in magnetic flux will maintain an induced current the coil.
Reason : The presence of large magnetic flux through a coil maintains a current in the coil if the circuit is continuous. [AIIMS 1999]
- Assertion : Magnetic flux can produce induced e.m.f.
Reason : Faraday established induced e.m.f. experimentally.
- Assertion : The induced e.m.f. and current will be same in two identical loops of copper and aluminium, when rotated with same speed in the same magnetic field.
Reason : Induced e.m.f. is proportional to rate of change of magnetic field while induced current depends on resistance of wire.
- Assertion : Inductance coil are made of copper.
Reason : Induced current is more in wire having less resistance.
- Assertion : Self-inductance is called the inertia of electricity.
Reason : Self-inductance is the phenomenon, according to which an opposing induced e.m.f. is produced in a coil as a result of change in current or magnetic flux linked in the coil.
- Assertion : When two coils are wound on each other, the mutual induction between the coils is maximum.
Reason : Mutual induction does not depend on the orientation of the coils. [AIIMS 2005]

- 10.** Assertion : Acceleration of a magnet falling through a long solenoid decreases.
Reason : The induced current produced in a circuit always flow in such direction that it opposes the change or the cause the produced it.
- 11.** Assertion : An aircraft flies along the meridian, the potential at the ends of its wings will be the same.
Reason : Whenever there is change in the magnetic flux e.m.f. induces.
- 12.** Assertion : A spark occur between the poles of a switch when the switch is opened.
Reason : Current flowing in the conductor produces magnetic field.
- 13.** Assertion : In the phenomenon of mutual induction, self induction of each of the coils persists.
Reason : Self induction arises when strength of current in same coil changes. In mutual induction, current is changing in both the individual coils.
- 14.** Assertion : Lenz's law violates the principle of conservation of energy.
Reason : Induced e.m.f., opposes always the change in magnetic flux responsible for its production.
- 15.** Assertion : The induced emf in a conducting loop of wire will be non zero when it rotates in a uniform magnetic field.
Reason : The emf is induced due to change in magnetic flux.
- 16.** Assertion : An induced emf is generated when magnet is withdrawn from the solenoid.
Reason : The relative motion between magnet and solenoid induces emf.
- 17.** Assertion : An artificial satellite with a metal surface is moving above the earth in a circular orbit. A current will be induced in satellite if the plane of the orbit is inclined to the plane of the equator.
Reason : The current will be induced only when the speed of satellite is more than 8 km/sec .
- 18.** Assertion : A bar magnet is dropped into a long vertical copper tube. Even taking air resistance as negligible, the magnet attains a constant terminal velocity. If the tube is heated, the terminal velocity gets increased.
Reason : The terminal velocity depends on eddy current produced in bar magnet.
- 19.** Assertion : A metal piece and a non-metal (stone) piece are dropped from the same height near earth's surface. Both will reach the earth's surface simultaneously.
Reason : There is no effect of earth's magnetic field on freely falling body.
- 20.** Assertion : A transformer cannot work on dc supply.
Reason : dc changes neither in magnitude nor in direction.
- 21.** Assertion : Soft iron is used as a core of transformer.
Reason : Area of hysteresis loop for soft iron is small.
- 22.** Assertion : An ac generator is based on the phenomenon of self-induction.
Reason : In single coil, we consider self-induction only.
- 23.** Assertion : An electric motor will maximum efficient when back e.m.f. is equal to applied e.m.f.
Reason : Efficiency of electric motor is depends only on magnitude of back e.m.f..
- 24.** Assertion : The back emf in a dc motor is maximum when the motor has just been switched on.
Reason : When motor is switched on it has maximum speed.

Answers

Faraday's and Lenz's Law

1	c	2	d	3	b	4	d	5	b
6	c	7	a	8	c	9	a	10	b
11	a	12	b	13	b	14	a	15	d
16	d	17	c	18	b	19	b	20	b
21	b	22	c	23	b	24	b	25	d
26	c	27	d	28	b	29	d	30	d
31	b	32	a	33	b	34	a	35	b
36	b	37	d	38	a	39	a	40	c
41	c	42	b	43	c	44	c	45	d
46	d	47	d	48	d	49	d	50	c
51	b	52	a	53	d	54	b	55	b
56	a	57	c	58	a	59	d	60	b
61	a	62	a	63	d	64	d	65	c
66	c	67	a	68	b				

Motional EMI

1	a	2	b	3	d	4	c	5	b
6	b	7	b	8	c	9	d	10	d
11	b	12	c	13	b	14	c	15	d
16	c	17	c	18	b	19	c	20	b
21	d	22	d	23	d	24	d	25	c
26	a	27	c	28	c	29	c	30	d
31	b	32	b	33	b				

Static EMI

1	d	2	d	3	b	4	a	5	d
6	d	7	c	8	c	9	c	10	c
11	a	12	b	13	b	14	d	15	d
16	b	17	a	18	b	19	b	20	b
21	a	22	d	23	c	24	c	25	b
26	c	27	b	28	c	29	c	30	d
31	b	32	a	33	b	34	c	35	d
36	a	37	a	38	c	39	c	40	a
41	d	42	a	43	a	44	b	45	b
46	abcd	47	c	48	b	49	d	50	c
51	c	52	b	53	a	54	a	55	a
56	a	57	a	58	d	59	a	60	b
61	d	62	b	63	a	64	d	65	a
66	d	67	c	68	c	69	c	70	b
71	a	72	b	73	c	74	b	75	a
76	c	77	c	78	c	79	c	80	d
81	b	82	c	83	a	84	b	85	a
86	a	87	a	88	c	89	b	90	d
91	d	92	a	93	d	94	c	95	b

96	c	97	a	98	c	99	d	100	d
101	b	102	c	103	b	104	a	105	c
106	d	107	c	108	c				

Application of EMI (Motor, Dynamo, Transformer ...)

1	b	2	d	3	c	4	c	5	a
6	d	7	a	8	a	9	c	10	a
11	b	12	b	13	a	14	b	15	b
16	c	17	d	18	b	19	d	20	c
21	d	22	d	23	d	24	a	25	d
26	b	27	a	28	c	29	c	30	a
31	a	32	c	33	b	34	c	35	a
36	b	37	a	38	c	39	a	40	c
41	b	42	a	43	d	44	d	45	b
46	b	47	a	48	a	49	d	50	a
51	c	52	b	53	b	54	b	55	a
56	b	57	c	58	a	59	a	60	a
61	d	62	b	63	d	64	c	65	b
66	b	67	c	68	d	69	b	70	c
71	a	72	b	73	c	74	c	75	b
76	c	77	c	78	a	79	a	80	c
81	a	82	b	83	a	84	a	85	b
86	a								

Critical Thinking Questions

1	d	2	a	3	acd	4	b	5	d
6	d	7	b	8	b	9	d	10	c
11	b	12	b	13	b	14	b	15	d
16	a	17	d	18	d	19	c	20	d
21	b	22	a	23	d	24	a	25	c
26	b	27	b	28	a	29	b	30	b
31	a	32	b	33	d	34	c	35	b
36	a	37	a	38	c	39	b	40	c
41	c	42	d	43	a	44	c	45	d
46	b	47	b	48	a				

Graphical Questions

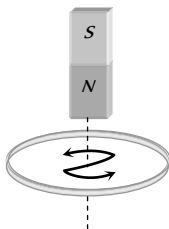
1	d	2	a	3	c	4	b	5	d
6	a	7	b	8	b	9	b	10	c
11	b	12	a	13	b	14	c	15	a
16	c	17	d	18	c	19	a		

Assertion and Reason

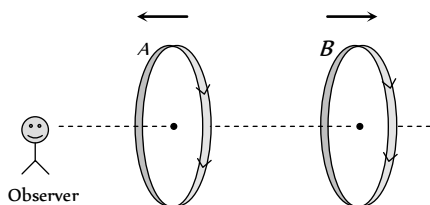
1	b	2	b	3	c	4	c	5	e
6	e	7	a	8	b	9	c	10	a
11	a	12	b	13	a	14	e	15	a
16	a	17	c	18	b	19	d	20	a
21	a	22	e	23	d	24	d		

Faraday's and Lenz's Law

1. (c) Because induced e.m.f. is given by $E = -N \frac{d\phi}{dt}$.
2. (d) The energy of the field increases with the magnitude of the field. Lenz's law infers that there is an opposite field created due to increase or decrease of magnetic flux around a conductor so as to hold the law of conservation of energy.
3. (b) We know that $e = \frac{d\phi}{dt}$
But $e = iR$ and $i = \frac{dq}{dt} \Rightarrow \frac{dq}{dt} R = \frac{d\phi}{dt} \Rightarrow dq = \frac{d\phi}{R}$
4. (d) Similar to Q.3
5. (b) Because there is no change in flux linked with coil
6. (c) As it is seen from the magnet side induced current will be anticlockwise.

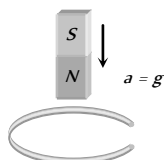


7. (a) $e = -\frac{d\phi}{dt} = \frac{-3B_0 A_0}{t}$
8. (c) $e = -\frac{d\phi}{dt} = -(16t + 3) = -67 \text{ units}$
9. (a) Induced current in both the coils assist the main current so current through each coil increases.



10. (b) When the magnet is allowed to fall vertically along the axis of loop with its north pole towards the ring. The upper face of the ring will become north pole in an attempt to oppose the approaching north pole of the magnet. Therefore the acceleration in the magnet is less than g .

Note : If coil is broken at any point then induced *emf* will be generated in it but no induced current will flow. In this condition the coil will not oppose the motion of magnet and the magnet will fall freely with acceleration g . (i.e. $a = g$)



11. (a) $\phi = BA = 10 \text{ weber}$
12. (b) The magnitude of induced e.m.f. is directly proportional to the rate of change of magnetic flux. Induced charge doesn't depend upon time.
13. (b)
14. (a) $I = \frac{e}{R} = \frac{-N(d\phi/dt)}{R} = \frac{10 \times 10^8 \times 10^{-4} \times 10^{-4} \times 10}{20} = 5 \text{ A}$
15. (d) Induced charge doesn't depend upon the speed of magnet.
16. (d) $|e| = N \left(\frac{\Delta B}{\Delta t} \right) A \cos \theta = 500 \times 1 \times (10 \times 10^{-2})^2 \cos 0 = 5 \text{ V}$
17. (c) When frequency is high, the galvanometer will not show deflection.
18. (b) $e = -\frac{N(B_2 - B_1)A \cos \theta}{\Delta t}$
 $= -\frac{500 \times (0 - 0.1) \times 100 \times 10^{-4} \cos 0}{0.1} = 5 \text{ V}$
19. (b) $e = -\frac{N(B_2 - B_1)A \cos \theta}{\Delta t}$
 $= -\frac{50(0.35 - 0.10) \times \pi(3 \times 10^{-2})^2 \times \cos 0^\circ}{2 \times 10^{-3}} = 17.7 \text{ V}$
20. (b) $|e| = A \cdot \frac{\Delta B}{\Delta t} = 2 \times \frac{(4 - 1)}{2} = 3 \text{ V}$
21. (b) $e = -\frac{NBA(\cos \theta_2 - \cos \theta_1)}{\Delta t}$
 $= -2000 \times 0.3 \times 70 \times 10^{-4} \frac{(\cos 180 - \cos 0)}{0.1}$
 $\Rightarrow e = 84 \text{ V}$
22. (c) The induced current will be in such a direction so that it opposes the change due to which it is produced.
23. (b)
24. (b)
25. (d) According to Lenz's law.
26. (c) $e = -N \left(\frac{\Delta B}{\Delta t} \right) A \cos \theta = -100 \times \frac{(6 - 1)}{2} \times (40 \times 10^{-4}) \cos 0$
 $\Rightarrow |e| = 1 \text{ V}$
27. (d)
28. (b)
29. (d)
30. (d) Emf induces in ring and it will oppose the motion. Hence due to the resistance of the ring all energy dissipates.
31. (b) $\Delta Q = \frac{NBA}{R} (\cos \theta_1 - \cos \theta_2)$
 $= \frac{500 \times 0.2 \times 0.1 (\cos 0 - \cos 180)}{50} = 0.4 \text{ C}$
32. (a) $\phi = NBA \cos \theta = 100 \times 0.2 \times 5 \times 10^{-4} \cos 60^\circ$
 $= 5 \times 10^{-3} \text{ Wb}$

33. (b) $\Delta Q = \frac{\Delta \phi}{R} = \frac{(10-2)}{2} = 4C$

34. (a)

35. (b)

36. (b) $\phi = \mu_0 n i A = 4\pi \times 10^{-7} \times \frac{3000}{1.5} \times 2 \times \pi (2 \times 10^{-2})^2$
 $= 6.31 \times 10^{-6} \text{ Wb}$

37. (d) $q = -\frac{N}{R} (B_2 - B_1) A \cos \theta$
 $32 \times 10^{-6} = -\frac{100}{(160+40)} (0-B) \times \pi \times (6 \times 10^{-3})^2 \times \cos 0^\circ$
 $\Rightarrow B = 0.565 \text{ T}$

38. (a) Faraday's laws involve conversion of mechanical energy into electric energy. This is in accordance with the law of conservation of energy.

39. (a) $e = -\frac{N(B_2 - B_1)A \cos \theta}{\Delta t}$
 $\Rightarrow 0.1 = \frac{-50 \times (0 - 2 \times 10^{-2}) \times 100 \times 10^{-4} \times \cos 0^\circ}{t}$
 $\Rightarrow t = 0.1 \text{ sec.}$

40. (c) $q = \frac{N}{R} d\phi \therefore q \propto d\phi$

41. (c) $i = \frac{|e|}{R} = \frac{N}{R} \cdot \frac{\Delta B}{\Delta t} A \cos \theta = \frac{20}{100} \times 1000 \times (25 \times 10^{-4}) \cos 0^\circ$
 $\Rightarrow i = 0.5 \text{ A}$

42. (b) According to Lenz's law.

43. (c)

44. (c) E.m.f. or current induces, only when flux linked with the coil changes.

45. (d) $e = -\frac{d\phi}{dt} = -\frac{d}{dt} (3t^2 + 4t + 9) = -(6t + 4)$
 $e = -[6(2) + 4] = -16 \Rightarrow |e| = 16 \text{ volt}$

46. (d) $e = -\frac{NBA(\cos \theta_2 - \cos \theta_1)}{\Delta t}$
 $= -\frac{800 \times 4 \times 10^{-5} \times 0.05 (\cos 90^\circ - \cos 0^\circ)}{0.1} = 0.016 \text{ V}$

47. (d)

48. (d)

49. (d) $e = -\frac{d\phi}{dt} = -(10t - 4) \Rightarrow (e)_{t=2} = -(10 \times 0.2 - 4) = 2 \text{ volt}$

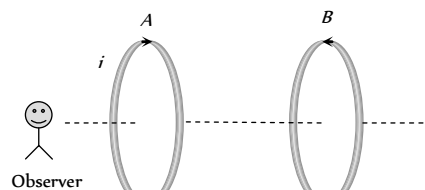
50. (c)

51. (b)

52. (a) If bar magnet is falling vertically through the hollow region of long vertical copper tube then the magnetic flux linked with the copper tube (due to 'non-uniform' magnetic field of magnet) changes and eddy currents are generated in the body of the tube by Lenz's law the eddy currents opposes the falling of the magnet which therefore experience a retarding force. The retarding force increases with increasing velocity of the magnet and finally equals the weight of the magnet. The

magnet then attains a constant final terminal velocity i.e. magnet ultimately falls with zero acceleration in the tube.

53. (d)



If current through A increases, crosses (X) linked with coil B increases, hence anticlockwise current induces in coil B. As shown in figure both the current produces repulsive effect.

54. (b) $e = -\frac{d\phi}{dt} = -\frac{d}{dt} (5t^3 - 100t + 300)$
 $= -(15t^2 - 100) \text{ at } t = 2 \text{ sec; } e = 40 \text{ V}$

55. (b) By using $e = -\frac{NBA(\cos \theta_2 - \cos \theta_1)}{\Delta t}$
 $e = -\frac{1000 \times 2 \times 10^{-5} \times 500 \times 10^{-4} (\cos 180^\circ - \cos 0^\circ)}{0.2}$
 $= 10^{-2} \text{ volt} = 10 \text{ mV}$

56. (a) Similar to Q. 52

57. (c)

58. (a) Induced charge $Q = -\frac{N}{R} (\phi_2 - \phi_1) = \frac{1}{100} (60 - 10) = 0.5C$

59. (d)

60. (b) $i = \frac{e}{R} = \frac{-N}{R} \frac{(\phi_2 - \phi_1)}{\Delta t} = \frac{-n(W_2 - W_1)}{5Rt}$

61. (a) Magnetic flux linked with the ring changes so current flows through it.

62. (a) $|e| = \frac{d\phi}{dt} = \frac{d}{dt} (5t^2 + 3t + 16) = (10t + 3)$

when $t = 3 \text{ sec, } e_3 = (10 \times 3 + 3) = 33 \text{ V}$

when $t = 4 \text{ sec, } e_4 = (10 \times 4 + 3) = 43 \text{ V}$

Hence emf induced in fourth second

$= e_4 - e_3 = 43 - 33 = 10 \text{ V}$

63. (d) $e = \frac{-NBA(\cos \theta_2 - \cos \theta_1)}{\Delta t}$
 $= -\frac{500 \times 4 \times 10^{-4} \times 0.1 (\cos 90^\circ - \cos 0^\circ)}{0.1} = 0.2 \text{ V}$

64. (d) $q = \frac{N}{R} (\Delta \phi) = \frac{1}{2} \times (10 - 2) = 4C$

65. (c) At low frequency of 1 to 2 Hz, oscillations may be observed as our eyes will be able to detect it.

66. (c) Since the magnetic field is uniform therefore there will be no change in flux hence no current will be induced.

67. (a) $\phi = BA$

$\Rightarrow \text{change in flux } d\phi = B \cdot dA = 0.05 (101 - 100) 10^{-4}$
 $= 5.10^{-6} \text{ Wb.}$

Now, charge $dQ = \frac{d\phi}{R} = \frac{5 \times 10^{-6}}{2} = 2.5 \times 10^{-6} \text{ C}$.

68. (b) $\Delta Q = \frac{\Delta\phi}{R} = \frac{n \times BA}{R}$
 $\Rightarrow B = \frac{\Delta Q \cdot R}{nA} = \frac{2 \times 10^{-4} \times 80}{40 \times 4 \times 10^{-4}} = 1 \text{ Wb/m}^2$

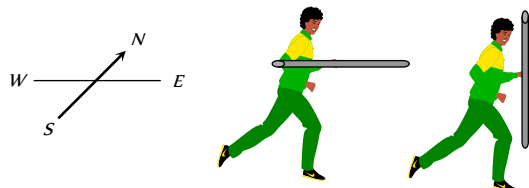
Motional EMI

1. (a) $\text{Emf} = e = e_0 \sin\theta$; e will be maximum when θ is 90° i.e. plane of the coil will be horizontal.
2. (b) Induced e.m.f. $= Blv = 0.3 \times 10^{-4} \times 10 \times 5$
 $= 1.5 \times 10^{-3} \text{ V} = 1.5 \text{ mV}$
3. (d) Conductor cuts the flux only when, if it moves in the direction of M .
4. (c) $e = B_v \cdot vl = 0.2 \times 10^{-4} \times \left(\frac{180 \times 1000}{3600}\right) \times 1 = 10^{-3} \text{ V}$
5. (b) $e = Blv = 3 \times 10^{-3} \times 10^2 = 0.3 \text{ volt}$
6. (b) This is the case of periodic EMI
7. (b) $e = B_v \cdot vl = 2 \times 10^{-4} \times \left(\frac{360 \times 1000}{3600}\right) \times 50 \Rightarrow e = 1 \text{ V}$
8. (c) $e = \frac{1}{2} B \omega r^2 = \frac{1}{2} \times 0.1 \times 2\pi \times 10 \times (0.1)^2 = \pi \times 10^{-2} \text{ V}$
9. (d) $e = \frac{1}{2} B \omega r^2 = \frac{1}{2} \times 0.2 \times 10^{-4} \times 5 \times (1)^2 = 50 \mu\text{V}$
10. (d) No flux change is taking place because magnetic field exists everywhere and is constant in time and space.
11. (b) If player is running with rod in vertical position towards east, then rod cuts the magnetic field of earth perpendicularly (magnetic field of earth is south to north).

Hence Maximum emf induced is

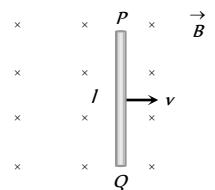
$$e = Blv = 4 \times 10^{-5} \times \frac{30 \times 1000}{3600} \times 3 = 1 \times 10^{-3} \text{ volt}$$

When he is running with rod in horizontal position, no field is cut by the rod, so $e = 0$.



12. (c) $e = NBA\omega$; $\omega = 2\pi f = 2\pi \times \frac{2000}{60}$
 $\therefore e = 50 \times 0.05 \times 80 \times 10^{-4} \times 2\pi \times \frac{2000}{60} = \frac{4\pi}{3}$

13. (b)
14. (c) According to Fleming's right hand rule, the direction of B will be perpendicular to the plane of paper and act downward.
15. (d) By Fleming's right hand rule.
16. (c) $e = Blv \Rightarrow e \propto v \propto gt$
17. (c) $e = Blv = 0.5 \times 2 \times 1 = 1 \text{ V}$
18. (b) A motional emf $e = Blv$ is induced in the rod, or we can say, a potential difference is induced between the two ends of the rod AB , with P at higher potential and Q at lower potential. Due to this potential difference, there is an electric field in the rod.



19. (c)
20. (b) $e = Blv \Rightarrow e = 0.7 \times 2 \times (10 \times 10^{-2}) = 0.14 \text{ V}$
21. (d) $e = Blv \Rightarrow e = 0.9 \times 7 \times 0.4 = 2.52 \text{ V}$
22. (d)
23. (d)
24. (d) $e = Bl^2 \pi v = 0.4 \times 10^{-4} \times (0.5)^2 \times (3.14) \times \frac{120}{60}$
 $= 6.28 \times 10^{-5} \text{ V}$
25. (c) $e = \frac{1}{2} Bl^2 \omega = \frac{1}{2} \times 0.3 \times (2)^2 \times 100 = 60 \text{ V}$
26. (a) $e = Blv = 5 \times 10^{-5} \times \frac{360 \times 1000}{3600} \times 20 = 0.1 \text{ V}$
27. (c)
28. (c) Peak value of $\text{emf} = e_0 = \omega NBA = 2\pi \nu NBA$
 $= 2\pi \times 50 \times 300 \times 4 \times 10^{-2} \times (25 \times 10^{-2} \times 10 \times 10^{-2})$
 $= 30 \pi \text{ volt}$
29. (c) $e = \frac{1}{2} Bl^2 \omega = Bl^2 \pi \nu$
 $\Rightarrow e = 0.5(20 \times 10^{-2})^2 \times 3.14 \times 100 = 6.28 \text{ V}$
30. (d)
31. (b) $e_0 = \omega NBA = (2\pi \nu) NB(\pi r^2) = 2 \times \pi^2 \nu N B r^2$
 $= 2 \times (3.14)^2 \times \frac{1800}{60} \times 4000 \times 0.5 \times 10^{-4} \times (7 \times 10^{-2})^2$
 $= 0.58 \text{ V}$
32. (b) Two emfs induces in the closed circuit each of value Blv . These two emfs are additive. So $E_{\text{net}} = 2Blv$.
33. (b) When a conductor lying along the magnetic north-south, moves eastwards it will cut vertical component of B_0 . So induced emf
 $e = vB_v l = v(B_0 \sin\delta) l = B_0 l v \sin\delta$.

Static EMI

1. (d) $e = -L \frac{di}{dt}$ but $e = 4V$ and $\frac{di}{dt} = \frac{0-1}{10^{-3}} = -1/10^{-3}$
 $\therefore \frac{-1}{10^{-3}}(-L) = 4 \Rightarrow L = 4 \times 10^{-3} \text{ henry}$
2. (d) $L = \frac{e}{di/dt} = \frac{5}{(3-2)/10^{-3}} = \frac{5}{1} \times 10^{-3} = 5 \text{ millihenry}$
3. (b) Energy stored $E = \frac{1}{2} Li^2 = \frac{1}{2} \times 50 \times 10^{-3} \times 4 = 0.1 J$
4. (a) Given $\frac{di}{dt} = 2 A / \text{sec.}$, $L = 5 H \therefore e = L \frac{di}{dt} = 5 \times 2 = 10 V$
5. (d) As we know $e = -\frac{d\phi}{dt} = -L \frac{di}{dt}$
 Work done against back e.m.f. e in time dt and current i is
 $dW = -eidt = L \frac{di}{dt} i dt = Li di \Rightarrow W = L \int_0^i i di = \frac{1}{2} Li^2$
6. (d) Induced emf $e = M \frac{di}{dt} = \frac{\mu_0 N_1 N_2 A}{l} \cdot \frac{di}{dt}$
 $= \frac{4\pi \times 10^{-7} \times 2000 \times 300 \times 1.2 \times 10^{-3}}{0.30} \times \frac{2 - (-2)}{0.25}$
 $= 48.2 \times 10^{-3} V = 48 mV$
7. (c) Self inductance $L = \mu_0 N^2 A / l = \mu_0 n^2 l A$
 Where n is the number of turns per unit length and N is the total number of turns and $N = nl$
 In the given question n is same. A is increased 4 times and l is increased 2 times and hence L will be increased 8 times.
8. (c) $M = -\frac{e_2}{di_1/dt} = -\frac{e_1}{di_2/dt}$
 Also $e_1 = -L_1 \frac{di_1}{dt}$, $e_2 = -L_2 \frac{di_2}{dt}$
 $M^2 = \frac{e_1 e_2}{\left(\frac{di_1}{dt}\right)\left(\frac{di_2}{dt}\right)} = L_1 L_2 \Rightarrow M = \sqrt{L_1 L_2}$
9. (c) Inductance is inherent property of electrical circuits. It will always be found in an electrical circuit whether we want it or not. The circuit in which a large emf is induced when the current in the circuit changes is said to have greater inductance. A straight wire carrying current with no iron part in the circuit will have lesser value of inductance while if the circuit contains a circular coil having many number of turns, the induced emf to oppose the cause will be greater and the circuit is therefore said to have greater value of inductance.
10. (c) Magnetic flux $\phi = LI$
 By analogy, since physical quantities mass (m) and linear velocity (v) are equivalent to electrical quantities inductance (L) and current (I) respectively. Thus magnetic flux $\phi = LI$ is equivalent to momentum $p = m \times v$.
11. (a) Energy stored $= \frac{1}{2} Li^2$, where Li is a magnetic flux.
12. (b) $L = \mu mi \Rightarrow \frac{L_2}{L_1} = \frac{\mu}{\mu_0}$ (n and i are same)
 $\Rightarrow L_2 = \mu_r L_1 = 900 \times 0.18 = 162 mH$
13. (b) $e = M \frac{di}{dt} = 0.2 \times 5 = 1 V$
14. (d) $e = -L \frac{di}{dt} \Rightarrow 2 = -L \left(\frac{8-2}{3 \times 10^{-2}} \right) \Rightarrow L = 0.01 H = 10 mH$
15. (d) $e = M \frac{di}{dt} = 1.25 \times 80 = 100 V$
16. (b) $\frac{L_B}{L_A} = \left(\frac{n_B}{n_A} \right)^2 \Rightarrow L_B = \left(\frac{500}{600} \right)^2 \times 108 = 75 mH$
17. (a) $L \propto N^2$ i.e. $\frac{L_1}{L_2} = \left(\frac{N_1}{N_2} \right)^2 \Rightarrow L_2 = L_1 \left(\frac{N_2}{N_1} \right)^2 = 4 L_1$
18. (b) $e = -L \frac{di}{dt} \Rightarrow 8 = L \frac{(4-2)}{0.05} \Rightarrow L = 0.2 H$
19. (b) $e = M \frac{di}{dt} \Rightarrow M = \frac{15000}{3} \times 0.001 = 5 H$
20. (b) $L = \frac{e}{di/dt} = \frac{12}{48/60} = 15 H$
21. (a) $B = \frac{\mu_0 Ni}{2r} = \frac{4\pi \times 10^{-7} \times 100 \times 2 \times \sqrt{\pi}}{2 \times 10^{-2}} = 0.022 \text{ wb} / m^2$
22. (d) $e = M \frac{di}{dt} = 0.09 \times \frac{20}{0.006} = 300 V$
23. (c)
24. (c) Inductors obey the laws of parallel and series combination of resistors.
25. (b) There will be self induction effect when soft iron core is inserted.
26. (c) $L = \mu_0 N^2 A / l$
27. (b)
28. (c) $e = -L \frac{di}{dt} \Rightarrow e = 5 \times \frac{1}{5} = 1 \text{ volt}$
29. (c) $e = L \frac{di}{dt} \Rightarrow L = \frac{\text{Volt-sec}}{\text{Ampere}}$
30. (d) $e = -L \frac{di}{dt} = -0.4 \times 10^{-3} \times \frac{250 \times 10^{-3}}{0.1} = -1 mV$
31. (b) In steady state current passing through solenoid
 $i = \frac{E}{R} = \frac{10}{10} = 1 A$
32. (a) $e = L \frac{di}{dt} \Rightarrow 2 = L \times \frac{6}{3 \times 10^{-3}} \Rightarrow L = 1 mH$

33. (b) From $i = i_0[1 - e^{-Rt/L}]$, where $i_0 = \frac{5}{5} = 1 \text{ amp}$

$$\therefore i = 1 \left(1 - e^{-\frac{5 \times 2}{10}} \right) = (1 - e^{-1}) \text{ amp}$$

34. (c) $M = \frac{\mu_0 N_1 N_2 A}{l}$

35. (d) $e = L \frac{di}{dt} = 60 \times 10^{-6} \cdot \frac{(1.5 - 1.0)}{0.1} = 3 \times 10^{-4} \text{ volt}$

36. (a) $\phi = Li \Rightarrow NBA = Li$

Since magnetic field at the centre of circular coil carrying

current is given by $B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi Ni}{r}$

$$\therefore N \cdot \frac{\mu_0}{4\pi} \cdot \frac{2\pi Ni}{r} \cdot \pi r^2 = Li \Rightarrow L = \frac{\mu_0 N^2 \pi r}{2}$$

Hence self inductance of a coil

$$= \frac{4\pi \times 10^{-7} \times 500 \times 500 \times \pi \times 0.05}{2} = 25 \text{ mH}$$

37. (a) Induced *e.m.f.* $e = M \frac{di}{dt} \Rightarrow 100 \times 10^{-3} = M \left(\frac{10}{0.1} \right)$

$$\therefore M = 10^{-3} \text{ H} = 1 \text{ mH}$$

38. (c) $\frac{\Delta i}{\Delta t} = \frac{10}{2} = 5 \text{ A/sec} \Rightarrow e = L \frac{\Delta i}{\Delta t} = 0.5 \times 5 = 2.5 \text{ volts}$

39. (c) Time in which the current will decay to $\frac{1}{e}$ of its steady value
is $t = \tau = \frac{L}{R} = \frac{50}{10} = 5 \text{ seconds}$

40. (a)

41. (d) $L \propto N^2$

42. (a) $e_2 = M \frac{di_1}{dt} \Rightarrow i_2 R_2 = M \frac{di_1}{dt} \Rightarrow 0.4 \times 5 = 0.5 \times \frac{di_1}{dt}$
 $\Rightarrow \frac{di_1}{dt} = 4 \text{ A/sec.}$

43. (a) $U = \frac{1}{2} Li^2 = \frac{1}{2} \times (50 \times 10^{-3}) \times (4)^2 = 400 \times 10^{-3} = 0.4 \text{ J}$

44. (b) $e = -L \left(\frac{di}{dt} \right) \Rightarrow 8 = -L \times \left(-\frac{2}{0.05} \right) \Rightarrow L = 0.2 \text{ H}$

45. (b) $U = \frac{1}{2} Li^2$ i.e. $\frac{U_2}{U_1} = \left(\frac{i_2}{i_1} \right)^2 = \left(\frac{1}{2} \right)^2 = \frac{1}{4} \Rightarrow U_2 = \frac{1}{4} U_1$

46. (a, b, c, d)

47. (c) $|e| = L \frac{di}{dt} \Rightarrow 220 = L \times \frac{10}{0.5} \Rightarrow L = 11 \text{ H}$

48. (b)

49. (d) $t = \tau = \frac{L}{R} = \frac{2.5}{0.5} = 5 \text{ sec}$

50. (c) $L = \mu_0 \frac{N^2}{l} A$. When N and l are doubled, L is also doubled.

51. (c) Energy $= \frac{1}{2} LI^2 = \frac{1}{2} \times 100 \times 10^{-3} \times 1^2 = 0.05 \text{ J}$

52. (b) $e = -M \frac{di}{dt} = -5 \times \frac{(-5)}{10^{-3}} = 25000 \text{ V}$

53. (a) $L \propto n$ (Number of turns), For straight conductor $n = 0$, hence $L = 0$.

54. (a) $\Delta \phi = L \Delta I \Rightarrow L = \frac{\Delta \phi}{\Delta I} = \frac{2 \times 10^{-2}}{0.01} = 2 \text{ H}$

55. (a) $e = L \frac{di}{dt} \Rightarrow 100 = L \times \frac{4}{0.1} \Rightarrow L = 2.5 \text{ H}$

56. (a) The inductances are in parallel $\Rightarrow L_{eq} = \frac{L}{3} = \frac{3}{3} = 1 \text{ H}$

57. (a) $|e| = M \frac{di}{dt} \Rightarrow 8 \times 10^{-3} = M \times 3 \Rightarrow A_1 = 2.66 \text{ mH}$

58. (d) $|e| = L \frac{di}{dt} \Rightarrow 10 = L \times \frac{10}{1} \Rightarrow L = 1 \text{ H}$

59. (a) $N\phi = Li \Rightarrow \phi = \frac{Li}{N} = \frac{8 \times 10^{-3} \times 5 \times 10^{-3}}{400} = 10^{-7} = \frac{\mu_0}{4\pi} \text{ wb}$

60. (b) According to Lenz's law.

61. (d) $N\phi = Li \Rightarrow \frac{Nd\phi}{dt} = \frac{Ldi}{dt} \Rightarrow NB \frac{dA}{dt} = \frac{Ldi}{dt}$

$$\Rightarrow \frac{1 \times 1 \times 5}{10^{-3}} = L \times \left(\frac{2-1}{2 \times 10^{-3}} \right) \Rightarrow L = 10 \text{ H}$$

62. (b) $L = \frac{\mu_0 N^2 A}{l} = \frac{4\pi \times 10^{-7} \times (500)^2 \times 20 \times 10^{-4}}{0.5} = 1.25 \text{ mH}$

63. (a) $L_S = L_1 + L_2 = 10 \text{ H}$ (i)

$$L_P = \frac{L_1 L_2}{L_1 + L_2} = 2.4 \text{ H} \quad \text{..... (ii)}$$

On solving (i) and (ii) $LL_P = 24$ (iii)

Also $(L_1 - L_2)^2 = (L_1 + L_2)^2 - 4L_1 L_2$

$$\Rightarrow (L_1 - L_2)^2 = (10)^2 - 4 \times 24 = 4 \Rightarrow L_1 - L_2 = 2 \text{ H}$$

64. (d) $e = L \frac{di}{dt} \Rightarrow 12 = L \times \frac{45}{60} \Rightarrow L = 16 \text{ H}$

65. (a) $|e| = L \frac{di}{dt} \Rightarrow 1 = \frac{L \times \{10 - (-10)\}}{0.5} \Rightarrow L = 25 \text{ mH}$

66. (d) $U = \frac{1}{2} Li^2 \Rightarrow U = \frac{1}{2} \times 5 \times \left(\frac{100}{10} \right)^2 = 250 \text{ J}$

67. (c) $\phi = Mi \Rightarrow M = \frac{1.2 \times 10^{-2}}{0.01} = 1.2 \text{ H}$

68. (c) $U = \frac{1}{2} Li^2 \Rightarrow U = \frac{1}{2} \times 40 \times 10^{-3} \times (2)^2 = 0.08 \text{ J}$

69. (c) $L \propto N^2$
70. (b) $N_2\phi_2 = Mi_1 \Rightarrow 9 \times 10^{-5} = M \times 3 \Rightarrow M = 3 \times 10^{-5} H$
71. (a) $|e| = L \frac{di}{dt} \Rightarrow 20 = L \times \frac{(18-2)}{0.05} \Rightarrow L = 62.5 mH$
72. (b) $|e| = L \frac{di}{dt} \Rightarrow |e| = 10 \times 10^{-6} \times \frac{1}{10} = 1 \mu V$
73. (c)
74. (b) $\phi_T = Li \Rightarrow L = \frac{10^{-5}}{5 \times 10^{-3}} = 2 mH$
75. (a) $L \propto N^2$
76. (c) $e = -L \frac{di}{dt}$, since current decreases so $\frac{di}{dt}$ is negative, hence
 $e = -5 \times (-2) = +10V$
77. (c) $e = L \frac{di}{dt} \Rightarrow e = 0.1 \times 200 = 20V$
78. (c) $e = M \frac{di}{dt} \Rightarrow e = 0.1 \times \frac{(20-0)}{0.02} = 100V$
79. (c) $L = \frac{\mu_0 N^2 A}{l} = \frac{4\pi \times 10^{-7} \times (1000)^2 \times 10 \times 10^{-4}}{1} = 1.256 mH$
80. (d) In secondary e.m.f. induces only when current through primary changes.
81. (b)
82. (c)
83. (a) $e = L \frac{di}{dt} \Rightarrow 8 = L \times \frac{(2-(-2))}{0.05} \Rightarrow L = 0.1H$
84. (b) $U = \frac{1}{2} Li^2 = \frac{1}{2} L \left(\frac{E}{R} \right)^2 = \frac{1}{2} \times 5 \times \left(\frac{100}{20} \right)^2 = 62.50J$
85. (a)
86. (a)
87. (a) $e = -L \frac{di}{dt} \Rightarrow 0.4 = -\frac{L(0.2-1)}{10} \Rightarrow L = 5 H$
88. (c) $|e| = L \frac{di}{dt} \Rightarrow 30 = L \times \frac{(6-0)}{0.3} \Rightarrow L = 1.5 H$
89. (b) $i = i_0 \left(1 - e^{-\frac{Rt}{L}} \right) \Rightarrow \frac{di}{dt} = -i_0 \left(-\frac{R}{L} \right) e^{-\frac{Rt}{L}} = \frac{i_0 R}{L} \cdot e^{-\frac{Rt}{L}}$
 At $t = 0$; $\frac{di}{dt} = \frac{i_0 R}{L} = \frac{E}{L} \Rightarrow 4 = \frac{E}{20} \Rightarrow E = 80 V$
90. (d) $N\phi = Li \Rightarrow 100 \times 10^{-5} = L \times 5 \Rightarrow L = 0.2 mH.$
91. (d) When the two coils are joined in series such that the winding of one is opposite to the other, then the emf produced in first coil is 180° out of phase of the emf produced in second coil.
 Thus, emf produced in first coil is negative and the emf produced in second coil is positive so, net inductance is
 $L = L_1 + L_2 = L + L \Rightarrow L = -\frac{\phi}{i} + \frac{\phi}{i} = 0$
92. (a) $e = M \frac{di}{dt} \Rightarrow 1.5 = M \times \frac{30}{0.1} \Rightarrow M = 0.05 H$
93. (d)
94. (c) Current in B_1 will promptly become zero while current in B_2 will slowly tend to zero.
95. (b)
96. (c) When battery disconnected current through the circuit start decreasing exponentially according to $i = i_0 e^{-Rt/L}$
 $\Rightarrow 0.37i_0 = i_0 e^{-Rt/L} \Rightarrow 0.37 = \frac{1}{e} = e^{-Rt/L} \Rightarrow t = \frac{L}{R}$
97. (a) Current at any instant of time t after closing an $L-R$ circuit is given by $I = I_0 \left[1 - e^{-\frac{R}{L}t} \right]$
 Time constant $t = \frac{L}{R}$
 $\therefore I = I_0 \left[1 - e^{-\frac{R}{L} \times \frac{L}{R}} \right] = I_0 (1 - e^{-1}) = I_0 \left(1 - \frac{1}{e} \right)$
 $= I_0 \left(1 - \frac{1}{2.718} \right) = 0.63I_0 = 63\% \text{ of } I_0$
98. (c) $i = \frac{V}{R} = \frac{10}{2} = 5A$
 $U = \frac{1}{2} Li^2 = \frac{1}{2} \times 2 \times 25 = 25 J$
99. (d)
100. (d) Time constant $= \frac{L}{R} = \frac{40}{8} = 5 \text{ sec.}$
101. (b) $t = \tau = \frac{L}{R} = \frac{60}{30} = 2 \text{ sec.}$
102. (c)
103. (b)
104. (a) $\nu_0 = \frac{1}{2\pi\sqrt{(0.25) \times (0.1 \times 10^{-6})}} = \frac{10^4}{9.93} = 1007 Hz$
105. (c) $\nu_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2 \times 3.14\sqrt{5 \times 10^{-4} \times 20 \times 10^{-6}}}$
 $\nu_0 = \frac{10^4}{6.28} = 1592 Hz$
106. (d) $i = i_0 \left(1 - e^{-\frac{Rt}{L}} \right) \Rightarrow \text{For } i = \frac{i_0}{2}, t = 0.693 \frac{L}{R}$
 $\Rightarrow t = 0.693 \times \frac{300 \times 10^{-3}}{2} = 0.1 \text{ sec}$

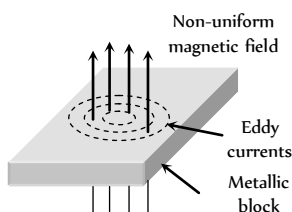
107. (c) $|e| = L \left| \frac{di}{dt} \right| = 0.5 \times \frac{10}{2} = 2.5 \text{ V}$

108. (c) Time period of LC circuit oscillations

$$T = 2\pi\sqrt{LC} \Rightarrow \text{dimensions of } \sqrt{LC} \text{ is Time.}$$

Application of EMI (Motor, Dynamo, Transformer...)

1. (b) Hot wire ammeter is not based on the phenomenon of electromagnetic induction.
2. (d)
3. (c) Direction of eddy currents is given by Lenz's rule.



4. (c) In a generator, e.m.f. is induced according as Lenz's rule.
5. (a)
6. (d)
7. (a) Circulation of eddy currents is prevented by use of laminated core.
8. (a)
9. (c)
10. (a)
11. (b) $e \propto \omega$
12. (b)
13. (a) Rotation of magnet in the dynamo creates the variable flux which in turn produces the induced current.
14. (b)
15. (b) With the increasing speed, ω increases. Thus current reduces due to increase in the back e.m.f.

Moreover $i = \frac{V - K\omega}{R}$. More ω will lead to the lesser current.

16. (c) Commutator converts ac into fluctuating dc.
17. (d) Only ac dynamo have slip rings.
18. (b) $e \propto \frac{d\phi}{dt}$; if $\phi \rightarrow$ maximum then $e \rightarrow$ minimum.

19. (d)

20. (c) Motor e.m.f. equation $E_b = V - I_a R_a$

At starting $E_b = 0$, so I_a will be maximum.

21. (d) $i = \frac{E - e}{R} \Rightarrow 1.5 = \frac{220 - e}{20} \Rightarrow e = 190 \text{ V}$

22. (d)

23. (d) $e_0 = \omega NBA = (2\pi\nu) NBA$

$$= 2 \times 3.14 \times 1000 \times 5000 \times 0.2 \times 0.25 = 157 \text{ kV}$$

24. (a) Back emf $\propto$ speed of motor.

25. (d)

26. (b)

27. (a) Efficiency $\eta = 50\%$ So $e = E/2$

$$\text{and } i = \frac{E - e}{R} \Rightarrow i = \frac{E - E/2}{R} = \frac{E}{2R}$$

$$\Rightarrow R = \frac{E}{2i} = \frac{60}{2 \times 10} = 3\Omega$$

28. (c)

29. (c) $\eta = \frac{e}{E} \times 100 \Rightarrow e = 0.3 E$

$$\text{Now, } i = \frac{E - e}{R} \Rightarrow 12 = \frac{50 - (0.3 \times 50)}{R} \Rightarrow R = 2.9\Omega$$

30. (a) $i = \frac{E - e}{R} = \frac{220 - 210}{2} = \frac{10}{2} = 5 \text{ A}$

31. (a)

32. (c) A transformer is a device to convert alternating current at high voltage into low voltage and *vice-versa*.

33. (b) We know that for step down transformer

$$V_p > V_s \text{ but } \frac{V_p}{V_s} = \frac{i_s}{i_p}; \therefore i_s > i_p$$

Current in the secondary coil is greater than the primary.

34. (c)

35. (a)

36. (b) Transformation ratio $k = \frac{N_s}{N_p} = \frac{V_s}{V_p}$

For step-up transformers, $N_s > N_p$ i.e. $V_s > V_p$, hence $k > 1$.

37. (a) $\frac{V_p}{V_s} = \frac{N_p}{N_s} \Rightarrow N_p = \left(\frac{220}{2200} \right) 2000 = 200$

38. (c) Provided no power losses, being assumed.

39. (a) $\frac{N_s}{N_p} = \frac{V_s}{V_p} \Rightarrow \frac{200}{100} = \frac{V_s}{120} \Rightarrow V_s = 240 \text{ V}$

$$\text{also } \frac{V_s}{V_p} = \frac{i_p}{i_s} \Rightarrow \frac{240}{120} = \frac{10}{i_s} \Rightarrow i_s = 5 \text{ A}$$

40. (c) $\frac{N_s}{N_p} = \frac{V_s}{V_p} \Rightarrow \frac{1}{20} = \frac{V_s}{2400} \Rightarrow V_s = 120 \text{ V}$

For 100% efficiency $V_s i_s = V_p i_p$

$$\Rightarrow 120 \times 80 = 2400 i_p \Rightarrow i_p = 4 \text{ A}$$

41. (b) $\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{500}{2500} = \frac{1}{5} \Rightarrow V_p = \frac{200}{5} = 40 \text{ V}$

$$\text{Also } i_p V_p = i_s V_s \Rightarrow i_p = i_s \frac{V_s}{V_p} = 8 \times 5 = 40 \text{ A}$$

42. (a) $\frac{N_s}{N_p} = \frac{V_s}{V_p} \Rightarrow \frac{250}{100} = \frac{V_s}{28/\sqrt{2}} \Rightarrow V_s = 50 \text{ V}$
43. (d) $\eta = \frac{V_s i_s}{V_p i_p} \times 100 = \frac{11 \times 90}{220 \times 5} \times 100 = 90 \%$
44. (d) Transformer doesn't work on dc
45. (b)
46. (b)
47. (a) For 100% efficient transformer
 $V_s i_s = V_p i_p \Rightarrow \frac{V_s}{V_p} = \frac{i_p}{i_s} = \frac{N_s}{N_p} \Rightarrow \frac{i_p}{4} = \frac{25}{100} \Rightarrow i_p = 1 \text{ A}$
48. (a)
49. (d) $\frac{N_s}{N_p} = \frac{i_p}{i_s} \Rightarrow i_s = i_p \times \frac{N_p}{N_s} = 2 \times \frac{100}{20} = 10 \text{ A}$
50. (a) $\frac{V_s}{V_p} = \frac{i_p}{i_s} \Rightarrow i_p = \frac{11000 \times 2}{220} = 100 \text{ A}$
51. (c) $\frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{i_s}{i_p}$. The transformer is step-down type, so primary coil will have more turns. Hence
 $\frac{5000}{500} = \frac{2200}{V_s} = \frac{i_s}{4} \Rightarrow V_s = 220 \text{ V}, i_s = 40 \text{ amp}$
52. (b) $i_s = \frac{P_s}{V_s} = \frac{4.4 \times 10^3}{11 \times 10^3} = 0.4 \text{ A}$
53. (b) $\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{22000}{220} = 100$
54. (b)
55. (a)
56. (b) $\frac{N_s}{N_p} = \frac{i_p}{i_s}$ or $\frac{25}{1} = \frac{i_p}{2} \Rightarrow i_p = 50 \text{ A}$
57. (c) $\frac{V_s}{V_p} = \frac{N_s}{N_p} \Rightarrow V_s = \frac{N_s}{N_p} \times V_p = \frac{10}{200} \times 240 = 12 \text{ volts}$
58. (a) $\frac{V_s}{V_p} = \frac{N_s}{N_p} \Rightarrow \frac{V_s}{20} = \frac{5000}{500} \Rightarrow V_s = 200 \text{ V}$
 Frequency remains unchanged.
59. (a) $\frac{V_s}{V_p} = \frac{N_s}{N_p} = k \Rightarrow \frac{V_s}{30} = \frac{3}{2} \Rightarrow V_s = 45 \text{ V}$
60. (a) $\frac{N_s}{N_p} = \frac{i_p}{i_s} \Rightarrow \frac{i_p}{i_s} = \frac{4}{5}$
61. (d) $V_p = 200 \text{ V}, V_s = 6 \text{ V}$
 $P_{out} = V_s i_s \Rightarrow 30 = 6 \times i_s \Rightarrow i_s = 5 \text{ A}$
 From $\frac{V_s}{V_p} = \frac{i_p}{i_s} \Rightarrow \frac{6}{200} = \frac{i_p}{5} \Rightarrow i_p = 0.15 \text{ A}$
62. (b) $\frac{E_p}{E_s} = \frac{N_p}{N_s} \Rightarrow \frac{200}{E_s} = \frac{100}{20} \Rightarrow E_s = 40 \text{ V}$
63. (d) Since all the losses are neglected.
 So $P_{out} = P_{in}$
64. (c)
65. (b) $\frac{i_p}{i_s} = \frac{N_s}{N_p} \Rightarrow \frac{i_p}{4} = \frac{1}{100} \Rightarrow i_p = 0.04 \text{ A}$
66. (b) $N_p : N_s = 1 : 10$ and $V_s = 0.5 \times 200 = 100 \text{ V}$
 $\frac{V_s}{V_p} = \frac{N_s}{N_p} \Rightarrow \frac{100}{V_p} = \frac{10}{1} \Rightarrow V_p = 10 \text{ V}$
 $\frac{i_p}{i_s} = \frac{N_s}{N_p} \Rightarrow \frac{i_p}{0.5} = \frac{10}{1}, i_p = 5 \text{ amp}$
67. (c)
68. (d) $\frac{V_p}{V_s} = \frac{i_s}{i_p} \Rightarrow \frac{220}{22000} = \frac{i_s}{5} \Rightarrow i_s = 0.05 \text{ amp}$
69. (b) $\frac{V_p}{V_s} = \frac{i_s}{i_p} \Rightarrow i_s = 4 \times \frac{140}{280} = 2 \text{ A}$
70. (c) $P_s = V_s i_s \Rightarrow 1000 = V_s \times 8 \Rightarrow V_s = \frac{1000}{8}$
 $\frac{V_p}{V_s} = \frac{N_p}{N_s} \Rightarrow \frac{(1000/8)}{500} = \frac{100}{N_s} \Rightarrow N_s = 400$
71. (a) Transformer works on ac only.
72. (b) $\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{2200}{220} = \frac{10}{1}$
73. (c) Transformation ratio $k = \frac{V_s}{V_p} \Rightarrow \frac{5}{3} = \frac{V_s}{60} \Rightarrow V_s = 100 \text{ V}$
74. (c) $\frac{N_s}{N_p} = \frac{V_s}{V_p} \Rightarrow \frac{N_s}{600} = \frac{2200}{220} \Rightarrow N_s = 6000$
75. (b) For 100% efficiency $V_s i_s = V_p i_p$
 $\Rightarrow 1100 \times 2 = 220 \times i_p \Rightarrow i_p = 10 \text{ A}$
76. (c)
77. (c) Amplitude of the current $i_0 = \frac{e_0}{R} = \frac{\omega N B A}{R} = \frac{2\pi \nu N B (\pi r^2)}{R}$
 $i_0 = \frac{2\pi \times 1 \times 10^{-2} \times \pi (0.3)^2}{\pi^2} = 6 \times 10^{-3} \text{ A} = 6 \text{ mA}$
78. (a) $\frac{N_s}{N_p} = \frac{i_p}{i_s} \Rightarrow \frac{2000}{500} = \frac{48}{i_s} \Rightarrow i_s = 12 \text{ A}$
79. (a) $U = \frac{1}{2} Li^2 = \frac{1}{2} \times 100 \times 10^{-3} \times (10)^2 = 5 \text{ J}$
80. (c) As $\frac{I_p}{I_s} = \frac{n_p}{n_s}$; i.e. $\frac{3}{I_s} = \frac{3}{2} \Rightarrow I_s = 2 \text{ A}$.
81. (a)
82. (b)
83. (a) $\eta = \frac{\text{output power}}{\text{input power}} = \frac{E_s I_s}{E_p I_p} \Rightarrow \frac{80}{100} = \frac{200 \times I_s}{4 \times 10^3}$
 $\Rightarrow I_s = \frac{80}{100} \times \frac{4 \times 1000}{200} = 16 \text{ A}$

Also, $E_p I_p = 4 \text{ KW} \Rightarrow I_p = \frac{4 \times 10^3}{100} = 40 \text{ A}$

84. (a) $\frac{N_p}{N_s} = \frac{I_s}{I_p} \Rightarrow I_p = \frac{N_s}{N_p} I_s = \frac{10}{1} \times 2 = 20 \text{ A}$

85. (b)

86. (a) $\eta = \frac{\text{Output}}{\text{Input}} \Rightarrow \frac{80}{100} = \frac{20 \times 20}{1000 \times i_l}$
 $\Rightarrow i_l = \frac{20 \times 20 \times 100}{1000 \times 80} = 3 \text{ A}$

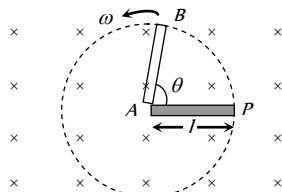
Critical Thinking Questions

1. (d) If electron is moving from left to right, the flux linked with the loop (which is into the page) will first increase and then decrease as the electron passes by. So the induced current in the loop will be first anticlockwise and will change direction as the electron passes by.

2. (a) If in time t the rod turns by an angle θ , the area generated by the rotation of rod will be

$$= \frac{1}{2} l \times \theta = \frac{1}{2} l^2 \theta$$

So the flux linked with the area generated by the rotation of rod



$$\phi = B \left(\frac{1}{2} l^2 \theta \right) \cos 0 = \frac{1}{2} B l^2 \theta = \frac{1}{2} B l^2 \omega t$$

And so $e = \frac{d\phi}{dt} = \frac{d}{dt} \left(\frac{1}{2} B l^2 \omega t \right) = \frac{1}{2} B l^2 \omega$

3. (a, c, d) From Faraday's Law, the induced voltage $V \propto L$

rate of change of current is constant $\left(V = -L \frac{di}{dt} \right)$

$$\therefore \frac{V_2}{V_1} = \frac{L_2}{L_1} = \frac{2}{8} = \frac{1}{4} \Rightarrow \frac{V_1}{V_2} = 4$$

Power given to the two coils is same, i.e.,

$$V_1 i_1 = V_2 i_2 \Rightarrow \frac{i_1}{i_2} = \frac{V_2}{V_1} = \frac{1}{4}$$

Energy stored $W = \frac{1}{2} L i^2$

$$\Rightarrow \frac{W_2}{W_1} = \left(\frac{L_2}{L_1} \right) \left(\frac{i_2}{i_1} \right)^2 = \left(\frac{1}{4} \right) (4)^2 = 4 \Rightarrow \frac{W_1}{W_2} = \frac{1}{4}$$

4. (b) $i = i_0 (1 - e^{-Rt/L})$

$$i_0 = \frac{E}{R} \text{ (Steady current) when } t = \infty$$

$$i_{\infty} = \frac{E}{R} (1 - e^{-\infty}) = \frac{5}{10} = 1.5$$

$$i_1 = 1.5 (1 - e^{-R/L}) = 1.5 (1 - e^{-2}) \Rightarrow \frac{i_{\infty}}{i_1} = \frac{1}{1 - e^{-2}} = \frac{e^2}{e^2 - 1}$$

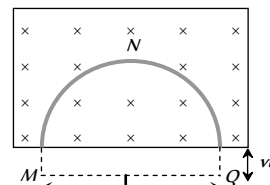
5. (d) Mutual inductance between two coil in the same plane with their centers coinciding is given by

$$M = \frac{\mu_0}{4\pi} \left(\frac{2\pi^2 R_2^2 N_1 N_2}{R_1} \right) \text{ henry.}$$

6. (d) Rate of decrease of area of the semicircular ring $-\frac{dA}{dt} = (2R) V$

According to Faraday's law of induction induced emf

$$e = -\frac{d\phi}{dt} = -B \frac{dA}{dt} = -B (2RV)$$



The induced current in the ring must generate magnetic field in the upward direction. Thus Q is at higher potential.

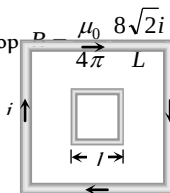
7. (b) Induced potential difference between two ends $= Blv = B_H lv$
 $= 3 \times 10^{-5} \times 2 \times 50 = 30 \times 10^{-3} \text{ volt} = 3 \text{ millivolt}$

By Fleming's right hand rule, end A becomes positively charged.

8. (b) Effective length between A and B remains same.
 9. (d) Circular loop behaves as a magnetic dipole whose one surface will be N-pole and another will be S-pole. Therefore magnetic lines a force emerges from N will meet at S. Hence total magnetic flux through x-y plane is zero.
 10. (c) If the current increases with time in loop A, then magnetic flux in B will increase. According to Lenz's law, loop -B is repelled by loop -A.

11. (b) $e = M \frac{di}{dt} = 0.005 \times \frac{d}{dt} (i_0 \sin \omega t) = 0.005 \times i_0 \omega \cos \omega t$
 $\therefore e_{\max} = 0.005 \times 10 \times 100\pi = 5\pi$

12. (b) Magnetic field produced due to large loop $B = \frac{\mu_0}{4\pi} \frac{8\sqrt{2}i}{L}$
 Flux linked with smaller loop $\phi = B(l^2) = \frac{\mu_0}{4\pi} \frac{8\pi l^2}{L}$
 $\therefore \phi = Mi \Rightarrow M = \frac{\phi}{i} = \frac{\mu_0}{4\pi} \cdot \frac{8\sqrt{2}l^2}{L} \Rightarrow M \propto \frac{l^2}{L}$



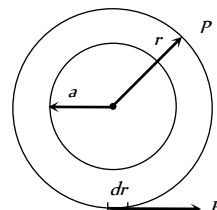
13. (b) Rate of work $= \frac{W}{t} = P = Fv$; also $F = Bil = B \left(\frac{Bvl}{R} \right) l$
 $\Rightarrow P = \frac{B^2 v^2 l^2}{R} = \frac{(0.5)^2 \times (2)^2 \times (1)^2}{6} = \frac{1}{6} \text{ W}$

14. (b) Construct a concentric circle of radius r . The induced electric field (E) at any point on the circle is equal to that at P. For this circle

$$\oint \vec{E} \cdot d\vec{l} = \left| \frac{d\phi}{dt} \right| = A \left| \frac{dB}{dt} \right|$$

$$\text{or } E \times (2\pi r) = \pi r^2 \cdot \left| \frac{dB}{dt} \right|$$

$$\Rightarrow E = \frac{r}{2} \left| \frac{dB}{dt} \right| \Rightarrow E \propto \frac{1}{r}$$



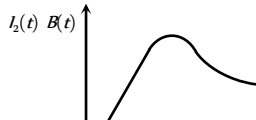
15. (d) Using k , k etc, as different constants.

$$I_1(t) = k_1 [1 - e^{-t/\tau}], B(t) = k_2 I_1(t)$$

$$I_2(t) = k_3 \frac{dB(t)}{dt} = k_4 e^{-t/\tau}$$

$$\therefore I_2(t) B(t) = k_5 [1 - e^{-t/\tau}] [e^{-t/\tau}]$$

This quantity is zero for $t=0$ and $t=\infty$ and positive for other value of t . It must, therefore, pass through a maximum.



16. (a) The mutual inductance between two coils depends on their degree of flux linkage, i.e., the fraction of flux linked with one coil which is also linked to the other coil. Here, the two coils in arrangement (a) are placed with their planes parallel. This will allow maximum flux linkage.

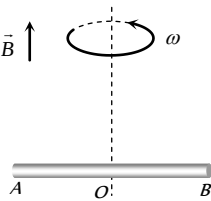
17. (d) Both AD and BC are straight conductors moving in a uniform magnetic field and emf will be induced in both. This will cause electric fields in both, but no net current flows in the circuit.

18. (d) Potential difference between

$$O \text{ and } A \text{ is } V_0 - V_A = \frac{1}{2} B l^2 \omega$$

$$O \text{ and } B \text{ is } V_0 - V_B = \frac{1}{2} B l^2 \omega$$

$$\text{so } V_A - V_B = 0$$



19. (c) $i = i_0 \left(1 - e^{-\frac{Rt}{L}} \right) \Rightarrow \frac{di}{dt} = \frac{d}{dt} i_0 - \frac{d}{dt} i_0 e^{-\frac{Rt}{L}}$

$$\Rightarrow \frac{di}{dt} = 0 - i_0 \left(-\frac{R}{L} \right) e^{-\frac{Rt}{L}} = \frac{i_0 R}{L} e^{-\frac{Rt}{L}}$$

$$\text{Initially, } t=0 \Rightarrow \frac{di}{dt} = \frac{i_0 \times R}{L} = \frac{E}{L} = \frac{5}{2} = 2.5 \text{ amp/sec.}$$

20. (d) When switch S is closed magnetic field lines passing through Q increases in the direction from right to left. So, according to Lenz's law induced current in Q i.e. I_{Q_1} will flow in such a direction so that the magnetic field lines due to I_{Q_2} passes from left to right through Q . This is possible when I_{Q_1} flows in anticlockwise direction as seen by E . Opposite is the case when switch S is opened i.e. I_{Q_2} will be clockwise as seen by E .

21. (b) Power $P = \frac{e^2}{R}$; hence $e = -\left(\frac{d\phi}{dt} \right)$ where $\phi = NBA$

$$\therefore e = -NA \left(\frac{dB}{dt} \right) \text{ Also } R \propto \frac{l}{r^2}$$

Where R = resistance, r = radius, l = Length

$$\therefore P \propto \frac{N^2 r^2}{l} \Rightarrow \frac{P_1}{P_2} = 1$$

22. (a) $H = \frac{V^2 t}{R}$ and $V = \frac{N(B_2 - B_1) A \cos \theta}{t}$

$$V = \frac{1 \times (1-2) \times 0.01 \times \cos 0^\circ}{10^{-3}} = 10 \text{ V}$$

$$\text{So, } H = \frac{(10)^2 \times 10^{-3}}{0.01} = 10 \text{ J}$$

23. (d) Peak current in the circuits $i_0 = \frac{12}{6} = 2 \text{ A}$

Current decreases from 2 A to 1 A i.e., becomes half in time

$$t = 0.693 \frac{L}{R} = 0.693 \times \frac{8.4 \times 10^{-3}}{6} = 1 \text{ millisecon.}$$

24. (a) Induced current in the circuit $i = \frac{Bvl}{R}$

$$\text{Magnetic force acting on the wire } F_m = Bil = B \left(\frac{Bvl}{R} \right) l$$

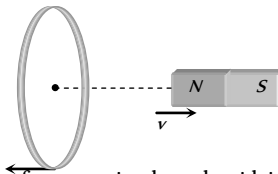
$$\Rightarrow F_m = \frac{B^2 v l^2}{R} \text{ External force needed to move the rod with constant velocity}$$

$$(F_m) = \frac{B^2 v l^2}{R} = \frac{(0.15)^2 \times (2) \times (0.5)^2}{3} = 3.75 \times 10^{-3} \text{ N}$$

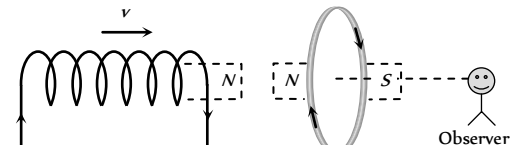
25. (c) According to Lenz's Law

26. (b) $\left(\frac{d\phi}{dt} \right)_{\text{In first case}} = e$

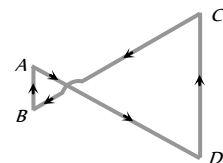
$$\left(\frac{d\phi}{dt} \right)_{\text{relative velocity } 2v} = 2 \left(\frac{d\phi}{dt} \right)_{\text{I case}} = 2e$$



27. (b) The direction of current in the solenoid is anti-clockwise as seen by observer. On displacing it towards the loop a current in the loop will be induced in a direction so as to oppose the approach of solenoid. Therefore the direction of induced current as observed by the observer will be clockwise.



28. (a) Inward magnetic field ($\times$) increasing. Therefore, induced current in both the loops should be anticlockwise. But as the area of loop on right side is more, induced emf in this will be more compared to the left side loop $\left(e = -\frac{d\phi}{dt} = -A \cdot \frac{dB}{dt} \right)$. Therefore net current in the complete loop will be in a direction shown below. Hence only option (a) is correct.

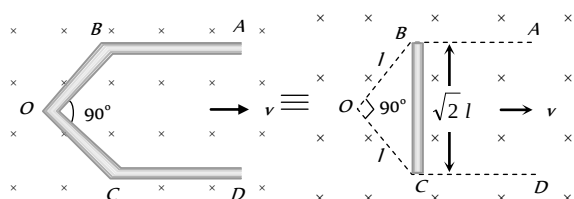


29. (b) Equivalent resistance of the given Wheatstone bridge circuit (balanced) is 3Ω so total resistance in circuit is $R = 3 + 1 = 4\Omega$. The emf induced in the loop $e = Bvl$.

$$\text{So induced current } i = \frac{e}{R} = \frac{Bvl}{R}$$

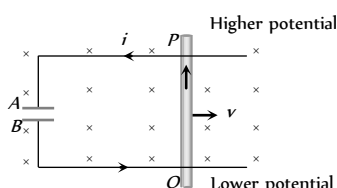
$$\Rightarrow 10^{-3} = \frac{2 \times v \times (10 \times 10^{-2})}{4} \Rightarrow v = 2 \text{ cm/sec.}$$

30. (b) There is no induced emf in the part AB and CD because they are moving along their length while emf induced between B and C i.e. between A and D can be calculated as follows



Induced emf between B and C = Induced emf between A and B
 $= Bv(\sqrt{2}l) = 1 \times 1 \times 1 \times \sqrt{2} = 1.41 \text{ volt.}$

31. (a) $Q = CV = C(Bv) = 10 \times 10^{-7} \times 4 \times 2 \times 1 = 80 \mu\text{C}$
 According to Fleming's right hand rule induced current flows from Q to P . Hence P is at higher potential and Q is at lower potential. Therefore A is positively charged and B is negatively charged.



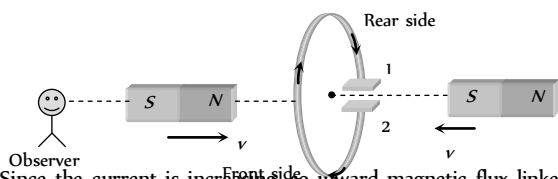
32. (b) If resistance is constant (10Ω) then steady current in the circuit
 $i = \frac{5}{10} = 0.5 \text{ A}$. But resistance is increasing it means current through the circuit start decreasing. Hence inductance comes in picture which induces a current in the circuit in the same direction of main current. So $i > 0.5 \text{ A}$.

33. (d) $P = \frac{e^2}{R}$; $e = -\frac{d}{dt}(BA) = A \frac{d}{dt}(B_0 e^{-t}) = AB_0 e^{-t}$
 $\Rightarrow P = \frac{1}{R}(AB_0 e^{-t})^2 = \frac{A^2 B_0^2 e^{-2t}}{R}$

At the time of starting $t = 0$ so $P = \frac{A^2 B_0^2}{R}$

$$\Rightarrow P = \frac{(\pi^2)^2 B_0^2}{R} = \frac{B_0^2 \pi^2 r^4}{R}$$

34. (c) When key k is pressed, current through the electromagnet start increasing i.e. flux linked with ring increases which produces repulsion effect.
 35. (b) By the movement of both the magnets, current will be anticlockwise, as seen from left side i.e. plate 1 will be positive and 2 will be negative.



36. (a) Since the current is increasing, so inward magnetic flux linked with the ring also increasing (as viewed from left side). Hence induced current in the ring is anticlockwise, so end x will be positive.

Induced emf $|e| = A \frac{dB}{dt} = A \frac{d}{dt}(B_0 + \alpha t) \Rightarrow |e| = A\alpha$

37. (a) Current in the inner coil $i = \frac{e}{R} = \frac{A_1}{R_1} \frac{dB}{dt}$

length of the inner coil $= 2\pi a$

so it's resistance $R_1 = 50 \times 10^{-3} \times 2\pi (a)$

$$\therefore i_1 = \frac{\pi a^2}{50 \times 10^{-3} \times 2\pi (a)} \times 0.1 \times 10^{-3} = 10^{-4} \text{ A}$$

According to lenz's law direction of i is clockwise.

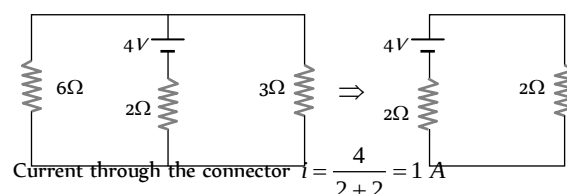
Induced current in outer coil $i_2 = \frac{e_2}{R_2} = \frac{A_2}{R_2} \frac{dB}{dt}$

$$\Rightarrow i_2 = \frac{\pi b^2}{50 \times 10^{-3} \times (2\pi b)} \times 0.1 \times 10^{-3} = 2 \times 10^{-4} \text{ A (CW)}$$

38. (c) Motional emf $e = Bvl \Rightarrow e = 2 \times 2 \times 1 = 4 \text{ V}$

This acts as a cell of emf $E = 4 \text{ V}$ and internal resistance $r = 2\Omega$.

This simple circuit can be drawn as follows



$\therefore$ magnetic force on connector $F_m = Bil = 2 \times 1 \times 1 = 2 \text{ N}$

(Towards left)

39. (b) Due to magnetic field, wire will experience an upward force
 $F = Bil = B \left(\frac{Bvl}{R} \right) l \Rightarrow F = \frac{B^2 vl^2}{R}$

If wire slides down with constant velocity then

$$F = mg \Rightarrow \frac{B^2 vl^2}{R} = mg \Rightarrow v = \frac{mgR}{B^2 l^2}$$

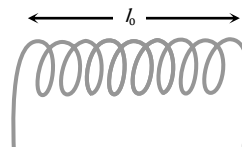
40. (c) By using $e = \frac{1}{2} Bl^2 \omega$

For part AO ; $e_{OA} = e_O - e_A = \frac{1}{2} Bl^2 \omega$

For part OC ; $e_{OC} = e_O - e_C = \frac{1}{2} B(3l)^2 \omega$

$\therefore e_A - e_C = 4 Bl^2 \omega$

41. (c) Suppose solenoid has N turns, each of radius r and length of wire is l



It's self inductance $L = \frac{\mu_0 N^2 A}{l_0} = \frac{\mu_0 N^2 \pi r^2}{l_0} \dots (i)$

Also length of the wire $l = N \times 2\pi r$

$$\Rightarrow N^2 r^2 = \frac{l^2}{4\pi^2} \dots (ii)$$

From equation (i) and (ii) $l = \sqrt{\frac{4\pi L l_0}{\mu_0}}$

42. (d) Magnetic field at the location of coil (2) produced due to coil (1)

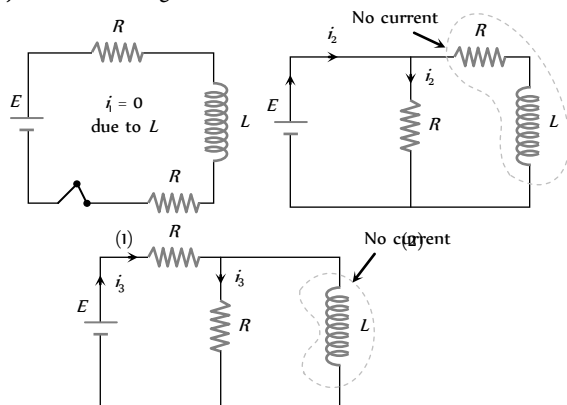
$$B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2M}{l^3}$$

Flux linked with coil (2)

$$\phi_2 = B_1 A_2 = \frac{\mu_0}{4\pi} \frac{2i(\pi a^2)}{l^3} \times (\pi a^2)$$

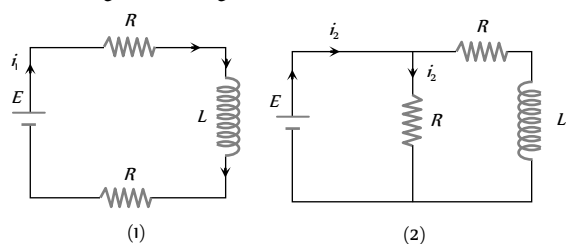
$$\text{Also } \phi_2 = Mi \Rightarrow M = \frac{\mu_0 \pi a^4}{2l^3}$$

43. (a) Just before closing the switch.



$$i_1 = 0, i_2 = \frac{E}{R}, i_3 = \frac{(3)E}{2R} \text{ so } i_2 > i_3 > i_1 \quad (i_1 = 0)$$

After a long time closing the switch



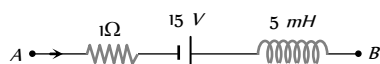
$$R_{eq} = 2R$$

$$R_{eq} = \frac{R}{2}$$

Hence $i_2 > i_3 > i_1$ (3)

44. (c) By using Kirchhoff's voltage law

$$V_A - iR + E - L \frac{di}{dt} = V_B \Rightarrow V_B - V_A = 15 \text{ volt.}$$



45. (d) The rate of increase of current

$$= \frac{di}{dt} = \frac{d}{dt} i_0 (1 - e^{-Rt/L}) = \frac{d}{dt} i_0 - \frac{d}{dt} i_0 e^{-Rt/L}$$

$$= 0 - i_0 e^{-Rt/L} \cdot \frac{d}{dt} \left(-\frac{Rt}{L} \right) = i_0 \frac{R}{L} e^{-Rt/L}$$

$$= \frac{50}{180} \times \frac{180}{5 \times 10^{-3}} \times e^{-(180 \times 0.001)/(5 \times 10^{-3})} = 10^4 \times e^{-36} \text{ A/sec}$$

46. (b) We know that $i = i_0 \left[1 - e^{-\frac{Rt}{L}} \right]$ or $\frac{3}{4} i_0 = i_0 \left[1 - e^{-t/\tau} \right]$

(where $\tau = \frac{L}{R}$ = time constant)

$$\frac{3}{4} = 1 - e^{-t/\tau} \text{ or } e^{-t/\tau} = 1 - \frac{3}{4} = \frac{1}{4}$$

$$e^{t/\tau} = 4 \text{ or } \frac{t}{\tau} = \ln 4$$

$$\Rightarrow \tau = \frac{t}{\ln 4} = \frac{4}{2 \ln 2} \Rightarrow \tau = \frac{2}{\ln 2} \text{ sec.}$$

47. (b) In a constant magnetic field conducting ring oscillates with a frequency of 100 Hz

i.e. $T = \frac{1}{100}$ s, in time $\frac{T}{2}$ flux links with coil changes from

BA to zero. $\Rightarrow$ Induced emf = $\frac{\text{change in flux}}{\text{time}}$

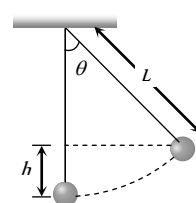
$$= \frac{BA}{T/2} = \frac{2BA}{T} = \frac{2B \times \pi r^2}{T} = \frac{2 \times 0.01 \times \pi \times 1^2}{1/100} = 4\pi V$$

Induced electric field along the circle, using Maxwell equation

$$\oint E \cdot dl = -\frac{d\phi}{dt} = A \frac{dB}{dt} = e$$

$$\Rightarrow E = \frac{1}{2\pi r} \times \left(\pi r^2 \times \frac{dB}{dt} \right) = \frac{e}{2\pi r} = \frac{4\pi}{2\pi r} = 2 \text{ V/m}$$

48. (a)



$$\Rightarrow h = L(1 - \cos \theta) \quad \dots\dots(i)$$

Maximum velocity at equilibrium is given by

$$\therefore v^2 = 2gh = 2g L(1 - \cos \theta) = 2g L \left(2 \sin^2 \frac{\theta}{2} \right)$$

$$\Rightarrow v = 2\sqrt{gL} \sin \frac{\theta}{2}$$

Thus, max. potential difference

$$V_{\max} = BvL = B \times 2\sqrt{gL} \sin \frac{\theta}{2} L = 2BL \sin \frac{\theta}{2} (gL)^{1/2}.$$

Graphical Questions

1. (d) At B , flux is maximum, so from $|e| = \frac{d\phi}{dt}$ at B $|e| = 0$
2. (b) As the magnet moves towards the coil, the magnetic flux increases (nonlinearly). Also there is a change in polarity of induced emf when the magnet passes on to the other side of the coil.
3. (c) Rate of decay of current between $t = 5 \text{ ms}$ to 6 ms

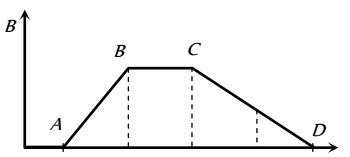
$$= \frac{di}{dt} = -(\text{Slope of the line } BC)$$

$$= -\left(\frac{5}{1 \times 10^{-3}}\right) = -5 \times 10^3 \text{ A/s. Hence induced emf}$$

$$e = -L \frac{di}{dt} = -4.6 \times (-5 \times 10^3) = 23 \times 10^3 \text{ V.}$$
4. (b) $e = -M \frac{di}{dt} = -1.5 \frac{(1-0)}{(T/4)} = -\frac{6}{T}$, $T = \frac{2\pi}{\omega} = \frac{2\pi}{200} = \frac{\pi}{100}$

$$\Rightarrow |e| = \frac{600}{\pi} = 190.9 \text{ V} \approx 191 \text{ V}$$
5. (d) When loop enters in field between the pole pieces, flux linked with the coil first increases (constantly) so a constant emf induces, when coil entered completely within the field, no flux change so $e = 0$.
 When coil exit out, flux linked with the coil decreases, hence again emf induces, but in opposite direction.
6. (a) $|dq| = \frac{d\phi}{R} = i dt = \text{Area under } i-t \text{ graph}$

$$\therefore d\phi = (\text{Area under } i-t \text{ graph}) R$$

$$= \frac{1}{2} \times 4 \times 0.1 \times (10) = 2 \text{ wb.}$$
7. (b) Induced emf $e = A \frac{dB}{dt}$
i.e. $e \propto \frac{dB}{dt}$ (= slope of $B-t$ graph)


In the given graph slope of $AB >$ slope of CD , slope in the 'a' region = slope in the 'c' region = 0, slope in the 'd' region = slope in the 'e' region $\neq 0$. That's why $b > (d = e) > (a = c)$
8. (b) $P = Fv = Bil \times v = B \left(\frac{Bvl}{R} \right) l \times v = \frac{B^2 v^2 l^2}{R} \Rightarrow P \propto v^2$

9. (b) As x increases so $\frac{dB}{dt}$ increases *i.e.* induced emf (e) is negative.
 When loop completely entered in the magnetic field, $\text{emf} = 0$
 When it exit out x increases but $\frac{dB}{dt}$ decreases *i.e.* e is positive.
10. (c) According to $i-t$ graph, in the first half current is increasing uniformly so a constant negative emf induces in the circuit.
 In the second half current is decreasing uniformly so a constant positive emf induces
 Hence graph (c) is correct
11. (b) $i = i_0 \left(1 - e^{-\frac{R}{L}t} \right)$
12. (a) $\frac{di}{dt}$ = slope of $i-t$ graph slope of graph (2) < slope of graph (1) so $\left(\frac{di}{dt} \right)_2 < \left(\frac{di}{dt} \right)_1$
 Also $L \propto \frac{1}{(di/dt)} \Rightarrow L_2 > L_1$
13. (b) $\phi = BA = B \times \pi r^2$

$$\therefore \phi \propto r^2 \Rightarrow \phi = kr^2 \quad (k = \text{constant})$$

$$\therefore e = \frac{d\phi}{dt} = k \cdot 2r \frac{dr}{dt}$$

From 0 - 1, r is constant, $\therefore \frac{dr}{dt} = 0$ hence, $e = 0$

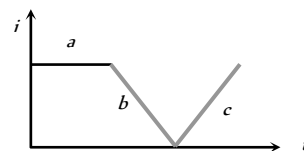
From 1 - 2, $r = \alpha t$, $\therefore \frac{dr}{dt} = \alpha$ hence $e \propto r \Rightarrow e \propto t$

From 2 - 3, again r is constant, $\therefore \frac{dr}{dt} = 0$ hence $e = 0$
14. (c) Emf induces during 'a' = 0
 emf induced during 'b' is constant throughout emf induced during 'c' is constant throughout magnitude of emf induced during 'b' is equal to the magnitude of emf induced during 'c'. But the direction opposite.
15. (a) $U = \frac{1}{2} Li^2$

$$\therefore \text{Rate} = \frac{dU}{dt} = Li \left(\frac{di}{dt} \right)$$

At $t = 0$, $i = 0 \therefore \text{rate} = 0$

At $t = \infty$, $i = i_0$ but $\frac{di}{dt} = 0$, therefore rate = 0
16. (c) At the time $t = 0$, e is max and is equal to E , but current i is zero.



As the time passes, current through the circuit increases but induced emf decreases.

17. (d) If at any instant, current through the circuit is i then applying Kirchhoff's voltage law, $iR + e = E \Rightarrow e = E - iR$. Therefore, graph between e and i will be a straight line having negative slope and having a positive intercept.
18. (c) When loop is entering in the field, magnetic flux (*i.e.* $\times$) linked with the loop increases so induced emf in it $e = Bvl = 0.6 \times 10^{-2} \times 5 \times 10^{-2} = 3 \times 10^{-4} \text{ V}$ (Negative).
- When loop completely entered in the field (after 5 sec) flux linked with the loop remains constant so $e = 0$.
- After 15 sec, loop begins to exit out, linked magnetic flux decreases so induced emf $e = 3 \times 10^{-4} \text{ V}$ (Positive).
19. (a)

Assertion and Reason

1. (b) When a metallic conductor is moved in a magnetic field; magnetic flux is varied. It disturbs the free electrons of the metal and set up an induced emf in it. As there are no free ends of the metal *i.e.* it will be closed in itself so there will be induced current.
 2. (b) The relation of induced emf is $e = \frac{L di}{dt}$ and current i is given by $i = \frac{e}{R} = \frac{1}{R} \cdot \frac{L di}{dt} \Rightarrow \frac{di}{dt} = i \frac{R}{L} = \frac{i}{L/R}$.
In order to decrease the rate of increase of current through solenoid. We have to increase the time constant $\frac{L}{R}$.
 3. (c) According to Faraday's laws, the conversion of mechanical energy into electrical energy. This is in accordance with the law of conservation of energy. It is also clearly known that in pure resistance, the emf is in phase with the current.
 4. (c) Presence of magnetic flux cannot produce current.
 5. (e) E.M.F. induces, when there is change in magnetic flux. Faraday did experiment in which, there is relative motion between the coil and magnet, the flux linked with the coil changes and e.m.f. induces.
 6. (e) Since both the loops are identical (same area and number of turns) and moving with a same speed in same magnetic field. Therefore same emf is induced in both the coils. But the induced current will be more in the copper loop as its resistance will be lesser as compared to that of the aluminium loop.
 7. (a) The inductance coils made of copper will have very small ohmic resistance. Due to change in magnetic flux a large induced current will be produced in such an inductance, which will offer appreciable opposition to the flow of current.
 8. (b) Self-inductance of a coil is its property virtue of which the coil opposes any change in the current flowing through it.
 9. (c) The manner in which the two coils are oriented, determines the coefficient of coupling between them.
- $$M = K^2 \cdot L_1 L_2$$
- When the two coils are wound on each other, the coefficient of coupling is maximum and hence mutual inductance between the coil is maximum.
10. (a) The induced current in the ring opposes the motion of falling magnet. Therefore, the acceleration of the falling magnet will be less than that due to gravity.
 11. (e) As the aircraft flies, magnetic flux changes through its wings due to the vertical component of the earth's magnetic field. Due to this, induced emf is produced across the wings of the aircraft. Therefore, the wings of the aircraft will not be at the same potential.
 12. (b) According to Lenz's law, induced emf are in a direction such as to attempt to maintain the original magnetic flux when a change occurs. When the switch is opened, the sudden drop in the magnetic field in the circuit induces an emf in a direction that attempts to keep the original current flowing. This can cause a spark as the current bridges the air gap between the poles of the switch. (The spark is more likely in circuits with large inductance).
 13. (b) Mutual inductance is the phenomenon according to which an opposing e.m.f. produce flux in a coil as a result of change in current or magnetic flux linked with a neighboring coil. But when two coils are inductively coupled, in addition to induced e.m.f. produced due to mutual induction, also induced e.m.f. is produced in each of the two coils due to self-induction.
 14. (e) Lenz's Law is based on conservation of energy and induced emf always opposes the cause of it *i.e.*, change in magnetic flux.
 15. (a) As the coil rotates, the magnetic flux linked with the coil (being $\vec{B} \cdot \vec{A}$) will change and emf will be induced in the loop.
 16. (a)
 17. (c) When the satellite moves in inclined plane with equatorial plane (including orbit around the poles), the value of magnetic field will change both in magnitude and direction. Due to this, the magnetic flux through the satellite will change and hence induced currents will be produced in the metal of the satellite. But no current will induced if satellite orbits in the equatorial plane because the magnetic flux does not change through the metal of the satellite in this plane.
 18. (b) When the tube is heated its resistance gets increased due to which eddy currents produced in copper tube becomes weak. Hence opposing force also gets reduced and the terminal velocity of magnet gets increased.
 19. (d) When a metal piece falls from a certain height then eddy currents are produced in it due to earth's magnetic field. Eddy current oppose the motion of piece. Hence metal piece falls with a smaller acceleration (as compared to g). But no eddy current are produced in non-metal piece, hence it drops with acceleration due to gravity. Therefore non-metal piece will reach the earth's surface earlier.
 20. (a) Transformer works on ac only, ac changes in magnitude as well as in direction.
 21. (a) Hysteresis loss in the core of transformer directly proportional to the hysteresis loop area of the core material. Since soft iron

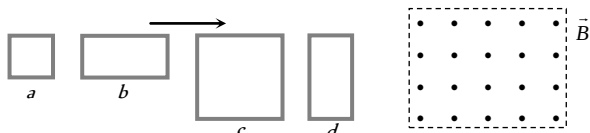
has narrow hysteresis loop area, that is why soft iron core is used in the transformer.

22. (e) *ac* generator is based on the principle of the electromagnetic induction. When a coil is rotated about an axis perpendicular to the direction of uniform magnetic field, an induced *emf* is produced across it.
23. (d) Efficiency of electric motor is maximum when the back *emf* set up in the armature is half the value of the applied battery *emf*.
24. (d) Backs *emf*. $e \propto \omega$. At start $\omega = 0$ so $e = 0$

Electromagnetic Induction

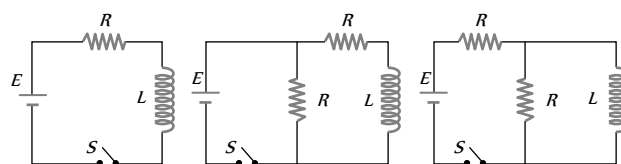
Self Evaluation Test - 23

1. The figure shows four wire loops, with edge lengths of either L or $2L$. All four loops will move through a region of uniform magnetic field $\vec{B}$ (directed out of the page) at the same constant velocity. Rank the four loops according to the maximum magnitude of the e.m.f. induced as they move through the field, greatest first

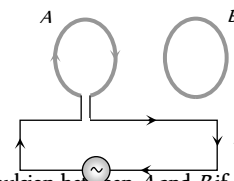


- (a) $(e_c = e_d) < (e_a = e_b)$ (b) $(e_c = e_d) > (e_a = e_b)$
 (c) $e_c > e_d > e_b > e_a$ (d) $e_c < e_d < e_b < e_a$
2. A circular coil and a bar magnet placed near by are made to move in the same direction. The coil covers a distance of 1 m in 0.5 sec and the magnet a distance of 2 m in 1 sec . The induced emf produced in the coil
- (a) Zero
 (b) 1 V
 (c) 0.5 V
 (d) Cannot be determined from the given information
3. A square coil $ABCD$ lying in x - y plane with its centre at origin. A long straight wire passing through origin carries a current $i = 2t$ in negative z -direction. The induced current in the coil is
- (a) Clockwise
 (b) Anticlockwise
 (c) Alternating
 (d) Zero
4. A short magnet is allowed to fall along the axis of a horizontal metallic ring. Starting from rest, the distance fallen by the magnet in one second may be
- (a) 4 m (b) 5 m
 (c) 6 m (d) 7 m
5. The horizontal component of the earth's magnetic field at a place is $3 \times 10^{-4}\text{ T}$ and the dip is $\tan^{-1}\left(\frac{4}{3}\right)$. A metal rod of length 0.25 m placed in the north-south position and is moved at a constant speed of 10 cm/s towards the east. The emf induced in the rod will be
- (a) Zero (b) $1\text{ }\mu\text{V}$
 (c) $5\text{ }\mu\text{V}$ (d) $10\text{ }\mu\text{V}$
6. A copper disc of radius 0.1 m rotates about its centre with 10 revolutions per second in a uniform magnetic field of 0.1 Tesla . The emf induced across the radius of the disc is
- (a) $\frac{\pi}{10}\text{ V}$ (b) $\frac{2\pi}{10}\text{ V}$
 (c) $10\pi\text{ mV}$ (d) $20\pi\text{ mV}$

7. A coil of Cu wire (radius- r , self inductance- L) is bent in two concentric turns each having radius $\frac{r}{2}$. The self inductance now
- (a) $2L$ (b) L
 (c) $4L$ (d) $L/2$
8. In which of the following circuit is the current maximum just after the switch S is closed



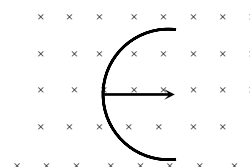
- (a) (i) (ii) (iii)
 (c) (iii) (d) Both (ii) and (iii)
9. A small coil is introduced between the poles of an electromagnet so that its axis coincides with the magnetic field direction. The number of turns is n and the cross sectional area of the coil is A . When the coil turns through 180° about its diameter, the charge flowing through the coil is Q . The total resistance of the circuit is R . What is the magnitude of the magnetic induction
- (a) $\frac{QR}{nA}$ (b) $\frac{2QR}{nA}$
 (c) $\frac{Qn}{2RA}$ (d) $\frac{QR}{2nA}$
10. Two circular coils A and B are facing each other as shown in figure. The current i through A can be altered



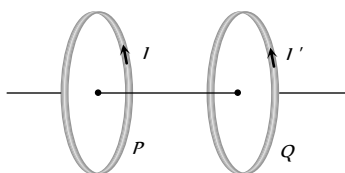
- (a) There will be repulsion between A and B if i is increased
 (b) There will be attraction between A and B if i is increased
 (c) There will be neither attraction nor repulsion when i is changed
 (d) Attraction or repulsion between A and B depends on the direction of current. It does not depend whether the current is increased or decreased
11. A conducting loop having a capacitor is moving outward from the magnetic field then which plate of the capacitor will be positive
- (a) Plate - A
 (b) Plate - B
 (c) Plate - A and Plate - B both
 (d) None



12. A straight wire of length L is bent into a semicircle. It is moved in a uniform magnetic field with speed v with diameter perpendicular to the field. The induced emf between the ends of the wire is
- (a) BLv
 (b) $2BLv$

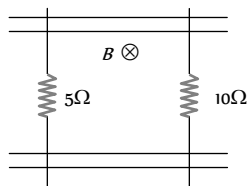


- (c) $2\pi BLv$
- (d) $\frac{2BvL}{\pi}$
13. If in a coil rate of change of area is $\frac{5 \text{ metre}^2}{\text{millisecond}}$ and current become 1 amp from 2 amp in $2 \times 10^{-3} \text{ sec}$. If magnetic field is 1 Tesla then self inductance of the coil is
- (a) 2 H (b) 5 H
- (c) 20 H (d) 10 H
14. In series with 20 ohm resistor a 5 henry inductor is placed. To the combination an e.m.f. of 5 volt is applied. What will be the rate of increase of current at $t = 0.25 \text{ sec}$
- (a) e (b) e'
- (c) e (d) None of these
15. Two coils P and Q are placed co-axially and carry current I and I' respectively



- (a) If $I = 0$ and P moves towards Q, a current in the same direction as I is induced in Q
- (b) If $I = 0$ and Q moves towards P, a current opposite in direction to that of I' is induced in P
- (c) When $I \neq 0$ and $I' \neq 0$ are in the same direction, then two coil tend to move apart
- (d) None of these
16. The phase difference between the flux linkage and the induced e.m.f. in a rotating coil in a uniform magnetic field
- (a) π (b) $\pi/2$
- (c) $\pi/4$ (d) $\pi/6$
17. A pair of parallel conducting rails lie at right angle to a uniform magnetic field of 2.0 T as shown in the fig. Two resistors 10 Ω and 5 Ω are to slide without friction along the rail. The distance between the conducting rails is 0.1 m. Then

- (a) Induced current = $\frac{1}{150} \text{ A}$
directed clockwise if 10 Ω resistor is pulled to the right

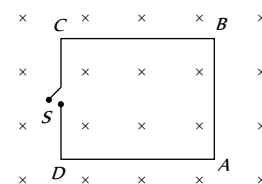


with speed 0.5 ms and 5 Ω resistor is held fixed

- (b) Induced current = $\frac{1}{300} \text{ A}$ directed anti-clockwise if 10 Ω resistor is pulled to the right with speed 0.5 ms and 5 Ω resistor is held fixed
- (c) Induced current = $\frac{1}{300} \text{ A}$ directed clockwise if 5 Ω resistor is pulled to the left at 0.5 ms and 10 Ω resistor is held at rest
- (d) Induced current = $\frac{1}{150} \text{ A}$ directed anti-clockwise if 5 Ω resistor is pulled to the left at 0.5 ms and 10 Ω resistor is held at rest

18. The magnetic field in the cylindrical region shown in figure increases at a constant rate of 20 mT/sec. Each side of the square loop ABCD has a length of 1 cm and resistance of 4 Ω . Find the current in the wire AB if the switch S is closed

- (a) $1.25 \times 10^{-7} \text{ A}$, (anti-clockwise)
- (b) $1.25 \times 10^{-7} \text{ A}$ (clockwise)
- (c) $2.5 \times 10^{-7} \text{ A}$ (anti clockwise)
- (d) $2.5 \times 10^{-7} \text{ A}$ (clockwise)



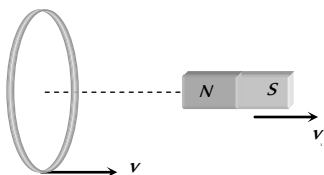
19. An aircraft with a wing-span of 40 m flies with a speed of 1080 km/h in the eastward direction at a constant altitude in the northern hemisphere, where the vertical component of earth's magnetic field is $1.75 \times 10^{-5} \text{ T}$. Then the emf that develops between the tips of the wings is
- (a) 0.5 V (b) 0.35 V
- (c) 0.21 V (d) 2.1 V
20. A hundred turns of insulated copper wire are wrapped around an iron cylinder of area $1 \times 10^{-2} \text{ m}^2$ and are connected to a resistor. The total resistance in the circuit is 10 ohms. If the longitudinal magnetic induction in the iron changes from 1 weber/m, in one direction to 1 Weber/m in the opposite direction, how much charge flows through the circuit
- (a) $2 \times 10^{-2} \text{ C}$ (b) $2 \times 10^{-3} \text{ C}$
- (c) $2 \times 10^{-4} \text{ C}$ (d) $2 \times 10^{-5} \text{ C}$

AS Answers and Solutions

(SET -23)

1. (b) Emf induces across the length of the wire which cuts the magnetic field. (Length of c = Length of d) > (Length of a = b). So $(e_c = e_d) > (e_a = e_b)$

2. (a) Speed of the magnet



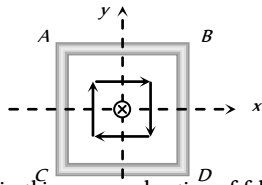
$$v_1 = \frac{2}{1} = 2 \text{ m/s}$$

Speed of the coil

$$v_2 = \frac{1}{0.5} = 2 \text{ m/s}$$

Relative speed between coil and magnet is zero, so there is no induced emf in the coil.

3. (d) Magnetic lines are tangential to the coil as shown in figure. Thus net magnetic flux passing through the coil is always zero or the induced current will be zero.



4. (a) We know that in this case acceleration of falling magnet will be lesser than g . If ' g ' would have been acceleration, then distance covered $= \frac{1}{2}gt^2 = 5\text{ m}$.

Now the distance covered will be less than 5 m hence only option (a) is correct.

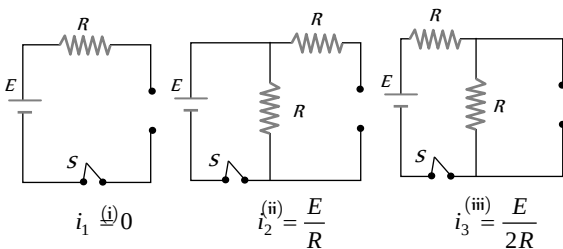
5. (d) Rod is moving towards east, so induced emf across its end will be $e = Bv\ell = (B_H \tan \phi)v\ell$

$$\therefore e = 3 \times 10^{-4} \times \frac{4}{3} \times (10 \times 10^{-2}) \times 0.25 = 10^{-5} \text{ V} = 10 \mu\text{V}$$

6. (c) The induced emf between centre and rim of the rotating disc is $E = \frac{1}{2}B\omega R^2 = \frac{1}{2} \times 0.1 \times 2\pi \times 10 \times (0.1)^2 = 10\pi \times 10^{-3} \text{ volt}$

7. (a) $\because L \propto N^2 r; \quad \frac{L_1}{L_2} = \left(\frac{N_1}{N_2}\right)^2 \times \frac{r_1}{r_2}$
 $\Rightarrow \frac{L}{L_2} = \left(\frac{1}{2}\right)^2 \times \left(\frac{r}{r/2}\right) = \frac{1}{2}; \quad L_2 = 2L$

8. (b) At $t = 0$ current through L is zero so it acts as open circuit. The given figures can be redrawn as follow.



Hence $i_1 > i_2 > i_3$.

9. (d) Induced charge $Q = -\frac{NBA}{R}(\cos \theta_2 - \cos \theta_1)$ ***

$$= -\frac{NBA}{R}(\cos 180^\circ - \cos 0^\circ) \Rightarrow B = \frac{QR}{2NA}$$

10. (a) With rise in current in coil A flux through B increases. According to Lenz's law repulsion occurs between A and B.

11. (a) Crosses ($\times$) linked with the loop are decreasing, so induced current in it is clockwise i.e. from $B \rightarrow A$. Hence electrons flow from plate A to B so plate A becomes positively charged.

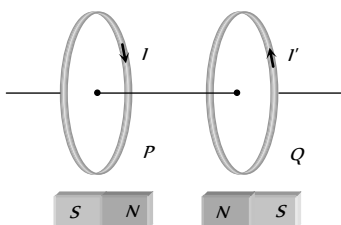
12. (d) Induced emf $e = Bv\ell \Rightarrow e = Bv(2R) = \frac{2BvL}{\pi}$

13. (d) $e = B \cdot \frac{dA}{dt} = L \frac{di}{dt} \Rightarrow 1 \times \frac{5}{10^{-3}} = L \times \frac{(2-1)}{2 \times 10^{-3}} \Rightarrow L = 10\text{ H}$

14. (c) $i = i_0(1 - e^{-Rt/L}) \Rightarrow \frac{di}{dt} = i_0 \left(\frac{R}{L}\right) e^{-Rt/L} = \frac{E}{L} e^{-Rt/L}$

On putting values $\frac{di}{dt} = \frac{1}{e} = e^{-1}$.

15. (b)



16. (b) $\phi = BA \cos \theta$ where θ is the angle between area and field

$$e = \frac{-d\phi}{dt} = -BA \sin \theta \cdot \frac{d\theta}{dt} = BA \left(\frac{d\theta}{dt}\right) \cos(90^\circ + \theta)$$

Here phase difference $= 90^\circ$.

17. (d) When 5Ω resistor is pulled left at 0.5 m/sec induced emf, in the said resistor $= e = vBl = 0.5 \times 2 \times 0.1 = 0.1\text{ V}$

Resistor 10Ω is at rest so induced emf in it ($e = vBl$) be zero.

Now net emf, in the circuit $= 0.1\text{ V}$

and equivalent resistance of the circuit

$$R = 15\Omega$$

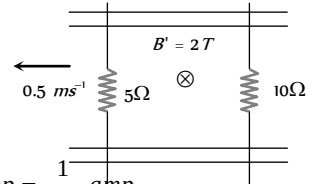
Hence current $i = \frac{0.1}{15} \text{ amp} = \frac{1}{150} \text{ amp}$

And its direction will be anti-clockwise (according to Lenz's law)

18. (a) $i = \frac{e}{R} = \frac{A}{R} \cdot \frac{dB}{dt} = \frac{(1 \times 10^{-2})^2}{16} \times 20 \times 10^{-3} = 1.25 \times 10^{-7} \text{ A}$
 (Anti-clockwise).

19. (c) $L = 40\text{ m}, v = 1080\text{ km/h} = 300\text{ m/sec}$ and $B = 1.75 \times 10^{-2} \text{ T} \Rightarrow$
 $e = Blv = 1.75 \times 10^{-5} \times 40 \times 300 = 0.21\text{ V}$

20. (a) $dQ = \frac{d\phi}{R} = \frac{nAdB}{R} = \frac{100 \times 1 \times 10^{-3} \times 2}{10} = 2 \times 10^{-2} \text{ C}$



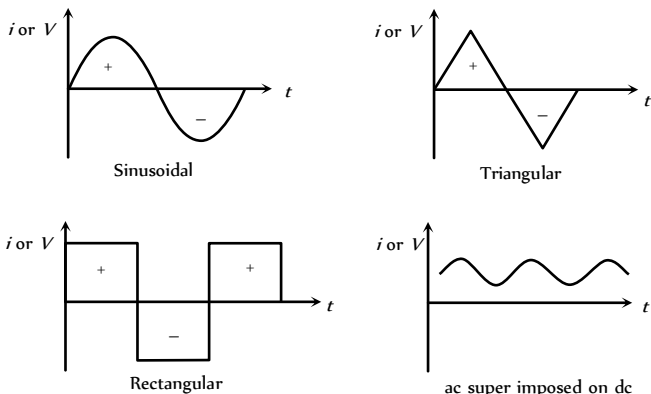


Chapter 24 Alternating Current

Alternating Quantities (i or V)

(1) An alternating quantity (current i or voltage V) is one whose magnitude changes continuously with time between zero and a maximum value and whose direction reverses periodically.

(2) Some graphical representation for alternating quantities



(3) **Equation for i and V :** Alternating current or voltage varying as sine function can be written as

$$i = i_0 \sin \omega t = i_0 \sin 2\pi \nu t = i_0 \sin \frac{2\pi}{T} t$$

$$\text{and } V = V_0 \sin \omega t = V_0 \sin 2\pi \nu t = V_0 \sin \frac{2\pi}{T} t$$

where i and V are instantaneous values of current and voltage,

i_0 and V_0 are peak values of current and voltage

ω = Angular frequency in rad/sec , ν = Frequency in Hz and T = time period

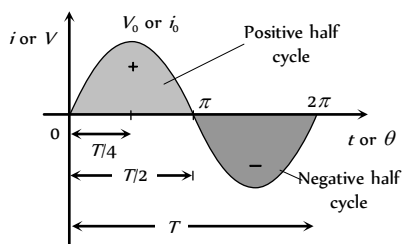


Fig. 24.2

(i) The time taken to complete one cycle of variations is called the periodic time or time period.

(ii) Alternating quantity is positive for half the cycle and negative for the rest half. Hence average value of alternating quantity (i or V) over a complete cycle is zero.

(iii) The value of alternating quantity is zero or maximum $2V$ times every second. The direction also changes $2V$ times every second.

(iv) Generally sinusoidal waveform is used as alternating current/voltage.

(v) At $t = \frac{T}{4}$ from the beginning, i or V reaches to their maximum value.

Important Values of Alternating Quantities

(1) **Peak value (i_0 or V_0)** : The maximum value of alternating quantity (i or V) is defined as peak value or amplitude.

(2) **Mean square value ($\overline{V^2}$ or $\overline{i^2}$)** : The average of square of instantaneous values in one cycle is called mean square value. It is always positive for one complete cycle. e.g. $\overline{V^2} = \frac{1}{T} \int_0^T V^2 dt = \frac{V_0^2}{2}$ or $\overline{i^2} = \frac{i_0^2}{2}$

(3) **Root mean square (r.m.s.) value** : Root of mean of square of voltage or current in an ac circuit for one complete cycle is called $r.m.s.$ value. It is denoted by V_r or i_r

$$i_{rms} = \sqrt{\frac{i_1^2 + i_2^2 + \dots}{n}} = \sqrt{\overline{i^2}} = \sqrt{\frac{\int_0^T i^2 dt}{\int_0^T dt}} = \frac{i_0}{\sqrt{2}} = 0.707 i_0 = 70.7\% i_0$$

$$\text{Similarly } V_{rms} = \frac{V_0}{\sqrt{2}} = 0.707 V_0 = 70.7\% \text{ of } V_0$$

$$\left[\langle \sin^2(\omega t) \rangle = \langle \cos^2(\omega t) \rangle = \frac{1}{2} \right]$$

(i) The *r.m.s.* value of alternating current is also called virtual value or effective value.

(ii) In general when values of voltage or current for alternating circuits are given, these are *r.m.s.* value.

(iii) ac ammeter and voltmeter are always measure *r.m.s.* value. Values printed on ac circuits are *r.m.s.* values.

(iv) In our houses ac is supplied at 220 V, which is the *r.m.s.* value of voltage. Its peak value is $\sqrt{2} \times 200 = 311\text{V}$.

(v) *r.m.s.* value of ac is equal to that value of dc, which when passed through a resistance for a given time will produce the same amount of heat as produced by the alternating current when passed through the same resistance for same time.

(4) **Mean or Average value (i_a or V_a)** : The average value of alternating quantity for one complete cycle is zero.

The average value of ac over half cycle ($t = 0$ to $T/2$)

$$i_{av} = \frac{\int_0^{T/2} i dt}{\int_0^{T/2} dt} = \frac{2i_0}{\pi} = 0.637i_0 = 63.7\% \text{ of } i,$$

$$\text{Similarly } V_{av} = \frac{2V_0}{\pi} = 0.637V_0 = 63.7\% \text{ of } V.$$

(5) **Peak to peak value** : It is equal to the sum of the magnitudes of positive and negative peak values

$$\therefore \text{Peak to peak value} = V_+ + V_- = 2V_+$$

$$= 2\sqrt{2} V_{rms} = 2.828 V_{rms}$$

(6) **Form factor and peak factor** : The ratio of *r.m.s.* value of ac to its average during half cycle is defined as form factor. The ratio of peak value and *r.m.s.* value is called peak factor

Phase

Physical quantity which represents both the instantaneous value and direction of alternating quantity at any instant is called its phase. It's a dimensionless quantity and its unit is radian.

If an alternating quantity is expressed as $X = X_0 \sin(\omega t \pm \phi_0)$ then the argument of $\sin(\omega t + \phi)$ is called its phase. Where ωt = instantaneous phase (changes with time) and ϕ_0 = initial phase (constant w.r.t. time)

Table 24.1 : Some important values

Nature of wave form	Wave form	r.m.s. value	average value	Form factor $R_f = \frac{\text{r.m.s. value}}{\text{Average value}}$	Peak factor $R_p = \frac{\text{Peak value}}{\text{r.m.s. value}}$
Sinusoidal		$\frac{i_0}{\sqrt{2}}$	$\frac{2}{\pi} i_0$	$\frac{\pi}{2\sqrt{2}} = 1.11$	$\sqrt{2} = 1.41$
Half wave rectified		$\frac{i_0}{2}$	$\frac{i_0}{\pi}$	$\frac{\pi}{2} = 1.57$	2
Full wave rectified		$\frac{i_0}{\sqrt{2}}$	$\frac{2i_0}{\pi}$	$\frac{\pi}{2\sqrt{2}}$	$\sqrt{2}$
Square or Rectangular		i_0	i_0	1	1

(1) **Phase difference** (Phase constant) : The difference between the phases of currents and voltage is called phase difference. If alternating voltage and current are given by $V = V_0 \sin(\omega t + \phi_1)$ and $i = i_0 \sin(\omega t + \phi_2)$ then phase difference $\phi = \phi_2 - \phi_1$ (relative to current) or $\phi = \phi_2 - \phi_1$ (relative to voltage)

(2) **Time difference** : If phase difference between alternating current and voltage is ϕ then time difference between them is given as

$$\text{T.D.} = \frac{T}{2\pi} \times \phi$$

(3) **Phasor diagram** : A diagram representing alternating current and alternating voltage (of same frequency) as vectors (phasors) with the phase angle between them is called a phasor diagram.

While drawing phasor diagram for a pure element (e.g. R , L or C) either of the current or voltage can be plotted along X -axis.

But when phasor diagram for a combination of elements is drawn then quantity which remains constant for the combination must be plotted along X -axis so we observe that

- (i) In series circuits current has to be plotted along X -axis.
- (ii) In parallel circuits voltage has to be plotted along X -axis.

Measurement of Alternating Quantities

Alternating current shows heating effect only, hence meters used for measuring ac are based on heating effect and are called hot wire meters (Hot wire ammeter and hot wire voltmeter)

Table 24.2 : Measurement of ac and dc

ac measurement	dc measurement
(1) All ac meters read <i>r.m.s.</i> value.	(1) All dc meters read average value
(2) All ac meters are based on heating effect of current.	(2) All dc meters are based on magnetic effect of current
(3) Deflection in hot wire meters $\theta \propto i_{rms}^2$	(3) Deflection in dc meters $\theta \propto i$
	
(non-linear scale)	(Linear scale)

Impedance, Reactance, Admittance and Susceptance

(1) **Impedance (Z)** : The opposition offered by ac circuits to the flow of ac through it is defined as its impedance. Its unit is $\text{ohm}(\Omega)$.

(2) **Reactance (X)** : The opposition offered by inductor or capacitor or both to the flow of ac through it is defined as reactance. It is of following two type

(i) **Inductive reactance (X_L)** : Offered by inductive circuit
 $X_L = \omega L = 2\pi\nu L$ $\nu_{dc} = 0$ so for dc, $X_L = 0$.

Capacitive reactance (X_C) : Offered by capacitive circuit
 $X_C = \frac{1}{\omega C} = \frac{1}{2\pi\nu C}$ for dc $X_C = \infty$.

(3) **Admittance (Y)** : $Z = \frac{V_0}{i_0} = \frac{V_{rms}}{i_{rms}}$ Reciprocal of impedance is known as admittance $\left(Y = \frac{1}{Z}\right)$. Its unit is mho

(4) **Susceptance (S)** : the reciprocal of reactance is defined as susceptance $\left(S = \frac{1}{X}\right)$. It is of two type

(i) inductive susceptance $S_L = \frac{1}{X_L} = \frac{1}{2\pi\nu L}$ and

(ii) Capacitive susceptance, $S_C = \frac{1}{X_C} = \omega C = 2\pi\nu C$.

Power in ac Circuits

In dc circuits power is given by $P = Vi$. But in ac circuits, since there is some phase angle between voltage and current, therefore power is defined as the product of voltage and that component of the current which is in phase with the voltage.

Thus $P = Vi \cos \phi$; where V and i are *r.m.s.* value of voltage and current.

(1) **Instantaneous power** : Suppose in a circuit $V = V_0 \sin \omega t$ and $i = i_0 \sin(\omega t + \phi)$ then $P_{\text{instantaneous}} = Vi = V_0 i_0 \sin \omega t \sin(\omega t + \phi)$

(2) **Average power (True power)** : The average of instantaneous power in an ac circuit over a full cycle is called average power. Its unit is *watt* i.e.

$$P_{av} = V_{rms} i_{rms} \cos \phi = \frac{V_0}{\sqrt{2}} \cdot \frac{i_0}{\sqrt{2}} \cos \phi = \frac{1}{2} V_0 i_0 \cos \phi = i_{rms}^2 R = \frac{V_{rms}^2 R}{Z^2}$$

(3) **Apparent or virtual power** : The product of apparent voltage and apparent current in an electric circuit is called apparent power. This is always positive $P_{app} = V_{rms} i_{rms} = \frac{V_0 i_0}{2}$

Power Factor

(1) It may be defined as cosine of the angle of lag or lead (i.e. $\cos \phi$)

(2) It is also defined as the ratio of resistance and impedance (i.e. $\frac{R}{Z}$)

$$(3) \text{ The ratio } \frac{\text{True power}}{\text{Apparent power}} = \frac{W}{VA} = \frac{kW}{kVA} = \cos \phi$$

Resistive Circuit (R-Circuit)

(1) Current : $i = i_0 \sin \omega t$

(2) Peak current : $i_0 = \frac{V_0}{R}$

(3) Phase difference between voltage and current : $\phi = 0$

(4) Power factor : $\cos \phi = 1$

(5) Power : $P = V_{rms} i_{rms} = \frac{V_0 i_0}{2}$

(6) Time difference : T.D. = 0

(7) Phasor diagram : Both are in same phase

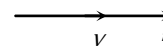
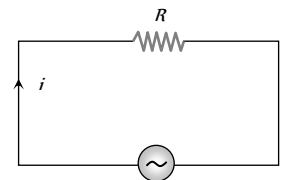


Fig. 24.4



$$V = V_0 \sin \omega t$$

Fig. 24.3

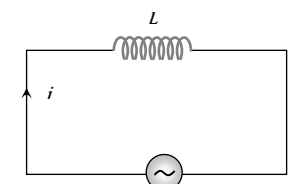
Inductive Circuit (L-Circuit)

(1) Current : $i = i_0 \sin\left(\omega t - \frac{\pi}{2}\right)$

(2) Peak current :

$$i_0 = \frac{V_0}{X_L} = \frac{V_0}{\omega L} = \frac{V_0}{2\pi\nu L}$$

(3) Phase difference between



$$V = V_0 \sin \omega t$$

Fig. 24.5

voltage and current $\phi = 90^\circ$ (or $+\frac{\pi}{2}$)

(4) Power factor : $\cos \phi = 0$

(5) Power : $P = 0$

(6) Time difference : T.D. = $\frac{T}{4}$

(7) Phasor diagram : Voltage leads the current by $\frac{\pi}{2}$

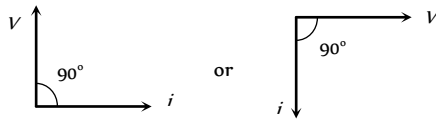


Fig. 24.6

Capacitive Circuit (C-Circuit)

(1) Current : $i = i_0 \sin\left(\omega t + \frac{\pi}{2}\right)$

(2) Peak current :

$$i_0 = \frac{V_0}{X_C} = V_0 \omega C = V_0 (2\pi \nu C)$$

(3) Phase difference between

voltage and current : $\phi = 90^\circ$ (or $-\frac{\pi}{2}$)

(4) Power factor : $\cos \phi = 0$

(5) Power : $P = 0$

(6) Time difference : TD = $\frac{T}{4}$

(7) Phasor diagram : Current leads the voltage by $\pi/2$

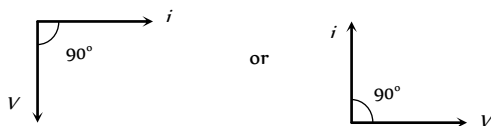
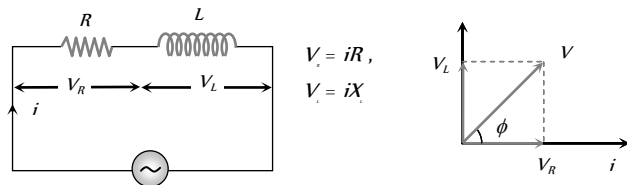


Fig. 24.8

Resistive, Inductive Circuit (RL-Circuit)



$$V = V_0 \sin \omega t$$

Fig. 24.9

(1) Applied voltage : $V = \sqrt{V_R^2 + V_L^2}$

(2) Impedance : $Z = \sqrt{R^2 + X_L^2} = \sqrt{R^2 + \omega^2 L^2} = \sqrt{R^2 + 4\pi^2 \nu^2 L^2}$

(3) Current : $i = i_0 \sin(\omega t - \phi)$

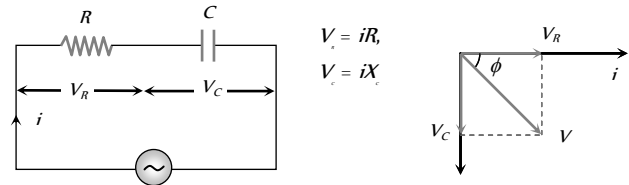
(4) Peak current $i_0 = \frac{V_0}{Z} = \frac{V_0}{\sqrt{R^2 + X_L^2}} = \frac{V_0}{\sqrt{R^2 + 4\pi^2 \nu^2 L^2}}$

(5) Phase difference : $\phi = \tan^{-1} \frac{X_L}{R} = \tan^{-1} \frac{\omega L}{R}$

(6) Power factor : $\cos \phi = \frac{R}{\sqrt{R^2 + X_L^2}}$

(7) Leading quantity : Voltage

Resistive, Capacitive Circuit (RC-Circuit)



$$V = V_0 \sin \omega t$$

Fig. 24.10

(1) Applied voltage : $V = \sqrt{V_R^2 + V_C^2}$

(2) Impedance : $Z = \sqrt{R^2 + X_C^2} = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$

(3) Current : $i = i_0 \sin(\omega t + \phi)$

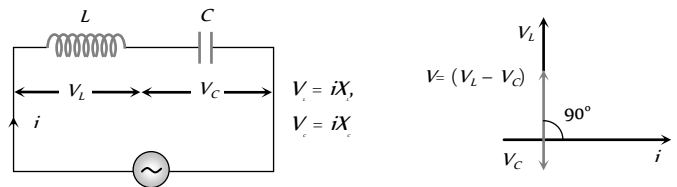
(4) Peak current : $i_0 = \frac{V_0}{Z} = \frac{V_0}{\sqrt{R^2 + X_C^2}} = \frac{V_0}{\sqrt{R^2 + \frac{1}{4\pi^2 \nu^2 C^2}}}$

(5) Phase difference : $\phi = \tan^{-1} \frac{X_C}{R} = \tan^{-1} \frac{1}{\omega CR}$

(6) Power factor : $\cos \phi = \frac{R}{\sqrt{R^2 + X_C^2}}$

(7) Leading quantity : Current

Inductive, Capacitive Circuit (LC-Circuit)



$$V = V_0 \sin \omega t$$

Fig. 24.11

(1) Applied voltage : $V = V_L - V_C$

(2) Impedance : $Z = X_L - X_C = X$

(3) Current : $i = i_0 \sin\left(\omega t \pm \frac{\pi}{2}\right)$

(4) Peak current : $i_0 = \frac{V_0}{Z} = \frac{V_0}{X_L - X_C} = \frac{V_0}{\omega L - \frac{1}{\omega C}}$

(5) Phase difference : $\phi = 90^\circ$

(6) Power factor : $\cos \phi = 0$

(7) Leading quantity : Either voltage or current

Series RLC-Circuit

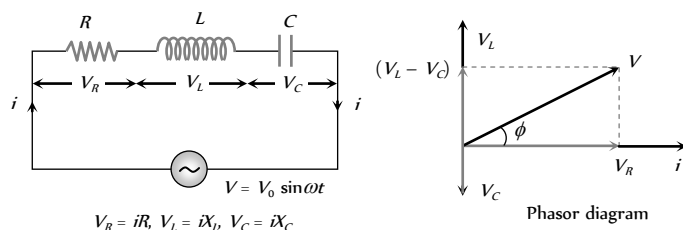


Fig. 24.12

(1) **Equation of current** : $i = i_0 \sin(\omega t \pm \phi)$; where $i_0 = \frac{V_0}{Z}$

(2) **Equation of voltage** : From phasor diagram

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

(3) **Impedance of the circuit** :

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

(4) **Phase difference** : From phasor diagram

$$\tan \phi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R} = \frac{\omega L - \frac{1}{\omega C}}{R} = \frac{2\pi \nu L - \frac{1}{2\pi \nu C}}{R}$$

(5) **If net reactance is inductive** : Circuit behaves as LR circuit

(6) **If net reactance is capacitive** : Circuit behave as CR circuit

(7) **If net reactance is zero** : Means $X = X_L - X_C = 0$

$\Rightarrow X_L = X_C$. This is the condition of resonance

(8) **At resonance** (series resonant circuit)

(i) $X_L = X_C \Rightarrow Z = R$ i.e. circuit behaves as resistive circuit

(ii) $V_L = V_C \Rightarrow V = V_L$ i.e. whole applied voltage appeared across the resistance

(iii) Phase difference : $\phi = 0 \Rightarrow \text{p.f.} = \cos \phi = 1$

(iv) Power consumption $P = V_L i_L = \frac{1}{2} V_0 i_0$

(v) Current in the circuit is maximum and it is $i_0 = \frac{V_0}{R}$

(vi) These circuit are used for voltage amplification and as selector circuits in wireless telegraphy.

(9) **Resonant frequency** (Natural frequency)

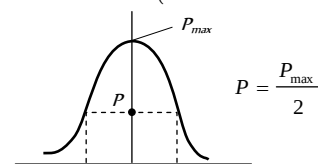
$$\text{At resonance } X_L = X_C \Rightarrow \omega_0 L = \frac{1}{\omega_0 C} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/sec}$$

$$\Rightarrow \nu_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz (or cps)}$$

(Resonant frequency doesn't depend upon the resistance of the circuit)

(10) **Half power frequencies and band width** : The frequencies at which the power in the circuit is half of the maximum power (The power at resonance), are called half power frequencies.

(i) The current in the circuit at half power frequencies (HPF) is $\frac{1}{\sqrt{2}}$ or 0.707 or 70.7% of maximum current (current at resonance).



(ii) There are two half power frequencies Fig. 24.13

(a) $\omega_1 \rightarrow$ called lower half power frequency. At this frequency the circuit is capacitive.

(b) $\omega_2 \rightarrow$ called upper half power frequency. It is greater than ω_0 . At this frequency the circuit is inductive.

(iii) Band width ($\Delta\omega$) : The difference of half power frequencies ω_1 and ω_2 is called band width ($\Delta\omega$) and $\Delta\omega = \omega_2 - \omega_1$. For series resonant circuit it can be proved $\Delta\omega = \left(\frac{R}{L}\right)$

(ii) **Quality factor (Q-factor) of series resonant circuit**

(i) The characteristic of a series resonant circuit is determined by the quality factor (Q - factor) of the circuit.

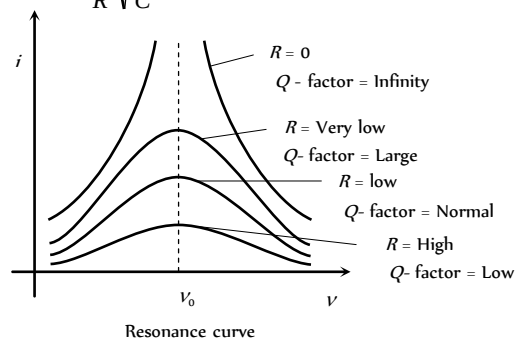
(ii) It defines sharpness of $i - \nu$ curve at resonance when Q - factor is large, the sharpness of resonance curve is more and vice-versa.

(iii) Q - factor also defined as follows

$$Q\text{-factor} = 2\pi \times \frac{\text{Max. energy stored}}{\text{Energy dissipation}} = \frac{2\pi}{T} \times \frac{\text{Max. energy stored}}{\text{Mean power dissipated}} = \frac{\text{Resonant frequency}}{\text{Band width}} = \frac{\omega_0}{\Delta\omega}$$

$$(iv) Q\text{-factor} = \frac{V_L}{V_R} \text{ or } \frac{V_C}{V_R} = \frac{\omega_0 L}{R} \text{ or } \frac{1}{\omega_0 C R}$$

$$\Rightarrow Q\text{-factor} = \frac{1}{R} \sqrt{\frac{L}{C}}$$



Parallel RLC Circuits Fig. 24.14

$$i_R = \frac{V_0}{R} = V_0 G$$

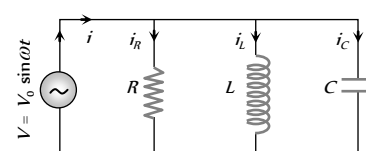


Fig. 24.15

$$i_L = \frac{V_0}{X_L} = V_0 S_L$$

$$i_C = \frac{V_0}{X_C} = V_0 S_C$$

(1) **Current and phase difference** : From phasor diagram current

$$i = \sqrt{i_R^2 + (i_C - i_L)^2} \quad \text{and phase difference}$$

$$\phi = \tan^{-1} \frac{(i_C - i_L)}{i_R} = \tan^{-1} \frac{(S_C - S_L)}{G}$$

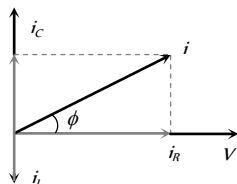


Fig. 24.16

(2) **Admittance (Y) of the circuit** : From equation of current

$$\frac{V_0}{Z} = \sqrt{\left(\frac{V_0}{R}\right)^2 + \left(\frac{V_0}{X_L} - \frac{V_0}{X_C}\right)^2}$$

$$\Rightarrow \frac{1}{Z} = Y = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_L} - \frac{1}{X_C}\right)^2} = \sqrt{G^2 + (S_L - S_C)^2}$$

(3) **Resonance** : At resonance (i) $i_C = i_L \Rightarrow i_{\min} = i_R$

$$(ii) \frac{V}{X_C} = \frac{V}{X_L} \Rightarrow S_C = S_L \Rightarrow \Sigma S = 0$$

$$(iii) Z_{\max} = \frac{V}{i_R} = R$$

$$(iv) \phi = 0 \Rightarrow \text{p.f.} = \cos \phi = 1 = \text{maximum}$$

$$(v) \text{Resonant frequency} \Rightarrow \nu = \frac{1}{2\pi\sqrt{LC}}$$

(4) **Parallel LC circuits** : If inductor has resistance (R) and it is connected in parallel with capacitor as shown

(i) **At resonance**

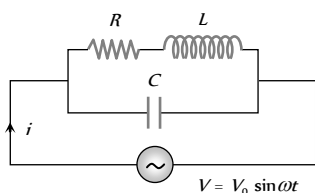


Fig. 24.17

$$(a) Z_{\max} = \frac{1}{Y_{\min}} = \frac{L}{CR}$$

$$(b) \text{Current through the circuit is minimum and } i_{\min} = \frac{V_0 CR}{L}$$

$$(c) S_L = S_C \Rightarrow \frac{1}{X_L} = \frac{1}{X_C} \Rightarrow X = \infty$$

$$(d) \text{Resonant frequency } \omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} \text{ rad/sec}$$

$$\nu_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} \text{ Hz (Condition for parallel resonance is } R < \sqrt{\frac{L}{C}})$$

$$(e) \text{Quality factor of the circuit} = \frac{1}{CR} \cdot \frac{1}{\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}}$$

resonance the quality factor of the circuit is equivalent to the current amplification of the circuit.

(ii) **If inductance has no resistance** : If $R = 0$ then circuit becomes parallel LC circuit as shown

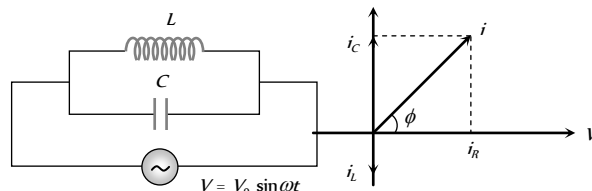


Fig. 24.18

$$\text{Condition of resonance : } i_C = i_L \Rightarrow \frac{V}{X_C} = \frac{V}{X_L}$$

$$\Rightarrow X_C = X_L. \text{ At resonance current } i \text{ in the circuit is zero and}$$

$$\text{impedance is infinite. Resonant frequency : } \nu_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$

Wattless Current

In an ac circuit $R = 0 \Rightarrow \cos \phi = 0$ so $P = 0$ i.e. in resistance less circuit the power consumed is zero. Such a circuit is called the wattless circuit and the current flowing is called the wattless current.

or

The component of current which does not contribute to the average power dissipation is called wattless current

(i) The average of wattless component over one cycle is zero

(ii) Amplitude of wattless current = $i \sin \phi$

$$\text{and r.m.s. value of wattless current} = i_{\text{rms}} \sin \phi = \frac{i_0}{\sqrt{2}} \sin \phi.$$

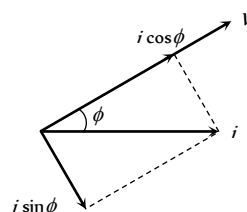


Fig. 24.19

It is quadrature (90°) with voltage.

Choke Coil

Choke coil (or ballast) is a device having high inductance and negligible resistance. It is used to control current in ac circuits and is used in fluorescent tubes. The power loss in a circuit containing choke coil is least.

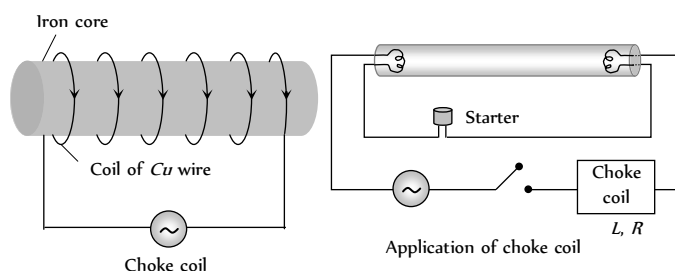


Fig. 24.20

- (1) It consist of a Cu coil wound over a soft iron laminated core.
- (2) Thick Cu wire is used to reduce the resistance (R) of the circuit.
- (3) Soft iron is used to improve inductance (L) of the circuit.
- (4) The inductive reactance or effective opposition of the choke coil is given by $X_L = \omega L = 2\pi\nu L$
- (5) For an ideal choke coil $r = 0$, no electric energy is wasted i.e. average power $P = 0$.
- (6) In actual practice choke coil is equivalent to a $R - L$ circuit.
- (7) Choke coil for different frequencies are made by using different substances in their core.

For low frequency L should be large thus iron core choke coil is used. For high frequency ac circuit, L should be small, so air cored choke coil is used.

Tips & Tricks

If ac is produced by a generator having a large number of poles then it's frequency

$$\nu = \frac{\text{Number of poles} \times \text{rotation per second}}{2} = \frac{P \times n}{2}$$

Where P is the number of poles; n is the rotational frequency of the coil.

✍ Alternating current in electric wires, bulbs etc. flows 50 times in one direction and 50 times in the opposite direction in 1 *second*. Since in one cycle the current becomes zero twice, hence a bulb lights up 100 times and is off 100 times in one second (50 cycles) but due to persistence of vision, it appears lighted continuously.

✍ ac is more dangerous than dc.

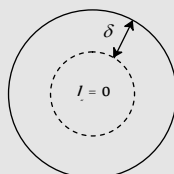
✍ The rate of change of ac is minimum at that instant when they are near their peak values.

✍ ac equipments such as electric motors, are more durable and convenient compared to dc equipments.

✍ Skin Effect

A direct current flows uniformly throughout the cross-section of the conductor. An alternating current, on the other hand, flows

mainly along the surface of the conductor. This effect is known as skin effect. the reason is that when alternating current flows through a conductor, the flux changes in the inner part of the conductor are higher. Therefore the inductance of the inner part is higher than that of the outer part. Higher the frequency of alternating current, more is the skin effect.



The depth upto which ac current flows through a wire is called skin depth (δ).

Alternating Current, Voltage and Power

1. The power is transmitted from a power house on high voltage ac because [CPMT 1984, 85]
 - (a) Electric current travels faster at higher *volts*
 - (b) It is more economical due to less power wastage
 - (c) It is difficult to generate power at low voltage
 - (d) Chances of stealing transmission lines are minimized

Ordinary Thinking

Objective Questions

2. The potential difference V and the current i flowing through an instrument in an ac circuit of frequency f are given by $V = 5 \cos \omega t$ volts and $I = 2 \sin \omega t$ amperes (where $\omega = 2\pi f$). The power dissipated in the instrument is [CPMT 1977, 80; MP PET 1999]
- (a) Zero (b) 10 W
(c) 5 W (d) 2.5 W
3. In an ac circuit, V and I are given by $V = 100 \sin(100t)$ volts, $I = 100 \sin\left(100t + \frac{\pi}{3}\right)$ mA. The power dissipated in circuit is [MP PET 1989; RPET 1999; MP PMT 1999, 2002]
- (a) 10 watt (b) 10 watt
(c) 2.5 watt (d) 5 watt
4. Alternating current can not be measured by dc ammeter because
- (a) ac cannot pass through dc ammeter
(b) Average value of complete cycle is zero
(c) ac is virtual
(d) ac changes its direction
5. The resistance of a coil for dc is in ohms. In ac, the resistance
- (a) Will remain same (b) Will increase
(c) Will decrease (d) Will be zero
6. If instantaneous current is given by $i = 4 \cos(\omega t + \phi)$ amperes, then the r.m.s. value of current is [RPET 2000]
- (a) 4 amperes (b) $2\sqrt{2}$ amperes
(c) $4\sqrt{2}$ amperes (d) Zero amperes
7. In an ac circuit, peak value of voltage is 423 volts. Its effective voltage is [JIPMER 1997]
- (a) 400 volts (b) 323 volts
(c) 300 volts (d) 340 volts
8. In an ac circuit $I = 100 \sin 200 \pi t$. The time required for the current to achieve its peak value will be [DPMT 2003]
- (a) $\frac{1}{100}$ sec (b) $\frac{1}{200}$ sec
(c) $\frac{1}{300}$ sec (d) $\frac{1}{400}$ sec
9. The peak value of an Alternating current is 6 amp, then r.m.s. value of current will be
- (a) 3 A (b) $3\sqrt{3}$ A
(c) $3\sqrt{2}$ A (d) $2\sqrt{3}$ A
10. A generator produces a voltage that is given by $V = 240 \sin 120t$, where t is in seconds. The frequency and r.m.s. voltage are [MP PET 1993; MP PMT 1990]
- (a) 60 Hz and 240 V (b) 19 Hz and 120 V
(c) 19 Hz and 170 V (d) 754 Hz and 70 V
11. If E_0 represents the peak value of the voltage in an ac circuit, the r.m.s. value of the voltage will be [CPMT 1972; MP PMT 1996]
- (a) $\frac{E_0}{\pi}$ (b) $\frac{E_0}{2}$ (c) $\frac{E_0}{\sqrt{\pi}}$ (d) $\frac{E_0}{\sqrt{2}}$
12. The peak value of 220 volts of ac mains is [CPMT 1990; MP PMT 1999; MP PET 2000; RPET 2001]
- (a) 155.6 volts (b) 220.0 volts
(c) 311.0 volts (d) 440 volts
13. A sinusoidal ac current flows through a resistor of resistance R . If the peak current is I_p , then the power dissipated is [MP PMT 1991]
- (a) $I_p^2 R \cos \theta$ (b) $\frac{1}{2} I_p^2 R$
(c) $\frac{4}{\pi} I_p^2 R$ (d) $\frac{1}{\pi} I_p^2 R$
14. A 40 Ω AC meter is connected to a 200 V, 50 Hz mains supply. The peak value of electric current flowing in the circuit is approximately [MP PET 1992]
- (a) 2.5 A (b) 5.0 A
(c) 7 A (d) 10 A
15. The frequency of ac mains in India is [CPMT 1987; NCERT 1974; MP PMT/PET 1988; RPMT 1997; RPET 2000]
- (a) 30 c/s or Hz (b) 50 c/s or Hz
(c) 60 c/s or Hz (d) 120 c/s or Hz
16. The r.m.s. value of an ac of 50 Hz is 10 amp. The time taken by the alternating current in reaching from zero to maximum value and the peak value of current will be [MP PET 1993; KCET 2003]
- (a) 2×10^{-2} sec and 14.14 amp
(b) 1×10^{-2} sec and 7.07 amp
(c) 5×10^{-2} sec and 7.07 amp
(d) 5×10^{-2} sec and 14.14 amp
17. The root mean square value of the alternating current is equal to
- (a) Twice the peak value
(b) Half the peak value
(c) $\frac{1}{\sqrt{2}}$ times the peak value
(d) Equal to the peak value
18. The peak value of an alternating e.m.f. E is given by $E = E_0 \cos \omega t$ is 10 volts and its frequency is 50 Hz. At time $t = \frac{1}{600}$ sec, the instantaneous e.m.f. is [MP PMT 1990; MP PET 2004]
- (a) 10 V (b) $5\sqrt{3}$ V
(c) 5 V (d) 1 V
19. If a current I given by $I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$ flows in an ac circuit across which an ac potential of $E = E_0 \sin \omega t$ has been applied, then the power consumption P in the circuit will be [CPMT 1986; Roorkee 1992; SCRA 1996; MP PMT 1994; RPET 2001; MP PET 2001, 02]
- (a) $P = \frac{E_0 I_0}{\sqrt{2}}$ (b) $P = \sqrt{2} E_0 I_0$

- (c) $P = \frac{E_0 I_0}{2}$ (d) $P = 0$
20. In an ac circuit, the instantaneous values of e.m.f. and current are $e = 200 \sin 314 t$ volt and $i = \sin\left(314t + \frac{\pi}{3}\right)$ ampere. The average power consumed in watt is
[NCERT 1990; RPMT 1997]
- (a) 200 (b) 100
(c) 50 (d) 25
21. An ac generator produced an output voltage $E = 170 \sin 377 t$ volts, where t is in seconds. The frequency of ac voltage is
[MP PET 1994]
- (a) 50 Hz (b) 110 Hz
(c) 60 Hz (d) 230 Hz
22. In general in an alternating current circuit [MP PMT 1994]
- (a) The average value of current is zero
(b) The average value of square of the current is zero
(c) Average power dissipation is zero
(d) The phase difference between voltage and current is zero
23. An alternating current is given by the equation $i = i_1 \cos \omega t + i_2 \sin \omega t$. The r.m.s. current is given by
[MP PMT 1994]
- (a) $\frac{1}{\sqrt{2}}(i_1 + i_2)$ (b) $\frac{1}{\sqrt{2}}(i_1 + i_2)^2$
(c) $\frac{1}{\sqrt{2}}(i_1^2 + i_2^2)^{1/2}$ (d) $\frac{1}{2}(i_1^2 + i_2^2)^{1/2}$
24. In an ac circuit, the current is given by $i = 5 \sin\left(100t - \frac{\pi}{2}\right)$ and the ac potential is $V = 200 \sin(100t)$ volt. Then the power consumption is
[CBSE PMT 1995; MH CET 1999; CPMT 2002]
- (a) 20 watts (b) 40 watts
(c) 1000 watts (d) 0 watt
25. An electric lamp is connected to 220 V, 50 Hz supply. Then the peak value of voltage is
[AFMC 1996]
- (a) 210 V (b) 211 V
(c) 311 V (d) 320 V
26. In a circuit, the value of the alternating current is measured by hot wire ammeter as 10 ampere. Its peak value will be
[MP PET 1996; AMU (Med.) 1999; KCET (Engg./Med.) 2000; CPMT 2003]
- (a) 10 A (b) 20 A
(c) 14.14 A (d) 7.07 A
27. The voltage of domestic ac is 220 volt. What does this represent
- (a) Mean voltage
(b) Peak voltage
(c) Root mean voltage
(d) Root mean square voltage
28. The r.m.s. voltage of domestic electricity supply is 220 volt. Electrical appliances should be designed to withstand an instantaneous voltage of
- (a) 220 V (b) 310 V
(c) 330 V (d) 440 V
29. The process by which ac is converted into dc is known as
- (b) Purification (b) Amplification
(c) Rectification (d) Current amplification
30. In an ac circuit with voltage V and current I , the power dissipated is
[CBSE PMT 1997]
- (a) VI
(b) $\frac{1}{2} VI$
(c) $\frac{1}{\sqrt{2}} VI$
(d) Depends on the phase between V and I
31. For an ac circuit $V = 15 \sin \omega t$ and $I = 20 \cos \omega t$ the average power consumed in this circuit is [RPET 1999]
- (a) 300 Watt (b) 150 Watt
(c) 75 Watt (d) zero
32. A bulb is connected first with dc and then ac of same voltage then it will shine brightly with
[RPET 2000]
- (a) AC
(b) DC
(c) Brightness will be in ratio 1/1.4
(d) Equally with both
33. An ac supply gives 30 V r.m.s. which passes through a 10Ω resistance. The power dissipated in it is [AMU (Med.) 2001]
- (a) $90\sqrt{2}$ W (b) 90 W
(c) $45\sqrt{2}$ W (d) 45 W
34. The frequency of an alternating voltage is 50 cycles/sec and its amplitude is 120 V. Then the r.m.s. value of voltage is
[BHU 1999; MH CET (Med.) 2001; KCET (Med.) 2001; MH CET 2003]
- (a) 101.3 V (b) 84.8 V
(c) 70.7 V (d) 56.5 V
35. A resistance of 20 ohms is connected to a source of an alternating potential $V = 220 \sin(100\pi t)$. The time taken by the current to change from its peak value to r.m.s. value is
[MP PET 2001]
- (a) 0.2 sec (b) 0.25 sec
(c) 25×10^{-3} sec (d) 2.5×10^{-3} sec
36. Voltage and current in an ac circuit are given by
 $V = 5 \sin\left(100\pi t - \frac{\pi}{6}\right)$ and $I = 4 \sin\left(100\pi t + \frac{\pi}{6}\right)$
[MP PMT 1996] [Kerala PET 2001]
- (a) Voltage leads the current by 30°
(b) Current leads the voltage by 30°
(c) Current leads the voltage by 60°
(d) Voltage leads the current by 60°

37. If an ac main supply is given to be 220 V. What would be the average e.m.f. during a positive half cycle [MH CET 2002]

(a) 198 V (b) 386 V
(c) 256 V (d) None of these

38. In an ac circuit, the *r.m.s.* value of current, I_{rms} is related to the peak current, I_0 by the relation [AFMC 2002]

(a) $I_{rms} = \frac{1}{\pi} I_0$ (b) $I_{rms} = \frac{1}{\sqrt{2}} I_0$
(c) $I_{rms} = \sqrt{2} I_0$ (d) $I_{rms} = \pi I_0$

39. An alternating voltage is represented as $E = 20 \sin 300t$. The average value of voltage over one cycle will be [MP PMT 2002]

(a) Zero (b) 10 volt
(c) $20\sqrt{2}$ volt (d) $\frac{20}{\sqrt{2}}$ volt

40. The ratio of peak value and *r.m.s* value of an alternating current is

(a) 1 (b) $\frac{1}{2}$
(c) $\sqrt{2}$ (d) $1/\sqrt{2}$

41. A 280 ohm electric bulb is connected to 200 V electric line. The peak value of current in the bulb will be [MP PET 2002]

(a) About one ampere (b) Zero
(c) About two ampere (d) About four ampere

42. An ac source is rated at 220 V, 50 Hz. The time taken for voltage to change from its peak value to zero is [Orissa JEE 2003]

(a) 50 sec (b) 0.02 sec
(c) 5 sec (d) 5×10^{-3} sec

43. If the value of potential in an ac, circuit is 10 V, then the peak value of potential is [CPMT 2003]

(a) $\frac{10}{\sqrt{2}}$ (b) $10\sqrt{2}$
(c) $20\sqrt{2}$ (d) $\frac{20}{\sqrt{2}}$

44. A lamp consumes only 50% of peak power in an a.c. circuit. What is the phase difference between the applied voltage and the circuit current [MP PMT 2004]

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

45. If an alternating voltage is represented as

$E = 141 \sin(628t)$, then the *rms* value of the voltage and the frequency are respectively [Kerala PET 2005]

(a) 141V, 628Hz (b) 100V, 50Hz
(c) 100V, 100Hz (d) 141V, 100Hz

46. The maximum value of a.c. voltage in a circuit is 707 V. Its *rms* value is [MP PET 2005]

(a) 70.7 V (b) 100 V
(c) 500 V (d) 707 V

ac Circuits

- Choke coil works on the principle of [MP PET/PMT 1988]
 - Transient current
 - Self induction
 - Mutual induction
 - Wattless current
- A choke coil has [RPET 1999; AIIMS 1999]
 - High inductance and low resistance
 - Low inductance and high resistance
 - High inductance and high resistance
 - Low inductance and low resistance
- Choke coil is used to control [CPMT 1984]
 - ac
 - dc
 - Both ac and dc
 - Neither ac nor dc
- Current in the circuit is wattless, if [MP PMT 2002]
 - Inductance in the circuit is zero
 - Resistance in the circuit is zero
 - Current is alternating
 - Resistance and inductance both are zero
- The phase angle between e.m.f. and current in *LCR* series ac circuit is [MP PMT/PET 1998]
 - 0 to $\pi/2$
 - $\pi/4$
 - $\pi/2$
 - π
- A choke coil is preferred to a rheostat in ac circuit as
 - It consumes almost zero power
 - It increases current
 - It increases power
 - It increases voltage
- An alternating e.m.f. is applied to purely capacitive circuit. The phase relation between e.m.f. and current flowing in the circuit is **or**
In a circuit containing capacitance only [MP PET 1996; AIIMS 1997]
 - e.m.f. is ahead of current by $\pi/2$
 - Current is ahead of e.m.f. by $\pi/2$
 - Current lags behind e.m.f. by π
 - Current is ahead of e.m.f. by π
- An ac source is connected to a resistive circuits. Which of the following is true [CPMT 1985]
 - Current leads the voltage and both are in same phase
 - Current lags behind the voltage and both are in same phase
 - Current and voltage are in same phase
 - Any of the above may be true depending upon the value of resistance
- The average power dissipated in a pure inductor of inductance L when an ac current is passing through it, is [CPMT 1974; RPMT 1997; MP PET 1999]
 - $\frac{1}{2} LI^2$
 - $\frac{1}{4} LI^2$
 - $2 LI^2$
 - Zero

(Inductance of the coil L and current I)

10. An alternating current of frequency ' f ' is flowing in a circuit containing a resistance R and a choke L in series. The impedance of this circuit is
- [CPMT 1978; MP PMT 1993; MP PET 1999; AIIMS 2000; Pb. PET 2004; RPET 2001, 03]
- (a) $R + 2\pi fL$ (b) $\sqrt{R^2 + 4\pi^2 f^2 L^2}$
(c) $\sqrt{R^2 + L^2}$ (d) $\sqrt{R^2 + 2\pi fL}$
11. A resonant ac circuit contains a capacitor of capacitance $10^{-6} F$ and an inductor of $10^{-4} H$. The frequency of electrical oscillations will be
- (a) $10^5 Hz$ (b) $10 Hz$
(c) $\frac{10^5}{2\pi} Hz$ (d) $\frac{10}{2\pi} Hz$
12. Power delivered by the source of the circuit becomes maximum, when [DCE 2004]
- (a) $\omega L = \omega C$ (b) $\omega L = \frac{1}{\omega C}$
(c) $\omega L = -\left(\frac{1}{\omega C}\right)^2$ (d) $\omega L = \sqrt{\omega C}$
13. An alternating voltage is connected in series with a resistance R and an inductance L . If the potential drop across the resistance is $200 V$ and across the inductance is $150 V$, then the applied voltage is
- (a) $350 V$ (b) $250 V$
(c) $500 V$ (d) $300 V$
14. An inductive circuit contains resistance of 10Ω and an inductance of $20 H$. If an ac voltage of $120 V$ and frequency $60 Hz$ is applied to this circuit, the current would be nearly
- (a) $0.32 amp$ (b) $0.016 amp$
(c) $0.48 amp$ (d) $0.80 amp$
15. Same current is flowing in two alternating circuits. The first circuit contains only inductance and the other contains only a capacitor. If the frequency of the e.m.f. of ac is increased, the effect on the value of the current will be [MP PET 1993]
- (a) Increases in the first circuit and decreases in the other
(b) Increases in both the circuits
(c) Decreases in both the circuits
(d) Decreases in the first circuit and increases in the other
16. A capacitor is a perfect insulator for
- (a) Alternating currents (b) Direct currents
(c) Both ac and dc (d) None of these
17. In a circuit containing an inductance of zero resistance, the e.m.f. of the applied ac voltage leads the current by [CPMT 1990]
- (a) 90° (b) 45°
(c) 30° (d) 0°
18. In a pure inductive circuit or In an ac circuit containing inductance only, the current [MP PMT 1993; CPMT 1996; Kerala PET 2002]
- (a) Leads the e.m.f. by 90°
(b) Lags behind the e.m.f. by 90°
(c) Sometimes leads and sometime lags behind the e.m.f.
- (d) Is in phase with the e.m.f.
19. A $20 volts$ ac is applied to a circuit consisting of a resistance and a coil with negligible resistance. If the voltage across the resistance is $12 V$, the voltage across the coil is [MP PMT 1989; RPMT 1997]
- (a) $16 volts$ (b) $10 volts$
(c) $8 volts$ (d) $6 volts$
20. A resistance of 300Ω and an inductance of $\frac{1}{\pi}$ henry are connected in series to a ac voltage of $20 volts$ and $200 Hz$ frequency. The phase angle between the voltage and current is
- (a) $\tan^{-1} \frac{4}{3}$ (b) $\tan^{-1} \frac{3}{4}$
(c) $\tan^{-1} \frac{3}{2}$ (d) $\tan^{-1} \frac{2}{5}$
21. The power factor of LCR circuit at resonance is [MP PMT 1991; RPMT 1999; RPET 2001; UPSEAT 1999]
- (a) 0.707 (b) 1
(c) Zero (d) 0.5
22. An inductance of $1 mH$ a condenser of $10 \mu F$ and a resistance of 50Ω are connected in series. The reactances of inductor and condensers are same. The reactance of either of them will be [CPMT 1990]
- (a) 100Ω (b) 30Ω
(c) 3.2Ω (d) 10Ω
23. The natural frequency of a $L-C$ circuit is equal to [CPMT 1978, 97]
- (a) $\frac{1}{2\pi} \sqrt{LC}$ (b) $\frac{1}{2\pi\sqrt{LC}}$
(c) $\frac{1}{2\pi} \sqrt{\frac{L}{C}}$ (d) $\frac{1}{2\pi} \sqrt{\frac{C}{L}}$
24. An alternating voltage $E = 200\sqrt{2} \sin(100t)$ is connected to a $1 microfarad$ capacitor through an ac ammeter. The reading of the ammeter shall be [NCERT 1984; MNR 1995; MP PET 1999; RPET 1999; UPSEAT 2000]
- (a) $10 mA$ (b) $20 mA$
(c) $40 mA$ (d) $80 mA$
25. An ac circuit consists of an inductor of inductance $0.5 H$ and a capacitor of capacitance $8 \mu F$ in series. The current in the circuit is maximum when the angular frequency of ac source is
- (a) $500 rad/sec$ (b) $2 \times 10^4 rad/sec$
(c) $4000 rad/sec$ (d) $5000 rad/sec$
26. The average power dissipation in a pure capacitance in ac circuit is [DPMT 1989]
- (a) $\frac{1}{2} CV^2$ (b) CV^2
(c) $\frac{1}{4} CV^2$ (d) Zero
27. In a region of uniform magnetic induction $B = 10^{-2} tesla$, a circular coil of radius $30 cm$ and resistance πohm is rotated about an axis which is perpendicular to the direction of B and which forms

- a diameter of the coil. If the coil rotates at 200 rpm the amplitude of the alternating current induced in the coil is
- (a) $4\pi \text{ mA}$ (b) 30 mA
(c) 6 mA (d) 200 mA
28. An inductive circuit contains a resistance of 10 ohm and an inductance of 2.0 henry. If an ac voltage of 120 volt and frequency of 60 Hz is applied to this circuit, the current in the circuit would be nearly [CPMT 1990; MP PET 2002]
- (a) 0.32 amp (b) 0.16 amp
(c) 0.48 amp (d) 0.80 amp
29. In a LCR circuit having $L = 8.0 \text{ henry}$, $C = 0.5 \mu\text{F}$ and $R = 100 \text{ ohm}$ in series. The resonance frequency in per second is
- (a) 600 radian (b) 600 Hz
(c) 500 radian (d) 500 Hz
30. In LCR circuit, the capacitance is changed from C to $4C$. For the same resonant frequency, the inductance should be changed from L to [MP PMT 1986; BHU 1998]
- (a) $2L$ (b) $L/2$
(c) $L/4$ (d) $4L$
31. A 120 volt ac source is connected across a pure inductor of inductance 0.70 henry. If the frequency of the source is 60 Hz, the current passing through the inductor is [MP PET 1994]
- (a) 4.55 amps (b) 0.355 amps
(c) 0.455 amps (d) 3.55 amps
32. The impedance of a circuit consists of 3 ohm resistance and 4 ohm reactance. The power factor of the circuit is [MP PMT 1994]
- (a) 0.4 (b) 0.6
(c) 0.8 (d) 1.0
33. L , C and R denote inductance, capacitance and resistance respectively. Pick out the combination which does not have the dimensions of frequency [MP PMT 1994]
- (a) $\frac{1}{RC}$ (b) $\frac{R}{L}$
(c) $\frac{1}{\sqrt{LC}}$ (d) $\frac{C}{L}$
34. The power factor of a good choke coil is [MP PMT 1994]
- (a) Nearly zero (b) Exactly zero
(c) Nearly one (d) Exactly one
35. If resistance of 100Ω , inductance of 0.5 henry and capacitance of $10 \times 10^{-6} \text{ F}$ are connected in series through 50 Hz ac supply, then impedance is [BHU 1995]
- (a) 1.876 (b) 18.76
(c) 189.72 (d) 101.3
36. An alternating current source of frequency 100 Hz is joined to a combination of a resistance, a capacitance and a coil in series. The potential difference across the coil, the resistance and the capacitor is 46, 8 and 40 volt respectively. The electromotive force of alternating current source in volt is [MP PET 1995]
- (a) 94 (b) 14
(c) $10 \sqrt{10}$ [CBSE PMT 1990] (d) 76
37. A 10 ohm resistance, 5 mH coil and 10 μF capacitor are joined in series. When a suitable frequency alternating current source is joined to this combination, the circuit resonates. If the resistance is halved, the resonance frequency [MP PET 1995]
- (a) Is halved (b) Is doubled
(c) Remains unchanged (d) Is quadrupled
38. L , C and R represent physical quantities inductance, capacitance and resistance respectively. The combination representing dimension of frequency is [CPMT 1990] [MP PET 1995; DCE 2001]
- (a) LC (b) $(LC)^{-1/2}$
(c) $\left(\frac{L}{C}\right)^{-1/2}$ (d) $\frac{C}{L}$
39. In a series circuit $R = 300 \Omega$, $L = 0.9 \text{ H}$, $C = 2.0 \mu\text{F}$ and $\omega = 1000 \text{ rad/sec}$. The impedance of the circuit is
- (a) 1300Ω (b) 900Ω
(c) 500Ω (d) 400Ω
40. In a L - R circuit, the value of L is $\left(\frac{0.4}{\pi}\right) \text{ henry}$ and the value of R is 30 ohm. If in the circuit, an alternating e.m.f. of 200 volt at 50 cycles per sec is connected, the impedance of the circuit and current will be [MP PET 1996; DPMT 2003]
- (a) $11.4 \Omega, 17.5 \text{ A}$ (b) $30.7 \Omega, 6.5 \text{ A}$
(c) $40.4 \Omega, 5 \text{ A}$ (d) $50 \Omega, 4 \text{ A}$
41. The reactance of a coil when used in the domestic ac power supply (220 volt, 50 cycles) is 100 ohm. The self inductance of the coil is nearly [MP PMT 1996]
- (a) 3.2 henry (b) 0.32 henry
(c) 2.2 henry (d) 0.22 henry
42. In a series LCR circuit, operated with an ac of angular frequency ω , the total impedance is [MP PET 1996]
- (a) $[R^2 + (L\omega - C\omega)^2]^{1/2}$
(b) $\left[R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2\right]^{1/2}$
(c) $\left[R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2\right]^{-1/2}$
(d) $\left[(R\omega)^2 + \left(L\omega - \frac{1}{C\omega}\right)^2\right]^{1/2}$
43. The reactance of a $25 \mu\text{F}$ capacitor at the ac frequency of 4000 Hz is
- (a) $\frac{5}{\pi} \text{ ohm}$ (b) $\sqrt{\frac{5}{\pi}} \text{ ohm}$
(c) 10 ohm (d) $\sqrt{10} \text{ ohm}$

44. The frequency for which a $5 \mu F$ capacitor has a reactance of $\frac{1}{1000} \text{ ohm}$ is given by [MP PET 1997]
- (a) $\frac{100}{\pi} \text{ MHz}$ (b) $\frac{1000}{\pi} \text{ Hz}$
(c) $\frac{1}{1000} \text{ Hz}$ (d) 1000 Hz
45. An e.m.f. $E = 4 \cos(1000t) \text{ volt}$ is applied to an LR-circuit of inductance 3 mH and resistance 4 ohms . The amplitude of current in the circuit is [MP PMT 1997]
- (a) $\frac{4}{\sqrt{7}} \text{ A}$ (b) 1.0 A
(c) $\frac{4}{7} \text{ A}$ (d) 0.8 A
46. In an ac circuit, a resistance of $R \text{ ohm}$ is connected in series with an inductance L . If phase angle between voltage and current be 45° , the value of inductive reactance will be [MP PMT/PET 1998]
- (a) $\frac{R}{4}$
(b) $\frac{R}{2}$
(c) R
(d) Cannot be found with the given data
47. A coil of inductance L has an inductive reactance of X_L in an AC circuit in which the effective current is I . The coil is made from a super-conducting material and has no resistance. The rate at which power is dissipated in the coil is [MP PMT 1999]
- (a) 0 (b) IX_L
(c) $I^2 X_L$ (d) IX_L^2
48. The phase difference between the current and voltage of LCR circuit in series combination at resonance is [CPMT 1999; Pb. PET 2001]
- (a) 0 (b) $\pi/2$
(c) π (d) $-\pi$
49. In a series resonant circuit, the ac voltage across resistance R , inductance L and capacitance C are 5 V , 10 V and 10 V respectively. The ac voltage applied to the circuit will be [KCET 1994]
- (a) 20 V (b) 10 V
(c) 5 V (d) 25 V
50. When 100 volt dc is applied across a coil, a current of 1 amp flows through it. When 100 volt ac at 50 cycle s^{-1} is applied to the same coil, only 0.5 ampere current flows. The impedance of the coil is
- (a) 100Ω (b) 200Ω
(c) 300Ω (d) 400Ω
51. The coefficient of induction of a choke coil is 0.1 H and resistance is 12Ω . If it is connected to an alternating current source of frequency 60 Hz , then power factor will be
- (a) 0.32 (b) 0.30
(c) 0.28 (d) 0.24
52. For series LCR circuit, wrong statement is [RPMT 1997]
- (a) Applied e.m.f. and potential difference across resistance are in same phase
(b) Applied e.m.f. and potential difference at inductor coil have phase difference of $\pi/2$
(c) Potential difference at capacitor and inductor have phase difference of $\pi/2$
(d) Potential difference across resistance and capacitor have phase difference of $\pi/2$
53. In a purely resistive ac circuit, the current [Roorkee 1992]
- (a) Lags behind the e.m.f. in phase
(b) Is in phase with the e.m.f.
(c) Leads the e.m.f. in phase
(d) Leads the e.m.f. in half the cycle and lags behind it in the other half
54. If an 8Ω resistance and 6Ω reactance are present in an ac series circuit then the impedance of the circuit will be [MP PMT 2003]
- (a) 20 ohm (b) 5 ohm
(c) 10 ohm (d) $14\sqrt{2} \text{ ohm}$
55. A 12 ohm resistor and a 0.21 henry inductor are connected in series to an ac source operating at 20 volts , 50 cycle/second . The phase angle between the current and the source voltage is
- (a) 30° (b) 40°
(c) 80° (d) 90°
56. What will be the phase difference between virtual voltage and virtual current, when the current in the circuit is wattless [RPET 1996]
- (a) 90° (b) 45°
(c) 180° (d) 60°
57. The resonant frequency of a circuit is f . If the capacitance is made 4 times the initial values, then the resonant frequency will become
- (a) $f/2$ (b) $2f$
(c) f (d) $f/4$
58. In the non-resonant circuit, what will be the nature of the circuit for frequencies higher than the resonant frequency [RPET 1996]
- (a) Resistive (b) Capacitive
(c) Inductive (d) None of the above
59. In an ac circuit, the potential difference across an inductance and resistance joined in series are respectively 16 V and 20 V . The total potential difference across the circuit is [Bihar MEE 1995] [AFMC 1998; BHU 1999]
- (a) 20.0 V (b) 25.6 V
(c) 31.9 V (d) 53.5 V
60. A 220 V , 50 Hz ac source is connected to an inductance of 0.2 H and a resistance of 20 ohm in series. What is the current in the circuit [MNR 1998; JIPMER 2001, 02]
- (a) 10 A (b) 5 A

- (c) 33.3 A (d) 3.33 A
61. An LCR circuit contains $R = 50 \Omega$, $L = 1 \text{ mH}$ and $C = 0.1 \mu\text{F}$. The impedance of the circuit will be minimum for a frequency of

[Bihar MEE 1995]

- (a) $\frac{10^5}{2\pi} \text{ s}^{-1}$ (b) $\frac{10^6}{2\pi} \text{ s}^{-1}$
- (c) $2\pi \times 10^5 \text{ s}^{-1}$ (d) $2\pi \times 10^6 \text{ s}^{-1}$
62. In a series LCR circuit, resistance $R = 10\Omega$ and the impedance $Z = 20\Omega$. The phase difference between the current and the voltage is

[KCET (Engg./Med.) 1999]

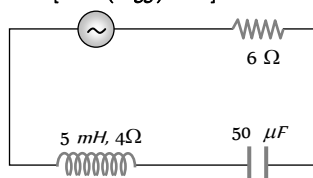
- (a) 30° (b) 45°
- (c) 60° (d) 90°
63. A series ac circuit consist of an inductor and a capacitor. The inductance and capacitance is respectively 1 henry and $25 \mu\text{F}$. If the current is maximum in circuit then angular frequency will be
- (a) 200 (b) 100
- (c) 50 (d) $200/2\pi$

64. An alternating e.m.f. of frequency $\nu = \frac{1}{2\pi\sqrt{LC}}$ is applied to a series LCR circuit. For this frequency of the applied e.m.f.

- (a) The circuit is at resonance and its impedance is made up only of a reactive part
- (b) The current in the circuit is in phase with the applied e.m.f. and the voltage across R equals this applied emf
- (c) The sum of the p.d.'s across the inductance and capacitance equals the applied e.m.f. which is 180° ahead of phase of the current in the circuit
- (d) The quality factor of the circuit is $\omega L/R$ or $1/\omega CR$ and this is a measure of the voltage magnification (produced by the circuit at resonance) as well as the sharpness of resonance of the circuit

65. In the circuit shown below, the ac source has voltage $V = 20 \cos(\omega t)$ volts with $\omega = 2000 \text{ rad/sec}$. the amplitude of the current will be nearest to

[AMU (Engg.) 2000]



- (a) 2 A
- (b) 3.3 A
- (c) $2/\sqrt{5} \text{ A}$
- (d) $\sqrt{5} \text{ A}$
66. The value of the current through an inductance of 1 H and of negligible resistance, when connected through an ac source of 200 V and 50 Hz , is

[AFMC 2000]

- (a) 0.637 A (b) 1.637 A
- (c) 2.637 A (d) 3.637 A
67. The quality factor of LCR circuit having resistance (R) and inductance (L) at resonance frequency (ω) is given by

[AFMC 2000; CBSE PMT 2000]

- (a) $\frac{\omega L}{R}$ (b) $\frac{R}{\omega L}$
- (c) $\left(\frac{\omega L}{R}\right)^{1/2}$ (d) $\left(\frac{\omega L}{R}\right)^2$

68. Power factor is maximum in an LCR circuit when

- (a) $X_L = X_C$ (b) $R = 0$
- (c) $X_L = 0$ (d) $X_C = 0$

69. In an ac circuit the reactance of a coil is $\sqrt{3}$ times its resistance, the phase difference between the voltage across the coil to the current through the coil will be

[KCET (Engg.) 2000]

- (a) $\pi/3$ (b) $\pi/2$
- (c) $\pi/4$ (d) $\pi/6$

70. The capacity of a pure capacitor is 1 farad . In dc circuits, its effective resistance will be

[MP PMT 2000]

- (a) Zero (b) Infinite
- (c) 1 ohm (d) $1/2 \text{ ohm}$

71. In an ac circuit, the current lags behind the voltage by $\pi/3$. The components in the circuit are

[MP PMT 2000]

- (a) R and L (b) R and C
- (c) L and C (d) Only R

72. The reactance of a coil when used in the domestic ac power supply (220 volts , $50 \text{ cycles per second}$) is 50 ohms . The inductance of the coil is

[Roorkee 1999]

[MP PMT 2000]

- (a) 2.2 henry (b) 0.22 henry
- (c) 1.6 henry (d) 0.16 henry

73. In an ac circuit, the power factor

[Roorkee 2000]

- (a) Is zero when the circuit contains an ideal resistance only
- (b) Is unity when the circuit contains an ideal resistance only
- (c) Is zero when the circuit contains an ideal inductance only
- (d) Is unity when the circuit contains an ideal inductance only

74. A resistance of 40 ohm and an inductance of 95.5 millihenry are connected in series in a 50 cycles/second ac circuit. The impedance of this combination is very nearly

[MP PET 2000]

- (a) 30 ohm (b) 40 ohm
- (c) 50 ohm (d) 60 ohm

75. For high frequency, a capacitor offers

[CPMT 1999; CBSE PMT 1999; AFMC 2001; Pb. PET 2001; J & K CET 2004]

- (a) More reactance (b) Less reactance
- (c) Zero reactance (d) Infinite reactance

76. The coil of choke in a circuit

[AIIMS 2001]

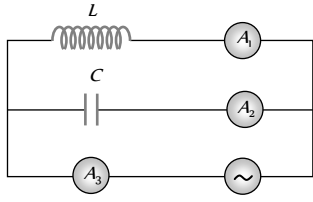
- (a) Increases the current
- (b) Decreases the current
- (c) Does not change the current
- (d) Has high resistance to dc circuit

77. In a circuit, the current lags behind the voltage by a phase difference of $\pi/2$. The circuit contains which of the following

- (a) Only R (b) Only L
- (c) Only C (d) R and C

78. The inductive reactance of an inductor of $\frac{1}{\pi} \text{ henry}$ at 50 Hz frequency is

[MP PET 2001, 02]

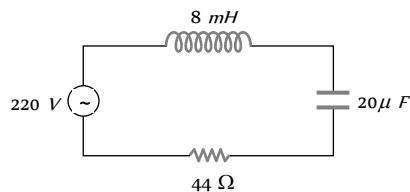
- (a) $\frac{50}{\pi} \text{ ohm}$ (b) $\frac{\pi}{50} \text{ ohm}$
(c) 100 ohm (d) 50 ohm
79. An oscillator circuit consists of an inductance of 0.5 mH and a capacitor of $20 \mu\text{F}$. The resonant frequency of the circuit is nearly
(a) 15.92 Hz (b) 159.2 Hz
(c) 1592 Hz (d) 15910 Hz
80. Reactance of a capacitor of capacitance $C \mu\text{F}$ for ac frequency $\frac{400}{\pi} \text{ Hz}$ is 25Ω . The value C is [MH CET 2002]
(a) $50 \mu\text{F}$ (b) $25 \mu\text{F}$
(c) $100 \mu\text{F}$ (d) $75 \mu\text{F}$
81. The power factor of an ac circuit having resistance (R) and inductance (L) connected in series and an angular velocity ω is
(a) $R/\omega L$ (b) $R/(R^2 + \omega^2 L^2)^{1/2}$
(c) $\omega L/R$ (d) $R/(R^2 - \omega^2 L^2)^{1/2}$
82. A circuit has a resistance of 11Ω , an inductive reactance of 25Ω and a capacitive resistance of 18Ω . It is connected to an ac source of 260 V and 50 Hz . The current through the circuit (in amperes) is
(a) 11 (b) 15
(c) 18 (d) 20
83. A 0.7 henry inductor is connected across a $120 \text{ V} - 60 \text{ Hz}$ ac source. The current in the inductor will be very nearly [MP PMT 2002]
(a) 4.55 amp (b) 0.355 amp
(c) 0.455 amp (d) 3.55 amp
84. There is a 5Ω resistance in an ac circuit. Inductance of 0.1 H is connected with it in series. If equation of ac e.m.f. is $5 \sin 50t$ then the phase difference between current and e.m.f. is
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{6}$
(c) $\frac{\pi}{4}$ (d) 0
85. An inductor of inductance L and resistor of resistance R are joined in series and connected by a source of frequency ω . Power dissipated in the circuit is [AIEEE 2002; RPET 2003]
(a) $\frac{(R^2 + \omega^2 L^2)}{V}$ (b) $\frac{V^2 R}{(R^2 + \omega^2 L^2)}$
(c) $\frac{V}{(R^2 + \omega^2 L^2)}$ (d) $\frac{\sqrt{R^2 + \omega^2 L^2}}{V^2}$
86. In an ac circuit of capacitance the current from potential is [CPMT 2003]
(a) Forward
(b) Backward
(c) Both are in the same phase
(d) None of these
87. A coil of 200Ω resistance and 1.0 H inductance is connected to an ac source of frequency $200/2\pi \text{ Hz}$. Phase angle between potential and current will be [MP PMT 2003]
(a) 30° (b) 90°
(c) 45° (d) 0
88. In an LCR circuit the pd between the terminals of the inductance is 60 V , between the terminals of the capacitor is 30 V and that between the terminals of resistance is 40 V . The supply voltage will be equal to [KCET 2004]
(a) 50 V (b) 70 V
(c) 130 V [Kerala PET 2002] (d) 10 V
89. Radio frequency choke uses core of [AFMC 2004]
(a) Air (b) Iron
(c) Air and iron (d) None of these
90. In an LCR circuit capacitance is changed from C to $2C$. For the resonant frequency to remain unchanged, the inductance should be change from L to [AIEEE 2004]
(a) $4L$ (b) $2L$
(c) $L/2$ (d) $L/4$
91. In an LCR series ac circuit, the voltage across each of the components, L , C and R is 50 V . The voltage across the LC combination will be [AIEEE 2004]
(a) 50 V (b) $50\sqrt{2} \text{ V}$
(c) 100 V (d) 0 V (zero)
92. A coil has $L = 0.04 \text{ H}$ and $R = 12 \Omega$. When it is connected to $220 \text{ V}, 50 \text{ Hz}$ supply the current flowing through the coil, in amperes [Kerala PMT 2002]
(a) 10.7 (b) 11.7
(c) 14.7 (d) 12.7 [Kerala PMT 2004]
93. The current in series LCR circuit will be maximum when ω is [Kerala PMT 2004]
(a) As large as possible
(b) Equal to natural frequency of LCR system
(c) $\sqrt{LC}$
(d) $\sqrt{1/LC}$ [RPET 2003]
94. An inductor L and a capacitor C are connected in the circuit as shown in the figure. The frequency of the power supply is equal to the resonant frequency of the circuit. Which ammeter will read zero ampere [DCE 2002]
- 
- (a) A_1 (b) A_2
(c) A_3 (d) None of these
95. Which of the following components of an LCR circuit, with ac supply, dissipates energy [DCE 2004]
(a) L (b) R
(c) C (d) All of these
96. In a circuit L , C and R are connected in series with an alternating voltage source of frequency f . The current leads the voltage by 45° . The value of C is [CBSE PMT 2005]
(a) $\frac{1}{2\pi f(2\pi fL + R)}$

- (b) $\frac{1}{\pi f(2\pi fL + R)}$
- (c) $\frac{1}{2\pi f(2\pi fL - R)}$
- (d) $\frac{1}{\pi f(2\pi fL - R)}$

97. In an A.C. circuit the current [CPMT 2005]

- (a) Always leads the voltage
- (b) Always lags behind the voltage
- (c) Is always in phase with the voltage
- (d) May lead or lag behind or be in phase with the voltage

98. For the series LCR circuit shown in the figure, what is the resonance frequency and the amplitude of the current at the resonating frequency [Kerala PET 2005]



- (a) 2500 rad-s^{-1} and $5\sqrt{2} \text{ A}$
- (b) 2500 rad-s^{-1} and 5 A
- (c) 2500 rad-s^{-1} and $\frac{5}{\sqrt{2}} \text{ A}$
- (d) 25 rad-s^{-1} and $5\sqrt{2} \text{ A}$

Critical Thinking

Objective Questions

1. When 100 volts dc is supplied across a solenoid, a current of 1.0 amperes flows in it. When 100 volts ac is applied across the same coil, the current drops to 0.5 ampere. If the frequency of ac source is 50 Hz, then the impedance and inductance of the solenoid are
- (a) 200Ω and 0.55 henry (b) 100Ω and 0.86 henry
- (c) 200Ω and 1.0 henry (d) 100Ω and 0.93 henry
2. In an LR-circuit, the inductive reactance is equal to the resistance R of the circuit. An e.m.f. $E = E_0 \cos(\omega t)$ applied to the circuit. The power consumed in the circuit is

[MP PMT 1997]

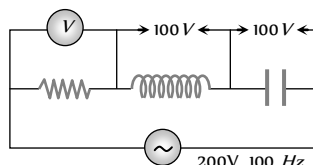
- (a) $\frac{E_0^2}{R}$ (b) $\frac{E_0^2}{2R}$
- (c) $\frac{E_0^2}{4R}$ (d) $\frac{E_0^2}{8R}$

3. One 10 V, 60 W bulb is to be connected to 100 V line. The required induction coil has self inductance of value ($f = 50 \text{ Hz}$)

- (a) 0.052 H (b) 2.42 H
- (c) 16.2 mH (d) 1.62 mH

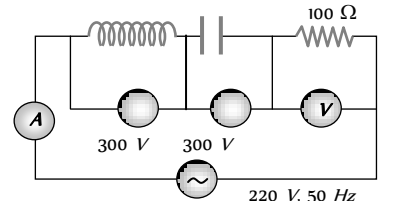
4. In the circuit given below, what will be the reading of the voltmeter

- (a) 300 V



- (b) 900 V
- (c) 200 V
- (d) 400 V

5. In the circuit shown below, what will be the readings of the voltmeter and ammeter [RPMT 1996]



- (a) 800 V, 2 A (b) 300 V, 2 A
- (c) 220 V, 2.2 A (d) 100 V, 2 A

6. A bulb and a capacitor are connected in series to a source of alternating current. If its frequency is increased, while keeping the voltage of the source constant, then

[Roorkee 1999]

- (a) Bulb will give more intense light
- (b) Bulb will give less intense light
- (c) Bulb will give light of same intensity as before
- (d) Bulb will stop radiating light

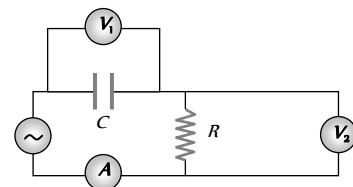
7. An alternating e.m.f. of angular frequency ω is applied across an inductance. The instantaneous power developed in the circuit has an angular frequency [Roorkee 1999]

- (a) $\frac{\omega}{4}$ (b) $\frac{\omega}{2}$
- (c) ω (d) 2ω

8. The voltage of an ac source varies with time according to the equation $V = 100 \sin 100\pi \cos 100\pi$ where t is in seconds and V is in volts. Then [MP PMT 1996; 2000]

- (a) The peak voltage of the source is 100 volts
- (b) The peak voltage of the source is 50 volts
- (c) The peak voltage of the source is $100/\sqrt{2}$ volts
- (d) The frequency of the source is 50 Hz

9. The diagram shows a capacitor C and a resistor R connected in series to an ac source. V_1 and V_2 are voltmeters and A is an ammeter

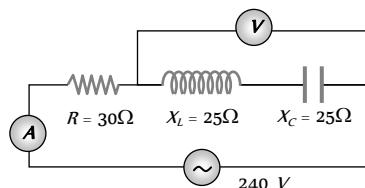


Consider now the following statements

- I. Readings in A and V_1 are always in phase
- II. Reading in V_1 is ahead in phase with reading in V_2 [RPET 1997]
- III. Readings in A and V are always in phase which of these statements are/is correct [AMU (Med.) 2001]
- (a) I only (b) II only
- (c) I and II only (d) II and III only

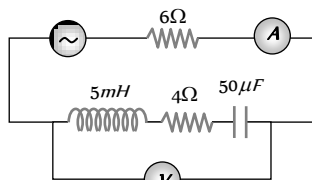
10. In the circuit shown in figure neglecting source resistance the voltmeter and ammeter reading will respectively, will be

[KCET (Engg.) 2001]

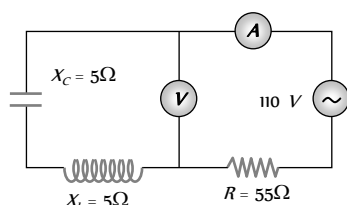


- (a) 0 V, 3 A (b) 150 V, 3 A
(c) 150 V, 6 A (d) 0 V, 8 A
11. The voltage of an ac supply varies with time (t) as $V = 120 \sin 100\pi t \cos 100\pi t$. The maximum voltage and frequency respectively are [MP PMT 2001; MP PET 2002]
- (a) 120 volts, 100 Hz (b) $\frac{120}{\sqrt{2}}$ volts, 100 Hz
(c) 60 volts, 200 Hz (d) 60 volts, 100 Hz
12. In the circuit shown in the figure, the ac source gives a voltage $V = 20 \cos(2000t)$. Neglecting source resistance, the voltmeter and ammeter reading will be

[KCET (Engg.) 2002]



- (a) 0 V, 0.47 A (b) 1.68 V, 0.47 A
(c) 0 V, 1.4 A (d) 5.6 V, 1.4 A
13. A telephone wire of length 200 km has a capacitance of $0.014 \mu F$ per km. If it carries an ac of frequency 5 kHz, what should be the value of an inductor required to be connected in series so that the impedance of the circuit is minimum
- (a) 0.35 mH (b) 35 mH
(c) 3.5 mH (d) Zero
14. In a certain circuit current changes with time according to $i = 2\sqrt{t}$. r.m.s. value of current between $t = 2$ to $t = 4$ s will be
- (a) 3 A (b) $3\sqrt{3}$ A
(c) $2\sqrt{3}$ A (d) $(2 - \sqrt{2})$ A
15. Match the following
- | Currents | r.m.s. values |
|---|---------------------------------|
| (1) $x_0 \sin \omega t$ | (i) x_0 |
| (2) $x_0 \sin \omega t \cos \omega t$ | (ii) $\frac{x_0}{\sqrt{2}}$ |
| (3) $x_0 \sin \omega t + x_0 \cos \omega t$ | (iii) $\frac{x_0}{(2\sqrt{2})}$ |
- (a) 1. (i), 2. (ii), 3. (iii) (b) 1. (ii), 2. (iii), 3. (i)
(c) 1. (i), 2. (iii), 3. (ii) (d) None of these
16. The reading of ammeter in the circuit shown will be



- (b) 2.4 A
(c) Zero
(d) 1.7 A

17. An ac source of angular frequency ω is fed across a resistor r and a capacitor C in series. The current registered is I . If now the frequency of source is changed to $\omega/3$ (but maintaining the same voltage), the current in then circuit is found to be halved. Calculate the ratio of reactance to resistance at the original frequency ω [Roorkee 1996]

- (a) $\sqrt{\frac{3}{5}}$ (b) $\sqrt{\frac{2}{5}}$
(c) $\sqrt{\frac{1}{5}}$ (d) $\sqrt{\frac{4}{5}}$

18. An LCR series circuit with a resistance of 100 ohm is connected to an ac source of 200 V (r.m.s.) and angular frequency 300 rad/s. When only the capacitor is removed, the current lags behind the voltage by 60° . When only the inductor is removed the current leads the voltage by 60° . The average power dissipated is

- (a) 50 W (b) 100 W
(c) 200 W (d) 400 W

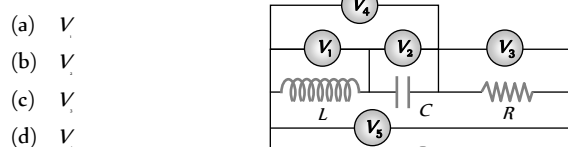
19. A virtual current of 4 A and 50 Hz flows in an ac circuit containing a coil. The power consumed in the coil is 240 W. If the virtual voltage across the coil is 100 V its inductance will be

- (a) $\frac{1}{3\pi}$ H (b) $\frac{1}{5\pi}$ H
(c) $\frac{1}{7\pi}$ H (d) $\frac{1}{9\pi}$ H

20. For a series RLC circuit $R = X_L = 2X_C$. The impedance of the circuit and phase difference (between) V and i will be

- (a) $\frac{\sqrt{5}R}{2}, \tan^{-1}(2)$ (b) $\frac{\sqrt{5}R}{2}, \tan^{-1}\left(\frac{1}{2}\right)$
(c) $\sqrt{5}X_C, \tan^{-1}(2)$ (d) $\sqrt{5}R, \tan^{-1}\left(\frac{1}{2}\right)$

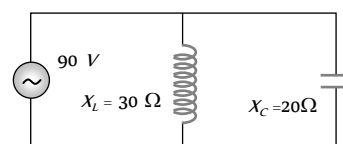
21. In the adjoining ac circuit the voltmeter whose reading will be zero at resonance is



- (a) V_1 (b) V_2
(c) V_3 (d) V_4

22. In the adjoining figure the impedance of the circuit will be

- (a) 120 ohm
(b) 50 ohm
(c) 60 ohm
(d) 90 ohm

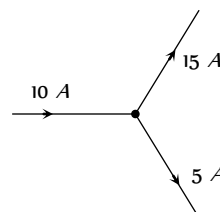


23. If $i = t^2$ $0 < t < T$ then r.m.s. value of current is

- (a) $\frac{T^2}{\sqrt{2}}$ (b) $\frac{T^2}{2}$
(c) $\frac{T^2}{\sqrt{5}}$ (d) None of these

24. Is it possible

- (a) Yes
(b) No

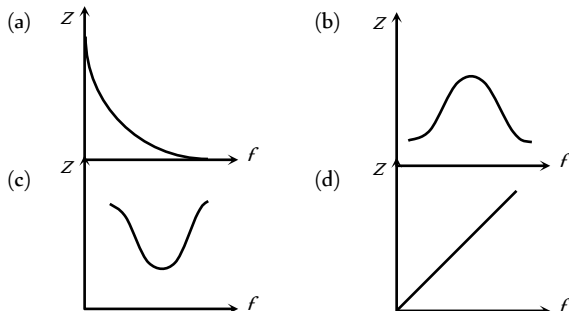


- (c) Cannot be predicted
(d) Insufficient data to reply

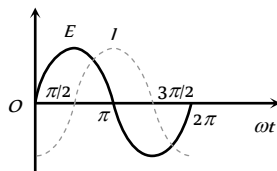
25. In a series circuit $C = 2\mu F$, $L = 1mH$ and $R = 10\Omega$, when the current in the circuit is maximum, at that time the ratio of the energies stored in the capacitor and the inductor will be
- (a) 1 : 1 (b) 1 : 2
(c) 2 : 1 (d) 1 : 5

Graphical Questions

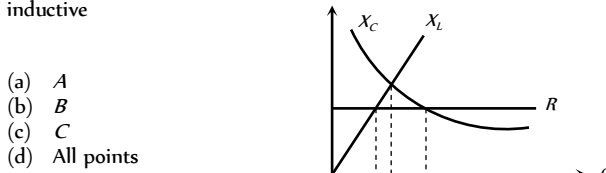
1. Which one of the following curves represents the variation of impedance (Z) with frequency f in series LCR circuit



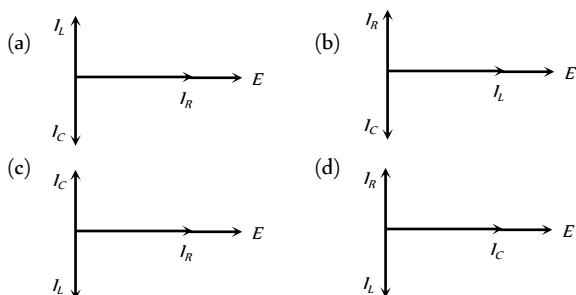
2. The variation of the instantaneous current (I) and the instantaneous emf (E) in a circuit is as shown in fig. Which of the following statements is correct



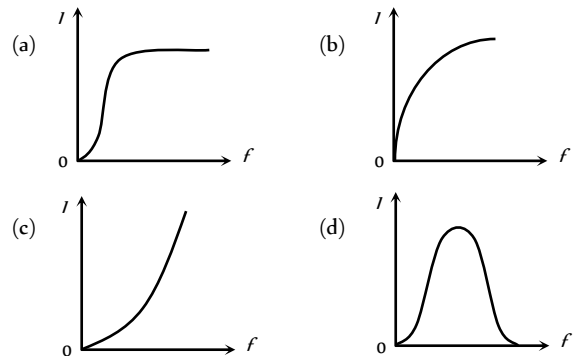
- (a) The voltage lags behind the current by $\pi/2$
(b) The voltage leads the current by $\pi/2$
(c) The voltage and the current are in phase
(d) The voltage leads the current by π
3. The figure shows variation of R , X_C and X_L with frequency f in a series L , C , R circuit. Then for what frequency point, the circuit is inductive



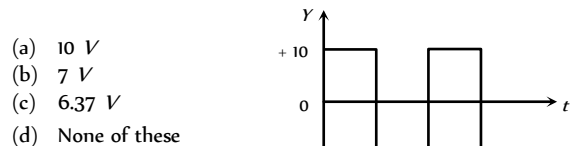
- (a) A
(b) B
(c) C
(d) All points
4. An alternating emf is applied across a parallel combination of a resistance R , capacitance C and an inductance L . If I , I_L , I_C are the currents through R , L and C respectively, then the diagram which correctly represents the phase relationship among I , I_L , I_C and source emf E is given by



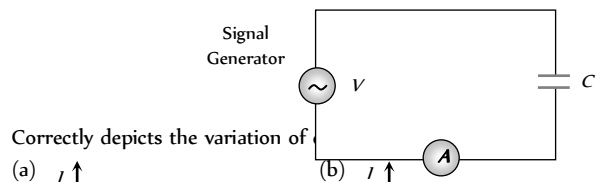
5. An ac source of variable frequency f is connected to an LCR series circuit. Which one of the graphs in figure. represents the variation of current of current I in the circuit with frequency f



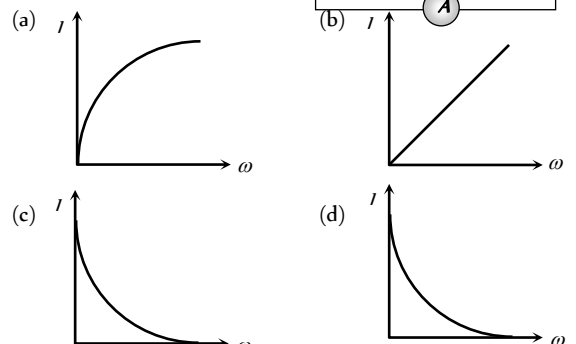
6. The r.m.s. voltage of the wave form shown is



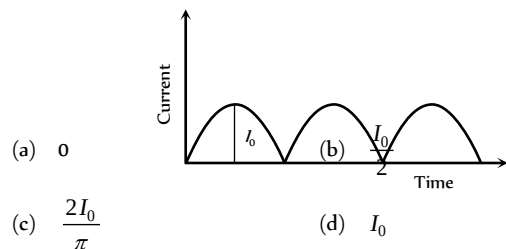
- (a) 10 V
(b) 7 V
(c) 6.37 V
(d) None of these
7. A constant voltage at different frequencies is applied across a capacitance C as shown in the figure. Which of the following graphs



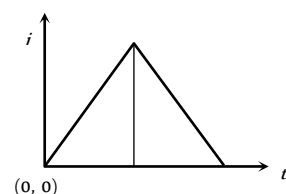
Correctly depicts the variation of



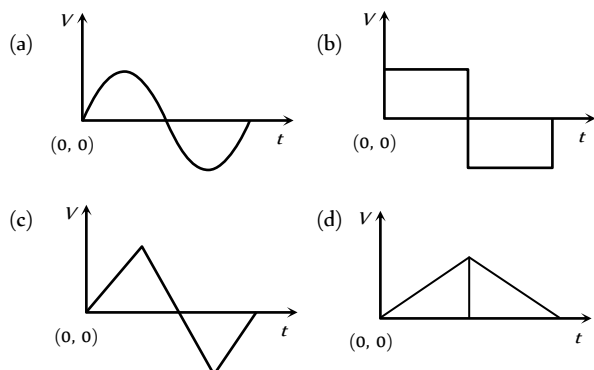
8. The output current versus time curve of a rectifier is shown in the figure. The average value of output current in this case is



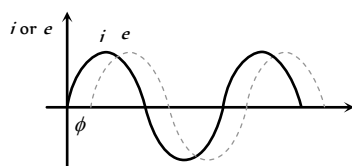
- (a) 0
(b) $\frac{2I_0}{\pi}$
(c) $\frac{I_0}{2}$
(d) I_0
9. The current ' i ' in an inductance coil varies with time ' t ' according to following graph



Which one of the following plots shows the variations of voltage in the coil
[CBSE PMT 1994]

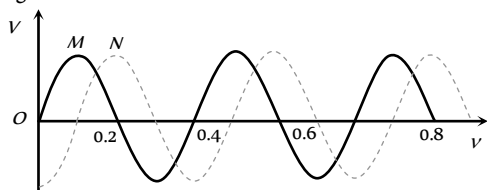


10. When an ac source of e.m.f. $e = E_0 \sin(100t)$ is connected across a circuit, the phase difference between the e.m.f. e and the current i in the circuit is observed to be $\pi/4$, as shown in the diagram. If the circuit consists possibly only of RC or LC in series, find the relationship between the two elements



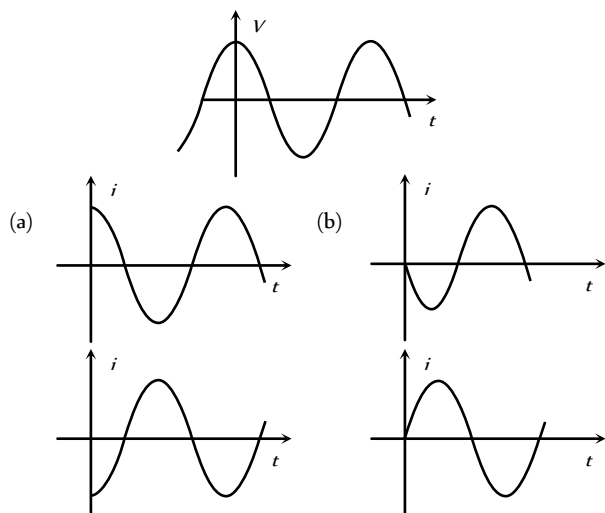
- (a) $R = 1k\Omega, C = 10\mu F$ (b) $R = 1k\Omega, C = 1\mu F$
(c) $R = 1k\Omega, L = 10H$ (d) $R = 1k\Omega, L = 1H$

11. Two sinusoidal voltages of the same frequency are shown in the diagram. What is the frequency, and the phase relationship between the voltages



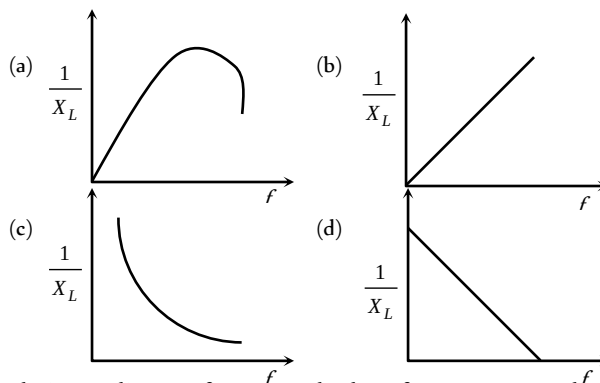
- Frequency in Hz Phase lead of v over V in radians
(a) 0.4 $-\pi/4$
(b) 2.5 $-\pi/2$
(c) 2.5 $+\pi/2$
(d) 2.5 $-\pi/4$

12. The voltage across a pure inductor is represented by the following diagram. Which one of the following diagrams will represent the current
[MP PMT 1995]

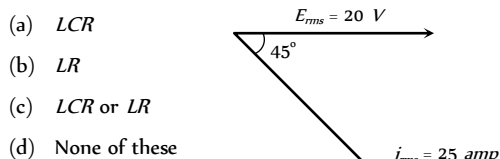


- (c) (d)

13. In pure inductive circuit, the curves between frequency f and reciprocal of inductive reactance $1/X_L$ is

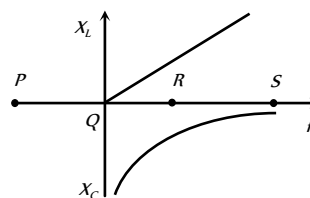


14. [IIT-JEE (Screening) 2003] The graph of current and voltage for a circuit is as shown. The components of the circuit will be



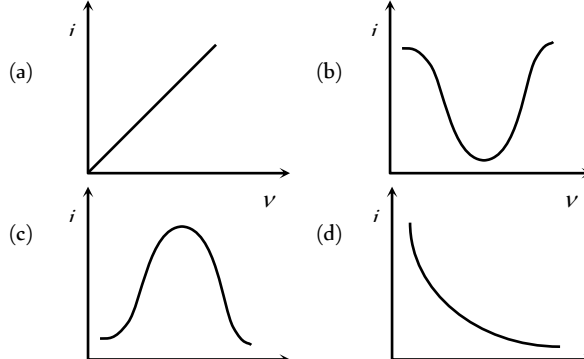
- (a) LCR (b) LR (c) LCR or LR (d) None of these

15. The resonance point in $X_L - f$ and $X_C - f$ curves is

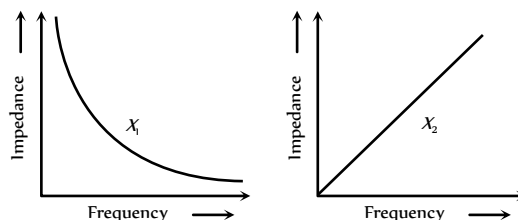


- (a) P (b) Q
(c) R (d) S

16. The $i - v$ curve for anti-resonant circuit is

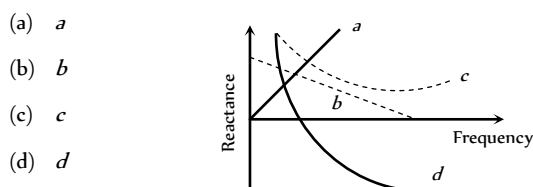


17. The graphs given below depict the dependence of two reactive impedances X and X' on the frequency of the alternating e.m.f. applied individually to them. We can then say that [Haryana CEE 1996; RPMT 2000]

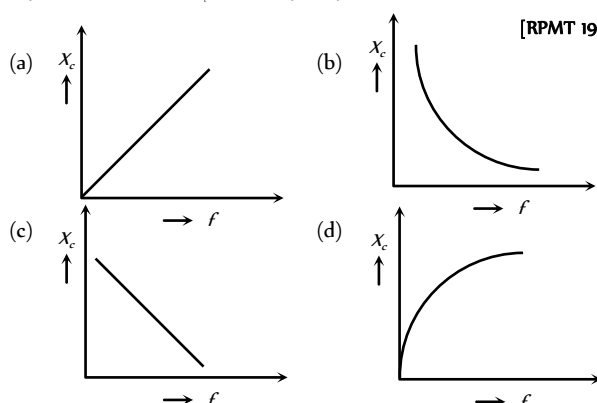


- (a) X_L is an inductor and X_C is a capacitor
 (b) X_L is a resistor and X_C is a capacitor
 (c) X_L is a capacitor and X_C is an inductor
 (d) X_L is an inductor and X_C is a resistor

18. Which of the following plots may represent the reactance of a series LC combination [MP PMT 1999]



19. Which of the following curves correctly represents the variation of capacitive reactance X_C with frequency f [RPMT 1996]



Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

1. Assertion : In series LCR circuit resonance can take place.
 Reason : Resonance takes place if inductance and capacitive reactances are equal and opposite.

[AIIMS 1998]

2. Assertion : The alternating current lags behind the e.m.f. by a phase angle of $\pi/2$, when ac flows through an inductor.

Reason : The inductive reactance increases as the frequency of ac source decreases.

3. Assertion : Capacitor serves as a block for dc and offers an easy path to ac.

Reason : Capacitive reactance is inversely proportional to frequency.

4. Assertion : When capacitive reactance is smaller than the inductive reactance in LCR circuit, e.m.f. leads the current.

Reason : The phase angle is the angle between the alternating e.m.f. and alternating current of the circuit.

5. Assertion : Chock coil is preferred over a resistor to adjust current in an ac circuit.

Reason : Power factor for inductance is zero.

6. Assertion : If the frequency of alternating current in an ac circuit consisting of an inductance coil is increased then current gets decreased.

Reason : The current is inversely proportional to frequency of alternating current.

7. Assertion : A bulb connected in series with a solenoid is connected to ac source. If a soft iron core is introduced in the solenoid, the bulb will glow brighter.

Reason : On introducing soft iron core in the solenoid, the inductance increases.

8. Assertion : An alternating current does not show any magnetic effect.

Reason : Alternating current varies with time.

9. Assertion : The dc and ac both can be measured by a hot wire instrument.

Reason : The hot wire instrument is based on the principle of magnetic effect of current.

10. Assertion : ac is more dangerous than dc

Reason : Frequency of ac is dangerous for human body.

11. Assertion : Average value of ac over a complete cycle is always zero.

Reason : Average value of ac is always defined over half cycle.

12. Assertion : The divisions are equally marked on the scale of ac ammeter.

Reason : Heat produced is directly proportional to the current.

13. Assertion : When ac circuit contains resistor only, its power is minimum.

Reason : Power of a circuit is independent of phase angle.

14. Assertion : An electric lamp connected in series with a variable capacitor and ac source, its brightness increases with increase in capacitance.

Reason : Capacitive reactance decrease with increase in capacitance of capacitor.

15. Assertion : An inductance and a resistance are connected in series with an ac circuit. In this circuit the current and the potential difference across the resistance lag behind potential difference across the inductance by an angle $\pi/2$.

Reason : In LR circuit voltage leads the current by phase angle which depends on the value of inductance and resistance both.

16. Assertion : A capacitor of suitable capacitance can be used in an ac circuit in place of the choke coil.

Reason : A capacitor blocks dc and allows ac only.

Answers

Alternating Current, Voltage and Power

1	b	2	a	3	c	4	b	5	b
6	b	7	c	8	d	9	c	10	c
11	d	12	c	13	b	14	c	15	b
16	d	17	c	18	b	19	d	20	c
21	c	22	a	23	c	24	d	25	c
26	c	27	d	28	b	29	c	30	d
31	d	32	d	33	b	34	b	35	d
36	c	37	a	38	b	39	a	40	c
41	a	42	d	43	b	44	b	45	c
46	c								

ac Circuits

1	b	2	a	3	a	4	b	5	a
6	a	7	b	8	c	9	d	10	b
11	c	12	b	13	b	14	b	15	d
16	b	17	a	18	b	19	a	20	a
21	b	22	d	23	b	24	b	25	a
26	d	27	c	28	b	29	c	30	c
31	c	32	b	33	d	34	a	35	c
36	c	37	c	38	b	39	c	40	d
41	b	42	b	43	a	44	a	45	d
46	c	47	a	48	a	49	c	50	b
51	b	52	c	53	b	54	c	55	c
56	a	57	a	58	b	59	b	60	d
61	a	62	c	63	a	64	bd	65	a
66	a	67	a	68	a	69	a	70	b
71	a	72	d	73	bc	74	c	75	b
76	b	77	b	78	c	79	c	80	a
81	b	82	d	83	c	84	c	85	b
86	a	87	c	88	a	89	a	90	c
91	d	92	d	93	d	94	c	95	b
96	a	97	d	98	b				

Critical Thinking Questions

1	a	2	c	3	a	4	c	5	c
6	a	7	d	8	b	9	b	10	d
11	d	12	d	13	a	14	c	15	b
16	c	17	a	18	d	19	b	20	b
21	d	22	c	23	c	24	a	25	d

Graphical Questions

1	c	2	b	3	c	4	c	5	d
6	a	7	b	8	c	9	b	10	a
11	b	12	d	13	c	14	c	15	c
16	b	17	c	18	d	19	b		

Assertion and Reason

1	a	2	c	3	a	4	b	5	a
6	a	7	e	8	b	9	c	10	a
11	b	12	d	13	d	14	a	15	b
16	b								

AS Answers and Solutions

Alternating Current, Voltage and Power

- (b) Power loss $\propto \frac{1}{(\text{Voltage})^2}$
- (a) $V = 5 \cos \omega t = 5 \sin \left(\omega t + \frac{\pi}{2} \right)$ and $i = 2 \sin \omega t$
Power = $V_{r.m.s.} \times i_{r.m.s.} \times \cos \phi = 0$
(Since $\phi = \frac{\pi}{2}$, therefore $\cos \phi = \cos \frac{\pi}{2} = 0$)
- (c) $P = V_{r.m.s.} \times i_{r.m.s.} \times \cos \phi = \frac{100}{\sqrt{2}} \times \frac{100 \times 10^{-3}}{\sqrt{2}} \times \cos \frac{\pi}{3}$
 $= \frac{10^4 \times 10^{-3}}{2} \times \frac{1}{2} = \frac{10}{4} = 2.5 \text{ watt}$
- (b) In dc ammeter, a coil is free to rotate in the magnetic field of a fixed magnet.
If an alternating current is passed through such a coil, the torque will reverse it's direction each time the current changes direction and the average value of the torque will be zero.
- (b) The coil having inductance L besides the resistance R . Hence for ac it's effective resistance $\sqrt{R^2 + X_L^2}$ will be larger than it's resistance R for dc.
- (b) $i_{r.m.s.} = \frac{i_0}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2} \text{ ampere}$
- (c) Effective voltage $V_{r.m.s.} = \frac{V_0}{\sqrt{2}} = \frac{423}{\sqrt{2}} = 300 \text{ V}$

8. (d) The current takes $\frac{T}{4}$ sec to reach the peak value.
- In the given question $\frac{2\pi}{T} = 200\pi \Rightarrow T = \frac{1}{100}$ sec
- $\therefore$ Time to reach the peak value $= \frac{1}{400}$ sec
9. (c) $i_{r.m.s} = \frac{6}{\sqrt{2}} = 3\sqrt{2}$ A
10. (c) $\nu = \frac{\omega}{2\pi} = \frac{120 \times 7}{2 \times 22} = 19$ Hz
- $V_{r.m.s.} = \frac{240}{\sqrt{2}} = 120\sqrt{2} \approx 170$ V
11. (d)
12. (c) Peak value $= 220\sqrt{2} = 311$ V
13. (b) Power $= I^2 R = \left(\frac{I_p}{\sqrt{2}}\right)^2 R = \frac{I_p^2 R}{2}$
14. (c) $i_{r.m.s.} = \frac{V_{r.m.s.}}{R} = \frac{200}{40} = 5$ A $\Rightarrow i_0 = i_{r.m.s.} \sqrt{2} = 7.07$ A
15. (b)
16. (d) Time taken by the current to reach the maximum value
- $t = \frac{T}{4} = \frac{1}{4\nu} = \frac{1}{4 \times 50} = 5 \times 10^{-3}$ sec
- and $i_0 = i_{rms} \sqrt{2} = 10\sqrt{2} = 14.14$ amp
17. (c)
18. (b) $E = E_0 \cos \omega t = E_0 \cos \frac{2\pi}{T}$
- $= 10 \cos \frac{2\pi \times 50 \times 1}{600} = 10 \cos \frac{\pi}{6} = 5\sqrt{3}$ volt.
19. (d) Phase angle $\phi = 90^\circ$, so power $P = V \cos \phi = 0$
20. (c) $V_{rms} = \frac{200}{\sqrt{2}}$, $i_{rms} = \frac{1}{\sqrt{2}}$
- $\therefore P = V_{rms} i_{rms} \cos \phi = \frac{200}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \cos \frac{\pi}{3} = 50$ watt
21. (c) $2\pi\nu = 377 \Rightarrow \nu = 60.03$ Hz
22. (a)
23. (c) $i_{rms} = \sqrt{\frac{i_1^2 + i_2^2}{2}} = \frac{1}{\sqrt{2}}(i_1^2 + i_2^2)^{1/2}$
24. (d) $P = V \cos \phi$
- Phase difference $\phi = \frac{\pi}{2} \Rightarrow P = \text{zero}$
25. (c) $V_0 = V_{rms} \times \sqrt{2} = 220 \times \sqrt{2} = 310$
26. (c) Hot wire ammeter reads rms value of current. Hence its peak value $= i_{rms} \times \sqrt{2} = 14.14$ amp
27. (d)
28. (b) Peak voltage $= \sqrt{2} \times 220 = 311$ V
29. (c)
30. (d) $\because P = V \cos \phi$, $\therefore P \propto \cos \phi$
31. (d) $P = V_{rms} I_{rms} \cos \phi$; since $\phi = 90^\circ$. So $P = 0$
32. (d) Brightness $\propto P_{consumed} \propto \frac{1}{R}$ for Bulb, $R_{ac} = R_{dc}$, so brightness will be equal in both the cases.
33. (b) $P = \frac{V_{rms}^2}{R} = \frac{(30)^2}{10} = 90$ W
34. (b) $V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{120}{1.414} = 84.8$ V
35. (d) Peak value to rms value means, current becomes $\frac{1}{\sqrt{2}}$ times.
- So from $i = i_0 \sin 100\pi t \Rightarrow \frac{1}{\sqrt{2}} \times i_0 = i_0 \sin 100\pi t$
- $\Rightarrow \sin \frac{\pi}{4} = \sin 100\pi t \Rightarrow t = \frac{1}{400}$ sec $= 2.5 \times 10^{-3}$ sec.
36. (c) Phase difference $\Delta\phi = \phi_2 - \phi_1 = \frac{\pi}{6} - \left(\frac{-\pi}{6}\right) = \frac{\pi}{3}$
37. (a) $V_{av} = \frac{2}{\pi} V_0 = \frac{2}{\pi} \times (V_{rms} \times \sqrt{2}) = \frac{2\sqrt{2}}{\pi} \cdot V_{rms}$
- $= \frac{2\sqrt{2}}{\pi} \times 220 = 198$ V
38. (b)
39. (a)
40. (c)
41. (a) $i_{rms} = \frac{200}{280} = \frac{5}{7}$ A. So $i_0 = i_{rms} \times \sqrt{2} = \frac{5}{7} \times \sqrt{2} \approx 1$ A.
42. (d) Required time $t = T/4 = \frac{1}{4 \times 50} = 5 \times 10^{-3}$ sec
43. (b) $V_0 = \sqrt{2} V_{rms} = 10\sqrt{2}$
44. (b) $P = \frac{1}{2} V_0 i_0 \cos \phi \Rightarrow P = P_{peak} \cdot \cos \phi$
- $\Rightarrow \frac{1}{2}(P_{peak}) = P_{peak} \cos \phi \Rightarrow \cos \phi = \frac{1}{2} \Rightarrow \phi = \frac{\pi}{3}$
45. (c) $E = 141 \sin(628t)$,
- $E_{rms} = \frac{E_0}{\sqrt{2}} = \frac{141}{1.41} = 100$ V and $2\pi f = 628$
- $\Rightarrow f = 100$ Hz
46. (c) $E_{rms} = \frac{E_0}{\sqrt{2}} = \frac{707}{1.41} = 500$ V

ac Circuits

1. (b)
2. (a)
3. (a) The choke coil can be used only in ac circuits, not in dc circuits, because for dc ($\omega = 0$) the inductive reactance $X_L = \omega L$ of the coil is zero, only the resistance of the coil remains effective which too is almost zero.

4. (b) Because power $= i^2 R$, if $R = 0$, then $P = 0$.
5. (a)
6. (a) A choke coil contains high inductance but negligible resistance, due to which power loss becomes appreciably small.
7. (b) For purely capacitive circuit $e = e_0 \sin \omega t$
 $i = i_0 \sin \left(\omega t + \frac{\pi}{2} \right)$ i.e. current is ahead of emf by $\frac{\pi}{2}$
8. (c)
9. (d)
10. (b) $Z = \sqrt{R^2 + X_L^2}$, $X_L = \omega L$ and $\omega = 2\pi f$
 $\therefore Z = \sqrt{R^2 + 4\pi^2 f^2 L^2}$
11. (c) $\nu = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10^{-6} \times 10^{-4}}} = \frac{10^5}{2\pi} \text{ Hz}$
12. (b)
13. (b) The applied voltage is given by $V = \sqrt{V_R^2 + V_L^2}$
 $V = \sqrt{(200)^2 + (150)^2} = 250 \text{ volt}$
14. (b) $i = \frac{V}{Z} = \frac{120}{\sqrt{R^2 + \omega^2 L^2}} = \frac{120}{\sqrt{100 + 4\pi^2 \times 60^2 \times 20^2}} = 0.016 \text{ A}$
15. (d) For the first circuit $i = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + \omega^2 L^2}}$
 $\therefore$ Increase in ω will cause a decrease in i .
 For the second circuit $i = \frac{V}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}}$
 $\therefore$ Increase in ω will cause an increase in i .
16. (b) $X_C = \frac{1}{\omega C} = \frac{1}{2\pi\nu C}$; For dc $\nu = 0$, $\therefore X_C = \infty$
17. (a) In a pure inductor (zero resistance), voltage leads the current by 90° i.e. $\pi/2$.
18. (b)
19. (a) The voltage across a L - R combination is given by
 $V^2 = V_R^2 + V_L^2$
 $V_L = \sqrt{V^2 - V_R^2} = \sqrt{400 - 144} = \sqrt{256} = 16 \text{ volt}$
20. (a) Phase angle $\tan \phi = \frac{\omega L}{R} = \frac{2\pi \times 200}{300} \times \frac{1}{\pi} = \frac{4}{3}$
 $\therefore \phi = \tan^{-1} \frac{4}{3}$
21. (b) At resonance, LCR circuit behaves as purely resistive circuit, for purely resistive circuit power factor $= 1$
22. (d) Given $\omega L = \frac{1}{\omega C} \Rightarrow \omega^2 = \frac{1}{LC}$
 or $\omega = \frac{1}{\sqrt{10^{-3} \times 10 \times 10^{-6}}} = \frac{1}{\sqrt{10^{-8}}} = 10^4$
 $X_L = \omega L = 10^4 \times 10^{-3} = 10 \Omega$
23. (b)
24. (b) Reading of ammeter $= i_{\text{rms}} = \frac{V_{\text{rms}}}{X_C} = \frac{V_0 \omega C}{\sqrt{2}}$
- $= \frac{200\sqrt{2} \times 100 \times (1 \times 10^{-6})}{\sqrt{2}} = 2 \times 10^{-2} \text{ A} = 20 \text{ mA}$
25. (a) Current will be maximum at the condition of resonance. So resonant frequency $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.5 \times 8 \times 10^{-6}}}$
 $= 500 \text{ rad/s}$
26. (d) Average power in ac circuits is given by $P = V_{\text{rms}} i_{\text{rms}} \cos \phi$
 For pure capacitive circuit $\phi = 90^\circ$ so $P = 0$
27. (c) Amplitude of ac $= i_0 = \frac{V_0}{R} = \frac{\omega NBA}{R} = \frac{(2\pi\nu)NB(\pi r^2)}{R}$
 $\Rightarrow i_0 = \frac{2\pi \times \frac{200}{60} \times 1 \times 10^{-2} \times \pi \times (0.3)^2}{\pi^2} = 6 \text{ mA}$
28. (b) $Z = \sqrt{R^2 + X_L^2} = \sqrt{10^2 + (2\pi \times 60 \times 2)^2} = 753.7$
 $\therefore i = \frac{120}{753.7} = 0.159 \text{ A}$
29. (c) Resonance frequency in radian/second is
 $\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{8 \times 0.5 \times 10^{-6}}} = 500 \text{ rad/sec}$
30. (c) $\omega = \frac{1}{\sqrt{L_1 C_1}} = \frac{1}{\sqrt{L_2 C_2}} \Rightarrow L_2 = \frac{L_1}{4}$
31. (c) $Z = X_L = 2\pi \times 60 \times 0.7$
 $\therefore i = \frac{120}{Z} = \frac{120}{2\pi \times 60 \times 0.7} = 0.455 \text{ ampere}$
32. (b) $Z = \sqrt{R^2 + X^2} = \sqrt{4^2 + 3^2} = 5$
 $\therefore \cos \phi = \frac{R}{Z} = \frac{3}{5} = 0.6$
33. (d)
34. (a) $\cos \phi = \frac{R}{Z}$. In choke coil $\phi \approx 90^\circ$ so $\cos \phi \approx 0$
35. (c) $Z = \sqrt{R^2 + (X_L - X_C)^2}$
 $= \sqrt{100^2 + \left(0.5 \times 100\pi - \frac{1}{10 \times 10^{-6} \times 100\pi} \right)^2} = 189.72 \Omega$
36. (c) $V_L = 46 \text{ volts}$, $V_C = 40 \text{ volts}$, $V_R = 8 \text{ volts}$
 E.M.F. of source $V = \sqrt{8^2 + (46 - 40)^2} = 10 \text{ volts}$
37. (c) Resonant frequency $= \frac{1}{2\pi\sqrt{LC}}$ does not depend on resistance.
38. (b) Frequency $= \frac{1}{2\pi\sqrt{LC}}$
 So the combination which represents dimension of frequency is $\frac{1}{\sqrt{LC}} = (LC)^{-1/2}$
39. (c) For series R - L - C circuit, $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$$= \sqrt{(300)^2 + \left(1000 \times 0.9 - \frac{10^6}{1000 \times 2}\right)^2} = 500 \Omega$$

40. (d) $Z = \sqrt{R^2 + X^2} = \sqrt{R^2 + (2\pi fL)^2}$

$$= \sqrt{(30)^2 + \left(2\pi \times 50 \times \frac{0.4}{\pi}\right)^2} = \sqrt{900 + 1600} = 50 \Omega$$

$$i = \frac{V}{Z} = \frac{200}{50} = 4 \text{ ampere}$$

41. (b) Reactance $= 2\pi fL \Rightarrow 100 \Omega = 2 \times \frac{22}{7} \times 50 \times L$

$$\therefore L = 0.32 \text{ henry}$$

42. (b)

43. (a) $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 4000 \times 25 \times 10^{-6}} = \frac{5}{\pi} \Omega$

44. (a) $X_C = \frac{1}{2\pi fC} \Rightarrow \frac{1}{1000} = \frac{1}{2\pi \times \nu \times 5 \times 10^{-6}}$

$$\Rightarrow \nu = \frac{100}{\pi} \text{ MHz}$$

45. (d) $i = \frac{V}{Z} = \frac{4}{\sqrt{4^2 + (1000 \times 3 \times 10^{-3})^2}} = 0.8 \text{ A}$

46. (c) $\tan \phi = \frac{X_L}{R} \Rightarrow \tan 45^\circ = \frac{X_L}{R} = 1 \Rightarrow X_L = R$

47. (a) For purely L-circuit $P = 0$

48. (a) At resonance LCR series circuit behaves as pure resistive circuit. For resistive circuit $\phi = 0^\circ$

49. (c) $V = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{(5)^2 + (10 - 10)^2} = 5 \text{ Volt}$

50. (b) When dc is supplied $R = \frac{V}{i} = \frac{100}{1} = 100 \Omega$

When ac is supplied $Z = \frac{V}{i} = \frac{100}{0.5} = 200 \Omega$

51. (b) $\cos \phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + \omega^2 L^2}}$

$$= \frac{12}{\sqrt{(12)^2 + 4 \times \pi^2 \times (60)^2 \times (0.1)^2}} \Rightarrow \cos \phi = 0.30$$

52. (c)

53. (b)

54. (c) Impedance $Z = \sqrt{R^2 + X^2} = \sqrt{(8)^2 + (6)^2} = 10 \Omega$

55. (c) $\tan \phi = \frac{\omega L}{R} = \frac{2\pi \times 50 \times 0.21}{12} = 5.5 \Rightarrow \phi = 80^\circ$

56. (a) If the current is wattless then power is zero. Hence phase difference $\phi = 90^\circ$

57. (a) $f = \frac{1}{2\pi\sqrt{LC}} \Rightarrow f \propto \frac{1}{\sqrt{C}}$

58. (b) In non resonant circuits

impedance $Z = \frac{1}{\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}}$, with rise in

frequency Z decreases i.e. current increases so circuit behaves as capacitive circuit.

59. (b) $V = \sqrt{V_R^2 + V_L^2} = \sqrt{(20)^2 + (16)^2} = \sqrt{656} = 25.6 \text{ V}$

60. (d) $i = \frac{220}{\sqrt{(20)^2 + (2 \times \pi \times 50 \times 0.2)^2}} = \frac{220}{66} = 3.33 \text{ A}$

61. (a) Impedance of LCR circuit will be minimum at resonant frequency so $\nu_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{1 \times 10^{-3} \times 0.1 \times 10^{-6}}}$

$$= \frac{10^5}{2\pi} \text{ Hz}$$

62. (c) $\cos \phi = \frac{R}{Z} = \frac{10}{20} = \frac{1}{2} \Rightarrow \phi = 60^\circ$

63. (a) Current in LC circuit becomes maximum when resonance occurs. So

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 25 \times 10^{-6}}} = \frac{1000}{5} = 200 \text{ rad/sec}$$

64. (b, d)

65. (a) $R = 6 + 4 = 10 \Omega$

$$X_L = \omega L = 2000 \times 5 \times 10^{-3} = 10 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2000 \times 50 \times 10^{-6}} = 10 \Omega$$

$$\therefore Z = \sqrt{R^2 + (X_L - X_C)^2} = 10 \Omega$$

$$\text{Amplitude of current} = i_0 = \frac{V_0}{Z} = \frac{20}{10} = 2 \text{ A}$$

66. (a) $i = \frac{V}{X_L} = \frac{200}{\omega L} = \frac{200}{2\pi \times 50 \times 1} = 0.637 \text{ A}$

67. (a)

68. (a) In LCR circuit; in the condition of resonance $X_L = X_C$ i.e. circuit behaves as resistive circuit. In resistive circuit power factor is maximum.

69. (a) $\tan \phi = \frac{X_L}{R} = \frac{\sqrt{3} R}{R} = \sqrt{3} \Rightarrow \phi = 60^\circ = \pi/3$

70. (b) $X_C = \frac{1}{2\pi fC} = \frac{1}{0} = \infty$

71. (a)

72. (d) $X_L = 2\pi fL \Rightarrow L = \frac{X_L}{2\pi f} = \frac{50}{2 \times 3.14 \times 50} = 0.16 \text{ H}$

73. (b, c)

74. (c) $Z = \sqrt{R^2 + (2\pi fL)^2}$

$$= \sqrt{(40)^2 + 4\pi^2 \times (50)^2 \times (95.5 \times 10^{-3})^2} = 50 \text{ ohm}$$

75. (b) $X_C = \frac{1}{2\pi fC} \Rightarrow X_C \propto \frac{1}{f}$

76. (b)

77. (b)

78. (c) $X_L = 2\pi\nu L = 2 \times \pi \times 50 \times \frac{1}{\pi} = 100 \Omega$

79. (c)
$$\nu_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2 \times 3.14 \sqrt{5 \times 10^{-4} \times 20 \times 10^{-6}}}$$
$$\nu_0 = \frac{10^4}{6.28} = 1592 \text{ Hz}$$

80. (a) $X_C = \frac{1}{2\pi\nu C} \Rightarrow C = \frac{1}{2\pi\nu X_C} = \frac{1}{2 \times \pi \times \frac{400}{\pi} \times 25} = 50 \mu F$

81. (b) $\cos \phi = \frac{R}{Z} = \frac{R}{(R^2 + \omega^2 L^2)^{1/2}}$

82. (d) $Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{(11)^2 + (25 - 18)^2} = 13 \Omega$
Current $i = \frac{260}{13} = 20 \text{ A}$

83. (c) $i = \frac{V}{X_L} = \frac{120}{2 \times 3.14 \times 60 \times 0.7} = 0.455 \text{ A}$

84. (c) $\cos \phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + \omega^2 L^2}} = \frac{5}{\sqrt{25 + (50)^2 \times (0.1)^2}}$
 $= \frac{5}{\sqrt{25 + 25}} = \frac{1}{\sqrt{2}} \Rightarrow \phi = \pi / 4$

85. (b) $P = V \cos \phi = V \left(\frac{V}{Z} \right) \left(\frac{R}{Z} \right) = \frac{V^2 R}{Z^2} = \frac{V^2 R}{(R^2 + \omega^2 L^2)}$

86. (a)

87. (c) $\tan \phi = \frac{X_L}{R} = \frac{2\pi\nu L}{R} = \frac{2\pi \times \frac{200}{2\pi} \times 1}{200} = 1 \Rightarrow \phi = 45^\circ$

88. (a) $V = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{(40)^2 + (60 - 30)^2} = 50 \text{ V}$

89. (a)

90. (c) $\nu_0 = \frac{1}{2\pi\sqrt{LC}}$

If C changes to $2C$ then for keeping ν_0 constant L must change to $L/2$.

91. (d) Net voltage across LC combination $= V_L - V_C = 0 \text{ V}$

92. (d) Impedance $Z = \sqrt{R^2 + 4\pi^2\nu^2 L^2}$
 $= \sqrt{(12)^2 + 4 \times (3.14)^2 \times (50)^2 \times (0.04)^2} = 17.37 \text{ A}$
Now current $i = \frac{V}{Z} = \frac{220}{17.37} = 12.7 \Omega$

93. (d) At resonant frequency current in series LCR circuit is maximum.

94. (c)

95. (b)

96. (a) $\tan \phi = \frac{X_C - X_L}{R} \Rightarrow \tan 45^\circ = \frac{\frac{1}{2\pi f C} - 2\pi f L}{R}$
 $\Rightarrow C = \frac{1}{2\pi f(2\pi f L + R)}$

97. (d)

98. (b) Resonance frequency

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{8 \times 10^{-3} \times 20 \times 10^{-6}}} = 2500 \text{ rad/sec}$$

$$\text{Resonance current} = \frac{V}{R} = \frac{220}{44} = 5 \text{ A}$$

Critical Thinking Questions

1. (a) For dc, $R = \frac{V}{i} = \frac{100}{1} = 100 \Omega$

For ac, $Z = \frac{V}{i} = \frac{100}{0.5} = 200 \Omega$

$$\therefore Z = \sqrt{R^2 + (\omega L)^2} \Rightarrow 200 = \sqrt{(100)^2 + 4\pi^2(50)^2 L^2}$$

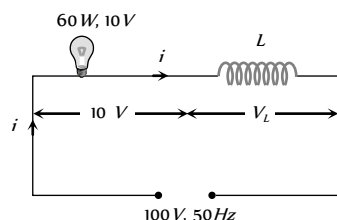
$$\therefore L = 0.55 H$$

2. (c) $P = E_{rms} i_{rms} \cos \phi = \frac{E_0}{\sqrt{2}} \times \frac{i_0}{\sqrt{2}} \times \frac{R}{Z}$

$$\Rightarrow \frac{E_0}{\sqrt{2}} \times \frac{E_0}{Z\sqrt{2}} \times \frac{R}{Z} \Rightarrow P = \frac{E_0^2 R}{2Z^2}$$

Given $X_L = R$ so, $Z = \sqrt{2}R \Rightarrow P = \frac{E_0^2}{4R}$

3. (a) Current through the bulb $i = \frac{P}{V} = \frac{60}{10} = 6 A$



$$V = \sqrt{V_R^2 + V_L^2}$$

$$(100)^2 = (10)^2 + V_L^2 \Rightarrow V_L = 99.5 \text{ Volt}$$

Also $V_L = iX_L = i \times (2\pi\nu L)$

$$\Rightarrow 99.5 = 6 \times 2 \times 3.14 \times 50 \times L \Rightarrow L = 0.052 H$$

4. (c) $V^2 = V_R^2 + (V_L - V_C)^2$

Since $V_L = V_C$ hence $V = V_R = 200 V$

5. (c) $V^2 = V_R^2 + (V_L - V_C)^2 \Rightarrow V_R = V = 220 V$

Also $i = \frac{220}{100} = 2.2 A$

6. (a) When a bulb and a capacitor are connected in series to an ac source, then on increasing the frequency the current in the circuit is increased, because the impedance of the circuit is decreased. So the bulb will give more intense light.

7. (d) The instantaneous values of emf and current in inductive circuit are given by $E = E_0 \sin \omega t$ and

$$i = i_0 \sin\left(\omega t - \frac{\pi}{2}\right) \text{ respectively.}$$

So, $P_{inst} = Ei = E_0 \sin \omega t \times i_0 \sin\left(\omega t - \frac{\pi}{2}\right)$

$$= E_0 i_0 \sin \omega t \left(\sin \omega t \cos \frac{\pi}{2} - \cos \omega t \sin \frac{\pi}{2} \right)$$

$$= E_0 i_0 \sin \omega t \cos \omega t$$

$$= \frac{1}{2} E_0 i_0 \sin 2\omega t \quad (\sin 2\omega t = 2 \sin \omega t \cos \omega t)$$

Hence, angular frequency of instantaneous power is 2ω .

8. (b) $V = 50 \times 2 \sin 100\pi t \cos 100\pi t = 50 \sin 200\pi t$

$$\Rightarrow V_0 = 50 \text{ Volts and } \nu = 100 \text{ Hz}$$

9. (b) In RC series circuit voltage across the capacitor leads the voltage across the resistance by $\frac{\pi}{2}$

10. (d) The voltage V_L and V_C are equal and opposite so voltmeter reading will be zero.

Also $R = 30 \Omega, X_L = X_C = 25 \Omega$

$$\text{So } i = \frac{V}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{V}{R} = \frac{240}{30} = 8 A$$

11. (d) $V = 120 \sin 100\pi t \cos 100\pi t \Rightarrow V = 60 \sin 200\pi t$

$$V_{\max} = 60 V \text{ and } \nu = 100 \text{ Hz}$$

12. (d) $Z = \sqrt{R^2 + (X_L - X_C)^2};$

$$R = 10 \Omega, X_L = \omega L = 2000 \times 5 \times 10^{-3} = 10 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2000 \times 50 \times 10^{-6}} = 10 \Omega \text{ i.e. } Z = 10 \Omega$$

$$\text{Maximum current } i_0 = \frac{V_0}{Z} = \frac{20}{10} = 2 A$$

$$\text{Hence } i_{rms} = \frac{2}{\sqrt{2}} = 1.4 A$$

$$\text{and } V_{rms} = 4 \times 1.41 = 5.64 V$$

13. (a) Capacitance of wire

$$C = 0.014 \times 10^{-6} \times 200 = 2.8 \times 10^{-6} F = 2.8 \mu F$$

For impedance of the circuit to be minimum $X_L = X_C \Rightarrow$

$$2\pi\nu L = \frac{1}{2\pi\nu C}$$

$$\Rightarrow L = \frac{1}{4\pi^2\nu^2 C} = \frac{1}{4(3.14)^2 \times (5 \times 10^3)^2 \times 2.8 \times 10^{-6}} \\ = 0.35 \times 10^{-3} H = 0.35 mH$$

14. (c) $i^2 = \frac{\int i^2 dt}{\int dt} = \frac{\int_2^4 (4t) dt}{\int_2^4 dt} = \frac{4 \int_2^4 t dt}{2} = 2 \left[\frac{t^2}{2} \right]_2^4 = [t^2]_2^4 = 12$

$$\Rightarrow i_{rms} = \sqrt{i^2} = \sqrt{12} = 2\sqrt{3} \text{ A}$$

15. (b) 1. $rms \text{ value} = \frac{x_0}{\sqrt{2}}$

2. $x_0 \sin \omega t \cos \omega t = \frac{x_0}{2} \sin 2\omega t \Rightarrow rms \text{ value} = \frac{x_0}{2\sqrt{2}}$

3. $x_0 \sin \omega t + x_0 \cos \omega t \Rightarrow rms \text{ value} = \sqrt{\left(\frac{x_0}{\sqrt{2}}\right)^2 + \left(\frac{x_0}{\sqrt{2}}\right)^2}$
 $= \sqrt{x_0^2} = x_0$

16. (c) Given $X_L = X_C = 5\Omega$, this is the condition of resonance. So $V_L = V_C$, so net voltage across L and C combination will be zero.

17. (a) At angular frequency ω , the current in RC circuit is given by

$$i_{rms} = \frac{V_{rms}}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} \quad \dots(i)$$

$$\text{Also } \frac{i_{rms}}{2} = \frac{V_{rms}}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} = \frac{V_{rms}}{\sqrt{R^2 + \frac{9}{\omega^2 C^2}}} \quad \dots(ii)$$

From equation (i) and (ii) we get

$$3R^2 = \frac{5}{\omega^2 C^2} \Rightarrow \frac{\omega C}{R} = \sqrt{\frac{3}{5}} \Rightarrow \frac{X_C}{R} = \sqrt{\frac{3}{5}}$$

18. (d) $\tan \phi = \frac{X_L}{R} = \frac{X_C}{R} \Rightarrow \tan 60^\circ = \frac{X_L}{R} = \frac{X_C}{R}$

$$\Rightarrow X_L = X_C = \sqrt{3} R$$

$$i.e. Z = \sqrt{R^2 + (X_L - X_C)^2} = R$$

$$\text{So average power } P = \frac{V^2}{R} = \frac{200 \times 200}{100} = 400 \text{ W}$$

19. (b) $R = \frac{P}{i_{rms}^2} = \frac{240}{16} = 15\Omega$

$$Z = \frac{V}{i} = \frac{100}{4} = 25\Omega$$

$$\text{Now } X_L = \sqrt{Z^2 - R^2} = \sqrt{(25)^2 - (15)^2} = 20\Omega$$

$$\therefore 2\pi \nu L = 20 \Rightarrow L = \frac{20}{2\pi \times 50} = \frac{1}{5\pi} \text{ Hz}$$

20. (b) $X_L = R, X_C = R/2$

$$\therefore \tan \phi = \frac{X_L - X_C}{R} = \frac{R - \frac{R}{2}}{R} = \frac{1}{2}$$

$$\Rightarrow \phi = \tan^{-1}(1/2)$$

$$\text{Also } Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \frac{R^2}{4}} = \frac{\sqrt{5}}{2} R$$

21. (d) At resonance net voltage across L and C is zero.

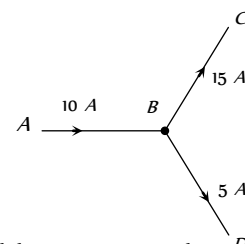
22. (c) $i_L = \frac{90}{30} = 3 \text{ A}, i_C = \frac{90}{20} = 4.5 \text{ A}$

$$\text{Net current through circuit } i = i_C - i_L = 1.5 \text{ A}$$

$$\therefore Z = \frac{V}{i} = \frac{90}{1.5} = 60\Omega$$

23. (c) $i_{rms} = \sqrt{\frac{1}{T} \int_0^T i^2 dt} = \frac{T^2}{\sqrt{5}}$

24. (a) Yes, in AC if branch AB has R , BC has a capacitor C , and BD has a pure inductance L



25. (d) Current will be maximum in the condition of resonance so

$$i_{\max} = \frac{V}{R} = \frac{V}{10} \text{ A}$$

$$\text{Energy stored in the coil } W_L = \frac{1}{2} Li_{\max}^2 = \frac{1}{2} L \left(\frac{E}{10}\right)^2$$

$$= \frac{1}{2} \times 10^{-3} \left(\frac{E^2}{100}\right) = \frac{1}{2} \times 10^{-5} E^2 \text{ joule}$$

$$\therefore \text{Energy stored in the capacitor}$$

$$W_C = \frac{1}{2} CE^2 = \frac{1}{2} \times 2 \times 10^{-6} E^2 = 10^{-6} E^2 \text{ joule}$$

$$\therefore \frac{W_C}{W_L} = \frac{1}{5}$$

Graphical Questions

1. (c) $Z = \sqrt{R^2 + \left(2\pi fL - \frac{1}{2\pi fC}\right)^2}$

$$\text{From above equation at } f = 0 \Rightarrow Z = \infty$$

$$\text{When } f = \frac{1}{2\pi\sqrt{LC}} \text{ (resonant frequency)} \Rightarrow Z = R$$

$$\text{For } f > \frac{1}{2\pi\sqrt{LC}} \Rightarrow Z \text{ starts increasing.}$$

$$i.e., \text{ for frequency } 0 - f, Z \text{ decreases}$$

$$\text{and for } f \text{ to } \infty, Z \text{ increases. This is justified by graph c.}$$

2. (b) At $t = 0$, phase of the voltage is zero, while phase of the current is $-\frac{\pi}{2}$ i.e., voltage leads by $\frac{\pi}{2}$

3. (c) At $A : X_C > X_L$

$$\text{At } B : X_C = X_L$$

$$\text{At } C : X_C < X_L$$

4. (c) I lags behind V by a phase of $\frac{\pi}{2}$, while V leads by a phase of $\frac{\pi}{2}$.

5. (d) As explained in solution (i) for frequency $0 \rightarrow f_r$, Z decreases hence $(i = V/Z)$, increases and for frequency $f_r \rightarrow \infty$, Z increases hence i decreases.

6. (a) $V_{rms} = \sqrt{\frac{1}{T} \int_0^T 10^2 dt} = 10 \text{ V}$

7. (b) For capacitive circuits $X_C = \frac{1}{\omega C}$

$$\therefore i = \frac{V}{X_C} = V\omega C \Rightarrow i \propto \omega$$

8. (c) $I_{av} = \frac{\int_0^{T/2} i dt}{\int_0^{T/2} dt} = \frac{\int_0^{T/2} I_0 \sin(\omega t) dt}{T/2}$

$$= \frac{2I_0}{T} \left[\frac{-\cos \omega t}{\omega} \right]_0^{T/2} = \frac{2I_0}{T} \left[-\frac{\cos\left(\frac{\omega T}{2}\right)}{\omega} + \frac{\cos 0^\circ}{\omega} \right]$$

$$= \frac{2I_0}{\omega T} [-\cos \pi + \cos 0^\circ] = \frac{2I_0}{2\pi} [1 + 1] = \frac{2I_0}{\pi}$$

9. (b) (i) For time interval $0 < t < T/2$

$$I = kt, \text{ where } k \text{ is the slope}$$

$$\text{For inductor as we know, induced voltage } V = -L \frac{di}{dt}$$

$$\Rightarrow V_1 = -KL$$

(2) For time interval $\frac{T}{2} < t < T$

$$I = -Kt \Rightarrow V_2 = KL$$

10. (a) As the current i leads the voltage by $\frac{\pi}{4}$, it is an RC circuit, hence

$$\tan \phi = \frac{X_C}{R} \Rightarrow \tan \frac{\pi}{4} = \frac{1}{\omega CR}$$

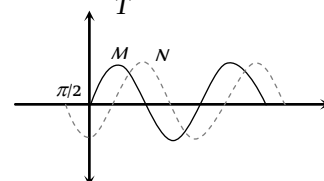
$$\Rightarrow \omega CR = 1 \text{ as } \omega = 100 \text{ rad/sec}$$

$$\Rightarrow CR = \frac{1}{100} \text{ sec}^{-1}.$$

From all the given options only option (a) is correct.

11. (b) From the graph shown below. It is clear that phase lead of N over M is $-\frac{\pi}{2}$. Since time period (i.e. taken to complete one cycle) = 0.4 sec .

$$\text{Hence frequency } \nu = \frac{1}{T} = 2.5 \text{ Hz}$$



12. (d) In purely inductive circuit voltage leads the current by 90° .

13. (c) $X_L = 2\pi fL \Rightarrow X_L \propto f \Rightarrow \frac{1}{X_L} \propto \frac{1}{f}$

i.e., graph between $\frac{1}{X_L}$ and f will be a hyperbola.

14. (c) From phasor diagram it is clear that current is lagging with respect to E_r . This may be happen in LCR or LR circuit.

15. (c) At resonance $X_L = X_C$

16. (b) For anti-resonant circuit current is minimum at resonant frequency and at frequencies other than resonant frequency current rises with frequency.

17. (c) We have $X_C = \frac{1}{C \times 2\pi f}$ and $X_L = L \times 2\pi f$

18. (d) Reactance $X = X_L - X_C = 2\pi fL - \frac{1}{2\pi fC}$

19. (b) $X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}$ i.e. $X_C \propto \frac{1}{f}$

Assertion and Reason

1. (a) At resonant frequency, $X_L = X_C \therefore Z = R$ (minimum) there for current in the circuit is maximum.
2. (c) When ac flows through an inductor current lags behind the emf , by phase of $\pi/2$, inductive reactance, $X_L = \omega L = \pi \cdot 2f \cdot L$, so when frequency increases correspondingly inductive reactance also increases.
3. (a) The capacitive reactance of capacitor is given by

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}$$

So this is infinite for dc ($f = 0$) and has a very small value for ac. Therefore a capacitor blocks dc.

4. (b) The phase angle for the LCR circuit is given by

$$\tan \phi = \frac{X_L - X_C}{R} = \frac{\omega L - 1/\omega C}{R}$$

Where X_L , X_C are inductive reactance and capacitive reactance respectively when $X_L > X_C$ then $\tan\phi$ is positive i.e. ϕ is positive (between 0 and $\pi/2$). Hence emf leads the current.

5. (a) If resistor is used in controlling ac supply, electrical energy will be wasted in the form of heat energy across the resistance wire. However, ac supply can be controlled with choke without any wastage of energy. This is because, power factor ($\cos\phi$) for resistance is unity and is zero for an inductance. [$P = EI \cos\phi$].
6. (a) When frequency of alternating current is increased, the effective resistance of the inductive coil increases. Current ($X_L = \omega L = 2\pi fL$) in the circuit containing inductor is given by $I = \frac{V}{X_L} = \frac{V}{2\pi fL}$. As inductive resistance of the inductor increases, current in the circuit decreases.
7. (e) On introducing soft iron core, the bulb will glow dimmer. This is because on introducing soft iron core in the solenoid, its inductance L increases, the inductive reactance, $X_L = \omega L$ increases and hence the current through the bulb decreases.
8. (b) Like direct current, an alternating current also produces magnetic field. But the magnitude and direction of the field goes on changing continuously with time.
9. (c) Both ac and dc produce heat, which is proportional to square of the current. The reversal of direction of current in ac is immaterial so far as production of heat is concerned.
10. (a) The effect of ac on the body depends largely on the frequency. Low frequency currents of 50 to 60 Hz (cycles/sec), which are commonly used, are usually more dangerous than high frequency currents and are 3 to 5 times more dangerous than dc of same voltage and amperage (current). The usual frequency of 50 cps (or 60 cps) is extremely dangerous as it corresponds to the fibrillation frequency of the myocardium. This results in ventricular fibrillation and instant death.
11. (b) The mean average value of alternating current (or emf) during a half, cycle is given by $I_m = 0.636 I_0$ (or $E_m = 0.636 E_0$)

During the next half cycle, the mean value of ac will be equal in magnitude but opposite in direction.

For this reason the average value of ac over a complete cycle is always zero. So the average value is always defined over a half cycle of ac.

12. (d) An ac ammeter is constructed on the basis of heating effect of the electric current. Since heat produced varies as square of current ($H = I^2 R$). Therefore the division marked on the scale of ac ammeter are not equally spaced.
13. (d) The power of a ac circuit is given by $P = EI \cos\phi$

where $\cos\phi$ is power factor and ϕ is phase angle. In case of circuit containing resistance only, phase angle is zero and power factor is equal to one. Therefore power is maximum in case of circuit containing resistor only.

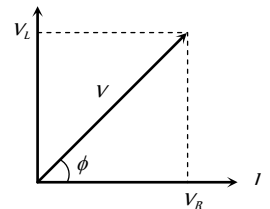
14. (a) Capacitive reactance $X_C = \frac{1}{\omega C}$. When capacitance (C)

increases, the capacitive reactance decreases. Due to decrease in its values, the current in the circuit will increase

$$\left(I = \frac{E}{\sqrt{R^2 + X_C^2}} \right) \text{ and hence brightness of source (or electric$$

lamp) will also increase.

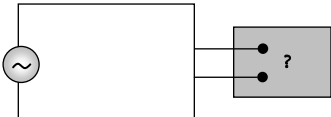
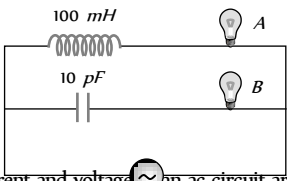
15. (b) As both the inductance and resistance are joined in series, hence current through both will be same. But in case of resistance, both the current and potential vary simultaneously, hence they are in same phase. In case of an inductance when current is zero, potential difference across it is maximum and when current reaches maximum (at $\omega t = \pi/2$), potential difference across it becomes zero i.e. potential difference leads the current by $\pi/2$ or current lags behind the potential difference by $\pi/2$. Phase angle in case of LR circuit is given as $\phi = \tan^{-1} \left(\frac{\omega L}{R} \right)$.



16. (b) We can use a capacitor of suitable capacitance as a choke coil, because average power consumed per cycle in an ideal capacitor is zero. Therefore, like a choke coil, a condenser can reduce ac without power dissipation.

Alternating Current

SET Self Evaluation Test -24

- A bulb and a capacitor are in series with an ac source. On increasing frequency how will glow of the bulb change
(a) The glow decreases (b) The glow increases
(c) The glow remain the same (d) The bulb quenches
 - The *r.m.s.* current in an ac circuit is 2 A. If the wattless current be $\sqrt{3}A$, what is the power factor
(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{2}}$
(c) $\frac{1}{2}$ (d) $\frac{1}{3}$
 - $\frac{2.5}{\pi} \mu F$ capacitor and 3000-ohm resistance are joined in series to an ac source of 200 volt and 50 sec^{-1} frequency. The power factor of the circuit and the power dissipated in it will respectively
(a) 0.6, 0.06 W (b) 0.06, 0.6 W
(c) 0.6, 4.8 W (d) 4.8, 0.6 W
 - The self inductance of a choke coil is 10 mH. When it is connected with a 10 V dc source, then the loss of power is 20 watt. When it is connected with 10 volt ac source loss of power is 10 watt. The frequency of ac source will be
(a) 50 Hz (b) 60 Hz
(c) 80 Hz (d) 100 Hz
 - In an LCR circuit $R = 100 \text{ ohm}$. When capacitance C is removed, the current lags behind the voltage by $\pi/3$. When inductance L is removed, the current leads the voltage by $\pi/3$. The impedance of the circuit is
(a) 50 ohm (b) 100 ohm
(c) 200 ohm (d) 400 ohm
 - A group of electric lamps having a total power rating of 1000 watt is supplied by an ac voltage $E = 200 \sin(310t + 60^\circ)$. Then the *r.m.s.* value of the circuit current is
(a) 10 A (b) $10\sqrt{2}$ A
(c) 20 A (d) $20\sqrt{2}$ A
 - Following figure shows an ac generator connected to a "block box" through a pair of terminals. The box contains possible R , L , C or their combination, whose elements and arrangements are not known to us. Measurements outside the box reveals that $e = 75 \sin(\sin \omega t) \text{ volt}$, $i = 1.5 \sin(\omega t + 45^\circ) \text{ amp}$ then, the wrong statement is
(a) There must be a capacitor in the box
(b) There must be an inductor in the box
(c) There must be a resistance in the box
(d) The power factor is 0.707
- 
- A resistor R , an inductor L and a capacitor C are connected in series to an oscillator of frequency n . If the resonant frequency is n_r , then the current lags behind voltage, when
(a) $n = 0$ (b) $n < n_r$
(c) $n = n_r$ (d) $n > n_r$
 - If power factor is $1/2$ in a series RL circuit $R = 100 \Omega$. ac mains is used then L is
(a) $\frac{\sqrt{3}}{\pi} \text{ Henry}$ (b) $\pi \text{ Henry}$
(c) $\frac{\pi}{\sqrt{3}} \text{ Henry}$ (d) None of these
 - What will be the self inductance of a coil, to be connected in a series with a resistance of $\pi\sqrt{3} \Omega$ such that the phase difference between the emf and the current at 50 Hz frequency is 30°
(a) 0.5 Henry (b) 0.03 Henry
(c) 0.05 Henry (d) 0.01 Henry
 - The phase difference between the voltage and the current in an ac circuit is $\pi/4$. If the frequency is 50 Hz then this phase difference will be equivalent to a time of
(a) 0.02 s (b) 0.25 s
(c) 2.5 ms (d) 25 ms
 - The instantaneous values of current and emf in an ac circuit are $I = 1/\sqrt{2} \sin 314 t \text{ amp}$ and $E = \sqrt{2} \sin(314 t - \pi/6) \text{ V}$ respectively. The phase difference between E and I will be
(a) $-\pi/6 \text{ rad}$ (b) $-\pi/3 \text{ rad}$
(c) $\pi/6 \text{ rad}$ (d) $\pi/3 \text{ rad}$
 - If A and B are identical bulbs which bulbs glows brighter
(a) A
(b) B
(c) Both equally bright
(d) Cannot say
- 
- The instantaneous values of current and voltage in an ac circuit are $i = 100 \sin 314 t \text{ amp}$ and $e = 200 \sin(314 t + \pi/3) \text{ V}$ respectively. If the resistance is 1Ω then the reactance of the circuit will be
(a) $-200\sqrt{3} \Omega$ (b) $\sqrt{3} \Omega$
(c) $-200/\sqrt{3} \Omega$ (d) $100\sqrt{3} \Omega$
 - What is the *r.m.s.* value of an alternating current which when passed through a resistor produces heat which is thrice of that produced by a direct current of 2 amperes in the same resistor
(a) 6 amp (b) 2 amp
(c) 3.46 amp (d) 0.66 amp

1. (b) This is because, when frequency ν is increased, the capacitive reactance $X_C = \frac{1}{2\pi\nu C}$ decreases and hence the current through the bulb increases.
2. (c) $i_{WL} = i_{rms} \sin\phi \Rightarrow \sqrt{3} = 2 \sin\phi \Rightarrow \sin\phi = \frac{\sqrt{3}}{2}$
 $\Rightarrow \phi = 60^\circ$ so p.f. $= \cos\phi = \cos 60^\circ = \frac{1}{2}$.
3. (c) $Z = \sqrt{R^2 + \left(\frac{1}{2\pi\nu C}\right)^2} = \sqrt{(3000)^2 + \left(\frac{1}{2\pi \times 50 \times \frac{2.5}{\pi} \times 10^{-6}}\right)^2}$
 $\Rightarrow Z = \sqrt{(3000)^2 + (4000)^2} = 5 \times 10^3 \Omega$
 So power factor $\cos\phi = \frac{R}{Z} = \frac{3000}{5 \times 10^3} = 0.6$ and power
 $P = V_{rms} i_{rms} \cos\phi = \frac{V_{rms}^2 \cos\phi}{Z} \Rightarrow P = \frac{(200)^2 \times 0.6}{5 \times 10^3} = 4.8 W$
4. (c) With dc : $P = \frac{V^2}{R} \Rightarrow R = \frac{(10)^2}{20} = 5 \Omega$;
 With ac : $P = \frac{V_{rms}^2 R}{Z^2} \Rightarrow Z^2 = \frac{(10)^2 \times 5}{10} = 50 \Omega^2$
 Also $Z^2 = R^2 + 4\pi^2 \nu^2 L^2$
 $\Rightarrow 50 = (5)^2 + 4(3.14)^2 \nu^2 (10 \times 10^{-3})^2 \Rightarrow \nu = 80 Hz$.
5. (b) When C is removed circuit becomes RL circuit hence
 $\tan \frac{\pi}{3} = \frac{X_L}{R}$ (i)
 When L is removed circuit becomes RC circuit hence
 $\tan \frac{\pi}{3} = \frac{X_C}{R}$ (ii)
 From equation (i) and (ii) we obtain $X = X$. This is the condition of resonance and in resonance $Z = R = 100 \Omega$.
8. (d) The current will lag behind the voltage when reactance of inductance is more than the reactance of condenser. Thus,
 $\omega L > \frac{1}{\omega C}$ or $\omega > \frac{1}{\sqrt{LC}}$
 or $n > \frac{1}{2\pi\sqrt{LC}}$ or $n > n_r$, where n_r = resonant frequency.
9. (a) $\cos\phi = \frac{1}{2} \Rightarrow \phi = 60^\circ$ $\tan 60^\circ = \frac{\omega L}{R} \Rightarrow L = \frac{\sqrt{3}}{\pi} H$
10. (d) $\tan\phi = \frac{X_L}{R} = \frac{2\pi\nu L}{R} \Rightarrow \tan 30^\circ = \frac{2\pi \times 50 \times L}{\pi\sqrt{3}} = 0.01 H$
11. (c) Time difference $= \frac{T}{2\pi} \times \phi = \frac{(1/50)}{2\pi} \times \frac{\pi}{4} = \frac{1}{400} s = 2.5 m-s$
12. (a) Phase difference relative to the current
 $\phi = (314 t - \frac{\pi}{6}) - (314 t) = -\frac{\pi}{6}$
13. (a) $\because (X) \gg (X)$
14. (b) $V_L = iZ \Rightarrow 200 = 100 Z \Rightarrow Z = 2\Omega$
 Also $Z^2 = R^2 + X_L^2 \Rightarrow (2)^2 = (1)^2 + X_L^2 \Rightarrow X_L = \sqrt{3}\Omega$
15. (c) Heat produced by ac $= 3 \times$ Heat produced by dc
 $\therefore i_{rms}^2 R t = 3 \times i^2 R t \Rightarrow i_{rms}^2 = 3 \times i^2$
 $\Rightarrow i_{rms} = 2\sqrt{3} = 3.46 A$
6. (b) $P = \frac{1}{2} V_0 i_0 \cos\phi \Rightarrow 1000 = \frac{1}{2} \times 200 \times i_0 \cos 60^\circ$
 $\Rightarrow i_0 = 20 A \Rightarrow i_{rms} = \frac{i_0}{\sqrt{2}} = \frac{20}{\sqrt{2}} = 10\sqrt{2} A$.
7. (b) Since voltage is lagging behind the current, so there must be no inductor in the box.



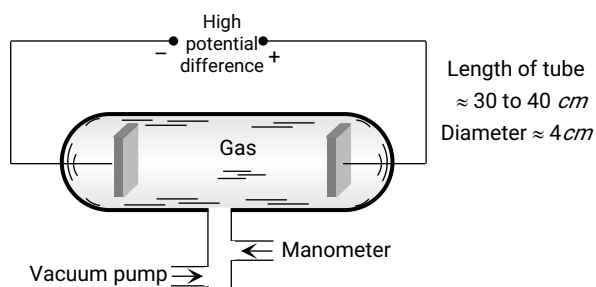
Chapter 25

Electron, Photon, Photoelectric Effect and X-rays

Electric Discharge Through Gases

At normal atmospheric pressure, the gases are poor conductor of electricity. If we establish a potential difference (of the order of 30 kV) between two electrodes placed in air at a distance of few cm from each other, electric conduction starts in the form of sparks.

The discharge of electricity through gases can be systematically studied with the help of discharge tube shown below



As the pressure inside the discharge tube is gradually reduced, the following is the sequence of phenomenon that are observed.

- (1) At normal pressure no discharge takes place.
- (2) At the pressure 10 mm of Hg , a zig-zag thin red spark runs from one electrode to other and cracking sound is heard.

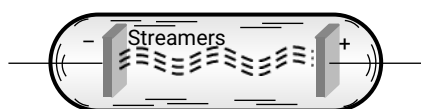


Fig. 25.2

(3) At the pressure 4 mm of Hg , an illumination is observed at the electrodes and the rest of the tube appears dark. This type of discharge is called dark discharge.

(4) When the pressure falls below 4 mm of Hg then the

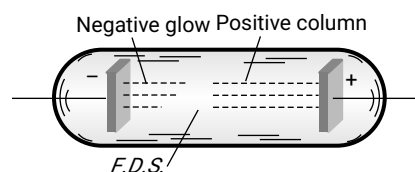
whole tube is filled with bright light called positive column and colour of light depends upon the nature of gas in the tube as shown in the following table.

Table 25.1 : Colour for different gases

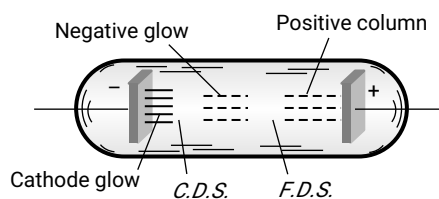
Gas	Air	H_2	N_2	Cl_2	CO_2	Neon
Colour	Purple red	Blue	Red	Green	Bluish white	Dark red

(5) At a pressure of 1.65 mm of Hg :

Sky colour light is produced at the cathode it is called as negative glow. Positive column shrinks towards the anode and the dark space between positive column and negative glow is called Faradays dark space (FDS).



(6) At a pressure of 0.8 mm of Hg : At this pressure, negative glow is detached from the cathode and moves towards the anode. The dark space created between cathode and negative glow is called as Crook's dark space. Length of positive column further reduced. A glow appear at cathode called cathode glow.



(7) At a pressure of 0.05 mm of Hg : The positive column splits into dark and bright disc of light called striations.

(8) At the pressure of 0.01 or 10^{-2} mm of Hg some invisible

particles move from cathode which on striking with the glass tube on the opposite side of cathode cause the tube to glow. These invisible rays emerging from cathode are called cathode rays.

(9) Finally when pressure drops to nearly 10^{-4} mm of Hg, there is no discharge in tube.

Cathode Rays

- (1) Cathode rays, discovered by Sir William Crooke
- (2) They are streams of fast moving electrons.
- (3) They can be produced by using a discharge tube containing gas at a low pressure of the order of 10^{-2} mm of Hg.
- (4) The cathode rays in the discharge tube are the electrons produced due to ionisation of gas and that emitted by cathode due to collision of positive ions.
- (5) Cathode rays travel in straight lines.
- (6) Cathode rays are emitted normally from the cathode surface. Their direction is independent of the position of the anode.
- (7) Cathode rays exert mechanical force on the objects they strike.
- (8) Cathode rays produce heat when they strikes a metal surface.
- (9) Cathode rays produce fluorescence.
- (10) When cathode rays strike a solid object, specially a metal of high atomic weight and high melting point X-rays are emitted from the objects.
- (11) Cathode rays are deflected by an electric field and also by a magnetic field.
- (12) Cathode rays ionise the gases through which they are passed.
- (13) Cathode rays can penetrate through thin foils of metal.
- (14) Cathode rays are found to have velocity ranging $\frac{1}{30}$ th

to $\frac{1}{10}$ th of velocity of light.

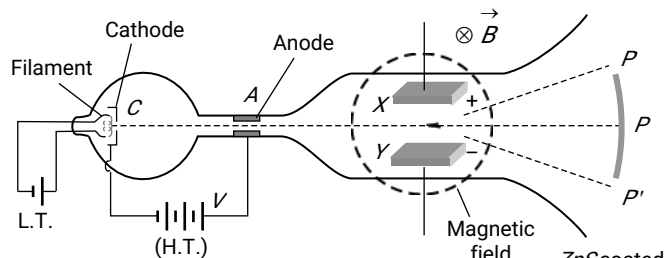
J.J. Thomson's Experiment

(1) It's working is based on the fact that if a beam of electron is subjected to the crossed electric field $\vec{E}$ and magnetic field $\vec{B}$, it experiences a force due to each field. In case the forces on the electrons in the electron beam due to these fields are equal and opposite, the beam remains undeflected.

(2) When no field is applied, the electron beam produces illuminations at point P .

(3) In the presence of any field (electric and magnetic) electron beam deflected up or down (illumination at P' or P'')

(4) If both the fields are applied simultaneously and adjusted such that electron beam passes undeflected and produces illumination at point P .



In this case; Electric force = Magnetic force $\Rightarrow eE = evB$
Fig. 25.5

$$\Rightarrow v = \frac{E}{B}; \quad v = \text{velocity of electron}$$

(5) As electron beam accelerated from cathode to anode its loss in potential energy appears as gain in the K.E. at the anode. If suppose V is the potential difference between cathode and anode then, loss in potential energy = eV

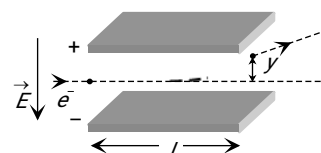
And gain in kinetic energy at anode will be K.E. = $\frac{1}{2}mv^2$

$$\text{i.e. } eV = \frac{1}{2}mv^2 \Rightarrow \frac{e}{m} = \frac{v^2}{2V} \Rightarrow \frac{e}{m} = \frac{E^2}{2VB^2}$$

Thomson found, $\frac{e}{m} = 1.77 \times 10^{11} \text{ C/kg}$.

If one includes the relativistic variation of mass with speed ($m = m_0 / \sqrt{1 - v^2/c^2}$), then specific charge of an electron decreases with the increase in its velocity.

(6) The deflection of an electron in a purely electric field is given by $y = \frac{1}{2} \left(\frac{eE}{m} \right) \frac{l^2}{v^2}$; where l = Length of each plate, y = deflection of electron in the field region, v = speed of the electron.



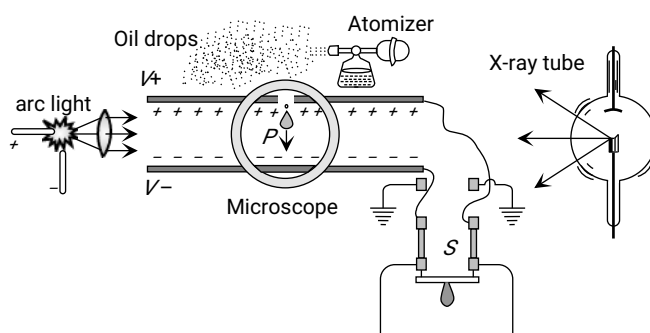
Millikans Oil Drop Experiment

(1) Millikan performed the pioneering oil drop experiment for the precise measurement of the charge on the electron.

(2) By applying suitable electric field across two metal plates, the charged oil droplets could be caused to rise or fall or even held stationary in the field of view for sufficiently long time. He found that the charge on an oil droplet was always an integral multiple of an elementary charge $1.602 \times 10^{-19} \text{ C}$.

(3) In this experiment charge on the drop is given by

$$q = \frac{6\pi\eta(v_1 + v_2)d}{v} \left[\frac{9\eta v_1}{2g(\rho - \sigma)} \right]^{1/2}$$

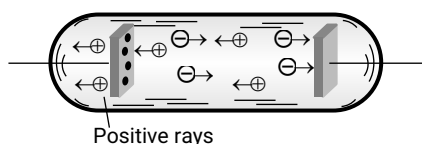


where η = Coefficient of viscosity of air, v_t = Terminal velocity of drop when no electric field is applied between the plates, v_e = Terminal velocity of drop when electric field is applied between the plates.

V = Potential difference between the plates, d = Separation between plates, ρ = density of oil, σ = Density of air.

Positive Rays

When potential difference is applied across the electrodes of a discharge tube (10^{-3} mm of Hg), electrons are emitted from the perforated cathode. As they move towards anode, they gain energy. These energetic electrons when collide with the atoms of the gas in the discharge tube, they ionize the atoms. The positive ions so formed at various places between cathode and anode, travel towards the cathode. Since during their motion, the positive ions when reach the cathode, some pass through the holes in the cathode and a faint luminous glow comes out from each hole on the backside of the cathode. It is called positive rays, which are coming out from the holes.



(1) Positive rays are positive ions having same mass if the experimental gas does not have isotopes. However if the gas has isotopes then positive rays are group of positive ions having different masses.

(2) They travel in straight lines and cast shadows of objects placed in their path. But the speed of the positive rays is much smaller than that of cathode rays.

(3) They are deflected by electric and magnetic fields but the deflections are small as compared to that for cathode rays.

(4) They show a spectrum of velocities. Different positive ions move with different velocities. Being heavy, their velocity is much less than that of cathode rays.

(5) q/m ratio of these rays depends on the nature of the gas in the tube (while in case of the cathode rays q/m is constant and doesn't depend on the nature of gas in the tube). q/m for hydrogen is maximum.

(6) They carry energy and momentum. The kinetic energy of positive rays is more than that of cathode rays.

(7) The value of charge on positive rays is an integral multiple of electronic charge.

(8) They cause ionisation (which is much more than that produced by cathode rays).

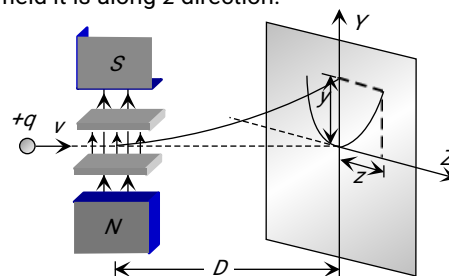
Thomson's Mass Spectrograph

It is used to measure atomic masses of various isotopes in

gas. This is done by measuring q/m of singly ionised positive ion of the gas.

(1) The positive ions are produced in the bulb at the left hand side. These ions are accelerated towards cathode. Some of the positive ions pass through the fine hole in the cathode. This fine ray of positive ions is subjected to electric field E and magnetic field B and then allowed to strike a fluorescent screen ($\vec{E} \parallel \vec{B}$ but $\vec{E}$ or $\vec{B} \perp \vec{v}$).

(2) If the initial motion of the ions is in $+x$ direction and electric and magnetic fields are applied along $+y$ axis then force due to electric field is in the direction of y -axis and due to magnetic field it is along z -direction.



The deflection due to electric field alone $y = \frac{qELD}{mv^2}$ (i)

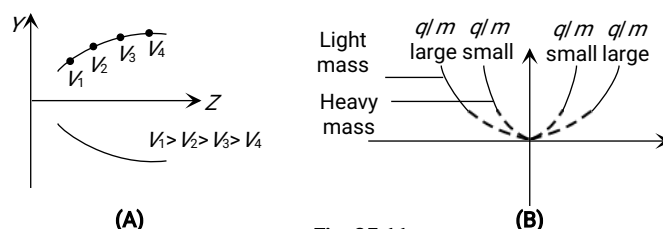
The deflection due to magnetic field alone $z = \frac{qBLD}{mv}$ (ii)

From equation (i) and (ii), $z^2 = k \left(\frac{q}{m} \right) y$

where $k = \frac{B^2 LD}{E}$; This is the equation of parabola. It

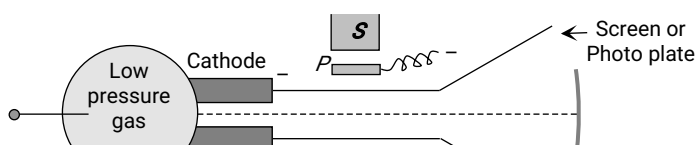
means all the charged particles moving with different velocities but of same q/m value will strike the screen placed in yz plane on a parabolic track as shown in the above figure.

(3) All the positive ions of same q/m moving with different velocity lie on the same parabola. Higher is the velocity lower is the value of y and z . The ions of different specific charge will lie on different parabola.



(4) The number of parabola tells the number of isotopes present in the given ionic beam.

Bainbridge Mass Spectrograph



4 Electron, Photon, Photoelectric Effect and X-Rays

In Bainbridge mass spectrograph, field particles of same velocity are selected by using a velocity selector and then they are subjected to a uniform magnetic field perpendicular to the velocity of the particles. The particles corresponding to different isotopes follow different circular paths as shown in the figure.

(1) **Velocity selector** : The positive ions having a certain velocity v gets isolated from all other velocity particles. In this chamber the electric and magnetic fields are so balanced that the particle moves undeflected. For this the necessary condition is $v = \frac{E}{B}$ and E , B and v should be mutually perpendicular to each other.

(2) **Analysing chamber** : In this chamber magnetic field B is applied perpendicular to the direction of motion of the particle. As a result the particles move along a circular path of radius

$$r = \frac{mE}{qBB'} \Rightarrow \frac{q}{m} = \frac{E}{BB' r} \text{ also } \frac{r_1}{r_2} = \frac{m_1}{m_2}$$

In this way the particles of different masses gets deflected on circles of different radii and reach on different points on the photo plate.

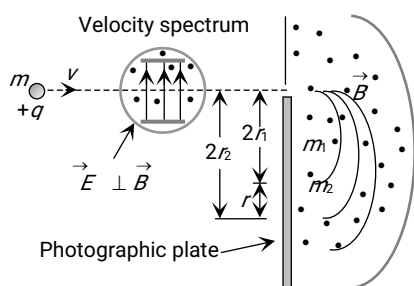


Fig. 25.12

$$\text{Separation between two rays} \\ = d = 2r_2 - 2r_1 = \frac{2v(m_2 - m_1)}{qB'}$$

Matter Waves (de-Broglie Waves)

According to de-Broglie a moving material particle sometimes acts as a wave and sometimes as a particle.

The wave associated with moving particle is called matter wave or de-Broglie wave and it propagates in the form of wave packets with group velocity.

(1) **de-Broglie wavelength** : According to de-Broglie theory, the wavelength of de-Broglie wave is given by

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda \propto \frac{1}{p} \propto \frac{1}{v} \propto \frac{1}{\sqrt{E}}$$

Where h = Planck's constant, m = Mass of the particle, v = Speed of the particle, E = Energy of the particle.

The smallest wavelength whose measurement is possible is that of γ -rays.

The wavelength of matter waves associated with the microscopic particles like electron, proton, neutron, α -particle etc. is of the order of 10^{-10} m.

(2) **de-Broglie wavelength associated with the charged particles** : The energy of a charged particle accelerated through potential difference V is $E = \frac{1}{2}mv^2 = qV$

$$\text{Hence de-Broglie wavelength } \lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}}$$

$$\lambda_{\text{Electron}} = \frac{12.27}{\sqrt{V}} \text{ \AA}, \quad \lambda_{\text{Proton}} = \frac{0.286}{\sqrt{V}} \text{ \AA}$$

$$\lambda_{\text{Deuteron}} = \frac{0.202}{\sqrt{V}} \text{ \AA}, \quad \lambda_{\alpha\text{-particle}} = \frac{0.101}{\sqrt{V}} \text{ \AA}$$

(3) **de-Broglie wavelength associated with uncharged particles** : For Neutron de-Broglie wavelength is given as

$$\lambda_{\text{Neutron}} = \frac{0.286 \times 10^{-10}}{\sqrt{E \text{ (in eV)}}} \text{ m} = \frac{0.286}{\sqrt{E \text{ (in eV)}}} \text{ \AA}$$

Energy of thermal neutrons at ordinary temperature

$$\because E = kT \Rightarrow \lambda = \frac{h}{\sqrt{2mkT}} ; \text{ where } T = \text{Absolute}$$

temperature, k = Boltzman's constant $= 1.38 \times 10^{-23} \text{ Joule/kelvin}$, So,

$$\lambda_{\text{Thermal neutron}} = \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 1.67 \times 10^{-27} \times 1.38 \times 10^{-23} T}} = \frac{30.83}{\sqrt{T}} \text{ \AA}$$

(4) **Ratio of wavelength of photon and electron** : The wavelength of a photon of energy E is given by $\lambda_{ph} = \frac{hc}{E}$

While the wavelength of an electron of kinetic energy K is given by $\lambda_e = \frac{h}{\sqrt{2mK}}$. Therefore, for the same energy,

$$\text{the ratio } \frac{\lambda_{ph}}{\lambda_e} = \frac{c}{E} \sqrt{2mK} = \sqrt{\frac{2mc^2 K}{E^2}}$$

Characteristics of Matter Waves

(1) Matter wave represents the probability of finding a particle in space.

(2) Matter waves are not electromagnetic in nature.

(3) de-Broglie or matter wave is independent of the charge on the material particle. It means, matter wave of de-Broglie wave is associated with every moving particle (whether charged or uncharged).

(4) Practical observation of matter waves is possible only when the de-Broglie wavelength is of the order of the size of the particles.

(5) Electron microscope works on the basis of de-Broglie waves.

(6) The phase velocity of the matter waves can be greater than the speed of the light.

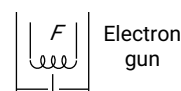
(7) Matter waves can propagate in vacuum, hence they are not mechanical waves.

(8) The number of de-Broglie waves associated with n^{th} orbital electron is n .

(9) Only those circular orbits around the nucleus are stable whose circumference is integral multiple of de-Broglie wavelength associated with the orbital electron.

Davison and Germer Experiment

(1) It is used to study the scattering of electron from a solid or to verify the wave nature of electron. A beam of electrons



emitted by electron gun is made to fall on nickel crystal cut along cubical axis at a particular angle. *Ni* crystal behaves like a three dimensional diffraction grating and it diffracts the electron beam obtained from electron gun.

(2) The diffracted beam of electrons is received by the detector which can be positioned at any angle by rotating it about the point of incidence. The energy of the incident beam of electrons can also be varied by changing the applied voltage to the electron gun.

(3) According to classical physics, the intensity of scattered beam of electrons at all scattering angle will be same but Davisson and Germer, found that the intensity of scattered beam of electrons was not the same but different at different angles of scattering. It is maximum for diffracting angle 50° at 54 volt potential difference.

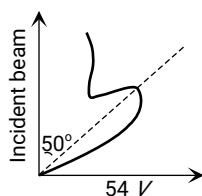


Fig. 25.14

(4) If the de-Broglie waves exist for electrons then these should be diffracted as X-rays. Using the Bragg's formula $2d \sin \theta = n\lambda$, we can determine the wavelength of these waves.

where d = distance between diffracting planes,
 $\theta = \frac{(180 - \phi)}{2}$ = glancing angle for incident beam = Bragg's angle.

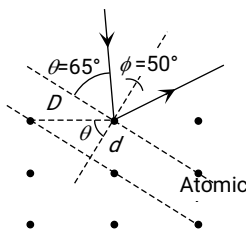


Fig. 25.15

The distance between diffracting planes in *Ni* crystal for this experiment is $d = 0.91 \text{ \AA}$ and the Bragg's angle = 65° . This gives for $n = 1$, $\lambda = 2 \times 0.91 \times 10^{-10} \sin 65^\circ = 1.65 \text{ \AA}$

Now the de-Broglie wavelength can also be determined by using the formula $\lambda = \frac{12.27}{\sqrt{V}} = \frac{12.27}{\sqrt{54}} = 1.67 \text{ \AA}$. Thus the de-Broglie hypothesis is verified.

(5) The Bragg's formula can be rewritten in the form containing interatomic distance D and angle ϕ

$$\therefore \theta = 90 - \frac{\phi}{2} \text{ and } d = D \cos \theta = D \sin \frac{\phi}{2}$$

$$\text{Using } \sin \theta = \cos \frac{\phi}{2}$$

$$2d \sin \theta = \lambda \Rightarrow 2(D \sin \frac{\phi}{2}) \cdot \cos \frac{\phi}{2} = \lambda \Rightarrow D \sin \phi = \lambda$$

Heisenberg Uncertainty Principle

(1) According to Heisenberg's uncertainty principle, it is impossible to measure simultaneously both the position and the momentum of the particle.

(2) Let Δx and Δp be the uncertainty in the simultaneous measurement of the position and momentum of the particle,

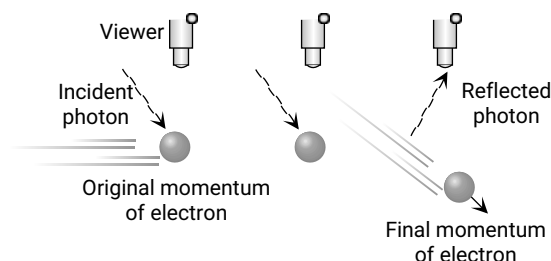
then $\Delta x \Delta p = \hbar$; where $\hbar = \frac{h}{2\pi}$ and $h = 6.63 \times 10^{-34} \text{ J-s}$ is

the Planck's constant. ($\frac{h}{2\pi} = 1.05 \times 10^{-34} \text{ J-s}$)

A more rigorous treatment gives $\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \left(\text{or } \frac{h}{4\pi} \right)$.

(3) If $\Delta x = 0$ then $\Delta p = \infty$ and if $\Delta p = 0$ then $\Delta x = \infty$

i.e., if we are able to measure the exact position of the particle (say an electron) then the uncertainty in the measurement of the linear momentum of the particle is infinite. Similarly, if we are able to measure the exact linear momentum of the particle i.e., $\Delta p = 0$, then we can not measure the exact position of the particle at that time.



An electron cannot be observed without changing its momentum

(4) Uncertainty principle successfully explains

(i) Non-existence of electrons in the nucleus

(ii) Finite size of spectral lines.

(5) The Heisenberg uncertainty principle is also applicable to energy and time, angular momentum and angular displacement. Hence $\Delta E \cdot \Delta t \geq \frac{h}{2\pi}$ and $\Delta L \cdot \Delta \theta \geq \frac{h}{2\pi}$

(6) If the radius of the nucleus is r then the probability of finding the electron inside the nucleus is $\Delta x = 2r$ and uncertainty

in momentum is $\Delta p = \frac{h}{4\pi r}$

Photon

According to Einstein's quantum theory light propagates in the bundles (packets or quanta) of energy, each bundle being called a photon and possessing energy.

(1) **Energy of photon** : Energy of photon is given by

$$E = h\nu = \frac{hc}{\lambda}; \quad \text{where } c = \text{Speed of light, } h = \text{Plank's}$$

constant = $6.6 \times 10^{-34} \text{ J-sec}$, ν = Frequency in Hz , λ = Wavelength of light.

$$\text{In electron volt } E(\text{eV}) = \frac{hc}{e\lambda} = \frac{12375}{\lambda(\text{\AA})} \approx \frac{12400}{\lambda(\text{\AA})}$$

(2) **Mass of photon** : Actually rest mass of the photon is zero. But its effective mass is given as

$$E = mc^2 = h\nu \Rightarrow m = \frac{E}{c^2} = \frac{h\nu}{c^2} = \frac{h}{c\lambda}. \text{ This mass is also}$$

known as kinetic mass of the photon

(3) **Momentum of the photon**

$$\text{Momentum } p = m \times c = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda}$$

(4) **Number of emitted photons** : The number of photons emitted per second from a source of monochromatic radiation of wavelength λ and power P is given as $n = \frac{P}{E} = \frac{P}{h\nu} = \frac{P\lambda}{hc}$; where E = energy of each photon

(5) **Intensity of light (I)** : Energy crossing per unit area normally per second is called intensity or energy flux

$$\text{i.e. } I = \frac{E}{At} = \frac{P}{A} \quad \left(\frac{E}{t} = P = \text{radiation power} \right)$$

At a distance r from a point source of power P intensity is given by $I = \frac{P}{4\pi r^2} \Rightarrow I \propto \frac{1}{r^2}$

(6) **Number of photons falling per second (n)** : If P is the power of radiation and E is the energy of a photon then $n = \frac{P}{E}$

Photo-Electric Effect

The photo-electric effect is the emission of electrons (called photo-electrons when light strikes a surface. To escape from the surface, the electron must absorb enough energy from the incident radiation to overcome the attraction of positive ions in the material of the surface.

The photoelectric effect was first observed by Heinrich Hertz and it was investigated in detail by Wilhelm Hallwachs and Philipp Lenard.

The photoelectric effect is based on the principle of conservation of energy.

(1) **Work function (or threshold energy) (W_0)** : The minimum energy of incident radiation, required to eject the electrons from metallic surface is defined as work function of that surface.

$$W_0 = h\nu_0 = \frac{hc}{\lambda_0} \text{ Joules}; \quad \nu_0 = \text{Threshold frequency};$$

λ_0 = Threshold wavelength

$$\text{Work function in electron volt } W_0(\text{eV}) = \frac{hc}{e\lambda_0} = \frac{12375}{\lambda_0(\text{\AA})}$$

Table 25.2 : Work function of several elements

Element	Work function (eV)	Element	Work function (eV)
Platinum	6.4	Aluminum	4.3
Gold	5.1	Silver	4.3
Nickel	5.1	Sodium	2.7
Carbon	5.0	Lithium	2.5
Silicon	4.8	Potassium	2.2
Copper	4.7	Cesium	1.9

(2) **Threshold frequency (ν_0)** : The minimum frequency of incident radiations required to eject the electron from metal surface is defined as threshold frequency.

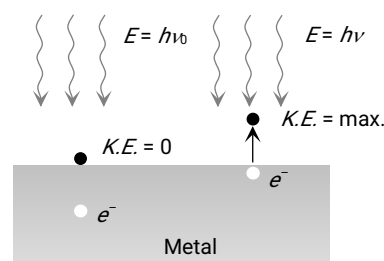
If incident frequency $\nu < \nu_0 \Rightarrow$ No photoelectron emission

For most metals the threshold frequency is in the ultraviolet (corresponding to wavelengths between 200 and 300 nm), but for potassium and cesium oxides it is in the visible spectrum (λ between 400 and 700 nm)

(3) **Threshold wavelength (λ_0)** : The maximum wavelength of incident radiations required to eject the electrons from a metallic surface is defined as threshold wavelength.

If incident wavelength $\lambda > \lambda_0 \Rightarrow$ No photoelectron emission

(4) **Einstein's photoelectric equation** : According to Einstein, photoelectric effect is the result of one to one inelastic collision between photon and electron in which photon is completely absorbed



Einstein's photoelectric equation is $E = W_0 + K_{\max}$

where $K_{\max} = \frac{1}{2} m v_{\max}^2$ = maximum kinetic energy of emitted electrons.

Experimental Setup for Photoelectric Effect

(1) Two conducting electrodes, the anode (Q) and cathode (P) are enclosed in an evacuated glass tube as shown

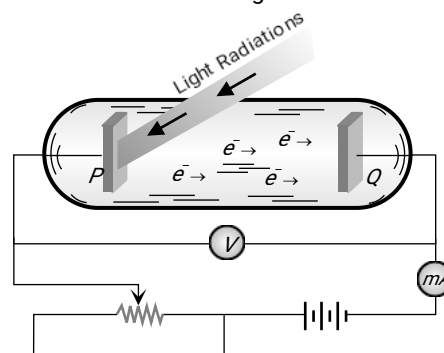


Fig. 25.18

(2) The battery or other source of potential difference creates an electric field in the direction from anode to cathode.

(3) Light of certain wavelength or frequency falling on the surface of cathode causes a current in the external circuit called photoelectric current.

(4) As potential difference increases, photo electric current also increases till saturation is reached.

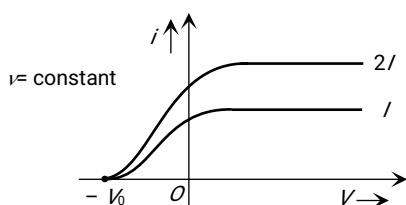
(5) When polarity of battery is reversed (*i.e.* plate *Q* is at negative potential *w.r.t.* plate *P*) electrons start moving back towards the cathode.

(6) At a particular negative potential of plate *Q* no electron will reach the plate *Q* and the current will become zero, this negative potential is called **stopping potential** denoted by V_0 . Maximum kinetic energy of photo electrons in terms of stopping potential will therefore be $K_{\max} = (e V_0) \text{ eV}$

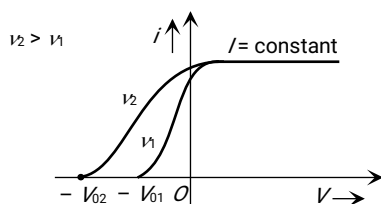
Effect of Intensity and Frequency of Light

(1) **Effect of intensity** : If the intensity of light is increased (while it's frequency is kept the same) the current levels off at a higher value, showing that more electrons are being emitted per unit time. But the stopping potential V_0 doesn't change *i.e.*

Intensity $\propto$ no. of incident photon $\propto$ no. of emitted photoelectron per time $\propto$ photo current



(2) **Effect of frequency** : If frequency of incident light increases, (keeping intensity is constant) stopping potential increases but there is no change in photoelectric current



Important Formulae for Photoelectric Effect

$$(1) h\nu = h\nu_0 + K_{\max} \quad \text{and} \quad K_{\max} = eV_0$$

$$(2) K_{\max} = eV_0 = h(\nu - \nu_0) \Rightarrow \frac{1}{2}mv_{\max}^2 = h(\nu - \nu_0)$$

$$(3) v_{\max} = \sqrt{\frac{2h(\nu - \nu_0)}{m}}$$

$$(4) K_{\max} = \frac{1}{2}mv_{\max}^2 = eV_0 = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right) = hc \left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0} \right)$$

$$(5) v_{\max} = \sqrt{\frac{2hc(\lambda_0 - \lambda)}{m\lambda\lambda_0}}$$

$$(6) v_0 = \frac{h}{e}(\nu - \nu_0) = \frac{hc}{e} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right) = 12375 \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$$

Compton Effect

(1) The scattering of a photon by an electron is called Compton effect.

(2) The energy and momentum is conserved.

(3) Scattered photon will have less energy (more wavelength) as compare to incident photon (less wavelength).

(4) The energy lost by the photon is taken by electron as kinetic energy.

(5) The change in wavelength due to Compton effect is called Compton shift. Compton shift

$$\lambda_f - \lambda_i = \Delta\lambda = \frac{h}{m_0 c} (1 - \cos \phi)$$

$$\text{If } \phi = 0^\circ, \Delta\lambda = 0$$

$$\phi = 90^\circ, \Delta\lambda = \frac{h}{m_0 c} = 0.24 \text{ nm}$$

$$\phi = 180^\circ, \Delta\lambda = \frac{2h}{m_0 c} = 0.48 \text{ nm} \quad (\text{called Compton wave length})$$

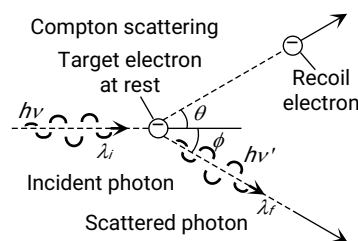


Fig. 25.21

X-Rays

(1) X-rays were discovered by scientist Rontgen that's why they are also called Rontgen rays.

(2) Rontgen discovered that when pressure inside a discharge tube is kept 10^{-3} mm of *Hg* and potential difference is kept 25 kV, then some unknown radiations (X-rays) are emitted by anode.

(3) There are three essential requirements for the production of X-rays.

(i) A source of electron

(ii) An arrangement to accelerate the electrons

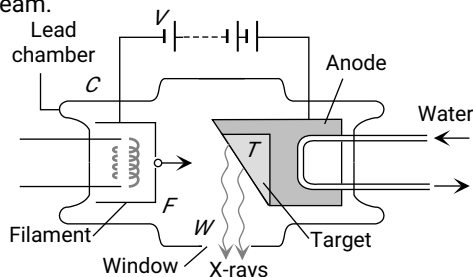
(iii) A target of suitable material of high atomic weight and high melting point on which these high speed electrons strike.

Coolidge X-Ray Tube

(1) It consists of a highly evacuated glass tube containing cathode and target (also known as filament type X-ray tube). The cathode consists of a tungsten filament. The filament is coated with oxides of barium or strontium to have an emission of electrons even at low temperature. The filament is surrounded by a molybdenum cylinder kept at negative potential *w.r.t.* the target.

(2) The target (It is a material of high atomic weight, high melting point and high thermal conductivity) made of tungsten or molybdenum is embedded in a copper block.

(3) The face of the target is set at 45° to the incident electron stream.



(4) The filament is heated by passing the current through it. A high potential difference ($\approx 10 \text{ kV}$ to 80 kV) is applied between the target and cathode to accelerate the electrons which are emitted by filament. The stream of highly energetic electrons are focussed on the target.

(5) Most of the energy of the electrons is converted into heat (above 98%) and only a fraction of the energy of the electrons (about 2%) is used to produce X-rays.

(6) During the operation of the tube, a huge quantity of heat is produced in this target, this heat is conducted through the copper anode to the cooling fins from where it is dissipated by radiation and convection.

(7) **Control of intensity of X-rays** : Intensity implies the number of X-ray photons produced from the target. The intensity of X-rays emitted is directly proportional to the electrons emitted per second from the filament and this can be increased by increasing the filament current. So *intensity of X-rays* $\propto$ *Filament current*

(8) **Control of quality or penetration power of X-rays** : Quality of X-rays implies the penetrating power of X-rays, which can be controlled by varying the potential difference between the cathode and the target.

For large potential difference, energy of bombarding electrons will be large and hence larger is the penetration power of X-rays.

Table 25.3 : Types of X-rays

Hard X-rays	Soft X-rays
More penetration power	Less penetration power
More frequency of the order of $\approx 10^{19} \text{ Hz}$	Less frequency of the order of $\approx 10^{16} \text{ Hz}$
Lesser wavelength range ($0.1 \text{ \AA} - 4 \text{ \AA}$)	More wavelength range ($4 \text{ \AA} - 100 \text{ \AA}$)

Properties of X-Rays

(1) X-rays are electromagnetic waves with wavelength range $0.1 \text{ \AA} - 100 \text{ \AA}$.

(2) The wavelength of X-rays is very small in comparison to the wavelength of light. Hence they carry much more energy (This is the only difference between X-rays and light)

(3) X-rays are invisible.

(4) They travel in a straight line with speed of light.

(5) X-rays are measured in Rontgen (measure of ionization power).

(6) X-rays carry no charge so they are not deflected in magnetic field and electric field.

$$(7) \lambda_{\text{Gamma rays}} < \lambda_{\text{X-rays}} < \lambda_{\text{UV rays}}$$

(8) They used in the study of crystal structure.

(9) They ionise gases

(10) X-rays do not pass through heavy metals and bones.

(11) They affect photographic plates.

(12) Long exposure to X-rays is injurious for human body.

(13) Lead is the best absorber of X-rays.

(14) For X-ray photography of human body parts, BaSO_4 is the best absorber.

(15) They produce photoelectric effect and Compton effect

(16) X-rays are not emitted by hydrogen atom.

(17) These cannot be used in Radar because they are not reflected by the target.

(18) They show all the important properties of light rays like; reflection, refraction, interference, diffraction and polarization etc.

Absorption of X-Rays

X-rays are absorbed when they incident on substance.

$$\text{Intensity of emergent X-rays } I = I_0 e^{-\mu x}$$

So intensity of absorbed X-rays

$$I' = I_0 - I = I_0 (1 - e^{-\mu x})$$

where x = thickness of absorbing medium, μ = absorption coefficient

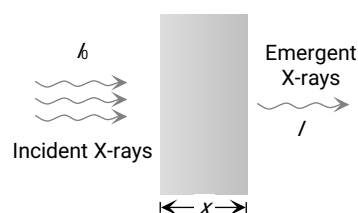


Fig. 25.23

$$\mu \propto \lambda^3; (\lambda = \text{Wavelength of X-ray})$$

$$\mu \propto \nu^3 (\nu = \text{Frequency of X-ray})$$

$$\mu \propto Z^4 (Z = \text{Atomic number of target})$$

Classification of X-Rays

In X-ray tube, when high speed electrons strikes the target, they penetrate the target. They lose their kinetic energy and come to rest inside the metal. The electron before finally being stopped makes several collisions with the atoms in the target. At each collision one of the following two types of X-rays may get formed.

(1) Continuous X-rays

(2) Characteristic X-rays

Continuous X-Rays

As an electron passes close to the positive nucleus of atom of the target, the electron is deflected from its path as shown in figure. This results in deceleration of the electron. The loss in energy of the electron during deceleration is emitted in the form of X-rays.

The X-ray photons so emitted form the continuous X-ray spectrum.

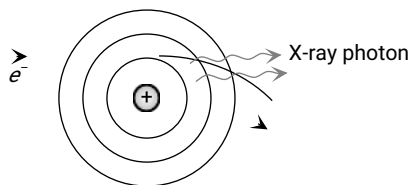


Fig. 25.24

(1) **Minimum wavelength**: When the electron loses whole of its energy in a single collision with the atom, an X-ray photon of maximum energy $h\nu_{\max}$ is emitted i.e.

$$\frac{1}{2}mv^2 = eV = h\nu_{\max} = \frac{hc}{\lambda_{\min}}$$

where v = velocity of electron before collision with target atom, V = potential difference through which electron is accelerated, c = speed of light = 3×10^8 m/s

$$\text{Maximum frequency of radiations (X-rays)} \quad \nu_{\max} = \frac{eV}{h}$$

Minimum wavelength = cut off wavelength of X-ray

$$\lambda_{\min} = \frac{hc}{eV} = \frac{12375}{V} \text{ \AA}$$

(2) **Intensity wavelength graph**: The continuous X-ray spectra consist of all the wavelengths over a given range. These wavelengths are of different intensities. Following figure shows the intensity variation of different wavelengths for various accelerating voltages applied to X-ray tube.

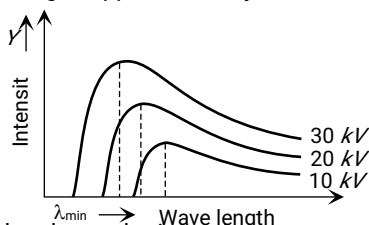


Fig. 25.25

For each voltage, the intensity curve starts at a particular minimum wavelength ($\lambda_{\min}$). It rises rapidly to a maximum and then drops gradually.

The wavelength at which the intensity is maximum depends on the accelerating voltage, being shorter for higher voltage and vice-versa.

Characteristic X-Rays

Few of the fast moving electrons having high velocity penetrate the surface atoms of the target material and knock out the tightly bound electrons even from the inner most shells of the atom. Now when the electron is knocked out, a vacancy is created at that place.

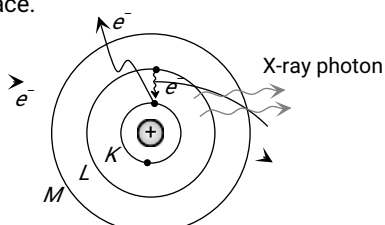


Fig. 25.26

To fill this vacancy electrons from higher shells jump to fill the created vacancies, we know that when an electron jumps from a higher energy orbit E_1 to lower energy orbit E_2 , it radiates energy ($E_1 - E_2$). Thus this energy difference is radiated in the form of X-rays of very small but definite wavelength which depends upon the target material. The X-ray spectrum consists of sharp lines and is called characteristic X-ray spectrum.

(1) **K, L, M, series**: If the electron striking the target ejects an electron from the K-shell of the atom, a vacancy is created in the K-shell. Immediately an electron from one of the outer shells, say L-shell jumps to the K-shell, emitting an X-ray photon of energy equal to the energy difference between the two shells. Similarly, if an electron from the M-shell jumps to the K-shell, X-ray photon of higher energy is emitted. The X-ray photons emitted due to the jump of electron from the L, M, N shells to the K-shells gives K_{α} , K_{β} , K_{γ} lines of the K-series of the spectrum.

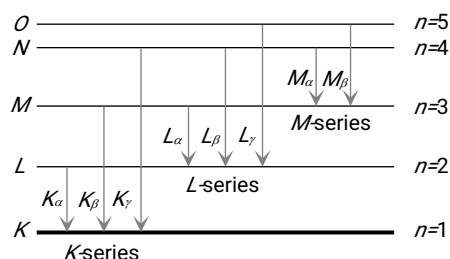


Fig. 25.27

If the electron striking the target ejects an electron from the L-shell of the target atom, an electron from the M, N, shells jumps to the L-shell so that X-rays photons of lesser energy are emitted.

These photons form the L-series of the spectrum. In a similar way the formation of M series, N series etc. may be explained.

(2) **Intensity-wavelength graph**: At certain sharply defined wavelengths, the intensity of X-rays is very large as marked K_{α} , K_{β} , as shown in figure. These X-rays are known as characteristic X-rays. At other wavelengths the intensity varies gradually and these X-rays are called continuous X-rays.

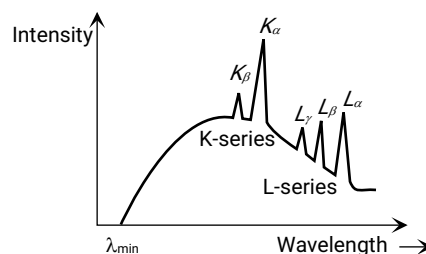


Fig. 25.28

Mosley's Law

Mosley studied the characteristic X-ray spectrum of a number of heavy elements and concluded that the spectra of different elements are very similar and with increasing atomic

number, the spectral lines merely shift towards higher frequencies.

He also gave the following relation $\sqrt{\nu} = a(Z - b)$

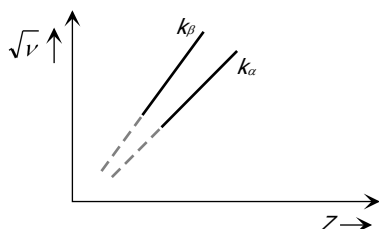


Fig. 25.29

where ν = Frequency of emitted line, Z = Atomic number of target, a = Proportionality constant, b = Screening constant or Shielding constant.

$(Z - b)$ = Effective atomic number

a and b doesn't depend on the nature of target. Different values of b are as follows

$b = 1$	for	K -series
$b = 7.4$	for	L -series
$b = 19.2$	for	M -series

(1) Mosley's law supported Bohr's theory

(2) It experimentally determined the atomic number (Z) of elements.

(3) This law established the importance of ordering of elements in periodic table by atomic number and not by atomic weight.

(4) Gaps in Moseley's data for $A = 43, 61, 72, 75$ suggested existence of new elements which were later discovered.

(5) The atomic numbers of Cu, Ag and Pt were established to be 29, 47 and 78 respectively.

(6) When a vacancy occurs in the K -shell, there is still one electron remaining in the K -shell. An electron in the L -shell will feel an effective charge of $(Z - 1)e$ due to $+Ze$ from the nucleus and $-e$ from the remaining K -shell electron, because L -shell orbit is well outside the K -shell orbit.

(7) Wave length of characteristic spectrum $\frac{1}{\lambda} = R(Z - b)^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ and energy of X-ray radiations.

$$\Delta E = h\nu = \frac{hc}{\lambda} = Rhc(Z - b)^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

(8) If transition takes place from $n_2 = 2$ to $n_1 = 1$ (K_α -line)

$$(i) a = \sqrt{\frac{3RC}{4}} = 2.47 \times 10^{15} \text{ Hz}$$

$$(ii) \nu_{K\alpha} = RC(Z - 1)^2 \left(1 - \frac{1}{2^2} \right) = \frac{3RC}{4}(Z - 1)^2$$

$$= 2.47 \times 10^{15} (Z - 1)^2 \text{ Hz}$$

(iii) In general the wavelength of all the K -lines are given by

$$\frac{1}{\lambda_K} = R(Z - 1)^2 \left(1 - \frac{1}{n^2} \right) \text{ where } n = 2, 3, 4, \dots$$

$$\text{While for } K_\alpha \text{ line } \lambda_{K\alpha} = \frac{1216}{(Z - 1)^2} \text{ \AA}$$

$$(iv) E_{K\alpha} = 10.2(Z - 1)^2 \text{ eV}$$

Uses of X-Rays

(i) In study of crystal structure : Structure of DNA was also determined using X-ray diffraction.

(ii) In medical science

(iii) In radiography

(iv) In radio therapy

(v) In engineering

(vi) In laboratories

(vii) In detective department

(viii) In art the change occurring in old oil paintings can be examined by X-rays.

Tips & Tricks

✍ Discovery of positive rays helps in discovering of isotopes.

✍ The de-Broglie wavelength of electrons in first Bohr orbit of an atom is equal to circumference of orbit.

✍ A particle having zero rest mass and non zero energy and momentum must travel with a speed equal to speed of light.

✍ de-Broglie wave length associates with gas molecules

$$\text{is given as } \lambda = \frac{h}{mv_{rms}} = \frac{h}{\sqrt{3mkT}} \quad (\text{Energy of gas molecules})$$

$$\text{at temperature } T \text{ is } E = \frac{3}{2} kT$$

✍ A photon is not a material particle. It is a quanta of energy.

✍ When a particle exhibits wave nature, it is associated with a wave packet, rather than a wave.

✍ By coating the metal surface with a layer of barium oxide or strontium oxide its work function is lowered.

✍ We must remember that intensity of incident light radiation is inversely proportional to the square of distance between source of light and photosensitive plate P i.e., $I \propto \frac{1}{d^2}$ so $I \propto i \propto \frac{1}{d^2}$

✍ The photoelectric current can be increased by filling some inert gas like Argon into the bulb. The photoelectrons emitted by cathode ionise the gas by collision and hence the current is increased.

✍ Compton effect shows that photon have momentum.

✍ Production of X-ray is the reverse phenomenon of photoelectric effect.

✍ The thickness of medium at which intensity of emergent X-rays becomes half *i.e.* $I' = \frac{I_0}{2}$ is called half value thickness

$(x_{1/2})$ and it is given as $x_{1/2} = \frac{0.693}{\mu}$.

✍ Continuous X-rays are produced due to the phenomenon called "Bremsstrahlung". It means slowing down or braking radiation.

✍ The wavelength of characteristic X-ray doesn't depend on accelerating voltage. It depends on the atomic number (Z) of the target material.

✍ In characteristic X-ray spectrum $\lambda_{K\alpha} < \lambda_{L\alpha} < \lambda_{M\alpha}$ and $V_{K\alpha} > V_{L\alpha} > V_{M\alpha}$ also $\lambda_{K\alpha} > \lambda_{K\beta} > \lambda_{K\gamma}$

✍ Nearly all metals emit photoelectrons when exposed to UV light. But alkali metals like lithium, sodium, potassium, rubidium and cesium emit photoelectrons even when exposed to visible light.

✍ Oxide coated filament in vacuum tubes is used to emit electrons at relatively lower temperature.

✍ Conduction of electricity in gases at low pressure takes place because colliding electrons acquire higher kinetic energy due to increase in mean free path.

✍ Kinetic energy of cathode rays depends on both voltage and work function of cathode.

✍ Photoelectric effect is due to the particle nature of light.

✍ Hydrogen atom does not emit X-rays because its energy levels are too close to each other.

✍ The essential difference between X-rays and γ -rays is that, γ -rays emit from nucleus while X-rays from outer part of atom.

✍ There is no time delay between emission of electron and incidence of photon *i.e.* the electrons are emitted out as soon as the light falls on metal surface.

✍ If light were wave (not photons) it will take about an year to eject a photoelectron out of the metal surface.

✍ Dose of X-ray is measured in terms of produced ions or free energy via ionisation.

✍ Safe dose for human body per week is one Röntgen (One Röntgen is the amount of X-rays which emits 2.5×10^4 J free energy through ionization of 1 gm air at NTP)

✍ The photoelectrons emitted from the metallic surface have different kinetic energies even when the incident photons have same energy. This happens because all the electrons do not exist in the surface layer.

Those coming from below the surface lose more energy in getting themselves free.

✍ Einstein was awarded Nobel prize for explaining the photoelectric effect.

✍ Uncertainty in the measurement of momentum of photon within the nucleus is $\Delta p = \frac{h}{2\pi d}$

where d = diameter of the nucleus and $\Delta x = d$ = uncertainty in the measurement of position of proton.

Ordinary Thinking

Objective Questions

Cathode Rays and Positive Rays

- In the Millikan's experiment, the distance between two horizontal plates is 2.5 cm and the potential difference applied is 250 V. The electric field between the plates will be
(a) 900 V/m (b) 10000 V/m
(c) 625 V/m (d) 6250 V/m
- The cathode rays have particle nature because of the fact that
[CPMT 1986; MNR 1986]
(a) They can propagate in vacuum
(b) They are deflected by electric and magnetic fields
(c) They produced fluorescence
(d) They cast shadows
- In Millikan's experiment for the determination of the charge on the electron, the reason for using the oil is
(a) It is a lubricant (b) Its density is higher
(c) It vapourises easily (d) It does not vapourise
- The mass of a particle is 400 times than that of an electron and the charge is double. The particle is accelerated by 5 V. Initially the particle remained in rest, then its final kinetic energy will be
[MP PMT 1990]
(a) 5 eV (b) 10 eV
(c) 100 eV (d) 2000 eV
- An electron (charge = 1.6×10^{-19} C) is accelerated through a potential of 100,000 V. The energy acquired by the electron is
[MP PET 1989]
(a) 1.6×10^{-24} J (b) 1.6×10^{-14} erg
(c) 0.53×10^{-17} J (d) 1.6×10^{-14} J
- While doing his experiment, Millikan one day observed the following charges on a single drop
(i) 6.563×10^{-19} C (ii) 8.204×10^{-19} C
(iii) 11.50×10^{-19} C (iv) 13.13×10^{-19} C
(v) 16.48×10^{-19} C (vi) 18.09×10^{-19} C

From this data the value of the elementary charge (e) was found to be **[MP PMT 1993]**

- (a) $1.641 \times 10^{-19} \text{ C}$ (b) $1.630 \times 10^{-19} \text{ C}$
 (c) $1.648 \times 10^{-19} \text{ C}$ (d) $1.602 \times 10^{-19} \text{ C}$
7. When electron beam passes through an electric field, they gain kinetic energy. If the same beam passes through magnetic field, then
 (a) Their energy increases
 (b) Their momentum increases
 (c) Their potential energy increases
 (d) Energy and momentum both remains unchanged
8. Which of the following law is used in the Millikan's method for the determination of charge
[DPMT 2002]
 (a) Ampere's law (b) Stoke's law
 (c) Fleming's left hand rule (d) Fleming's right hand rule
9. The mass of the electron varies with
 (a) The size of the cathode ray tube
 (b) The variation of ' g '
 (c) Velocity
 (d) Size of the electron
10. When the speed of electrons increases, then the value of its specific charge **[MP PMT 1994]**
 (a) Increases
 (b) Decreases
 (c) Remains unchanged
 (d) Increases upto some velocity and then begins to decrease
11. An electron is accelerated through a potential difference of 1000 *volts*. Its velocity is nearly **[MP PMT 1985; Pb. PET 2003]**
 (a) $3.8 \times 10^7 \text{ m/s}$ (b) $1.9 \times 10^6 \text{ m/s}$
 (c) $1.9 \times 10^7 \text{ m/s}$ (d) $5.7 \times 10^7 \text{ m/s}$
12. In an electron gun the control grid is given a negative potential relative to cathode in order to **[NCERT 1988]**
 (a) Decelerate electrons
 (b) Repel electrons and thus to control the number of electrons passing through it
 (c) To select electrons of same velocity and to converge them along the axis
 (d) To decrease the kinetic energy of electrons

13. The ratio of momenta of an electron and an α -particle which are accelerated from rest by a potential difference of 100 V is
- (a) 1 (b) $\sqrt{\frac{2m_e}{m_\alpha}}$
- (c) $\sqrt{\frac{m_e}{m_\alpha}}$ (d) $\sqrt{\frac{m_e}{2m_\alpha}}$
14. When subjected to a transverse electric field, cathode rays move
- (a) Down the potential gradient
(b) Up the potential gradient
(c) Along a hyperbolic path
(d) Along a circular path
15. The fact that electric charges are integral multiples of the fundamental electronic charge was proved experimentally by
- (a) Planck (b) J.J. Thomson
(c) Einstein (d) Millikan
16. In Millikan oil drop experiment, a charged drop of mass $1.8 \times 10^{-14} \text{ kg}$ is stationary between its plates. The distance between its plates is 0.90 cm and potential difference is 2.0 kV. The number of electrons on the drop is
- [MP PMT 1994, 2003; MP PET 1997]
- (a) 500 (b) 50
(c) 5 (d) 0
17. The charge on electron was discovered by
- [BHU 1995; RPMT 1999; DCE 2004]
- (a) J.J. Thomson (b) Neil Bohr
(c) Millikan (d) Chadwick
18. From the following, what charges can be present on oil drops in Millikan's experiment [MP PET 1995]
- (a) Zero, equal to the magnitude of charge on α -particle
(b) $2e, 1.6 \times 10^{-18} \text{ C}$,
(c) $1.6 \times 10^{-19} \text{ C}, 2.5e$
(d) $1.5e, e$
- (Here e is the electronic charge)
19. A narrow electron beam passes undeviated through an electric field $E = 3 \times 10^4 \text{ volt/m}$ and an overlapping magnetic field $B = 2 \times 10^{-3} \text{ Weber/m}^2$. If electric field and magnetic field are mutually perpendicular. The speed of the electrons is
- (a) 60 m/s (b) $10.3 \times 10^7 \text{ m/s}$
(c) $1.5 \times 10^7 \text{ m/s}$ (d) $0.67 \times 10^{-7} \text{ m/s}$
20. In Thomson's method of determining e/m of electrons [MP PMT 1997]
- (a) Electric and magnetic fields are parallel to electrons beam
(b) Electric and magnetic fields are perpendicular to each other and perpendicular to electrons beam
(c) Magnetic field is parallel to the electrons beam
(d) Electric field is parallel to the electrons beam
21. Cathode rays enter into a uniform magnetic field perpendicular to the direction of the field. In the magnetic field their path will be
- (a) Straight line (b) Circle
(c) Parabolic (d) Ellipse
22. The specific charge of an electron is [MP PMT/PET 1998; J&K CET 2004; Pb. PET 2002; MH CET 1999]
- [MP PET 1994]
- (a) $1.6 \times 10^{-19} \text{ coulomb}$
(b) $4.8 \times 10^{-10} \text{ statcoulomb}$
(c) $1.76 \times 10^{11} \text{ coulomb/kg}$
(d) $1.76 \times 10^{-11} \text{ coulomb/kg}$
23. An electron is moving with constant velocity along x -axis. If a uniform electric field is applied along y -axis, then its path in the $x-y$ plane will be [MP PMT 1999]
- (a) A straight line (b) A circle
(c) A parabola (d) An ellipse
24. Cathode rays are similar to visible light rays in that [SCRA 1994]
- (a) They both can be deflected by electric and magnetic fields
(b) They both have a definite magnitude of wavelength
(c) They both can ionise a gas through which they pass
(d) They both can expose a photographic plate
25. Which one of the following devices makes use of the electrons to strike certain substances to produce fluorescence [SCRA 1994]
- (a) Thermionic valve (b) Photoelectric cell
(c) Cathode ray oscilloscope (d) Electron gun
26. An oxide coated filament is useful in vacuum tubes because essentially [SCRA 1994]
- (a) It has high melting point
(b) It can withstand high temperatures
(c) It has good mechanical strength
(d) It can emit electrons at relatively lower temperatures
27. Gases begin to conduct electricity at low pressure because [CBSE PMT 1994]
- (a) At low pressure, gases turn to plasma [MP PET 1995]
(b) Colliding electrons can acquire higher kinetic energy due to increased mean free path leading to ionisation of atoms
(c) Atoms break up into electrons and protons
(d) The electrons in atoms can move freely at low pressure
28. A beam of electrons is moving with constant velocity in a region having electric and magnetic fields of strength 20 Vm^{-1} and 0.5 T at right angles to the direction of motion of the electrons. What is the velocity of the electrons [CBSE PMT 1996]

- (a) 20 ms^{-1} (b) 40 ms^{-1}
(c) 8 ms^{-1} (d) 5.5 ms^{-1}
29. Kinetic energy of emitted cathode rays is dependent on [CPMT 1996]
(a) Only voltage
(b) Only work function
(c) Both (a) and (b)
(d) It does not depend upon any physical quantity
30. The radius of the orbital of electron in the hydrogen atom $0.5 \text{ \AA}$. The speed of the electron is $2 \times 10^6 \text{ m/s}$. Then the current in the loop due to the motion of the electron is [RPMT 1996]
(a) 1 mA (b) 1.5 mA
(c) 2.5 mA (d) $1.5 \times 10^{-2} \text{ mA}$
31. The kinetic energy of an electron which is accelerated through a potential of 100 volts is [MP PET 1986; CBSE PMT 1997; AIIMS 1998]
(a) $1.602 \times 10^{-17} \text{ J}$ (b) 418.6 calories
(c) $1.16 \times 10^4 \text{ K}$ (d) $6.626 \times 10^{-34} \text{ W-sec}$
32. When a proton is accelerated with 1 volt potential difference, then its kinetic energy is [CPMT 1997; CBSE PMT 1999; RPET 2003]
(a) $\frac{1}{1840} \text{ eV}$ (b) 1840 eV
(c) 1 eV (d) 1840 c eV
33. Energy of electrons can be increased by allowing them [JIPMER 1997]
(a) To fall through electric potential
(b) To move in high magnetic field
(c) To fall from great heights
(d) To pass through lead blocks
34. Cathode rays and canal rays produced in a certain discharge tube are deflected in the same direction if [SCRA 1998]
(a) A magnetic field is applied normally
(b) An electric field is applied normally
(c) An electric field is applied tangentially
(d) A magnetic field is applied tangentially
35. In a Millikan's oil drop experiment the charge on an oil drop is calculated to be $6.35 \times 10^{-19} \text{ C}$. The number of excess electrons on the drop is [MNR 1998]
(a) 3.9 (b) 4
(c) 4.2 (d) 6
36. Cathode rays consist of [DCE 1999]
(a) Photons (b) Electrons
(c) Protons (d) α -particles
37. A metal plate gets heated, when cathode rays strike against, it due to [CPMT 2000; Pb. PET 2000]
(a) Kinetic energy of cathode rays
(b) Potential energy of cathode rays
(c) Linear velocity of cathode rays
(d) Angular velocity of cathode rays
38. Cathode rays are [RPET 2000]
(a) Positive rays (b) Neutral rays
(c) He rays (d) Electron waves
39. An electron of charge ' e ' coulomb passes through a potential difference of $V \text{ volts}$. Its energy in ' joules ' will be [MP PET 2000]
(a) V/e (b) eV
(c) e/V (d) V
40. An electron is accelerated through a potential difference of 200 volts . If e/m for the electron be $1.6 \times 10^{11} \text{ coulomb/kg}$, the velocity acquired by the electron will be [MP PET 2000]
(a) $8 \times 10^6 \text{ m/s}$ (b) $8 \times 10^5 \text{ m/s}$
(c) $5.9 \times 10^6 \text{ m/s}$ (d) $5.9 \times 10^5 \text{ m/s}$
41. Which is not true with respect to the cathode rays [Kerala PET 2001]
(a) A stream of electrons
(b) Charged particles
(c) Move with speed same as that of light
(d) Can be deflected by magnetic fields
42. In Milikan's experiment, an oil drop having charge q gets stationary on applying a potential difference V in between two plates separated by a distance ' d '. The weight of the drop is
(a) qVd (b) $q \frac{d}{V}$
(c) $\frac{q}{Vd}$ (d) $q \frac{V}{d}$
43. Electron volt is a unit of [MP PMT 2001]
(a) Potential (b) Charge
(c) Power (d) Energy
44. In Thomson experiment of finding e/m for electrons, beam of electron is replaced by that of muons (particle with same charge as of electrons but mass 208 times that of electrons). No deflection condition in this case satisfied if [Orissa (Engg.) 2002]
(a) B is increased 208 times
(b) E is increased 208 times
(c) B is increased 14.4 times
(d) None of these
45. The colour of the positive column in a gas discharge tube depends on [Kerala (Engg.) 2002]
(a) The type of glass used to construct the tube
(b) The gas in the tube
(c) The applied voltage
(d) The material of the cathode
46. Cathode rays are produced when the pressure is of the order of
(a) 2 cm of Hg (b) 0.1 cm of Hg
(c) 0.01 mm of Hg (d) $1 \text{ }\mu\text{m of Hg}$

47. The speed of an electron having a wavelength of 10^{-10} m is [AIIMS 2002]
 (a) $7.25 \times 10^6 \text{ m/s}$ (b) $6.26 \times 10^6 \text{ m/s}$
 (c) $5.25 \times 10^6 \text{ m/s}$ (d) $4.24 \times 10^6 \text{ m/s}$
48. Which of the following is not the property of a cathode ray [CBSE PMT 2002]
 (a) It casts shadow
 (b) It produces heating effect
 (c) It produces fluorescence
 (d) It does not deflect in electric field
49. In a Thomson set-up for the determination of e/m , electrons accelerated by 2.5 kV enter the region of crossed electric and magnetic fields of strengths $3.6 \times 10^4 \text{ Vm}^{-1}$ and $1.2 \times 10^{-3} \text{ T}$ respectively and go through undeflected. The measured value of e/m of the electron is equal to [AMU 2002]
 (a) $1.0 \times 10^{11} \text{ C-kg}^{-1}$ (b) $1.76 \times 10^{11} \text{ C-kg}^{-1}$
 (c) $1.80 \times 10^{11} \text{ C-kg}^{-1}$ (d) $1.85 \times 10^{11} \text{ C-kg}^{-1}$
50. The ratio of specific charge of an α -particle to that of a proton is
 (a) 2 : 1 (b) 1 : 1
 (c) 1 : 2 (d) 1 : 3
51. In Bainbridge mass spectrograph a potential difference of 1000 V is applied between two plates distant 1 cm apart and magnetic field in $B = 17$. The velocity of undeflected positive ions in m/s from the velocity selector is [RPMT 1998]
 (a) 10^7 m/s (b) 10^4 m/s
 (c) 10^5 m/s (d) 10^2 m/s
52. When cathode rays (tube voltage $\sim 10 \text{ kV}$) collide with the anode of high atomic weight then we get [MP PET 1985]
 (a) Positive rays (b) X-rays
 (c) Gamma rays (d) Canal rays
53. In Thomson's experiment if the value of q/m is the same for all positive ions striking the photographic plate, then the trace would be [RPMT 1986]
 (a) Straight line (b) Parabolic
 (c) Circular (d) Elliptical
54. In a discharge tube at 0.02 mm , there is a formation of [CBSE PMT 1996]
 (a) FDS (b) CDS
 (c) Both space (d) None of these
55. Electric field and magnetic field in Thomson mass spectrograph are applied [RPMT 1998]
 (a) Simultaneously, perpendicular
 (b) Perpendicular but not simultaneously
 (c) Parallel but not simultaneously
 (d) Parallel simultaneously
56. The current conduction in a discharged tube is due to
 (a) Electrons only
 (b) +ve ions and electrons
 (c) -ve ions and electrons
 (d) +ve ions, -ve ions and electrons
57. In Milikan's oil drop experiment, a charged drop falls with terminal velocity V . If an electric field E is applied in vertically upward direction then it starts moving in upward direction with terminal velocity $2V$. If magnitude of electric field is decreased to $\frac{E}{2}$, then terminal velocity will become [CBSE PMT 1999]
 (a) $\frac{V}{2}$ (b) V
 (c) $\frac{3V}{2}$ (d) $2V$
58. An electron is accelerated through a p.d. of 45.5 volt . The velocity acquired by it is (in ms^{-1}) [AIIMS 2004]
 (a) 4×10^6 (b) 4×10^4
 (c) 10^6 (d) Zero
59. A cathode ray tube emits 1.8×10^{14} electrons per second, when heated. When 400 V is applied to anode all the emitted electrons reach the anode. The charge on electron is $1.6 \times 10^{-19} \text{ C}$. The maximum anode current is [MP PMT 2004]
 (a) $2.7 \mu\text{A}$ (b) $29 \mu\text{A}$
 (c) $72 \mu\text{A}$ (d) 29 mA
60. Order of q/m ratio of proton, α -particle and electron is [AFMC 2004]
 (a) $e > p > \alpha$ (b) $p > \alpha > e$
 (c) $e > \alpha > p$ (d) None of these
61. A charge of magnitude $3e$ and mass $2m$ is moving in an electric field $\vec{E}$. The acceleration imparted to the charge is [DCE 2004]
 (a) $2Ee/3m$ (b) $3Ee/2m$
 (c) $2m/3Ee$ (d) $3m/2Ee$
62. An electron initially at rest, is accelerated through a potential difference of 200 volt , so that it acquires a velocity $8.4 \times 10^6 \text{ m/s}$. The value of e/m of electron will be [DPMT 2003]
 (a) $2.76 \times 10^{12} \text{ C/kg}$ (b) $1.76 \times 10^{11} \text{ C/kg}$
 (c) $0.76 \times 10^{12} \text{ C/kg}$ (d) None of these
63. An α particle is accelerated through a p.d. of 10^6 volt then K.E. of particle will be [Pb. PET 2003]
 (a) 8 MeV (b) 4 MeV
 (c) 2 MeV (d) 1 MeV
64. Positive rays consists of [RPMT 1996, 2003]
 (a) Electrons (b) Neutrons
 (c) Positive ions (d) Electro magnetic waves

65. O^{++} , C^+ , He^{++} and H^+ ions are projected on the photographic plate with same velocity in a mass spectrograph. Which one will strike farthest [RPMT 2003]
- (a) O^{++} (b) C^+
(c) He^{++} (d) H^+
66. An electron beam is moving between two parallel plates having electric field $1.125 \times 10^{-6} N/m$. A magnetic field $3 \times 10^{-10} T$ is also applied so that beam of electrons do not deflect. The velocity of the electron is [MH CHT 2004]
- (a) 4225 m/s (b) 3750 m/s
(c) 2750 m/s (d) 3200 m/s
67. Positive rays was discovered by [RPMT 1998]
- (a) Thomson (b) Goldstem
(c) W. Crookes (d) Rutherford
68. An electron is moving in electric field and magnetic field it will gain energy from [DCE 1998]
- (a) Electric field (b) Magnetic field
(c) Both of these (d) None of these
69. If an electron oscillates at a frequency of 1 GHz it gives [DCE 1999]
- (a) X-rays (b) Mirowaves
(c) Infrared rays (d) None of these
70. In an electron gun, the electrons are accelerated by the potential V . If e is the charge and m is the mass of an electron, then the maximum velocity of these electrons will be
- (a) $\frac{2eV}{m}$ (b) $\sqrt{\frac{2eV}{m}}$
(c) $\sqrt{\frac{2m}{eV}}$ (d) $\frac{V^2}{2em}$
71. Which of the following have highest specific charge [BHU 2005]
- (a) Positron (b) Proton
(c) He (d) None of these
72. In Millikan's oil drop experiment, an oil drop of mass $16 \times 10^{-6} kg$ is balanced by an electric field of $10^6 V/m$. The charge in coulomb on the drop, assuming $g = 10 m/s^2$ is
- (a) 6.2×10^{-11} (b) 16×10^{-9}
(c) 16×10^{-11} (d) 16×10^{-13}
3. The de-Broglie wavelength associated with the particle of mass m moving with velocity v is [CBSE PMT 1992]
- (a) h/mv (b) mv/h
(c) mh/v (d) m/hv
4. A photon, an electron and a uranium nucleus all have the same wavelength. The one with the most energy [MP PMT 1992]
- (a) Is the photon
(b) Is the electron
(c) Is the uranium nucleus
(d) Depends upon the wavelength and the properties of the particle.
5. A particle which has zero rest mass and non-zero energy and momentum must travel with a speed [MP PMT 1992; DPMT 2001; Kerala PMT 2004]
- (a) Equal to c , the speed of light in vacuum
(b) Greater than c
(c) Less than c
(d) Tending to infinity
6. When the kinetic energy of an electron is increased, the wavelength of the associated wave will
- (a) Increase
(b) Decrease
(c) Wavelength does not depend on the kinetic energy
(d) None of the above
7. [MP PMT 1987, 96; BHU 1995; MNR 1998]
If the de-Broglie wavelengths for a proton and for a α -particle are equal, then the ratio of their velocities will be
- (a) 4 : 1 (b) 2 : 1
(c) 1 : 2 (d) 1 : 4
8. The de-Broglie wavelength λ associated with an electron having kinetic energy E is given by the expression [MP PMT 1990; CPMT 1996]
- (a) $\frac{h}{\sqrt{2mE}}$ (b) $\frac{2h}{mE}$
(c) $2mhE$ (d) $\frac{2\sqrt{2mE}}{h}$
9. [UP SEAT 2005]
Dual nature of radiation is shown by [MP PET 1991]
- (a) Diffraction and reflection
(b) Refraction and diffraction
(c) Photoelectric effect alone
(d) Photoelectric effect and diffraction
10. For the Bohr's first orbit of circumference $2\pi r$, the de-Broglie wavelength of revolving electron will be [MP PMT 1987]
- (a) $2\pi r$ (b) πr
(c) $\frac{1}{2\pi r}$ (d) $\frac{1}{4\pi r}$
11. An electron of mass m when accelerated through a potential difference V has de-Broglie wavelength λ . The de-Broglie

Matter Waves

1. The idea of matter waves was given by
- (a) Davisson and Germer (b) de-Broglie
(c) Einstein (d) Planck
2. Wave is associated with matter
- (a) When it is stationary
(b) When it is in motion with the velocity of light only
(c) When it is in motion with any velocity
(d) None of the above

wavelength associated with a proton of mass M accelerated through the same potential difference will be

[CBSE PMT 1995; EAMCET 2001; J & K CET 2004]

(a) $\lambda \frac{m}{M}$ (b) $\lambda \sqrt{\frac{m}{M}}$

(c) $\lambda \frac{M}{m}$ (d) $\lambda \sqrt{\frac{M}{m}}$

12. What will be the ratio of de-Broglie wavelengths of proton and α -particle of same energy

[RPET 1991, 96; DCE 2002; Kerala PET 2005]

(a) 2 : 1 (b) 1 : 2

(c) 4 : 1 (d) 1 : 4

13. What is the de-Broglie wavelength of the α -particle accelerated through a potential difference V [RPMT 1996]

(a) $\frac{0.287}{\sqrt{V}} \text{ \AA}$ (b) $\frac{12.27}{\sqrt{V}} \text{ \AA}$

(c) $\frac{0.101}{\sqrt{V}} \text{ \AA}$ (d) $\frac{0.202}{\sqrt{V}} \text{ \AA}$

14. de-Broglie hypothesis treated electrons as

[BHU 2000]

- (a) Particles (b) Waves
(c) Both 'a' and 'b' (d) None of these

15. The energy that should be added to an electron, to reduce its de-Broglie wavelengths from 10^{-10} m to $0.5 \times 10^{-10} \text{ m}$, will be [KCET (Engg./Med.) 2000]

- (a) Four times the initial energy
(b) Thrice the initial energy
(c) Equal to the initial energy
(d) Twice the initial energy

16. The de-Broglie wavelength of an electron having 80 eV of energy is nearly

($1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$, Mass of electron = $9 \times 10^{-31} \text{ kg}$)

Plank's constant = $6.6 \times 10^{-34} \text{ J-sec}$)

[EAMCET (Engg.) 2001]

- (a) $140 \text{ \AA}$ (b) $0.14 \text{ \AA}$
(c) $14 \text{ \AA}$ (d) $1.4 \text{ \AA}$

17. If particles are moving with same velocity, then maximum de-Broglie wavelength will be for [CBSE PMT 2002]

- (a) Neutron (b) Proton
(c) β -particle (d) α -particle

18. If an electron and a photon propagate in the form of waves having the same wavelength, it implies that they have the same [CBSE PMT 1995; DCE 2001; AIIMS 2003]

- (a) Energy (b) Momentum
(c) Velocity (d) Angular momentum

19. The de-Broglie wavelength is proportional to [RPET 2003]

(a) $\lambda \propto \frac{1}{v}$ (b) $\lambda \propto \frac{1}{m}$

(c) $\lambda \propto \frac{1}{p}$ (d) $\lambda \propto p$

20. Particle nature and wave nature of electromagnetic waves and electrons can be shown by [AIIMS 2000]

- (a) Electron has small mass, deflected by the metal sheet
(b) X-ray is diffracted, reflected by thick metal sheet
(c) Light is refracted and defracted
(d) Photoelectricity and electron microscopy

21. The de-Broglie wavelength of a particle moving with a velocity $2.25 \times 10^6 \text{ m/s}$ is equal to the wavelength of photon. The ratio of kinetic energy of the particle to the energy of the photon is (velocity of light is $3 \times 10^8 \text{ m/s}$)

[EAMCET (Med.) 2003]

- (a) $1/8$ (b) $3/8$
(c) $5/8$ (d) $7/8$

22. According to de-Broglie, the de-Broglie wavelength for electron in an orbit of hydrogen atom is 10^{-10} m . The principle quantum number for this electron is [RPMT 2003]

- (a) 1 (b) 2
(c) 3 (d) 4

23. The speed of an electron having a wavelength of 10^{-10} m is

[Manipal 1997; AIIMS 2002]

- (a) $7.25 \times 10^6 \text{ m/s}$ (b) $6.26 \times 10^6 \text{ m/s}$
(c) $5.25 \times 10^6 \text{ m/s}$ (d) $4.24 \times 10^6 \text{ m/s}$

24. The kinetic energy of electron and proton is 10^{-32} J . Then the relation between their de-Broglie wavelengths is

[CPMT 1999]

- (a) $\lambda_p < \lambda_e$ (b) $\lambda_p > \lambda_e$
(c) $\lambda_p = \lambda_e$ (d) $\lambda_p = 2\lambda_e$

25. The de-Broglie wavelength of a particle accelerated with 150 volt potential is 10^{-10} m . If it is accelerated by 600 volts p.d. , its wavelength will be [RPET 1988]

- (a) $0.25 \text{ \AA}$ (b) $0.5 \text{ \AA}$
(c) $1.5 \text{ \AA}$ (d) $2 \text{ \AA}$

26. The de-Broglie wavelength associated with a hydrogen molecule moving with a thermal velocity of 3 km/s will be

- (a) $1 \text{ \AA}$ (b) $0.66 \text{ \AA}$
(c) $6.6 \text{ \AA}$ (d) $66 \text{ \AA}$

27. When the momentum of a proton is changed by an amount P , the corresponding change in the de-Broglie wavelength is found to be 0.25% . Then, the original momentum of the proton was

- (a) $100 P$
(c) $400 P$ (d) $4 P$

28. The de-Broglie wavelength of a neutron at 27°C is λ . What will be its wavelength at 927°C [DPMT 2002]

- (a) $\lambda/2$ (b) $\lambda/3$
(c) $\lambda/4$ (d) $\lambda/9$

29. An electron and proton have the same de-Broglie wavelength. Then the kinetic energy of the electron is [Kerala PMT 2004]
- Zero
 - Infinity
 - Equal to the kinetic energy of the proton
 - Greater than the kinetic energy of the proton
30. For moving ball of cricket, the correct statement about de-Broglie wavelength is [RPMT 2001]
- It is not applicable for such big particle
 - $\frac{h}{\sqrt{2mE}}$
 - $\sqrt{\frac{h}{2mE}}$
 - $\frac{h}{2mE}$
31. Photon and electron are given same energy ($10^{-20} J$). Wavelength associated with photon and electron are λ_{ph} and λ_{el} then correct statement will be [RPMT 2001]
- $\lambda_{ph} > \lambda_{el}$
 - $\lambda_{ph} < \lambda_{el}$
 - $\lambda_{ph} = \lambda_{el}$
 - $\frac{\lambda_{el}}{\lambda_{ph}} = C$
32. The kinetic energy of an electron with de-Broglie wavelength of 0.3 nanometer is [UPSEAT 2004]
- 0.168 eV
 - 16.8 eV
 - 1.68 eV
 - 2.5 eV
33. A proton and an α -particle are accelerated through a potential difference of 100 V. The ratio of the wavelength associated with the proton to that associated with an α -particle is
- $\sqrt{2} : 1$
 - $2 : 1$
 - $2\sqrt{2} : 1$
 - $\frac{1}{2\sqrt{2}} : 1$
34. The wavelength of de-Broglie wave is $2\mu m$, then its momentum is ($h = 6.63 \times 10^{-34} J\cdot s$) [DCE 2004]
- $3.315 \times 10^{-28} kg\cdot m/s$
 - $1.66 \times 10^{-28} kg\cdot m/s$
 - $4.97 \times 10^{-28} kg\cdot m/s$
 - $9.9 \times 10^{-28} kg\cdot m/s$
35. de-Broglie wavelength of a body of mass 1 kg moving with velocity of 2000 m/s is [Pb. PMT 2003]
- $3.32 \times 10^{-7} \text{ \AA}$
 - $1.5 \times 10^{-7} \text{ \AA}$
 - $0.55 \times 10^{-7} \text{ \AA}$
 - None of these
36. The kinetic energy of an electron is 5 eV. Calculate the de-Broglie wavelength associated with it ($h = 6.6 \times 10^{-34} J\cdot s$, $m_e = 9.1 \times 10^{-31} kg$)
- 5.47 $\text{\AA}$
 - 10.9 $\text{\AA}$
 - 2.7 $\text{\AA}$
 - None of these
37. The wavelength associated with an electron accelerated through a potential difference of 100 V is nearly [RPMT 2003]
- 100 $\text{\AA}$
 - 123 $\text{\AA}$
 - 1.23 $\text{\AA}$
 - 0.123 $\text{\AA}$
38. The de-Broglie wavelength λ [RPMT 2004]
- is proportional to mass
 - is proportional to impulse
 - Inversely proportional to impulse
 - does not depend on impulse
39. Davisson and Germer experiment proved [RPET 2002; DCE 2004]
- Wave nature of light
 - Particle nature of light
 - Both (a) and (b)
 - Neither (a) nor (b)
40. If the kinetic energy of a free electron doubles, its de-Broglie wavelength changes by the factor [AIEEE 2005]
- $\frac{1}{\sqrt{2}}$
 - $\sqrt{2}$
 - $\frac{1}{2}$
 - 2
41. The energy that should be added to an electron to reduce its de Broglie wavelength from one nm to 0.5 nm is [KCET 2005]
- Four times the initial energy
 - Equal to the initial energy
 - Twice the initial energy
 - Thrice the initial energy
42. de-Broglie wavelength of a body of mass m and kinetic energy E is given by [BCECE 2005]
- $\lambda = \frac{h}{mE}$
 - $\lambda = \frac{\sqrt{2mE}}{h}$
 - $\lambda = \frac{h}{2mE}$
 - $\lambda = \frac{h}{\sqrt{2mE}}$
43. The wavelength of the matter wave is independent of [Kerala PMT 2005]
- Mass
 - Velocity
 - Momentum
 - Charge

Photon and Photoelectric Effect

1. The momentum of a photon is $3.3 \times 10^{-29} kg - m/sec$. Its frequency will be [CPMT 1980; MP PET 1992; DPMT 1999]
- $3 \times 10^3 Hz$
 - $6 \times 10^3 Hz$
 - $7.5 \times 10^{12} Hz$
 - $1.5 \times 10^{13} Hz$
2. The energy of a photon of wavelength λ is given by [CPMT 1974; CBSE PMT 1992; DCE 1998; BHU 2000; DPMT 2001]
- $h\lambda$
 - $ch\lambda$
 - λ / hc
 - hc / λ
3. The momentum of a photon is $2 \times 10^{-16} gm\cdot cm/sec$. Its energy is [Pb. PMT 2004]
- $0.61 \times 10^{-26} erg$
 - $2.0 \times 10^{-26} erg$
 - $6 \times 10^{-6} erg$
 - $6 \times 10^{-8} erg$
4. The rest mass of the photon is [MP PET 1994; CPMT 1996; RPMT 1999; JIPMER 2002]
- 0
 - ∞

- (c) Between 0 and ∞
(d) Equal to that of an electron
5. The momentum of the photon of wavelength $5000 \text{ \AA}$ will be [CPMT 1987]
(a) $1.3 \times 10^{-27} \text{ kg} \cdot \text{m/sec}$ (b) $1.3 \times 10^{-28} \text{ kg} \cdot \text{m/sec}$
(c) $4 \times 10^{-29} \text{ kg} \cdot \text{m/sec}$ (d) $4 \times 10^{-18} \text{ kg} \cdot \text{m/sec}$
6. The momentum of a photon of energy $h\nu$ will be [DCE 2000]
(a) $h\nu$ (b) $h\nu/c$
(c) $h\nu c$ (d) h/ν
7. A photon in motion has a mass [MP PMT 1992]
(a) $c/h\nu$ (b) h/ν
(c) $h\nu$ (d) $h\nu/c^2$
8. If the momentum of a photon is p , then its frequency is [MP PET 1989]
(a) $\frac{ph}{c}$ (b) $\frac{pc}{h}$
(c) $\frac{mh}{c}$ (d) $\frac{mc}{h}$
- Where m is the rest mass of the photon
9. An AIR station is broadcasting the waves of wavelength 300 metres. If the radiating power of the transmitter is 10 kW, then the number of photons radiated per second is [MP PET 1989; RPMT 2000]
(a) 1.5×10^{29} (b) 1.5×10^{31}
(c) 1.5×10^{33} (d) 1.5×10^{35}
10. The energy of a photon is $E = h\nu$ and the momentum of photon $p = \frac{h}{\lambda}$, then the velocity of photon will be [CPMT 1991]
(a) E/p (b) Ep
(c) $\left(\frac{E}{p}\right)^2$ (d) $3 \times 10^8 \text{ m/s}$
11. The approximate wavelength of a photon of energy 2.48 eV is
(a) 500 Å (b) 5000 Å
(c) 2000 Å (d) 1000 Å
12. An important spectral emission line has a wavelength of 21 cm. The corresponding photon energy is [MP PMT 1993]
(a) $5.9 \times 10^{-4} \text{ eV}$ (b) $5.9 \times 10^{-6} \text{ eV}$
(c) $5.9 \times 10^{-8} \text{ eV}$ (d) $11.8 \times 10^{-6} \text{ eV}$
($h = 6.62 \times 10^{-34} \text{ Js}$; $c = 3 \times 10^8 \text{ m/s}$)
13. The momentum of a photon in an X-ray beam of 10^{-10} metre wavelength is [MP PET 1996]
(a) $1.5 \times 10^{-23} \text{ kg} \cdot \text{m/sec}$ (b) $6.6 \times 10^{-24} \text{ kg} \cdot \text{m/sec}$
(c) $6.6 \times 10^{-44} \text{ kg} \cdot \text{m/sec}$ (d) $2.2 \times 10^{-52} \text{ kg} \cdot \text{m/sec}$
14. The energy of a photon of light with wavelength 5000 Å is approximately 2.5 eV. This way the energy of an X-ray photon with wavelength 1 Å would be [MP PET 1997]
(a) 2.5/5000 eV (b) $2.5/(5000)^2 \text{ eV}$
(c) $2.5 \times 5000 \text{ eV}$ (d) $2.5 \times (5000)^2 \text{ eV}$
15. Energy of a quanta of frequency 10^{15} Hz and $h = 6.6 \times 10^{-34} \text{ J} \cdot \text{sec}$ will be [RPMT 1997]
(a) $6.6 \times 10^{-19} \text{ J}$ (b) $6.6 \times 10^{-12} \text{ J}$
(c) $6.6 \times 10^{-49} \text{ J}$ (d) $6.6 \times 10^{-41} \text{ J}$
16. Momentum of a photon of wavelength λ is [CBSE PMT 1993; JIPMER 2001, 02]
(a) $\frac{h}{\lambda}$ (b) Zero
(c) $\frac{h\lambda}{c^2}$ (d) $\frac{h\lambda}{c}$
17. Wavelength of a 1 keV photon is $1.24 \times 10^{-9} \text{ m}$. What is the frequency of 1 MeV photon [CBSE PMT 1993; MP PET 2005]
(a) $1.24 \times 10^{15} \text{ Hz}$ (b) $2.4 \times 10^{20} \text{ Hz}$
(c) $1.24 \times 10^{18} \text{ Hz}$ (d) $2.4 \times 10^{23} \text{ Hz}$
18. What is the momentum of a photon having frequency $1.5 \times 10^{13} \text{ Hz}$ [BHU 1997]
(a) $3.3 \times 10^{-29} \text{ kg m/s}$ (b) $3.3 \times 10^{-34} \text{ kg m/s}$
(c) $6.6 \times 10^{-34} \text{ kg m/s}$ (d) $6.6 \times 10^{-30} \text{ kg m/s}$
19. The energy of a photon of light of wavelength 450 nm is [BHU 1997; JIPMER 2000]
(a) $4.4 \times 10^{-19} \text{ J}$ (b) $2.5 \times 10^{-19} \text{ J}$
(c) $1.25 \times 10^{-17} \text{ J}$ (d) $2.5 \times 10^{-17} \text{ J}$
20. Frequency of photon having energy 66 eV is [CPMT PMT 1997]
(a) $8 \times 10^{-15} \text{ Hz}$ (b) $12 \times 10^{-15} \text{ Hz}$
(c) $16 \times 10^{15} \text{ Hz}$ (d) None of these
21. Which of the following statement is not correct [AFMC 1999]
(a) Photographic plates are sensitive to infrared rays
(b) Photographic plates are sensitive to ultraviolet rays
(c) Infra-red rays are invisible but can cast shadows like visible light
(d) Infrared photons have more energy than photons of visible light [MP PMT 1987]
22. If we express the energy of a photon in KeV and the wavelength in angstroms, then energy of a photon can be calculated from the relation [AMU (Engg.) 1999]
(a) $E = 12.4 h \nu$ (b) $E = 12.4 h/\lambda$
(c) $E = 12.4/\lambda$ (d) $E = h\nu$
23. The frequency of a photon, having energy 100 eV is ($h = 6.610^{-34} \text{ J} \cdot \text{sec}$) [AFMC 2000]
(a) $2.42 \times 10^{26} \text{ Hz}$ (b) $2.42 \times 10^{16} \text{ Hz}$
(c) $2.42 \times 10^{12} \text{ Hz}$ (d) $2.42 \times 10^9 \text{ Hz}$
24. A photon of wavelength 4400 Å is passing through vacuum. The effective mass and momentum of the photon are respectively
(a) $5 \times 10^{-36} \text{ kg}$, $1.5 \times 10^{-27} \text{ kg} \cdot \text{m/s}$
(b) $5 \times 10^{-35} \text{ kg}$, $1.5 \times 10^{-26} \text{ kg} \cdot \text{m/s}$

- (c) Zero, $1.5 \times 10^{-26} \text{ kg} \cdot \text{m} / \text{s}$
 (d) $5 \times 10^{-36} \text{ kg}, 1.67 \times 10^{-43} \text{ kg} \cdot \text{m} / \text{s}$
25. Which of the following is true for photon [RPET 2001]
 (a) $E = \frac{hc}{\lambda}$ (b) $E = \frac{1}{2} mu^2$
 (c) $p = \frac{E}{2v}$ (d) $E = \frac{1}{2} mc^2$
26. Which of the following is incorrect statement regarding photon [MH CET (Med.) 2001]
 (a) Photon exerts no pressure
 (b) Photon energy is $h\nu$
 (c) Photon rest mass is zero
 (d) None of these
27. If a photon has velocity c and frequency ν , then which of following represents its wavelength [AIEEE 2002]
 (a) $\frac{hc}{E}$ (b) $\frac{h\nu}{c}$
 (c) $\frac{h\nu}{c^2}$ (d) $h\nu$
28. The mass of a photo electron is [MP PMT 2002]
 (a) $9.1 \times 10^{-27} \text{ kg}$ (b) $9.1 \times 10^{-29} \text{ kg}$
 (c) $9.1 \times 10^{-31} \text{ kg}$ (d) $9.1 \times 10^{-34} \text{ kg}$
29. Energy of photon whose frequency is 10^{12} MHz , will be [MH CET 2002]
 (a) $4.14 \times 10^3 \text{ keV}$ (b) $4.14 \times 10^2 \text{ eV}$
 (c) $4.14 \times 10^3 \text{ MeV}$ (d) $4.14 \times 10^3 \text{ eV}$
30. There are n_1 photons of frequency γ_1 in a beam of light. In an equally energetic beam, there are n_2 photons of frequency γ_2 . Then the correct relation is [KCET 2003]
 (a) $\frac{n_1}{n_2} = 1$ (b) $\frac{n_1}{n_2} = \frac{\gamma_1}{\gamma_2}$
 (c) $\frac{n_1}{n_2} = \frac{\gamma_2}{\gamma_1}$ (d) $\frac{n_1}{n_2} = \frac{\gamma_1^2}{\gamma_2^2}$
31. Einstein's photoelectric equation states that $E_k = h\nu - \phi$. In this equation E_k refers to [CPMT 1982; MP PMT 1997]
 (a) Kinetic energy of all the emitted electrons
 (b) Mean kinetic energy of the emitted electrons
 (c) Maximum kinetic energy of the emitted electrons
 (d) Minimum kinetic energy of the emitted electrons
32. Kinetic energy with which the electrons are emitted from the metal surface due to photoelectric effect is [CPMT 1973]
 (a) Independent of the intensity of illumination
 (b) Independent of the frequency of light
 (c) Inversely proportional to the intensity of illumination
 (d) Directly proportional to the intensity of illumination
33. The threshold wavelength for photoelectric emission from a material is $5200 \text{ \AA}$. Photo-electrons will be emitted when this material is illuminated with monochromatic radiation from a [IIT JEE 1982; MP PMT 1992; MP PET 1992; UPSEAT 2001; KCET 2004; J & K CET 2004; BHU 2004]
 (a) 50 watt infrared lamp
 (b) 1 watt infrared lamp
 (c) 50 watt ultraviolet lamp
 (d) 1 watt ultraviolet lamp
 (e) Both (c) and (d)
34. Threshold frequency for a metal is 10^{15} Hz . Light of $\lambda = 4000 \text{ \AA}$ falls on its surface. Which of the following statements is correct
 (a) No photoelectric emission takes place
 (b) Photo-electrons come out with zero speed
 (c) Photo-electrons come out with 10^7 m/sec speed
 (d) Photo-electrons come out with 10^8 m/sec speed
35. Photo cells are used for the
 (a) Reproduction of pictures from the cinema film
 (b) Reproduction of sound from the cinema film
 (c) Automatic switching of street light
 (d) (b) and (c) both
36. Einstein got Nobel prize on which of the following works [DCE 1995]
 (a) Mass-energy relation
 (b) Special theory of relativity
 (c) Photoelectric equation
 (d) (a) and (b) both
37. The photo-electrons emitted from a surface of sodium metal are such that [MP PMT 1992]
 (a) They all are of the same frequency
 (b) They have the same kinetic energy
 (c) They have the same de Broglie wavelength
 (d) They have their speeds varying from zero to a certain maximum
38. A metal surface of work function 1.07 eV is irradiated with light of wavelength 332 nm . The retarding potential required to stop the escape of photo-electrons is [MP PMT 1992]
 (a) 4.81 eV (b) 3.74 eV
 (c) 2.66 eV (d) 1.07 eV
39. In a photo cell, the photo-electrons emission takes place
 (a) After 10 sec on incident of light rays
 (b) After 10 sec on incident of light rays
 (c) After 10 sec on incident of light rays
 (d) After 10 sec on incident of light rays
40. When light falls on a metal surface, the maximum kinetic energy of the emitted photo-electrons depends upon [MP PMT 1989; MP PET 1992, 93]
 (a) The time for which light falls on the metal
 (b) Frequency of the incident light
 (c) Intensity of the incident light
 (d) Velocity of the incident light
41. Photo-electrons are emitted in the photoelectric effect from a metal surface [MP PET 1992]
 (a) Only if the frequency of the incident radiation is above a certain threshold value

- (b) Only if the temperature of the surface is high
(c) At a rate that is independent of the nature of the metal
(d) With a maximum velocity proportional to the frequency of the incident radiation
42. The work function of a metal is 4.2 eV , its threshold wavelength will be [BHU 2003; CPMT 2004]
(a) $4000 \text{ \AA}$ (b) $3500 \text{ \AA}$
(c) $2955 \text{ \AA}$ (d) $2500 \text{ \AA}$
43. The number of photo-electrons emitted per second from a metal surface increases when [EAMCET (Med.) 1995; CBSE PMT 1993; MP PMT 1994, 2002; MH CET 1999; KCET 2003]
(a) The energy of incident photons increases
(b) The frequency of incident light increases
(c) The wavelength of the incident light increases
(d) The intensity of the incident light increases
44. The work function of metal is 1 eV . Light of wavelength $3000 \text{ \AA}$ is incident on this metal surface. The velocity of emitted photo-electrons will be [MP PMT 1990]
(a) 10 m/sec (b) $1 \times 10^3 \text{ m/sec}$
(c) $1 \times 10^4 \text{ m/sec}$ (d) $1 \times 10^6 \text{ m/sec}$
45. The retarding potential for having zero photo-electron current
(a) Is proportional to the wavelength of incident light
(b) Increases uniformly with the increase in the wavelength of incident light
(c) Is proportional to the frequency of incident light
(d) Increases uniformly with the increase in the frequency of incident light wave
46. In a dark room of photography, generally red light is used. The reason is
(a) Most of the photographic films are not sensitive to red light
(b) The frequency for red light is low and hence the energy $h\nu$ of photons is less
(c) (a) and (b) both
(d) None of the above
47. The work function of a metal is $1.6 \times 10^{-19} \text{ J}$. When the metal surface is illuminated by the light of wavelength $6400 \text{ \AA}$, then the maximum kinetic energy of emitted photo-electrons will be (Planck's constant $h = 6.4 \times 10^{-34} \text{ Js}$) [MP PMT 1989]
(a) $14 \times 10^{-19} \text{ J}$ (b) $2.8 \times 10^{-19} \text{ J}$
(c) $1.4 \times 10^{-19} \text{ J}$ (d) $1.4 \times 10^{-19} \text{ eV}$
48. Ultraviolet radiations of 6.2 eV falls on an aluminium surface (work function 4.2 eV). The kinetic energy in joules of the fastest electron emitted is approximately [MNR 1987; MP PET 1990; CBSE PMT 1993; Pb. PMT 2001; BVP 2003; Pb. PET 2004]
(a) 3.2×10^{-21} (b) 3.2×10^{-19}
(c) 3.2×10^{-17} (d) 3.2×10^{-15}
49. The work function for tungsten and sodium are 4.5 eV and 2.3 eV respectively. If the threshold wavelength λ for sodium is $5460 \text{ \AA}$, the value of λ for tungsten is [MP PET 1990]
(a) $5893 \text{ \AA}$ (b) $10683 \text{ \AA}$
(c) $2791 \text{ \AA}$ (d) $528 \text{ \AA}$
50. A photon of energy 3.4 eV is incident on a metal having work function 2 eV . The maximum K.E. of photo-electrons is equal to
(a) 1.4 eV (b) 1.7 eV
(c) 5.4 eV (d) 6.8 eV
51. The work function of a metallic surface is 5.01 eV . The photo-electrons are emitted when light of wavelength $2000 \text{ \AA}$ falls on it. The potential difference applied to stop the fastest photo-electrons is [$h = 4.14 \times 10^{-15} \text{ eV sec}$] [MP PET 1991; DPMT 1999]
(a) 1.2 volts (b) 2.24 volts
(c) 3.6 volts (d) 4.8 volts
52. The photoelectric threshold wavelength for a metal surface is $6600 \text{ \AA}$. The work function for this is [MP PET 1991]
(a) 1.87 V (b) 1.87 eV
(c) 18.7 eV (d) 0.18 eV
53. Photoelectric effect was successfully explained first by
(a) Planck (b) Hallwath
(c) Hertz (d) Einstein [MP PMT/PET 1988]
54. The spectrum of radiation $1.0 \times 10^{14} \text{ Hz}$ is in the infrared region. The energy of one photon of this in joules will be [MP PET 1982]
(a) 6.62×10^{-48} (b) 6.62×10^{-20}
(c) $\frac{6.62}{3} \times 10^{-28}$ (d) $3 \times 6.62 \times 10^{-28}$
55. A radio transmitter operates at a frequency of 880 kHz and a power of 10 kW . The number of photons emitted per second are [CBSE PMT 1990; MP PMT 1990]
(a) 1.72×10^{31} (b) 1327×10^{34}
(c) 13.27×10^{34} (d) 0.075×10^{-34}
56. A photo cell is receiving light from a source placed at a distance of 1 m . If the same source is to be placed at a distance of 2 m , then the ejected electron [MNR 1986; UPSEAT 2000, 01]
(a) Moves with one-fourth energy as that of the initial energy
(b) Moves with one-fourth of momentum as that of the initial momentum
(c) Will be half in number
(d) Will be one-fourth in number
57. In a photoelectric experiment for $4000 \text{ \AA}$ incident radiation, the potential difference to stop the ejection is 2 V . If the incident light is changed to $3000 \text{ \AA}$, then the potential required to stop the ejection of electrons will be [MP PET 1995]
(a) 2 V (b) Less than 2 V
(c) Zero (d) Greater than 2 V

58. Light of wavelength $4000 \text{ \AA}$ is incident on a sodium surface for which the threshold wave length of photo – electrons is $5420 \text{ \AA}$. The work function of sodium is
[MP PMT 1993; Pb. PMT 2002]
- (a) 4.58 eV (b) 2.29 eV
(c) 1.14 eV (d) 0.57 eV
59. Photo cell is a device to [MP PET 1993]
- (a) Store photons
(b) Measure light intensity
(c) Convert photon energy into mechanical energy
(d) Store electrical energy for replacing storage batteries
60. If the work function for a certain metal is $3.2 \times 10^{-19} \text{ joule}$ and it is illuminated with light of frequency $8 \times 10^{14} \text{ Hz}$. The maximum kinetic energy of the photo-electrons would be
[MP PET 1993]
- (a) $2.1 \times 10^{-19} \text{ J}$ (b) $8.5 \times 10^{-19} \text{ J}$
(c) $5.3 \times 10^{-19} \text{ J}$ (d) $3.2 \times 10^{-19} \text{ J}$
($h = 6.63 \times 10^{-34} \text{ Js}$)
61. Stopping potential for photoelectrons [MP PET 1994]
- (a) Does not depend on the frequency of the incident light
(b) Does not depend upon the nature of the cathode material
(c) Depends on both the frequency of the incident light and nature of the cathode material
(d) Depends upon the intensity of the incident light
62. The maximum wavelength of radiation that can produce photoelectric effect in a certain metal is 200 nm . The maximum kinetic energy acquired by electron due to radiation of wavelength 100 nm will be [MP PMT 1994]
- (a) 12.4 eV (b) 6.2 eV
(c) 100 eV (d) 200 eV
63. When the light source is kept 20 cm away from a photo cell, stopping potential 0.6 V is obtained. When source is kept 40 cm away, the stopping potential will be [MP PMT 1994]
- (a) 0.3 V (b) 0.6 V
(c) 1.2 V (d) 2.4 V
64. The minimum energy required to remove an electron is called [AFMC 1995; DPMT 2001]
- (a) Stopping potential (b) Kinetic energy
(c) Work function (d) None of these
65. Light of wavelength $4000 \text{ \AA}$ falls on a photosensitive metal and a negative 2 V potential stops the emitted electrons. The work function of the material (in eV) is approximately
($h = 6.6 \times 10^{-34} \text{ Js}$, $e = 1.6 \times 10^{-19} \text{ C}$, $c = 3 \times 10^8 \text{ ms}^{-1}$)
[MP PMT 1995; MH CET 2004]
- (a) 1.1 (b) 2.0
(c) 2.2 (d) 3.1
66. Assuming photoemission to take place, the factor by which the maximum velocity of the emitted photoelectrons changes when the wavelength of the incident radiation is increased four times, is
- (a) 4 (b) $\frac{1}{4}$
(c) 2 (d) $\frac{1}{2}$
67. Work function of a metal is 2.51 eV . Its threshold frequency is
- (a) $5.9 \times 10^{14} \text{ cycle/sec}$ (b) $6.5 \times 10^{14} \text{ cycle/sec}$
(c) $9.4 \times 10^{14} \text{ cycle/sec}$ (d) $6.08 \times 10^{14} \text{ cycle/sec}$
68. Energy conversion in a photoelectric cell takes place from [AFMC 1993; MP PET 1996; MP PMT 1996]
- (a) Chemical to electrical (b) Magnetic to electrical
(c) Optical to electrical (d) Mechanical to electrical
69. Which one of the following is true in photoelectric emission [MP PMT 1996; JIPMER 2001, 02]
- (a) Photoelectric current is directly proportional to the amplitude of light of a given frequency
(b) Photoelectric current is directly proportional to the intensity of light of a given frequency at moderate intensities
(c) Above the threshold frequency, the maximum K.E. of photoelectrons is inversely proportional to the frequency of incident light
(d) The threshold frequency depends upon the wavelength of incident light
70. When a point source of light is at a distance of one metre from a photo cell, the cut off voltage is found to be V . If the same source is placed at 2 m distance from photo cell, the cut off voltage will be
- (a) V (b) $V/2$
(c) $V/4$ (d) $V/\sqrt{2}$
71. The work function of a photoelectric material is 3.3 eV . The threshold frequency will be equal to [UPSEAT 1999]
- (a) $8 \times 10^4 \text{ Hz}$ (b) $8 \times 10^{56} \text{ Hz}$
(c) $8 \times 10^{10} \text{ Hz}$ (d) $8 \times 10^{14} \text{ Hz}$
72. If the work function of a metal is ' ϕ ' and the frequency of the incident light is ' ν ', there is no emission of photoelectron if
- (a) $\nu < \frac{\phi}{h}$ (b) $\nu = \frac{\phi}{h}$
(c) $\nu > \frac{\phi}{h}$ (d) $\nu > = < \frac{\phi}{h}$
73. A photoelectric cell is illuminated by a point source of light 1 m away. When the source is shifted to 2 m then [CBSE PMT 2003]
- (a) Number of electrons emitted is half the initial number
(b) Each emitted electron carries half the initial energy
(c) Number of electrons emitted is a quarter of the initial number
(d) Each emitted electron carries one quarter of the initial energy
74. Light of wavelength λ strikes a photo-sensitive surface and electrons are ejected with kinetic energy E . If the kinetic energy is to be increased to $2E$, the wavelength must be changed to λ' where
- (a) $\lambda' = \frac{\lambda}{2}$ (b) $\lambda' = 2\lambda$
(c) $\frac{\lambda}{2} < \lambda' < \lambda$ (d) $\lambda' > \lambda$
75. If in a photoelectric experiment, the wavelength of incident radiation is reduced from $6000 \text{ \AA}$ to $4000 \text{ \AA}$ then [Haryana CEE 1996]
- (a) Stopping potential will decrease
(b) Stopping potential will increase

- (c) Kinetic energy of emitted electrons will decrease
(d) The value of work function will decrease
76. The photoelectric work function for a metal surface is 4.125 eV . The cut-off wavelength for this surface is [CBSE PMT 1999; KCET 2001]
- (a) $4125 \text{ \AA}$ (b) $2062.5 \text{ \AA}$
(c) $3000 \text{ \AA}$ (d) $6000 \text{ \AA}$
77. As the intensity of incident light increases [CPMT 1999; CBSE PMT 1999; MH CET (Med.) 2000; KCET (Engg./Med.) 2001; Pb. PET 2001]
- (a) Photoelectric current increases
(b) Photoelectric current decreases
(c) Kinetic energy of emitted photoelectrons increases
(d) Kinetic energy of emitted photoelectrons decreases
78. Light of wavelength $5000 \text{ \AA}$ falls on a sensitive plate with photoelectric work function of 1.9 eV . The kinetic energy of the photoelectron emitted will be [CBSE PMT 1998]
- (a) 0.58 eV (b) 2.48 eV
(c) 1.24 eV (d) 1.16 eV
79. Which of the following is dependent on the intensity of incident radiation in a photoelectric experiment [AIIMS 1998]
- (a) Work function of the surface
(b) Amount of photoelectric current
(c) Stopping potential will be reduced
(d) Maximum kinetic energy of photoelectrons
80. The work function of a substance is 4.0 eV . The longest wavelength of light that can cause photoelectron emission from this substance is approximately [IIT JEE 1998; UPSEAT 2002, 03; AIEEE 2004]
- (a) 540 nm (b) 400 nm
(c) 310 nm (d) 220 nm
81. The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy 6 eV fall on it is 4 eV . The stopping potential in volts is [IIT JEE 1997 Re-Exam]
- (a) 2 (b) 4
(c) 6 (d) 10
82. Work function of a metal is 2.1 eV . Which of the waves of the following wavelengths will be able to emit photoelectrons from its surface [Bihar MEE 1995]
- (a) $4000 \text{ \AA}$, $7500 \text{ \AA}$ (b) $5500 \text{ \AA}$, $6000 \text{ \AA}$
(c) $4000 \text{ \AA}$, $6000 \text{ \AA}$ (d) None of these
83. If mean wavelength of light radiated by 100 W lamp is $5000 \text{ \AA}$, then number of photons radiated per second are [RPET 1997]
- (a) 3×10^{23} (b) 2.5×10^{22}
(c) 2.5×10^{20} (d) 5×10^{17}
84. The frequency of the incident light falling on a photosensitive metal plate is doubled, the kinetic energy of the emitted photoelectrons is
- (a) Double the earlier value (b) Unchanged
(c) More than doubled (d) Less than doubled
85. When light of wavelength 300 nm (nanometer) falls on a photoelectric emitter, photoelectrons are liberated. For another emitter, however light of 600 nm wavelength is sufficient for creating photoemission. What is the ratio of the work functions of the two emitters [CBSE PMT 1993; JIPMER 2000]
- (a) 1 : 2 (b) 2 : 1
(c) 4 : 1 (d) 1 : 4
86. Threshold wavelength for photoelectric effect on sodium is $5000 \text{ \AA}$. Its work function is [CBSE PMT 1993]
- (a) 15 J (b) $16 \times 10^{-14} \text{ J}$
(c) $4 \times 10^{-19} \text{ J}$ (d) $4 \times 10^{-81} \text{ J}$
87. The cathode of a photoelectric cell is changed such that the work function changes from W_1 to W_2 ($W_2 > W_1$). If the current before and after change are I_1 and I_2 , all other conditions remaining unchanged, then (assuming $h\nu > W$) [CBSE PMT 1992]
- (a) $I_1 = I_2$ (b) $I_1 < I_2$
(c) $I_1 > I_2$ (d) $I_1 < I_2 < 2I_1$
88. A beam of light of wavelength λ and with illumination L falls on a clean surface of sodium. If N photoelectrons are emitted each with kinetic energy E , then [BHU 1994]
- (a) $N \propto L$ and $E \propto L$ (b) $N \propto L$ and $E \propto \frac{1}{\lambda}$
(c) $N \propto \lambda$ and $E \propto L$ (d) $N \propto \frac{1}{\lambda}$ and $E \propto \frac{1}{L}$
89. Which of the following statements is correct [CBSE PMT 1997]
- (a) The current in a photocell increases with increasing frequency of light
(b) The photocurrent is proportional to applied voltage
(c) The photocurrent increases with increasing intensity of light
(d) The stopping potential increases with increasing intensity of incident light
90. What is the stopping potential when the metal with work function 0.6 eV is illuminated with the light of 2 eV [BHU 1998; MH CET 2003]
- (a) 2.6 V (b) 3.6 V
(c) 0.8 V (d) 1.4 V
91. When yellow light is incident on a surface, no electrons are emitted while green light can emit. If red light is incident on the surface, then [MNR 1998; MP PET 2000; MH CET 2000]
- (a) No electrons are emitted
(b) Photons are emitted
(c) Electrons of higher energy are emitted
(d) Electrons of lower energy are emitted
92. The photoelectric threshold wavelength of a certain metal is $3000 \text{ \AA}$. If the radiation of $2000 \text{ \AA}$ is incident on the metal [MNR 1998; KCET 1994]
- (a) Electrons will be emitted
(b) Positrons will be emitted
(c) Protons will be emitted
(d) Electrons will not be emitted
93. A photocell stops emission if it is maintained at 2 V negative potential. The energy of most energetic photoelectron is [JIPMER 1999]

- (a) $2eV$ (b) $2J$
(c) $2kJ$ (d) $2keV$
94. The work functions for sodium and copper are $2eV$ and $4eV$. Which of them is suitable for a photocell with $4000 \text{ \AA}$ light
(a) Copper (b) Sodium
(c) Both (d) Neither of them
95. For intensity I of a light of wavelength $5000 \text{ \AA}$ the photoelectron saturation current is $0.40 \mu A$ and stopping potential is $1.36 V$, the work function of metal is
[RPET 1999]
(a) $2.47 eV$ (b) $1.36 eV$
(c) $1.10 eV$ (d) $0.43 eV$
96. The work function of aluminium is $4.2 eV$. If two photons, each of energy $3.5 eV$ strike an electron of aluminium, then emission of electrons will be
[AFMC 1999]
(a) Possible
(b) Not possible
(c) Data is incomplete
(d) Depend upon the density of the surface
97. In photoelectric effect if the intensity of light is doubled then maximum kinetic energy of photoelectrons will become
[RPMT 1999]
(a) Double (b) Half
(c) Four time (d) No change
98. Energy required to remove an electron from aluminium surface is $4.2 eV$. If light of wavelength $2000 \text{ \AA}$ falls on the surface, the velocity of the fastest electron ejected from the surface will be
(a) $8.4 \times 10^5 m/sec$ (b) $7.4 \times 10^5 m/sec$
(c) $6.4 \times 10^5 m/sec$ (d) $8.4 \times 10^6 m/sec$
99. Mercury violet light ($\lambda = 4558 \text{ \AA}$) is falling on a photosensitive material ($\phi = 2.5 eV$). The speed of the ejected electrons is in ms^{-1} , about
[AMU (Engg.) 1999]
(a) 3×10^5 (b) 2.65×10^5
(c) 4×10^4 (d) 3.65×10^7
100. The work functions of metals A and B are in the ratio $1 : 2$. If light of frequencies f and $2f$ are incident on the surfaces of A and B respectively, the ratio of the maximum kinetic energies of photoelectrons emitted is (f is greater than threshold frequency of A , $2f$ is greater than threshold frequency of B)
(a) $1 : 1$ (b) $1 : 2$
(c) $1 : 3$ (d) $1 : 4$
101. Light of frequency ν is incident on a substance of threshold frequency ν_0 ($\nu_0 < \nu$). The energy of the emitted photo-electron will be
(a) $h(\nu - \nu_0)$ (b) h/ν
(c) $he(\nu - \nu_0)$ (d) h/ν_0
102. The stopping potential (V_0)
[BHU 2000]
(a) Depends upon the angle of incident light
(b) Depends upon the intensity of incident light
(c) Depends upon the surface nature of the substance
(d) Is independent of the intensity of the incident light
103. If work function of metal is $3 eV$ then threshold wavelength will be
[RPMT 2000]
(a) $4125 \text{ \AA}$ (b) $4000 \text{ \AA}$
(c) $4500 \text{ \AA}$ [RPET 1999] (d) $5000 \text{ \AA}$
104. When wavelength of incident photon is decreased then
[RPET 2000]
(a) Velocity of emitted photo-electron decreases
(b) Velocity of emitted photoelectron increases
(c) Velocity of photoelectron do not change
(d) Photo electric current increases
105. Quantum nature of light is explained by which of the following phenomenon
[RPET 2000]
(a) Huygen wave theory
(b) Photoelectric effect
(c) Maxwell electromagnetic theory
(d) de-Broglie theory
106. When a metal surface is illuminated by light of wavelengths $400 nm$ and $250 nm$, the maximum velocities of the photoelectrons ejected are v and $2v$ respectively. The work function of the metal is ($h =$ Planck's constant, $c =$ velocity of light in air)
(a) $2hc \times 10^6 J$ (b) $1.5hc \times 10^6 J$
(c) $hc \times 10^6 J$ (d) $0.5 hc \times 10^6 J$
107. $4 eV$ is the energy of the incident photon and the work function is $2eV$. What is the stopping potential
[AMU 1999] [DCE 2000; AIIMS 2004]
(a) $2V$ (b) $4V$
(c) $6V$ (d) $2\sqrt{2}V$
108. Light of frequency ν is incident on a certain photoelectric substance with threshold frequency ν_0 . The work function for the substance is
(a) $h\nu$ (b) $h\nu_0$
(c) $h(\nu - \nu_0)$ (d) $h(\nu + \nu_0)$
109. If threshold wavelength for sodium is $6800 \text{ \AA}$ then the work function will be
[RPET 2001]
(a) $1.8eV$ (b) $2.5eV$
(c) $2.1eV$ (d) $1.4eV$
110. If intensity of incident light is increased in PEE then which of the following is true
[RPET 2001]
(a) Maximum $K.E.$ of ejected electron will increase
(b) Work function will remain unchanged
(c) Stopping potential will decrease
(d) Maximum $K.E.$ of ejected electron will decrease
111. Light of frequency $8 \times 10^{15} Hz$ is incident on a substance of photoelectric work function $6.125 eV$. The maximum kinetic energy of the emitted photoelectrons is
[RPET 2000; AFMC 2001]
(a) $17 eV$ (b) $22 eV$
(c) $27 eV$ (d) $37 eV$
112. The photoelectric threshold wavelength for potassium (work function being $2eV$) is
[CPMT 2001]
(a) $310 nm$ (b) $620 nm$
(c) $1200 nm$ (d) $2100 nm$

113. Photons of energy 6 eV are incident on a metal surface whose work function is 4 eV . The minimum kinetic energy of the emitted photoelectrons will be [MP PET 2001]
 (a) 0 eV (b) 1 eV
 (c) 2 eV (d) 10 eV
114. According to photon theory of light which of the following physical quantities associated with a photon do not/does not change as it collides with an electron in vacuum [AMU (Engg.) 2001]
 (a) Energy and momentum (b) Speed and momentum
 (c) Speed only (d) Energy only
115. The lowest frequency of light that will cause the emission of photoelectrons from the surface of a metal (for which work function is 1.65 eV) will be [JIPMER 2002]
 (a) $4 \times 10^{10}\text{ Hz}$ (b) $4 \times 10^{11}\text{ Hz}$
 (c) $4 \times 10^{14}\text{ Hz}$ (d) $4 \times 10^{-10}\text{ Hz}$
116. Light of two different frequencies whose photons have energies 1 eV and 2.5 eV respectively, successively illuminates a metal of work function 0.5 eV . The ratio of maximum kinetic energy of the emitted electron will be [AIEEE 2002]
 (a) $1 : 5$ (b) $1 : 4$
 (c) $1 : 2$ (d) $1 : 1$
117. Sodium and copper have work functions 2.3 eV and 4.5 eV respectively. Then the ratio of their threshold wavelengths is nearest to [AIEEE 2002]
 (a) $1 : 2$ (b) $4 : 1$
 (c) $2 : 1$ (d) $1 : 4$
118. Photon of 5.5 eV energy fall on the surface of the metal emitting photoelectrons of maximum kinetic energy 4.0 eV . The stopping voltage required for these electrons are [Orissa (Engg.) 2002; DPMT 2004]
 (a) 5.5 V (b) 1.5 V
 (c) 9.5 V (d) 4.0 V
119. A caesium photocell, with a steady potential difference of 60 V across, is illuminated by a bright point source of light 50 cm away. When the same light is placed 1 m away the photoelectrons emitted from the cell [KCET 2002]
 (a) Are one quarter as numerous
 (b) Are half as numerous
 (c) Each carry one quarter of their previous momentum
 (d) Each carry one quarter of their previous energy
120. A radio transmitter radiates 1 kW power at a wavelength 198.6 metres . How many photons does it emit per second [Kerala (Engg.) 2002]
 (a) 10^{10} (b) 10^{20}
 (c) 10^{30} (d) 10^{40}
121. The number of photons of wavelength 540 nm emitted per second by an electric bulb of power 100 W is (taking $h = 6 \times 10^{-34}\text{ J-sec}$) [Kerala (Engg.) 2002; Pb. PET 2001]
 (a) 100 (b) 1000
 (c) 3×10^{20} (d) 3×10^{18}
122. When radiation is incident on a photoelectron emitter, the stopping potential is found to be 9 volts . If e/m for the electron is $1.8 \times 10^{11}\text{ C kg}^{-1}$ the maximum velocity of the ejected electrons is [Kerala (Engg.) 2002]
 (a) $6 \times 10^5\text{ ms}^{-1}$ (b) $8 \times 10^5\text{ ms}^{-1}$
 (c) $1.8 \times 10^6\text{ ms}^{-1}$ (d) $1.8 \times 10^5\text{ ms}^{-1}$
123. Two identical metal plates show photoelectric effect by a light of wavelength λ_A falls on plate A and λ_B on plate B ($\lambda_A = 2\lambda_B$). The maximum kinetic energy is [CPMT 2002]
 (a) $2K_A = K_B$ (b) $K_A < K_B/2$
 (c) $K_A = 2K_B$ (d) $K_A = K_B/2$
124. The threshold wavelength for photoelectric effect of a metal is $6500\text{ \AA}$. The work function of the metal is approximately [MP PMT 2002]
 (a) 2 eV (b) 1 eV
 (c) 0.1 eV (d) 3 eV
125. When ultraviolet rays are incident on metal plate, then photoelectric effect does not occurs. It occurs by the incidence of [CBSE PMT 2002; DCE 1997;
 (a) X-rays (b) Radio wave
 (c) Infrared rays (d) Green house effect
126. Light of frequency 4ν is incident on the metal of the threshold frequency ν . The maximum kinetic energy of the emitted photoelectrons is [MP PET 2002]
 (a) $3h\nu_0$ (b) $2h\nu_0$
 (c) $\frac{3}{2}h\nu_0$ (d) $\frac{1}{2}h\nu_0$
127. By photoelectric effect, Einstein, proved [MP PET 2003]
 (a) $E = h\nu$ (b) $K.E. = \frac{1}{2}mv^2$
 (c) $E = mc^2$ (d) $E = \frac{Rhc^2}{n^2}$
128. The work function of sodium is 2.3 eV . The threshold wavelength of sodium will be [BHU 2003]
 (a) $2900\text{ \AA}$ (b) $2500\text{ \AA}$
 (c) $5380\text{ \AA}$ (d) $2000\text{ \AA}$
129. Which of the following shown particle nature of light [AFMC 2003; CBSE PMT 2001]
 (a) Refraction (b) Interference
 (c) Polarization (d) Photoelectric effect
130. Two identical photo-cathodes receive light of frequencies f_1 and f_2 . If the velocities of the photo electrons (of mass m) coming out are respectively v_1 and v_2 , then [AIEEE 2003]
 (a) $v_1 - v_2 = \left[\frac{2h}{m} (f_1 - f_2) \right]^{1/2}$ (b) $v_1^2 - v_2^2 = \frac{2h}{m} (f_1 - f_2)$
 (c) $v_1 + v_2 = \left[\frac{2h}{m} (f_1 + f_2) \right]^{1/2}$ (d) $v_1^2 + v_2^2 = \frac{2h}{m} (f_1 + f_2)$
131. Consider the two following statements A and B and identify the correct choice given in the answers;

- (A) In photovoltaic cells the photoelectric current produced is not proportional to the intensity of incident light.
 (B) In gas filled photoemissive cells, the velocity of photoelectrons depends on the wavelength of the incident radiation.
 (a) Both A and B are true (b) Both A and B are false
 (c) A is true but B is false (d) A is false B is true
132. When radiation of wavelength λ is incident on a metallic surface, the stopping potential is 4.8 volts. If the same surface is illuminated with radiation of double the wavelength, then the stopping potential becomes 1.6 volts. Then the threshold wavelength for the surface is [EAMCET (Engg.) 2003]
 (a) 2λ (b) 4λ
 (c) 6λ (d) 8λ
133. The frequency and work function of an incident photon are ν and ϕ_0 . If ν_0 is the threshold frequency then necessary condition for the emission of photo electron is [RPET 2003]
 (a) $\nu < \nu_0$ (b) $\nu = \frac{\nu_0}{2}$
 (c) $\nu \geq \nu_0$ (d) None of these
134. Light of wavelength $1824 \text{ \AA}$, incident on the surface of a metal, produces photo-electrons with maximum energy 5.3 eV . When light of wavelength $1216 \text{ \AA}$ is used, the maximum energy of photoelectrons is 8.7 eV . The work function of the metal surface is
 (a) 3.5 eV (b) 13.6 eV
 (c) 6.8 eV (d) 1.5 eV
135. If the energy of a photon corresponding to a wavelength of $6000 \text{ \AA}$ is $3.32 \times 10^{-19} \text{ J}$, the photon energy for a wavelength of $4000 \text{ \AA}$ will be [DPMT 2004]
 (a) 1.4 eV (b) 4.9 eV
 (c) 3.1 eV (d) 1.6 eV
136. If the wavelength of light is $4000 \text{ \AA}$, then the number of waves in 1 mm length will be [J & K CET 2004]
 (a) 25 (b) 0.25
 (c) 0.25×10^4 (d) 25×10^4
137. The velocity of photon is proportional to (where ν is frequency)
 (a) $\frac{\nu^2}{2}$ (b) $\frac{1}{\sqrt{\nu}}$
 (c) $\sqrt{\nu}$ (d) ν
138. If the work function of a photometer is 6.825 eV . Its threshold wavelength will be ($c = 3 \times 10^8 \text{ m/s}$) [Pb. PET 2000; BHU 2004]
 (a) $1200 \text{ \AA}$ (b) $1800 \text{ \AA}$
 (c) $2400 \text{ \AA}$ (d) $3600 \text{ \AA}$
139. A photon of energy 8 eV is incident on a metal surface of threshold frequency $1.6 \times 10^{15} \text{ Hz}$, then the maximum kinetic energy of photoelectrons emitted is ($h = 6.6 \times 10^{-34} \text{ Js}$)
 (a) 4.8 eV (b) 2.4 eV
 (c) 1.4 eV (d) 0.8 eV
140. If the energy of the photon is increased by a factor of 4, then its momentum [UPSEAT 2004]
 (a) Does not change
 (b) Decreases by a factor of 4
 (c) Increases by a factor of 4
 (d) Decreases by a factor of 2
141. The ratio of the energy of a photon with $\lambda = 150 \text{ nm}$ to that with $\lambda = 300 \text{ nm}$ is [DCE 2003]
 (a) 2 [EAMCET (Engg.) 2003] (b) $1/4$
 (c) 4 (d) $1/2$
142. Photo-electric effect can be explained by [DCE 2003]
 (a) Corpuscular theory of light (b) Wave nature of light
 (c) Bohr's theory (d) Quantum theory of light
143. In photoelectric effect, the K.E. of electrons emitted from the metal surface depends upon [DCE 2003]
 (a) Intensity of light
 (b) Frequency of incident light
 (c) Velocity of incident light
 (d) Both intensity and velocity of light
144. The photoelectric effect can be understood on the basis of [Pb. PET 2004]
 (a) The principle of superposition
 (b) The electromagnetic theory of light
 (c) The special theory of relativity
 (d) Line spectrum of the atom
145. If the threshold wavelength for sodium is $5420 \text{ \AA}$, then the work function of sodium is [RPMT 2003]
 (a) 4.58 eV (b) 2.28 eV
 (c) 1.14 eV [MP PMT 2004] (d) 0.23 eV
146. The work function of a metal is [RPMT 2004]
 (a) The energy for the electron to enter into the metal
 (b) The energy for producing X-ray
 (c) The energy for the electron to come out from metal surface
 (d) None of these
147. The minimum wavelength of photon is $5000 \text{ \AA}$, its energy will be
 (a) 2.5 eV (b) 50 V
 (c) 5.48 eV (d) 7.48 eV
148. Which of one is correct [DCE 1998]
 (a) $E^2 = p^2 c^2$ (b) $E^2 = p^2 c$
 (c) $E^2 = pc^2$ [Pb. PMT 2004] (d) $E^2 = p^2 / c^2$
149. The work function for metals A, B and C are respectively 1.92 eV , 2.0 eV and 5 eV . According to Einstein's equation, the metals which will emit photo electrons for a radiation of wavelength $4100 \text{ \AA}$ is/are [CBSE PMT 2001]
 (a) None of these (b) A only
 (c) A and B only (d) All the three metals
150. A photosensitive metallic surface has work function $h\nu_0$. If photons of energy $2h\nu_0$ fall on this surface the electrons come out with a maximum velocity of $4 \times 10^6 \text{ m/s}$. When the photon energy is increases to $5h\nu_0$ then maximum velocity of photo electron will be [CBSE PMT 2001]
 (a) $2 \times 10^6 \text{ m/s}$ [Pb. PET 2002] (b) $2 \times 10^7 \text{ m/s}$
 (c) $8 \times 10^5 \text{ m/s}$ (d) $8 \times 10^6 \text{ m/s}$
151. A photocell is illuminated by a small bright source placed 1 m away. When the same source of light is placed $\frac{1}{2} \text{ m}$ away, the number of electrons emitted by photo cathode would [CBSE PMT 2001; AIEEE 2005]
 (a) Decrease by a factor of 2 (b) Increase by a factor of 2
 (c) Decrease by a factor of 4 (d) Increase by a factor of 4
152. The magnitude of saturation photoelectric current depends upon [AFMC 2005]

- (a) Frequency (b) Intensity
(c) Work function (d) Stopping potential
153. For photoelectric emission, tungsten requires light of $2300 \text{ \AA}$. If light of $1800 \text{ \AA}$ wavelength is incident then emission [AFMC 2005]
(a) Takes place
(b) Don't take place
(c) May or may not take place
(d) Depends on frequency
154. The light rays having photons of energy 1.8 eV are falling on a metal surface having a work function 1.2 eV . What is the stopping potential to be applied to stop the emitting electrons
(a) 3 eV (b) 1.2 eV
(c) 0.6 eV (d) 1.4 eV
155. The incident photon involved in the photoelectric effect experiment.
(a) Completely disappears
(b) Comes out with an increased frequency
(c) Comes out with a decreased frequency
(d) Comes out without change in frequency
156. A photon of energy 8 eV is incident on metal surface of threshold frequency $1.6 \times 10^{15} \text{ Hz}$. The maximum kinetic energy of the photoelectrons emitted (in eV) (Take $h = 6 \times 10^{-34} \text{ Js}$).
(a) 1.6 (b) 6
(c) 2 (d) 1.2
- (d) Detect fault in radio receiving circuits
7. Hydrogen atom does not emit X-rays because [NCERT 1979; CPMT 1980, 90; RPET 1999]
(a) Its energy levels are too close to each other
(b) Its energy levels are too apart
(c) It is too small in size
(d) It has a single electron
8. X-rays were discovered by [NCERT 1977; BHU 2005]
(a) Becquerel (b) Roentgen
(c) Marie Curie (d) Von Laue [BHU 2005]
9. X-rays are [CPMT 1975; EAMCET 1995; RPET 2000; SCRA 1994]
(a) Stream of electrons [EAMCET 2005]
(b) Stream of positively charged particles
(c) Electromagnetic radiations of high frequency
(d) Stream of uncharged particles
10. The voltage applied across an X-rays tube is nearly [CPMT 1983]
(a) 10 V (b) 100 V
(c) 10^4 V (d) 10^6 V [RPET 2005]
11. The characteristic X-ray radiation is emitted, when [CPMT 1975, 80, 90; RPET 1999]
(a) The electrons are accelerated to a fixed energy
(b) The source of electrons emits a monoenergetic beam
(c) The bombarding electrons knock out electrons from the inner shell of the target atoms and one of the outer electrons falls into this vacancy
(d) The valence electrons in the target atoms are removed as a result of the collision

X-Rays

1. An X-ray tube is operated at 50 kV . The minimum wavelength produced is [CPMT 1996]
(a) $0.5 \text{ \AA}$ (b) $0.75 \text{ \AA}$
(c) $0.25 \text{ \AA}$ (d) $1 \text{ \AA}$
2. Which of the following wavelength falls in X-ray region [CPMT 1975; MP PMT 1984]
(a) $10000 \text{ \AA}$ (b) $1000 \text{ \AA}$
(c) $1 \text{ \AA}$ (d) $10^{-4} \text{ \AA}$
3. A metal block is exposed to beams of X-ray of different wavelength. X-rays of which wavelength penetrate most [NCERT 1980; JIPMER 2002]
(a) $2 \text{ \AA}$ (b) $4 \text{ \AA}$
(c) $6 \text{ \AA}$ (d) $8 \text{ \AA}$
4. X-rays and gamma rays are both electromagnetic waves. Which of the following statements is true [NCERT 1973]
(a) In general X-rays have larger wavelength than of gamma rays
(b) X-rays have smaller wavelength than that of gamma rays
(c) Gamma rays have smaller frequency than that of X-rays
(d) Wavelength and frequency of X-rays are both larger than that of gamma rays
5. In producing X-rays a beam of electrons accelerated by a potential difference V is made to strike a metal target. For what value of V , X-rays will have the lowest wavelength of $0.3094 \text{ \AA}$ [CPMT 1982; NCERT 1986, 87]
(a) 10 kV (b) 20 kV
(c) 30 kV (d) 40 kV
6. In radio therapy, X-rays are used to [CPMT 1972; BHU 2005]
(a) Detect bone fractures
(b) Treat cancer by controlled exposure
(c) Detect heart diseases
12. Molybdenum is used as a target element for production of X-rays because it is [CPMT 1980; RPET 1999]
(a) A heavy element and can easily absorb high velocity electrons
(b) A heavy element with a high melting point
(c) An element having high thermal conductivity
(d) Heavy and can easily deflect electrons
13. Mosley's law relates the frequencies of line X-rays with the following characteristics of the target element [CPMT 1980; NCERT 1985]
(a) Its density
(b) Its atomic weight
(c) Its atomic number
(d) Interplaner spacing of the atomic planes
14. Compton effect is associated with [CPMT 1971]
(a) α - rays (b) β - rays
(c) X-rays (d) Positive rays
15. X-rays are in nature similar to
(a) Beta rays (b) Gamma rays
(c) de-Broglie waves (d) Cathode rays
16. If the cathode-anode potential difference in an X-ray tube be 10^6 V , then the maximum energy of X-ray photon can be
(a) 10^6 J (b) 10^6 MeV
(c) 10^6 MeV (d) 10^6 KeV

17. The shortest wavelength of X-rays emitted from an X-ray tube depends on the
[MP PMT 1987; CPMT 1988, 92; IIT 1982]
- Current in the tube
 - Voltage applied to the tube
 - Nature of gas in the tube
 - Atomic number of target material
18. The wavelength of X-rays is of the order of
[CPMT 1983; MP PMT 1987; KCET 1994; JIPMER 1997]
- Centimetre
 - Micron (10^{-6} m)
 - Angstrom (10^{-10} m)
 - Metre
19. X-rays and γ -rays of the same energies may be distinguished by
- Their velocity
 - Their ionising power
 - Their intensity
 - Method of production
20. When a beam of accelerated electrons hits a target, a continuous X-ray spectrum is emitted from the target. Which of the following wavelength is absent in the X-ray spectrum, if the X-ray tube is operating at 40,000 volts
[MP PMT 1993; NCERT 1984; MNR 1995; RPMT 2002]
- 0.25 Å
 - 0.5 Å
 - 1.5 Å
 - 1.0 Å
21. For continuous X-rays produced wavelength is
- Inversely proportional to the energy of the electrons hitting the target
 - Inversely proportional to the intensity of the electron beam
 - Proportional to intensity of the electron beam
 - Proportional to target temperature
22. An X-ray has a wavelength of 0.010 Å. Its momentum is
[AFMC 1980; RPMT 1995; Pb. PMT 2004]
- 2.126×10^{-27} kg-m/sec
 - 6.626×10^{-27} kg-m/sec
 - 3.456×10^{-27} kg-m/sec
 - 3.313×10^{-27} kg-m/sec
23. X-rays are not used for radar purpose because
- They are not reflected by the target
 - They are not electromagnetic waves
 - They are completely absorbed by the air
 - They sometimes damage the target
24. A direct X-ray photograph of the intestines is not generally taken by the radiologists because
[CPMT 1986, 88]
- Intestines would burst on exposure to X-rays
 - The X-rays would not pass through the intestines
 - The X-rays will pass through the intestines without causing a good shadow for any useful diagnosis
 - A very small exposure of X-rays causes cancer in the intestines
25. The patient is asked to drink $BaSO_4$ for examining the stomach by X-rays because X-rays are
- Reflected by heavy atoms
 - Refracted by heavy atoms
 - Less absorbed by heavy atoms
 - More absorbed by heavy atoms
26. X-rays can be used to study crystal structure, if the wavelength lies in the range
- 2 Å to 0.1 Å
 - 10 Å to 5 Å
 - 50 Å to 10 Å
 - 100 Å to 50 Å
27. When the accelerating voltage applied on the electrons increased beyond a critical value
[CPMT 1975]
- Only the intensity of the various wavelengths is increased
 - Only the wavelength of characteristic relation is affected
 - The spectrum of white radiation is unaffected
 - The intensities of characteristic lines relative to the white spectrum are increased but there is no change in their wavelength
28. The X-ray beam coming from an X-ray tube will be
[IIT 1985; SCRA 1996; MP PET 1999]
- Monochromatic
 - Having all wavelengths smaller than a certain maximum wavelength
 - Having all wavelengths larger than a certain minimum wavelength
 - Having all wavelengths lying between a minimum and a maximum wavelength
29. The continuous X-rays spectrum produced by an X-ray machine at constant voltage has
[DPMT 1999]
- A maximum wavelength
 - A minimum wavelength
 - A single wavelength
 - A minimum frequency
30. The penetrating power of X-rays increases with the
[MP PMT 1984]
- Increase in its velocity
 - Increase in its frequency
 - Increase in its intensity
 - Decrease in its velocity
31. If λ_1 and λ_2 are the wavelengths of characteristic X-rays and gamma rays respectively, then the relation between them is
- $\lambda_1 = \frac{1}{\lambda_2}$
 - $\lambda_1 = \lambda_2$
 - $\lambda_1 > \lambda_2$
 - $\lambda_1 < \lambda_2$
32. The wavelength λ of the K_α line of characteristic X-ray spectra varies with atomic number approximately
[MP PMT 1987]
- $\lambda \propto Z$
 - $\lambda \propto \sqrt{Z}$
 - $\lambda \propto \frac{1}{Z^2}$
 - $\lambda \propto \frac{1}{\sqrt{Z}}$
33. The minimum frequency ν of continuous X-rays is related to the applied potential difference V as
- $\nu \propto \sqrt{V}$
 - $\nu \propto V$
 - $\nu \propto V^{3/2}$
 - $\nu \propto V^2$
34. If V be the accelerating voltage, then the maximum frequency of continuous X-rays is given by
[NCERT 1971; CPMT 1991; MP PET 2000; RPMT 2001; MP PMT 2002]
- $\frac{eh}{V}$
 - $\frac{hV}{e}$
 - $\frac{eV}{h}$
 - $\frac{h}{eV}$
35. The minimum wavelength of X-rays produced by electrons accelerated by a potential difference of volts is equal to
[CPMT 1986, 88, 91; RPMT 1997; RPMT 1997, 98; MP PET 1997, 98; MP PMT 1996, 98, 2003; UPSEAT 2005]

- (a) $\frac{eV}{hc}$ (b) $\frac{eh}{cV}$
(c) $\frac{hc}{eV}$ (d) $\frac{cV}{eh}$
36. The potential difference applied to an X-ray tube is increased. As a result, in the emitted radiation
[IIT 1988; ISM Dhanbad 1994; AIIMS 1997; MP PMT 1995, 2004]
(a) The intensity increases
(b) The minimum wavelength increases
(c) The intensity decreases
(d) The minimum wavelength decreases
37. A potential difference of 42,000 volts is used in an X-ray tube to accelerate electrons. The maximum frequency of the X-radiations produced is [MP PMT 1993]
(a) 10^{19} Hz (b) 10^{18} Hz
(c) 10^{16} Hz (d) 10^{20} Hz
($1\text{ eV} = 1.6 \times 10^{-19} \text{ J}$ and $h = 6.63 \times 10^{-34} \text{ J-sec}$)
38. Which of the following is accompanied by the characteristic X-ray emission [MP PET 1993]
(a) α -particle emission (b) Electron emission
(c) Positron emission (d) K-electron capture
39. X-rays are known to be electromagnetic radiations. Therefore the X-ray photon has [MP PET 1993]
(a) Electric charge
(b) Magnetic moment
(c) Both electric charge and magnetic moment
(d) Neither electric charge nor magnetic moment
40. X-rays of which of the following wavelengths are hardest
(a) 4 Å (b) 1 Å
(c) 0.1 Å (d) 2 Å
41. X-ray beam can be deflected by [CPMT 2000; BHU 2001; Pb. PMT 2002]
(a) Magnetic field (b) Electric field
(c) Both (a) and (b) (d) None of these
42. X-rays are produced due to [CPMT 1985; JIPMER 2002]
(a) Break up of molecules
(b) Changing in atomic energy level
(c) Changing in nuclear energy level
(d) Radioactive disintegration
43. X-rays region lies between [CPMT 1990]
(a) Short radiowave and visible region
(b) Visible and ultraviolet region
(c) Gamma rays and ultraviolet region
(d) Short radiowave and long radiowave
44. The structure of solid crystals is investigated by using [CPMT 1992; NCERT 1975; CBSE PMT 1992]
(a) Cosmic rays (b) X-rays
(c) Infrared radiations (d) γ -rays
45. In an X-rays tube, the intensity of the emitted X-rays beam is increased by [MNR 1992; RPMT 1996; UPSEAT 2000]
(a) Increasing the filament current
(b) Decreasing the filament current
(c) Increasing the target potential
(d) Decreasing the target potential
46. The binding energy of the innermost electron in tungsten is 40 keV. To produce characteristic X-rays using a tungsten target in an X-rays tube the potential difference V between the cathode and the anti-cathode should be [IIT 1985]
(a) $V < 40 \text{ kV}$ (b) $V \leq 40 \text{ kV}$
(c) $V > 40 \text{ kV}$ (d) $V \geq 40 \text{ kV}$
47. In above question the energy of the characteristic X-rays given out is
(a) Less than 40 keV (b) More than 40 keV
(c) Equal to 40 keV (d) $\geq 40 \text{ keV}$
48. The wavelength of most energetic X-rays emitted when a metal target is bombarded by 40 keV electrons, is approximately
($h = 6.62 \times 10^{-34} \text{ J-sec}$; $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$; $c = 3 \times 10^8 \text{ m/s}$) [MNR 1991; MP PMT 1999; UPSEAT 2000; Pb. PET 2004]
(a) 300 Å (b) 10 Å
(c) 4 Å (d) 0.31 Å
49. X-rays which can penetrate through longer distances in substance are called [EAMCET 1983]
(a) Soft X-rays (b) Continuous X-rays
(c) Hard X-rays (d) None of the above
50. An X-ray machine has an accelerating potential difference of 25,000 volts. By calculation the shortest wavelength will be obtained as
($h = 6.62 \times 10^{-34} \text{ J-sec}$; $e = 1.6 \times 10^{-19} \text{ coulomb}$) [MP PET 1994]
(a) 0.25 Å (b) 0.50 Å
(c) 1.00 Å (d) 2.50 Å
51. For the production of X-rays of wavelength 0.1 Å the minimum potential difference will be [MP PMT 1994; RPMT 1995]
(a) 12.4 kV (b) 24.8 kV
(c) 124 kV (d) 248 kV
52. Mosley measured the frequency (f) of the characteristic X-rays from many metals of different atomic number (Z) and represented his results by a relation known as Mosley's law. This law is (a, b are constants) [MP PMT 1994; RPMT 1996]
(a) $f = a(Z-b)^2$ (b) $Z = a(f-b)^2$
(c) $f^2 = a(Z-b)$ (d) $f = a(Z-b)^{1/2}$
53. Penetrating power of X-rays depends on [MP PMT 1994]
(a) Current flowing in the filament
(b) Applied potential difference
(c) Nature of the target
(d) All the above
54. The energy of a photon of characteristic X-rays from a Coolidge tube comes from [MP PET 1995]
(a) The kinetic energy of the striking electron
(b) The kinetic energy of the free electrons of the target
(c) The kinetic energy of the ions of the target
(d) An electronic transition of the target atom
55. An X-ray tube operates on 30 kV. What is the minimum wavelength emitted
($h = 6.6 \times 10^{-34} \text{ Js}$, $e = 1.6 \times 10^{-19} \text{ Coulomb}$, $c = 3 \times 10^8 \text{ ms}$) [MP PMT 1995; DPMT 2001, 03]
(a) 0.133 Å (b) 0.4 Å

- (c) $1.2 \text{ \AA}$ (d) $6.6 \text{ \AA}$
56. The wavelength of the most energetic X-ray emitted when a metal target is bombarded by 100 KeV electrons is approximately
(a) $12 \text{ \AA}$ (b) 4
(c) $0.31 \text{ \AA}$ (d) $0.124 \text{ \AA}$
57. An electron beam in an X-ray tube is accelerated through a potential difference of 50000 volts . These are then made to fall on a tungsten target. The shortest wavelength of the X-ray emitted by the tube is
(a) $2.5 \text{ \AA}$ (b) 0.25 nm
(c) 0.25 cm (d) 0.025 nm
58. For harder X-rays [MP PET 1997]
(a) The wavelength is higher
(b) The intensity is higher
(c) The frequency is higher
(d) The photon energy is lower
59. When cathode rays strike a metal target of high melting point with very high velocity, then [MP PMT 1997; AIIMS 1999]
(a) X-rays are produced
(b) Ealpha-rays are produced
(c) TV waves are produced
(d) Ultrasonic waves are produced
60. Penetrating power of X-rays can be increased by [MP PMT 1997, 2000]
(a) Increasing the potential difference between anode and cathode
(b) Decreasing the potential difference between anode and cathode
(c) Increasing the cathode filament current
(d) Decreasing the cathode filament current
61. K_{α} characteristic X-ray refers to the transition [MP PMT 1999]
(a) $n = 2$ to $n = 1$ (b) $n = 3$ to $n = 2$
(c) $n = 3$ to $n = 1$ (d) $n = 4$ to $n = 2$
62. X-rays are produced in X-ray tube operating at a given accelerating voltage. The wavelength of the continuous X-rays has values from
(a) 0 to ∞
(b) $\lambda_{\min}$ to ∞ , where $\lambda_{\min} > 0$
(c) 0 to $\lambda_{\max}$ where $\lambda_{\max} < \infty$
(d) $\lambda_{\min}$ to $\lambda_{\max}$, where $0 < \lambda_{\min} < \lambda_{\max} < \infty$
63. The wavelength of X-rays is [EAMCET (Med.) 1995]
(a) $2000 \text{ \AA}$ (b) $2 \text{ \AA}$
(c) 1 mm (d) 1 cm
64. The ratio of the energy of an X-ray photon of wavelength $1 \text{ \AA}$ to that of visible light of wavelength $5000 \text{ \AA}$ is [EAMCET (Med.) 1995]
(a) $1:5000$ (b) $5000:1$
(c) $1:25 \times 10$ (d) 25×10
65. According to Mosley's law, the frequency of a spectral line in X-ray spectrum varies as [EAMCET (Med.) 1995; Pb. PMT 1999]
(a) Atomic number of the element
(b) Square of the atomic number of the element
(c) Square root of the atomic number of the element
(d) Fourth power of the atomic number of the element
66. For the structural analysis of crystals, X-rays are used because
(a) X-rays have wavelength of the order of interatomic spacing
(b) X-rays are highly penetrating radiations
(c) Wavelength of X-rays is of the order of nuclear size
(d) X-rays are coherent radiations [MP PET 1996]
67. The essential distinction between X-rays and γ -rays is that [BHU 1994; RPMT 1991; JIPMER 2001, 02]
(a) γ -rays have smaller wavelength than X-rays
(b) γ -rays originate from nucleus while X-rays emanate from outer part of the atom [MP PET 1997]
(c) γ -rays have greater ionizing power than X-rays
(d) γ -rays are more penetrating than X-rays
68. The minimum wavelength of the X-rays produced by electrons accelerated through a potential difference of $V \text{ volts}$ is directly proportional to [CBSE PMT 1996]
(a) $\sqrt{V}$ (b) V^2
(c) $1/\sqrt{V}$ (d) $1/V$
69. What determines the hardness of the X-rays obtained from the Coolidge tube [RPMT 1996]
(a) Current in the filament
(b) Pressure of air in the tube
(c) Nature of target
(d) Potential difference between cathode and target
70. The most penetrating radiation out of the following is [CBSE PMT 1997]
(a) X-rays (b) β -rays
(c) α -particles (d) γ -rays
71. On increasing the number of electrons striking the anode of an X-ray tube, which one of the following parameters of the resulting X-rays would increase [IIT 1998; BVP 2003] [SCRA 1998; DPMT 2000]
(a) Penetration power (b) Frequency
(c) Wavelength (d) Intensity
72. What kV potential is to be applied on X-ray tube so that minimum wavelength of emitted X-rays may be $1 \text{ \AA}$ ($h = 6.625 \times 10^{-34} \text{ J-sec}$)
(a) 12.42 kV (b) 12.84 kV
(c) 11.98 kV (d) 10.78 kV
73. X-rays cannot be deflected by means of an ordinary grating due to [Pb. PMT 1995]
(a) Large wavelength (b) High speed
(c) Short wavelength (d) None of these
74. Consider the following two statements A and B and identify the correct choice in the given answer
A: The characteristic X-ray spectrum depends on the nature of the material of the target.
B: The short wavelength limit of continuous X-ray spectrum varies inversely with the potential difference applied to the X-rays tube [EAMCET (Med.) 2000]
(a) A is true and B is false (b) A is false and B is true
(c) Both A and B are true (d) Both A and B are false
75. The wavelength of an X-ray photon of wavelength $1.65 \text{ \AA}$ is [IIT 1992; JIPMER 2000]
($h = 6.6 \times 10^{-34} \text{ J-sec}$, $c = 3 \times 10^8 \text{ ms}^{-1}$, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$) [EAMCET (Engg.) 2000]

- (a) 3.5 keV (b) 5.5 keV
(c) 7.5 keV (d) 9.5 keV
76. If $\lambda = 10 \text{ \AA}$, then it corresponds to [DCE 2000]
(a) Infra-red (b) Microwave
(c) Ultra-violet (d) X-rays
77. Bragg's law for X-rays is [UPSEAT 2001]
(a) $d \sin \theta = 2n\lambda$ (b) $2d \sin \theta = n\lambda$
(c) $n \sin \theta = 2\lambda d$ (d) None of these
78. The X-rays produced in a Coolidge tube of potential difference 40 V have minimum wavelength of [MH CET (Med.) 2001]
(a) $3.09 \times 10^{-8} \text{ m}$ (b) $5.09 \times 10^{-8} \text{ m}$
(c) $4.09 \times 10^{-8} \text{ m}$ (d) $1.09 \times 10^{-8} \text{ m}$
79. For the production of X-rays, the target should be made of
(a) Steel (b) Copper
(c) Aluminum (d) Tungsten
80. Intensity of X-rays depends upon the number of [SCRA 1998; DPMT 2000; AFMC 2001]
(a) Electrons (b) Protons
(c) Neutrons (d) Positrons
81. In an X-ray tube electrons bombarding the target produce X-rays of minimum wavelength 1 Å. What must be the energy of bombarding electrons [KCET 2001]
(a) 13375 eV (b) 12375 eV
(c) 14375 eV (d) 15375 eV
82. If energy of K-shell electron is - 40000 eV and if 60000 V potential is applied at Coolidge tube then which of the following X-ray will get form [RPET 2001]
(a) Continuous
(b) White X-rays
(c) Continuous and all series of characteristic
(d) None of these
83. For production of characteristic K_{β} X-rays, the electron transition is [MP PET 2001]
(a) $n = 2$ to $n = 1$ (b) $n = 3$ to $n = 2$
(c) $n = 3$ to $n = 1$ (d) $n = 4$ to $n = 2$
84. Penetrating power of X-rays does not depend on [MP PET 2001]
(a) Wavelength (b) Energy
(c) Potential difference (d) Current in the filament
85. The potential difference applied to an X-ray tube is 5 kV and the current through it is 3.2 mA. Then the number of electrons striking the target per second is [IIT-JEE (Screening) 2002]
(a) 2×10^{16} (b) 5×10^{16}
(c) 1×10^{17} (d) 4×10^{15}
86. For the production of characteristic K_{γ} X-ray, the electron transition is [BHU 2002]
(a) $n = 2$ to $n = 1$ (b) $n = 3$ to $n = 2$
(c) $n = 3$ to $n = 1$ (d) $n = 4$ to $n = 1$
87. When X rays pass through a strong uniform magnetic field, Then they [MP PET 2002; RPMT 2002, 03]
(a) Do not get deflected at all
(b) Get deflected in the direction of the field
(c) Get deflected in the direction opposite to the field
(d) Get deflected in the direction perpendicular to the field
88. If the potential difference applied across X-ray tube is V volts, then approximately minimum wavelength of the emitted X-rays will be [RPMT 1995; CBSE PMT 1996]
(a) $\frac{1227}{\sqrt{V}} \text{ \AA}$ (b) $\frac{1240}{V} \text{ \AA}$
(c) $\frac{2400}{V} \text{ \AA}$ (d) $\frac{12400}{V} \text{ \AA}$
89. What is the difference between soft and hard X-rays [MP PMT 2002; AIIMS 2002]
(a) Velocity (b) Intensity
(c) Frequency (d) Polarization
90. X-ray will travel minimum distance in [MP PET 2003]
(a) Air (b) Iron
(c) Wood (d) Water
91. The minimum wavelength of X-ray emitted by X-rays tube is 0.4125 Å. The accelerating voltage is [BHU 2003; CPMT 2004; MP PMT 2005]
(a) 30 kV (b) 50 kV
(c) 80 kV (d) 60 kV
92. Characteristic X-rays are produced due to [AIIMS 2003]
(a) Transfer of momentum in collision of electrons with target atoms
(b) Transition of electrons from higher to lower electronic orbits in an atom
(c) Heating of the target
(d) Transfer of energy in collision of electrons with atoms in the target
93. X-rays when incident on a metal [BCECE 2003; RPMT 2003]
(a) Exert a force on it (b) Transfer energy to it
(c) Transfer pressure to it (d) All of the above
94. The minimum wavelength of X-rays produced in a Coolidge tube operated at potential difference of 40 kV is [BCECE 2003; RPET 2002, 03]
(a) 0.31 Å (b) 3.1 Å
(c) 31 Å (d) 311 Å
95. The potential difference between the cathode and the target in a Coolidge tube is 100 kV. The minimum wavelength of the X-rays emitted by the tube is [Pb. PMT 2004]
(a) 0.66 Å (b) 9.38 Å
(c) 0.246 Å (d) 0.123 Å
96. X-rays are produced by accelerating electrons by voltage V and let they strike a metal of atomic number Z. The highest frequency of X-rays produced is proportional to [UPSEAT 2004]
(a) V (b) Z
(c) $(Z - 1)$ (d) $(Z - 1)^2$
97. If the operating potential of an X-ray tube is 50 kV, the velocity of X-rays coming out of it [RPMT 2003]
(a) $4 \times 10^4 \text{ m/s}$ (b) $3 \times 10^8 \text{ m/s}$
(c) 10^8 m/s (d) 3 m/s

98. If the voltage of X-ray tube is doubled, the intensity of X-rays will become [RPMT 2003]
 (a) Half (b) Unchanged
 (c) Double (d) Four times
99. If the minimum wavelength obtained in an X-ray tube is $2.5 \times 10^{-10} \text{ m}$, the operating potential of the tube will be [RPMT 2003]
 (a) 2 kV (b) 3 kV
 (c) 4 kV (d) 5 kV
100. The wavelength of X-rays decreases, when [RPMT 2002]
 (a) Temperature of target is increased
 (b) Intensity of electron beam is increased
 (c) K.E. of electrons striking the target is increased
 (d) K.E. of electrons striking the target is decreased
101. X-rays are produced in laboratory by [RPMT 1998]
 (a) Radiation
 (b) Decomposition of the atom
 (c) Bombardment of high energy electron on heavy metal
 (d) None of these
102. In vacuum an electron of energy 10 keV hits tungsten target, then emitted radiation will be [RPMT 2001]
 (a) Cathode rays (b) X-rays
 (c) Infrared rays (d) Visible spectrum
103. X-rays of $\lambda = 1 \text{ \AA}$ have frequency [DCE 1998]
 (a) $3 \times 10^8 \text{ Hz}$ (b) $3 \times 10^{18} \text{ Hz}$
 (c) $3 \times 10^{10} \text{ Hz}$ (d) $3 \times 10^{15} \text{ Hz}$
104. Solid targets of different elements are bombarded by highly energetic electron beams. The frequency (f) of the characteristic X-rays emitted from different targets varies with atomic number Z as [AIIMS 2005]
 (a) $f \propto \sqrt{Z}$ (b) $f \propto Z^2$
 (c) $f \propto Z$ (d) $f \propto Z^{3/2}$
105. Compton effect shows that [DPMT 1995]
 (a) X-rays are waves
 (b) X-rays have high energy
 (c) X-rays can penetrate matter
 (d) Photons have momentum
106. An X-ray tube with a copper target emits Cu K_α line of wavelength 1.50 Å. What should be the minimum voltage through which electrons are to be accelerated to produce this wavelength of X rays
 $(h = 6.63 \times 10^{-34} \text{ J-sec}, c = 3 \times 10^8 \text{ m/s})$ [Orissa JEE 1996]
 (a) 8280 V (b) 828 V
 (c) 82800 V (d) 8.28 V
107. In X-ray spectrum wavelength λ of line K_α depends on atomic number Z as [RPMT 1995; DCE 2002]
 (a) $\lambda \propto Z^2$ (b) $\lambda \propto (Z-1)^2$
 (c) $\lambda \propto \frac{1}{(Z-1)}$ (d) $\lambda \propto \frac{1}{(Z-1)^2}$
108. Absorption of X-ray is maximum in which of the following different sheets [RPMT 1995]
 (a) Copper (b) Gold
- (c) Beryllium (d) Lead
109. The wavelength of K_α line in copper is 1.54 Å. The ionisation energy of K electron in copper in Joule is [EAMCET 1984]
 (a) 11.2×10^{-27} (b) 12.9×10^{-16}
 (c) 1.7×10^{-15} (d) 10×10^{-16}
110. The wavelength of K_α line for an element of atomic number 43 is λ . Then the wavelength of K_α line for an element of atomic number 29 is
 (a) $\frac{43}{29} \lambda$ (b) $\frac{42}{28} \lambda$
 (c) $\frac{9}{4} \lambda$ (d) $\frac{4}{9} \lambda$
111. In X-ray experiment K_α , K_β denotes [DCE 2005]
 (a) Characteristic
 (b) Continuous wavelength
 (c) α , β -emissions respectively
 (d) None of these

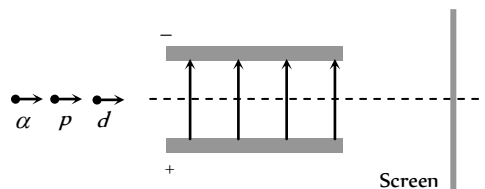
G Critical Thinking

Objective Questions

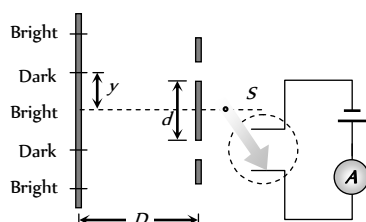
- A $1\mu A$ beam of protons with a cross-sectional area of 0.5 sq. mm is moving with a velocity of $3 \times 10^4\text{ ms}^{-1}$. Then charge density of beam is [CPMT 2002]
 - $6.6 \times 10^{-4}\text{ C/m}^3$
 - $6.6 \times 10^{-5}\text{ C/m}^3$
 - $6.6 \times 10^{-6}\text{ C/m}^3$
 - None of these
- A particle of mass M at rest decays into two particles of masses m and m_1 , having non-zero velocities. The ratio of the de-Broglie wavelengths of the particles, λ_1 / λ_2 is [IIT-JEE 1999; KCET 2003]
 - m_1 / m_2
 - m_2 / m_1
 - 1.0
 - $\sqrt{m_2} / \sqrt{m_1}$
- A photon and an electron have equal energy E . $\lambda_{\text{photon}} / \lambda_{\text{electron}}$ is proportional to [UPSEAT 2003; IIT-JEE (Screening) 2004]
 - $\sqrt{E}$
 - $1/\sqrt{E}$
 - $1/E$
 - Does not depend upon E
- When photon of energy 4.25 eV strike the surface of a metal A , the ejected photoelectrons have maximum kinetic energy $T_1\text{ eV}$ and de-Broglie wavelength λ_A . The maximum kinetic energy of photoelectrons liberated from another metal B by photon of energy 4.70 eV is $T_B = (T_A - 1.50)\text{ eV}$. If the de-Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$, then
 - The work function of A is 2.25 eV
 - The work function of B is 4.20 eV
 - $T_A = 2.00\text{ eV}$
 - $T_B = 2.75\text{ eV}$
- An image of the sun is formed by a lens of focal length of 30 cm on the metal surface of a photoelectric cell and a photoelectric current I is produced. The lens forming the image is then replaced by another of the same diameter but of focal length 15 cm . The photoelectric current in this case is
 - $\frac{I}{2}$
 - I
 - $2I$
 - $4I$
- When an inert gas is filled in the place vacuum in a photo cell, then
 - Photo-electric current is decreased
 - Photo-electric current is increased
 - Photo-electric current remains the same
 - Decrease or increase in photo-electric current does not depend upon the gas filled
- A photon of $1.7 \times 10^{-13}\text{ Joules}$ is absorbed by a material under special circumstances. The correct statement is [MP PET 1999; JIPMER 2000]
 - Electrons of the atom of absorbed material will go the higher energy states
 - Electron and positron pair will be created
 - Only positron will be produced
 - Photoelectric effect will occur and electron will be produced
- The maximum velocity of an electron emitted by light of wavelength λ incident on the surface of a metal of work function ϕ , is [MP PMT/PET 1998]
 - $\left[\frac{2(hc + \lambda\phi)}{m\lambda} \right]^{1/2}$
 - $\frac{2(hc - \lambda\phi)}{m}$
 - $\left[\frac{2(hc - \lambda\phi)}{m\lambda} \right]^{1/2}$
 - $\left[\frac{2(h\lambda - \phi)}{m} \right]^{1/2}$

Where h = Planck's constant, m = mass of electron and c = speed of light.
- When a point source of monochromatic light is at a distance of 0.2 m from a photoelectric cell, the cut-off voltage and the saturation current are 0.6 volt and 18 mA respectively. If the same source is placed 0.6 m away from the photoelectric cell, then [IIT JEE 1992; MP PMT 1999]
 - The stopping potential will be 0.2 V
 - The stopping potential will be 0.6 V
 - The saturation current will be 6 mA
 - The saturation current will be 18 mA
- In a photoemissive cell with executing wavelength λ , the fastest electron has speed v . If the exciting wavelength is changed to $3\lambda/4$, the speed of the fastest emitted electron will be
 - $v(3/4)^{1/2}$
 - $v(4/3)^{1/2}$
 - Less than $v(4/3)^{1/2}$
 - Greater than $v(4/3)^{1/2}$
- Ultraviolet light of wavelength 300 nm and intensity 1.0 watt/m falls on the surface of a photosensitive material. If 1% of the incident photons produce photoelectrons, then the number of photoelectrons emitted from an area of 1.0 cm^2 of the surface is nearly
 - $9.61 \times 10^{14}\text{ per sec}$
 - $4.12 \times 10^{13}\text{ per sec}$
 - $1.51 \times 10^{12}\text{ per sec}$ [IIT-JEE 1994]
 - $2.13 \times 10^{11}\text{ per sec}$
- Photoelectric emission is observed from a metallic surface for frequencies ν_1 and ν_2 of the incident light rays ($\nu_1 > \nu_2$). If the maximum values of kinetic energy of the photoelectrons emitted in the two cases are in the ratio of $1:k$, then the threshold frequency of the metallic surface is [EAMCET (Engg.) 2001]
 - $\frac{\nu_1 - \nu_2}{k - 1}$
 - $\frac{k\nu_1 - \nu_2}{k - 1}$
 - $\frac{k\nu_2 - \nu_1}{k - 1}$ [Mafatpal MEE-1995]
 - $\frac{\nu_2 - \nu_1}{k}$
- Light from a hydrogen discharge tube is incident on the cathode of a photoelectric cell the work function of the cathode surface is 4.2 eV . In order to reduce the photo-current to zero the voltage of the anode relative to the cathode must be made [MP PMT 1997]
 - -4.2 V
 - -9.4 V
 - -17.8 V
 - $+9.4\text{ V}$
- Work function of lithium and copper are respectively 2.3 eV and 4.0 eV . Which one of the metal will be useful for the photoelectric cell working with visible light? ($h = 6.6 \times 10^{-34}\text{ J-s}$, $c = 3 \times 10^8\text{ m/s}$)
 - Lithium
 - Copper
 - Both
 - None of these
- X-rays of wavelength $0.1\text{ \AA}$ allowed to fall on a metal get scattered. The wavelength of scattered radiation is $0.111\text{ \AA}$. If $h = 6.624 \times 10^{-34}\text{ J-s}$ and $m_e = 9 \times 10^{-31}\text{ kg}$, then the direction of the scattered photons will be
 - $\cos^{-1}(0.547)$
 - $\cos^{-1}(0.4484)$

- (c) $\cos(0.5)$ (d) $\cos(0.3)$
16. The largest distance between the interatomic planes of a crystal is 10 cm. The upper limit for the wavelength of X-rays which can be usefully studied with this crystal is [CPMT 1984]
- (a) 1 Å (b) 2 Å
(c) 10 Å (d) 20 Å
17. An X-ray tube is operating at 50 kV and 20 mA. The target material of the tube has a mass of 1.0 kg and specific heat $495 \text{ J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$. One percent of the supplied electric power is converted into X-rays and the entire remaining energy goes into heating the target. Then
- (a) A suitable target material must have a high melting temperature
(b) A suitable target material must have low thermal conductivity
(c) The average rate of rise of temperature of target would be $2 \text{ } ^\circ\text{C/s}$
(d) The minimum wavelength of the X-rays emitted is about $0.25 \times 10^{-10} \text{ m}$
18. The wavelength of K_α X-rays produced by an X-ray tube is 0.76 Å. The atomic number of the anode material of the tube is
- (a) 20 (b) 60
(c) 40 (d) 80
19. X-ray beam of intensity I_0 passes through an absorption plate of thickness d . If absorption coefficient of material of plate is μ , the correct statement regarding the transmitted intensity I of X-ray is
- (a) $I = I_0(1 - e^{-\mu d})$ (b) $I = I_0 e^{-\mu d}$
(c) $I = I_0(1 - e^{-\mu/d})$ (d) $I = I_0 e^{-\mu/d}$
20. The K_α X-ray emission line of tungsten occurs at $\lambda = 0.021 \text{ nm}$. The energy difference between K and L levels in this atom is about
- (a) 0.51 MeV (b) 1.2 MeV
(c) 59 KeV (d) 13.6 eV
21. Electrons with energy 80 keV are incident on the tungsten target of an X-ray tube. K shell electrons of tungsten have ionization energy 72.5 keV. X-rays emitted by the tube contain only
- (a) A continuous X-ray spectrum (Bremsstrahlung) with a minimum wavelength of $\sim 0.155 \text{ Å}$
(b) A continuous X-ray spectrum (Bremsstrahlung) with all wavelengths
(c) The characteristic X-rays spectrum of tungsten
(d) A continuous X-ray spectrum (Bremsstrahlung) with a minimum wavelength of $\sim 0.155 \text{ Å}$ and the characteristic X-ray spectrum of tungsten
22. The X-ray wavelength of L_α line of platinum ($Z=78$) is 1.30 Å. The X-ray wavelength of L_α line of Molybdenum ($Z=42$) is
- (a) 5.41 Å (b) 4.20 Å
(c) 2.70 Å (d) 1.35 Å
23. The ratio of de-Broglie wavelengths of molecules of hydrogen and helium which are at temperature 27°C and 127°C respectively is
- (a) $\frac{1}{2}$ (b) $\sqrt{\frac{3}{8}}$
- (c) $\sqrt{\frac{8}{3}}$ (d) 1
24. A silver ball of radius 4.8 cm is suspended by a thread in the vacuum chamber. UV light of wavelength 200 nm is incident on the ball for some times during which a total energy of $1 \times 10^{-7} \text{ J}$ falls on the surface. Assuming on an average one out of 10 photons incident is able to eject electron. The potential on sphere will be
- (a) 1 V (b) 2 V
(c) 3 V (d) Zero
25. A photon of wavelength 6630 Å is incident on a totally reflecting [IIT 1995] surface. The momentum delivered by the photon is equal to
- (a) $6.63 \times 10^{-28} \text{ kg-m/sec}$ (b) $2 \times 10^{-28} \text{ kg-m/sec}$
(c) $10^{-28} \text{ kg-m/sec}$ (d) None of these
26. The ratio of de-Broglie wavelength of a α -particle to that of a proton being subjected to the same magnetic field so that the radii of their path are equal to each other assuming the field induction vector $\vec{B}$ is perpendicular to the velocity vectors of the α -particle and the proton is
- (a) 1 (b) $\frac{1}{4}$
[IIT 1996]
(c) $\frac{1}{2}$ (d) 2
27. K_α wavelength emitted by an atom of atomic number $Z = 11$ is λ . Find the atomic number for an atom that emits K_α radiation with wavelength $\frac{\lambda}{4}$. [IIT-JEE (Screening) 2005]
[MP PET 1999]
- (a) $Z = 6$ (b) $Z = 4$
(c) $Z = 11$ (d) $Z = 44$
28. The potential energy of a particle of mass m is given by
- $$U(x) = \begin{cases} E_0; & 0 \leq x \leq 1 \\ \text{[IIT 1997 Cancelled]} > 1 \end{cases}$$
- λ_1 and λ_2 are the de-Broglie wavelengths of the particle, when $0 \leq x \leq 1$ and $x > 1$ respectively. If the total energy of particle is $2E_0$, the ratio $\frac{\lambda_1}{\lambda_2}$ will be [IIT-JEE (Screening) 2000]
- (a) 2 (b) 1
(c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$
29. Rest mass energy of an electron is 0.51 MeV. If this electron is moving with a velocity 0.8 c (where c is velocity of light in vacuum), then kinetic energy of the electron should be. [IIT-JEE (Mains) 2005]
[IIT-JEE (Eng.) 2000]
- (a) 0.28 MeV (b) 0.34 MeV
(c) 0.39 MeV (d) 0.46 MeV
30. A proton, a deuteron and an α -particle having the same momentum, enters a region of uniform electric field between the parallel plates of a capacitor. The electric field is perpendicular to the initial path of the particles. Then the ratio of deflections suffered by them is



- (a) 1 : 2 : 8 (b) 1 : 2 : 4
(c) 1 : 1 : 2 (d) None of these
31. In order to coincide the parabolas formed by singly ionised ions in one spectrograph and doubly ionized ions in the other Thomson's mass spectrograph, the electric fields and magnetic fields are kept in the ratios 1 : 2 and 3 : 2 respectively. Then the ratio of masses of the ions is
(a) 3 : 4 (b) 1 : 3
(c) 9 : 4 (d) None of these
32. Let λ_α , λ_β and λ'_α denote the wavelengths of the X-rays of the K_α , K_β and L_α lines in the characteristic X-rays for a metal
(a) $\lambda_\alpha > \lambda'_\alpha > \lambda_\beta$ (b) $\lambda'_\alpha > \lambda_\beta > \lambda_\alpha$
(c) $\frac{1}{\lambda_\beta} = \frac{1}{\lambda_\alpha} + \frac{1}{\lambda'_\alpha}$ (d) $\frac{1}{\lambda_\alpha} + \frac{1}{\lambda_\beta} = \frac{1}{\lambda'_\alpha}$
33. The minimum intensity of light to be detected by human eye is 10^{-10} W/m^2 . The number of photons of wavelength $5.6 \times 10^{-7} \text{ m}$ entering the eye, with pupil area 10^{-6} m^2 , per second for vision will be nearly
(a) 100 (b) 200
(c) 300 (d) 400
34. In X-ray tube when the accelerating voltage V is halved, the difference between the wavelength of K_α line and minimum wavelength of continuous X-ray spectrum
(a) Remains constant
(b) Becomes more than two times
(c) Becomes half
(d) Becomes less than two times
35. In a photocell bichromatic light of wavelength 2475 Å and 6000 Å are incident on cathode whose work function is 4.8 eV. If a uniform magnetic field of $3 \times 10^4 \text{ Tesla}$ exists parallel to the plate, the radius of the path describe by the photoelectron will be (mass of electron = $9 \times 10^{-31} \text{ kg}$)
(a) 1 cm (b) 5 cm
(c) 10 cm (d) 25 cm
36. Two metallic plates A and B, each of area $5 \times 10^{-4} \text{ m}^2$ are placed parallel to each other at a separation of 1 cm. Plate B carries a positive charge of 33.7 pC. A monochromatic beam of light, with photons of energy 5 eV each, starts falling on plate A at $t = 0$, so that 10^{10} photons fall on it per square meter per second. Assume that one photoelectron is emitted for every 10^{10} incident photons. Also assume that all the emitted photoelectrons are collected by plate B and the work function of plate A remains constant at the value 2 eV. Electric field between the plates at the end of 10 seconds is
(a) $2 \times 10^4 \text{ N/C}$ (b) 10^4 N/C
(c) $5 \times 10^4 \text{ N/C}$ (d) Zero
37. In the following arrangement $y = 1.0 \text{ mm}$, $d = 0.24 \text{ mm}$ and $D = 1.2 \text{ m}$. The work function of the material of the emitter is 2.2 eV. The stopping potential V needed to stop the photo current will be

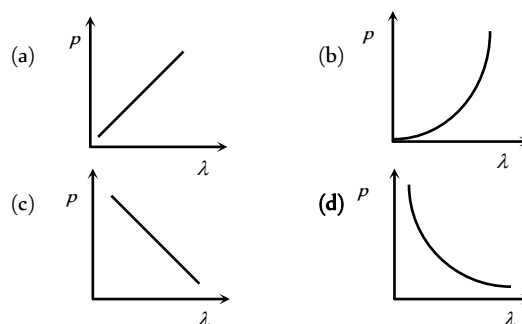


- (a) 0.9 V (b) 0.5 V
(c) 0.4 V (d) 0.1 V

38. The eye can detect 5×10^6 photons per square metre per sec of green light ($\lambda = 5000 \text{ Å}$) while the ear can detect $10^{-13} \text{ (W/m}^2\text{)}$. The factor by which the eye is more sensitive as a power detector than the ear is close to
(a) 5 (b) 10
(c) 10 (d) 15
39. A photon collides with a stationary hydrogen atom in ground state inelastically. Energy of the colliding photon is 10.2 eV. After a time interval of the order of micro second another photon collides with same hydrogen atom inelastically with an energy of 15 eV. What will be observed by the detector [IIT-JEE (Screening) 2005]
(a) 2 photon of energy 10.2 eV
(b) 2 photon of energy of 1.4 eV
(c) One photon of energy 10.2 eV and an electron of energy 1.4 eV
(d) One photon of energy 10.2 eV and another photon of 1.4 eV

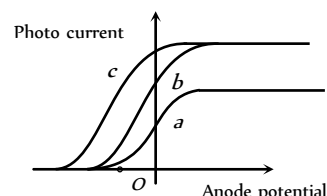
Graphical Questions

1. The curve drawn between velocity and frequency of photon in vacuum will be a [MP PET 2000]
(a) Straight line parallel to frequency axis
(b) Straight line parallel to velocity axis
(c) Straight line passing through origin and making an angle of 45° with frequency axis
(d) Hyperbola
2. Which of the following figure represents the variation of particle momentum and the associated de-Broglie wavelength



3. The figure shows the variation of photocurrent with anode potential for a photo-sensitive surface for three different radiations. Let I_a , I_b and I_c be the intensities and f_a , f_b and f_c be the frequencies for the curves a, b and c respectively [IIT-JEE (Screening) 2004]

- (a) $f_a = f_b$ and $I_a \neq I_b$
(b) $f_a = f_c$ and $I_a = I_c$

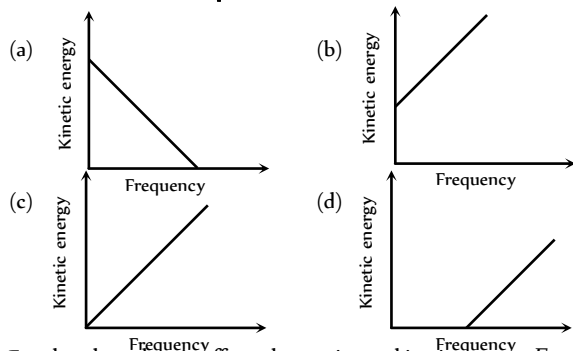


(c) $f_a = f_b$ and $I_a = I_b$

(d) $f_a = f_b$ and $I_a = I_b$

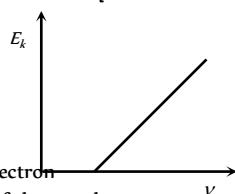
4. According to Einstein's photoelectric equation, the graph between the kinetic energy of photoelectrons ejected and the frequency of incident radiation is

[MP PMT 1994; CBSE PMT 1996; CBSE PMT 2004]



5. For the photoelectric effect, the maximum kinetic energy E_k of the emitted photoelectrons is plotted against the frequency ν of the incident photons as shown in the figure. The slope of the curve gives

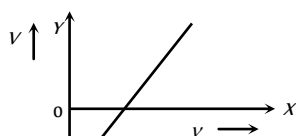
[CPMT 1987; MP PET 2001; DPMT 2002]



- (a) Charge of the electron
(b) Work function of the metal
(c) Planck's constant
(d) Ratio of the Planck's constant to electronic charge
6. The stopping potential V for photoelectric emission from a metal surface is plotted along Y-axis and frequency ν of incident light along X-axis. A straight line is obtained as shown. Planck's constant is given by

[CPMT 1987;

Similar to MP PMT 2000; Kerala PET 2001]



- (a) Slope of the line
(b) Product of slope on the line and charge on the electron
(c) Product of intercept along Y-axis and mass of the electron
(d) Product of Slope and mass of electron
7. In an experiment on photoelectric effect the frequency f of the incident light is plotted against the stopping potential V_0 . The work function of the photoelectric surface is given by (e is electronic charge)

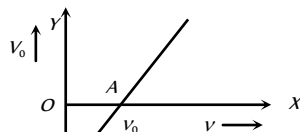
[CPMT 1987]

(a) $OB \times e$ in eV

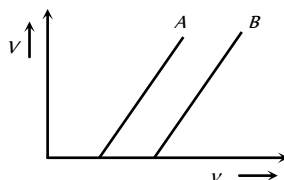
(b) OB in volt

(c) OA in eV

(d) The slope of the line AB



8. The stopping potential as a function of the frequency of the incident radiation is plotted for two different photoelectric surfaces A and B . The graphs show that work function of A is



(a) Greater than that of B

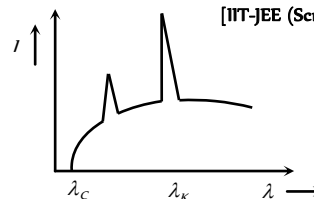
(b) Smaller than that of B

(c) Equal to that of B

(d) No inference can be drawn about their work functions from the given graphs

9. The intensity of X-rays from a Coolidge tube is plotted against wavelength as shown in the figure. The minimum wavelength found is λ_c and the wavelength of the K_α line is λ_k . As the accelerating voltage is increased

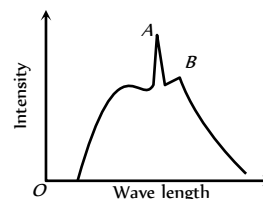
[IIT-JEE (Screening) 2001]



- (a) $(\lambda_k - \lambda_c)$ increases (b) $(\lambda_k - \lambda_c)$ decreases
(c) λ_k increases (d) λ_k decreases

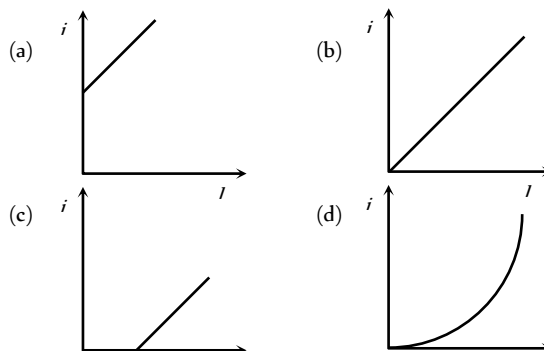
10. The figure represents the observed intensity of X-rays emitted by an X-ray tube as a function of wavelength. The sharp peaks A and B denote

[CBSE PMT 1995]

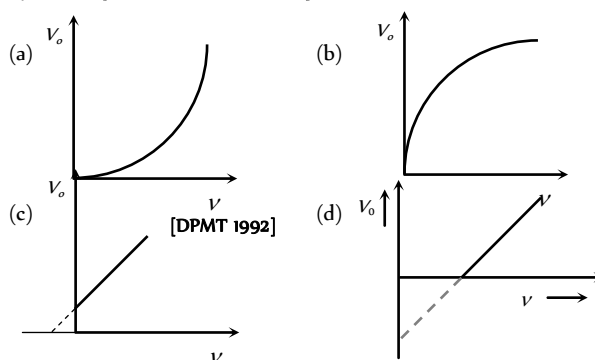


- (a) Band spectrum (b) Continuous spectrum
(c) Characteristic radiations (d) White radiations

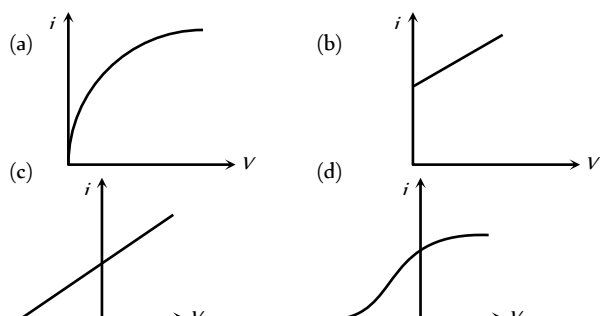
11. The graph between intensity of light falling on a metallic plate (I) with the current (i) generated is [DCE 2001]



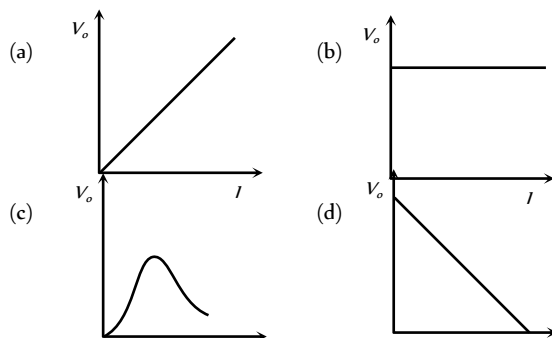
12. For a photoelectric cell the graph showing the variation of cut of voltage (V) with frequency (ν) of incident light is best represented by [DCE 2001; MP PET 2003]



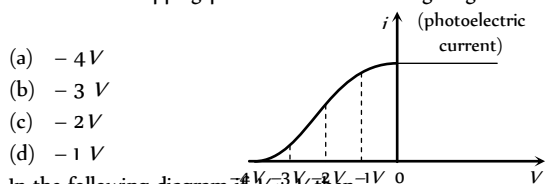
13. The curve between current (i) and potential difference (V) for a photo cell will be



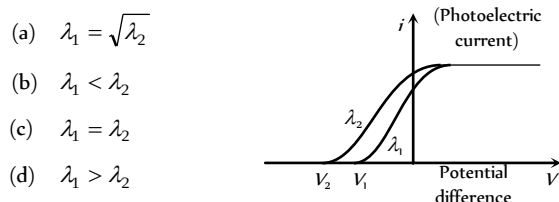
14. The correct curve between the stopping potential (V) and intensity of incident light (I) is



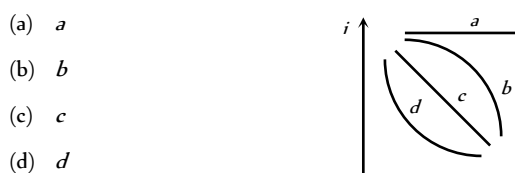
15. The value of stopping potential in the following diagram



16. In the following diagram if $V_1 > V_2$ then

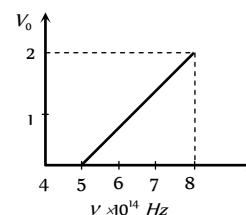


17. A point source of light is used in an experiment on photoelectric effect. Which of the following curves best represents the variation of photo current (i) with distance (d) of the source from the emitter



18. According to Einstein's photoelectric equation, the plot of the kinetic energy of the emitted photo electrons from a metal versus the frequency, of the incident radiation gives a straight line whose slope
- (a) Is the same for all metals and independent of the intensity of the radiation
- (b) Depends on the intensity of the radiation
- (c) Depends both on the intensity of the radiation and the metal used
- (d) Depends on the nature of the metals used

19. The stopping potential (V_0) versus frequency (ν) plot of a substance is shown in figure the threshold wave length is

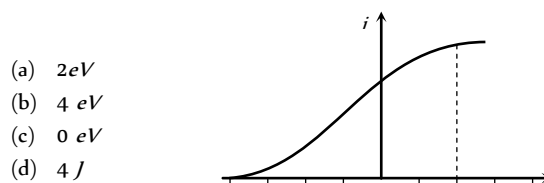


- (a) $5 \times 10^{14} m$
- (b) $6000 \text{ \AA}$
- (c) $5000 \text{ \AA}$
- (d) Can not be estimated from given data

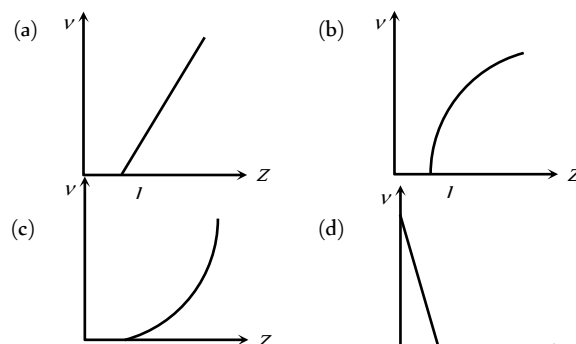
20. Figure represents a graph of kinetic energy (K) of photoelectrons (in eV) and frequency (ν) for a metal used as cathode in photoelectric experiment. The work function of metal is



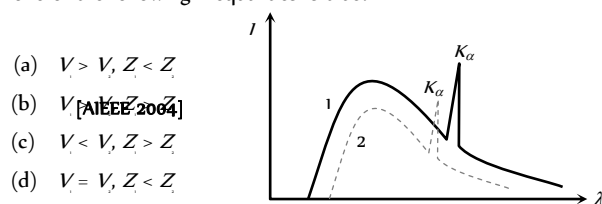
21. Figure represents the graph of photo current I versus applied voltage (V). The maximum energy of the emitted photoelectrons is



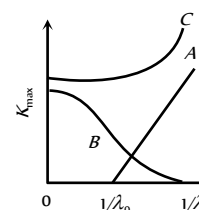
22. The graph that correctly represents the relation of frequency ν of a particular characteristic X-ray with the atomic number Z of the material is



23. The intensity I distribution of X-rays from two coolidge tubes operated on different voltages V_1 and V_2 and using different target materials of atomic numbers Z_1 and Z_2 is shown in the figure. Which one of the following inequalities is true?

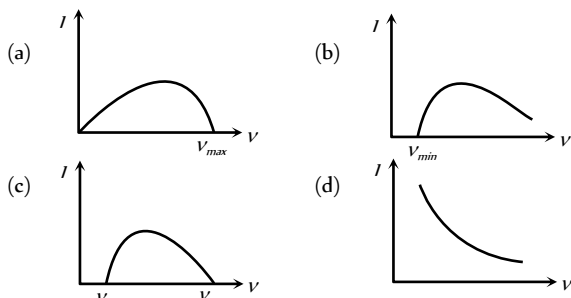


24. The correct graph between the maximum energy of a photoelectron and the inverse of wavelength of the incident radiation is given by the curve

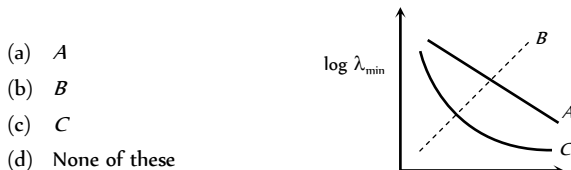


- (a) A
(b) B
(c) C
(d) None of the above

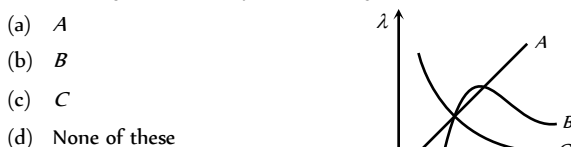
25. The continuous x-ray spectrum obtained from a Coolidge tube is of the form



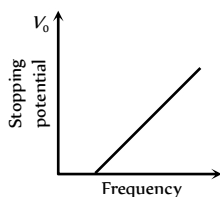
26. The dependence of the short wavelength limit $\lambda_{\min}$ on the accelerating potential V is represented by the curve of figure



27. The variation of wavelength λ of the K_{α} line with atomic number Z of the target is shown by the following curve of

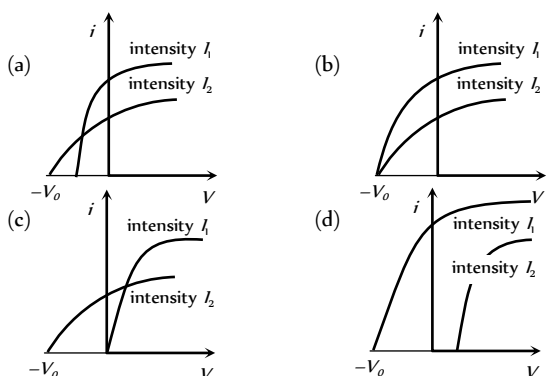


28. In the graph given below. If the slope is $4.12 \times 10^{-15} \text{ V-sec}$, then value of 'h' should be

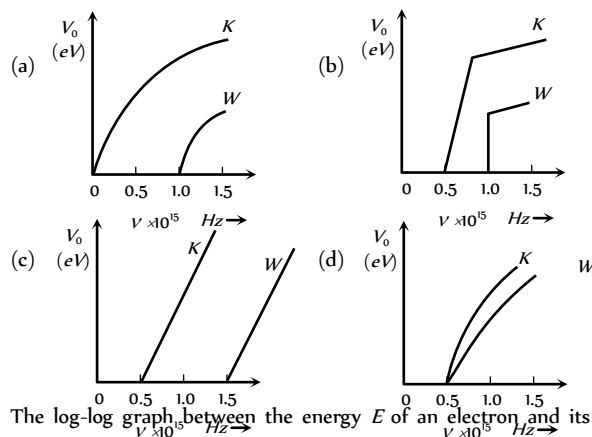


- (a) $6.6 \times 10^{-31} \text{ J-sec}$
(b) $6.6 \times 10^{-34} \text{ J-sec}$
(c) $9.1 \times 10^{-31} \text{ J-sec}$
(d) None of these

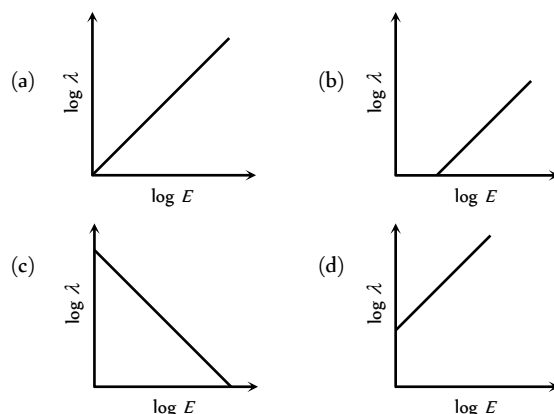
29. The curves (a), (b), (c) and (d) show the variation between the applied potential difference (V) and the photoelectric current (i), at two different intensities of light ($I_1 > I_2$). In which figure is the correct variation shown



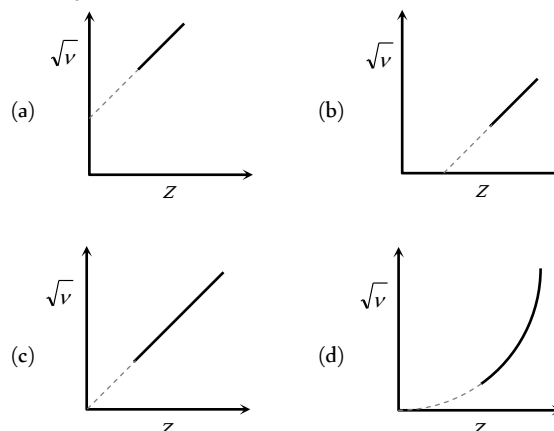
30. The figure showing the correct relationship between the stopping potential V_s and the frequency ν of light for potassium and tungsten is



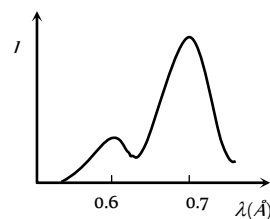
31. The log-log graph between the energy E of an electron and its de-Broglie wavelength λ will be



32. The graph between the square root of the frequency of a specific line of characteristic spectrum of X-rays and the atomic number of the target will be



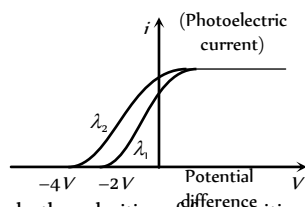
33. In the diagram a graph between the intensity of X-rays emitted by a molybdenum target and the wavelength is shown, when electrons of 30 keV are incident on the target. In the graph one peak is of K_{α} line and the other peak is of K_{β} line



- (a) First peak is of K_{α} line at $0.6 \text{ \AA}$
 (b) Highest peak is of K_{α} line at $0.7 \text{ \AA}$
 (c) If the energy of incident particles is increased, then the peaks will shift towards left
 (d) If the energy of incident particles is increased, then the peaks will shift towards right

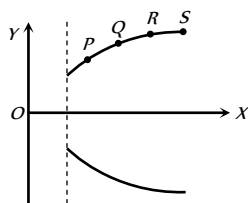
34. The maximum value of stopping potential in the following diagram is

- (a) -4 V
 (b) -1 V
 (c) -3 V
 (d) -2 V

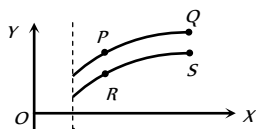


35. In a parabola spectrograph, the velocities of four positive ions P, Q, R and S are v_1, v_2, v_3 and v_4 respectively

- (a) $v_1 > v_2 > v_3 > v_4$
 (b) $v_1 < v_2 < v_3 < v_4$
 (c) $v_1 = v_2 = v_3 = v_4$
 (d) $v_1 \ll v_2 > v_3 < v_4$



36. In Thomson spectrograph experiment, four positive ions P, Q, R and S are situated on Y - X curve as shown in the figure



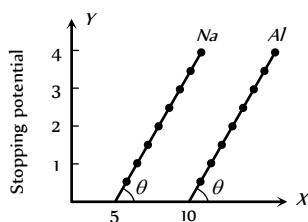
- (a) The specific charge of R and S are same
 (b) The masses of P and S are same
 (c) The specific charges of Q and R are same
 (d) The velocities of R and S are same

37. The slope of frequency of incident light and stopping potential graph for a given surface will be [MP PET 1999;

MP PMT 2000; JIPMER 2001, 02; UPSEAT 2003]

- (a) h
 (b) h/e
 (c) eh
 (d) e

38. From the figure describing photoelectric effect we may infer correctly that [KCET 2005]



- (a) Na and Al both have the same threshold frequency

- (b) Maximum kinetic energy for both the metals depend linearly on the frequency
 (c) The stopping potentials are different for Na and Al for the same change in frequency
 (d) Al is a better photo sensitive material than Na

Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

- Assertion : The energy (E) and momentum (p) of a photon are related by $p = E/c$.
 Reason : The photon behaves like a particle. [AIIMS 2005]
- Assertion : Photoelectric effect demonstrates the wave nature of light.
 Reason : The number of photoelectrons is proportional to the frequency of light. [AIIMS 2004]
- Assertion : When the speed of an electron increases its specific charge decreases.
 Reason : Specific charge is the ratio of the charge to mass. [AIIMS 2001]
- Assertion : X -ray travel with the speed of light.
 Reason : X -rays are electromagnetic rays. [AIIMS 2001]
- Assertion : Mass of moving photon varies inversely as the wavelength.
 Reason : Energy of the particle = Mass $\times$ (Speed of light)² [AIIMS 2000]
- Assertion : Kinetic energy of photo electrons emitted by a photosensitive surface depends upon the intensity of incident photon.
 Reason : The ejection of electrons from metallic surface is possible with frequency of incident photon below the threshold frequency. [AIIMS 1999]
- Assertion : Separation of isotope is possible because of the difference in electron numbers of isotope.
 Reason : Isotope of an element can be separated by using a mass spectrometer. [AIIMS 1999]
- Assertion : The specific charge of positive rays is not constant.
 Reason : The mass of ions varies with speed. [AIIMS 1999]
- Assertion : Photosensitivity of a metal is high if its work function is small.
 Reason : Work function = hf_0 where f_0 is the threshold frequency. [AIIMS 1997]
- Assertion : The de-Broglie wavelength of a molecule varies inversely as the square root of temperature.

- Reason : The root mean square velocity of the molecule depends on the temperature.
[AIIMS 1997]
11. Assertion : An electron is not deflected on passing through certain region of space. This observation confirms that there is no magnetic field in that region.
Reason : The deflection of electron depends on angle between velocity of electron and direction of magnetic field.
12. Assertion : Electric conduction in gases is possible at normal pressure.
Reason : The electric conduction in gases depends only upon the potential difference between the electrodes.
13. Assertion : Light is produced in gases in the process of electric discharge through them at high pressure.
Reason : At high pressure electrons of gaseous atoms collide and reach an excited state.
14. Assertion : If different gases are filled turn by turn at the same pressure in the discharge tube the discharge in them takes place at the same potential.
Reason : The discharge depends only on the pressure of discharge tube and not on the ionisation potential of gas.
15. Assertion : An electric field is preferred in comparison to magnetic field for detecting the electron beam in a television picture tube.
Reason : Electric field requires low voltage.
16. Assertion : The specific charge for positive rays is a characteristic constant.
Reason : The specific charge depends on charge and mass of positive ions present in positive rays.
17. Assertion : In Millikan's experiment for the determination of charge on an electron, oil drops of any size can be used.
Reason : Millikan's experiment determines the charge on electron, by simply measuring the terminal velocity.
18. Assertion : In the process of photoelectric emission, all the emitted photoelectrons have the same kinetic energy.
Reason : The photon transfers its whole energy to the electron of the atom in photoelectric effect.
19. Assertion : In photoelectric effect, on increasing the intensity of light, both the number of electrons emitted and kinetic energy of each of them get increased but photoelectric current remains unchanged.
Reason : The photoelectric current depends only on wavelength of light.
20. Assertion : Though light of a single frequency (monochromatic) is incident on a metal, the energies of emitted photoelectrons are different.
Reason : The energy of electrons emitted from inside the metal surface is lost in collision with the other atoms in the metal.
21. Assertion : The threshold frequency of photoelectric effect supports the particle nature of sunlight.
Reason : If frequency of incident light is less than the threshold frequency, electrons are not emitted from metal surface.
22. Assertion : In photoemissive cell inert gas is used.
Reason : Inert gas in the photoemissive cell gives greater current.
23. Assertion : X-rays cannot be diffracted by means of grating.
Reason : X-rays do not obey Bragg's law.
24. Assertion : X-rays can penetrate through the flesh but not through the bones.
Reason : The penetrating power of X-rays depends on voltage.
25. Assertion : Intensity of X-rays can be controlled by adjusting the filament current and voltage.
Reason : The intensity of X-rays does not depend on number of X-ray photons emitted per second from the target.
26. Assertion : Anode of Coolidge tube gets heated up at time of emission of X-rays.
Reason : The anode of Coolidge tube is made of a material of high melting point.
27. Assertion : Penetrating power of X-rays increases with the increasing wavelength.
Reason : The penetrating power of X-rays increases with the frequency of X-rays.
28. Assertion : X-rays are used for studying the structure of crystals.
Reason : The distance between the atoms of crystals is of the order of wavelength of X-rays.
29. Assertion : The phenomenon of X-ray production is basically inverse of photoelectric effect.
Reason : X-rays are electromagnetic waves.
30. Assertion : Soft and hard X-rays differ in frequency as well as velocity.
Reason : The penetrating power of hard X-rays is more than the penetrating power of soft X-rays.

Answers

Cathode Rays and Positive Rays

1	b	2	b	3	d	4	b	5	d
6	a	7	d	8	b	9	c	10	b
11	c	12	b	13	d	14	b	15	d
16	c	17	c	18	b	19	c	20	b
21	b	22	c	23	c	24	d	25	c
26	d	27	b	28	b	29	c	30	a
31	a	32	c	33	a	34	a	35	b
36	b	37	a	38	d	39	b	40	a
41	c	42	d	43	d	44	c	45	b
46	c	47	a	48	d	49	c	50	c
51	c	52	b	53	b	54	b	55	d
56	d	57	c	58	a	59	b	60	a



1424 Electron, Photon, Photoelectric Effect and X-Rays

61	b	62	b	63	c	64	c	65	b
66	b	67	a	68	a	69	d	70	b
71	a	72	c						

Matter Waves

1	b	2	c	3	a	4	a	5	a
6	b	7	a	8	a	9	d	10	a
11	b	12	a	13	c	14	b	15	b
16	d	17	c	18	b	19	c	20	d
21	b	22	c	23	a	24	a	25	b
26	b	27	c	28	a	29	d	30	b
31	a	32	b	33	c	34	a	35	a
36	a	37	c	38	c	39	d	40	a
41	d	42	d	43	d				

Photon and Photoelectric Effect

1	d	2	d	3	c	4	a	5	a
6	b	7	d	8	b	9	b	10	a
11	b	12	b	13	b	14	c	15	a
16	a	17	b	18	a	19	a	20	c
21	d	22	c	23	b	24	a	25	a
26	a	27	a	28	c	29	d	30	c
31	c	32	a	33	e	34	a	35	d
36	c	37	d	38	c	39	d	40	b
41	a	42	c	43	d	44	d	45	d
46	c	47	c	48	b	49	c	50	a
51	a	52	b	53	d	54	b	55	a
56	d	57	d	58	b	59	b	60	a
61	c	62	b	63	b	64	c	65	a
66	d	67	d	68	c	69	b	70	a
71	d	72	a	73	c	74	c	75	b
76	c	77	a	78	a	79	b	80	c
81	b	82	d	83	c	84	c	85	b
86	c	87	a	88	b	89	c	90	d
91	a	92	a	93	a	94	b	95	c
96	b	97	d	98	a	99	b	100	b
101	a	102	d	103	a	104	b	105	b
106	a	107	a	108	b	109	a	110	b
111	c	112	b	113	a	114	c	115	c
116	b	117	c	118	d	119	a	120	c
121	c	122	c	123	b	124	a	125	a
126	a	127	a	128	c	129	d	130	b
131	d	132	b	133	c	134	d	135	c
136	c	137	d	138	b	139	c	140	c
141	a	142	d	143	b	144	d	145	b

146	c	147	a	148	a	149	c	150	d
151	d	152	b	153	a	154	c	155	a
156	a								

X-Rays

1	c	2	c	3	a	4	a	5	d
6	b	7	a	8	b	9	c	10	c
11	c	12	b	13	c	14	c	15	b
16	c	17	b	18	c	19	d	20	a
21	a	22	b	23	a	24	c	25	d
26	a	27	d	28	c	29	b	30	b
31	c	32	c	33	b	34	c	35	c
36	d	37	a	38	d	39	d	40	c
41	d	42	b	43	c	44	b	45	a
46	c	47	a	48	d	49	c	50	b
51	c	52	a	53	b	54	d	55	b
56	d	57	d	58	c	59	a	60	a
61	a	62	b	63	b	64	b	65	b
66	a	67	b	68	d	69	d	70	d
71	d	72	a	73	c	74	c	75	c
76	d	77	b	78	a	79	d	80	a
81	b	82	c	83	c	84	d	85	a
86	d	87	a	88	d	89	c	90	b
91	a	92	b	93	d	94	a	95	d
96	d	97	b	98	b	99	d	100	c
101	c	102	b	103	b	104	b	105	d
106	a	107	d	108	d	109	b	110	c
111	a								

Critical Thinking Questions

1	b	2	c	3	b	4	abc	5	d
6	b	7	b	8	c	9	b	10	d
11	c	12	b	13	b	14	a	15	a
16	d	17	acd	18	c	19	b	20	c
21	d	22	a	23	c	24	c	25	b
26	c	27	a	28	c	29	b	30	a
31	c	32	c	33	c	34	d	35	b
36	a	37	a	38	a	39	c		

Graphical Questions

1	a	2	d	3	a	4	d	5	c
6	b	7	a	8	b	9	a	10	c
11	b	12	d	13	d	14	b	15	a

16	d	17	d	18	a	19	b	20	c
21	b	22	c	23	a	24	a	25	a
26	a	27	c	28	b	29	b	30	c
31	c	32	b	33	b	34	a	35	a
36	a	37	b	38	b				

Assertion and Reason

1	a	2	d	3	b	4	a	5	b
6	d	7	e	8	b	9	b	10	a
11	e	12	d	13	d	14	d	15	d
16	b	17	e	18	e	19	d	20	a
21	b	22	a	23	c	24	b	25	c
26	b	27	e	28	a	29	b	30	e

AS Answers and Solutions

Cathode Rays and Positive Rays

- (b) Electric field $= \frac{V}{d} = \frac{250}{2.5 \times 10^{-2}} = 10000 \text{ V/m}$.
- (b)
- (d) In Millikan's experiment, drops of non-volatile liquid (cloak oil) are used to prevent evaporation.
- (b) $E = eV = 2e \times 5 = 10 \text{ eV}$
- (d) $E = eV = 1.6 \times 10^{-19} \times 10^5 = 1.6 \times 10^{-14} \text{ J}$
- (a) Any charge in the universe is given by
 $q = ne \Rightarrow e = \frac{q}{n}$ (where n is an integer)
 $q_1 : q_2 : q_3 : q_4 : q_5 : q_6 :: n_1 : n_2 : n_3 : n_4 : n_5 : n_6$
 $6.563 : 8.204 : 11.5 : 13.13 : 16.48 : 18.09$
 $:: n_1 : n_2 : n_3 : n_4 : n_5 : n_6$
 Divide by 6.563
 $1 : 1.25 : 1.75 : 2.0 : 2.5 : 2.75 :: n_1 : n_2 : n_3 : n_4 : n_5 : n_6$
 Multiplied by 4
 $4 : 5 : 7 : 8 : 10 : 11 :: n_1 : n_2 : n_3 : n_4 : n_5 : n_6$
 $e = \frac{q_1 + q_2 + q_3 + q_4 + q_5 + q_6}{n_1 + n_2 + n_3 + n_4 + n_5 + n_6} = \frac{73.967 \times 10^{-19}}{45}$
 $= 1.641 \times 10^{-19} \text{ C}$
 (Note : If you take 45.0743 in place of 45, you will get the exact value)
- (d) Because magnetic force always points perpendicular to the particle velocity. That is why velocity remains unchanged thereby keeping energy $\left(\frac{1}{2}mv^2\right)$ and momentum (mv) unchanged.

- (b)
- (c) Mass is basically a constant term for any physical application at low velocity. But in accordance with Einstein's theory of relativity, at higher speeds the mass of the particle change according to formula

$$m = \frac{m_0}{\sqrt{1 - (v^2/c^2)}}$$
- (b) Refer Q.No. 9. Here the velocity of electron increases, so as per Einstein's equation mass of the electron increases, hence the specific charge $\frac{e}{m}$ decreases.
- (c) If the voltage given is V , then the energy of electron

$$\frac{1}{2}mv^2 = eV \Rightarrow v = \sqrt{\frac{2eV}{m}}$$

$$= \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 1000}{9.1 \times 10^{-31}}} = 1.875 \times 10^7 \approx 1.9 \times 10^7 \text{ m/s}$$
- (b)
- (d) Momentum $p = mv$ and $v = \sqrt{\frac{2QV}{m}}$

$$\Rightarrow p = \sqrt{2QmV} \Rightarrow p \propto \sqrt{Qm} \Rightarrow \frac{p_e}{p_\alpha} = \sqrt{\frac{e \times m_e}{2e \times m_\alpha}} = \sqrt{\frac{m_e}{2m_\alpha}}$$
- (b) In an electric field, a force opposite to the direction of electric field acts on negatively charged particles (i.e. from lower potential to higher potential).
- (d)
- (c) $QE = mg \Rightarrow Q = \frac{mg}{E} \Rightarrow n = \frac{mgd}{Ve}$

$$\Rightarrow n = \frac{1.8 \times 10^{-14} \times 10 \times 0.9 \times 10^{-2}}{2 \times 10^{-3} \times 1.6 \times 10^{-19}} = 5$$
- (c)
- (b) In Millikan's experiment, the charges present on the oil drops are the integral multiples, so $2e$ and $10e(1.6 \times 10^{-18} \text{ C})$ charges are present.
- (c) $eE = evB \Rightarrow v = \frac{E}{B} = \frac{3 \times 10^4}{2 \times 10^{-3}} = 1.5 \times 10^7 \text{ m/s}$
- (b)
- (b) Charged particles trace a circular path in a perpendicular magnetic field.
- (c) $\frac{e}{m} = \frac{1.6 \times 10^{-19}}{9.1 \times 10^{-31}} = 1.76 \times 10^{11} \text{ C/kg}$
- (c)
- (d) Light consists of photons and cathode rays consists of electrons. However both effect the photographic plate.
- (c)
- (d)
- (b) For ionisation, high energy electrons are required.
- (b) $v = \frac{E}{B} = \frac{20}{0.5} = 40 \text{ m/sec}$.
- (c) Higher the voltage, higher is the KE . Higher the work function, smaller is the KE .

30. (a) Time period of revolution of electron $T = \frac{2\pi}{\omega} = \frac{2\pi}{v}$

Hence corresponding electric current $i = \frac{e}{T} = \frac{ev}{2\pi r}$

$\Rightarrow i = \frac{1.6 \times 10^{-19} \times 2 \times 10^6}{2 \times 3.14 \times 0.5 \times 10^{-10}} = 1 \text{ mA.}$

31. (a) $K = Q.V = 1.6 \times 10^{-19} \times 100 = 1.6 \times 10^{-17} \text{ Joules}$

32. (c) $K = Q.V = 1e \times 1 \text{ Volt} = 1 \text{ eV}$

33. (a) Kinetic energy $\propto$ Potential difference

34. (a) In discharge tube cathode rays (a beam of negative particles) and canal rays (positive rays) moves opposite to each other. They will experience a magnetic force in the same direction, if a normal magnetic field is switched on

35. (b) $n = \frac{Q}{e} = \frac{6.35 \times 10^{-19}}{1.6 \times 10^{-19}} \approx 4$

36. (b)

37. (a) When cathode rays strike the metal plate, they transfer their energy to plate.

38. (d) Cathode rays are beam of electrons.

39. (b) $K = QV = e \times V = eV$

40. (a) $\frac{1}{2}mv^2 = QV \Rightarrow v = \sqrt{\frac{2QV}{m}} = \sqrt{2\left(\frac{e}{m}\right)V}$
 $\Rightarrow v = \sqrt{2 \times 1.6 \times 10^{11} \times 200} = 8 \times 10^6 \text{ m/s.}$

41. (c) Speed of the cathode rays is $10^7 \text{ m/sec} - 3 \times 10^7 \text{ m/s}$

42. (d) $QE = mg \Rightarrow mg = \frac{QV}{d}$

43. (d)

44. (c) In the condition of no deflection $\frac{e}{m} = \frac{E^2}{2VB^2} \Rightarrow$ If m is increased by 208 times then B should be increased $\sqrt{208} = 14.4$ times

45. (b) The colour of the positive column in a discharge tube depends on the type of gas e.g. For air, colour is purple red, for H_2 , colour is Blue etc.

46. (c)

47. (a) $v = \frac{p}{m} = \frac{h}{m\lambda} = \frac{6.6 \times 10^{-34}}{9.1 \times 10^{-31} \times 10^{-10}} = 7.25 \times 10^6 \text{ m/s}$

48. (d) Cathode rays are stream of negative charged particle, so they deflect in electric field.

49. (c) $\frac{e}{m} = \frac{E^2}{2VB^2} = \frac{(3.6 \times 10^4)^2}{2 \times 2.5 \times 10^3 \times (1.2 \times 10^{-3})^2}$
 $= 1.8 \times 10^{11} \text{ C/kg.}$

50. (c) Specific charge $= \frac{q}{m}$; Ratio $= \frac{\left(\frac{q}{m}\right)_\alpha}{\left(\frac{q}{m}\right)_p} = \frac{q_\alpha}{q_p} \times \frac{m_p}{m_\alpha} = \frac{1}{2}$

51. (c) $v = \frac{E}{B}$; where $E = \frac{V}{d} = \frac{1000}{1 \times 10^{-2}} = 10^5 \text{ V/m}$
 $\Rightarrow v = \frac{10^5}{1} = 10^5 \text{ m/s.}$

52. (b)

53. (b)

54. (b)

55. (d) In Thomson's mass spectrograph $\vec{E} \parallel \vec{B}$

56. (d)

57. (c) In the absence of electric field (i.e. $E = 0$)

$mg = 6\pi\eta rv$ $D_1 = 6\pi\eta rv$... (i)

In the presence of Electric field

$mg + QE = 6\pi\eta r(2v)$ $D_2 = 6\pi\eta r(2v)$... (ii)

When Electric field to reduced to $E/2$

$mg + Q(E/2) = 6\pi\eta r(v')$ $D_3 = 6\pi\eta r(2v)$... (iii)

After solving (i), (ii) and (iii)

We get $v' = \frac{3}{2}v$

58. (a) $v = \sqrt{\frac{2eV}{m}} = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 45.5}{9.1 \times 10^{-31}}} = 4 \times 10^6 \text{ m/s}$

59. (b) $i = \frac{Q}{t} = \frac{ne}{t} = 1.8 \times 10^{14} \times 1.6 \times 10^{-19} = 28.8 \times 10^{-6} \text{ A}$
 $= 29 \mu\text{A}$

60. (a) $\because m_e < m_p < m_\alpha \Rightarrow \left(\frac{q}{m}\right)_e > \left(\frac{q}{m}\right)_p > \left(\frac{q}{m}\right)_\alpha$

61. (b) Acceleration $a = \frac{QE}{m} = \frac{(3e)E}{2m}$

62. (b) $\frac{1}{2}mv^2 = eV \Rightarrow \frac{e}{m} = \frac{v^2}{2V} = \frac{(8.4 \times 10^6)^2}{2 \times 200} = 1.76 \times 10^{11} \frac{\text{C}}{\text{kg}}$

63. (c) $K = Q.\Delta V = (2e) \times 10^6 \text{ V} = 2 \times 10^6 \text{ eV} = 2 \text{ MeV}$

64. (c) Positive rays consist of positive ions.

65. (b) $2r = \frac{2mv}{qB} \Rightarrow 2r \propto \frac{m}{q} \Rightarrow \frac{m}{q}$ is maximum for C^+

66. (b) $v = \frac{E}{B} = \frac{1.125 \times 10^{-6}}{3 \times 10^{-10}} = 3750 \text{ m/s}$

67. (a) Positive rays was discovered by J.J. Thomson.

68. (a)

69. (d) If electron oscillate with a frequency of 1 GHz, it does not radiate any energy, which corresponds a definite wavelength. It only radiate when it jump from one orbit to another orbit.
70. (b) $eV = \frac{1}{2}mv^2 \Rightarrow v^2 = \frac{2eV}{m} \Rightarrow v = \sqrt{\frac{2eV}{m}}$
71. (a)
72. (c) $eE = mg \Rightarrow e = \frac{mg}{E} = \frac{16 \times 10^{-6} \times 10}{10^6} = 16 \times 10^{-11} \text{ C}$

Matter Waves

1. (b)
2. (c) According to de-Broglie hypothesis.
3. (a) $\lambda = \frac{h}{p} = \frac{h}{mv}$
4. (a) $\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}; \therefore E = \frac{h^2}{2m\lambda^2}$
 λ is same for all, so $E \propto \frac{1}{m}$. Hence energy will be maximum for particle with lesser mass.
5. (a) Particle is photon and it travels with the velocity equal to light in vacuum.
6. (b) $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}; \therefore \lambda \propto \frac{1}{\sqrt{E}}$ (h and $m = \text{constant}$)
7. (a) $\lambda = \frac{h}{m_1 v_1} = \frac{h}{m_2 v_2}; \therefore \frac{v_1}{v_2} = \frac{m_2}{m_1} = \frac{4}{1}$
8. (a) $\frac{1}{2}mv^2 = E \Rightarrow mv = \sqrt{2mE}; \therefore \lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$
9. (d) $\left\{ \begin{array}{l} \text{Photoelectric effect} \rightarrow \text{Particlenature} \\ \text{Diffraction} \rightarrow \text{Wave nature} \end{array} \right\}$ Dual nature
10. (a) $mvr = \frac{nh}{2\pi}$ According to Bohr's theory
 $\Rightarrow 2\pi r = n \left(\frac{h}{mv} \right) = n\lambda$ for $n = 1, \lambda = 2\pi r$
11. (b) $\lambda = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda \propto \frac{1}{\sqrt{m}}$ ($E = \text{same}$)
12. (a) $\lambda = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda \propto \frac{1}{\sqrt{m}} \Rightarrow \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha}{m_p}} = \frac{2}{1}$
13. (c) $\lambda = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2m_\alpha Q_\alpha V}}$
 On putting $Q_\alpha = 2 \times 1.6 \times 10^{-19} \text{ C}$
 $m_\alpha = 4m_p = 4 \times 1.67 \times 10^{-27} \text{ kg} \Rightarrow \lambda = \frac{0.101}{\sqrt{V}} \text{ \AA}$
14. (b)
15. (b) $\lambda = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda \propto \frac{1}{\sqrt{E}} \Rightarrow \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{E_2}{E_1}}$
 $\Rightarrow \frac{10^{-10}}{0.5 \times 10^{-10}} = \sqrt{\frac{E_2}{E_1}} \Rightarrow E_2 = 4E_1$
 Hence added energy $= E_2 - E_1 = 3E_1$

16. (d) $\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9 \times 10^{-31} \times 80 \times 1.6 \times 10^{-19}}} = 1.4 \text{ \AA}$
17. (c) $\lambda = \frac{h}{mv} \Rightarrow \lambda \propto \frac{1}{m}$
18. (b) If an electron and a photon propagates in the from of waves having the same wavelength, it implies that they have same momentum. This is according to de-Broglie equation, $p \propto \frac{1}{\lambda}$
19. (c) $\lambda = \frac{h}{p} \Rightarrow \lambda \propto \frac{1}{p}$
20. (d) In photoelectric effect particle nature of electron is shown. While in electron microscope, beam of electron is considered as electron wave.
21. (b) $K_{\text{particle}} = \frac{1}{2}mv^2$ also $\lambda = \frac{h}{mv}$
 $\Rightarrow K_{\text{particle}} = \frac{1}{2} \left(\frac{h}{\lambda v} \right) \cdot v^2 = \frac{vh}{2\lambda} \dots (i)$
 $K_{\text{photon}} = \frac{hc}{\lambda} \dots (ii)$
 $\therefore \frac{K_{\text{particle}}}{K_{\text{photon}}} = \frac{v}{2c} = \frac{2.25 \times 10^8}{2 \times 3 \times 10^8} = \frac{3}{8}$
22. (c) $2\pi r n = \lambda \Rightarrow n = \frac{\lambda}{2\pi r} = \frac{10^{-9}}{2 \times 3.14 \times 5.13 \times 10^{-11}} = 3$
23. (a) By using $\lambda_{\text{electron}} = \frac{h}{m_e v} \Rightarrow v = \frac{h}{m_e \lambda_e}$
 $= \frac{6.6 \times 10^{-34}}{9.1 \times 10^{-31} \times 10^{-10}} = 7.25 \times 10^6 \text{ m/s}$
24. (a) By using $\lambda = \frac{h}{\sqrt{2mE}}$ $E = 10^{-6} \text{ J} = \text{Constant}$ for both particles. Hence $\lambda \propto \frac{1}{\sqrt{m}}$ Since $m_p > m_e$ so $\lambda_p < \lambda_e$.
25. (b) By using $\lambda \propto \frac{1}{\sqrt{V}} \Rightarrow \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{V_2}{V_1}}$
 $\Rightarrow \frac{10^{-10}}{\lambda_2} = \sqrt{\frac{600}{150}} = 2 \Rightarrow \lambda_2 = 0.5 \text{ \AA}$
26. (b) $\lambda = \frac{h}{mv_{\text{rms}}} \Rightarrow \lambda = \frac{6.6 \times 10^{-34}}{2 \times 1.67 \times 10^{-27} \times 3 \times 10^3} = 0.66 \text{ \AA}$
27. (c) $\lambda \propto \frac{1}{p} \Rightarrow \frac{\Delta p}{p} = -\frac{\Delta \lambda}{\lambda} \Rightarrow \left| \frac{\Delta p}{p} \right| = \left| \frac{\Delta \lambda}{\lambda} \right|$
 $\Rightarrow \frac{p_0}{p} = \frac{0.25}{100} = \frac{1}{400} \Rightarrow p = 400 p_0$
28. (a) $\lambda_{\text{neutron}} \propto \frac{1}{\sqrt{T}} \Rightarrow \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{T_2}{T_1}}$
 $\Rightarrow \frac{\lambda}{\lambda_2} = \sqrt{\frac{(273+927)}{(273+27)}} = \sqrt{\frac{1200}{300}} = 2 \Rightarrow \lambda_2 = \frac{\lambda}{2}$
29. (d) $\lambda = \frac{h}{\sqrt{2mE}} \Rightarrow E \propto \frac{1}{\sqrt{m}}$ ($\lambda = \text{constant}$)
 $\therefore m_e < m_p$ so $E_e > E_p$

30. (b)
31. (a) Wavelength of photon will be greater than that of electron because mass of photon is less than that of electron
 $\Rightarrow \lambda_{ph} > \lambda_e$
32. (b) $\lambda = \frac{h}{\sqrt{2mE}} \Rightarrow E = \frac{h^2}{2m\lambda^2}$
 $= \frac{(6.6 \times 10^{-34})^2}{2 \times 9.1 \times 10^{-31} \times (0.3 \times 10^{-9})^2} = 2.65 \times 10^{-18} \text{ J}$
 $= 16.8 \text{ eV}$
33. (c) $\lambda = \frac{h}{\sqrt{2mQV}} \Rightarrow \lambda \propto \frac{1}{\sqrt{mQ}} \Rightarrow \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha Q_\alpha}{m_p Q_p}}$
 $= \sqrt{\frac{4m_p \times 2Q_p}{m_p \times Q_p}} = 2\sqrt{2}$
34. (a) $\lambda = \frac{h}{p} \Rightarrow p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{2 \times 10^{-6}}$
 $= 3.31 \times 10^{-28} \text{ kg-m / sec}$
35. (a) $\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34}}{1 \times 2000} = 3.3 \times 10^{-37} \text{ m} = 3.3 \times 10^{-27} \text{ \AA}$
36. (a) $\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 5 \times 1.6 \times 10^{-19}}}$
 $= 5.469 \times 10^{-10} \text{ m} = 5.47 \text{ \AA}$
37. (c) $\lambda = \frac{h}{\sqrt{2mQV}} = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 100}}$
 $= 1.23 \text{ \AA}$
38. (c) The De-Broglie wavelength is $\lambda = \frac{h}{|p|} = \frac{h}{|I|} \Rightarrow \lambda \propto \frac{1}{|I|}$
39. (d) Davission and Germer proved the wave nature of electron by performing an experiment.
40. (a) $\lambda = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda \propto \frac{1}{\sqrt{E}}$
41. (d) $\lambda = \frac{h}{\sqrt{2mE}}; \frac{\lambda'}{\lambda} = \sqrt{\frac{E}{E'}} \Rightarrow \frac{E}{E'} = \left(\frac{0.5}{1}\right)^2 \Rightarrow E' = \frac{E}{0.25} = 4E$
 The energy should be added to decrease wavelength.
 $= E' - E = 3E$
42. (d)
43. (d)
6. (b) $p = \frac{E}{c} = \frac{h\nu}{c}$
7. (d) $E = h\nu = mc^2 \Rightarrow m = \frac{h\nu}{c^2}$
8. (b) $p = \frac{E}{c} = \frac{h\nu}{c} \Rightarrow \nu = \frac{pc}{h}$
9. (b) $P = \frac{W}{t} = \frac{nhc}{\lambda t} \Rightarrow \left(\frac{n}{t}\right) = \frac{P\lambda}{hc} = \frac{10 \times 10^3 \times 300}{6.6 \times 10^{-34} \times 3 \times 10^8}$
 $= 1.5 \times 10^{31}$
10. (a) Momentum of photon $p = \frac{E}{c}$
 $\Rightarrow$ Velocity of photon $c = \frac{E}{p}$
11. (b) By using $E(\text{eV}) = \frac{12375}{\lambda(\text{\AA})}$
 $\Rightarrow \lambda = \frac{12375}{2.48} = 4989.9 \text{ \AA} \approx 5000 \text{ \AA}$
12. (b) $E = \frac{hc}{\lambda} = \frac{3 \times 10^8 \times 6.62 \times 10^{-34}}{0.21 \times 1.6 \times 10^{-19}} = 5.9 \times 10^{-6} \text{ eV}$
13. (b) Momentum of photon
 $p = \frac{h}{\lambda} = \frac{6.6 \times 10^{-34}}{10^{-10}} = 6.6 \times 10^{-24} \text{ kg-m/sec.}$
14. (c) $E \propto \frac{1}{\lambda} \Rightarrow \frac{2.5}{E'} = \frac{1}{5000} \Rightarrow E' = (2.5) \times 5000 \text{ eV}$
15. (a) $E = h\nu = 6.6 \times 10^{-34} \times 10^{15} = 6.6 \times 10^{-19} \text{ J}$
16. (a) Since $h\nu = mc^2$, hence $p = mc = \frac{h\nu}{c} = \frac{h}{\lambda}$
17. (b) $E = h\nu \Rightarrow \nu = \frac{E}{h} = \frac{1 \times 10^6 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}} = 2.4 \times 10^{20} \text{ Hz}$
18. (a) $p = \frac{h\nu}{c} = \frac{6.6 \times 10^{-34} \times 1.5 \times 10^{13}}{3 \times 10^8} = 3.3 \times 10^{-29} \text{ kg-m / sec}$
19. (a) $E = \frac{hc}{\lambda} = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{450 \times 10^{-9}} = 4.4 \times 10^{-19} \text{ J}$
20. (c) $E = h\nu \Rightarrow \nu = \frac{E}{h} = \frac{66 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}} = 16 \times 10^{15} \text{ Hz}$
21. (d) $E \propto \frac{1}{\lambda}$; also $\lambda_{\text{infrared}} > \lambda_{\text{visible}}$ so $E_{\text{infrared}} < E_{\text{visible}}$
22. (c) Energy of photon $E = \frac{hc}{\lambda}$ (Joules) $= \frac{hc}{e\lambda}$ (eV)
 $\Rightarrow E_{(\text{eV})} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times \lambda(\text{\AA})} = \frac{12375}{\lambda(\text{\AA})}$
 $\Rightarrow E(\text{keV}) = \frac{12.37}{\lambda(\text{\AA})} \approx \frac{12.4}{\lambda}$
23. (b) $E = h\nu \Rightarrow 100 \times 1.6 \times 10^{-19} = 6.6 \times 10^{-34} \times \nu$
 $\Rightarrow \nu = 2.42 \times 10^{16} \text{ Hz}$
24. (a) $p = \frac{h}{\lambda} = \frac{6.6 \times 10^{-34}}{4400 \times 10^{-10}} = 1.5 \times 10^{-27} \text{ kg.m / s}$

Photon and Photoelectric Effect

1. (d) $p = \frac{h\nu}{c} \Rightarrow \nu = \frac{pc}{h} = \frac{3.3 \times 10^{-29} \times 3 \times 10^8}{6.6 \times 10^{-34}} = 1.5 \times 10^{13} \text{ Hz}$
2. (d)
3. (c) $p = \frac{E}{c} \Rightarrow E = p \times c = 2 \times 10^{-16} \times (3 \times 10^{10}) = 6 \times 10^{-6} \text{ erg.}$
4. (a)
5. (a) $p = \frac{h}{\lambda} = \frac{6.6 \times 10^{-34}}{(5000 \times 10^{-10})} = 1.3 \times 10^{-27} \text{ kg-m / s}$
23. (b) $E = h\nu \Rightarrow 100 \times 1.6 \times 10^{-19} = 6.6 \times 10^{-34} \times \nu$
 $\Rightarrow \nu = 2.42 \times 10^{16} \text{ Hz}$
24. (a) $p = \frac{h}{\lambda} = \frac{6.6 \times 10^{-34}}{4400 \times 10^{-10}} = 1.5 \times 10^{-27} \text{ kg.m / s}$

$$\text{and mass } m = \frac{p}{c} = \frac{1.5 \times 10^{-27}}{3 \times 10^8} = 5 \times 10^{-36} \text{ kg}$$

25. (a)

26. (a)

27. (a) $E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E}$

28. (c)

29. (d) $E(\text{eV}) = \frac{h\nu}{e} = \frac{6.0 \times 10^{-34} \times 10^{12} \times 10^6}{1.6 \times 10^{-19}} = 4.14 \times 10^3 \text{ eV}$

30. (c) $E = nh\nu \Rightarrow \nu \propto \frac{1}{n} \Rightarrow \frac{n_1}{n_2} = \frac{\nu_2}{\nu_1}$

31. (c) According to Einstein's photoelectric equation.

32. (a) Kinetic energy of photoelectrons depends on the frequency of incident radiations and is independent of the intensity of illumination.

 33. (e) In this case, for photoelectric emission the wavelength of incident radiations must be less than $5200 \text{ \AA}$. Wavelength of ultraviolet radiations is less than this value ($5200 \text{ \AA}$) but wavelength of infrared radiations is higher than this value.

 34. (a) Frequency of light of wavelength ($\lambda = 4000 \text{ \AA}$) is

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{4000 \times 10^{-10}} = 0.75 \times 10^{15} \text{ which is less than}$$

the given threshold frequency. Hence no photoelectric emission takes place.

35. (d) Refer to the application of photo-cell.

36. (c) Albert Einstein was awarded Nobel Prize in 1921 for discovering the photoelectric effect.

37. (d)

38. (c) Energy of incident light $E(\text{eV}) = \frac{12375}{3320} = 3.72 \text{ eV}$

$$(332 \text{ nm} = 3320 \text{ \AA})$$

 According to the relation $E = W_0 + eV_0$

$$\Rightarrow V_0 = \frac{(E - W_0)}{e} = \frac{3.72 \text{ eV} - 1.07 \text{ eV}}{e} = 2.65 \text{ Volt}$$

39. (d)

40. (b) $K_{\max} = (h\nu - W_0)$; ν = frequency of incident light.

41. (a) Refer to threshold frequency.

42. (c) $W_0(\text{eV}) = \frac{12375}{\lambda_0} \Rightarrow \lambda_0 = \frac{12375}{4.2} \approx 2955 \text{ \AA}$

 43. (d) Intensity $\propto$ (No. of photons) $\propto$ (No. of photoelectrons)

44. (d) $E = W_0 + K_{\max}$; $E = \frac{12375}{3000} = 4.125 \text{ eV}$

$$\Rightarrow K_{\max} = E - W_0 = 4.125 \text{ eV} - 1 \text{ eV} = 3.125 \text{ eV}$$

$$\Rightarrow \frac{1}{2} m v_{\max}^2 = 3.125 \times 1.6 \times 10^{-19} \text{ J}$$

$$\Rightarrow v_{\max} = \sqrt{\frac{2 \times 3.125 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}} = 1 \times 10^6 \text{ m/s}$$

45. (d) Retarding potential $V_0 = \frac{h}{e}(\nu - \nu_0)$

46. (c)

47. (c) $K_{\max} = \frac{hc}{\lambda} - W_0 = \frac{6.4 \times 10^{-34} \times 3 \times 10^8}{6400 \times 10^{-10}} - 1.6 \times 10^{-19}$

$$= 1.4 \times 10^{-19} \text{ J}$$

48. (b) $K_{\max}(\text{eV}) = E(\text{eV}) - W_0(\text{eV}) = 6.2 - 4.2 = 2 \text{ eV}$
 $\therefore K_{\max}(\text{Joules}) = 2 \times 1.6 \times 10^{-19} \text{ J} = 3.2 \times 10^{-19} \text{ J}$

49. (c) Since $W_0 = \frac{hc}{\lambda_0}$; $\therefore \frac{(W_0)_T}{(W_0)_{Na}} = \frac{\lambda_{Na}}{\lambda_T}$ or

$$\lambda_T = \frac{\lambda_{Na} \times (W_0)_{Na}}{(W_0)_T} = \frac{5460 \times 2.3}{4.5} = 2791 \text{ \AA}$$

50. (a) $K_{\max} = (E - W_0) = (3.4 - 2) \text{ eV} = 1.4 \text{ eV}$

51. (a) Energy of incident light $E = \frac{12375}{2000} = 6.18 \text{ eV}$

 According to relation $E = W_0 + eV_0$

$$\Rightarrow V_0 = \frac{(E - W_0)}{e} = \frac{(6.18 \text{ eV} - 5.01 \text{ eV})}{e} = 1.17 \text{ V} \approx 1.2 \text{ V}$$

52. (b) $W_0 = \frac{12375}{6600} = 1.87 \text{ eV}$

53. (d)

54. (b) $E = h\nu = 6.64 \times 10^{-34} \times 1.0 \times 10^{14} = 6.62 \times 10^{-20} \text{ J}$

55. (a) Number of photons emitted per second

$$n = \frac{P}{h\nu} = \frac{10 \times 10^3}{6.6 \times 10^{-34} \times 880 \times 10^3} = 1.72 \times 10^{31}$$

56. (d) Number of ejected electrons $\propto$ (Intensity) $\propto \frac{1}{(\text{Distance})^2}$

 Therefore an increment of distance two times will reduce the number of ejected electrons to $\frac{1}{4}$ th of the previous one.

57. (d) According to Einstein's photoelectric equation

$$E = W_0 + K_{\max} \Rightarrow V_0 = \frac{hc}{e} \left[\frac{1}{\lambda} - \frac{1}{\lambda_0} \right]$$

 Hence if λ decreases V_0 increases.

58. (b) $W_0 = \frac{12375}{\lambda_0(\text{\AA})} = \frac{12375}{5420} = 2.28 \text{ eV}$

59. (b) Number of electrons can be measured which are directly proportional to the intensity of radiation.

60. (a) $K_{\max} = h\nu - W_0 = 6.6 \times 10^{-34} \times 8 \times 10^{14} - 3.2 \times 10^{-19}$

$$= 2.1 \times 10^{-19} \text{ J}$$

61. (c)

62. (b) $K_{\max}(\text{eV}) = 12375 \left[\frac{1}{\lambda(\text{\AA})} - \frac{1}{\lambda_0(\text{\AA})} \right]$

$$= 12375 \left[\frac{1}{1000} - \frac{1}{2000} \right] = 6.2 \text{ eV}$$

63. (b) Stopping potential does not depend on the relative distance between the source and the cell.

64. (c)

65. (a) Energy of incident light $E(\text{eV}) = \frac{12375}{4000} = 3.09 \text{ eV}$
Stopping potential is -2 V so $K_{\text{max}} = 2 \text{ eV}$
Hence by using $E = W_0 + K_{\text{max}}$; $W_0 = 1.09 \text{ eV} \approx 1.1 \text{ eV}$
66. (d) $\frac{hc}{\lambda} = W_0 + \frac{1}{2}mv_{\text{max}}^2$
Assuming W_0 to be negligible in comparison to $\frac{hc}{\lambda}$
i.e. $v_{\text{max}}^2 \propto \frac{1}{\lambda} \Rightarrow v_{\text{max}} \propto \frac{1}{\sqrt{\lambda}}$.
(On increasing wavelength λ to 4λ , v_{max} becomes half).
67. (d) $W_0 = h\nu_0 \Rightarrow \nu_0 = \frac{W_0}{h} = \frac{2.51 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}} = 6.08 \times 10^{14} \text{ Cycle / sec.}$
68. (c)
69. (b)
70. (a) By changing distance of source, photoelectric current changes. But there is no change in stopping potential.
71. (d) $\nu_0 = \frac{W_0}{h} = \frac{3.3 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}} = 8 \times 10^{14} \text{ Hz}$
72. (a) For no emission of photoelectron, energy of incident light < Work function $\Rightarrow h\nu < \phi \Rightarrow \nu < \frac{\phi}{h}$
73. (c) Number of electrons emitted $\propto \text{intensity} \propto \frac{1}{(\text{distance})^2}$
 $\Rightarrow \frac{n_1}{n_2} = \left(\frac{d_2}{d_1}\right)^2 = \left(\frac{2}{1}\right)^2 = 4 \Rightarrow n_2 = \frac{n_1}{4}$
74. (c) $E = \frac{hc}{\lambda} - W_0$ and $2E = \frac{hc}{\lambda'} - W_0$
 $\Rightarrow \frac{\lambda'}{\lambda} = \frac{E + W_0}{2E + W_0} \Rightarrow \lambda' = \lambda \left(\frac{1 + W_0/E}{2 + W_0/E} \right)$
Since $\frac{(1 + W_0/E)}{(2 + W_0/E)} > \frac{1}{2}$ so $\lambda' > \frac{\lambda}{2}$
75. (b) Stopping potential $V_0 = \frac{hc}{e} \left[\frac{1}{\lambda} - \frac{1}{\lambda_0} \right]$. As λ decreases so V_0 increases.
76. (c) $W_0(\text{eV}) = \frac{12375}{\lambda_0(\text{\AA})} \Rightarrow \lambda_0 = \frac{12375}{4.125} = 3000 \text{\AA}$
77. (a) Intensity increases means more photons of same energy will emit more electrons of same energy, hence only photoelectric current increases.
78. (a) $E = W_0 + K_{\text{max}}$; $E = \frac{12375}{5000} = 2.475 \text{ eV}$
 $\therefore K_{\text{max}} = E - W_0 = 2.475 - 1.9 = 0.57 \text{ eV}$
79. (b)
80. (c) $\lambda_0 = \frac{hc}{W_0} = \frac{12400}{4} = 3100 \text{\AA} = 310 \text{ nm}$
81. (b) $K_{\text{max}} = (|V_s|) \text{ eV} \Rightarrow |V_s| = 4 \text{ V}$
82. (d) Threshold wavelength $\lambda_0 = \frac{12375}{2.1} = 5892.8 \text{\AA}$
83. (c) $P = \frac{nhc}{\lambda t} \Rightarrow \frac{n}{t} = \frac{P\lambda}{hc} = \frac{100 \times 5000 \times 10^{-10}}{6.6 \times 10^{-34} \times 3 \times 10^8} = 2.50 \times 10^{20}$
84. (c) $E = W_0 + K_{\text{max}} \Rightarrow K_{\text{max}} = E - W_0 = h\nu - W_0$
 $\Rightarrow K_1 = h\nu - W_0$ and $K_2 = 2h\nu - W_0 \Rightarrow K_2 > 2K_1$
85. (b) Work function $= \frac{hc}{\lambda_0}$; where λ_0 is threshold wavelength.
 $\therefore \frac{W_{01}}{W_{02}} = \frac{\lambda_{02}}{\lambda_{01}} = \frac{2}{1}$
86. (c) $W_0 = \frac{hc}{\lambda_0} = \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{5000 \times 10^{-10}} \text{ J} = 4 \times 10^{-19} \text{ J}$
87. (a) The work function has no effect on current so long as $h\nu > W_0$. The photoelectric current is proportional to the intensity of light. Since there is no change in the intensity of light, therefore $I_1 = I_2$.
88. (b) Number of photons emitted is proportional to the intensity.
Also $\frac{hc}{\lambda} = W_0 + E$.
89. (c) Photoelectric current $\propto$ Intensity of light
90. (d) $V_0 = \frac{(E - W_0)}{e} = \frac{(2 \text{ eV} - 0.6 \text{ eV})}{e} = 1.4 \text{ V}$
91. (a) $\lambda_r > \lambda_y > \lambda_g$. Here threshold wavelength $< \lambda_y$.
92. (a) For electron emission $\lambda_{\text{incident}} < \lambda_0$
93. (a) $K_{\text{max}} = (|V_0|) \text{ eV} = 2 \text{ eV}$.
94. (b) Threshold wavelength for Na, $\lambda_{\text{Na}} = \frac{12375}{2} = 6187.5 \text{\AA}$
Also $\lambda_{\text{Cu}} = \frac{12375}{4} = 3093.75$
Since $\lambda_{\text{Na}} > 4000 \text{\AA}$; So Na is suitable.
95. (c) By using $E = W_0 + K_{\text{max}}$
 $E = \frac{12375}{5000} = 2.475 \text{ eV}$ and $K_{\text{max}} = eV_0 = 1.36 \text{ eV}$
So $2.475 = W_0 + 1.36 \Rightarrow W_0 = 1.1 \text{ eV}$.
96. (b) For emission of electrons incident energy of each photon must be greater than work function (threshold energy).
97. (d) K_{max} of photoelectrons doesn't depends upon intensity of incident light.
98. (a) By using $E = W_0 + \frac{1}{2}mv_{\text{max}}^2$ where $E = \frac{12375}{2000} = 6.18 \text{ eV}$
 $\Rightarrow 6.18 \text{ eV} = 4.2 \text{ eV} + \frac{1}{2}mv_{\text{max}}^2 \Rightarrow 1.98 \text{ eV} = \frac{1}{2}mv_{\text{max}}^2$
 $\Rightarrow 1.98 \times 1.6 \times 10^{-19} = \frac{1}{2} \times 9.1 \times 10^{-31} \times v_{\text{max}}^2$
 $\Rightarrow v_{\text{max}} = 8.4 \times 10^5 \text{ m/s}$
99. (b) By using $E = W_0 + \frac{1}{2}mv_{\text{max}}^2$; where $E = \frac{12375}{4558} = 2.71 \text{ eV}$

$$\Rightarrow 2.71 \text{ eV} = 2.5 \text{ eV} + \frac{1}{2} \times 9.1 \times 10^{-31} \times v_{\max}^2$$

$$\Rightarrow 0.21 \times 1.6 \times 10^{-19} = \frac{1}{2} \times 9.1 \times 10^{-31} \times v_{\max}^2$$

$$\Rightarrow v_{\max} = 2.65 \times 10^5 \text{ m/s}$$

100. (b) $E = W_0 + K_{\max}$ (i)

$$\Rightarrow hf = W_A + K_A \quad \text{.....(ii)}$$

$$\text{and } 2hf = W_B + K_B = 2W_A + K_B \quad \left(\because \frac{W_A}{W_B} = \frac{1}{2} \right)$$

Dividing equation (i) by (ii)

$$\frac{1}{2} = \frac{W_A + K_A}{2W_A + K_B} \Rightarrow \frac{K_A}{K_B} = \frac{1}{2}$$

101. (a)

102. (d) Stopping potential depends upon the energy of photon

103. (a) $\lambda_0 = \frac{12375}{W_0(\text{eV})} = \frac{12375}{3} = 4125 \text{ \AA}$

104. (b) With decrease in wavelength of incident photons, energy of photoelectrons increases.

105. (b)

106. (a) By using $\frac{hc}{\lambda} = W_0 + \frac{1}{2}mv^2$

$$\Rightarrow \frac{hc}{400 \times 10^{-9}} = W_0 + \frac{1}{2}mv^2 \quad \text{.....(i)}$$

$$\text{and } \frac{hc}{250 \times 10^{-9}} = W_0 + \frac{1}{2}m(2v)^2 \quad \text{.....(ii)}$$

On solving (i) and (ii)

$$\frac{1}{2}mv^2 = \frac{hc}{3} \left[\frac{1}{250 \times 10^{-9}} - \frac{1}{400 \times 10^{-9}} \right] \quad \text{.....(iii)}$$

From equation (i) and (iii) $W_0 = 2hc \times 10^6 \text{ J}$.

107. (a) $E = W_0 + eV_0 \Rightarrow 4\text{eV} = 2\text{eV} + eV_0 \Rightarrow V_0 = 2 \text{ volt}$

108. (b)

109. (a) $W_0 = \frac{12375}{6800} = 1.8\text{eV}$

110. (b) With the increase in intensity of light photoelectric current increases, but Kinetic energy of ejected electron, stopping potential and work function remains unchanged.

111. (c) $E = h\nu = 6.6 \times 10^{-34} \times 8 \times 10^{15} = 5.28 \times 10^{-18} \text{ J} = 33\text{eV}$ By using $E = W_0 + K_{\max} \Rightarrow K_{\max} = E - W_0$
 $= 33 - 6.125 = 27\text{eV}$

112. (b) $\lambda = \frac{12375}{W_0} = \frac{12375}{2} = 6187.5 \text{ \AA} = 620 \text{ nm}$

113. (a) Minimum kinetic energy is always zero.

114. (c) Speed of photon is $3 \times 10^8 \text{ m/s}$ in vacuum.

115. (c) Minimum frequency : $W_0 = h\nu_0$

$$\Rightarrow \nu_0 = \frac{W_0}{h} = \frac{1.65 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}} = 4 \times 10^{14} \text{ Hz}$$

116. (b) By using $E = W_0 + K_{\max} \Rightarrow K_{\max} = E - W_0$

$$\text{Hence, } K_1 = 1 - 0.5 = 0.5$$

$$\text{and } K_2 = 2.5 - 0.5 = 2 \Rightarrow \frac{K_1}{K_2} = \frac{1}{4}.$$

117. (c) $W_0 \propto \frac{1}{\lambda} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{(W_0)_2}{(W_0)_1} = \frac{4.5}{2.3} = \frac{2}{1}.$

118. (d) $K_{\max} = eV_0 \Rightarrow eV_0 = 4\text{eV} \Rightarrow V_0 = 4\text{V}$

119. (a) Number of photo electrons

$$(N) \propto \text{Intensity} \propto \frac{1}{d^2} \Rightarrow \frac{N_1}{N_2} = \left(\frac{d_2}{d_1} \right)^2$$

$$\Rightarrow \frac{N_1}{N_2} = \left(\frac{100}{50} \right)^2 = \frac{4}{1} \Rightarrow N_2 = \frac{N_1}{4}.$$

120. (c) $P = \frac{W}{t} = \frac{nhc}{\lambda t} \Rightarrow 10^3 = \frac{n \times 6.6 \times 10^{-34} \times 3 \times 10^8}{198.6 \times 1}$
 $\Rightarrow n = 10^{30}.$

121. (c) $p = \frac{nhc}{\lambda t} \Rightarrow 100 = \frac{n \times 6 \times 10^{-34} \times 3 \times 10^8}{540 \times 10^{-9} \times 1} \Rightarrow n = 3 \times 10^{20}$

122. (c) $\frac{1}{2}mv_{\max}^2 = eV_0 \Rightarrow v_{\max} = \sqrt{2 \left(\frac{e}{m} \right) V_0}$
 $= \sqrt{2 \times 1.8 \times 10^{11} \times 9} = 1.8 \times 10^6 \text{ m/s}.$

123. (b) $\frac{hc}{\lambda} = W_0 + K_{\max} \Rightarrow \frac{hc}{\lambda_A} = W_0 + K_A \quad \text{...(i)}$

$$\text{and } \frac{hc}{\lambda_B} = W_0 + K_B \quad \text{.....(ii)}$$

$$\text{Subtracting (i) from (ii), } hc \left[\frac{1}{\lambda_B} - \frac{1}{\lambda_A} \right] = K_B - K_A$$

$$\Rightarrow hc \left[\frac{1}{\lambda_B} - \frac{1}{2\lambda_B} \right] = K_B - K_A \Rightarrow \frac{hc}{2\lambda_B} = K_B - K_A \quad \text{...(iii)}$$

From (ii) and (iii), $2K_B - 2K_A = W_0 + K_B$

$$\Rightarrow K_B - 2K_A = W_0$$

$$\Rightarrow K_A = \frac{K_B}{2} - \frac{W_0}{2} \text{ which gives } K_A < \frac{K_B}{2}.$$

124. (a) $\lambda_0 = \frac{12375}{6500} = 1.9 \text{ eV} \approx 2\text{eV}.$

125. (a) $\lambda_{X\text{-ray}} < \lambda_{UV\text{-ray}}$

126. (a) $E = h\nu_0 + K_{\max} \Rightarrow h(4\nu_0) = h\nu_0 + K_{\max} \Rightarrow K_{\max} = 3h\nu_0.$

127. (a)

128. (c) $W_0 = \frac{12375}{2.3} = 5380 \text{ \AA}.$

129. (d)

130. (b) Using Einstein photoelectric equation $E = W_0 + K_{\max}$

$$hf_1 = W_0 + \frac{1}{2}mv_1^2 \quad \text{.....(i)}$$

$$hf_2 = W_0 + \frac{1}{2}mv_2^2 \quad \text{.....(ii)}$$

$$\Rightarrow h(f_1 - f_2) = \frac{1}{2} m(v_1^2 - v_2^2) \Rightarrow (v_1^2 - v_2^2) = \frac{2h}{m}(f_1 - f_2)$$

131. (d)

132. (b) By using $\frac{hc}{e} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right) = V_0$

$$\Rightarrow \frac{hc}{e} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right) = 4.8 \quad \dots(i)$$

$$\text{and } \frac{hc}{e} \left(\frac{1}{2\lambda} - \frac{1}{\lambda_0} \right) = 1.6 \quad \dots(ii)$$

$$\text{From equation (i) by (ii), } \frac{\left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)}{\left(\frac{1}{2\lambda} - \frac{1}{\lambda_0} \right)} = \frac{4.8}{1.6} \Rightarrow \lambda_0 = 4\lambda.$$

133. (c)

134. (d) $E = W_0 + K_{\max}$. From the given data E is 6.78 eV (for $\lambda = 1824 \text{ \AA}$) or 10.17 eV (for $\lambda = 1216 \text{ \AA}$)

$$\therefore W_0 = E - K_{\max} = 6.78 - 5.3 = 1.48 \text{ eV}$$

or

$$W_0 = 10.17 - 8.7 = 1.47 \text{ eV}.$$

135. (c) $E = \frac{hc}{\lambda} \Rightarrow \frac{E_1}{E_2} = \frac{\lambda_1}{\lambda_2} \Rightarrow \frac{3.32 \times 10^{-19}}{E_2} = \frac{4000}{6000}$

$$\Rightarrow E_2 = 4.98 \times 10^{-19} \text{ J} = 3.1 \text{ eV}.$$

136. (c) Number of waves $= \frac{10^{-3}}{4000 \times 10^{-10}} = 0.25 \times 10^4$

137. (d) Velocity of photon $c = \nu\lambda$

138. (b) $\lambda_0 = \frac{12375}{6.825} = 1813 \text{ \AA} \approx 1800 \text{ \AA}$

139. (c) Work function $W_0 = h\nu_0 = 6.6 \times 10^{-34} \times 1.6 \times 10^{15}$
 $= 1.056 \times 10^{-18} \text{ J} = 6.6 \text{ eV}$

$$\text{From } E = W_0 + K_{\max} \Rightarrow K_{\max} = E - W_0 = 1.4 \text{ eV}$$

140. (c) $P = \frac{h}{\lambda}, E = \frac{hc}{\lambda} \Rightarrow E = Pc.$

141. (a) $E = \frac{hc}{\lambda} \Rightarrow \frac{E_1}{E_2} = \frac{300}{150} = \frac{2}{1}$

142. (d)

143. (b) If frequency of incident light increases, kinetic energy of photoelectron also increases.

144. (d) Photoelectric effect can be explained on the basis of spectrum of an atom.

145. (b) $W_0 = \frac{12375}{\lambda_0} = \frac{12375}{5420} = 2.28 \text{ eV}$

146. (c)

147. (a) $E = \frac{12375}{\lambda} = \frac{12375}{5000} = 2.47 \text{ eV} \approx 2.5 \text{ eV}$

148. (a) Momentum $p = \frac{E}{c} \Rightarrow E^2 = p^2 c^2$

149. (c) Energy of incident radiations (in eV) $= \frac{12375}{4100} = 3.01 \text{ eV}$

Work function of metal A and B are less than 3.01 eV , so A and B will emit photo electrons.

150. (d) From $E = W_0 + \frac{1}{2} m v_{\max}^2$

$$\Rightarrow 2h\nu_0 = h\nu_0 + \frac{1}{2} m v_1^2 \Rightarrow h\nu_0 = \frac{1}{2} m v_1^2 \quad \dots(i)$$

$$\text{and } 5h\nu_0 = h\nu_0 + \frac{1}{2} m v_2^2 \Rightarrow 4h\nu_0 = \frac{1}{2} m v_2^2 \quad \dots(ii)$$

$$\text{Dividing equation (ii) by (i) } \left(\frac{v_2}{v_1} \right)^2 = \frac{4}{1}$$

$$\Rightarrow v_2 = 2v_1 = 2 \times 4 \times 10^6 = 8 \times 10^6 \text{ m/s}$$

151. (d) Number of photoelectrons $\propto \frac{1}{(\text{Distance})^2}$.

152. (b) The value of saturation current depends on intensity. It is independent of stopping potential

153. (a) In tungsten, photoemission takes place with a light of wavelength $2300 \text{ \AA}$. As emission of electron is inversely proportional to wavelength, all the wavelengths smaller than $2300 \text{ \AA}$ will cause emission of electrons.

154. (c) Stopping potential $= 1.8 \text{ eV} - 1.2 \text{ eV} = 0.6 \text{ eV}$.

155. (a)

156. (a) $K.E. = h\nu - h\nu_0 = 8 \text{ eV} - \left(\frac{6 \times 10^{-34} \times 1.6 \times 10^{15}}{1.6 \times 10^{-19}} \text{ eV} \right)$
 $= 8 \text{ eV} - 6 \text{ eV} = 2 \text{ eV}$

X-Rays

1. (c) $\lambda_{\min} = \frac{12375}{50 \times 10^3} \text{ \AA} = 0.247 = 0.25 \text{ \AA}$.

2. (c) X-rays are electromagnetic waves of wavelength ranging from 0.1 to $100 \text{ \AA}$.

3. (a) Penetrating power is greater for lower wavelength.

4. (a)

5. (d) From the formula

$$V = \frac{12375}{\lambda_{\min}} = \frac{12375}{0.3094} = 39.99 \text{ kV} \approx 40 \text{ kV}$$

6. (b) Refer to the application of X-rays.

7. (a)

8. (b)

9. (c)

10. (c) The voltage applied across the X-ray tube is of the range of $10 \text{ kV} - 80 \text{ kV}$.

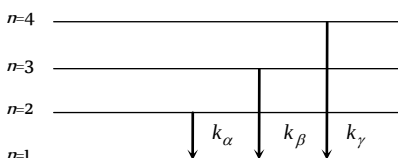
11. (c)

12. (b) In X-ray tube, target must be heavy element with high melting point.

13. (c) $\nu \propto (Z - b)^2 \Rightarrow \nu = a(Z - b)^2$
 $Z = \text{atomic number of element (a, b are constant)}.$

14. (c)

15. (b) X-rays and gamma rays are electromagnetic waves.
16. (c) Since $\lambda_{\min} = \frac{12375}{V} \text{ \AA} = \frac{12375}{10^5} \text{ \AA} = 0.123 \text{ \AA}$
- $$E_{\max} = \frac{hc}{\lambda_{\min}};$$
- On putting the values. $E_{\max} \cong 10^{-1} \text{ MeV}$.
17. (b) $\lambda_{\min} = \frac{hc}{eV}$. where h , c and e are constants. Hence
- $$\lambda_{\min} \propto \frac{1}{V}$$
18. (c) Range of X-rays is $0.1 \text{ \AA}$ to $100 \text{ \AA}$.
19. (d) The production of X-rays is an atomic property whereas the production of γ -rays is a nuclear property.
20. (a) $\lambda_{\min} = \frac{12375}{40,000} = 0.30 \text{ \AA}$ Hence wavelength less than $0.30 \text{ \AA}$ is not possible.
21. (a) $\lambda_{\min} = \frac{hc}{eV}$
22. (b) $p = \frac{h}{\lambda} = \frac{6.6 \times 10^{-34}}{0.01 \times 10^{-10}} = 6.6 \times 10^{-22} \text{ kg} \cdot \text{m} / \text{sec}$.
23. (a) X-rays are absorbed by the target; they are not reflected by the target.
24. (c)
25. (d)
26. (a)
27. (d)
28. (c)
29. (b) Continuous spectrum of X-rays consists of radiations of all possible wavelength range having a definite short wavelength limit.
30. (b) $\frac{E}{t} = P = \frac{h\nu}{t}$
- i.e. Penetrating power $\propto$ energy $\propto$ Frequency
31. (c) In general X-rays have larger wavelength than that of gamma rays.
32. (c) According to Mosley's law $\nu = a(Z-b)^2$ and $\nu \propto \frac{1}{\lambda}$
33. (b) $E = h\nu = eV \Rightarrow \nu \propto V$
34. (c) $E = eV = h\nu_{\max} \Rightarrow \nu_{\max} = \frac{eV}{h}$
35. (c) $E = eV = h\nu_{\max} = \frac{hc}{\lambda_{\min}} \Rightarrow \lambda_{\min} = \frac{hc}{eV}$
36. (d) $\lambda_{\min} = \frac{hc}{eV}$ or $\lambda_{\min} \propto \frac{1}{V}$ On increasing potential, $\lambda_{\min}$ decreases.
37. (a) $h\nu_o = eV \therefore \nu_o = \frac{eV}{h} = \frac{1.6 \times 10^{-19} \times 42000}{6.63 \times 10^{-34}} = 10^{19} \text{ Hz}$
38. (d) Nucleus of heavy atom captures electron of k -orbit. This is a radioactive process, so vacancy of this electron is filled by an outer electron and x-rays are produced.
39. (d) Because they are electromagnetic waves.
40. (c) $\nu_{\max} \propto \frac{1}{\lambda_{\min}}$ Hard X-rays have high frequency and low wavelength.
41. (d) X-rays are electromagnetic in nature so they remain unaffected in electric and magnetic field.
42. (b)
43. (c)
44. (b) X-rays have high energy. They penetrate into the solid crystal and used to find out the internal structure.
45. (a) By changing the filament current with the help of rheostat, thermionic emission intensity of X-rays can be changed.
46. (c) Applied voltage must be greater than binding energy.
47. (a)
48. (d) $\lambda = \frac{12375}{(40 \times 10^3)} = 0.309 \text{ \AA} \approx 0.31 \text{ \AA}$
49. (c)
50. (b) $\lambda_{\min} = \frac{hc}{eV} = \frac{12375}{V} \text{ \AA} = 0.495 \text{ \AA} \approx 0.5 \text{ \AA}$
51. (c) $\lambda_{\min} = \frac{hc}{eV} = \frac{12375}{V} \text{ \AA}; \therefore V = \frac{12375}{\lambda \text{ in \AA}} = 124 \text{ kV}$
52. (a) Mosley's law is $f = a(Z-b)^2$
53. (b) The potential difference across the filament and target determines the energy and thence the penetrating power of X-rays.
54. (d) The energy of X-ray photon obtained from a Coolidge tube by an electronic transition of target atom such as K_{α} line is obtained from transition from L orbit in K orbit.
55. (b) $\lambda_{\min} = \frac{12375}{V} = \frac{12375}{30 \times 10^3} = 0.4 \text{ \AA}$
56. (d) $\lambda_{\min} = \frac{12375}{100 \times 10^3} \text{ \AA} = 0.124 \text{ \AA}$
57. (d) $\lambda_{\min} = \frac{12375}{50000} = 0.025 \text{ nm}$
58. (c)
59. (a) Refer theory
60. (a) With the increase in potential difference between anode and cathode energy of striking electrons increases which in turn increases the energy (penetration power) of X-rays.
61. (a)
62. (b)
63. (b) The wavelength range of X-ray is $0.1 \text{ \AA} - 100 \text{ \AA}$.
64. (b) Energy $E = h\nu = h \frac{c}{\lambda} \therefore \frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} = \frac{5000}{1}$
65. (b)
66. (a) Interatomic spacing in a crystal acts as a diffraction grating.
67. (b) The wavelength of the γ -rays is shorter. However the main distinguishing feature is the nature of emission.
68. (d) $h\nu_{\max} = eV \Rightarrow \frac{hc}{\lambda_{\min}} = eV \therefore \lambda_{\min} \propto \frac{1}{V}$

69. (d) Hard X-rays are of higher energy and the energy of X-rays depends on the potential difference between the cathode and the target.
70. (d) Penetration is directly proportional to the energy of radiations.
71. (d) Greater the number of electrons striking the anode, larger is the number of X-ray photons emitted.
72. (a) $\lambda_{\min} = \frac{12375}{V} \text{ \AA} \Rightarrow V = \frac{12375}{1} = 12375 \text{ V}$
 $= 12.375 \text{ kV} \approx 12.42 \text{ kV}$
73. (c)
74. (c)
75. (c) $E(\text{eV}) = \frac{12375}{1.65} = 7500 \text{ eV} = 7.5 \text{ keV}$.
76. (d)
77. (b)
78. (a) $\lambda_{\min} = \frac{12375}{40} \text{ \AA} = 3.09 \times 10^{-8} \text{ m}$
79. (d) Target should be of high atomic number and high melting point
80. (a) Intensity of X-rays depends upon the number of electron striking the target.
81. (b) $E(\text{eV}) = \frac{hc}{e\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 1 \times 10^{-10}} = 12375 \text{ eV}$
82. (c) When applied voltage is greater than energy of K-electron, continuous and all characteristic X-rays are emitted.
83. (c)
- 
84. (d) When current through the filament increases, number of emitted electrons also increases. Hence intensity of X-ray increases but no effect on penetration power.
85. (a) $i = \frac{Ne}{t} \Rightarrow \frac{N}{t} = \frac{i}{e} = \frac{3.2 \times 10^{-3}}{1.6 \times 10^{-19}} = 2 \times 10^{16} / \text{sec}$
86. (d)
87. (a) Because X-rays are electromagnetic (Neutral) in nature.
88. (d) $\lambda_{\min} = \frac{hc}{eV} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times V} = \frac{12375}{V} \approx \frac{12400}{V} \text{ \AA}$
89. (c) Frequency of hard X-rays is greater than that of soft X-rays.
90. (b)
91. (a) $\lambda_{\min} = \frac{12375}{V} \text{ \AA} \Rightarrow V = \frac{12375}{0.4125} = 30 \text{ kV}$
92. (b)
93. (d)
94. (a) $\lambda_{\min} = \frac{12375}{40 \times 10^3} = 0.309 \text{ \AA} \approx 0.31 \text{ \AA}$
95. (d) $\lambda_{\min} = \frac{hc}{eV} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 100 \times 10^3} = 0.123 \text{ \AA}$
96. (d) According to Mosley's law $\nu \propto (Z - b)^2$
 For k_α line, $b = 1$, and it has maximum frequency so $\nu_{\max} \propto (Z - 1)^2$
97. (b) The velocity of X-rays is always equal to that of light.
98. (b)
99. (d) $\lambda_{\min} = \frac{12375}{V} \text{ \AA} \Rightarrow V = \frac{12375}{2.5} = 4950 \text{ V} \approx 5 \text{ kV}$
100. (c) $\lambda_{\min} = \frac{hc}{eV(\text{energy})}$; when KE (or eV) increases, λ decreases.
101. (c)
102. (b) When a high energy electron incident on heavy metal, it produces X-rays.
103. (b) $\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{1 \times 10^{-10}} = 3 \times 10^{18} \text{ Hz}$
104. (b) $\lambda \propto \frac{1}{Z^2} \Rightarrow \frac{c}{\nu} \propto \frac{1}{Z^2} \Rightarrow \nu \propto Z^2$
105. (d)
106. (a) $eV = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1.5 \times 10^{-10}}$
 $\Rightarrow V = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 1.5 \times 10^{-10}} = 8280 \text{ Volt}$
107. (d)
108. (d)
109. (b) Required ionisation energy
 $= \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.54 \times 10^{-10}} \text{ J} = 12.9 \times 10^{-16} \text{ J}$
110. (c) $\lambda \propto \frac{1}{(Z-1)^2} \Rightarrow \frac{\lambda_2}{\lambda_1} = \left(\frac{Z_1 - 1}{Z_2 - 1} \right)^2$
 $\Rightarrow \frac{\lambda_2}{\lambda} = \left(\frac{43 - 1}{29 - 1} \right)^2 = \left(\frac{42}{28} \right)^2 \Rightarrow \lambda_2 = \frac{9}{4} \lambda$
- III. (a)

Critical Thinking Questions

1. (b) For one second, distance = Velocity = $3 \times 10^4 \text{ m/sec}$ and
 $Q = i \times t = 10^{-6} \text{ C}$. Charge density = $\frac{\text{Charge}}{\text{Volume}}$
 $= \frac{10^{-6}}{3 \times 10^4 \times 0.5 \times 10^{-6}} = 6.6 \times 10^{-5} \text{ C/m}^3$.
2. (c) By law of conservation of momentum
 $0 = m_1 \vec{v}_1 + m_2 \vec{v}_2 \Rightarrow m_1 \vec{v}_1 = -m_2 \vec{v}_2$
 - ve sign indicates that both the particles are moving in opposite direction. Now de-Broglie wavelengths
 $\lambda_1 = \frac{h}{m_1 v_1}$ and $\lambda_2 = \frac{h}{m_2 v_2}$; $\therefore \frac{\lambda_1}{\lambda_2} = \frac{m_2 v_2}{m_1 v_1} = 1$
3. (b) $\lambda_{\text{photon}} = \frac{hc}{E}$ and $\lambda_{\text{proton}} = \frac{h}{\sqrt{2mE}}$

$$\Rightarrow \frac{\lambda_{\text{photon}}}{\lambda_{\text{electron}}} = c \sqrt{\frac{2m}{E}} \Rightarrow \frac{\lambda_{\text{photon}}}{\lambda_{\text{electron}}} \propto \frac{1}{\sqrt{E}}$$

4. (a,b,c) $K_{\text{max}} = E - W_{\text{A}}$

$$\therefore T_{\text{A}} = 4.25 - (W_{\text{A}}) \quad \dots(i)$$

$$T_{\text{B}} = (T_{\text{A}} - 1.5) = 4.70 - (W_{\text{A}}) \quad \dots(ii)$$

Equation (i) and (ii) gives $(W_{\text{A}}) - (W_{\text{A}}) = 1.95 \text{ eV}$

$$\text{De Broglie wave length } \lambda = \frac{h}{\sqrt{2mK}} \Rightarrow \lambda \propto \frac{1}{\sqrt{K}}$$

$$\Rightarrow \frac{\lambda_{\text{B}}}{\lambda_{\text{A}}} = \sqrt{\frac{K_{\text{A}}}{K_{\text{B}}}} \Rightarrow 2 = \sqrt{\frac{T_{\text{A}}}{T_{\text{A}} - 1.5}} \Rightarrow T_{\text{A}} = 2 \text{ eV}$$

From equation (i) and (iii)

$$W_{\text{A}} = 2.25 \text{ eV and } W_{\text{B}} = 4.20 \text{ eV.}$$

5. (d)

6. (b) In the presence of inert gas photoelectrons emitted by cathode ionise the gas by collision and hence the current increases.

7. (b) For electron and positron pair production, minimum energy is 1.02 MeV.

$$\text{Energy of photon is given } 1.7 \times 10^J = \frac{1.7 \times 10^{-13}}{1.6 \times 10^{-19}}$$

$$= 1.06 \text{ MeV.}$$

Since energy of photon is greater than 1.02 MeV,

So electron, positron pair will be created.

8. (c) According to Einstein's photoelectric equation

$$\frac{hc}{\lambda} = \phi + \frac{1}{2}mv^2 \Rightarrow v = \left[\frac{2(hc - \lambda\phi)}{m\lambda} \right]^{1/2}$$

9. (b) Cut off voltage is independent of intensity and hence remains the same. Since distance becomes 3 times, so intensity (I)

becomes $\frac{I}{9}$. Hence photo current also decreases by this factor

$$\text{i.e. becomes } \frac{18}{9} = 2 \text{ mA.}$$

10. (d) $h\nu - W_0 = \frac{1}{2}mv_{\text{max}}^2 \Rightarrow \frac{hc}{\lambda} - \frac{hc}{\lambda_0} = \frac{1}{2}mv_{\text{max}}^2$

$$\Rightarrow hc \left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0} \right) = \frac{1}{2}mv_{\text{max}}^2 \Rightarrow v_{\text{max}} = \sqrt{\frac{2hc}{m} \left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0} \right)}$$

When wavelength is λ and velocity is v , then

$$v = \sqrt{\frac{2hc}{m} \left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0} \right)} \quad \dots (i)$$

When wavelength is $\frac{3\lambda}{4}$ and velocity is v' then

$$v' = \sqrt{\frac{2hc}{m} \left[\frac{\lambda_0 - (3\lambda/4)}{(3\lambda/4) \times \lambda_0} \right]} \quad \dots(ii)$$

Divide equation (ii) by (i), we get

$$\frac{v'}{v} = \sqrt{\frac{[\lambda_0 - (3\lambda/4)] \times \lambda\lambda_0}{\frac{3}{4}\lambda\lambda_0 \times \lambda_0 - \lambda}} \quad \dots(iii)$$

$$v' = v \left(\frac{4}{3} \right)^{1/2} \sqrt{\frac{[\lambda_0 - (3\lambda/4)]}{\lambda_0 - \lambda}} \text{ i.e. } v' > v \left(\frac{4}{3} \right)^{1/2}$$

11. (c) Intensity of light

$$I = \frac{\text{Watt}}{\text{Area}} = \frac{nhc}{A\lambda} \Rightarrow \text{Number of photon } n = \frac{IA\lambda}{hc}$$

$$\therefore \text{Number of photo electron} = \frac{1}{100} \times \frac{IA\lambda}{hc}$$

$$= \frac{1}{100} \frac{1 \times 10^{-4} \times 300 \times 10^{-9}}{6.6 \times 10^{-34} \times 3 \times 10^8} = 1.5 \times 10^{12}$$

12. (b) By using $h\nu - h\nu_0 = K_{\text{max}}$

$$\Rightarrow h(\nu_1 - \nu_0) = K_1 \quad \dots(i)$$

$$\text{And } h(\nu_2 - \nu_0) = K_2 \quad \dots(ii)$$

$$\Rightarrow \frac{\nu_1 - \nu_0}{\nu_2 - \nu_0} = \frac{K_1}{K_2} = \frac{1}{K}, \text{ Hence } \nu_0 = \frac{K\nu_1 - \nu_2}{K - 1}$$

13. (b) $E = W_0 + eV_0$

For hydrogen atom, $E = +13.6 \text{ eV}$

$$\therefore +13.6 = 4.2 + eV_0$$

$$\Rightarrow V_0 = \frac{(13.6 - 4.2)eV}{e} = 9.4 \text{ V}$$

Potential at anode = -9.4 V

14. (a) From $\lambda_0 = \frac{12375}{W_0}$

The maximum wavelength of light required for the photoelectron emission, $(\lambda_0)_{\text{Li}} = \frac{12375}{2.3} = 5380 \text{ \AA}$.

$$\text{Similarly } (\lambda_0)_{\text{Cu}} = \frac{12375}{4} = 3094 \text{ \AA}.$$

Since the wavelength 3094 \AA does not in the visible region, but it is in the ultraviolet region. Hence to work with visible light, lithium metal will be used for photoelectric cell.

15. (a) Direction of scattered photon $\cos\phi = 1 - \frac{\Delta\lambda m_e c}{h}$

Here $\Delta\lambda = 0.011 \text{ \AA}$

$$\therefore \cos\phi = 1 - \frac{0.011 \times 10^{-10} \times 9.1 \times 10^{-31} \times 3 \times 10^8}{6.624 \times 10^{-34}} = 1 - 0.453 = 0.547$$

$$\therefore \phi = \cos^{-1}(0.547)$$

16. (d) Bragg's law, $2d \sin\theta = n\lambda$ or $\lambda = \frac{2d \sin\theta}{n}$

For maximum wavelength, $n_{\text{min}} = 1$, $(\sin\theta)_{\text{max}} = 1$

$$\therefore \lambda_{\text{max}} = 2d \text{ or } \lambda_{\text{max}} = 2 \times 10^{-7} \text{ cm} = 20 \text{ \AA}$$

17. (a,c,d) $P = VI = 50 \times 10^3 \times 20 \times 10^{-3} = 1000 \text{ W}$

Power converted into heat = 990 W

$$ms\Delta T = 990 \Rightarrow \Delta T = 2^\circ \text{ C/sec}$$

$$\text{Now } \frac{hc}{\lambda_{\text{min}}} = eV \Rightarrow \lambda_{\text{min}} = \frac{hc}{eV} = 0.248 \times 10^{-10} \text{ m}$$

18. (c) The wavelength of X-ray lines is given by Rydberg

Formula $\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

For K_α line, $n_1 = 1$ and $n_2 = 2$

$$\therefore \frac{1}{\lambda} = RZ^2 \left(\frac{3}{4} \right) \Rightarrow Z = \left(\frac{4}{3R\lambda} \right)^{1/2}$$

$$= \left[\frac{4}{3(1.097 \times 10^7 \text{ m}^{-1})(0.76 \times 10^{-10} \text{ m})} \right]^{1/2} = 39.99 \approx 40$$

19. (b) If intensity of X-ray is decreased by dI when it passes through a length dx of absorbing material then, the amount of observed intensity is $\mu I dx$.

Thus, $-dI = \mu I dx$ or $\frac{dI}{I} + \mu dx = 0$

On solving this equation $I = I_0 e^{-\mu x}$ ($x=d$)

20. (c) $E_K - E_L = \frac{hc}{\lambda} = \frac{(6.6 \times 10^{-34}) (3 \times 10^8)}{(0.021 \times 10^{-9}) (1.6 \times 10^{-19})} \text{ eV} = 59 \text{ keV}$

21. (d) Minimum wavelength of continuous X-ray spectrum is given by

$$\lambda_{\min} (\text{in } \text{\AA}) = \frac{12375}{E(\text{eV})} = \frac{12375}{80 \times 10^3} \approx 0.155$$

Also the energy of the incident electrons (80 KeV) is more than the ionization energy of the K-shell electrons (i.e. 72.5 KeV). Therefore characteristic X-ray spectrum will also be obtained because energy of incident electron is enough to knock out the electron from K or L shells.

22. (a) The wave length of L_α line is given by

$$\frac{1}{\lambda} = R(z-7.4)^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \Rightarrow \lambda \propto \frac{1}{(z-7.4)^2}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{(z_2-7.4)^2}{(z_1-7.4)^2} \Rightarrow \frac{1.30}{\lambda_2} = \frac{(42-7.4)^2}{(78-7.4)^2} \Rightarrow \lambda_2 = 5.41 \text{ \AA}$$

23. (c) de-Broglie wavelength $\lambda = \frac{h}{mv_{rms}}$, rms velocity of a gas particle at the given temperature (T) is given as

$$\frac{1}{2} mv_{rms}^2 = \frac{3}{2} kT \Rightarrow v_{rms} = \sqrt{\frac{3kT}{m}} \Rightarrow mv_{rms} = \sqrt{3mkT}$$

$$\therefore \lambda = \frac{h}{mv_{rms}} = \frac{h}{\sqrt{3mkT}}$$

$$\Rightarrow \frac{\lambda_H}{\lambda_{He}} = \sqrt{\frac{m_{He} T_{He}}{m_H T_H}} = \sqrt{\frac{4(273+127)}{2(273+27)}} = \sqrt{\frac{8}{3}}$$

24. (c) $n = \frac{E\lambda}{hc} = \frac{1 \times 10^{-7} \times 200 \times 10^{-9}}{6.6 \times 10^{-34} \times 3 \times 10^8} = 1 \times 10^{11}$

$$\text{Number of electrons ejected} = \frac{10^{11}}{10^3} = 10^8$$

$$\therefore V = \frac{q}{4\pi\epsilon_0 r} = \frac{(10^8 \times 1.6 \times 10^{-19}) \times 9 \times 10^9}{4.8 \times 10^{-2}} = 3 \text{ V}$$

25. (b) The momentum of the incident radiation is given as $p = \frac{h}{\lambda}$.

When the light is totally reflected normal to the surface the direction of the ray is reversed. That means it reverses the direction of its momentum without changing its magnitude

$$\therefore \Delta p = 2p = \frac{2h}{\lambda} = \frac{2 \times 6.6 \times 10^{-34}}{6630 \times 10^{-10}} = 2 \times 10^{-19} \text{ kg-m/sec.}$$

26. (c) When a charged particle (charge q , mass m) enters perpendicularly in a magnetic field (B) then, radius of the path described by it $r = \frac{mv}{qB} \Rightarrow mv = qBr$.

Also de-Broglie wavelength $\lambda = \frac{h}{mv}$

$$\Rightarrow \lambda = \frac{h}{qBr} \Rightarrow \frac{\lambda_\alpha}{\lambda_p} = \frac{q_p r_p}{q_\alpha r_\alpha} = \frac{1}{2}$$

27. (a) $\sqrt{f_1} = \sqrt{\frac{v}{\lambda_1}} = a(11-1)$ and $\sqrt{f_2} = \sqrt{\frac{v}{\lambda_2}} = a(Z-1)$

By dividing, $\sqrt{\frac{\lambda_2}{\lambda_1}} = \frac{10}{Z-1} \Rightarrow \sqrt{\frac{4}{1}} = \frac{10}{Z-1} \Rightarrow Z = 6$

28. (c) $K.E. = 2 E_i - E_f = E_i$ (for $0 \leq x \leq 1$) $\Rightarrow \lambda_1 = \frac{h}{\sqrt{2mE_0}}$

$$K.E. = 2 E_i$$
 (for $x > 1$) $\Rightarrow \lambda_2 = \frac{h}{\sqrt{4mE_0}} \Rightarrow \frac{\lambda_1}{\lambda_2} = \sqrt{2}$

29. (b) Given $mc = 0.51 \text{ MeV}$ and $v = 0.8 c$
K.E. of the electron = $mc - mc$

$$\text{But } m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{m_0}{\sqrt{1 - \left(\frac{0.8c}{c} \right)^2}} = \frac{m_0}{\sqrt{0.36}} = \frac{m_0}{0.6}$$

$$\text{Now, } mc^2 = \frac{0.51}{0.6} \text{ MeV} = 0.85 \text{ MeV}$$

$$\therefore K.E. = (0.85 - 0.51) \text{ MeV} = 0.34 \text{ MeV.}$$

30. (a) The deflection suffered by charged particle in an electric field is

$$y = \frac{qELD}{mu^2} = \frac{qELD}{p^2/m} \quad (p=mu)$$

$$\Rightarrow y \propto \frac{qm}{p^2} \Rightarrow y : y' : y_\alpha = \frac{q_p m_p}{p_p^2} : \frac{q_d m_d}{p_d^2} : \frac{q_\alpha m_\alpha}{p_\alpha^2}$$

Since $p_\alpha = p$, (given)

$$m : m' : m_\alpha = 1 : 2 : 4 \text{ and } q : q' : q_\alpha = 1 : 1 : 2$$

$$\Rightarrow y : y' : y_\alpha = 1 \times 1 : 1 \times 2 : 2 \times 4 = 1 : 2 : 8$$

31. (c) Using $Z^2 = k \left(\frac{q}{m} \right) y$; where $k = \frac{B^2 LD}{E}$. For parabolas to

coincide in the two photographs, the $\frac{kq}{m}$ should be same for

$$\text{the two cases. Thus, } \frac{B_1^2 L D e}{E_1 m_1} = \frac{B_2^2 L D (2e)}{E_2 m_2}$$

$$\Rightarrow \frac{m_1}{m_2} = \left(\frac{B_1}{B_2} \right)^2 \times \left(\frac{E_2}{E_1} \right) \times \frac{1}{2} = \frac{9}{4} \times \frac{2}{1} \times \frac{1}{2} = \frac{9}{4}$$

32. (c) According to the energy diagram of X-ray spectra

$$\therefore \Delta E = \frac{hc}{\lambda} \Rightarrow \lambda \propto \frac{1}{\Delta E}$$

(ΔE = Energy radiated when e jumps from, higher energy orbit to lower energy orbit)

$$\because (\Delta E)_{k\beta} > (\Delta E)_{k\alpha} > (\Delta E)_{L\alpha} \therefore \lambda'_\alpha > \lambda_\alpha > \lambda_\beta$$

$$\text{Also } (\Delta E)_{k\beta} = (\Delta E)_{k\alpha} + (\Delta E)_{L\alpha}$$

$$\Rightarrow \frac{hc}{\lambda_\beta} = \frac{hc}{\lambda_\alpha} + \frac{hc}{\lambda'_\alpha} \Rightarrow \frac{1}{\lambda_\beta} = \frac{1}{\lambda_\alpha} + \frac{1}{\lambda'_\alpha}$$

33. (c) By using $I = \frac{P}{A}$; where P = radiation power

$$\Rightarrow P = I \times A \Rightarrow \frac{nhc}{t\lambda} = IA \Rightarrow \frac{n}{t} = \frac{IA\lambda}{hc}$$

Hence number of photons entering per sec the eye

$$\left(\frac{n}{t}\right) = \frac{10^{-10} \times 10^{-6} \times 5.6 \times 10^{-7}}{6.6 \times 10^{-34} \times 3 \times 10^8} = 300.$$

34. (d) $\Delta\lambda = \lambda_{K\alpha} - \lambda_{\min}$ When V is halved $\lambda_{\min}$ becomes two times but $\lambda_{K\alpha}$ remains the same.

$$\therefore \Delta\lambda' = \lambda_{K\alpha} - 2\lambda_{\min} = 2(\Delta\lambda) - \lambda_{K\alpha}$$

$$\therefore \Delta\lambda' < 2(\Delta\lambda)$$

35. (b) Energy of photons corresponding to light of wave length λ_c =

$$2475 \text{ \AA} \text{ is } E_1 = \frac{12375}{2475} = 5 \text{ eV.}$$

and that corresponding to $\lambda_c = 6000 \text{ \AA}$ is

$$E_2 = \frac{12375}{6000} = 2.06 \text{ eV}$$

As $E_1 < W$ and $E_2 > W$

Photoelectric emission is possible with λ_c only. Maximum kinetic energy of emitted photoelectrons $K = E - W = 5 - 4.8 = 0.2 \text{ eV}$.

Photo electrons experiences magnetic force and move along a circular path of radius

$$r = \frac{\sqrt{2mk}}{QB} = \frac{\sqrt{2 \times 9 \times 10^{-31} \times 0.2 \times 1.6 \times 10^{-19}}}{1.6 \times 10^{-19} \times 3 \times 10^{-5}}$$

$$= 0.05 \text{ m} = 5 \text{ cm.}$$

36. (a) Number of photoelectrons emitted up to $t = 10 \text{ sec}$ are

$$n = \frac{(\text{Number of photons per unit area}) \times (\text{Area} \times \text{Time})}{10^6}$$

$$= \frac{1}{10^6} [(10)^{16} \times (5 \times 10^{-4}) \times (10)] = 5 \times 10^7$$

At time $t = 10 \text{ sec}$

$$\text{Charge on plate A; } q_1 = +ne = 5 \times 10^7 \times 1.6 \times 10^{-19} \\ = 8 \times 10^{-12} \text{ C} = 8 \text{ pC}$$

and charge on plate B; $q_2 = 33.7 - 8 = 25.7 \text{ pC}$

Electric field between the plates

$$E = \frac{(q_B - q_A)}{2\epsilon_0 A} = \frac{(25.7 - 8) \times 10^{-12}}{2 \times 8.85 \times 10^{-12} \times 5 \times 10^{-4}} = 2 \times 10^3 \frac{\text{N}}{\text{C}}$$

37. (a) As we know in Young's double slit experiment fringe width = separation between two consecutive fringe or dark fringes

$$= \beta = \frac{\lambda D}{d}$$

$$\text{Here } \beta = 2y \Rightarrow 2y = \frac{\lambda D}{d} \Rightarrow \lambda = \frac{2yd}{D}$$

$$\Rightarrow \lambda = \frac{2 \times 1 \times 10^{-3} \times 0.24 \times 10^{-3}}{1.2} = 4 \times 10^{-7} \text{ m} = 4000 \text{ \AA}$$

Energy of light incident on photo plate

$$E (\text{eV}) = \frac{12375}{4000} = 3.1 \text{ eV}$$

According to Einstein photoelectric equation

$$E = W_0 + eV_0 \Rightarrow V_0 = \frac{(E - W_0)}{e} = \frac{(3 - 2.2)}{e} \text{ eV} \approx 0.9 \text{ V}$$

$$38. (a) E = \frac{12375}{5000} = 2.475 \text{ eV} \approx 4 \times 10^{-19} \text{ J}$$

So the minimum intensity to which the eye can respond

$$I_{\text{Eye}} = (\text{Photon flux}) \times (\text{Energy of a photon})$$

$$\Rightarrow I_{\text{Eye}} = (5 \times 10^4) \times (4 \times 10^{-19}) \approx 2 \times 10^{-14} \text{ (W/m}^2\text{)}$$

Now as lesser the intensity required by a detector for detection, more sensitive it will be

$$\frac{S_{\text{Eye}}}{S_{\text{Ear}}} = \frac{I_{\text{Ear}}}{I_{\text{Eye}}} = \frac{10^{-13}}{2 \times 10^{-14}} = 5 \text{ i.e. as intensity (power)}$$

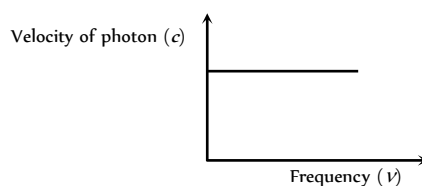
detector, the eye is five times more sensitive than ear.

39. (c) Due to 10.2 eV photon one photon of energy 10.2 eV will be detected.

Due to 15 eV photon the electron will come out of the atom with energy $(15 - 13.6) = 1.4 \text{ eV}$.

Graphical Questions

1. (a) Velocity of photon (i.e. light) does not depend upon frequency. Hence the graph between velocity of photon and frequency will be as follows



2. (d) De-Broglie wavelength $\lambda = \frac{h}{p} \Rightarrow \lambda \propto \frac{1}{p}$

i.e. graph will be a rectangular hyperbola.

3. (a) The stopping potential for curves a and b is same.

$$\therefore f_a = f_b$$

Also saturation current is proportional to intensity

$$\therefore I_a < I_b$$

4. (d) According to Einstein equation

$$h\nu = h\nu_0 + K_{\max} \Rightarrow K_{\max} = h\nu - h\nu_0 \text{ on comparing it with } y = mx + c, \text{ it is clear to say that,}$$

This is the equation of straight line having positive slope (h) and negative intercept ($h\nu_0$) on KE axis.

5. (c) Comparing Einstein's equation

$$K_{\max} = h\nu - h\nu_0, \text{ with } y = mx + c, \text{ we get slope, } m = h$$

6. (b) $K_{\max} = h\nu - h\nu_0 \Rightarrow eV_0 = h\nu - h\nu_0 \Rightarrow V_0 = \frac{h}{e}\nu - \frac{h\nu_0}{e}$
Comparing this equation with $y = mx + c$, we get slope
 $m = \frac{h}{e} \Rightarrow h = m \times e$.
7. (a) Using Einstein's equation, $V_0 = \left(\frac{h}{e}\right)\nu - \frac{W_0}{e}$
Comparing this equation with $y = mx + c$
We get intercept on $-V_0$ axis = $\frac{W_0}{e}$
 $\Rightarrow OB = \frac{W_0}{e} \Rightarrow W_0 = OB \times e$
8. (b) From the given graph it is clear that if we extend the given graph for A and B , intercept of the line A on V axis will be smaller as compared to line B means work function of A is smaller than that of B .
9. (a) Wavelength λ_k is independent of the accelerating voltage (V), while the minimum wavelength λ_c is inversely proportional to V . Therefore as V increases, λ_k remains unchanged whereas λ_c decreases or $\lambda_k - \lambda_c$ will increase.
10. (c) In X-ray spectra, depending on the accelerating voltage and the target element, we may find sharp peaks super imposed on continuous spectrum. These are at different wavelengths for different elements. They form characteristic X-ray spectrum.
11. (b) Photo current (i) directly proportional to light intensity (I) falling on a photosensitive plate. $\Rightarrow i \propto I$
12. (d) According to Einstein's equation
$$h\nu = W_0 + K_{\max} \Rightarrow V_0 = \left(\frac{h}{e}\right)\nu - \frac{W_0}{e}$$

This is the equation of straight line having positive slope (h/e) and intercept on $-V_0$ axis, equals to $\frac{W_0}{e}$
13. (d) In photocell, at a particular negative potential (stopping potential V) of anode, photoelectric current is zero,
At the potential difference between cathode and anode increases current through the circuit increases but after some time constant current (saturation current) flows through the circuit even if potential difference still increasing.
14. (b) Stopping potential does not depend upon intensity of incident light (I).
15. (a) Stopping potential is that negative potential for which photo electric current is zero.
16. (d) $\because V_0 = \left(\frac{h}{e}\right)\nu - \left(\frac{W_0}{e}\right)$. From the graph $V_2 > V_1$
 $\Rightarrow \frac{h\nu_2}{e} - \frac{W_0}{e} > \frac{h\nu_1}{e} - \frac{W_0}{e} \Rightarrow \nu_2 > \nu_1$
 $\Rightarrow \lambda_1 > \lambda_2$ (as $\lambda \propto \frac{1}{\nu}$)
17. (d) $I \propto \frac{1}{d^2}$ and photo current $i \propto I \Rightarrow i \propto \frac{1}{d^2}$
18. (a) $h\nu = h\nu_0 + KE_{\max} \Rightarrow KE_{\max} = h\nu - h\nu_0$
On comparing this equation with $y = mx + c$ we get
 $m = h = \text{Universal constant}$
19. (b) $\lambda_0 = \frac{c}{\nu_0} = \frac{3 \times 10^8}{5 \times 10^{14}} = 6 \times 10^{-7} \text{ m} = 6000 \text{ \AA}$
20. (c) Work function is the intercept on KE axis i.e. $2eV$.
21. (b) From the graph stopping potential $V_f - V$
Also $K_{\max} = (V_f - V)e = 4eV$.
22. (c) By Moseley's law, $\sqrt{\nu} = a(Z - b)$ or, $\nu = a^2(Z - b)^2$
Comparing with the equation of a parabola, $y^2 = 4ax$ it conforms to graph c .
23. (a) $\lambda_{\min} = \frac{hc}{eV} \Rightarrow \lambda \propto \frac{1}{V}$
 $\therefore \lambda_2 > \lambda_1$ (see graph) $\Rightarrow V_1 > V_2$
 $\sqrt{\nu} = a(Z - b)$ Moseley's law
 $\nu \propto (Z - 1)^2 \Rightarrow \lambda \propto \frac{1}{(Z - 1)^2} \quad \left(\because \nu \propto \frac{1}{\lambda}\right)$
 $\lambda_1 > \lambda_2$ (see graph for characteristic lines) $\Rightarrow Z_2 > Z_1$.
24. (a) $K_{\max} = h\nu - h\nu_0 = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$ i.e. graph between $K_{\max}$ and $\frac{1}{\lambda}$ will be straight line having slope (hc) and intercept $\frac{hc}{\lambda_0}$ on $-KE$ axis.
25. (a) ν varies from 0 to $\nu_{\max}$.
26. (a) $\lambda_{\min} = \frac{hc}{eV} \Rightarrow \log \lambda_{\min} = \log \frac{hc}{e} - \log V$
 $\Rightarrow \log \lambda_{\min} = -\log V + \log \frac{hc}{e}$
This is the equation of straight line having slope (-1) and intercept $\log \frac{hc}{e}$ on $+\log_e \lambda_{\min}$ axis.
27. (c) For K_{α} line $\nu \propto (Z - 1)^2 \Rightarrow \lambda \propto \frac{1}{(Z - 1)^2}$
i.e. the graph between λ and z will be (c).
28. (b) Slope of $V_0 - \nu$ curve = $\frac{h}{e}$
 $\Rightarrow h = \text{Slope} \times e = 1.6 \times 10^{-19} \times 4.12 \times 10^6$
 $= 6.6 \times 10^{-34} \text{ J-sec}$.
29. (b) $I_1 > I_2$ (given) $\Rightarrow i > i'$ ($\because i \propto I$)
and stopping potential does not depend upon intensity. So its value will be same (V_0).
30. (c) Slope of $V_f - \nu$ curve for all metals be same $\left(\frac{h}{e}\right)$ i.e. curves should be parallel.
31. (c) $\lambda = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2m}} \cdot \frac{1}{\sqrt{E}}$. Taking $\log$ of both sides
 $\log \lambda = \log \frac{h}{\sqrt{2m}} + \log \frac{1}{\sqrt{E}} \Rightarrow \log \lambda = \log \frac{h}{\sqrt{2m}} - \frac{1}{2} \log E$
 $\Rightarrow \log \lambda = -\frac{1}{2} \log E + \log \frac{h}{\sqrt{2m}}$
This is the equation of straight line having slope $(-1/2)$ and positive intercept on $\log \lambda$ axis.

32. (b) $\sqrt{v} \propto (Z - b)$
33. (b) Peak of K_{α} is greater than peak of K_{β} line.
34. (a) $\vdash 4V \vdash 2V$
35. (a) $\because x \propto \frac{1}{v^2}$. The ion whose deflection is less, its velocity will be more. From the curve $x_1 < x_2 < x_3 < x_4$, therefore $v_1 > v_2 > v_3 > v_4$.
36. (a) All the positive ions of same specific charge moving with different velocity lie on the same parabola.
37. (b) The equation of curve between V and v is $\frac{h\nu}{e} - \frac{h\nu_0}{e} = V_0$.
This is equation of a straight line with slope $= \frac{h}{e}$.
38. (b) Stopping potential equals to maximum kinetic energy. Since stopping potential is varying linearly with the frequency. There fore max. KE for both the metals also vary linearly with frequency.

Assertion and Reason

1. (a) Momentum of a photon is given by $p = \frac{h}{\lambda}$
Also the photon is a form of energy packets behaves as a particle having energy $E = \frac{hc}{\lambda}$. So $p = \frac{E}{c}$
2. (d) Photoelectric effect demonstrates the particle nature of light. Number of emitted photoelectrons depends upon the intensity of light.
3. (b) Charge does not change with speed but mass varies with the speed as per relation $m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$. Hence specific charge e/m decreases with increase in speed.
4. (a) X-rays lies in electromagnetic spectrum.
5. (b) Mass of moving photon $m = \frac{h\nu}{c^2} = \frac{h}{c\lambda}$ and $E = mc^2$.
6. (d) According Einstein equation $KE = h\nu - h\nu_0$; i.e., KE depends upon the frequency. Photoelectron emitted only if incident frequency more than threshold frequency.
7. (e) The atomic number (number of electrons or protons) remains same in isotope. Isotope of an element can be separated on account of their different atomic weight by using mass spectrograph.
8. (b) The specific charge (e/m) of the positive rays is not universal constant because these rays may consists of ions of different element.
9. (b) Less work function means less energy is required for ejecting out the electrons.
10. (a) de-Broglie wavelength associated with gas molecules varies as $\lambda \propto \frac{1}{\sqrt{T}}$
11. (e) If electron is moving parallel to the magnetic field, then the electron is not deflected i.e., if electron is not deflected we cannot be sure that there is no magnetic field in that region.
12. (d) At normal pressure positive ions and electrons liberated by ionisation of gas atoms, due to cosmic rays are very small in number and they collide constantly with the gas atoms which are present in large numbers, and hence are unable to move a

long distance under the electric field and soon get recombined i.e., flow of ions in the gas does not take place.

13. (d) Light is produced in gases in the process of electric discharge at low pressure. When accelerated electrons collide with atoms of the gas, atoms get excited. The excited atoms return to their normal state and in this process light radiations are emitted.
14. (d) The discharge depends on both pressure of discharge tube and ionisation potential of gas. Since the ionisation potential of different gases are different, hence the discharge in different gases takes place at different potential.
15. (d) If electric field is used for detecting the electron beam, then very high voltage will have to be applied or very long tube will have to be taken.
16. (b) Specific charge of a positive ion corresponding to one gas is fixed but it is different for different gases.
17. (e) In Millikan's experiment oil drops should be of microscopic sizes. If much bigger oil drops are used, then a very high electric field will be required to balance it which is not possible to achieve practically.

Further, the apparent weight of the liquid $\frac{4}{3}\pi r^3 g$

$$(\rho_{\text{liquid}} - \sigma_{\text{air}}) = 6\pi\eta r v.$$

If a is large, v will be large and the experimental errors will be high.

18. (e) Only the photoelectrons emitted from the surface of the metal have maximum kinetic energy. Those emitted from inside the metal loses part of their energy in collision with the other atoms inside the metal.
19. (d) On increasing the intensity of incident light, the current in photoelectric cell will increase. The energy of the photons ($h\nu$) will, however not increase with increase in intensity, and hence the kinetic energy of the emitted electrons will not increase.
20. (a) When a light of single frequency falls on the electron of inner layer of metal, then this electron comes out of the metal surface after a large number of collisions with atoms of its upper layer.
21. (b) There is no emission of photoelectrons till the frequency of incident light is less than a minimum frequency, however intense light it may be. In photoelectric effect, it is a single particle collision. Intensity is $h\nu \times N$, where $h\nu$ is the individual energy of the photon and N is the total number of photon. In the wave theory, the intensity is proportional, not only to ν^2 but also to the amplitude squared. For the same frequency, increase in intensity only increase the number of photons (in the quantum theory of Einstein).
22. (a) The photoemissive cell may be evacuated contain an inert gas at low pressure. An inert gas in the cell gives greater current but causes a time lag in the response of the cell to very rapid changes of radiation which may make it unsuitable for some purpose.
23. (c) Wavelength of X-rays is very small ($\approx \text{\AA}$). Hence they are not diffracted by means of ordinary grating. X-rays follows the Bragg's law.
24. (b) The penetrating power of X-rays depends upon the voltage applied across the tube producing X-rays. X-rays can pass through matter of lighter elements such as flesh (which is composed of oxygen, hydrogen and carbon) but cannot pass through substances made of heavier elements like bones (which are made of phosphorus and calcium).
25. (c) Intensity of X-rays (I) is proportional to the filament current and also to the square of the voltage. It is well known that

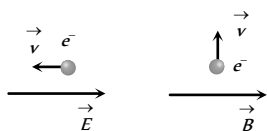
intensity of X -rays depends on the number of photons emitted per second from target.

26. (b) When fast moving electrons strike the atoms of the target, then most of their kinetic energy is used in increasing the thermal agitation of the atoms of the target and only a small part is radiated in the form of X -rays. So the temperature of the target rises.
27. (e) Higher is the wavelength of X -ray, lesser is the frequency and penetration power.
28. (a) The distance between the atoms of crystals is of the order of wavelength of X -rays. When they fall on a crystal, they are diffracted. The diffraction pattern is helpful in the study of crystal structure.
29. (b) In photoelectric effect, the photon falling on some matter is absorbed by the matter and its energy is transferred to an electron of the matter. In X -ray production, photons are produced which get energy from energetic electrons ionising the inner shells of the target which in turn cause a cascade of emission lines.
30. (e) Soft and hard X -rays differ only in frequency. But both types of X -ray travel with speed of light.

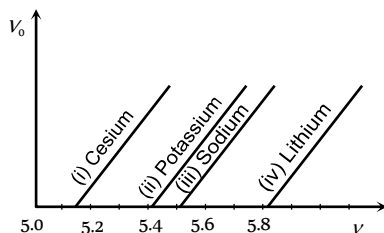
Electron, Photon, Photoelectric Effect and X-rays

Self Evaluation Test -25

- Which of the following will have the least value of $\frac{q}{m}$
 - Electron
 - Proton
 - α -particle
 - β -particle
- When green light is incident on the surface of metal, it emits photo-electrons but there is no such emission with yellow colour light. Which one of the colour can produce emission of photo-electrons
 - Orange
 - Red
 - Indigo
 - None of the above
- An electron is moving through a field. It is moving (i) opposite an electric field (ii) perpendicular to a magnetic field as shown. For each situation the de-Broglie wave length of electron



- Increasing, increasing (i)
 - Increasing, decreasing (ii)
 - Decreasing, same
 - Same, Same
- The figure shows different graphs between stopping potential (V_0) and frequency (ν) for photosensitive surface of cesium, potassium, sodium and lithium. The plots are parallel. Correct ranking of the targets according to their work function greatest first will be

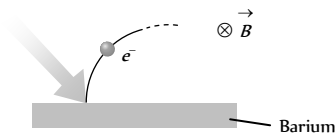


- (i) > (ii) > (iii) > (iv)
 - (i) > (iii) > (ii) > (iv)
 - (iv) > (iii) > (ii) > (i)
 - (i) = (iii) > (ii) = (iv)
- The K_{α} X-rays arising from a cobalt ($z = 27$) target have a wavelength of 179 pm . The K_{α} X-rays arising from a nickel target ($z = 28$) is
 - $> 179 \text{ pm}$
 - $< 179 \text{ pm}$
 - $= 179 \text{ pm}$
 - None of these
 - If a voltage applied to an X-ray tube is increased to 1.5 times the minimum wavelength ($\lambda_{\min}$) of an X-ray continuous spectrum shifts by $\Delta\lambda = 26 \text{ pm}$. The initial voltage applied to the tube is
 - $\approx 10 \text{ kV}$
 - $\approx 16 \text{ kV}$
 - $\approx 50 \text{ kV}$
 - $\approx 75 \text{ kV}$
 - Light of wavelength $2475 \text{ \AA}$ is incident on barium. Photoelectrons emitted describe a circle of radius 100 cm by a magnetic field of flux

density $\frac{1}{\sqrt{17}} \times 10^{-5} \text{ Tesla}$. Work function of the barium is (Given

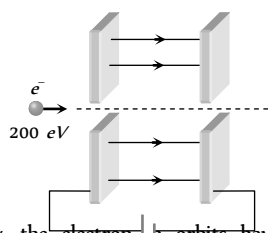
$$\frac{e}{m} = 1.7 \times 10^{11})$$

- 1.8 eV
- 2.1 eV
- 4.5 eV
- 3.3 eV



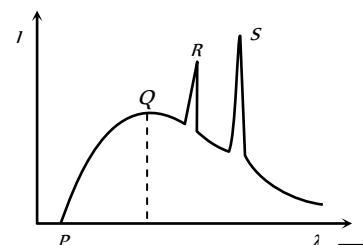
- Five elements A, B, C, D and E have work functions $1.2 \text{ eV}, 2.4 \text{ eV}, 3.6 \text{ eV}, 4.8 \text{ eV}$ and 6 eV respectively. If light of wavelength $4000 \text{ \AA}$ is allowed to fall on these elements, then photoelectrons are emitted by
 - A, B and C
 - A, B, C, D and E
 - A and B
 - Only E
- If light of wavelength λ_1 is allowed to fall on a metal, then kinetic energy of photoelectrons emitted is E_1 . If wavelength of light changes to λ_2 then kinetic energy of electrons changes to E_2 . Then work function of the metal is
 - $\frac{E_1 E_2 (\lambda_1 - \lambda_2)}{\lambda_1 \lambda_2}$
 - $\frac{E_1 \lambda_1 - E_2 \lambda_2}{(\lambda_1 - \lambda_2)}$
 - $\frac{E_1 \lambda_1 - E_2 \lambda_2}{(\lambda_2 - \lambda_1)}$
 - $\frac{\lambda_1 \lambda_2 E_1 E_2}{(\lambda_2 - \lambda_1)}$
- If maximum velocity with which an electron can be emitted from a photo cell is $4 \times 10^8 \text{ cm/sec}$, the stopping potential is (mass of electron = $9 \times 10^{-31} \text{ kg}$)
 - 30 volt
 - 45 volt
 - 59 volt
 - Information is insufficient
- Three particles having their charges in the ratio of $1 : 3 : 5$ produce the same spot on the screen in Thomson's experiment. Their masses are in the ratio of
 - $5 : 3 : 1$
 - $3 : 1 : 5$
 - $1 : 3 : 5$
 - $5 : 1 : 3$
- If the momentum of an electron is changed by Δp , then the de-Broglie wavelength associated with it changes by 0.50% . The initial momentum of the electron will be
 - $\frac{\Delta p}{200}$
 - $\frac{\Delta p}{199}$
 - $199 \Delta p$
 - $400 \Delta p$
- If 10000 V is applied across an X-ray tube, what will be the ratio of de-Broglie wavelength of the incident electrons to the shortest wavelength of X-ray produced ($\frac{e}{m}$ for electron is $1.8 \times 10^{11} \text{ C kg}^{-1}$)

- (a) 1 (b) 0.1
(c) 0.2 (d) 0.3
14. Two large parallel plates are connected with the terminal of 100 V power supply. These plates have a fine hole at the centre. An electron having energy 200 eV is so directed that it passes through the holes. When it comes out its de-Broglie wavelength is



- (a) 1.22 Å
(b) 1.75 Å
(c) 2 Å
(d) None of these
15. According to Bohr's theory, the electron orbits have definite energy values, then according to uncertainty principle, the life time of an excited state will be
- (a) Zero (b) Finite
(c) 10^{-16} sec (d) Infinite
16. Monochromatic light of wavelength 3000 Å is incident on a surface area 4 cm². If intensity of light is 150 mW/m², then rate at which photons strike the target is
- (a) 3×10^{16} /sec (b) 9×10^{16} /sec
(c) 7×10^{16} /sec (d) 6×10^{16} /sec
17. For characteristic X-ray of some material
- (a) $E(K_\gamma) < E(K_\beta) < E(K_\alpha)$ (b) $E(K_\alpha) < E(L_\alpha) < E(M_\alpha)$
(c) $\lambda(K_\gamma) < \lambda(K_\beta) < \lambda(K_\alpha)$ (d) $\lambda(M_\alpha) < \lambda(L_\alpha) < \lambda(K_\alpha)$

18. The maximum velocity of electrons emitted from a metal surface is V . When frequency of light falling on it is f . The maximum velocity when frequency becomes $4f$ is
- (a) $2V$ (b) $> 2V$
(c) $< 2V$ (d) Between $2V$ and $4V$
19. If the potential difference between the anode and cathode of the X-ray tube is increases



- (a) The peaks at R and S would move to shorter wavelength
(b) The peaks at R and S would remain at the same wavelength
(c) The cut off wavelength at P would decrease
(d) (b) and (c) both are correct
20. The collector plate in an experiment on photoelectric effect is kept vertically above the emitter plate. Light source is put on and a saturation photo current is recorded. An electric field is switched on which has a vertically downward direction
- (a) The photo current will increase
(b) The kinetic energy of the electrons will increase
(c) The stopping potential will decrease
(d) The threshold wavelength will increase

AS Answers and Solutions

(SET -25)

1. (c) Mass of α -particle is maximum so $\left(\frac{q}{m}\right)_\alpha$ is least.
2. (c) Wave length of green light is threshold wave length. Hence for emission of electron, wave length of incident light $<$ wavelength of green light.
3. (c) $\lambda = \frac{h}{mv}$. Since v is increasing in case (i), but it is not changing in case (ii). Hence, in the first case de-Broglie wavelength will change, but in second case, it remains the same.
4. (c) The graph between V_c and v cut the v -axis at v_0 . For the given graphs $(v_0)_{(iv)} > (v_0)_{(ii)} > (v_0)_{(i)} > (v_0)_{(j)}$
 $\therefore (W)_{(iv)} > (W)_{(ii)} > (W)_{(i)} > (W)_{(j)}$.

$$5. \quad (b) \quad \lambda_{k\alpha} \propto \frac{1}{(Z-1)^2} \Rightarrow \frac{\lambda_{Ni}}{\lambda_{Co}} = \left(\frac{Z_{Co}-1}{Z_{Ni}-1} \right)^2 = \left(\frac{27-1}{28-1} \right)^2$$

$$\Rightarrow \lambda_{Ni} = \left(\frac{26}{27} \right)^2 \times \lambda_{Co} = \left(\frac{26}{27} \right)^2 \times 179 = 165.9 \text{ pm} < 179 \text{ pm}.$$

$$6. \quad (b) \quad \lambda_{\min} = \frac{hc}{eV} \Rightarrow \lambda_1 = \frac{hc}{eV_1} \text{ and } \lambda_2 = \frac{hc}{eV_2}$$

$$\therefore \Delta\lambda = \lambda_2 - \lambda_1 = \frac{hc}{e} \left[\frac{1}{V_2} - \frac{1}{V_1} \right]. \text{ Given } V_1 = 1.5 \text{ V}$$

on solving we get $V_2 = 16000 \text{ volt} = 16 \text{ kV}$.

$$7. \quad (c) \quad \text{Radius of circular path described by a charged particle in a magnetic field is given by } r = \frac{\sqrt{2mK}}{qB}; \text{ where } K = \text{Kinetic}$$

$$\text{energy of electron } \Rightarrow K = \frac{q^2 B^2 r^2}{2m} = \left(\frac{e}{m} \right) \frac{eB^2 r^2}{2}$$

$$= \frac{1}{2} \times 1.7 \times 10^{11} \times 1.6 \times 10^{-19} \times \left(\frac{1}{\sqrt{17}} \times 10^{-5} \right)^2 \times (1)^2$$

$$= 8 \times 10^{-20} \text{ J} = 0.5 \text{ eV}$$

By using $E = W_0 + K_{\max}$

$$\Rightarrow W_0 = E - K_{\max} = \left(\frac{12375}{2475} \right) \text{ eV} - 0.5 \text{ eV} = 4.5 \text{ eV}$$

$$8. \quad (c) \quad E = \frac{12375}{4000} = 3.09 \text{ eV} \text{ Photoelectrons emits if energy of incident light } > \text{ work function.}$$

$$9. \quad (c) \quad E = W_0 + K_{\max} \Rightarrow \frac{hc}{\lambda_1} = W_0 + E_1 \text{ and } \frac{hc}{\lambda_2} = W_0 + E_2$$

$$\Rightarrow hc = W_0 \lambda_1 + E_1 \lambda_1 \text{ and } hc = W_0 \lambda_2 + E_2 \lambda_2$$

$$\Rightarrow W_0 \lambda_1 + E_1 \lambda_1 = W_0 \lambda_2 + E_2 \lambda_2 \Rightarrow W_0 = \frac{E_1 \lambda_1 - E_2 \lambda_2}{(\lambda_2 - \lambda_1)}.$$

$$10. \quad (b) \quad v_{\max} = 4 \times 10^6 \text{ cm/sec} = 4 \times 10^4 \text{ m/sec.}$$

$$\therefore K_{\max} = \frac{1}{2} m v_{\max}^2 = \frac{1}{2} \times 9 \times 10^{-31} \times (4 \times 10^4)^2$$

$$= 7.2 \times 10^{-17} \text{ J} = 45 \text{ eV.}$$

$$\text{Hence, stopping potential } |V_0| = \frac{K_{\max}}{e} = \frac{45 \text{ eV}}{e} = 45 \text{ volt.}$$

$$11. \quad (c) \quad \text{Since spot is same, hence } \frac{e}{m} \text{ should be same i.e.,}$$

As $q : q : q = 1 : 3 : 5$. Hence $m : m : m = 1 : 3 : 5$

$$12. \quad (c) \quad \lambda = \frac{h}{p} \Rightarrow \lambda - \frac{0.5}{100} \lambda = \frac{h}{p + \Delta p} \Rightarrow \frac{199\lambda}{200} = \frac{h}{p + \Delta p} = \frac{199}{200} \frac{h}{p}$$

$$\Rightarrow p + \Delta p = \frac{200}{199} p \Rightarrow p = 199 \Delta p$$

$$13. \quad (b) \quad \text{For the incident electron } \frac{1}{2} m v^2 = eV \text{ or } p^2 = 2m eV$$

$$\therefore \text{de-Broglie wavelength } \lambda_1 = \frac{h}{p} = \frac{h}{\sqrt{2m eV}}$$

$$\text{Shortest X-ray wavelength } \lambda_2 = \frac{hc}{eV}$$

$$\therefore \frac{\lambda_1}{\lambda_2} = \frac{1}{c} \sqrt{\left(\frac{V}{2} \right) \left(\frac{e}{m} \right)} = \frac{\sqrt{\frac{10^4}{2} \times 1.8 \times 10^{11}}}{3 \times 10^8} = 0.1$$

$$14. \quad (a) \quad \text{Energy of the electron, when it comes out from the second plate} = 200 \text{ eV} - 100 \text{ eV} = 100 \text{ eV}$$

Hence accelerating potential difference = 100 V

$$\lambda_{\text{Electron}} = \frac{12.27}{\sqrt{V}} = \frac{12.27}{\sqrt{100}} = 1.23 \text{ \AA}$$

$$15. \quad (d) \quad \text{According to Bohr's theory } \Delta E = 0, \text{ since } \Delta E \cdot \Delta t \geq h$$

$$\Rightarrow \Delta t \rightarrow \infty$$

$$16. \quad (b) \quad \frac{n}{t} = \frac{I A \lambda}{hc} = \frac{150 \times 10^{-3} \times 4 \times 10^{-4} \times 3 \times 10^{-7}}{6.6 \times 10^{-34} \times 3 \times 10^8} = 9 \times 10^{13} \frac{1}{\text{sec}}$$

$$17. \quad (c) \quad \because E(K_\gamma) > E(K_\beta) > E(K_\alpha) \Rightarrow \lambda(K_\gamma) < \lambda(K_\beta) < \lambda(K_\alpha)$$

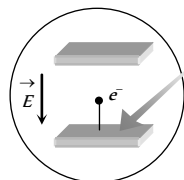
$$18. \quad (b) \quad \because E = W_0 + \frac{1}{2} m v_{\max}^2 \Rightarrow v_{\max} = \sqrt{\frac{2(hf - W_0)}{m}}$$

If frequency becomes $4f$ then

$$V' = \sqrt{\frac{2(h \times 4f - W_0)}{m}} = 2 \sqrt{\frac{2(hf - \frac{W_0}{4})}{m}} \Rightarrow V' > 2V$$

$$19. \quad (d) \quad \text{Peaks on the graph represent characteristic X-ray spectrum. Every peak has a certain wavelength, which depends upon the transition of electron inside the atom of the target. While } \lambda_{\min} \text{ depends upon the accelerating voltage (As. } \lambda_{\min} \propto 1/V \text{).}$$

$$20. \quad (b) \quad \text{In electric field photoelectron will experience force and accelerate opposite to the field so it's K.E. increases (i.e. stopping potential will increase), no change in photoelectric current, and threshold wavelength.}$$





Chapter 26

Atomic and Nuclear Physics

Thomson's Atomic Model

J.J. Thomson gave the first idea regarding structure of atom. According to this model.

(1) An atom is a solid sphere in which entire and positive charge and it's mass is uniformly distributed and in which negative charge (*i.e.* electron) are embedded like seeds in watermelon.

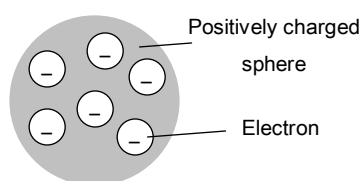


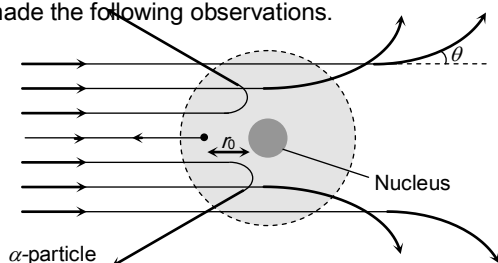
Fig. 26.1

(2) This model explained successfully the phenomenon of thermionic emission, photoelectric emission and ionization.

(3) The model fail to explain the scattering of α - particles and it cannot explain the origin of spectral lines observed in the spectrum of hydrogen and other atoms.

α -Scattering Experiment

'Geiger and Marsden (students of Rutherford) studied the scattering of α -particles by gold foil on the advice of Rutherford and made the following observations.



(1) Most of the α -particles pass through the foil straight away undeflected.

(2) Some of them are deflected through small angles.

(3) A few α -particles (1 in 1000) are deflected through the angle more than 90° .

(4) A few α -particles (very few) returned back *i.e.* deflected by 180° .

(5) Number of scattered particles : $N \propto \frac{1}{\sin^4(\theta/2)}$

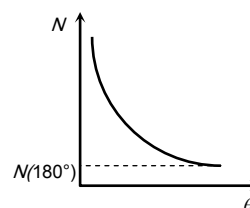


Fig. 26.3

(6) If t is the thickness of the foil and N is the number of α -particles scattered in a particular direction (*i.e.* $\theta = \text{constant}$), it was observed that $\frac{N}{t} = \text{constant} \Rightarrow \frac{N_1}{N_2} = \frac{t_1}{t_2}$

(7) **Distance of closest approach** (Nuclear dimension) :

The minimum distance from the nucleus up to which the α -particle approach, is called the distance of closest approach (r_0). At this distance the entire initial kinetic energy has been converted into potential energy so

$$\frac{1}{2}mv^2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{(Ze)2e}{r_0} \Rightarrow r_0 = \frac{Ze^2}{mv^2\pi\epsilon_0} = \frac{4kZe^2}{mv^2}$$

(8) **Impact parameter** (b) : The perpendicular distance of the velocity vector ($\vec{v}$) of the α -particle from the centre of the nucleus when it is far away from the nucleus is known as impact parameter. It is given as

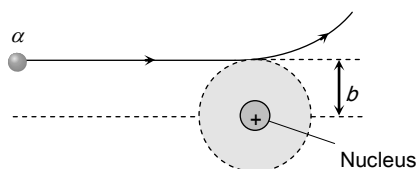


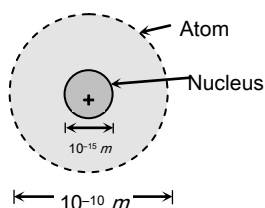
Fig. 26.4

$$b = \frac{Ze^2 \cot(\theta/2)}{4\pi\epsilon_0 \left(\frac{1}{2}mv^2\right)} \Rightarrow b \propto \cot(\theta/2)$$

For large b , α particles will go undeviated and for small b the α -particle will suffer large scattering.

Rutherford's Atomic Model

After α -particles scattering experiment, following conclusions were made by Rutherford as regard as atomic structure :



Size of the nucleus = 1 Fermi = 10^{-15} m

Size of the atom $1 \text{ \AA} = 10^{-10} \text{ m}$

Fig. 26.5

(1) Most of the mass (at least 99.95%) and all of the charge of an atom concentrated in a very small region is called atomic nucleus.

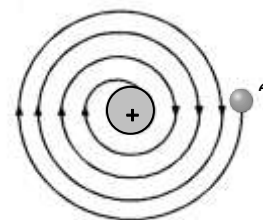
(2) Nucleus is positively charged and its size is of the order of

$10^{-15} \text{ m} \approx 1 \text{ Fermi}$. The nucleus occupies only about 10^{-12} of the total volume of the atom or less.

(3) In an atom there is maximum empty space and the electrons revolve around the nucleus in the same way as the planets revolve around the sun.

Failure of Rutherford's Model

(1) **Stability of atom** : It could not explain stability of atom because according to classical electrodynamics theory an accelerated charged particle should continuously radiate energy. Thus an electron moving in an circular path around the nucleus should lose energy and move into smaller and smaller orbits and it should ultimately fall into the nucleus.



Instability of atom

Fig. 26.6

(2) According to this model the spectrum of atom must be continuous where as practically it is a line spectrum.

(3) It did not explain the distribution of electrons outside the nucleus.

Bohr's Atomic Model

Bohr proposed a model for hydrogen atom which is also applicable for some lighter atoms in which a single electron revolves around a stationary nucleus of positive charge Ze (called hydrogen like atom)

Bohr's model is based on the following postulates.

(1) He postulated that an electron in an atom can move around the nucleus in certain circular stable orbits without emitting radiations.

(2) Bohr found that the magnitude of the electron's

Angular momentum is quantized *i.e.* $L = mv_n r_n = n \left(\frac{h}{2\pi} \right)$

where $n = 1, 2, 3, \dots$ each value of n corresponds to a permitted value of the orbit radius.

r_n = Radius of n^{th} orbit, v_n = corresponding speed

(3) The radiation of energy occurs only when an electron jumps from one permitted orbit to another.

When electron jumps from higher energy orbit (E_2) to lower energy orbit (E_1) then difference of energies of these orbits *i.e.* $E_2 - E_1$ emits in the form of photon. But if electron goes from E_1 to E_2 it absorbs the same amount of energy.

Draw Backs of Bohr's Atomic Model

(1) It is valid only for one electron atoms, *e.g.* : $H, He^+, Li^{+2}, Na^{+1}$ etc.

(2) Orbits were taken as circular but according to Sommerfeld these are elliptical.

(3) Intensity of spectral lines could not be explained.

(4) Nucleus was taken as stationary but it also rotates on its own axis.

(5) It could not be explained the minute structure in spectrum line.

(6) This does not explain the Zeeman effect (splitting up of spectral lines in magnetic field) and Stark effect (splitting up in electric field)

(7) This does not explain the doublets in the spectrum of some of the atoms like sodium ($5890 \text{ \AA}$ & $5896 \text{ \AA}$)

Bohr's Orbits (for Hydrogen and H_2 -like Atoms)

(1) **Radius of orbit** : For an electron around a stationary nucleus the electrostatics force of attraction provides the necessary centripetal force

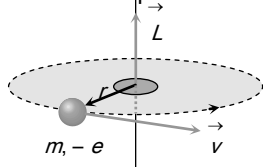


Fig. 26.7

$$\text{i.e. } \frac{1}{4\pi\epsilon_0} \frac{(Ze)e}{r^2} = \frac{mv^2}{r} \quad \dots (i)$$

$$\text{also } mvr = \frac{nh}{2\pi} \quad \dots (ii)$$

From equation (i) and (ii) radius of n^{th} orbit

$$r_n = \frac{n^2 h^2}{4\pi^2 k Z m e^2} = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2} = 0.53 \frac{n^2}{Z} \text{ \AA} \quad (k = \frac{1}{4\pi\epsilon_0})$$

$$\Rightarrow r_n \propto \frac{n^2}{Z}$$

(2) **Speed of electron** : From the above relations, speed of electron in n^{th} orbit can be calculated as

$$v_n = \frac{2\pi k Z e^2}{nh} = \frac{Z e^2}{2\epsilon_0 n h} = \left(\frac{c}{137} \right) \cdot \frac{Z}{n} = 2.2 \times 10^6 \frac{Z}{n} \text{ m/sec}$$

where (c = speed of light $3 \times 10^8 \text{ m/s}$)

Table 26.1 : Some other quantities for revolution of electron in n^{th} orbit

Quantity	Formula	Dependency on n and Z
(1) Angular speed	$\omega_n = \frac{v_n}{r_n} = \frac{\pi m z^2 e^4}{2 \epsilon_0^2 n^3 h^3}$	$\omega_n \propto \frac{Z^2}{n^3}$
(2) Frequency	$\nu_n = \frac{\omega_n}{2\pi} = \frac{m z^2 e^4}{4 \epsilon_0^2 n^3 h^3}$	$\nu_n \propto \frac{Z^2}{n^3}$
(3) Time period	$T_n = \frac{1}{\nu_n} = \frac{4 \epsilon_0^2 n^3 h^3}{m z^2 e^4}$	$T_n \propto \frac{n^3}{Z^2}$
(4) Angular momentum	$L_n = m v_n r_n = n \left(\frac{h}{2\pi} \right)$	$L_n \propto n$
(5) Corresponding current	$i_n = e \nu_n = \frac{m z^2 e^5}{4 \epsilon_0^2 n^3 h^3}$	$i_n \propto \frac{Z^2}{n^3}$
(6) Magnetic moment	$M_n = i_n A = i_n (\pi r_n^2)$ (where $\mu_0 = \frac{eh}{4\pi m}$ = Bohr magneton)	$M_n \propto n$
(7) Magnetic field	$B = \frac{\mu_0 i_n}{2r_n} = \frac{\pi m^2 z^3 e^7 \mu_0}{8 \epsilon_0^3 n^5 h^5}$	$B \propto \frac{Z^3}{n^5}$

Energy

(1) **Potential energy** : An electron possesses some potential energy because it is found in the field of nucleus potential energy of electron in n^{th} orbit of radius r_n is given by

$$U = k \cdot \frac{(Ze)(-e)}{r_n} = -\frac{kZe^2}{r_n}$$

(2) **Kinetic energy** : Electron possesses kinetic energy because of its motion. Closer orbits have greater kinetic energy than outer ones.

$$\text{As we know } \frac{mv^2}{r_n} = \frac{k(Ze)(e)}{r_n^2}$$

$$\Rightarrow \text{Kinetic energy } K = \frac{kZe^2}{2r_n} = \frac{|U|}{2}$$

(3) **Total energy** : Total energy (E) is the sum of potential energy and kinetic energy *i.e.* $E = K + U$

$$\Rightarrow E = -\frac{kZe^2}{2r_n} \text{ also } r_n = \frac{n^2 h^2 \epsilon_0}{\pi m z e^2}$$

$$\text{Hence } E = -\left(\frac{me^4}{8\epsilon_0^2 h^2}\right) \cdot \frac{z^2}{n^2} = -\left(\frac{me^4}{8\epsilon_0^2 h^3}\right) ch \frac{z^2}{n^2}$$

$$= -Rch \frac{Z^2}{n^2} = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

$$\text{where } R = \frac{me^4}{8\epsilon_0^2 h^3} = \text{Rydberg's constant} = 1.09 \times 10^7 \text{ per } m.$$

(4) **Ionisation energy and potential** : The energy required to ionise an atom is called ionisation energy. It is the energy required to make the electron jump from the present orbit to the infinite orbit.

$$\text{Hence } E_{\text{ionisation}} = E_{\infty} - E_n = 0 - \left(-13.6 \frac{Z^2}{n^2}\right) = +\frac{13.6Z^2}{n^2} \text{ eV}$$

For H_2 -atom in the ground state

$$E_{\text{ionisation}} = \frac{+13.6(1)^2}{n^2} = 13.6 \text{ eV}$$

The potential through which an electron needs to be accelerated so that it acquires energy equal to the ionisation energy is called ionisation potential. $V_{\text{ionisation}} = \frac{E_{\text{ionisation}}}{e}$

(5) **Excitation energy and potential** : When energy is given to an electron from an external source, it jumps to a higher energy level. This phenomenon is called excitation.

The minimum energy required to excite an atom is called excitation energy of the particular excited state and corresponding potential is called exciting potential.

$$E_{\text{Excitation}} = E_{\text{Final}} - E_{\text{Initial}} \text{ and } V_{\text{Excitation}} = \frac{E_{\text{excitation}}}{e}$$

(6) **Binding energy (B.E.)** : Binding energy of a system is defined as the energy released when its constituents are brought from infinity to form the system. It may also be defined as the energy needed to separate its constituents to large distances. If an electron and a proton are initially at rest and brought from large distances to form a hydrogen atom, 13.6 eV energy will be released. The binding energy of a hydrogen atom is therefore 13.6 eV.

(7) **Energy level diagram** : The diagrammatic description of the energy of the electron in different orbits around the nucleus is called energy level diagram.

Table 26.2 : Energy level diagram of hydrogen/hydrogen like atom

atom				
-----	$n = \infty$	Infinite	Infinite	0 eV
-----	$n = 4$	Fourth	Third	- 0.85 eV
-----	$n = 3$	Third	Second	- 1.51 eV
-----	$n = 2$	Second	First	- 3.4 eV
-----	$n = 1$	First	Ground	- 13.6 eV
	Principle quantum number	Orbit	Excited state	Energy for H_2 - atom

Transition of Electron

When an electron makes transition from a higher energy level having energy $E_2(n_2)$ to a lower energy level having energy $E_1(n_1)$ then a photon of frequency ν is emitted

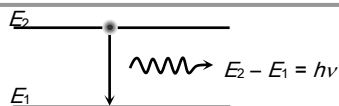


Fig. 26.8

(1) Energy of emitted radiation

$$\Delta E = E_2 - E_1 = \frac{-RchZ^2}{n_2^2} - \left(\frac{-RchZ^2}{n_1^2} \right)$$

$$= 13.6Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

(2) Frequency of emitted radiation

$$\Delta E = h\nu \Rightarrow \nu = \frac{\Delta E}{h} = \frac{E_2 - E_1}{h} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

(3) Wave number/wavelength

Wave number is the number of waves in unit length

$$\bar{\nu} = \frac{1}{\lambda} = \frac{\nu}{c} \Rightarrow \frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = \frac{13.6Z^2}{hc} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

(4) Number of spectral lines : If an electron jumps from higher energy orbit to lower energy orbit it emits radiations with various spectral lines.

If electron falls from orbit n_2 to n_1 then the number of spectral lines emitted is given by

$$N_E = \frac{(n_2 - n_1 + 1)(n_2 - n_1)}{2}$$

If electron falls from n^{th} orbit to ground state (*i.e.* $n_2 = n$ and $n_1 = 1$) then number of spectral lines emitted

$$N_E = \frac{n(n-1)}{2}$$

(5) Recoiling of an atom : Due to the transition of electron, photon is emitted and the atom is recoiled

Recoil momentum of atom = momentum of photon

$$= \frac{h}{\lambda} = hRZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Also recoil energy of atom = $\frac{p^2}{2m} = \frac{h^2}{2m\lambda^2}$ (where m = mass of recoil atom)

Hydrogen Spectrum and Spectral Series

When hydrogen atom is excited, it returns to its normal unexcited (or ground state) state by emitting the energy it had absorbed earlier. This energy is given out by the atom in the form of radiations of different wavelengths as the electron jumps down from a higher to a lower orbit. Transition from different orbits cause different wavelengths, these constitute spectral series which are characteristic of the atom emitting them. When observed through a spectroscope, these radiations are imaged as sharp and straight vertical lines of a single colour.

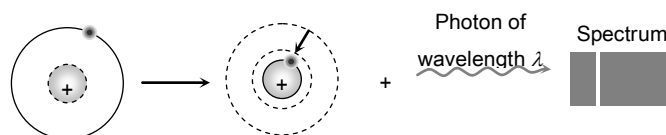


Fig. 26.9 : Emission spectra

The spectral lines arising from the transition of electron forms a spectra series.

(1) Mainly there are five series and each series is named after its discover as Lyman series, Balmer series, Paschen series, Brackett series and Pfund series.

(2) According to the Bohr's theory the wavelength of the radiations emitted from hydrogen atom is given by

$$\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \Rightarrow \lambda = \frac{n_1^2 n_2^2}{(n_2^2 - n_1^2)R} = \frac{n_1^2}{\left(1 - \frac{n_1^2}{n_2^2} \right)R}$$

where n_2 = outer orbit (electron jumps from this orbit), n_1 = inner orbit (electron falls in this orbit)

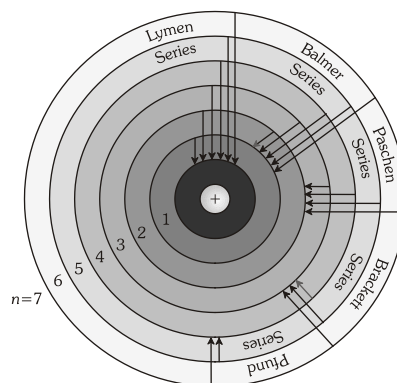


Fig. 26.10

(3) First line of the series is called first member, for this line wavelength is maximum ($\lambda_{\max}$)

For maximum wavelength if $n_1 = n$ then $n_2 = n + 1$

$$\text{So } \lambda_{\max} = \frac{n^2(n+1)^2}{(2n+1)R}$$

(4) Last line of the series is called series limit, for this line wavelength is minimum ($\lambda_{\min}$)

For minimum wavelength $n_2 = \infty, n_1 = n$ So $\lambda_{\min} = \frac{n^2}{R}$

(5) The ratio of first member and series limit can be calculated as $\frac{\lambda_{\max}}{\lambda_{\min}} = \frac{(n+1)^2}{(2n+1)}$

Table 26.3 : Different spectral series

Spectral series	Transition	$\lambda_{\max}$	$\lambda_{\min}$	$\frac{\lambda_{\max}}{\lambda_{\min}}$	Region
1. Lyman series	$n_2 = 2, 3, 4 \dots \infty$ $n_1 = 1$	$\frac{4}{3R}$	$\frac{1}{R}$	$\frac{4}{3}$	Ultraviolet region
2. Balmer series	$n_2 = 3, 4, 5 \dots \infty$ $n_1 = 2$	$\frac{36}{5R}$	$\frac{4}{R}$	$\frac{9}{5}$	Visible region
3. Paschen series	$n_2 = 4, 5, 6 \dots \infty$ $n_1 = 3$	$\frac{144}{7R}$	$\frac{9}{R}$	$\frac{16}{7}$	Infrared region
4. Brackett series	$n_2 = 5, 6, 7 \dots \infty$ $n_1 = 4$	$\frac{400}{9R}$	$\frac{16}{R}$	$\frac{25}{9}$	Infrared region
5. Pfund series	$n_2 = 6, 7, 8 \dots \infty$ $n_1 = 5$	$\frac{900}{11R}$	$\frac{25}{R}$	$\frac{36}{11}$	Infrared region

Quantum Numbers

An atom contains large number of shells and subshells. These are distinguished from one another on the basis of their size, shape and orientation (direction) in space. The parameters are expressed in terms of different numbers called quantum number.

Quantum numbers may be defined as a set of four number with the help of which we can get complete information about all the electrons in an atom. It tells us the address of the electron *i.e.* location, energy, the type of orbital occupied and orientation of that orbital.

(1) **Principal Quantum number (n)** : This quantum number determines the main energy level or shell in which the electron is present. The average distance of the electron from the nucleus and the energy of the electron depends on it.

$$E_n \propto \frac{1}{n^2} \quad \text{and} \quad r_n \propto n^2 \quad (\text{in } H\text{-atom})$$

The principal quantum number takes whole number values, $n = 1, 2, 3, 4, \dots, \infty$

(2) **Orbital quantum number (l) or azimuthal quantum number (l)** : This represents the number of subshells present in the main shell. These subsidiary orbits within a shell will be denoted as 1, 2, 3, 4 ... or *s, p, d, f* ... This tells the shape of the subshells.

The orbital angular momentum of the electron is given as $L = \sqrt{l(l+1)} \frac{h}{2\pi}$ (for a particular value of n).

For a given value of n the possible values of l are $l = 0, 1, 2, \dots$ upto $(n-1)$

(3) **Magnetic quantum number (m_l)** : An electron due to its angular motion around the nucleus generates an electric field. This electric field is expected to produce a magnetic field. Under the influence of external magnetic field, the electrons of a subshell can orient themselves in certain preferred regions of space around the nucleus called orbitals.

The magnetic quantum number determines the number of preferred orientations of the electron present in a subshell.

The angular momentum quantum number m can assume all integral value between $-l$ to $+l$ including zero. Thus m_l can be $-1, 0, +1$ for $l = 1$. Total values of m_l associated with a particular value of l is given by $(2l+1)$.

(4) **Spin (magnetic) quantum number (m_s)** : An electron in atom not only revolves around the nucleus but also spins about its own axis. Since an electron can spin either in clockwise direction or in anticlockwise direction. Therefore for any particular value of magnetic quantum number, spin quantum number can have two values, *i.e.* $m_s = \frac{1}{2}$ (Spin up) or $m_s = -\frac{1}{2}$ (Spin down)

This quantum number helps to explain the magnetic properties of the substance.

Table 26.4 : Quantum states of the hydrogen atom

n	l	m_l	Spectroscopic notation	Shell
1	0	0	1 s	K
2	0	0	2 s	L
2	1	-1, 0, 1	2 p	
3	0	0	3 s	M
3	1	-1, 0, 1	3 p	
3	2	-2, -1, 0, 1, 2	3 d	N
4	0	0	4 s	

Electronic Configurations of Atoms

The distribution of electrons in different orbitals of an atom is called the electronic configuration of the atom. The filling of electrons in orbitals is governed by the following rules.

(1) **Pauli's exclusion principle** : "It states that no two electrons in an atom can have all the four quantum number (n , l , m_l and m_s) the same."

It means each quantum state of an electron must have a different set of quantum numbers n , l , m_l and m_s . This principle sets an upper limit on the number of electrons that can occupy a shell.

$N_{\max}$ in one shell = $2n^2$; Thus $N_{\max}$ in $K, L, M, N \dots$ shells are 2, 8, 18, 32,

(2) **Aufbau principle** : Electrons enter the orbitals of lowest energy first.

As a general rule, a new electron enters an empty orbital for which $(n + l)$ is minimum. In case the value $(n + l)$ is equal for two orbitals, the one with lower value of n is filled first.

Thus the electrons are filled in subshells in the following order (memorize)

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p,

(3) **Hund's Rule** : When electrons are added to a subshell where more than one orbital of the same energy is available, their spins remain parallel. They occupy different orbitals until each one of them has at least one electron. Pairing starts only when all orbitals are filled up.

Pairing takes place only after filling 3, 5 and 7 electrons in p , d and f orbitals, respectively.

Nucleus

(1) Rutherford's α -scattering experiment established that the mass of atom is concentrated with small positively charged region at the centre which is called 'nucleus'.

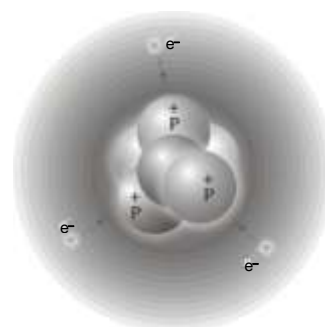


Fig. 26.11

(2) The stability or instability of a particular nucleus is determined by the competition between the attractive nuclear force among the protons and neutrons and the repulsive electrical interactions among the protons. Unstable nuclei *decay*, transforming themselves spontaneously into other structure by a variety of decay processes.

(3) We could not survive without the 3.90×10^{26} *watt* output of one near by fusion reactor, our sun.

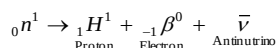
(4) Nuclei are made up of proton and neutron. The number of protons in a nucleus (called the atomic number or proton number) is represented by the symbol Z . The number of neutrons (neutron number) is represented by N . The total number of neutrons and protons in a nucleus is called it's mass number A so $A = Z + N$.

(5) Neutrons and proton, when described collectively are called **nucleons**. A single nuclear species having specific values of both Z and N is called a nuclide.

(6) Nuclides are represented as ${}_Z X^A$; where X denotes the chemical symbol of the element.

Neutron

Neutron is a fundamental particle which is essential constituent of all nuclei except that of hydrogen atom. It was discovered by Chadwick. A free neutron outside the nucleus is unstable and decays into proton and electron.



(1) The charge of neutron : It is neutral

(2) The mass of neutron : 1.6750×10^{-27} *kg*

(3) It's spin angular momentum : $\frac{1}{2} \times \left(\frac{h}{2\pi} \right) J - s$

(4) It's magnetic moment : 9.57×10^{-27} *Jl Tesla*

(5) It's half life : 12 minutes

(6) Penetration power : High

(7) Types : Neutrons are of two types slow neutron and fast neutron, both are fully capable of penetrating a nucleus and causing artificial disintegration.

Thermal Neutrons

Fast neutrons can be converted into slow neutrons by certain materials called moderator's (Paraffin wax, heavy water, graphite) when fast moving neutrons pass through a moderator, they collide with the molecules of the moderator, as a result of

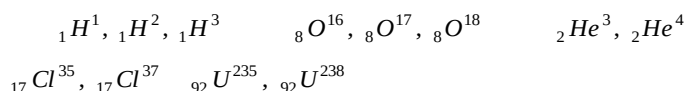
this, the energy of moving neutron decreases while that of the molecules of the moderator increases. After sometime they both attains same energy. The neutrons are then in thermal equilibrium with the molecules of the moderator and are called thermal neutrons.

Energy of thermal neutron is about 0.025 *eV* and speed is about 2.2 *km/s*.

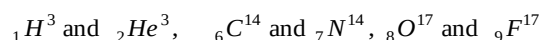
Types of Nuclei

The nuclei have been classified on the basis of the number of protons (atomic number) or the total number of nucleons (mass number) as follows

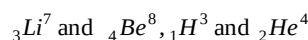
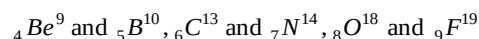
(1) **Isotopes** : The atoms of element having same atomic number but different mass number are called isotopes. All isotopes have the same chemical properties. The isotopes of some elements are the following



(2) **Isobars** : The nuclei which have the same mass number (A) but different atomic number (Z) are called isobars. Isobars occupy different positions in periodic table so all isobars have different chemical properties. Some of the examples of isobars are



(3) **Isotones** : The nuclei having equal number of neutrons are called isotones. For them both the atomic number (Z) and mass number (A) are different, but the value of ($A - Z$) is same. Some examples are



(4) **Mirror nuclei** : Nuclei having the same mass number A but with the proton number (Z) and neutron number ($A - Z$) interchanged (or whose atomic number differ by 1) are called mirror nuclei for example.

${}_1H^3$ and ${}_2He^3$, ${}_3Li^7$ and ${}_4Be^7$

Size of Nucleus

(1) **Nuclear radius** : Experimental results indicates that the nuclear radius is proportional to $A^{1/3}$, where A is the mass number of nucleus *i.e.* $R \propto A^{1/3} \Rightarrow R = R_0 A^{1/3}$, where $R_0 = 1.2 \times 10^{-15} m = 1.2 fm$.

(2) **Nuclear volume** : The volume of nucleus is given by

$$V = \frac{4}{3} \pi R^3 = \frac{4}{3} \pi R_0^3 A \Rightarrow V \propto A$$

(3) **Nuclear density** : Mass per unit volume of a nucleus is called nuclear density.

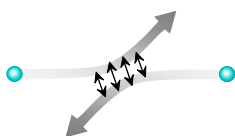
$$\text{Nuclear density}(\rho) = \frac{\text{Mass of nucleus}}{\text{Volume of nucleus}} = \frac{mA}{\frac{4}{3} \pi (R_0 A^{1/3})^3}$$

where m = Average of mass of a nucleon (= mass of proton + mass of neutron = $1.66 \times 10^{-27} kg$) and mA = Mass of nucleus

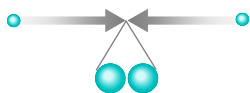
$$\Rightarrow \rho = \frac{3m}{4\pi R_0^3} = 2.38 \times 10^{17} kg / m^3$$

Nuclear Force

Forces that keep the nucleons bound in the nucleus are called nuclear forces.



(A) At low speeds, electromagnetic repulsion



(B) At high speeds, nuclei come close enough for the strong

Fig. 26.12

(1) Nuclear forces are short range forces. These do not exist at large distances greater than $10^{-15} m$.

(2) Nuclear forces are the strongest forces in nature.

(3) These are attractive force and causes stability of the nucleus.

(4) These forces are charge independent.

(5) Nuclear forces are non-central force.

(6) **Nuclear forces are exchange forces** : According to scientist Yukawa the nuclear force between the two nucleons is the result of the exchange of particles called mesons between the nucleons.

π - mesons are of three types – Positive π meson (π^+), negative π meson (π^-), neutral π meson (π^0)

The force between neutron and proton is due to exchange of charged meson between them *i.e.*

$$p \rightarrow \pi^+ + n, \quad n \rightarrow p + \pi^-$$

The forces between a pair of neutrons or a pair of protons are the result of the exchange of neutral meson (π^0) between them *i.e.* $p \rightarrow p' + \pi^0$ and $n \rightarrow n' + \pi^0$

Thus exchange of π meson between nucleons keeps the nucleons bound together. It is responsible for the nuclear forces.

Dog-Bone analogy

The above interactions can be explained with the dog bone analogy according to which we consider the two interacting nucleons to be two dogs having a common bone clenched in between their teeth very firmly. Each one of these dogs wants to take the bone and hence they cannot be separated easily. They seem to be bound to each other with a strong attractive force (which is the bone) the meson plays the role of the common bone in between two nucleons



Fig. 26.13

Atomic Mass Unit (amu)

(1) In nuclear physics, a convenient unit of mass is the unified atomic mass unit abbreviated u .

(2) The amu is defined as $\frac{1}{12}$ th mass of a ${}_B C^{12}$ at on.

(3) $1 amu$ (or $1 u$) = $1.6605402 \times 10^{-27} kg$.

(4) Masses of electron, proton and neutrons :

Mass of electron (m_e) = $9.1 \times 10^{-31} kg$ = $0.0005486 amu$,

Mass of proton (m_p) = $1.6726 \times 10^{-27} kg$ = $1.007276 amu$

Mass of neutron (m_n) = $1.6750 \times 10^{-27} kg$ = $1.00865 amu$, Mass of hydrogen atom ($m_e + m_p$) = $1.6729 \times 10^{-27} kg$ = $1.0078 amu$

(5) The energy associated with a nuclear process is usually large, of the order of MeV .

(6) According to Einstein, mass and energy are inter convertible. The Einstein's mass energy relationship is given by $E = mc^2$

If $m = 1 amu$, $c = 3 \times 10^8 m/sec$ then $E = 931 MeV$ i.e. $1 amu$ is equivalent to $931 MeV$ or $1 amu$ (or $1 u$) = $931 MeV$

$$(1 u) c^2 = 931 MeV \Rightarrow 1u = 931 \frac{MeV}{c^2} \text{ or } c^2 = 931 \frac{MeV}{u}$$

Table 26.5 : Neutral atomic masses for some light nuclides

Element and isopore	Atomic mass (u)
Hydrogen (${}_1^1 H$)	1.007825
Deuterium (${}_1^2 H$)	2.014102
Tritium (${}_1^3 H$)	3.016049
Helium (${}_2^3 He$)	3.016029
Helium (${}_2^4 He$)	4.002603
Lithium (${}_3^7 Li$)	7.016004
Beryllium (${}_4^9 Be$)	9.012182
Carbon (${}_6^{12} C$)	12.000000
Nitrogen (${}_7^{14} N$)	14.003074
Oxygen (${}_8^{16} O$)	15.994915

Pair Production and Pair-Annihilation

When an energetic γ -ray photon falls on a heavy substance. It is absorbed by some nucleus of the substance and an electron and a positron are produced. This phenomenon is

called pair production and may be represented by the following equation

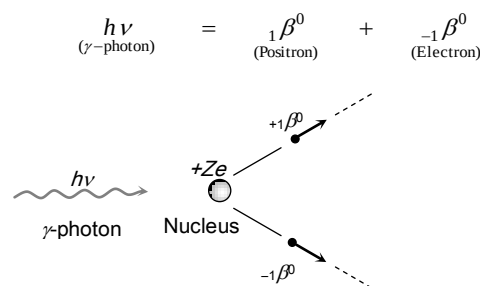


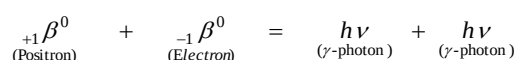
Fig. 26.14

The rest-mass energy of each of positron and electron is

$$\begin{aligned} E_0 &= m_0 c^2 = (9.1 \times 10^{-31} kg) \times (3.0 \times 10^8 m/s)^2 \\ &= 8.2 \times 10^{-14} J = 0.51 MeV \end{aligned}$$

Hence, for pair-production it is essential that the energy of γ -photon must be at least $2 \times 0.51 = 1.02 MeV$. If the energy of γ -photon is less than this, it would cause photo-electric effect or Compton effect on striking the matter.

The converse phenomenon pair-annihilation is also possible. Whenever an electron and a positron come very close to each other, they annihilate each other by combining together and two γ -photons (energy) are produced. This phenomenon is called pair annihilation and is represented by the following equation.



Nuclear Stability

Among about 1500 known nuclides, less than 260 are stable. The others are unstable that decay to form other nuclides by emitting α , β -particles and γ - EM waves. (This process is called radioactivity). The stability of nucleus is determined by many factors. Few such factors are given below :

(1) **Neutron-proton ratio** $\left(\frac{N}{Z} \text{Ratio}\right)$: The chemical

properties of an atom are governed entirely by the number of protons (Z) in the nucleus, the stability of an atom appears to

depend on both the number of protons and the number of neutrons.

(i) For lighter nuclei, the greatest stability is achieved when the number of protons and neutrons are approximately equal ($N \approx Z$) i.e. $\frac{N}{Z} = 1$

(ii) Heavy nuclei are stable only when they have more neutrons than protons. Thus heavy nuclei are neutron rich compared to lighter nuclei (for heavy nuclei, more is the number of protons in the nucleus, greater is the electrical repulsive force between them. Therefore more neutrons are added to provide the strong attractive forces necessary to keep the nucleus stable.)

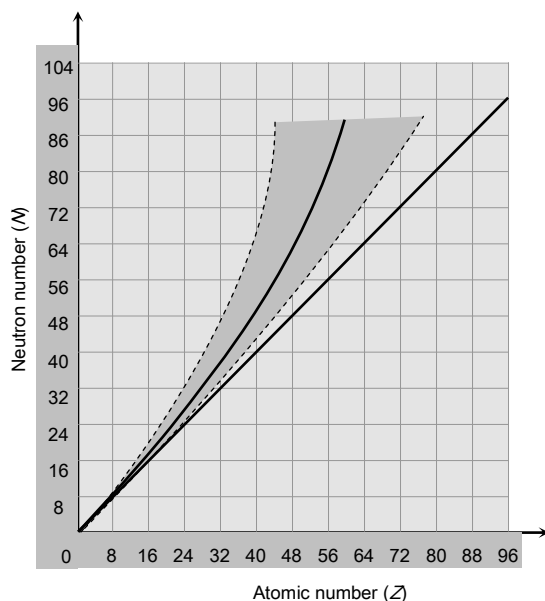


Fig. 26.15

(iii) Figure shows a plot of N versus Z for the stable nuclei. For mass number upto about $A = 40$. For larger value of Z the nuclear force is unable to hold the nucleus together against the electrical repulsion of the protons unless the number of neutrons exceeds the number of protons. At Bi ($Z = 83$, $A = 209$), the

neutron excess in $N - Z = 43$. There are no stable nuclides with $Z > 83$.

(2) **Even or odd numbers of Z or N** : The stability of a nuclide is also determined by the consideration whether it contains an even or odd number of protons and neutrons.

(i) It is found that an even-even nucleus (even Z and even N) is more stable (60% of stable nuclide have even Z and even N).

(ii) An even-odd nucleus (even Z and odd N) or odd-even nuclide (odd Z and even N) is found to be lesser stable while the odd-odd nucleus is found to be less stable.

(iii) Only five stable odd-odd nuclides are known : ${}^1_1H^2$, ${}^3_3Li^6$, ${}^5_5Be^{10}$, ${}^7_7N^{14}$ and ${}^{75}_{75}Ta^{180}$

(3) **Binding energy per nucleon** : The stability of a nucleus is determined by value of its binding energy per nucleon. In general higher the value of binding energy per nucleon, more stable the nucleus is

Mass Defect and Binding Energy

(1) **Mass defect (Δm)** : It is found that the mass of a nucleus is always less than the sum of masses of its constituent nucleons in free state. This difference in masses is called mass defect. Hence mass defect

$$\Delta m = \text{Sum of masses of nucleons} - \text{Mass of nucleus}$$

$$= \{Zm_p + (A - Z)m_n\} - M = \{Zm_p + Zm_e + (A - Z)m_z\} - M'$$

where m_p = Mass of proton, m_n = Mass of each neutron, m_e = Mass of each electron

M = Mass of nucleus, Z = Atomic number, A = Mass number, M' = Mass of atom as a whole.

(2) **Packing fraction** : Mass defect per nucleon is called packing fraction

Packing fraction (f) = $\frac{\Delta m}{A} = \frac{M - A}{A}$ where M = Mass of nucleus, A = Mass number

Packing fraction measures the stability of a nucleus. Smaller the value of packing fraction, larger is the stability of the nucleus.

(i) Packing fraction may be of positive, negative or zero value.

(ii) At $A = 16$, $f \rightarrow$ Zero

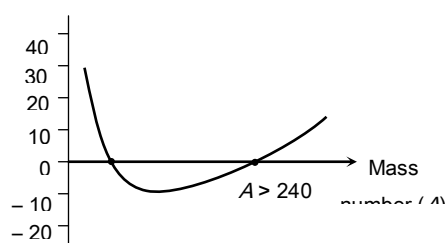


Fig. 26.16

(3) **Binding energy (B.E.)** : The neutrons and protons in a stable nucleus are held together by nuclear forces and energy is needed to pull them infinitely apart (or the same energy is released during the formation of the nucleus). This energy is called the binding energy of the nucleus.

or

The binding energy of a nucleus may be defined as the energy equivalent to the mass defect of the nucleus.

If Δm is mass defect then according to Einstein's mass energy relation

$$\text{Binding energy} = \Delta m \cdot c^2 = \{[m_p Z + m_n(A - Z)] - M\} \cdot c^2$$

(This binding energy is expressed in *joule*, because Δm is measured in *kg*)

If Δm is measured in *amu* then binding energy = $\Delta m \text{ amu} = [m_p Z + m_n(A - Z)] - M \text{ amu} = \Delta m \times 931 \text{ MeV}$

(4) **Binding energy per nucleon** : The average energy required to release a nucleon from the nucleus is called binding energy per nucleon.

Binding energy per nucleon

$$= \frac{\text{Total binding energy}}{\text{Mass number (i.e. total number of nucleons)}} = \frac{\Delta m \times 931}{A} \frac{\text{MeV}}{\text{Nucleon}}$$

Binding energy per nucleon $\propto$ Stability of nucleus

Binding Energy Curve

It is the graph between binding energy per nucleon and total number of nucleons (i.e. mass number A)

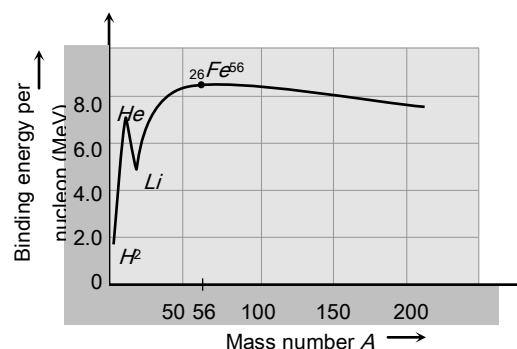


Fig. 26.17

(1) Some nuclei with mass number $A < 20$ have large binding energy per nucleon than their neighbour nuclei. For example ${}_2\text{He}^4$, ${}_4\text{Be}^8$, ${}_6\text{C}^{12}$, ${}_8\text{O}^{16}$ and ${}_{10}\text{Ne}^{20}$. These nuclei are more stable than their neighbours.

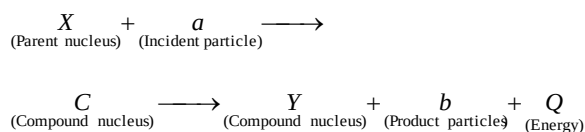
(2) The binding energy per nucleon is maximum for nuclei of mass number $A = 56$ (${}_{26}\text{Fe}^{56}$). It's value is 8.8 MeV per nucleon.

(3) For nuclei having $A > 56$, binding energy per nucleon gradually decreases for uranium ($A = 238$), the value of binding energy per nucleon drops to 7.5 MeV .

Nuclear Reactions

The process by which the identity of a nucleus is changed when it is bombarded by an energetic particle is called nuclear

reaction. The general expression for the nuclear reaction is as follows.



Here X and a are known as reactants and Y and b are known as products. This reaction is known as (a, b) reaction and can be represented as $X(a, b)Y$

(1) **Q value or energy of nuclear reaction** : The energy absorbed or released during nuclear reaction is known as Q -value of nuclear reaction.

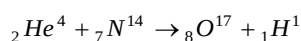
$$\begin{aligned} Q\text{-value} &= (\text{Mass of reactants} - \text{mass of products})c^2 \text{ Joules} \\ &= (\text{Mass of reactants} - \text{mass of products}) \text{ amu} \end{aligned}$$

If $Q < 0$, The nuclear reaction is known as endothermic. (The energy is absorbed in the reaction)

If $Q > 0$, The nuclear reaction is known as exothermic (The energy is released in the reaction)

(2) Law of conservation in nuclear reactions

(i) Conservation of mass number and charge number : In the following nuclear reaction



Mass number (A) $\rightarrow$ Before the reaction After the reaction

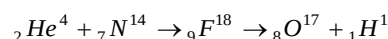
$$4 + 14 = 18 \qquad 17 + 1 = 18$$

$$\text{Charge number } (Z) \rightarrow 2 + 7 = 9 \qquad 8 + 1 = 9$$

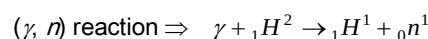
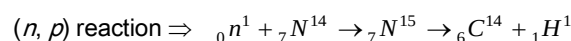
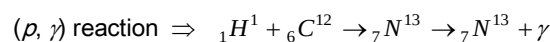
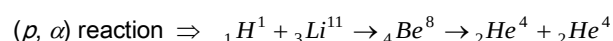
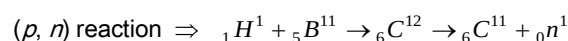
(ii) Conservation of momentum : Linear momentum/angular momentum of particles before the reaction is equal to the linear/angular momentum of the particles after the reaction. That is $\Sigma p = 0$

(iii) Conservation of energy : Total energy before the reaction is equal to total energy after the reaction. Term Q is added to balance the total energy of the reaction.

(3) **Common nuclear reactions** : The nuclear reactions lead to artificial transmutation of nuclei. Rutherford was the first to carry out artificial transmutation of nitrogen to oxygen in the year 1919.



It is called (α, p) reaction. Some other nuclear reactions are given as follows.



Nuclear Fission

(1) The process of splitting of a heavy nucleus into two lighter nuclei of comparable masses (after bombardment with a energetic particle) with liberation of energy is called nuclear fission.

(2) The phenomenon of nuclear fission was discovered by scientist Ottobahn and F. Strassman and was explained by N. Bohr and J.A. Wheeler on the basis of liquid drop model of nucleus.

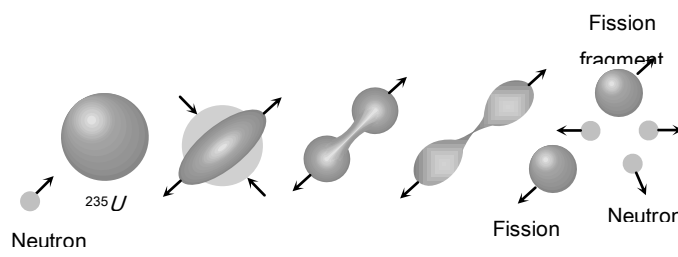
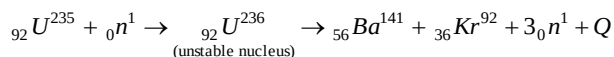


Fig. 26.18

(3) Fission reaction of U^{235}



(4) The energy released in U^{235} fission is about 200 MeV or 0.8 MeV per nucleon.

(5) By fission of ${}_{92}U^{235}$, on an average 2.5 neutrons are liberated. These neutrons are called fast neutrons and their energy is about 2 MeV (for each). These fast neutrons can escape from the reaction so as to proceed the chain reaction they are need to slow down.

(6) Fission of U^{235} occurs by slow neutrons only (of energy about 1 eV) or even by thermal neutrons (of energy about 0.025 eV).

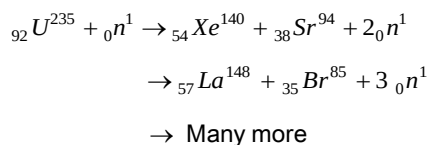
(7) 50 kg of U^{235} on fission will release $\approx 4 \times 10^{15}$ J of energy. This is equivalence to 20,000 tones of TNT explosion. The nuclear bomb dropped at Hiroshima had this much explosion power.

(8) The mass of the compound nucleus must be greater than the sum of masses of fission products.

(9) The $\frac{\text{Binding energy}}{A}$ of compound nucleus must be less than that of the fission products.

(10) It may be pointed out that it is not necessary that in each fission of uranium, the two fragments ${}_{56}Ba$ and ${}_{36}Kr$ are formed but they may be any stable isotopes of middle weight atoms.

(11) Same other U^{235} fission reactions are



(12) The neutrons released during the fission process are called prompt neutrons.

(13) Most of energy released appears in the form of kinetic energy of fission fragments.

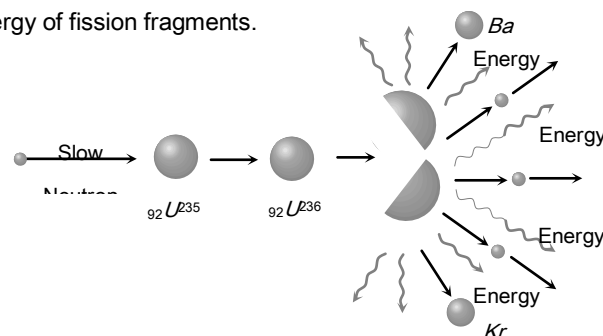


Fig. 26.19

Chain Reaction

In nuclear fission, three neutrons are produced along with the release of large energy. Under favourable conditions, these neutrons can cause further fission of other nuclei, producing large number of neutrons. Thus a chain of nuclear fissions is established which continues until the whole of the uranium is consumed.

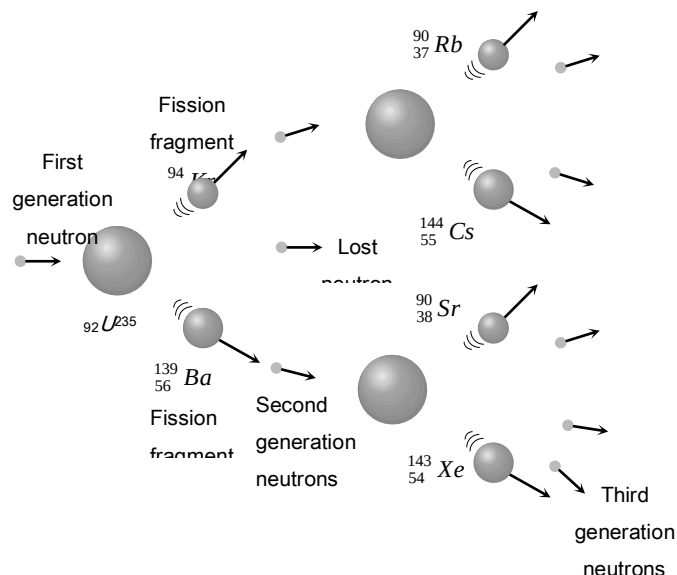


Fig. 26.20

In the chain reaction, the number of nuclei undergoing fission increases very fast. So, the energy produced takes a tremendous magnitude very soon.

Difficulties in Chain Reaction

In chain reaction following difficulties are observed

(1) **Absorption of neutrons by U^{238}** : the major part in natural uranium is the isotope U^{238} (99.3%), the isotope U^{235} is very little (0.7%). It is found that U^{238} is fissionable with fast neutrons, whereas U^{235} is fissionable with slow neutrons. Due to the large percentage of U^{238} , there is more possibility of collision of neutrons with U^{238} . It is found that the neutrons get slowed on colliding with U^{238} , as a result of it further fission of U^{238} is not possible (Because they are slow and they are absorbed by U^{238}). This stops the chain reaction.

Removal : (i) To sustain chain reaction ${}_{92}U^{235}$ is separated from the ordinary uranium. Uranium so obtained (${}_{92}U^{235}$) is known as enriched uranium, which is fissionable with the fast and slow neutrons and hence chain reaction can be sustained.

(ii) If neutrons are slowed down by any method to an energy of about 0.3 eV, then the probability of their absorption by U^{238} becomes very low, while the probability of their fissioning U^{235} becomes high. This job is done by moderators. Which reduce the speed of neutron rapidly graphite and heavy water are the example of moderators.

(2) **Critical size** : The neutrons emitted during fission are very fast and they travel a large distance before being slowed down. If the size of the fissionable material is small, the neutrons emitted will escape the fissionable material before they are slowed down. Hence chain reaction cannot be sustained.

Removal : The size of the fissionable material should be large than a critical size.

The chain reaction once started will remain steady, accelerate or retard depending upon, a factor called neutron reproduction factor (k). It is defined as follows.

$$k = \frac{\text{Rate of production of neutrons}}{\text{Rate of loss of neutrons}}$$

If $k = 1$, the chain reaction will be steady. The size of the fissionable material used is said to be the critical size and its mass, the critical mass.

If $k > 1$, the chain reaction accelerates, resulting in an explosion. The size of the material in this case is super critical. (Atom bomb)

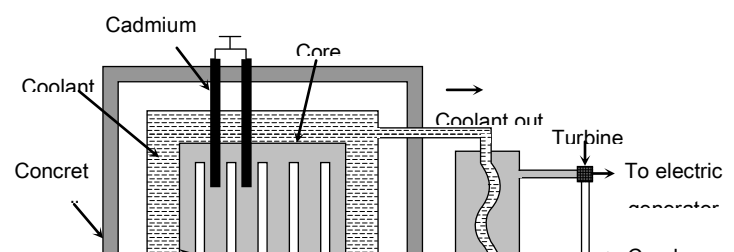
If $k < 1$, the chain reaction gradually comes to a halt. The size of the material used is said to be sub-critical.

Table 26.6 : Types of chain reaction

Controlled chain reaction	Uncontrolled chain reaction
Controlled by artificial method	No control over this type of nuclear reaction
All neutrons are absorbed except one	More than one neutron takes part into reaction
It's rate is slow	Fast rate
Reproduction factor $k = 1$	Reproduction factor $k > 1$
Energy liberated in this type of reaction is always less than explosive energy	A large amount of energy is liberated in this type of reaction
Chain reaction is the principle of nuclear reactors	Uncontrolled chain reaction is the principle of atom bomb.

Nuclear Reactor

A nuclear reactor is a device in which nuclear fission can be carried out through a sustained and a controlled chain reaction. It is also called an atomic pile. It is thus a source of controlled energy which is utilised for many useful purposes.



(1) **Fissionable material (Fuel)** : The fissionable material used in the reactor is called the fuel of the reactor. Uranium isotope (U^{235}) Thorium isotope (Th^{232}) and Plutonium isotopes (Pu^{239} , Pu^{240} and Pu^{241}) are the most commonly used fuels in the reactor.

(2) **Moderator** : Moderator is used to slow down the fast moving neutrons. Most commonly used moderators are graphite and heavy water (D_2O).

(3) **Control Material** : Control material is used to control the chain reaction and to maintain a stable rate of reaction. This material controls the number of neutrons available for the fission. For example, cadmium rods are inserted into the core of the reactor because they can absorb the neutrons. The neutrons available for fission are controlled by moving the cadmium rods in or out of the core of the reactor.

(4) **Coolant** : Coolant is a cooling material which removes the heat generated due to fission in the reactor. Commonly used coolants are water, CO_2 nitrogen *etc.*

(5) **Protective shield** : A protective shield in the form a concrete thick wall surrounds the core of the reactor to save the

persons working around the reactor from the hazardous radiations.

(6) Uses of nuclear reactor

- (i) In electric power generation.
- (ii) To produce radioactive isotopes for their use in medical science, agriculture and industry.
- (iii) In manufacturing of Pu^{239} which is used in atom bomb.
- (iv) They are used to produce neutron beam of high intensity which is used in the treatment of cancer and nuclear research.

Nuclear Fusion

(1) In nuclear fusion two or more than two lighter nuclei combine to form a single heavy nucleus. The mass of single nucleus so formed is less than the sum of the masses of parent nuclei. This difference in mass results in the release of tremendous amount of energy

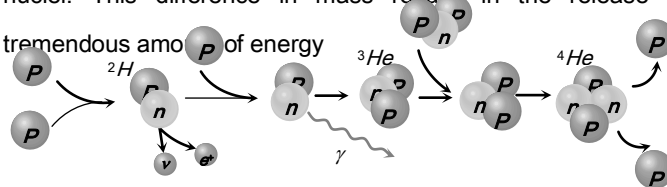
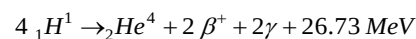
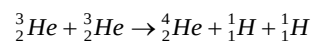
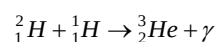
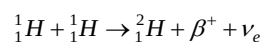


Fig. 26.22

(2) For fusion high pressure ($\approx 10^6 \text{ atm}$) and high temperature (of the order of 10^7 K to 10^8 K) is required and so the reaction is called thermonuclear reaction.

(3) Here are three examples of energy-liberating fusion reactions, written in terms of the neutral atoms. Together the reactions make up the process called the proton-proton chain.



(4) The proton-proton chain takes place in the interior of the sun and other stars. Each gram of the sun's mass contains about 4.5×10^{23} protons. If all of these protons were fused into helium, the energy released would be about 130,000 *kWh*. If the sun were to continue to radiate at its present rate, it would take about 75×10^9 years to exhaust its supply of protons.

(5) For the same mass of the fuel, the energy released in fusion is much larger than in fission.

(6) **Plasma** : The temperature of the order of 10^8 K required for thermonuclear reactions leads to the complete ionisation of the atom of light elements. The combination of bare nuclei and electron cloud is called plasma. The enormous gravitational field of the sun confines the plasma in the interior of the sun.

The main problem to carry out nuclear fusion in the laboratory is to contain the plasma at a temperature of 10^8 K. No solid container can tolerate this much temperature. If this problem of containing plasma is solved, then the large quantity of deuterium present in sea water would be able to serve as inexhaustible source of energy.

Table 26.7 : Nuclear bomb (Based on uncontrolled nuclear reactions)

Atom bomb	Hydrogen bomb
Based on fission process it involves the fission of ^{235}U	Based on fusion process. Mixture of deuterium and tritium is used in it
In this critical size is important	There is no limit to critical size
Explosion is possible at normal temperature and pressure	High temperature and pressure are required
Less energy is released compared to hydrogen bomb	More energy is released as compared to atom bomb so it is more dangerous than atom bomb

Radioactivity

The phenomenon of spontaneous emission of radiations by heavy elements is called radioactivity. The elements which show this phenomenon are called radioactive elements.

(1) Radioactivity was discovered by Henry Becquerel in uranium salt in the year 1896.

(2) After the discovery of radioactivity in uranium, Pierre Curie and Madame Curie discovered a new radioactive element called radium (which is 10^6 times more radioactive than uranium)

(3) Some examples of radioactive substances are : Uranium, Radium, Thorium, Polonium, Neptunium *etc.*

(4) Radioactivity of a sample cannot be controlled by any physical (pressure, temperature, electric or magnetic field) or chemical changes.

(5) All the elements with atomic number (Z) > 82 are naturally radioactive.

(6) The conversion of lighter elements into radioactive elements by the bombardment of fast moving particles is called artificial or induced radioactivity.

(7) Radioactivity is a nuclear event and not atomic. Hence electronic configuration of atom doesn't have any relationship with radioactivity.

Nuclear Radiations

According to Rutherford's experiment when a sample of radioactive substance is put in a lead box and allow the emission of radiation through a small hole only. When the radiation enters into the external electric field, they split into three parts (α -rays, β -rays and γ -rays)

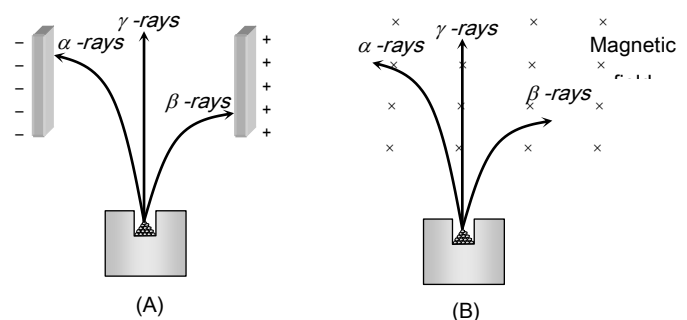


Fig. 26.23

(1) **α -decay** : Nearly 90% of the 2500 known nuclides are radioactive ; they are not stable but decay into other nuclides

(i) When unstable nuclides decay into different nuclides, they usually emit alpha (α) or beta (β) particles.

(ii) Alpha emission occurs principally with nuclei that are too large to be stable. When a nucleus emits an alpha particle, its N and Z values each decrease by two and A decreases by four.

(iii) Alpha decay is possible whenever the mass of the original neutral atom is greater than the sum of the masses of the final neutral atom and the neutral helium- atom.

(2) **β -decay** : There are different simple type of β -decay β^- , β^+ and electron capture.

(i) A beta minus particle (β^-) is an electron. Emission of β^- involves transformation of a neutron into a proton, an electron and a third particle called an antineutrino ($\bar{\nu}$).

(ii) β^- decay usually occurs with nuclides for which the neutron to proton ratio ($\frac{N}{Z}$ ratio) is too large for stability.

(iii) In β^- decay, N decreases by one, Z increases by one and A doesn't change.

(iv) β^- decay can occur whenever the neutral atomic mass of the original atom is larger than that of the final atom.

(v) Nuclides for which N/Z is too small for stability can emit a positron, the electron's antiparticle, which is identical to the electron but with positive charge. The basic process called beta plus β^+ decay

$$p \rightarrow n + \beta^+ + \nu \quad (\nu = \text{neutrino})$$

(vi) β^+ decay can occur whenever the neutral atomic mass of the original atom is at least two electron masses larger than that of the final atom

(vii) The mass of ν and $\bar{\nu}$ is zero. The spin of both is $\frac{1}{2}$ in units of $\frac{h}{2\pi}$. The charge on both is zero. The spin of neutrino is antiparallel to it's momentum while that of antineutrino is parallel to it's momentum.

(viii) There are a few nuclides for which β^+ emission is not energetically possible but in which an orbital electron (usually in the k -shell) can combine with a proton in the nucleus to form a neutron and a neutrino. The neutron remains in the nucleus and the neutrino is emitted.

$$p + \beta^+ \rightarrow n + \nu$$

(3) **γ -decay** : The energy of internal motion of a nucleus is quantized. A typical nucleus has a set of allowed energy levels, including a *ground state* (state of lowest energy) and several excited states. Because of the great strength of nuclear interactions, excitation energies of nuclei are typically of the order of the order of 1 MeV, compared with a few eV for atomic energy levels. In ordinary physical and chemical transformations the nucleus always remains in its ground state. When a nucleus is placed in an excited state, either by bombardment with high-energy particles or by a radioactive transformation, it can decay to the ground state by emission of one or more photons called gamma rays or gamma-ray photons, with typical energies of 10 keV to 5 MeV. This process is called gamma (γ) decay.

All the known conservation laws are obeyed in γ -decay.

The intensity of γ -decay after passing through x thickness of a material is given by $I = I_0 e^{-\mu x}$ (μ = absorption co-efficient)

Radioactive Disintegration

(1) **Law of radioactive disintegration** : According to Rutherford and Soddy law for radioactive decay is as follows.

"At any instant the rate of decay of radioactive atoms is proportional to the number of atoms present at that instant" *i.e.*

$$-\frac{dN}{dt} \propto N \Rightarrow \frac{dN}{dt} = -\lambda N. \text{ It can be proved that } N = N_0 e^{-\lambda t}$$

$$\text{In terms of mass } M = M_0 e^{-\lambda t}$$

where N = Number of atoms remains undecayed after time t , N_0 = Number of atoms present initially (*i.e.* at $t = 0$), M = Mass of radioactive nuclei at time t , M_0 = Mass of radioactive nuclei at time $t = 0$, $N_0 - N$ = Number of disintegrated nucleus in time t

$$\frac{dN}{dt} = \text{rate of decay, } \lambda = \text{Decay constant or disintegration}$$

constant or radioactivity constant or Rutherford Soddy's

constant or the probability of decay per unit time of a nucleus.

Table 26.8 : Properties of α , β and γ -rays

Features	α - particles	β - particles	γ - rays
1. Identity	Helium nucleus or doubly ionised helium atom (${}_2\text{He}^4$)	Fast moving electron ($-\beta^0$ or β^-)	Photons (E.M. waves)
2. Charge	$+2e$	$-e$	Zero
3. Mass $4 m_p$ (m_p = mass of proton = 1.87×10^{-27})	$4 m_p$	m_e	Massless
4. Speed	$\approx 10^7 \text{ m/s}$	1% to 99% of speed of light	Speed of light
5. Range of kinetic energy	4 MeV to 9 MeV	All possible values between a minimum certain value to 1.2 MeV	Between a minimum value to 2.23 MeV
6. Penetration power (γ, β, α)	1 (Stopped by a paper)	100 (100 times of α)	10,000 (100 times of β upto 30 cm of iron (or Pb) sheet)
7. Ionisation power ($\alpha > \beta > \gamma$)	10,000	100	1
8. Effect of electric or magnetic field	Deflected	Deflected	Not deflected
9. Energy spectrum	Line and discrete	Continuous	Line and discrete
10. Mutual interaction with matter	Produces heat	Produces heat	Produces, photo-electric effect, Compton effect, pair production
11. Equation of decay	${}_Z X^A \xrightarrow{\alpha\text{-decay}} {}_{Z-2} Y^{A-4} + {}_2\text{He}^4$ ${}_Z X^A \xrightarrow{n\alpha} {}_Z Y^{A'}$ $\Rightarrow n_\alpha = \frac{A - A'}{4}$	${}_Z X^A \rightarrow {}_{Z+1} Y^A + {}_{-1} e^0 + \bar{\nu}$ ${}_Z X^A \xrightarrow{n\beta} {}_Z X^A$ $\Rightarrow n_\beta = (2n_\alpha - Z + Z')$	${}_Z X^A \rightarrow {}_Z X^A + \gamma$

(2) **Activity** : It is defined as the rate of disintegration (or count rate) of the substance (or the number of atoms of any material decaying per second) *i.e.*

$$A = -\frac{dN}{dt} = \lambda N = \lambda N_0 e^{-\lambda t} = A_0 e^{-\lambda t}$$

where A_0 = Activity of $t = 0$, A = Activity after time t

Units of activity (Radioactivity)

It's units are *Becquerel* (Bq), *Curie* (Ci) and *Rutherford* (Rd)

1 *Becquerel* = 1 *disintegration/sec*,

1 Rutherford = 10^6 dis/sec, 1 Curie = 3.7×10^{11} dis/sec

(3) **Half life ($T_{1/2}$)** : Time interval in which the mass of a radioactive substance or the number of its atom reduces to half of its initial value is called the half life of the substance.

i.e. if $N = \frac{N_0}{2}$

then $t = T_{1/2}$

Hence from $N = N_0 e^{-\lambda t}$

$$\frac{N_0}{2} = N_0 e^{-\lambda(T_{1/2})} \Rightarrow T_{1/2} = \frac{\log_e 2}{\lambda} = \frac{0.693}{\lambda}$$

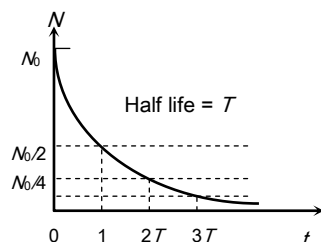


Fig. 26.24

Table 26.9 : Fraction of active/decayed atom at different time

Time (t)	Remaining fraction of active atoms (N/N_0) probability of survival	Fraction of atoms decayed ($N_0 - N$) / N_0 probability of decay
$t = 0$	1 (100%)	0
$t = T_{1/2}$	$\frac{1}{2}$ (50%)	$\frac{1}{2}$ (50%)
$t = 2(T_{1/2})$	$\frac{1}{4}$ (25%)	$\frac{3}{4}$ (75%)
$t = 3(T_{1/2})$	$\frac{1}{8}$ (12.5%)	$\frac{7}{8}$ (87.5%)
$t = 10(T_{1/2})$	$\left(\frac{1}{2}\right)^{10} \approx 0.1\%$	$\approx 99.9\%$
$t = n(N_{1/2})$	$\left(\frac{1}{2}\right)^n$	$\left\{1 - \left(\frac{1}{2}\right)^n\right\}$

(4) **Mean (or average) life (τ)** : The time for which a radioactive material remains active is defined as mean (average) life of that material.

(i) or it is defined as the sum of lives of all atoms divided by the total number of atoms

i.e. $\tau = \frac{\text{Sum of the lives of all the atoms}}{\text{Total number of atoms}} = \frac{1}{\lambda}$

(ii) From $N = N_0 e^{-\lambda t} \Rightarrow \frac{\ln \frac{N}{N_0}}{t} = -\lambda$ slope of the line shown

in the graph i.e. the magnitude of inverse of slope of $\ln \frac{N}{N_0}$ vs t curve is known as mean life (τ).

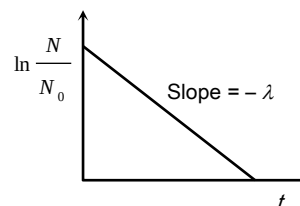


Fig. 26.25

(iii) From $N = N_0 e^{-\lambda t}$, if $t = \frac{1}{\lambda}$

$$\Rightarrow N = N_0 e^{-1} = N_0 \left(\frac{1}{e}\right) = 0.37 N_0 = 37\% \text{ of } N_0.$$

i.e. mean life is the time interval in which number of undecayed atoms (N) becomes $\frac{1}{e}$ times or 0.37 times or 37% of original number of atoms. or

It is the time in which number of decayed atoms ($N_0 - N$) becomes $\left(1 - \frac{1}{e}\right)$ times or 0.63 times or 63% of original number of atoms.

(iv) From $T_{1/2} = \frac{0.693}{\lambda} \Rightarrow \frac{1}{\lambda} = \tau = \frac{1}{0.693} \cdot (T_{1/2}) = 1.44(T_{1/2})$

i.e. mean life is about 44% more than that of half life. Which gives us $\tau > T_{1/2}$

Radioactive Series

(1) If the isotope that results from a radioactive decay is itself radioactive then it will also decay and so on.

(2) The sequence of decays is known as radioactive decay series. Most of the radio-nuclides found in nature are members of four radioactive series. These are as follows

Table 26.10 : Four radioactive series

Mass number	Series (Nature)	Parent	Stable end product	Integer n
$4n$	Thorium (natural)	${}_{90}\text{Th}^{232}$	${}_{82}\text{Pb}^{208}$	52
$4n + 1$	Neptunium	${}_{93}\text{Np}^{237}$	${}_{83}\text{Bi}^{209}$	52

	(Artificial)			
$4n + 2$	Uranium (Natural)	${}_{92}\text{U}^{238}$	${}_{82}\text{Pb}^{206}$	51
$4n + 3$	Actinium (Natural)	${}_{89}\text{Ac}^{227}$	${}_{82}\text{Pb}^{207}$	51

(3) The $4n + 1$ series starts from ${}_{94}\text{Pu}^{241}$ but commonly known as neptunium series because neptunium is the longest lived member of the series.

(4) The $4n + 3$ series actually starts from ${}_{92}\text{U}^{235}$.

Successive Disintegration and Radioactive Equilibrium

Suppose a radioactive element A disintegrates to form another radioactive element B which then disintegrates to still another element C , such decays are called successive disintegration.

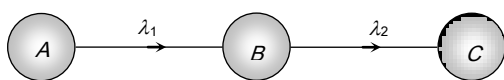


Fig. 26.26

Rate of disintegration of $A = \frac{dN_1}{dt} = -\lambda_1 N_1$ (which is also the rate of formation of B)

$$\text{Rate of disintegration of } B = \frac{dN_2}{dt} = -\lambda_2 N_2$$

$\therefore$ Net rate of formation of B = Rate of disintegration of A – Rate of disintegration of B

$$= \lambda_1 N_1 - \lambda_2 N_2$$

Equilibrium

In radioactive equilibrium, the rate of decay of any radioactive product is just equal to its rate of production from the previous member.

$$\text{i.e. } \lambda_1 N_1 = \lambda_2 N_2 \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{N_2}{N_1} = \frac{\tau_2}{\tau_1} = \frac{(T_{1/2})_1}{(T_{1/2})_2}$$

Uses of Radioactive Isotopes



(1) In medicine

- (i) For testing blood-chromium - 51
- (ii) For testing blood circulation - Na - 24
- (iii) For detecting brain tumor- Radio mercury - 203
- (iv) For detecting fault in thyroid gland - Radio iodine - 131
- (v) For cancer - cobalt - 60
- (vi) For blood - Gold - 189
- (vii) For skin diseases - Phosphorous - 31

(2) In Archaeology

- (i) For determining age of archaeological sample (carbon dating) C^{14}

- (ii) For determining age of meteorites - K^{40}
- (iii) For determining age of earth-Lead isotopes

(3) In agriculture

- (i) For protecting potato crop from earthworm- CO^{60}
- (ii) For artificial rains - Ag / (iii) As fertilizers - P^{32}

(4) **As tracers** - (Tracer) : Very small quantity of radioisotopes present in a mixture is known as tracer

- (i) Tracer technique is used for studying biochemical reaction in tracer and animals.

(5) In industries

- (i) For detecting leakage in oil or water pipe lines (ii) For determining the age of planets.

Tips & Tricks

✍ According to Bohr theory the momentum of an e^- revolving in second orbit of H_2 atom will be $\frac{h}{\pi}$

✍ For an electron in the n^{th} orbit of hydrogen atom in Bohr

1466 Atomic and Nuclear Physics

model, circumference of orbit $= n\lambda$; where λ = de-Broglie wavelength.

✍ Rch = Rydberg's energy $\approx 2.17 \times 10^{-18} \text{ J} \approx 13.6 \text{ eV}$.

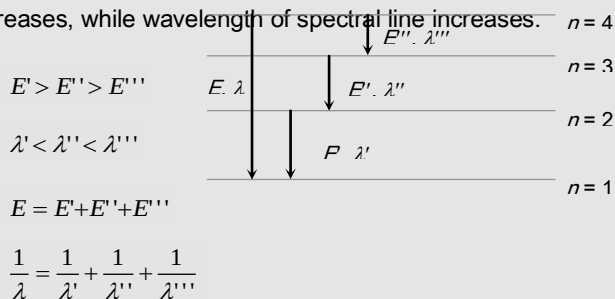
✍ For hydrogen atom principle quantum number

$$n = \sqrt{\frac{13.6}{(\text{B.E.})}}$$

✍ In an H_2 atom when e^- makes a transition from an excited state to the ground state its kinetic energy increases while potential and total energy decreases.

✍ The maximum number of electrons in a subshell with orbital quantum number l is $2(2l + 1)$.

✍ With the increase in principal quantum number the energy difference between the two successive energy level decreases, while wavelength of spectral line increases.



✍ Rydberg constant is different for different elements

$R(=1.09 \times 10^7 \text{ m}^{-1})$ is the value of Rydberg constant when the nucleus is considered to be infinitely massive as compared to the revolving electron. In other words, the nucleus is considered to be stationary.

In case, the nucleus is not infinitely massive or stationary, then the value of Rydberg constant is given as

$$R' = \frac{R}{1 + \frac{m}{M}}$$

where m is the mass of electron and M is the mass of nucleus.

✍ Atomic spectrum is a line spectrum

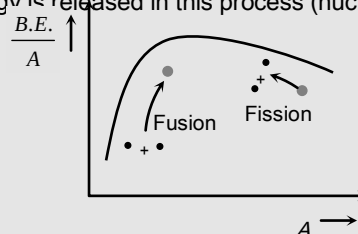
Each atom has its own characteristic allowed orbits depending upon the electronic configuration. Therefore photons emitted during transition of electrons from one

allowed orbit to inner allowed orbit are of some definite energy only. They do not have a continuous graduation of energy. Therefore the spectrum of the emitted light has only some definite lines and therefore atomic spectrum is line spectrum.

✍ Just as dots of light of only three colours combine to form almost every conceivable colour on T.V. screen, only about 100 distinct kinds of atoms combine to form all the materials in the universe.

✍ Density of a nucleus is maximum at its centre and decreases as we move outwards from the nucleus.

✍ When two very light nuclei combine to form a relatively heavy nucleus, then binding energy per nucleon increases. Thus, energy is released in this process (nuclear fusion).



✍ It may be noted that Plutonium is the best fuel as compared to other fissionable material. It is because fission in Plutonium can be initiated by both slow and fast neutrons. Moreover it can be obtained from U^{238} .

✍ Nuclear reactor is firstly devised by fermi.

✍ Apsara was the first Indian nuclear reactor.

✍ A type of reactor that can produce more fissile fuel than it consumes is the breeder reactor.

✍ To achieve fusion in laboratory a device is used to confine the plasma, called **Tokamak**.

✍ A test tube full of base nuclei will weigh heavier than the

earth.

✍ The nucleus of hydrogen contains only one proton. Therefore we may say that the proton is the nucleus of hydrogen atom.

✍ If the relative abundance of isotopes in an element has a ratio $n_1 : n_2$ whose atomic masses are m_1 and m_2 then atomic mass of the element is $M = \frac{n_1 m_1 + n_2 m_2}{n_1 + n_2}$

✍ No radioactive substance emits both α and β particles simultaneously. Also γ -rays are emitted after the emission of α or β -particles.

✍ β -particles are not orbital electrons they come from nucleus. The neutron in the nucleus decays into proton and an electron. This electron is emitted out of the nucleus in the form of β -rays.

✍ Activity per *gm* of a substance is known as specific activity. The specific activity of 1 *gm* of radium – 226 is 1 *Curie*.

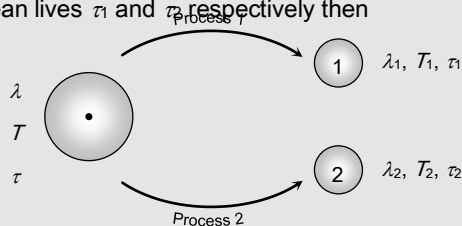
✍ 1 *millicurie* = 37 *Rutherford*

✍ The activity of a radioactive substance decreases as the number of undecayed nuclei decreases with time.

✍ Activity $\propto \frac{1}{\text{Half life}}$

✍ Half life and mean life of a substance doesn't change with time or with pressure, temperature *etc.*

✍ If a nuclide can decay simultaneously by two different process which have decay constant λ_1 and λ_2 , half life T_1 and T_2 and mean lives τ_1 and τ_2 respectively then



$$\Rightarrow \lambda = \lambda_1 + \lambda_2$$

$$\Rightarrow T = \frac{T_1 T_2}{T_1 + T_2}$$

✍ There are at least three varieties of neutrinos, each with its corresponding antineutrino; one is associated with beta decay and the other two are associated with the decay of two unstable particles, the muon and the tau particles.

✍ Are all fusion reaction exoergic ?

Fusion reaction between sufficiently light nuclei are exoergic because the $\frac{B.E.}{A}$ increases. If the nuclei are too massive, however $\frac{B.E.}{A}$ decreases and fusion is endoergic (*i.e.* it takes in energy rather than releasing it)

✍ The Zeeman effect is the splitting of atomic energy levels and the associated spectrum lines when the atoms are placed in a magnetic field. This effect confirms experimentally the quantization of angular momentum.

Ordinary Thinking

Objective Questions

Atomic Structure

1. If in nature there may not be an element for which the principal quantum number $n > 4$, then the total possible number of elements will be

[IIT 1983; MP PET 1999; RPMT 1999; RPET 2001]

- (a) 60 (b) 32
(c) 4 (d) 64

2. In the Bohr's hydrogen atom model, the radius of the stationary orbit is directly proportional to (n = principle quantum number) [MNR 1988; SCRA 1994; CBSE PMT 1996; AIIMS 1999; DCE 2002]

- (a) n^{-1} (b) n
(c) n^{-2} (d) n^2

3. In the n orbit, the energy of an electron $E_n = -\frac{13.6}{n^2} \text{ eV}$ for hydrogen atom. The energy required to take the electron from first orbit to second orbit will be

[MP PMT 1987; CPMT 1991, 97; RPMT 1999; DCE 2001; Kerala PMT 2004]

- (a) 10.2 eV (b) 12.1 eV
(c) 13.6 eV (d) 3.4 eV

4. In the following atoms and molecules for the transition from $n = 2$ to $n = 1$, the spectral line of minimum wavelength will be produced by [IIT 1983]

- (a) Hydrogen atom (b) Deuterium atom
(c) Uni-ionized helium (d) di-ionized lithium

5. The Lyman series of hydrogen spectrum lies in the region [MNR 1993; MP PMT 1995; UPSEAT 2002]

- (a) Infrared (b) Visible
(c) Ultraviolet (d) Of X - rays

6. The size of an atom is of the order of [CPMT 1990; MP PMT 1984; KCET 1994]

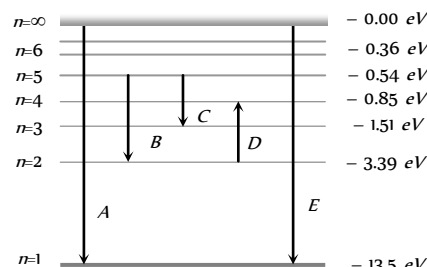
- (a) 10^{-8} m (b) 10^{-10} m
(c) 10^{-12} m (d) 10^{-14} m

7. Which one of the series of hydrogen spectrum is in the visible region [RPMT 1999; MP PET 1990; MP PMT 1994; AFMC 1998; CBSE PMT 1990; MH CET 2004]

- (a) Lyman series (b) Balmer series
(c) Paschen series (d) Bracket series

8. The energy levels of the hydrogen spectrum is shown in figure. There are some transitions A, B, C, D and E . Transition A, B and C respectively represent

[CPMT 1986, 88]



- (a) First member of Lyman series, third spectral line of Balmer series and the second spectral line of Paschen series
(b) Ionization potential of hydrogen, second spectral line of Balmer series and third spectral line of Paschen series
(c) Series limit of Lyman series, third spectral line of Balmer series and second spectral line of Paschen series
(d) Series limit of Lyman series, second spectral line of Balmer series and third spectral line of Paschen series

9. In the above figure D and E respectively represent

[CPMT 1986, 88]

- (a) Absorption line of Balmer series and the ionization potential of hydrogen
(b) Absorption line of Balmer series and the wavelength lesser than lowest of the Lyman series
(c) Spectral line of Balmer series and the maximum wavelength of Lyman series
(d) Spectral line of Lyman series and the absorption of greater wavelength of limiting value of Paschen series

10. The Rutherford α -particle experiment shows that most of the α -particles pass through almost unscattered while some are scattered through large angles. What information does it give about the structure of the atom [AFMC 1997]

- (a) Atom is hollow
(b) The whole mass of the atom is concentrated in a small centre called nucleus
(c) Nucleus is positively charged
(d) All the above

11. Which of the following is true [MP PET 1993]

- (a) Lyman series is a continuous spectrum
(b) Paschen series is a line spectrum in the infrared
(c) Balmer series is a line spectrum in the ultraviolet
(d) The spectral series formula can be derived from the Rutherford model of the hydrogen atom

12. The energy required to knock out the electron in the third orbit of a hydrogen atom is equal to [DPMT 1987]

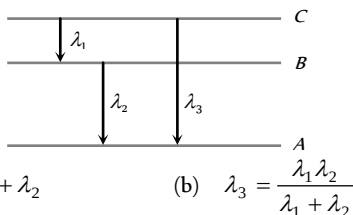
- (a) 13.6 eV (b) $+\frac{13.6}{9} \text{ eV}$
(c) $-\frac{13.6}{3} \text{ eV}$ (d) $-\frac{3}{13.6} \text{ eV}$

13. An electron has a mass of $9.1 \times 10^{-31} \text{ kg}$. It revolves round the nucleus in a circular orbit of radius $0.529 \times 10^{-10} \text{ metre}$ at a speed of $2.2 \times 10^6 \text{ m/s}$. The magnitude of its linear momentum in this motion is

[AFMC 1988]

[Roorkee 1993]

- (a) $1.1 \times 10^{-34} \text{ kg-m/s}$ (b) $2.0 \times 10^{-24} \text{ kg-m/s}$
 (c) $4.0 \times 10^{-24} \text{ kg-m/s}$ (d) $4.0 \times 10^{-31} \text{ kg-m/s}$
14. In a beryllium atom, if a_0 be the radius of the first orbit, then the radius of the second orbit will be in general
 [CBSE PMT 1992; Roorkee 1993; BHU 1998]
 (a) na_0 (b) a_0
 (c) $n^2 a_0$ (d) $\frac{a_0}{n^2}$
15. The ionization potential for second He electron is
 (a) 13.6 eV (b) 27.2 eV
 (c) 54.4 eV (d) 100 eV
16. The energy required to remove an electron in a hydrogen atom from $n = 10$ state is
 [MP PMT 1993]
 (a) 13.6 eV (b) 1.36 eV
 (c) 0.136 eV (d) 0.0136 eV
17. Every series of hydrogen spectrum has an upper and lower limit in wavelength. The spectral series which has an upper limit of wavelength equal to $18752 \text{ \AA}$ is [MP PMT 1993]
 (a) Balmer series (b) Lyman series
 (c) Paschen series (d) Pfund series
- (Rydberg constant $R = 1.097 \times 10^7 \text{ per metre}$)
18. The kinetic energy of the electron in an orbit of radius r in hydrogen atom is ($e = \text{electronic charge}$) [MP PMT 1987]
 (a) $\frac{e^2}{r^2}$ (b) $\frac{e^2}{2r}$
 (c) $\frac{e^2}{r}$ (d) $\frac{e^2}{2r^2}$
19. Ionization potential of hydrogen atom is 13.6 V.
 Hydrogen atoms in the ground state are excited by monochromatic radiation of photon energy 12.1 eV. The spectral lines emitted by hydrogen atoms according to Bohr's theory will be
 [CBSE PMT 1996; MP PMT 1999; AMU (Med.) 2002]
 (a) One (b) Two
 (c) Three (d) Four
20. Energy levels A, B, C of a certain atom corresponding to increasing values of energy i.e. $E_A < E_B < E_C$. If $\lambda_1, \lambda_2, \lambda_3$ are the wavelengths of radiations corresponding to the transitions C to B, B to A and C to A respectively, which of the following statements is correct
 [AIIMS 1995; CBSE PMT 1990, 2005]

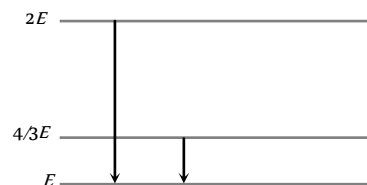


- (a) $\lambda_3 = \lambda_1 + \lambda_2$ (b) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$
 (c) $\lambda_1 + \lambda_2 + \lambda_3 = 0$ (d) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$

21. The angular momentum of electron in n orbit is given by

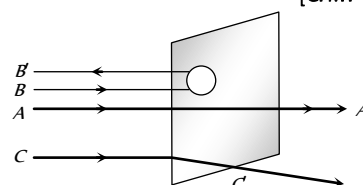
- (a) nh (b) $\frac{h}{2\pi m}$
 (c) $n \frac{h}{2\pi}$ (d) $n^2 \frac{h}{2\pi}$

22. The ratio of the energies of the hydrogen atom in its first to second excited state is [CPMT 1978]
 (a) 1/4 (b) 4/9
 (c) 9/4 (d) 4
23. An electron jumps from the 4th orbit to the 2nd orbit of hydrogen atom. Given the Rydberg's constant $R = 10^5 \text{ cm}^{-1}$. The frequency in Hz of the emitted radiation will be
 (a) $\frac{3}{16} \times 10^5$ (b) $\frac{3}{16} \times 10^{15}$
 (c) $\frac{9}{16} \times 10^{15}$ (d) $\frac{3}{4} \times 10^{15}$
24. The ionisation potential of hydrogen atom is 13.6 volt. The energy required to remove an electron in the $n = 2$ state of the hydrogen atom is [NCERT 1983; MP PET 2005]
 (a) 27.2 eV (b) 13.6 eV
 (c) 6.8 eV (d) 3.4 eV
25. The ionisation energy of 10 times ionised sodium atom is [DPMT 1991]
 (a) 13.6 eV (b) $13.6 \times 11 \text{ eV}$
 (c) $\frac{13.6}{11} \text{ eV}$ (d) $13.6 \times (11)^2 \text{ eV}$
26. If the wavelength of the first line of the Balmer series of hydrogen is $6561 \text{ \AA}$, the wavelength of the second line of the series should be [CPMT 1984; DPMT 2004]
 (a) 13122 Å (b) 3280 Å
 (c) 4860 Å (d) 2187 Å
27. The following diagram indicates the energy levels of a certain atom when the system moves from $2E$ level to E , a photon of wavelength λ is emitted. The wavelength of photon produced during its transition from $\frac{4E}{3}$ level to E is [CPMT 1989]



- (a) $\lambda/3$ (b) $3\lambda/4$
 (c) $4\lambda/3$ (d) 3λ
28. A beam of fast moving alpha particles were directed towards a thin film of gold. The parts A' , B' and C' of the transmitted and reflected beams corresponding to the incident parts A, B and C of the beam, are shown in the adjoining diagram. The number of alpha particles in

[CPMT 1986, 88; RPET 2000]



- (a) B' will be minimum and in C' maximum
 (b) A' will be maximum and in B' minimum
 (c) A' will be minimum and in B' maximum
 (d) C' will be minimum and in B' maximum
29. According to Bohr's theory the radius of electron in an orbit described by principal quantum number n and atomic number Z is proportional to [CPMT 1988]

- (a) $Z^2 n^2$ (b) $\frac{Z^2}{n^2}$
 (c) $\frac{Z^2}{n}$ (d) $\frac{n^2}{Z}$

30. The radius of electron's second stationary orbit in Bohr's atom is R . The radius of the third orbit will be

[EAMCET 1992; DPMT 1999]

- (a) $3R$ (b) $2.25R$
 (c) $9R$ (d) $\frac{R}{3}$

31. If m is mass of electron, v its velocity, r the radius of stationary circular orbit around a nucleus with charge Ze , then from Bohr's first postulate, the kinetic energy $K = \frac{1}{2}mv^2$ of the electron in C.G.S. system is equal to

[NCERT 1977]

- (a) $\frac{1}{2} \frac{Ze^2}{r}$ (b) $\frac{1}{2} \frac{Ze^2}{r^2}$
 (c) $\frac{Ze^2}{r}$ (d) $\frac{Ze}{r^2}$

32. Consider an electron in the n orbit of a hydrogen atom in the Bohr model. The circumference of the orbit can be expressed in terms of the de Broglie wavelength λ of that electron as

- (a) $(0.259)n\lambda$ (b) $\sqrt{n}\lambda$
 (c) $(13.6)\lambda$ (d) $n\lambda$

33. In any Bohr orbit of the hydrogen atom, the ratio of kinetic energy to potential energy of the electron is [MP PET 1994]

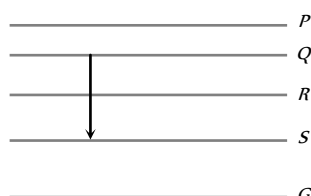
- (a) $1/2$ (b) 2
 (c) $-1/2$ (d) -2

34. The spectral series of the hydrogen spectrum that lies in the ultraviolet region is the

[CPMT 1990; MP PET 1994; MP PMT 2000]

- (a) Balmer series (b) Pfund series
 (c) Paschen series (d) Lyman series

35. Figure shows the energy levels P, Q, R, S and G of an atom where G is the ground state. A red line in the emission spectrum of the atom can be obtained by an energy level change from Q to S . A blue line can be obtained by following energy level change



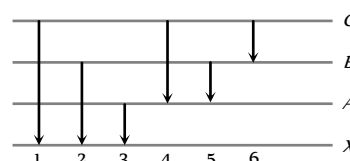
- (a) P to Q (b) Q to R
 (c) R to S (d) R to G

36. A hydrogen atom (ionisation potential 13.6 eV) makes a transition from third excited state to first excited state. The energy of the photon emitted in the process is [MNR 1995]

- (a) 1.89 eV (b) 2.55 eV
 (c) 12.09 eV (d) 12.75 eV

37. The figure indicates the energy level diagram of an atom and the origin of six spectral lines in emission (e.g. line no. 5 arises from the transition from level B to A). The following spectral lines will also occur in the absorption spectrum

[CBSE PMT 1995]



- (a) 1, 4, 6 (b) 4, 5, 6
 (c) 1, 2, 3 (d) 1, 2, 3, 4, 5, 6

38. When a hydrogen atom is raised from the ground state to an excited state [CBSE PMT 1995; AMU (Med.) 1999]

- (a) P.E. increases and K.E. decreases
 (b) P.E. decreases and K.E. increases
 (c) Both kinetic energy and potential energy increase
 (d) Both K.E. and P.E. decrease

39. An electron makes a transition from orbit $n = 4$ to the orbit $n = 2$ of a hydrogen atom. The wave number of the emitted radiations ($R = \text{Rydberg's constant}$) will be [CBSE PMT 1990]

[CBSE PMT 1995]

- (a) $\frac{16}{3R}$ (b) $\frac{2R}{16}$
 (c) $\frac{3R}{16}$ (d) $\frac{4R}{16}$

40. In Bohr model of the hydrogen atom, the lowest orbit corresponds to [Manipal MEE 1995]

- (a) Infinite energy (b) The maximum energy
 (c) The minimum energy (d) Zero energy

41. The ratio of the kinetic energy to the total energy of an electron in a Bohr orbit is [Roorkee 1995; BHU 2002]

- (a) -1 (b) 2
 (c) $1:2$ (d) None of these

42. An electron in the $n = 1$ orbit of hydrogen atom is bound by 13.6 eV . If a hydrogen atom is in the $n = 3$ state, how much energy is required to ionize it [MP PMT 1994]

- (a) 13.6 eV (b) 4.53 eV
 (c) 3.4 eV (d) 1.51 eV

43. Which of the following statements about the Bohr model of the hydrogen atom is false [MP PMT 1995]
- Acceleration of electron in $n = 2$ orbit is less than that in $n = 1$ orbit
 - Angular momentum of electron in $n = 2$ orbit is more than that in $n = 1$ orbit
 - Kinetic energy of electron in $n = 2$ orbit is less than that in $n = 1$ orbit
 - Potential energy of electron in $n = 2$ orbit is less than that in $n = 1$ orbit
44. If an electron jumps from 1st orbital to 3rd orbital, then it will
- Absorb energy
 - Release energy
 - No gain of energy
 - None of these
45. The ratio of the frequencies of the long wavelength limits of Lyman and Balmer series of hydrogen spectrum is [KCEE 1996]
- 27 : 5
 - 5 : 27
 - 4 : 1
 - 1 : 4
46. Which of the following transitions in a hydrogen atom emits photon of the highest frequency [MP PET 1996; DPMT 2001]
- $n = 1$ to $n = 2$
 - $n = 2$ to $n = 1$
 - $n = 2$ to $n = 6$
 - $n = 6$ to $n = 2$
47. In terms of Rydberg's constant R , the wave number of the first Balmer line is [MP PMT 1996]
- R
 - $3R$
 - $\frac{5R}{36}$
 - $\frac{8R}{9}$
48. If the ionisation potential of helium atom is 24.6 volt, the energy required to ionise it will be [MP PMT 1996]
- 24.6 eV
 - 24.6 V
 - 13.6 V
 - 13.6 eV
49. Which of the transitions in hydrogen atom emits a photon of lowest frequency ($n =$ quantum number) [BHU 1999]
- $n = 2$ to $n = 1$
 - $n = 4$ to $n = 3$
 - $n = 3$ to $n = 1$
 - $n = 4$ to $n = 2$
50. According to Bohr's theory, the expressions for the kinetic and potential energy of an electron revolving in an orbit is given respectively by
- $+\frac{e^2}{8\pi\epsilon_0 r}$ and $-\frac{e^2}{4\pi\epsilon_0 r}$
 - $+\frac{8\pi\epsilon_0 e^2}{r}$ and $-\frac{4\pi\epsilon_0 e^2}{r}$
 - $-\frac{e^2}{8\pi\epsilon_0 r}$ and $-\frac{e^2}{4\pi\epsilon_0 r}$
 - $+\frac{e^2}{8\pi\epsilon_0 r}$ and $+\frac{e^2}{4\pi\epsilon_0 r}$
51. In a hydrogen atom, which of the following electronic transitions would involve the maximum energy change [MP PET 1997]
- From $n = 2$ to $n = 1$
 - From $n = 3$ to $n = 1$
 - From $n = 4$ to $n = 2$
 - From $n = 3$ to $n = 2$
52. In the lowest energy level of hydrogen atom, the electron has the angular momentum [MP PET 1997; BCECE 2003]
- π/h
 - h/π
 - $h/2\pi$
 - $2\pi/h$
53. The minimum energy required to excite a hydrogen atom from its ground state is [EAMCET (Engg.) 1995; MP PMT 1997; CPMT 1999; DCE 1999]
- 13.6 eV
 - 13.6 eV
 - 3.4 eV
 - 10.2 eV
54. Ratio of the wavelengths of first line of Lyman series and first line of Balmer series is [AFMC 1996]
- 1 : 3
 - 27 : 5
 - 5 : 27
 - 4 : 9
55. The Rydberg constant R for hydrogen is [MP PMT/PET 1998]
- $R = -\left(\frac{1}{4\pi\epsilon_0}\right) \cdot \frac{2\pi^2 me^2}{ch^2}$
 - $R = \left(\frac{1}{4\pi\epsilon_0}\right) \cdot \frac{2\pi^2 me^4}{ch^2}$
 - $R = \left(\frac{1}{4\pi\epsilon_0}\right)^2 \cdot \frac{2\pi^2 me^4}{c^2 h^2}$
 - $R = \left(\frac{1}{4\pi\epsilon_0}\right)^2 \cdot \frac{2\pi^2 me^4}{ch^3}$
56. The wavelength of the first line of Balmer series is 6563 Å. The Rydberg constant for hydrogen is about [MP PMT/PET 1998]
- 1.09×10^7 per m
 - 1.09×10^8 per m
 - 1.09×10^9 per m
 - 1.09×10^5 per m
57. According to Bohr's theory the moment of momentum of an electron revolving in second orbit of hydrogen atom will be [MP PET 1999; KCET 2003]
- $2\pi\hbar$
 - $\pi\hbar$
 - $\frac{h}{\pi}$
 - $\frac{2h}{\pi}$
58. The velocity of an electron in the second orbit of sodium atom (atomic number = 11) is v . The velocity of an electron in its fifth orbit will be [MP PET 1999]
- v
 - $\frac{22}{5}v$
 - $\frac{5}{2}v$
 - $\frac{2}{5}v$
59. The absorption transitions between the first and the fourth energy states of hydrogen atom are 3. The emission transitions between these states will be [MP PET 1999]
- 3
 - 4
 - 5
 - 6
60. The ratio of longest wavelength and the shortest wavelength observed in the five spectral series of emission spectrum of hydrogen is [MP PET 1999]
- $\frac{4}{3}$
 - $\frac{525}{376}$
 - 25
 - $\frac{900}{11}$

61. In the Bohr model of a hydrogen atom, the centripetal force is furnished by the coulomb attraction between the proton and the electron. If a_0 is the radius of the ground state orbit, m is the mass, e is the charge on the electron and ϵ_0 is the vacuum permittivity, the speed of the electron is [CBSE PMT 1998]
- (a) 0 (b) $\frac{e}{\sqrt{\epsilon_0 a_0 m}}$
(c) $\frac{e}{\sqrt{4\pi\epsilon_0 a_0 m}}$ (d) $\frac{\sqrt{4\pi\epsilon_0 a_0 m}}{e}$
62. The electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$, where n_1 and n_2 are the principal quantum numbers of the two states. Assume the Bohr model to be valid. The time period of the electron in the initial state is eight times that in the final state. The possible values of n_1 and n_2 are [IIT 1998; KCET 2005]
- (a) $n_1 = 4, n_2 = 2$ (b) $n_1 = 8, n_2 = 2$
(c) $n_1 = 8, n_2 = 1$ (d) $n_1 = 6, n_2 = 3$
63. As per Bohr model, the minimum energy (in eV) required to remove an electron from the ground state of doubly ionized Li atom ($Z = 3$) is [IIT 1997 Re-Exam; MH CET 2000]
- (a) 1.51 (b) 13.6
(c) 40.8 (d) 122.4
64. Which one of these is non-divisible [KCET 1994]
- (a) Nucleus (b) Photon
(c) Proton (d) Atom
65. In Bohr's model of hydrogen atom, let PE represents potential energy and TE the total energy. In going to a higher level
- (a) PE decreases, TE increases
(b) PE increases, TE increases
(c) PE decreases, TE decreases
(d) PE increases, TE decreases
66. According to Bohr's model, the radius of the second orbit of helium atom is [Bihar MEE 1995]
- (a) $0.53 \text{ \AA}$ (b) $1.06 \text{ \AA}$
(c) $2.12 \text{ \AA}$ (d) $0.265 \text{ \AA}$
67. The fact that photons carry energy was established by [ISM Dhanbad 1994]
- (a) Doppler's effect (b) Compton's effect
(c) Bohr's theory (d) Diffraction of light
68. An ionic atom equivalent to hydrogen atom has wavelength equal to $1/4$ of the wavelengths of hydrogen lines. The ion will be
- (a) He^+ (b) Li^{++}
(c) Ne^{9+} (d) Na^{10+}
69. The extreme wavelengths of Paschen series are
- (a) $0.365 \mu\text{m}$ and $0.565 \mu\text{m}$ (b) $0.818 \mu\text{m}$ and $1.89 \mu\text{m}$
(c) $1.45 \mu\text{m}$ and $4.04 \mu\text{m}$ (d) $2.27 \mu\text{m}$ and $7.43 \mu\text{m}$
70. The third line of Balmer series of an ion equivalent to hydrogen atom has wavelength of 108.5 nm . The ground state energy of an electron of this ion will be [RPET 1997]
- (a) 3.4 eV (b) 13.6 eV
(c) 54.4 eV (d) 122.4 eV
71. An electron in the $n = 1$ orbit of hydrogen atom is bound by 13.6 eV energy is required to ionize it is [MP PMT 2003]
- (a) 13.6 eV (b) 6.53 eV
(c) 5.4 eV (d) 1.51 eV
72. Ionization energy of hydrogen is 13.6 eV . If $h = 6.6 \times 10^{-34} \text{ J-sec}$, the value of R will be of the order of [RPMT 1997]
- (a) 10^{10} m^{-1} (b) 10^7 m^{-1}
(c) 10^4 m^{-1} (d) 10^{-7} m^{-1}
73. To explain his theory, Bohr used [CBSE PMT 1993; MP PET 2002]
- (a) Conservation of linear momentum
(b) Conservation of angular momentum
(c) Conservation of quantum frequency
(d) Conservation of energy
74. The ionisation energy of hydrogen atom is 13.6 eV . Following Bohr's theory, the energy corresponding to a transition between the 3rd and the 4th orbit is [CBSE PMT 1992; DPMT 2000; RPMT 1999; AMU (Med.) 2001]
- (a) 3.40 eV (b) 1.51 eV
(c) 0.85 eV (d) 0.66 eV
75. Hydrogen atoms are excited from ground state of the principal quantum number 4. Then the number of spectral lines observed will be [CBSE PMT 1993]
- (a) 3 (b) 6
(c) 5 (d) 2
76. Hydrogen atom emits blue light when it changes from $n = 4$ energy level to the $n = 2$ level. Which colour of light would the atom emit when it changes from the $n = 5$ level to the $n = 2$ level
- (a) Red (b) Yellow
(c) Green (d) Violet
77. In Rutherford scattering experiment, what will be the correct angle for α scattering for an impact parameter $b = 0$ [RPET 1997] [CBSE PMT 1994; JIPMER 2000]
- (a) 90° (b) 270°
(c) 0° (d) 180°
78. The radius of hydrogen atom in its ground state is $5.3 \times 10^{-11} \text{ m}$. After collision with an electron it is found to have a radius of

$21.2 \times 10^{-11} \text{ m}$. What is the principal quantum number n of the final state of the atom

[CBSE PMT 1994; CPMT 2001; MH CET 2000]

- (a) $n = 4$ (b) $n = 2$
(c) $n = 16$ (d) $n = 3$

79. The splitting of line into groups under the effect of electric or magnetic field is called [AFMC 1995]

- (a) Zeeman's effect (b) Bohr's effect
(c) Heisenberg's effect (d) Magnetic effect

80. The energy of a hydrogen atom in its ground state is -13.6 eV . The energy of the level corresponding to the quantum number $n = 2$ (first excited state) in the hydrogen atom is [CBSE PMT 1996; CBSE PMT 1997, 2001; MP PET 2000; AFMC 2000, 01, 02; BCECE 2003]

- (a) -2.72 eV (b) -0.85 eV
(c) -0.54 eV (d) -3.4 eV

81. The first line of Balmer series has wavelength $6563 \text{ \AA}$. What will be the wavelength of the first member of Lyman series

[RPMT 1996]

- (a) $1215.4 \text{ \AA}$ (b) $2500 \text{ \AA}$
(c) $7500 \text{ \AA}$ (d) $600 \text{ \AA}$

82. The wavelength of Lyman series is [BHU 1997]

- (a) $\frac{4}{3 \times 10967} \text{ cm}$ (b) $\frac{3}{4 \times 10967} \text{ cm}$
(c) $\frac{4 \times 10967}{3} \text{ cm}$ (d) $\frac{3}{4} \times 10967 \text{ cm}$

83. When hydrogen atom is in its first excited level, its radius is its ground state radius [CBSE PMT 1997]

- (a) Half (b) Same
(c) Twice (d) Four times

84. Hydrogen atom excites energy level from fundamental state to $n = 3$. Number of spectrum lines according to Bohr, is

[CPMT 1997]

- (a) 4 (b) 3
(c) 1 (d) 2

85. Number of spectral lines in hydrogen atom is [CPMT 1997]

- (a) 3 (b) 6
(c) 15 (d) Infinite

86. In Bohr's model, the atomic radius of the first orbit is r_0 , then the radius of the third orbit is

[AIIMS 1997; CPMT 2001; KCET (Engg./Med.) 1999; Pb. PMT 2004]

- (a) $\frac{r_0}{9}$ (b) r_0
(c) $9r_0$ (d) $3r_0$

87. The wavelength of the energy emitted when electron come from fourth orbit to second orbit in hydrogen is 20.397 cm . The wavelength of energy for the same transition in He^+ is

[AIIMS 1997; JIPMER 2000]

- (a) 5.099 cm^{-1} (b) 20.497 cm^{-1}
(c) 40.994 cm^{-1} (d) 81.988 cm^{-1}

88. Minimum excitation potential of Bohr's first orbit in hydrogen atom is

[BHU 1998; JIPMER 2001, 02; Pb. PMT 2004]

- (a) 13.6 V (b) 3.4 V
(c) 10.2 V (d) 3.6 V

89. Which of the following statements are true regarding Bohr's model of hydrogen atom

- (I) Orbiting speed of electron decreases as it shifts to discrete orbits away from the nucleus
(II) Radii of allowed orbits of electron are proportional to the principal quantum number
(III) Frequency with which electrons orbits around the nucleus in discrete orbits is inversely proportional to the principal quantum number
(IV) Binding force with which the electron is bound to the nucleus increases as it shifts to outer orbits

Select correct answer using the codes given below

Codes :

[SCRA 1998]

- (a) I and III (b) II and IV
(c) I, II and III (d) II, III and IV

90. The wavelength of radiation emitted is λ_0 when an electron jumps from the third to the second orbit of hydrogen atom. For the electron jump from the fourth to the second orbit of the hydrogen atom, the wavelength of radiation emitted will be

- (a) $\frac{16}{25} \lambda_0$ (b) $\frac{20}{27} \lambda_0$
(c) $\frac{27}{20} \lambda_0$ (d) $\frac{25}{16} \lambda_0$

91. For electron moving in n orbit of H -atom the angular velocity is proportional to [RPET 1999]

- (a) n (b) $1/n$
(c) n (d) $1/n$

92. The energy of electron in first excited state of H -atom is -3.4 eV its kinetic energy is

[RPET 1999; CBSE PMT 2005]

- (a) -3.4 eV (b) $+3.4 \text{ eV}$
(c) -6.8 eV (d) 6.8 eV

93. The energy required to excite an electron from the ground state of hydrogen atom to the first excited state, is

[Pb. PMT 1999]

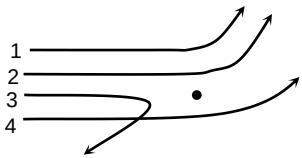
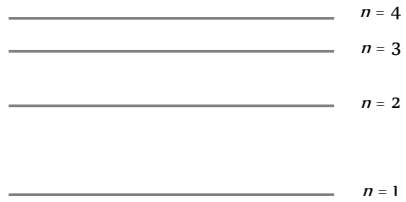
- (a) $1.602 \times 10^{-14} \text{ J}$ (b) $1.619 \times 10^{-16} \text{ J}$
(c) $1.632 \times 10^{-18} \text{ J}$ (d) $1.656 \times 10^{-20} \text{ J}$

94. Which of the following phenomena suggests the presence of electron energy levels in atoms [JIPMER 1999]

- (a) Radio active decay
(b) Isotopes
(c) Spectral lines
(d) α -particles scattering

95. Which of the following spectral series in hydrogen atom give spectral line of $4860 \text{ \AA}$ [Roorkee 1999]

- (a) Lyman (b) Balmer
(c) Paschen (d) Brackett
96. If scattering particles are 56° for 90° angle then this will be at 60° angle [RPMT 2000]
(a) 224 (b) 256
(c) 98 (d) 108
97. When an electron in hydrogen atom is excited, from its 4th to 5th stationary orbit, the change in angular momentum of electron is (Planck's constant: $h = 6.6 \times 10^{-34} \text{ J-s}$) [AFMC 2000; Pb. PET 2001]
(a) $4.16 \times 10^{-34} \text{ J-s}$ (b) $3.32 \times 10^{-34} \text{ J-s}$
(c) $1.05 \times 10^{-34} \text{ J-s}$ (d) $2.08 \times 10^{-34} \text{ J-s}$
98. Energy of electron in a orbit of H-atom is [RPET 2000]
(a) Positive (b) Negative
(c) Zero (d) Nothing can be said
99. The concept of stationary orbits was proposed by [Pb. PMT 2000]
(a) Neil Bohr (b) J.J. Thomson
(c) Rutherford (d) I. Newton
100. In a hydrogen atom, the distance between the electron and proton is $2.5 \times 10^{-11} \text{ m}$. The electrical force of attraction between them will be [Pb. PMT 2000]
(a) $2.8 \times 10^{-7} \text{ N}$ (b) $3.7 \times 10^{-7} \text{ N}$
(c) $6.2 \times 10^{-7} \text{ N}$ (d) $9.1 \times 10^{-7} \text{ N}$
101. If λ_{max} is 6563 Å, then wave length of second line for Balmer series will be [RPMT 2000]
(a) $\lambda = \frac{16}{3R}$ (b) $\lambda = \frac{36}{5R}$
(c) $\lambda = \frac{4}{3R}$ (d) None of the above
102. What will be the angular momentum of a electron, if energy of this electron in H-atom is 1.5 eV (in J-sec) [RPMT 2000]
(a) 1.05×10^{-34} (b) 2.1×10^{-34}
(c) 3.15×10^{-34} (d) -2.1×10^{-34}
103. Who discovered spin quantum number [RPMT 2000]
(a) Unlenbeck and Goudsmit
(b) Nell's Bohr
(c) Zeeman
(d) Sommerfield
104. The time of revolution of an electron around a nucleus of charge Ze in n Bohr orbit is directly proportional to [MP PET 2003]
(a) n (b) $\frac{n^3}{Z^2}$
(c) $\frac{n^2}{Z}$ (d) $\frac{Z}{n}$
105. In Bohr's model, if the atomic radius of the first orbit is r_0 , then the radius of the fourth orbit is [CBSE PMT 2000]
(a) r_0 (b) $4r_0$
(c) $r_0/16$ (d) $16r_0$
106. If R is the Rydberg's constant for hydrogen the wave number of the first line in the Lyman series will be [KCET 2000]
(a) $\frac{R}{4}$ (b) $\frac{3R}{4}$
(c) $\frac{R}{2}$ (d) $2R$
107. In hydrogen atom, if the difference in the energy of the electron in $n = 2$ and $n = 3$ orbits is E , the ionization energy of hydrogen atom is [EAMCET (Med.) 2000]
(a) $13.2 E$ (b) $7.2 E$
(c) $5.6 E$ (d) $3.2 E$
108. The first member of the Paschen series in hydrogen spectrum is of wavelength 18,800 Å. The short wavelengths limit of Paschen series is [EAMCET (Med.) 2000]
(a) 1215 Å (b) 6560 Å
(c) 8225 Å (d) 12850 Å
109. The ratio of the largest to shortest wavelengths in Lyman series of hydrogen spectra is [EAMCET (Med.) 2000]
(a) $\frac{25}{9}$ (b) $\frac{17}{6}$
(c) $\frac{9}{5}$ (d) $\frac{4}{3}$
110. In Bohr model of hydrogen atom, the ratio of periods of revolution of an electron in $n = 2$ and $n = 1$ orbits is [EAMCET (Engg.) 2000]
(a) 2 : 1 (b) 4 : 1
(c) 8 : 1 (d) 16 : 1
111. The ratio of the longest to shortest wavelengths in Brackett series of hydrogen spectra is [EAMCET (Engg.) 2000]
(a) $\frac{25}{9}$ (b) $\frac{17}{6}$
(c) $\frac{9}{5}$ (d) $\frac{4}{3}$
112. The electron in a hydrogen atom makes a transition from an excited state to the ground state. Which of the following statements is true
(a) Its kinetic energy increases and its potential and total energies decrease
(b) Its kinetic energy decreases, potential energy increases and its total energy remains the same

- (c) Its kinetic and total energies decrease and its potential energy increases
(d) Its kinetic, potential and total energies decreases
113. The ratio of minimum to maximum wavelength in Balmer series is
(a) 5 : 9 (b) 5 : 36
(c) 1 : 4 (d) 3 : 4
114. The radius of the Bohr orbit in the ground state of hydrogen atom is $0.5 \text{ \AA}$. The radius of the orbit of the electron in the third excited state of He^+ will be [MP PMT 2000]
(a) $8 \text{ \AA}$ (b) $4 \text{ \AA}$
(c) $0.5 \text{ \AA}$ (d) $0.25 \text{ \AA}$
115. The ratio of the speed of the electron in the first Bohr orbit of hydrogen and the speed of light is equal to (where e , h and c have their usual meanings) [MP PMT 2000]
(a) $2\pi hc/e^2$ (b) $e^2 h/2\pi c$
(c) $e^2 c/2\pi h$ (d) $2\pi e^2/hc$
116. According to the Rutherford's atomic model, the electrons inside the atom are [KCET (Med.) 2000]
(a) Stationary (b) Not stationary
(c) Centralized (d) None of these
117. The energy of hydrogen atom in its ground state is -13.6 eV . The energy of the level corresponding to the quantum number n is equal 5 is [KCET (Engg./Med.) 2001]
(a) -5.40 eV (b) -2.72 eV
(c) -0.85 eV (d) -0.54 eV
118. According to classical theory, the circular path of an electron in Rutherford atom is [BHU 2001]
(a) Spiral (b) Circular
(c) Parabolic (d) Straight line
119. Rutherford's α -particle experiment showed that the atoms have
(a) Proton (b) Nucleus
(c) Neutron (d) Electrons
120. Orbital acceleration of electron is [RPET 2001]
(a) $\frac{n^2 h^2}{4\pi^2 m^2 r^3}$ (b) $\frac{n^2 h^2}{2\pi^2 r^3}$
(c) $\frac{4n^2 h^2}{\pi^2 m^2 r^3}$ (d) $\frac{4n^2 h^2}{4\pi^2 m^2 r^3}$
121. Which of the following is true for number of spectral lines in going from Lyman series to Pfund series [RPET 2001]
(a) Increases
(b) Decreases
(c) Unchanged
(d) May decrease or increase
122. The wavelength of yellow line of sodium is $5896 \text{ \AA}$. Its wave number will be [MP PET 2001]
(a) $50883 \times 10^6 \text{ per second}$
(b) 16961 per cm
(c) 17581 per cm
(d) 50883 per cm
123. Radius of the first orbit of the electron in a hydrogen atom is $0.53 \text{ \AA}$. So, the radius of the third orbit will be [Kerala (Engg.) 2001]
(a) $2.12 \text{ \AA}$ (b) $4.77 \text{ \AA}$
(c) $1.06 \text{ \AA}$ (d) $1.59 \text{ \AA}$
124. The first line in the Lyman series has wavelength λ . The wavelength of the first line in Balmer series is [MH CET (Med.) 2001]
(a) $\frac{2}{9} \lambda$ (b) $\frac{9}{2} \lambda$
(c) $\frac{5}{27} \lambda$ (d) $\frac{27}{5} \lambda$
125. In hydrogen atom which quantity is integral multiple of $\frac{h}{2\pi}$ [DCE 2001]
(a) Angular momentum (b) Angular velocity
(c) Angular acceleration (d) Momentum
126. In the following transitions, which one has higher frequency [UPSEAT 2001]
(a) $3 \rightarrow 2$ (b) $4 \rightarrow 3$
(c) $4 \rightarrow 2$ (d) $3 \rightarrow 1$
127. The diagram shows the path of four α -particles of the same energy being scattered by the nucleus of an atom simultaneously. Which of these are/is not physically possible [AMU (Med.) 2001]
- 
- (a) 3 and 4 (b) 2 and 3
(c) 1 and 4 (d) 4 only
128. An electron jumps from 5th orbit to 4th orbit of hydrogen atom. Taking the Rydberg constant as 10^7 per metre . What will be the frequency of radiation emitted [AFMC 2001] [Pb. PMT 2001]
(a) $6.75 \times 10^{12} \text{ Hz}$ (b) $6.75 \times 10^{14} \text{ Hz}$
(c) $6.75 \times 10^{13} \text{ Hz}$ (d) None of these
129. For principal quantum number $n = 3$, the possible values of orbital quantum number ' l ' are [MP PET 2001; MP PMT 2001]
(a) 1, 2, 3 (b) 0, 1, 2, 3
(c) 0, 1, 2 (d) -1, 0, +1
130. Four lowest energy levels of H-atom are shown in the figure. The number of possible emission lines would be [MP PMT 2001]
- 

- (a) 3 (b) 4
(c) 5 (d) 6
131. The order of the size of nucleus and Bohr radius of an atom respectively are [MP PET 2001; MP PMT 2001]
(a) $10^{-14} m, 10^{-10} m$ (b) $10^{-10} m, 10^{-8} m$
(c) $10^{-20} m, 10^{-16} m$ (d) $10^{-8} m, 10^{-6} m$
132. Energy of an electron in an excited hydrogen atom is -3.4 eV . Its angular momentum will be: $h = 6.626 \times 10^{-34} \text{ J-s}$ [UPSEAT 1999; Kerala PET 2002]
(a) $1.11 \times 10^{34} \text{ J sec}$ (b) $1.51 \times 10^{-31} \text{ J sec}$
(c) $2.11 \times 10^{-34} \text{ J sec}$ (d) $3.72 \times 10^{-34} \text{ J sec}$
133. The ratio of the wavelengths for $2 \rightarrow 1$ transition in Li, He and H is
(a) $1 : 2 : 3$ (b) $1 : 4 : 9$
(c) $4 : 9 : 36$ (d) $3 : 2 : 1$
134. The wavelength of light emitted from second orbit to first orbits in a hydrogen atom is [Pb. PMT 2002]
(a) $1.215 \times 10^{-7} m$ (b) $1.215 \times 10^{-5} m$
(c) $1.215 \times 10^{-4} m$ (d) $1.215 \times 10^{-3} m$
135. Energy of the electron in n orbit of hydrogen atom is given by $E_n = -\frac{13.6}{n^2} \text{ eV}$. The amount of energy needed to transfer electron from first orbit to third orbit is [MH CET 2002; Kerala PMT 2002]
(a) 13.6 eV (b) 3.4 eV
(c) 12.09 eV (d) 1.51 eV
136. The ratio of speed of an electron in ground state in Bohrs first orbit of hydrogen atom to velocity of light in air is [MH CET 2002]
(a) $\frac{e^2}{2\epsilon_0 hc}$ (b) $\frac{2e^2 \epsilon_0}{hc}$
(c) $\frac{e^3}{2\epsilon_0 hc}$ (d) $\frac{2\epsilon_0 hc}{e^2}$
137. Whenever a hydrogen atom emits a photon in the Balmer series [KCET 2002]
(a) It need not emit any more photon
(b) It may emit another photon in the Paschen series
(c) It must emit another photon in the Lyman series
(d) It may emit another photon in the Balmer series
138. The de-Broglie wavelength of an electron in the first Bohr orbit is
(a) Equal to one fourth the circumference of the first orbit
(b) Equal to half the circumference of the first orbit
(c) Equal to twice the circumference of the first orbit
(d) Equal to the circumference of the first orbit
139. In hydrogen atom, when electron jumps from second to first orbit, then energy emitted is [AIEEE 2002]
(a) -13.6 eV (b) -27.2 eV
(c) -6.8 eV (d) None of these
140. Minimum energy required to takeout the only one electron from ground state of He^+ is [CPMT 2002]
(a) 13.6 eV (b) 54.4 eV
(c) 27.2 eV (d) 6.8 eV
141. The frequency of H line of Balmer series in H_2 atom is ν_0 . The frequency of line emitted by singly ionised He atom is [CPMT 2002]
(a) $2\nu_0$ (b) $4\nu_0$
(c) $\nu_0/2$ (d) $\nu_0/4$
142. When the electron in the hydrogen atom jumps from 2^{nd} orbit to 1^{st} orbit, the wavelength of radiation emitted is λ . When the electrons jump from 3^{rd} orbit to 1^{st} orbit, the wavelength of emitted radiation would be [UPSEAT 2005; MP PMT 2002]
(a) $\frac{27}{32} \lambda$ (b) $\frac{32}{27} \lambda$
(c) $\frac{2}{3} \lambda$ (d) $\frac{3}{2} \lambda$
143. The possible quantum number for $3d$ electron are [MP PMT 2002]
(a) $n = 3, l = 1, m_l = +1, m_s = -\frac{1}{2}$
(b) $n = 3, l = 2, m_l = +2, m_s = -\frac{1}{2}$
(c) $n = 3, l = 1, m_l = -1, m_s = +\frac{1}{2}$
(d) $n = 3, l = 0, m_l = +1, m_s = -\frac{1}{2}$
144. The radius of the first (lowest) orbit of the hydrogen atom is a_0 . The radius of the second (next higher) orbit will be [MP PET 2002; MP PMT 2004]
(a) $4a_0$ (b) $6a_0$
(c) $8a_0$ (d) $10a_0$
145. Which of the following transition will have highest emission wavelength [BHU 2003]
(a) $n = 2$ to $n = 1$ (b) $n = 1$ to $n = 2$
(c) $n = 2$ to $n = 5$ (d) $n = 5$ to $n = 2$
146. When the wave of hydrogen atom comes from infinity into the first orbit then the value of wave number is [RPET 2003]
(a) 109700 cm^{-1} (b) 1097 cm^{-1}
(c) 109 cm^{-1} (d) None of these
147. [KCET 2002] With the increase in principle quantum number, the energy difference between the two successive energy levels [RPET 2003]
(a) Increases
(b) Decreases
(c) Remains constant
(d) Sometimes increases and sometimes decreases
148. In which of the following systems will the radius of the first orbit ($n = 1$) be minimum

[Kerala PET 2002; CBSE PMT 2003]

[Pb. PMT 2004]

- (a) Single ionized helium
(b) Deuterium atom
(c) Hydrogen atom
(d) Doubly ionized lithium
149. If the binding energy of the electron in a hydrogen atom is 13.6 eV, the energy required to remove the electron from the first excited state of Li^{++} is [AIEEE 2003]
(a) 122.4 eV (b) 30.6 eV
(c) 13.6 eV (d) 3.4 eV
150. Which of the following is quantised according to Bohr's theory of hydrogen atom [MP PMT 2004]
(a) Linear momentum of electron
(b) Angular momentum of electron
(c) Linear velocity of electron
(d) Angular velocity of electron
151. The shortest wavelength in the Lyman series of hydrogen spectrum is 912 Å corresponding to a photon energy of 13.6 eV. The shortest wavelength in the Balmer series is about [MP PMT 2004]
(a) 3648 Å (b) 8208 Å
(c) 1228 Å (d) 6566 Å
152. Energy E of a hydrogen atom with principal quantum number n is given by $E = \frac{-13.6}{n^2}$ eV. The energy of a photon ejected when the electron jumps from $n = 3$ state to $n = 2$ state of hydrogen is approximately [CBSE PMT 2004]
(a) 1.5 eV (b) 0.85 eV
(c) 3.4 eV (d) 1.9 eV
153. The Bohr model of atoms [CBSE PMT 2004]
(a) Assumes that the angular momentum of electrons is quantized
(b) Uses Einstein's photo-electric equation
(c) Predicts continuous emission spectra for atoms
(d) Predicts the same emission spectra for all types of atoms
154. The colour of the second line of Balmer series is [J & K CET 2004]
(a) Blue (b) Yellow
(c) Red (d) Violet
155. Which state of triply ionised Beryllium (Be^{+++}) has the same orbital radius as that of the ground state of hydrogen [KCET 2004]
(a) $n = 4$ (b) $n = 3$
(c) $n = 2$ (d) $n = 1$
156. The ratio of areas within the electron orbits for the first excited state to the ground state for hydrogen atom is [BCECE 2004]
(a) 16 : 1 (b) 18 : 1
(c) 4 : 1 (d) 2 : 1
157. The kinetic energy of an electron revolving around a nucleus will be
(a) Four times of P.E. (b) Double of P.E.
(c) Equal to P.E. (d) Half of its P.E.
158. Taking Rydberg's constant $R_H = 1.097 \times 10^7 m$ first and second wavelength of Balmer series in hydrogen spectrum is
(a) 2000 Å, 3000 Å (b) 1575 Å, 2960 Å
(c) 6529 Å, 4280 Å (d) 6552 Å, 4863 Å
159. The kinetic energy of electron in the first Bohr orbit of the hydrogen atom is [Pb. PET 2000]
(a) - 6.5 eV (b) - 27.2 eV
(c) 13.6 eV (d) - 13.6 eV
160. In the spectrum of hydrogen atom, the ratio of the longest wavelength in Lyman series to the longest wavelength in the Balmer series is [UPSEAT 2004]
(a) 5/27 (b) 1/93
(c) 4/9 (d) 3/2
161. In Bohr's model of hydrogen atom, which of the following pairs of quantities are quantized [UPSEAT 2004]
(a) Energy and linear momentum
(b) Linear and angular momentum
(c) Energy and angular momentum
(d) None of the above
162. The energy of the highest energy photon of Balmer series of hydrogen spectrum is close to [UPSEAT 2004]
(a) 13.6 eV (b) 3.4 eV
(c) 1.5 eV (d) 0.85 eV
163. Energy of an electron in n orbit of hydrogen atom is $\left(k = \frac{1}{4\pi\epsilon_0} \right)$
(a) $-\frac{2\pi^2 k^2 m e^4}{n^2 h^2}$ (b) $-\frac{4\pi^2 m k e^2}{n^2 h^2}$
(c) $-\frac{n^2 h^2}{2\pi k m e^4}$ (d) $-\frac{n^2 h^2}{4\pi^2 k m e^2}$
164. Which one of the relation is correct between time period and number of orbits while an electron is revolving in a orbit [DPMT 2003]
(a) n^2 (b) $\frac{1}{n^2}$
(c) n^3 (d) $\frac{1}{n}$
165. An electron changes its position from orbit $n = 4$ to the orbit $n = 2$ of an atom. The wavelength of the emitted radiation's is ($R =$ Rydberg's constant) [BHU 2004]
(a) $\frac{16}{R}$ (b) $\frac{16}{3R}$
(c) $\frac{16}{5R}$ (d) $\frac{16}{7R}$
166. If the energy of a hydrogen atom in n th orbit is E_n , then energy in the n th orbit of a singly ionized helium atom will be [BCECE 2004]
(a) $4E_n$ (b) $E_n / 4$
(c) $2E_n$ (d) $E_n / 2$

167. What is the ratio of wavelength of radiations emitted when an electron in hydrogen atom jump from fourth orbit to second orbit and from third orbit to second orbit

[MH CET 2004]

- (a) 27 : 25 (b) 20 : 27
(c) 20 : 25 (d) 25 : 27

168. The energy of electron in the n th orbit of hydrogen atom is expressed as $E_n = \frac{-13.6}{n^2} \text{ eV}$. The shortest and longest wavelength of Lyman series will be [Pb. PET 2003]

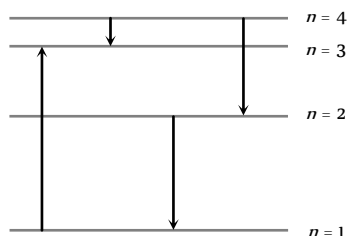
- (a) 910 Å, 1213 Å (b) 5463 Å, 7858 Å
(c) 1315 Å, 1530 Å (d) None of these

169. The ground state energy of hydrogen atom is -13.6 eV . What is the potential energy of the electron in this state

[AIIMS 2005]

- (a) 0 eV (b) -27.2 eV
(c) 1 eV (d) 2 eV

170. The diagram shows the energy levels for an electron in a certain atom. Which transition shown represents the emission of a photon with the most energy [AIEEE 2005]



- (a) I (b) II
(c) III (d) IV

171. As the electron in Bohr orbit of Hydrogen atom passes from state $n = 2$ to $n = 1$, the kinetic energy K and potential energy U change as [MP PET 2005]

- (a) K two-fold, U four-fold
(b) K four-fold, U two-fold
(c) K four-fold, U also four-fold
(d) K two-fold, U also two-fold

172. The magnetic moment (μ) of a revolving electron around the nucleus varies with principal quantum number n as

[AIIMS 2005]

- (a) $\mu \propto n$ (b) $\mu \propto 1/n$
(c) $\mu \propto n^2$ (d) $\mu \propto 1/n^2$

173. Bohr's atom model assumes [KCET 2005]

- (a) The nucleus is of infinite mass and is at rest
(b) Electrons in a quantized orbit will not radiate energy
(c) Mass of electron remains constant
(d) All the above conditions

174. Radius of first Bohr orbit is r . What is the radius of 2nd Bohr orbit?

- (a) $8r$ (b) $2r$
(c) $4r$ (d) $2\sqrt{2}r$

1. Which of the following particles are constituents of the nucleus [CBSE PMT 1999]

- (a) Protons and electrons (b) Protons and neutrons
(c) Neutrons and electrons (d) Neutrons and positrons

2. The particles which can be added to the nucleus of an atom without changing its chemical properties are called

[NCERT 1979]

- (a) Electrons (b) Protons
(c) Neutrons (d) None of the above

3. The neutron was discovered by

[MP PMT 1992; RPMT 1996]

- (a) Marie Curie (b) Pierre Curie
(c) James Chadwick (d) Rutherford

4. The mass number of a nucleus is

[IIT 1986; ISM Dhanbad 1994;

MP PMT 1997; CBSE PMT 2003; MH CET (Med.) 2001]

- (a) Always less than its atomic number
(b) Always more than its atomic number
(c) Always equal to its atomic number
(d) Sometimes more than and sometimes equal to its atomic number

5. The energy equivalent of 1 kilogram of matter is about

[MP PET/PMT 1988; MNR 1987]

- (a) 10^{-15} J (b) 1 J
(c) 10^{-12} J (d) 10^{17} J

6. Nuclear binding energy is equivalent to [MP PET/PMT 1988]

- (a) Mass of proton (b) Mass of neutron
(c) Mass of nucleus (d) Mass defect of nucleus

7. If the binding energy of the deuterium is 2.23 MeV .

The mass defect given in $a.m.u$ is [MP PET 1993]

- (a) -0.0024 (b) -0.0012
(c) 0.0012 (d) 0.0024

8. Which of the following has the mass closest in value to that of the positron [AFMC 1993]

- (a) Proton (b) Electron
(c) Photon (d) Neutrino

(1 $a.m.u = 931 \text{ MeV}$)

9. Size of nucleus is of the order of

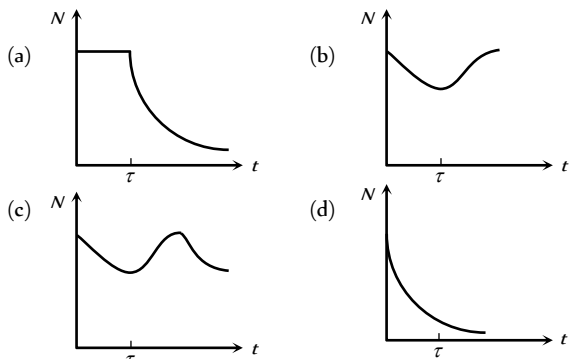
[CPMT 1983; MP PET 2002, 03]

- (a) 10^{-10} m (b) 10^{-15} m
(c) 10^{-12} m (d) 10^{-19} m

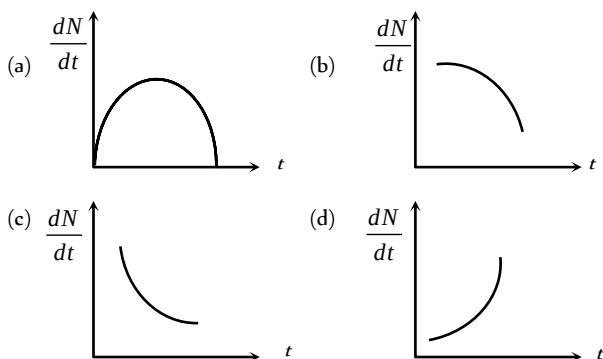
[BHU 2005]

4. A radioactive sample consists of two distinct species having equal number of atoms initially. The mean life time of one species is τ and that of the other is 5τ . The decay products in both cases are stable. A plot is made of the total number of radioactive nuclei as a function of time. Which of the following figures best represents the form of this plot

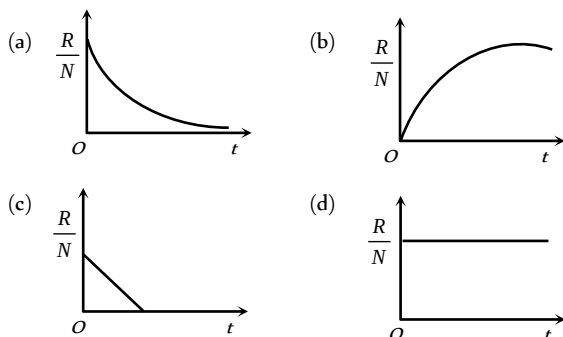
[IIT-JEE (Screening) 2001]



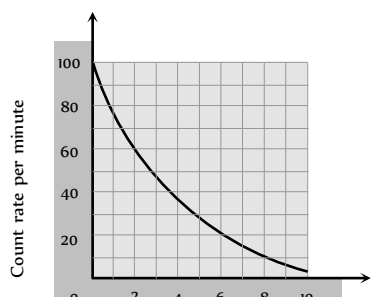
5. Radioactive element decays to form a stable nuclide, then the rate of decay of reactant $\left(\frac{dN}{dt}\right)$ will vary with time (t) as shown in figure



6. A radioactive sample has N_0 active atoms at $t = 0$. If the rate of disintegration at any time is R and the number of atoms is N , then the ratio R/N varies with time as



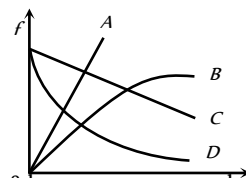
7. The count rate of 10g of radioactive material was measured at different times and this has been shown in the figure. The half life of material and the total counts (approximately) in the first half life period, respectively are [CPMT 1986]



- (a) $4h, 9000$ (b) $3h, 14000$
(c) $3h, 235$ (d) $3h, 50$

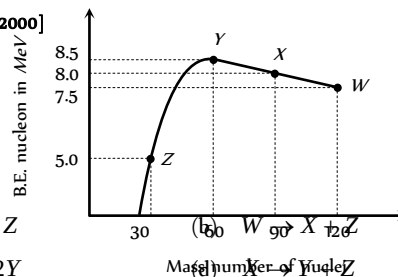
8. The fraction f of radioactive material that has decayed in time t , varies with time t . The correct variation is given by the curve

- (a) A
(b) B
(c) C
(d) D



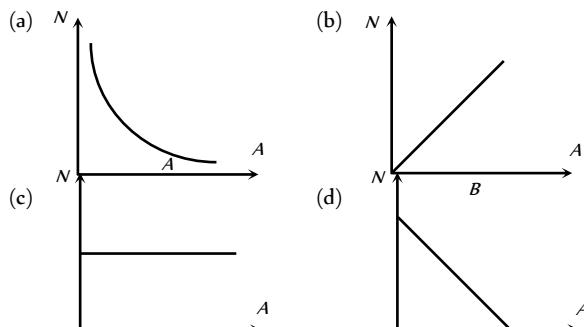
9. Binding energy per nucleon versus mass number curve for nuclei is shown in the figure. W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is

[DCE 2000]

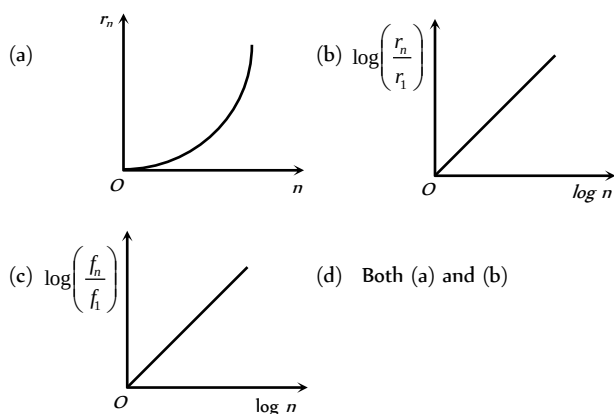


- (a) $Y \rightarrow 2Z$
(c) $W \rightarrow 2Y$

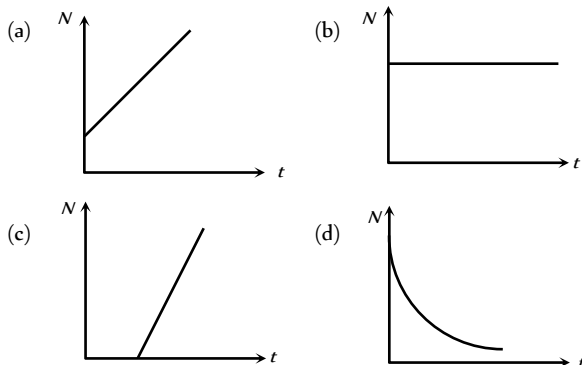
10. The plot of the number (N) of decayed atoms versus activity (A) of a radioactive substance is



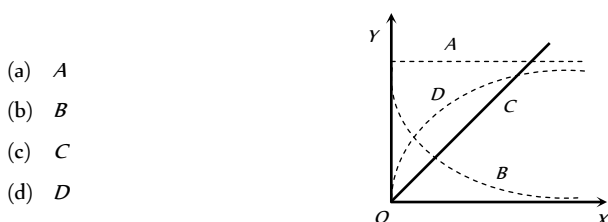
11. If in hydrogen atom, radius of n^{th} Bohr orbit is r_n^D , frequency of revolution of electron in n^{th} orbit is f_n choose the correct option



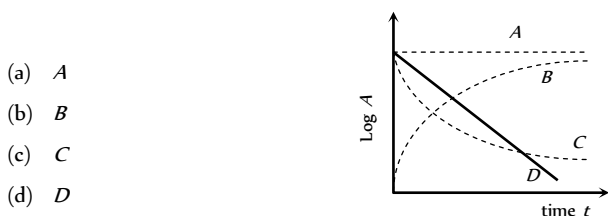
12. The graph between the instantaneous concentration (N) of a radioactive element and time (t) is



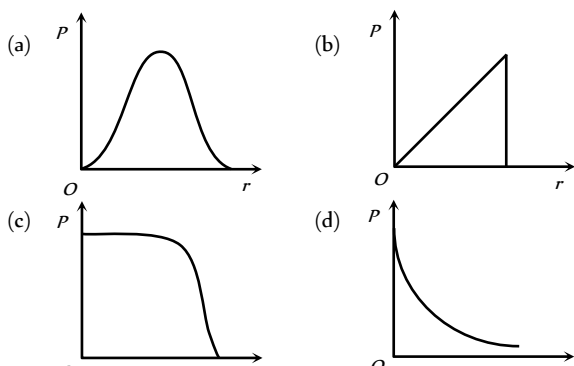
13. In Fig. X represents time and Y represent activity of a radioactive sample. Then the activity of sample, varies with time according to the curve



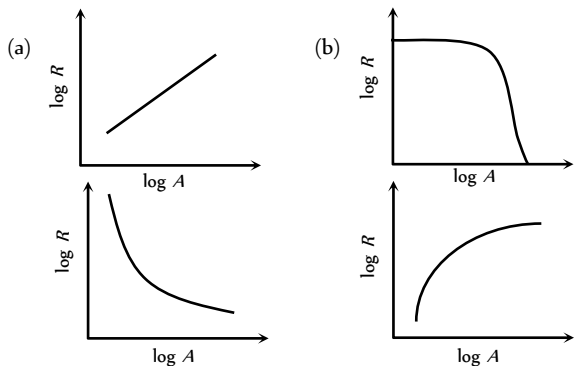
14. The graph which represents the correct variation of logarithm of activity ($\log A$) versus time, in figure is



15. The charge density in a nucleus varies with distance from the centre of the nucleus according to the curve in Fig.



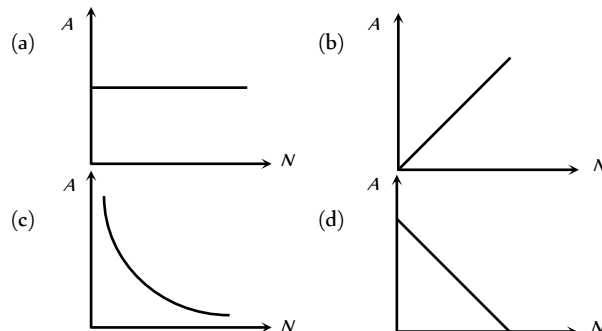
16. The graph between $\log R$ and $\log A$ where R is the nuclear radius and A is the mass number is



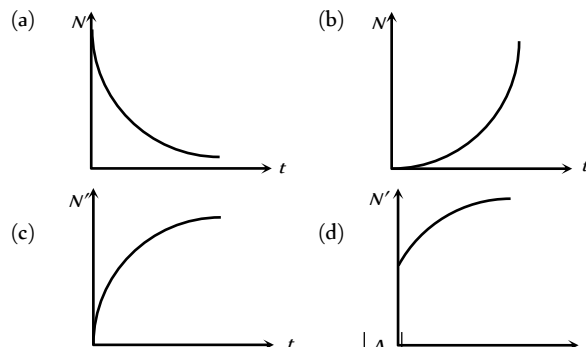
(c)

(d)

17. The curve between the activity A of a radioactive sample and the number of active atoms N is

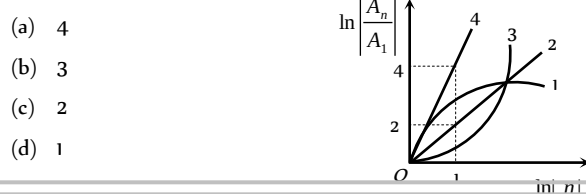


18. The graph between number of decayed atoms N' of a radioactive element and time t is



19. The figure shows a graph between $\ln \left| \frac{A_n}{A_1} \right|$ and $\ln |n|$, where A_n is

the area enclosed by the n th orbit in a hydrogen like atom. The correct curve is



Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

1. Assertion : It is not possible to use ^{35}Cl as the fuel for fusion energy.

Reason : The binding energy of ^{35}Cl is too small.

[AIIMS 2005]

2. Assertion : ^{90}Sr from the radioactive fall out from a nuclear bomb ends up in the bones of human beings through the milk consumed by them. It causes impairment of the production of red blood cells.
Reason : The energetics β -particles emitted in the decay of ^{90}Sr damage the bone marrow. [AIIMS 2004]
3. Assertion : Neutrons penetrate matter more readily as compared to protons.
Reason : Neutrons are slightly more massive than protons. [AIIMS 2003]
4. Assertion : Bohr had to postulate that the electrons in stationary orbits around the nucleus do not radiate.
Reason : According to classical physics all moving electrons radiate. [AIIMS 2003]
5. Assertion : Radioactive nuclei emit β^{-1} particles.
Reason : Electrons exist inside the nucleus. [AIIMS 2003]
6. Assertion : ${}_Z^AX^A$ undergoes 2α -decays. 2β -decays and 2γ -decays and the daughter product is ${}_{Z-2}Y^{A-8}$.
Reason : In α -decay the mass number decreases by 4 and atomic number decreases by 2. In β -decay the mass number remains unchanged, but atomic number increases by 1 only. [AIIMS 2001]
7. Assertion : Density of all the nuclei is same.
Reason : Radius of nucleus is directly proportional to the cube root of mass number. [AIIMS 2000]
8. Assertion : Isobars are the element having same mass number but different atomic number.
Reason : Neutrons and protons are present inside nucleus. [AIIMS 1997]
9. Assertion : The force of repulsion between atomic nucleus and α -particle varies with distance according to inverse square law.
Reason : Rutherford did α -particle scattering experiment.
10. Assertion : The positively charged nucleus of an atom has a radius of almost 10^{-15}m .
Reason : In α -particle scattering experiment, the distance of closest approach for α -particles is $\simeq 10^{-15}\text{m}$.
11. Assertion : According to classical theory, the proposed path of an electron in Rutherford atom model will be parabolic.
Reason : According to electromagnetic theory an accelerated particle continuously emits radiation.
12. Assertion : Electrons in the atom are held due to coulomb forces.
Reason : The atom is stable only because the centripetal force due to Coulomb's law is balanced by the centrifugal force.
13. Assertion : The electron in the hydrogen atom passes from energy level $n = 4$ to the $n = 1$ level. The maximum and minimum number of photon that can be emitted are six and one respectively.
Reason : The photons are emitted when electron make a transition from the higher energy state to the lower energy state.
14. Assertion : Hydrogen atom consists of only one electron but its emission spectrum has many lines.
Reason : Only Lyman series is found in the absorption spectrum of hydrogen atom whereas in the emission spectrum, all the series are found.
15. Assertion : It is essential that all the lines available in the emission spectrum will also be available in the absorption spectrum.
Reason : The spectrum of hydrogen atom is only absorption spectrum.
16. Assertion : For the scattering of α -particles at a large angles, only the nucleus of the atom is responsible.
Reason : Nucleus is very heavy in comparison to electrons.
17. Assertion : All the radioactive elements are ultimately converted in lead.
Reason : All the elements above lead are unstable.
18. Assertion : Amongst alpha, beta and gamma rays, α -particle has maximum penetrating power.
Reason : The alpha particle is heavier than beta and gamma rays.
19. Assertion : The ionising power of β -particle is less compared to α -particles but their penetrating power is more.
Reason : The mass of β -particle is less than the mass of α -particle.
20. Assertion : The mass of β -particles when they are emitted is higher than the mass of electrons obtained by other means.
Reason : β -particle and electron, both are similar particles.
21. Assertion : Radioactivity of 10^8 undecayed radioactive nuclei of half life of 50 days is equal to that of 1.2×10^8 number of undecayed nuclei of some other material with half life of 60 days.
Reason : Radioactivity is proportional to half-life.
22. Assertion : Fragments produced in the fission of U^{235} are radioactive.
Reason : The fragments have abnormally high proton to neutron ratio.
23. Assertion : Electron capture occurs more often than positron emission in heavy elements.
Reason : Heavy elements exhibit radioactivity.
24. Assertion : The mass of a nucleus can be either less than or more than the sum of the masses of nucleons present in it.
Reason : The whole mass of the atom is considered in the nucleus.

Answers

Atomic Structure

1	a	2	d	3	a	4	d	5	c
6	b	7	b	8	c	9	a	10	d
11	b	12	b	13	b	14	c	15	c
16	c	17	c	18	b	19	c	20	b
21	c	22	c	23	c	24	d	25	d
26	c	27	d	28	b	29	d	30	b
31	a	32	d	33	c	34	d	35	d
36	b	37	c	38	a	39	c	40	c
41	a	42	d	43	d	44	a	45	a
46	a	47	c	48	a	49	b	50	a
51	b	52	c	53	d	54	c	55	d
56	a	57	c	58	d	59	d	60	d
61	c	62	ad	63	d	64	b	65	b
66	b	67	c	68	a	69	b	70	c
71	a	72	b	73	b	74	d	75	b
76	d	77	d	78	b	79	a	80	d
81	a	82	a	83	b	84	b	85	d
86	c	87	a	88	c	89	a	90	b
91	d	92	b	93	c	94	c	95	b
96	a	97	c	98	b	99	a	100	b
101	a	102	c	103	a	104	b	105	d
106	b	107	b	108	c	109	d	110	c
111	a	112	a	113	a	114	b	115	d
116	b	117	d	118	a	119	b	120	a
121	b	122	b	123	b	124	d	125	a
126	d	127	d	128	c	129	c	130	d
131	a	132	c	133	c	134	a	135	c
136	a	137	c	138	d	139	d	140	b
141	b	142	a	143	b	144	a	145	d
146	a	147	b	148	d	149	b	150	b
151	a	152	d	153	a	154	a	155	c
156	d	157	d	158	d	159	c	160	a
161	c	162	b	163	a	164	c	165	b
166	a	167	b	168	a	169	b	170	c
171	c	172	a	173	d	174	c		

Nucleus, Nuclear Reaction

1	b	2	c	3	c	4	d	5	d
6	d	7	d	8	b	9	b	10	c
11	a	12	b	13	d	14	c	15	c
16	c	17	c	18	c	19	d	20	c
21	b	22	d	23	a	24	b	25	c
26	b	27	c	28	c	29	a	30	a
31	b	32	a	33	d	34	c	35	c
36	b	37	a	38	a	39	d	40	c
41	b	42	a	43	c	44	a	45	c

46	c	47	d	48	b	49	a	50	b
51	d	52	b	53	d	54	c	55	a
56	a	57	d	58	ad	59	c	60	a
61	c	62	b	63	a	64	a	65	d
66	c	67	a	68	b	69	b	70	d
71	b	72	d	73	bc	74	d	75	c
76	d	77	d	78	c	79	b	80	b
81	c	82	d	83	c	84	b	85	d
86	b	87	a	88	d	89	d	90	a
91	c	92	b	93	a	94	c	95	a
96	b	97	c	98	d	99	d	100	b
101	c	102	b	103	a	104	d	105	c
106	b	107	a	108	b	109	d	110	c
111	c	112	c	113	b	114	b	115	d
116	a	117	b	118	a	119	c	120	b
121	d	122	d	123	a	124	a	125	c
126	c	127	b	128	d	129	c	130	a
131	c	132	a	133	a	134	b	135	b
136	c	137	a	138	c	139	b	140	a
141	b	142	b	143	b	144	d	145	d
146	a	147	b	148	b	149	d	150	a
151	c	152	b	153	d	154	c	155	c
156	a	157	b	158	a	159	c	160	c
161	a	162	a	163	b	164	b	165	c
166	b	167	d	168	d	169	a	170	b
171	b	172	a	173	c	174	b	175	a
176	b	177	a	178	c	179	b		

Radioactivity

1	a	2	a	3	d	4	c	5	a
6	c	7	c	8	d	9	c	10	c
11	b	12	c	13	c	14	c	15	a
16	c	17	a	18	c	19	b	20	a
21	a	22	c	23	a	24	d	25	d
26	d	27	c	28	b	29	a	30	c
31	c	32	c	33	d	34	c	35	c
36	b	37	b	38	d	39	d	40	d
41	a	42	b	43	c	44	d	45	b
46	b	47	d	48	d	49	b	50	a
51	b	52	c	53	a	54	d	55	c
56	d	57	b	58	d	59	d	60	b
61	a	62	d	63	a	64	d	65	b
66	a	67	b	68	c	69	d	70	c
71	d	72	a	73	a	74	d	75	c
76	d	77	d	78	c	79	a	80	d
81	d	82	b	83	a	84	a	85	b

86	c	87	d	88	d	89	b	90	a
91	b	92	d	93	c	94	c	95	a
96	d	97	d	98	a	99	b	100	c
101	a	102	d	103	b	104	b	105	b
106	d	107	a	108	d	109	c	110	b
111	c	112	c	113	d	114	d	115	c
116	b	117	a	118	a	119	d	120	a
121	c	122	d	123	a	124	d	125	d
126	d	127	c	128	d	129	c	130	b
131	d	132	b	133	c	134	a	135	a
136	b	137	ac	138	b	139	c	140	c
141	d	142	c	143	a	144	d	145	c
146	b	147	d	148	b	149	b	150	c
151	c	152	a	153	b	154	b	155	d
156	b	157	c	158	c	159	d	160	c
161	a	162	d	163	c	164	c	165	d
166	d	167	c	168	c	169	b	170	d
171	b	172	c	173	b	174	a	175	c
176	d								

16	a	17	c	18	d	19	b	20	b
21	c	22	c	23	b	24	e		

Critical Thinking Questions

1	c	2	c	3	b	4	a	5	a
6	a	7	a	8	d	9	b	10	d
11	a	12	a	13	a	14	d	15	c
16	d	17	c	18	d	19	c	20	d
21	b	22	c	23	d	24	a	25	d
26	d	27	c	28	c	29	cd	30	a
31	a	32	c	33	a	34	a	35	b
36	b	37	b	38	c	39	b	40	a
41	a	42	b	43	c	44	c	45	d
46	a	47	b	48	d	49	a	50	b
51	d	52	a	53	b	54	b	55	a
56	a	57	b	58	b	59	b	60	a
61	a	62	a	63	b	64	c	65	a

Graphical Questions

1	a	2	a	3	c	4	d	5	c
6	d	7	b	8	b	9	c	10	d
11	d	12	d	13	b	14	d	15	c
16	a	17	b	18	c	19	a		

Assertion and Reason

1	c	2	a	3	b	4	b	5	c
6	a	7	a	8	b	9	b	10	a
11	e	12	c	13	b	14	b	15	d

AS Answers and Solutions

Atomic Structure

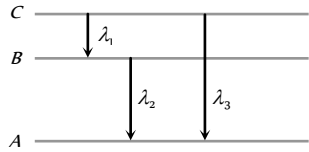
- (a) For $n=1$, maximum number of states $= 2n^2 = 2$ and for $n = 2, 3, 4$, maximum number of states would be 8, 18, 32 respectively. Hence number of possible elements $= 2 + 8 + 18 + 32 = 60$.
- (d) Bohr radius $r = \frac{\epsilon_0 n^2 h^2}{\pi Z m e^2}$; $\therefore r \propto n^2$
- (a)
$$n=2 \quad \text{-----} \quad E_2 = -\frac{13.6}{(2)^2} = -3.4 \text{ eV}$$

$$n=1 \quad \text{-----} \quad E_1 = -13.6 \text{ eV}$$

$$E_{1 \rightarrow 2} = -3.4 - (-13.6) = +10.2 \text{ eV}$$
- (d) $\frac{1}{\lambda} = RZ^2 \left(\frac{1}{1^2} - \frac{1}{2^2} \right)$
 For di-ionised lithium the value of Z is maximum.
- (c) Lyman series lies in the UV region.
- (b) The size of the atom is of the order of $1\text{\AA} = 10^{-10} \text{ m}$.
- (b) Balmer series lies in the visible region.
- (c) Transition A ($n = \infty$ to 1): Series limit of Lyman series
 Transition B ($n = 5$ to $n = 2$): Third spectral line of Balmer series
 Transition C ($n = 5$ to $n = 3$): Second spectral line of Paschen series
- (a) D is excitation of electron from 2^{nd} orbit corresponding to absorption line in Balmer series and E is the energy released to bring the electron from ∞ to ground state *i.e.* ionisation potential.
- (d)
- (b) Paschen series lies in the infrared region.
- (b) Energy required to knock out the electron in the n orbit $= +\frac{13.6}{n^2} \text{ eV} \Rightarrow E_3 = +\frac{13.6}{9} \text{ eV}$.
- (b) Linear momentum $= mv = 9.1 \times 10^{-31} \times 2.2 \times 10^6$
 $= 2.0 \times 10^{-24} \text{ kg-m/s}$
- (c) $r \propto n^2 \Rightarrow r_n = n^2 a_0$ ($\because r_1 = a_0$)
- (c) For the ionization of second He electron. He^+ will act as hydrogen like atom.
 Hence ionization potential
 $= Z^2 \times 13.6 \text{ volt} = (2)^2 \times 13.6 = 54.4 \text{ V}$
- (c) Energy required $= \frac{13.6}{n^2} = \frac{13.6}{10^2} = 0.136 \text{ eV}$

- (c) $\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \Rightarrow \frac{1}{n_1^2} - \frac{1}{n_2^2} = \frac{1}{R\lambda}$
 $= \frac{1}{1.097 \times 10^7 \times 18752 \times 10^{-10}} = 0.0486 = \frac{7}{144}$. But
 $\frac{1}{3^2} - \frac{1}{4^2} = \frac{7}{144} \Rightarrow n_1 = 3 \text{ and } n_2 = 4$ (Paschen series)
- (b) Potential energy of electron in n orbit of radius r in H -atom

$$U = -\frac{e^2}{r} \text{ (in CGS)}$$

$$\therefore \text{K.E.} = \frac{1}{2} |P.E.| \Rightarrow K = \frac{e^2}{2r}$$
- (c) Final energy of electron $= -13.6 + 12.1 = -1.51 \text{ eV}$. which corresponds to third level *i.e.* $n = 3$. Hence number of spectral lines emitted $= \frac{n(n-1)}{2} = \frac{3(3-1)}{2} = 3$
- (b) Let the energy in A, B and C state be E_A, E_B and E_C , then from the figure


$$(E_C - E_B) + (E_B - E_A) = (E_C - E_A) \text{ or } \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} = \frac{hc}{\lambda_3}$$

$$\Rightarrow \lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$$
- (c) According to Bohr's second postulate.
- (c) First excited state *i.e.* second orbit ($n = 2$)
 Second excited state *i.e.* third orbit ($n = 3$)

$$\therefore E = -\frac{13.6}{n^2} \Rightarrow \frac{E_2}{E_3} = \left(\frac{3}{2} \right)^2 = \frac{9}{4}$$
- (c) $\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \frac{3R}{16} \Rightarrow \lambda = \frac{16}{3R} = \frac{16}{3} \times 10^{-5} \text{ cm}$
 Frequency $n = \frac{c}{\lambda} = \frac{3 \times 10^{10}}{\frac{16}{3} \times 10^{-5}} = \frac{9}{16} \times 10^{15} \text{ Hz}$
- (d) Energy required to remove electron in the $n = 2$ state $= +\frac{13.6}{(2)^2} = +3.4 \text{ eV}$
- (d) $(E)_\infty = Z^2 (E_{\text{ion}})_H = (11)^2 13.6 \text{ eV}$
- (c) The wavelength of spectral line in Balmer series is given by

$$\frac{1}{\lambda} = R \left[\frac{1}{2^2} - \frac{1}{n^2} \right]$$

 For first line of Balmer series, $n = 3$

$$\Rightarrow \frac{1}{\lambda_1} = R \left[\frac{1}{2^2} - \frac{1}{3^2} \right] = \frac{5R}{36}; \text{ For second line } n = 4.$$

$$\Rightarrow \frac{1}{\lambda_2} = R \left[\frac{1}{2^2} - \frac{1}{4^2} \right] = \frac{3R}{16}$$

$$\therefore \frac{\lambda_2}{\lambda_1} = \frac{20}{27} \Rightarrow \lambda_1 = \frac{20}{27} \times 6561 = 4860 \text{ \AA}$$

27. (d) $2E - E = \frac{hc}{\lambda} \Rightarrow E = \frac{hc}{\lambda}$

$$\frac{4E}{3} - E = \frac{hc}{\lambda'} \Rightarrow \frac{E}{3} = \frac{hc}{\lambda'} \therefore \frac{\lambda'}{\lambda} = 3 \Rightarrow \lambda' = 3\lambda$$

28. (b) Because atom is hollow and whole mass of atom is concentrated in a small centre called nucleus.

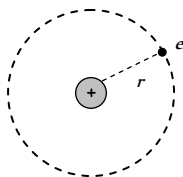
29. (d) $r = \frac{\epsilon_0 n^2 h^2}{\pi Z m e^2}; \therefore r \propto \frac{n^2}{Z}$

30. (b) $r \propto n^2 \Rightarrow \frac{r_{(n=2)}}{r_{(n=3)}} = \frac{4}{9} \Rightarrow r_{(n=3)} = \frac{9}{4} R = 2.25 R$

31. (a) In the revolution of electron, coulomb force provides the necessary centripetal force

$$\Rightarrow \frac{ze^2}{r^2} = \frac{mv^2}{r} \Rightarrow mv^2 = \frac{ze^2}{r}$$

$$\therefore \text{K.E.} = \frac{1}{2} mv^2 = \frac{ze^2}{2r}$$



32. (d) According to Bohr's theory $mvr = n \frac{h}{2\pi}$

$$\Rightarrow \text{Circumference } 2\pi r = n \left(\frac{h}{mv} \right) = n\lambda$$

33. (c) $K.E. = \frac{kZe^2}{2r}$ and $P.E. = -\frac{kZe^2}{r}; \therefore \frac{K.E.}{P.E.} = -\frac{1}{2}$.

34. (d) Lyman series lies in the UV region.

35. (d) If E is the energy radiated in transition

$$\text{then } E_{R \rightarrow G} > E_{Q \rightarrow S} > E_{R \rightarrow S} > E_{Q \rightarrow R} > E_{P \rightarrow Q}$$

For getting blue line energy radiated should be maximum

$$\left(E \propto \frac{1}{\lambda} \right). \text{ Hence (d) is the correct option.}$$

36. (b) Energy released $= 13.6 \left[\frac{1}{(2)^2} - \frac{1}{(4)^2} \right] = 2.55 \text{ eV}$

37. (c) The absorption lines are obtained when the electron jumps from ground state ($n = 1$) to the higher energy states. Thus only 1, 2 and 3 lines will be obtained.

38. (a) $P.E. \propto -\frac{1}{r}$ and $K.E. \propto \frac{1}{r}$

As r increases so K.E. decreases but P.E. increases.

39. (c) Wave number $\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] = R \left[\frac{1}{4} - \frac{1}{16} \right] = \frac{3R}{16}$

40. (c) In hydrogen atom, the lowest orbit ($n = 1$) corresponds to minimum energy (-13.6 eV).

41. (a) K.E. = - (T.E.)

42. (d) Required energy $E_3 = \frac{+13.6}{3^2} = 1.51 \text{ eV}$

43. (d) As n increases P.E. also increases.

44. (a) When an electron jumps from the orbit of lower energy ($n=1$) to the orbit of higher energy ($n=3$), energy is absorbed.

45. (a) For Lyman series

$$\nu_{\text{Lyman}} = \frac{c}{\lambda_{\text{max}}} = Rc \left[\frac{1}{(1)^2} - \frac{1}{(2)^2} \right] = \frac{3RC}{4}$$

For Balmer series

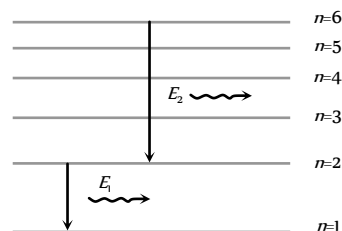
$$\nu_{\text{Balmer}} = \frac{c}{\lambda_{\text{max}}} = Rc \left[\frac{1}{(2)^2} - \frac{1}{(3)^2} \right] = \frac{5RC}{36}$$

$$\therefore \frac{\nu_{\text{Lyman}}}{\nu_{\text{Balmer}}} = \frac{27}{5}$$

46. (a) $\because E_1 > E_2$

$$\therefore \nu_1 > \nu_2$$

i.e. photons of higher frequency will be emitted if transition takes place from $n = 2$ to 1.



47. (c) Wave number

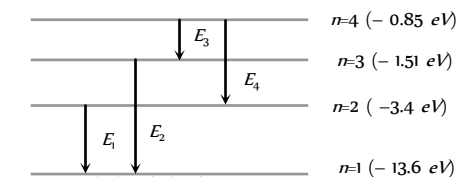
$$= \frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

For first Balmer line $n_1 = 2, n_2 = 3$

$$\therefore \text{Wave number} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = R \left(\frac{9-4}{9 \times 4} \right) = \frac{5R}{36}$$

48. (a) Energy required to ionise helium atom = 24.6 eV

49. (b) From diagram



$$E_1 = -13.6 - (-3.4) = -10.2 \text{ eV}$$

$$E_2 = -13.6 - (-1.51) = -12.09 \text{ eV}$$

$$E_3 = -1.51 - (-0.85) = -0.66 \text{ eV}$$

$$E_4 = -3.4 - (-1.51) = -1.89 \text{ eV}$$

E_3 is least i.e. frequency is lowest.

50. (a) $P.E. = -\frac{ke^2}{r} = -\frac{e^2}{4\pi\epsilon_0 r}; \text{ K.E.} = -\frac{1}{2} (P.E.) = \frac{e^2}{8\pi\epsilon_0 r}$

51. (b) Similar to Q. 49

52. (c) $mvr = \frac{nh}{2\pi}$, for $n=1$ it is $\frac{h}{2\pi}$

53. (d) Minimum energy required to excite from ground state

$$= 13.6 \left[\frac{1}{1^2} - \frac{1}{2^2} \right] = 10.2 \text{ eV}$$

54. (c) $\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

For first line of Lyman series $n = 1$ and $n_1 = 2$
For first line of Balmer series $n = 2$ and $n_1 = 3$

$$\text{So, } \frac{\lambda_{\text{Lyman}}}{\lambda_{\text{Balmer}}} = \frac{5}{27}$$

$$55. \quad (d) \quad R = \frac{2\pi^2 k^2 e^4 m}{ch^3} = \left(\frac{1}{4\pi\epsilon_0}\right)^2 \frac{2\pi^2 m e^4}{ch^3}$$

$$56. \quad (a) \quad \frac{1}{\lambda} = R \left[\frac{1}{4} - \frac{1}{9} \right] = \frac{5R}{36}$$

$$\therefore R = \frac{36}{5\lambda} = \frac{36}{5 \times 6563 \times 10^{-10}} = 1.09 \times 10^7 \text{ m}^{-1}$$

$$57. \quad (c) \quad \text{Angular momentum } L = n \left(\frac{h}{2\pi} \right)$$

$$\text{For this case } n=2, \text{ hence } L = 2 \times \frac{h}{2\pi} = \frac{h}{\pi}$$

$$58. \quad (d) \quad v \propto \frac{1}{n} \Rightarrow \frac{v_5}{v_2} = \frac{2}{5} \Rightarrow v_5 = \frac{2}{5} v_2 = \frac{2}{5} v$$

$$59. \quad (d) \quad \text{By using } N_E = \frac{n(n-1)}{2} \Rightarrow N_E = \frac{4(4-1)}{2} = 6$$

60. (d) Shortest wavelength comes from $n_1 = \infty$ to $n_2 = 1$ and longest wavelength comes from $n_1 = 6$ to $n_2 = 5$ in the given

$$\text{case. Hence } \frac{1}{\lambda_{\min}} = R \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right) = R$$

$$\frac{1}{\lambda_{\max}} = R \left(\frac{1}{5^2} - \frac{1}{6^2} \right) = R \left(\frac{36-25}{25 \times 36} \right) = \frac{11}{900} R$$

$$\therefore \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{900}{11}$$

$$61. \quad (c) \quad \frac{mv^2}{a_0} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{a_0^2} \Rightarrow v = \frac{e}{\sqrt{4\pi\epsilon_0 a_0 m}}$$

$$62. \quad (a, d) \quad T \propto n^3. \text{ Given } T_{n_1} = 8 T_{n_2}, \text{ hence } n_1 = 2n_2$$

Therefore, option (a) and (d) both are correct.

$$63. \quad (d) \quad E = -Z^2 \times 13.6 \text{ eV} = -9 \times 13.6 \text{ eV} = -122.4 \text{ eV}$$

So ionisation energy = + 122.4 eV.

$$64. \quad (b)$$

65. (b) As n increases P.E. increases and K.E. decreases.

$$66. \quad (b) \quad r = \frac{n^2}{Z} (r_0); \Rightarrow r_{(n=2)} = \frac{(2)^2}{2} \times 0.53 = 1.06 \text{ \AA}$$

$$67. \quad (c)$$

$$68. \quad (a) \quad \bar{v} \propto \frac{1}{\lambda} \propto Z^2 \Rightarrow \lambda Z^2 = \text{constant} \Rightarrow \lambda = \frac{\lambda}{4} Z^2 \Rightarrow Z = 2$$

$$69. \quad (b) \quad \text{In Paschen series } \frac{1}{\lambda_{\max}} = R \left[\frac{1}{(3)^2} - \frac{1}{(4)^2} \right]$$

$$\Rightarrow \lambda_{\max} = \frac{144}{7R} = \frac{144}{7 \times 1.1 \times 10^7} = 1.89 \times 10^{-6} \text{ m} = 1.89 \text{ \mu m}$$

$$\text{Similarly } \lambda_{\min} = \frac{9}{R} = \frac{9}{1.1 \times 10^7} = 0.818 \text{ \mu m}$$

$$70. \quad (c) \quad \text{For third line of Balmer series } n_1 = 2, n_2 = 5$$

$$\therefore \frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \text{ gives } Z^2 = \frac{n_1^2 n_2^2}{(n_2^2 - n_1^2) \lambda R}$$

On putting values $Z = 2$

$$\text{From } E = -\frac{13.6 Z^2}{n^2} = -\frac{13.6(2)^2}{(1)^2} = -54.4 \text{ eV}$$

$$71. \quad (a) \quad \text{Ionization energy} = \text{Binding energy.}$$

$$72. \quad (b) \quad E = -Rch \Rightarrow R = -\frac{E}{ch} = -\frac{13.6 \times 1.6 \times 10^{-19}}{3 \times 10^8 \times 6.6 \times 10^{-34}}$$

$$= 1.098 \times 10^7 \text{ per m}$$

$$73. \quad (b) \quad \text{Bohr postulated that the angular momentum of the electron is conserved.}$$

$$74. \quad (d) \quad E_3 = -\frac{13.6}{9} = -1.51 \text{ eV}; E_4 = -\frac{13.6}{16} = -0.85 \text{ eV}$$

$$\therefore E_4 - E_3 = 0.66 \text{ eV}$$

$$75. \quad (b) \quad \text{Number of spectral lines } N_E = \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$$

76. (d) In the transition from orbit $5 \rightarrow 2$, more energy is liberated as compared to transition from $4 \rightarrow 2$.

$$77. \quad (d) \quad \text{Impact parameter } b \propto \cot \frac{\theta}{2}$$

Here $b = 0$, hence $\theta = 180^\circ$

$$78. \quad (b) \quad r \propto n^2 \text{ i.e. } \frac{r_f}{r_i} = \left(\frac{n_f}{n_i} \right)^2$$

$$\Rightarrow \frac{21.2 \times 10^{-11}}{5.3 \times 10^{-11}} = \left(\frac{n}{1} \right)^2 \Rightarrow n^2 = 4 \Rightarrow n = 2$$

$$79. \quad (a)$$

$$80. \quad (d) \quad E_n = \frac{-13.6}{n^2} = \frac{-13.6}{4} = -3.4 \text{ eV}$$

$$81. \quad (a) \quad \frac{1}{\lambda_{\text{Balmer}}} = R \left[\frac{1}{2^2} - \frac{1}{3^2} \right] = \frac{5R}{36}, \quad \frac{1}{\lambda_{\text{Lyman}}} = R \left[\frac{1}{1^2} - \frac{1}{2^2} \right] = \frac{3R}{4}$$

$$\therefore \lambda_{\text{Lyman}} = \lambda_{\text{Balmer}} \times \frac{5}{27} = 1215.4 \text{ \AA}$$

$$82. \quad (a) \quad \frac{1}{\lambda} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]. \text{ For Lyman series } n=1 \text{ and } n=2, 3, 4,$$

$$\text{When } n=2, \text{ we get } \lambda = \frac{4}{3R_H} = \frac{4}{3 \times 10967} \text{ cm}$$

$$83. \quad (b) \quad r \propto n^2. \text{ For ground state } n=1 \text{ and for first excited state } n=2.$$

$$84. \quad (b) \quad \text{No. of lines } N_E = \frac{n(n-1)}{2} = \frac{3(3-1)}{2} = 3$$

85. (d) Infinitely large transitions are possible (in principle) for the hydrogen atom.

$$86. \quad (c) \quad r_n \propto n^2$$

$$87. \quad (a) \quad E \left(= \frac{hc}{\lambda} \right) \propto \frac{Z^2}{n^2} \Rightarrow \lambda \propto \frac{1}{Z^2}$$

Hence $\lambda_{He^+} = \frac{20.397}{4} = 5.099 \text{ cm}$

88. (c) Excitation potential = $\frac{\text{Excitation energy}}{e}$

Minimum excitation energy corresponds to excitation from $n = 1$ to $n = 2$

$\therefore$ Minimum excitation energy in hydrogen atom $= -3.4 - (-13.6) = +10.2 \text{ eV}$

so minimum excitation potential = 10.2 eV .

89. (a) Orbital speed varies inversely as the radius of the orbit. Energy increases with the increase in quantum number.

90. (b) $\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \Rightarrow \frac{1}{\lambda_{3 \rightarrow 2}} = R \left[\frac{1}{(2)^2} - \frac{1}{(3)^2} \right] = \frac{5R}{36}$

and $\frac{1}{\lambda_{4 \rightarrow 2}} = R \left[\frac{1}{(2)^2} - \frac{1}{(4)^2} \right] = \frac{3R}{16}$

$\therefore \frac{\lambda_{4 \rightarrow 2}}{\lambda_{3 \rightarrow 2}} = \frac{20}{27} \Rightarrow \lambda_{4 \rightarrow 2} = \frac{20}{27} \lambda_0$

91. (d)

92. (b) Kinetic energy = |Total energy|

93. (c) Energy to excite the e^- from $n = 1$ to $n = 2$

First excited state $n = 2$ (-3.4 eV)

Ground state $n = 1$ (-13.6 eV)

(For H_2 - atom)
 $E = -3.4 - (-13.6) = 10.2 \text{ eV} = 10.2 \times 1.6 \times 10^{-19}$

$= 1.632 \times 10^{-18} \text{ J}$

94. (c)

95. (b)

96. (a) According to scattering formula

$N \propto \frac{1}{\sin^4(\theta/2)} \Rightarrow \frac{N_2}{N_1} = \left[\frac{\sin(\theta_1/2)}{\sin(\theta_2/2)} \right]^4$

$\Rightarrow \frac{N_2}{N_1} = \left[\frac{\sin \frac{90^\circ}{2}}{\sin \frac{60^\circ}{2}} \right]^4 = \left[\frac{\sin 45^\circ}{\sin 30^\circ} \right]^4$

$\Rightarrow N_2 = (\sqrt{2})^4 \times N_1 = 4 \times 56 = 224$

97. (c) Change in the angular momentum

$\Delta L = L_2 - L_1 = \frac{n_2 h}{2\pi} - \frac{n_1 h}{2\pi} \Rightarrow \Delta L = \frac{h}{2\pi} (n_2 - n_1)$

$= \frac{6.6 \times 10^{-34}}{2 \times 3.14} (5 - 4) = 1.05 \times 10^{-34} \text{ J-S}$

98. (b) $E_n = -\frac{13.6}{n^2} \text{ eV}$

99. (a)

100. (b) $F = \frac{9 \times 10^9 \times 1.6 \times 10^{-19} \times 1.6 \times 10^{-19}}{(2.5 \times 10^{-11})^2} = 3.7 \times 10^{-7} \text{ N}$

101. (a) For Balmer series $\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$ where $n = 3, 4, 5$

For second line $n = 4$

So $\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \frac{3}{16} R \Rightarrow \lambda = \frac{16}{3R}$

102. (c) Energy of electron in H atom $E_n = \frac{-13.6}{n^2} \text{ eV}$

$\Rightarrow -1.5 = \frac{-13.6}{n^2} \Rightarrow n^2 = \frac{13.6}{1.5} = 3$

Now angular momentum

$p = n \frac{h}{2\pi} = \frac{3 \times 6.6 \times 10^{-34}}{2 \times 3.14} = 3.15 \times 10^{-34} \text{ J} \times \text{sec}$

103. (a)

104. (b) $T = \frac{2\pi r}{v}$; $r =$ radius of n orbit $= \frac{n^2 h^2}{\pi m Z e^2}$

$v =$ speed of e^- in n orbit $= \frac{ze^2}{2\epsilon_0 n h}$

$\therefore T = \frac{4\epsilon_0^2 n^3 h^3}{m Z^2 e^4} \Rightarrow T \propto \frac{n^3}{Z^2}$

105. (d) $r_n \propto n^2 \Rightarrow \frac{r_4}{r_1} = \left(\frac{4}{1} \right)^2 = \frac{16}{1} \Rightarrow r_4 = 16 r_1 \Rightarrow r_4 = 16 r_0$

106. (b) For Lyman series

$\bar{\nu} = \frac{1}{\lambda} = R \left(\frac{1}{1^2} - \frac{1}{n^2} \right)$ here $n = 2, 3, 4, 5, \dots$

For first line

$\bar{\nu} = R \left(\frac{1}{1^2} - \frac{1}{2^2} \right) \Rightarrow \bar{\nu} = R \left(1 - \frac{1}{4} \right) = \frac{3R}{4}$

107. (b) Energy $E = K \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$ ($K = \text{constant}$)

$n_1 = 2$ and $n_2 = 3$, so $E = K \left[\frac{1}{2^2} - \frac{1}{3^2} \right] = K \left[\frac{5}{36} \right]$

For removing an electron $n_1 = 1$ to $n_2 = \infty$

Energy $E_1 = K[1] = \frac{36}{5} E = 7.2 E$

$\therefore$ Ionization energy = $7.2 E$

108. (c) For Paschen series $\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{3^2} - \frac{1}{n^2} \right]$; $n = 4, 5, 6, \dots$

For first member of Paschen series $n = 4$

$\frac{1}{\lambda_1} = R \left[\frac{1}{3^2} - \frac{1}{4^2} \right] \Rightarrow \frac{1}{\lambda_1} = \frac{7R}{144}$

$\Rightarrow R = \frac{144}{7\lambda_1} = \frac{144}{7 \times 18800 \times 10^{-10}} = 1.1 \times 10^{-7}$

For shortest wave length $n = \infty$

So $\frac{1}{\lambda} = R \left[\frac{1}{3^2} - \frac{1}{\infty^2} \right] = \frac{R}{9}$

$$\Rightarrow \lambda = \frac{9}{R} = \frac{9}{1.1 \times 10^{-7}} = 8.225 \times 10^{-7} \text{ m} = 8225 \text{ \AA}$$

109. (d) For Lyman series $\frac{1}{\lambda_{\max}} = R \left[\frac{1}{1^2} - \frac{1}{2^2} \right] = \frac{3}{4} R$ and

$$\frac{1}{\lambda_{\min}} = R \left[\frac{1}{1^2} - \frac{1}{\infty^2} \right] = \frac{R}{1} \Rightarrow \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{4}{3}$$

110. (c) $T \propto n^3 \Rightarrow \frac{T_2}{T_1} = \frac{2^3}{1^3} = \frac{8}{1}$

111. (a) For Bracket series $\frac{1}{\lambda_{\max}} = R \left[\frac{1}{4^2} - \frac{1}{5^2} \right] = \frac{9}{25 \times 16} R$

$$\text{and } \frac{1}{\lambda_{\min}} = R \left[\frac{1}{4^2} - \frac{1}{\infty^2} \right] = \frac{R}{16} \Rightarrow \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{25}{9}$$

112. (a) For hydrogen and hydrogen like atoms $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$

$$U_n = 2E_n = -27.2 \frac{Z^2}{n^2} \text{ eV and } K_n = |E_n| = 13.6 \frac{Z^2}{n^2} \text{ eV}$$

From these three relations we can see that as n decreases, K will increase but E and U will decrease.

113. (a) $\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \Rightarrow \frac{\lambda_{\min}}{\lambda_{\max}} = \left[\frac{\frac{1}{2^2} - \frac{1}{3^2}}{\frac{1}{2^2} - \frac{1}{\infty}} \right] = \frac{5}{9}$

114. (b) By using $r_n = r_0 \frac{n^2}{Z}$; Where r_0 = Radius of the Bohr orbit in the ground state atom. So for He^+ third excited state $n = 4, Z = 2, r_0 = 0.5 \text{ \AA} \Rightarrow r_4 = 0.5 \times \frac{4^2}{2} = 4 \text{ \AA}$

115. (d) Speed of electron in n orbit (in CGS) $v_n = \frac{2\pi Ze^2}{nh} (k=1)$

For first orbit $H_2; n=1$ and $Z=1$

$$\text{So } v = \frac{2\pi e^2}{h} \Rightarrow \frac{v}{c} = \frac{2\pi e^2}{hc}$$

116. (b)

117. (d) $E_n = -\frac{13.6}{n^2} \text{ eV} \Rightarrow E_5 = -\frac{13.6}{5^2} = -\frac{13.6}{25} = -0.54 \text{ eV}$

118. (a)

119. (b)

120. (a) $mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi mr} \Rightarrow \frac{v^2}{r} = \frac{n^2 h^2}{4\pi^2 m^2 r^3}$

121. (b) Maximum number of spectral lines are observed in Lyman series.

122. (b) Wave number $\bar{\nu} = \frac{1}{\lambda} = \frac{1}{5896 \times 10^{-8}} = 16961 \text{ per cm}$

123. (b) $r_n \propto n^2 \Rightarrow \frac{r_3}{r_1} = \frac{3^2}{1} \Rightarrow r_3 = 9r_1 = 9 \times 0.53 = 4.77 \text{ \AA}$

124. (d) For first line in Lyman series $\lambda_{L_1} = \frac{4}{3R}$ (i)

For first line in Balmer series $\lambda_{B_1} = \frac{36}{5R}$ (ii)

From equation (i) and (ii)

$$\frac{\lambda_{B_1}}{\lambda_{L_1}} = \frac{27}{5} \Rightarrow \lambda_{B_1} = \frac{27}{5} \lambda_{L_1} \Rightarrow \lambda_{B_1} = \frac{27}{5} \lambda$$

125. (a)

126. (d) $3-1$ transition has higher energy so it has higher frequency $\left(\nu = \frac{E}{h} \right)$

127. (d) α -particles cannot be attracted by the nucleus.

128. (c) By using $\nu = RC \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

$$\Rightarrow \nu = 10^7 \times (3 \times 10^8) \left[\frac{1}{4^2} - \frac{1}{5^2} \right] = 6.75 \times 10^7 \text{ Hz}$$

129. (c) For M shell ($n=3$), orbital quantum number $l = 0, 1, 2$.

130. (d) Number of possible emission lines $= \frac{n(n-1)}{2}$

$$\text{Where } n=4; \text{ Number} = \frac{4(4-1)}{2} = 6.$$

131. (a) Diameter of nucleus is of the order of 10^{-14} m and radius of first Bohr orbit of hydrogen atom $r = 0.53 \times 10^{-10} \text{ m}$.

132. (c) The electron is in the second orbit ($n=2$)

$$\text{Hence } L = \frac{nh}{2\pi} = \frac{2h}{2\pi} = \frac{6.6 \times 10^{-34}}{\pi} = 2.11 \times 10^{-34} \text{ J-sec}$$

133. (c) $\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \Rightarrow \lambda \propto \frac{1}{Z^2}$

$$\lambda_{Li^{++}} : \lambda_{He^{+}} : \lambda_H = 4 : 9 : 36$$

134. (a) Energy radiated $E = 10.2 \text{ eV} = 10.2 \times 1.6 \times 10^{-19} \text{ J}$

$$\Rightarrow E = \frac{hc}{\lambda} \Rightarrow \lambda = 1.215 \times 10^{-7} \text{ m}$$

135. (c) For $n=1, E_1 = -\frac{13.6}{(1)^2} = -13.6 \text{ eV}$

$$\text{and for } n=3, E_3 = -\frac{13.6}{(3)^2} = -1.51 \text{ eV}$$

So required energy

$$= E_3 - E_1 = -1.51 - (-13.6) = 12.09 \text{ eV}$$

136. (a) Similar to Q. 115

137. (c) Since in spectral series of hydrogen atom, Lyman series lies lower Balmer series.

138. (d) $mvr_n = \frac{nh}{2\pi} \Rightarrow pr_n = \frac{nh}{2\pi} \Rightarrow \frac{h}{\lambda} \times r_n = \frac{nh}{2\pi}$

$$\Rightarrow \lambda = \frac{2\pi r_n}{n}, \text{ for first orbit } n=1 \text{ so } \lambda = 2\pi r_1$$

= circumference of first orbit

139. (d) $E_{n_1 \rightarrow n_2} = -13.6 \left[\frac{1}{n_2^2} - \frac{1}{n_1^2} \right]; n_1 = 2 \text{ \& } n_2 = 1$

$$\Rightarrow E_{II} \rightarrow E_I = -13.6 \times \frac{3}{4} = -10.2 \text{ eV}$$

140. (b) $E_n = -\frac{13.6 Z^2}{n^2} \text{ eV} \Rightarrow E_1 = -\frac{13.6 \times (2)^2}{(1)^2} = -54.4 \text{ eV}$

141. (b) $v \propto Z^2 \Rightarrow \frac{v_{H_2}}{v_{He}} = \left(\frac{1}{2}\right)^2 = \frac{1}{4} \Rightarrow v_{He} = 4v_{H_2} = 4V$

142. (a) $\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

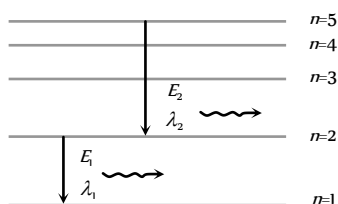
First condition $\frac{1}{\lambda} = R \left[\frac{1}{1^2} - \frac{1}{2^2} \right] \Rightarrow R = \frac{4}{3\lambda}$

Second condition $\frac{1}{\lambda'} = R \left[\frac{1}{1^2} - \frac{1}{3^2} \right]$
 $\Rightarrow \lambda' = \frac{9}{8R} \Rightarrow \lambda' = \frac{9}{8 \times \frac{4}{3\lambda}} = \frac{27\lambda}{32}$

143. (b)

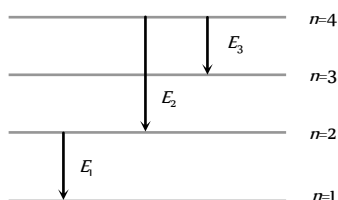
144. (a) $r_n \propto n^2$

145. (d) $\because E_2 < E_1 \Rightarrow \lambda_2 > \lambda_1$



146. (a) Wave number $\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]; n_2 = \infty$ and $n_1 = 1$
 $\Rightarrow \bar{\nu} = R = 1.097 \times 10^7 m^{-1} = 109700 cm^{-1}$

147. (b) $E_1 > E_2 > E_3$

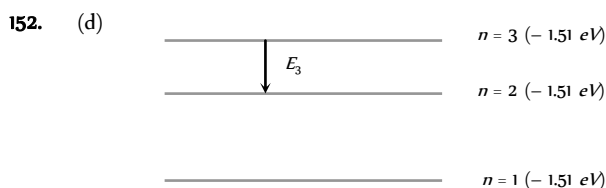


148. (d) $r \propto \frac{1}{Z}$, for double ionized lithium $Z(=3)$ will be maximum.
 So r will be minimum

149. (b) $E_n = \frac{13.6}{n^2} \times Z^2$. For first excited state $n = 2$ and for $Li^{++}, z = 3 \Rightarrow E = \frac{13.6}{4} \times 9 = 30.6 eV$

150. (b)

151. (a) In Lyman series $(\lambda_{min})_L = \frac{1}{R}$ and $(\lambda_{min})_B = \frac{4}{R}$
 $\Rightarrow (\lambda_{min})_B = 4 \times (\lambda_{min})_L = 4 \times 912 = 3648 \text{ \AA}$



$E_{3 \rightarrow 2} = -3.4 - (-1.51) = -1.89 eV \Rightarrow |E_{3 \rightarrow 2}| \approx 1.9 eV$

153. (a)

154. (a)

155. (c) Radius of n orbit for any hydrogen like atom

$r_n = r_0 \left(\frac{n^2}{Z} \right)$ (r_0 = radius of first orbit of H_2 -atom)

If $r_n = r_0 \Rightarrow n = \sqrt{Z}$. For $Be, Z = 4 \Rightarrow n = 2$.

156. (d) $r_n \propto n^4 \Rightarrow A_n \propto n^4 \Rightarrow \frac{A_1}{A_0} = \left(\frac{2}{1}\right)^4 = \frac{16}{1}$

157. (d)

158. (d) $\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$. For first wavelength, $n_1 = 2, n_2 = 3$
 $\Rightarrow \lambda_1 = 6563 \text{ \AA}$. For second wavelength, $n_1 = 2, n_2 = 4$
 $\Rightarrow \lambda_2 = 4861 \text{ \AA}$

159. (c) K.E. = - (Total energy) = - (-13.6 eV) = +13.6 eV

160. (a) In Lyman series $\lambda_{max} = \frac{4}{3R}$

In Balmer series $\lambda_{max} = \frac{36}{5R}$. So required ratio = $\frac{5}{27}$

161. (c)

162. (b) $E = 13.6 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$. For highest energy in Balmer series
 $n_1 = 2$ and $n_2 = \infty \Rightarrow E = 13.6 \left[\frac{1}{(2)^2} - \frac{1}{(\infty)^2} \right] = 3.4 eV$

163. (a)

164. (c) $T \propto n^3$

165. (b) $\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] = R \left[\frac{1}{(2)^2} - \frac{1}{(4)^2} \right] \Rightarrow \lambda = \frac{16}{3R}$

166. (a) $E_n \propto Z^2 \Rightarrow \frac{(E_n)_{He}}{(E_n)_H} = \frac{Z_{He}^2}{Z_H^2} = 4 \Rightarrow (E_n)_{He} = 4 \times (E_n)_H$

167. (b) By using $\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

168. (a) $\frac{1}{\lambda_{max}} = R \left[\frac{1}{(1)^2} - \frac{1}{(2)^2} \right] \Rightarrow \lambda_{max} = \frac{4}{3R} \approx 1213 \text{ \AA}$

and $\frac{1}{\lambda_{min}} = R \left[\frac{1}{(1)^2} - \frac{1}{\infty} \right] \Rightarrow \lambda_{min} = \frac{1}{R} \approx 910 \text{ \AA}$

169. (b) P.E. = $2 \times$ Total energy = $2 \times (-13.6) = -27.2 eV$

170. (c) Emitted energy $\Delta E = \frac{hc}{\lambda} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$.

171. (c) $U = 2E, K = -E$ and $E = -\frac{13.6}{n^2} = eV$.

172. (a)

173. (d)

174. (c) $r \propto n^2$

Nucleus, Nuclear Reaction

1. (b)

2. (c) Neutrons are neutral particles.

3. (c) James Chadwick discovered the neutron.

4. (d) In hydrogen, atomic number and mass number are equal.

5. (d) $E = mc^2 = 1 \times (3 \times 10^8)^2 = 9 \times 10^{16} \approx 10^{17} J$

6. (d) $B.E. = \Delta m \text{ amu} = \Delta m \times 931 \text{ MeV}$.

7. (d) Mass defect $\Delta m = \frac{2.23}{931} = 0.0024$.

8. (b) Positron is the antiparticle of electron.

9. (b)

10. (c)

11. (a) $B.E. = \Delta mc^2 = [2(1.0087 + 1.0073) - 4.0015] = 28.4 \text{ MeV}$

12. (b) $\frac{\text{Binding energy}}{\text{Nucleon}} = \frac{0.0303 \times 931}{4} \approx 7$

13. (d) Energy / day $= 200 \times 10^6 \times 24 \times 3600$
 $= 2 \times 2.4 \times 3.6 \times 10^{12} = 1728 \times 10^{10} J$

14. (c) $E = \Delta mc^2 = 10^{-6} \times (3 \times 10^8)^2 = 9 \times 10^{10} J$

15. (c)

16. (c) Mass of ${}_1H^2 = 2.01478 \text{ a.m.u.}$

Mass of ${}_2He^4 = 4.00388 \text{ a.m.u.}$

Mass of two deuterium $= 2 \times 2.01478 = 4.02956$

Energy equivalent to ${}_2H^2$

$= 4.02956 \times 1.112 \text{ MeV} = 4.48 \text{ MeV}$

Energy equivalent to ${}_2H^4$

$= 4.00388 \times 7.047 \text{ MeV} = 28.21 \text{ MeV}$

Energy released $= 28.21 - 4.48 = 23.73 \text{ MeV} = 24 \text{ MeV}$

17. (c) Energy released while forming a nucleus is known as binding energy (by definition).

18. (c) Nuclear force is stronger than coulomb force.

19. (d)

20. (c)

21. (b) $Q = 4(x - x)$

22. (d)

23. (a) Rest energy of an electron $= m_e c^2$

Here $m_e = 9.1 \times 10^{-31} \text{ kg}$ and $c = \text{velocity of light}$

$\therefore \text{Rest energy} = 9.1 \times 10^{-31} \times (3 \times 10^8)^2 \text{ joule}$

$= \frac{9.1 \times 10^{-31} \times (3 \times 10^8)^2}{1.6 \times 10^{-19}} \text{ eV} = 510 \text{ keV}$

24. (b) ${}_Z X^A = {}_{88}Ra^{226}$

Number of protons $= Z = 88$

Number of neutrons $= A - Z = 226 - 88 = 138$.

25. (c) Out side the Nucleus, neutron is unstable (life $\approx 932 \text{ sec}$).

26. (b) The order of magnitude of mass and volume of uranium nucleus will be

$m \approx A(1.67 \times 10^{-27} \text{ kg})$ (A is atomic number)

$V = \frac{4}{3} \pi r^3 \approx \frac{4}{3} \pi [(1.25 \times 10^{-15} \text{ m}) A^{1/3}]^3$

$\approx (8.2 \times 10^{-45} \text{ m}^3) A$

Hence, $\rho = \frac{m}{V} = \frac{A(1.67 \times 10^{-27} \text{ kg})}{(8.2 \times 10^{-45} \text{ m}^3) A}$

$\approx 2.0 \times 10^{17} \text{ kg/m}^3$.

27. (c) We have $r \propto A^{1/3} \Rightarrow \frac{r_2}{r_1} = \left(\frac{A_2}{A_1}\right)^{1/3} = \left(\frac{206}{4}\right)^{1/3}$

$\therefore r_2 = 3 \left(\frac{206}{4}\right)^{1/3} = 11.6 \text{ Fermi}$.

28. (c) Nucleus does not contains electron.

29. (a) Let the percentage of B^{10} atoms be x , then Average atomic weight

$= \frac{10x + 11(100 - x)}{100} = 10.81 \Rightarrow x = 19 \therefore \frac{N_{B^{10}}}{N_{B^{11}}} = \frac{19}{81}$

30. (a)

31. (b)

32. (a) Nuclear force is charge independent, it also acts between two neutrons.

33. (d) $p \rightarrow \pi^+ + n$, $n \rightarrow p + \pi^-$ and $n \rightarrow n' + \pi^0$

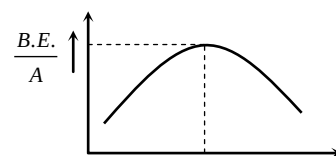
34. (c) Helium nucleus $\rightarrow {}_2He^4$.

Number of protons $= Z = 2$

Number of Neutrons $= A - Z = 2$.

35. (c)

36. (b) Binding energy per nucleon increases with atomic number and is maximum for iron. After that it decrease.



37. (a) For isotopes Z is same and A is different. Therefore the number of neutrons $A - Z$ will also be different.

38. (a) This is due to mass defect because a part of mass is used in keeping the neutrons and protons bound as α -particle.

39. (d) B.E. of $Li^7 = 39.20 \text{ MeV}$ and $He^4 = 28.24 \text{ MeV}$

Hence binding energy of $2He^4 = 56.48 \text{ MeV}$

Energy of reaction $= 56.48 - 39.20 = 17.28 \text{ MeV}$.

40. (c) $r \propto (A)^{1/3}$

41. (b) $r \propto A^{1/3}$

42. (a) $E = mc^2 = (1 \times 10^{-3})(3 \times 10^8)^2 = 9 \times 10^{13} J$.

43. (c) $\Delta E = 8.5 \times 234 - 7.6 \times 236 = 195.4 \text{ MeV} = 200 \text{ MeV}$.

44. (a) $N = M - Z = \text{Total no. of nucleons} - \text{no. of protons}$.

45. (c)

46. (c) Both coulomb and nuclear force act inside the nucleus.

47. (d) For stability in case of lighter nuclei $\frac{N}{Z} = 1$ and for heavier nuclei $\frac{N}{Z} > 1$.

48. (b) Nuclear forces are charge independent.

49. (a) Actual mass of the nucleus is always less than total mass of nucleons so

$$M < (NM_n + Zm_p).$$
50. (b) Mass of H nucleus = mass of proton = 1 amu energy equivalent to 1 amu is 931 MeV so correct option is (b).
51. (d) $R \propto A^{1/3} \Rightarrow R \propto A.$
52. (b)
53. (d) Number of neutrons = $A - Z = 23 - 11 = 12.$
54. (c) For ${}_6C^{12}$, $p = 6$, $e = 6$, $n = 6$
 For ${}_6C^{14}$, $p = 6$, $e = 6$, $n = 8$
55. (a)
56. (a) The mass of nucleus formed is always less than the sum of the masses of the constituent protons and neutrons i.e.

$$m < (A - Z)m_n + Zm_p.$$
57. (d) $E = \Delta mc^2 \Rightarrow E = \frac{0.3}{1000} \times (3 \times 10^8)^2 = 2.7 \times 10^{13} J$

$$= \frac{2.7 \times 10^{13}}{3.6 \times 10^6} = 7.5 \times 10^6 kWh.$$
58. (a, d)
59. (c) ${}_5B^{10} + {}_0n^1 \rightarrow {}_3Li^7 + {}_2He^4.$
60. (a)
61. (c) ${}_{92}U^{235}$ is normally fissionable.
62. (b)
63. (a)
64. (a) In atom bomb nuclear fission takes place with huge temperature.
65. (d) The given equation is ${}_2He^4 + {}_Z X^A \rightarrow {}_{Z+2} Y^{A+3} + A$
 Applying charge and mass conservation

$$4 + A = A + 3 + x \Rightarrow x = 1 \Rightarrow 2 + z = z + 2 + n \Rightarrow n = 0$$

 Hence A is a neutron.
66. (c) Energy of stars is due to the fusion of light hydrogen nuclei into He. In this process much energy is released.
67. (a) ${}_1H^2 + {}_1H^2 \rightarrow {}_2He^4 + 24 MeV.$
68. (b) Energy $\propto c^2$; $\therefore$ Decrease in energy $\propto \frac{4}{9}.$
69. (b) Fusion reaction requires a very high temperature
 $\approx (10^8 K).$
70. (d) ${}_4Be^9 + {}_2He^4 \rightarrow {}_6C^{12} + {}_0n^1.$
71. (b)
72. (d)
73. (b, c)
74. (d)
75. (c) Cadmium rods absorb the neutrons so they are used to control the chain reaction process.
76. (d)
77. (d) No energy and mass enters or goes out of the system of the reaction and no external force is assumed to act.
78. (c)
79. (b)
80. (b) Energy of γ -ray photon = $0.5 + 0.5 + 0.78 = 1.78 MeV.$
81. (c)
82. (d)
83. (c) When fast moving neutrons pass through a moderator, they collide with the molecules of the moderator. As a result of this the neutrons are in thermal equilibrium with the surrounding molecules of moderator. These neutrons are called thermal neutrons.
84. (b) $E_b + E_c > E_d$
85. (d) Because sound waves require medium to travel through and there is no medium (air) on moon's surface.
86. (b) Heavy water is used as moderators in nuclear reactions to slow down the neutrons.
87. (a) $m = \frac{E}{c^2} = \frac{931 \times 1.6 \times 10^{-13}}{(3 \times 10^8)^2} = 1.66 \times 10^{-27} kg.$
88. (d) $E = \Delta mc^2$, $\Delta m = \frac{0.1}{100} = 10^{-3} kg$

$$\therefore E = 10^{-3} \times (3 \times 10^8)^2 = 10^{-3} \times 9 \times 10^{16} = 9 \times 10^{13} J.$$
89. (d) Energy released by γ -rays for pair production must be greater than 1.02 MeV.
90. (a) ${}_8O^{18} + {}_1H^1 \rightarrow {}_9F^{18} + {}_0n^1$
91. (c) Power = 1000 kW = $10^6 J/s$

$$\text{Rate of nuclear fission} = \frac{10^6}{200 \times 1.6 \times 10^{-13}} = 3.125 \times 10^9.$$
92. (b) $A = 238 - 4 = 234$ and $Z = 92 - 2 = 90.$
93. (a) $P = n \left(\frac{E}{t} \right) \Rightarrow 1000 = \frac{n \times 200 \times 10^6 \times 1.6 \times 10^{-19}}{t}$

$$\Rightarrow \frac{n}{t} = 3.125 \times 10^{13}.$$
94. (c) Due to the production of neutrons, a chain of nuclear fission is established which continues until the whole of the source substance is consumed.
95. (a) ${}_{92}U^{235} + {}_0n^1 \rightarrow {}_{38}Sr^{90} + {}_{54}Xe^{143} + 3{}_0n^1$
96. (b) ${}_1H^2 + {}_1H^2 \rightarrow {}_2He^4 + Q.$
97. (c) Fast neutrons can escape from the reaction. So as to proceed the chain reaction. Slow neutrons are best.
98. (d) ${}_1H^2 + {}_1H^2 \rightarrow {}_1H^3 + {}_1H^1$
99. (d)
100. (b) $\Delta m = 1 - 0.993 = 0.007 gm$

$$\therefore E = (\Delta m)c^2 = (0.007 \times 10^{-3})(3 \times 10^8)^2 = 63 \times 10^{10} J.$$
101. (c) ${}_{85}X^{297} \rightarrow {}_{77}Y^{281} + 4({}_2He^4)$
102. (b) $x+1 = 24 + 4 \Rightarrow x = 27.$
103. (a)
104. (d) ${}_6C^{11} \rightarrow {}_5B^{11} + \beta^+ + \gamma$ because $\beta^+ = {}_1e^0$
105. (c)
106. (b) $\frac{\text{Energy}}{\text{Fission}} = 200 MeV = 200 \times 10^6 \times 1.6 \times 10^{-19} J$

$$\text{Fission rate} = \frac{5}{200 \text{ MeV}} = 1.56 \times 10^{11} \text{ fission/sec.}$$

107. (a)
108. (b) Energy is released in the sun due to fusion.
109. (d)
110. (c) In nuclear fission, neutrons are released.
111. (c) ${}_1H^1 + {}_1H^1 + {}_1H^2 \rightarrow {}_2He^4 + {}_{+1}e^0 + \text{energy.}$
112. (c)
113. (b) ${}_0n^1 = {}_1p^1 + {}_{-1}e^0 + \bar{\nu}$
Antineutrino is required for conservation of spin.
114. (b)
115. (d) Fusion is the main process of energy production in the sun.
116. (a)
117. (b) Mass of proton = mass of antiproton
 $= 1.67 \times 10^{-27} \text{ kg} = 1 \text{ amu}$
Energy equivalent to $1 \text{ amu} = 931 \text{ MeV}$
So energy equivalent to $2 \text{ amu} = 2 \times 931 \text{ MeV}$
 $= 1862 \times 10^6 \times 1.6 \times 10^{-19} = 2.97 \times 10^{-10} \text{ J} = 3 \times 10^{-10} \text{ J}.$
118. (a) In fusion reaction, two lighter nuclei combines.
119. (c)
120. (b) Hydrogen bomb is based on nuclear fusion.
121. (d) ${}_{92}U^{235} + {}_0n^1 \rightarrow {}_{92}U^{236}$ and
 ${}_{92}U^{236} \rightarrow {}_{56}Ba^{144} + {}_{36}Kr^{89} + 3{}_0n^1 + Q.$
122. (d) Fusion reaction of deuterium is
 ${}_1H^2 + {}_1H^2 \rightarrow {}_2He^3 + {}_0n^1 + 3.27 \text{ MeV}$
So $E = \frac{6.02 \times 10^{23} \times 10^3 \times 3.27 \times 1.6 \times 10^{-13}}{2 \times 2} = 7.8 \times 10^{13} \text{ J}$
 $= 8 \times 10^{13} \text{ J}.$
123. (a)
124. (a)
125. (c)
126. (c)
127. (b)
128. (d) Energy released in the fission of one nucleus = 200 MeV
 $= 200 \times 10^6 \times 1.6 \times 10^{-19} \text{ J} = 3.2 \times 10^{-11} \text{ J}$
 $P = 16 \text{ KW} = 16 \times 10^3 \text{ Watt}$
Now, number of nuclei required per second
 $n = \frac{P}{E} = \frac{16 \times 10^3}{3.2 \times 10^{-11}} = 5 \times 10^{14}.$
129. (c)
130. (a) Number of fissions per second
 $= \frac{\text{Power output}}{\text{Energy released per fission}}$
 $= \frac{3.2 \times 10^6}{200 \times 10^6 \times 1.6 \times 10^{-19}} = 1 \times 10^{17}$

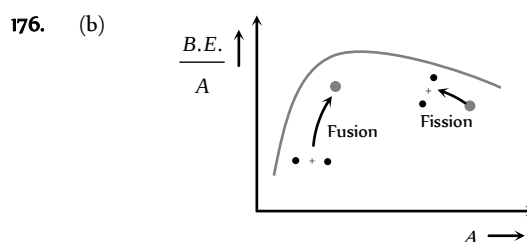
$$\Rightarrow \text{Number of fission per minute} = 60 \times 10^{17} = 6 \times 10^{18}$$

131. (c)
132. (a) $X(n, \alpha) {}_3Li^7 \Rightarrow {}_Z X^A + {}_0n^1 \rightarrow 3Li^7 + {}_2He^4$
 $Z = 3 + 2 = 5$ and $A = 7 + 4 - 1 = 10$
 $\therefore {}_5X^{10} = {}_5B^{10}$
133. (a)
134. (b) Mass of electron = mass of positron = $9.1 \times 10^{-31} \text{ kg}$
Energy released $E = (2m)c^2$
 $= 2 \times 9.1 \times 10^{-31} \times (3 \times 10^8)^2 = 1.6 \times 10^{-13} \text{ J}.$
135. (b) ${}_1H^2 + {}_1H^2 \rightarrow {}_2He^4 + \text{energy}$
Binding energy of a $({}_1H^2)$ deuterium nuclei
 $= 2 \times 1.1 = 2.2 \text{ MeV}$
Total binding energy of two deuterium nuclei
 $= 2.2 \times 2 = 4.4 \text{ MeV}$
Binding energy of a $({}_2He^4)$ nuclei = $4 \times 7 = 28 \text{ MeV}$
So, energy released in fusion = $28 - 4.4 = 23.6 \text{ MeV}$
136. (c) Mass of a uranium nucleus
 $= 92 \times 1.6725 \times 10^{-27} + 143 \times 1.6747 \times 10^{-27}$
 $= 393.35 \times 10^{-27} \text{ kg}$
Number of nuclei in the given mass
 $= \frac{1}{393.35 \times 10^{-27}} = 2.542 \times 10^{24}$
Energy released = $200 \times 2.542 \times 10^{24} \text{ MeV}$
 $= 5.08 \times 10^{26} \text{ MeV} = 8.135 \times 10^{13} \text{ J} = 8.2 \times 10^{13} \text{ J}$
137. (a)
138. (c)
139. (b) In a material medium, when a positron meets an electron both the particles annihilate leading to the emission of two γ ray photons. This process forms the basis of an important diagnostic procedure called PET.
140. (a) Total mass of reactants
 $= (2.0141) \times 2 = 4.0282 \text{ amu}$
Total mass of products = 4.0024 amu
Mass defect = $4.0282 \text{ amu} - 4.0024 \text{ amu}$
 $= 0.0258 \text{ amu}$
 $\therefore$ Energy released $E = 931 \times 0.0258 = 24 \text{ MeV}$
141. (b) ${}_7N^{14} + {}_2He^4 \rightarrow {}_8O^{17} + {}_1H^1$
142. (b)
143. (b)
144. (d) ${}_8O^{16} + {}_1H^2 \rightarrow {}_7N^{14} + {}_2He^4$

145. (d)
146. (a)
147. (b) Nuclear fusion takes place in stars which results in joining of nuclei accompanied by release of tremendous amount of energy.
148. (b)
149. (d) $B.E.$ per nucleon $\propto$ stability.
150. (a) Nuclei of different elements having the same mass number are called isotones *e.g.*, ${}_4\text{Be}^9$ and ${}_5\text{B}^{10}$
151. (c)
152. (b)
153. (d) Packing fraction $= \frac{M - A}{A}$
154. (c) $B = [ZM_p + NM_n - M(N, Z)]c^2$
 $\Rightarrow M(M, Z) = ZM_p + NM_n - B/c^2$
155. (c)
156. (a) In nuclear reactor, nuclear fission can be carried out through a sustained and a controlled chain reaction.
157. (b) ${}_6\text{C}^{12} + {}_0n^1 \rightarrow {}_7\text{N}^{13} + {}_{-1}\beta^0$
158. (a) ${}_0n^1 + {}_{92}\text{U}^{235} \rightarrow {}_{56}\text{Ba}^{144} + {}_{36}\text{Kr}^{89} + 3{}_0n^1$
159. (c) The energy released in sun and hydrogen bomb are due to nuclear fusion.
160. (c)
161. (a)
162. (a) ${}_{84}\text{Po}^{210} \rightarrow {}_{82}\text{X}^{206} + {}_2\text{He}^4$
 Using conservation of linear moments
 $206v' + 4v = 0 \Rightarrow v' = -\frac{4v}{206} \Rightarrow |v'| = \frac{4v}{206}$
163. (b) Power $P = \frac{\text{Energy}}{\text{time}} = \frac{mc^2}{t}$, $= 1 \times 10 \times (3 \times 10)^{\cdot}$
 $= 9 \times 10 \cdot W = 9 \times 10 \text{ kW}$.
164. (b)
165. (c)
166. (b) The elements high on the $B.E.$ versus mass number plot are very tightly bound and hence, are stable. And the elements those are lower on this plot, are less tightly bound and hence, are unstable.
 Since helium nucleus shows a peak on this plot so, it is very stable.
167. (d)
168. (d) $E = \Delta mc^2 = 1 \times (3 \times 10^8)^2 = 9 \times 10^{10} \text{ J}$
 $\Rightarrow E = \frac{9 \times 10^{16}}{1.6 \times 10^{-19}} = 5.625 \times 10^{35} \text{ eV} = 5.625 \times 10^{29} \text{ MeV}$.
169. (a) $E = \Delta mc^2 = 0.5 \times 10^{-3} \times (3 \times 10^8)^2 = 4.5 \times 10^{13} \text{ J}$

$$\Rightarrow E = \frac{4.5 \times 10^{13}}{3.6 \times 10^6} = 1.25 \times 10^7 \text{ kWh.}$$

170. (b) ${}_0n^1 + {}_{92}\text{U}^{235} \rightarrow {}_{36}\text{Kr}^{94} + {}_{56}\text{Ba}^{139} + 3{}_0n^1$
171. (b)
172. (a) Number of protons $= 2 + 2 + 6 + 2 + 6 = 18$
 Number of neutrons $= 40 - 18 = 22$.
173. (c) Neutrons are unstable and having mean life time of 32 sec, decaying by emitting an electron and antineutrino to become proton.
174. (b) During fusion binding energy of daughter nucleus is always greater than the total energy of the parent nuclei so energy released $= c - (a + b) = c - a - b$
175. (a) These nuclei having different Z and A but equal $(A - Z)$ are called isotones.



177. (a)
178. (c) $r \propto A^{1/3} \Rightarrow \frac{r_1}{r_2} = \left(\frac{A_1}{A_2}\right)^{1/3}$
 $\Rightarrow \frac{3.6}{r_2} = \left(\frac{27}{125}\right)^{1/3} = \frac{3}{5} \Rightarrow r_2 = 6 \text{ Fermi}$
179. (b)

Radioactivity

1. (a)
2. (a) By formula $N = N_0 \left(\frac{1}{2}\right)^{t/T}$ or $10^4 = 8 \times 10^4 \left(\frac{1}{2}\right)^{t/3}$
 or $\left(\frac{1}{8}\right) = \left(\frac{1}{2}\right)^{t/3}$ or $\left(\frac{1}{2}\right)^3 = \left(\frac{1}{2}\right)^{t/3} \Rightarrow 3 = \frac{t}{3}$
 Hence $t = 9 \text{ years}$.
3. (d) Fraction $= \frac{N}{N_0} = \left(\frac{1}{2}\right)^{\frac{6400}{1600}} = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$
4. (c) Negative β -decay is expressed by the equation
 $n = p^+ + e^- + \bar{\nu}$
5. (a) No radioactive substance emit both α and β particles simultaneously. Some substances emit α -particles and some other emits β -particles. γ -rays are emitted along with both α and β -particles.
6. (c) γ -rays are highly penetrating.

7. (c) Average life $\frac{1}{\lambda} = \frac{1600}{0.693} = 2308 \approx 2319$ years.
8. (d) Fraction of atoms remains after five half lives

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/T} = \left(\frac{1}{2}\right)^{5T/T} = \frac{1}{32}$$

$$\Rightarrow \text{Percentage atom remains} = \frac{1}{32} \times 100 = 3.125\%$$
9. (c) β -rays emitted from nucleus and they carry negative charge.
10. (c)
11. (b) ${}_Z X^A \xrightarrow{-1\beta^0} {}_{Z+1} Y^A \xrightarrow{2\text{He}^4(\alpha)} {}_{Z-1} K^{A-4} \xrightarrow{-0\gamma^0} {}_{Z-1} K^{A-4}$
12. (c) $N_t = N_0 \left(\frac{1}{2}\right)^{t/T} = 50000 \left(\frac{1}{2}\right)^{10/5} = 12500$
13. (c) ${}_Z X^A \xrightarrow{-1\beta^0} {}_{Z+1} X^A$
14. (c) $N = N_0 \left(\frac{1}{2}\right)^{t/T} \Rightarrow \frac{N_0}{64} = N_0 \left(\frac{1}{2}\right)^{30/T} \Rightarrow T = \frac{30}{6} = 5$ sec
15. (a)
16. (c)
17. (a) Average life $T = \frac{\text{Sum of all lives of all the atoms}}{\text{Total number of atoms}} = \frac{1}{\lambda}$
 $\Rightarrow T\lambda = 1$
18. (c) Fraction remains after n half lives $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^{t/T}$

$$\therefore \frac{N}{N_0} = \left(\frac{1}{2}\right)^{T/2} = \left(\frac{1}{2}\right)^{1/2} = \frac{1}{\sqrt{2}}$$
19. (b)
20. (a) Penetration power of γ is 100 times of β , while that of β is 100 times of α .
21. (a) $\frac{N_0}{32} = N_0 \left(\frac{1}{2}\right)^{60/T} \Rightarrow 5 = \frac{60}{T} \Rightarrow T = 12$ days
22. (c) By using $N = N_0 \left(\frac{1}{2}\right)^{t/T}$; where $N = \left(1 - \frac{7}{8}\right) N_0 = \frac{1}{8} N_0$

$$\text{So } \frac{1}{8} N_0 = N_0 \left(\frac{1}{2}\right)^{t/T} \Rightarrow \left(\frac{1}{2}\right)^3 = \left(\frac{1}{2}\right)^{t/5} \Rightarrow t = 15 \text{ days.}$$
23. (a) ${}_{72} A^{180} \xrightarrow{\alpha} {}_{70} A_1^{176} \xrightarrow{\beta} {}_{71} A_2^{176} \xrightarrow{\alpha} {}_{69} A_3^{172} \xrightarrow{\gamma} {}_{69} A_4^{172}$
24. (d)
25. (d) Half life of a substance doesn't depends upon Amount, temperature and pressure. It depends upon the nature of the substance.
26. (d) $T = \frac{0.6931 \times 1}{\lambda} = \frac{0.6931 \times 1}{4.28 \times 10^{-4}} \text{ year} = 1620 \text{ years}$
27. (c) In fusion two lighter nuclei combines, it is not the radioactive decay.
28. (b) $n_\alpha = \frac{A - A'}{4} = \frac{232 - 208}{4} = 6$
 and $n_\beta = (2n_\alpha - Z + Z') = (2 \times 6 - 90 + 82) = 4$
29. (a) Remaining amount

$$= 16 \times \left(\frac{1}{2}\right)^{32/2} = 16 \times \left(\frac{1}{2}\right)^{16} = \left(\frac{1}{2}\right)^{12} < 1 \text{ mg}$$
30. (c) $\frac{N}{N_0} = \left(\frac{1}{2}\right)^{15/5} = \frac{1}{8} \Rightarrow \text{Decayed fraction} = 1 - \frac{1}{8} = \frac{7}{8}$
31. (c)
32. (c) By using $n_\alpha = \frac{A - A'}{4}$ and $n_\beta = 2n_\alpha - Z + Z'$

$$\Rightarrow A' = A - 4n_\alpha = 236 - 4 \times 3 = 224$$
 and $Z' = (n_\beta - 2n_\alpha + Z) = (1 - 2 \times 3 + 88) = 83$
33. (d) Uncertain, because it is infinite. No radioactive element can be disintegrated fully.
34. (c) $\frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/140} \Rightarrow \frac{1}{16} = \left(\frac{1}{2}\right)^{t/140}$

$$\Rightarrow \frac{t}{140} = 4 \Rightarrow t = 560 \text{ days}$$
35. (c) $\frac{C_{14}}{C_{12}} = \frac{1}{4} = \left(\frac{1}{2}\right)^{t/5700} \Rightarrow \frac{t}{5700} = 2 \Rightarrow t = 11400 \text{ years}$
36. (b) Ionising property depends upon the charge and mass.
37. (b) $R = \frac{dN}{dt} \propto N \Rightarrow \frac{R_2}{R_1} = \frac{N_2}{N_1}$
 But $\frac{N_2}{N_1} = \left(\frac{1}{2}\right)^{t_{1/2}} \Rightarrow \frac{25}{200} = \frac{1}{8} = \left(\frac{1}{2}\right)^3 \Rightarrow \frac{t}{t_{1/2}} = 3$

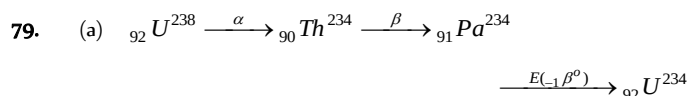
$$\therefore t_{1/2} = \frac{t}{3} = \frac{3}{3} = 1 \text{ hour} = 60 \text{ minutes}$$
38. (d) $t_{1/2} = \frac{0.6931}{0.01} = 69.31 \text{ seconds.}$
39. (d) Because radioactivity is a spontaneous phenomenon.
40. (d) Undecayed isotope $= 1 - \frac{7}{8} = \frac{1}{8}$

$$\therefore \frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/T} \Rightarrow \left(\frac{1}{8}\right) = \left(\frac{1}{2}\right)^{t/15} \Rightarrow \frac{t}{15} = 3$$
 or $t = 45 \text{ hours}$
41. (a) Mean life $= \frac{\text{Half life}}{0.6931} = \frac{10}{0.6931} = 14.4 \text{ hours}$

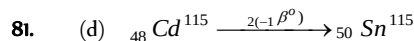
42. (b) 20 gm substance reduces to 10 gm (i.e. becomes half in 4 min.
So $T_{1/2} = 4 \text{ min}$. Again $M = M_0 \left(\frac{1}{2}\right)^{t/T_{1/2}}$
 $\Rightarrow 10 = 20 \left(\frac{1}{2}\right)^{t/4} \Rightarrow \frac{1}{2} = \left(\frac{1}{2}\right)^{t/4} \Rightarrow t = 4 \text{ min}$
43. (c) $N = N_0 \left(\frac{1}{2}\right)^{t/T_{1/2}} \Rightarrow 1 = 16 \left(\frac{1}{2}\right)^{t/T_{1/2}} \Rightarrow T_{1/2} = \frac{1}{2} \text{ hour}$
44. (d)
45. (b) β -decay from nuclei based on this process only.
46. (b) $A = A_0 \left(\frac{1}{2}\right)^{t/T_{1/2}} \Rightarrow 5 = A_0 \left(\frac{1}{2}\right)^{30/T_{1/2}} = \frac{A_0}{16} \Rightarrow A_0 = 80 \text{ sec}^{-1}$
47. (d) $n_\alpha = \frac{A - A'}{4} = \frac{200 - 168}{4} = 8$
 $n_\beta = 2n_\alpha - Z + Z' = 2 \times 8 - 90 + 80 = 6$
48. (d) Similar to Q. 47
49. (b) $N = N_0 \left(\frac{1}{2}\right)^{t/T_{1/2}}$. Hence fraction of atoms decayed
 $= 1 - \frac{N}{N_0} = 1 - \left(\frac{1}{2}\right)^{t/T_{1/2}} = 1 - \left(\frac{1}{2}\right)^{3 \times 60/T_{1/2}} = \frac{7}{8}$
In percentage it is $\frac{7}{8} \times 100 = 87.5\%$
50. (a) C-14 is carbon dating substance.
51. (b) $\frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/T} \Rightarrow \left(\frac{1}{16}\right) = \left(\frac{1}{2}\right)^{t/T} \Rightarrow \left(\frac{1}{2}\right)^4 = \left(\frac{1}{2}\right)^{t/T} \Rightarrow T = 0.5 \text{ hour} = 30 \text{ minutes}$
52. (c) $\frac{dN}{dt} = -\lambda N \Rightarrow n = -\lambda N$ (Given $\frac{dN}{dt} = n$)
 $\therefore \lambda = -\frac{n}{N} \therefore \text{Half life} = \frac{0.693}{\lambda} = \frac{0.693}{n} = \frac{0.693}{N} \text{ sec}$
53. (a) ${}_{92}\text{X}^{235} \xrightarrow{\alpha} {}_{90}\text{X}^{231} \xrightarrow{-1e^0} {}_{91}\text{Y}^{231}$
54. (d) ${}_{7}\text{N}^{13} \rightarrow {}_{6}\text{C}^{13} + {}_{+1}\text{e}^0$
55. (c) $A = A_0 \left(\frac{1}{2}\right)^{t/T_{1/2}} \Rightarrow 100 = 1600 \left(\frac{1}{2}\right)^{t/T_{1/2}} \Rightarrow T_{1/2} = 2 \text{ sec}$
Again at $t = 6 \text{ sec}$, $A = 1600 \left(\frac{1}{2}\right)^{6/2} = 200 \text{ counts/sec}$
56. (d) ${}_{92}\text{U}^{238} \rightarrow {}_{92}\text{Th}^{234} + {}_2\text{He}^4$
57. (b)
58. (d)
59. (d)
60. (b) By using $N = N_0 e^{-\lambda t} \Rightarrow \frac{N_0}{2} = N_0 e^{-\lambda T_{1/2}} \Rightarrow 2 = e^{\lambda T_{1/2}}$
By taking log both the side
 $\log_e 2 = \lambda T_{1/2} \Rightarrow \lambda T_{1/2} = 0.693$
61. (a) Number of half lives in 20 min = $n = \frac{20}{5} = 4$
Fraction of material remains after four half lives = $\frac{1}{16}$
Hence fraction that decays = $1 - \frac{1}{16} = \frac{15}{16} = 93.75\%$
62. (d) In the given case, 12 days = 3 half lives Number of atoms left after 3 half lives.
 $= 6.4 \times 10^{10} \times \frac{1}{2^3} = 0.8 \times 10^{10}$
63. (a) Decay constant remains unchanged in a chemical reaction.
64. (d) $n_\alpha = \frac{A - A'}{4} = \frac{238 - 206}{4} = 8$
65. (b) ${}_Z\text{X}^A \xrightarrow{\alpha} {}_{Z-2}\text{Y}^{A-4} \xrightarrow{2\beta} {}_Z\text{X}^{A-4}$
66. (a) Both the β -rays and the cathode rays are made up of electrons.
 γ -rays are EM waves, α -particles are doubly ionized helium atoms and protons and neutrons have approximately the same mass.
67. (b) ${}_{10}^{22}\text{Ne} \rightarrow {}_2^4\text{He} + {}_8^{18}\text{O}$; hence X is carbon.
68. (c) For 80 minutes, number of half lives of sample
 $A = n_A = \frac{80}{20} = 4$ and number of half lives of sample
 $B = n_B = \frac{80}{40} = 2$. Also by using $N = N_0 \left(\frac{1}{2}\right)^n$
 $\Rightarrow N \propto \frac{1}{2^n} \Rightarrow \frac{N_A}{N_B} = \frac{2^{n_B}}{2^{n_A}} = \frac{2^2}{2^4} = \frac{1}{4}$
69. (d) ${}_n\text{X}^m \xrightarrow{\alpha} {}_{n-2}\text{X}^{m-4} \xrightarrow{-\beta} {}_{n-1}\text{X}^{m-4}$
70. (c) Half-life $T_{1/2} = \frac{0.693}{\lambda} = \frac{0.693}{1.07 \times 10^{-4}} = 6476 \text{ years}$
71. (d) Number of nuclei decreases exponentially
 $N = N_0 e^{-\lambda t}$ and Rate of decay $\left(-\frac{dN}{dt}\right) = \lambda N$
Therefore, decay process lasts up to $t = \infty$.
Therefore, a given nucleus may decay at any time after $t = 0$.
72. (a) To becomes $\frac{1}{4}$ th, it requires time of two half lives
i.e., $t = 2(T_{1/2}) = 2 \times 5800 = 2 \times 58 \text{ centuries}$
73. (a) Carbon dating
74. (d) ${}_7\text{X}^{15} + {}_2\text{He}^4 \rightarrow {}_1\text{P}^1 + {}_8\text{Y}^{18}$
75. (c)
76. (d) ${}_{92}\text{U}^{238} \xrightarrow{\alpha} {}_{90}\text{X}^{234} \xrightarrow{\beta^-} {}_{91}\text{Y}^{234}$
77. (d)

78. (c) After three half lives (i.e., 30 days) it remains $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$, so it

will remain $\frac{1}{10}$ th, approximately in 33 days.



80. (d) By using $A = A_0 \left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}} \Rightarrow \frac{A}{A_0} = \left(\frac{1}{2}\right)^{9/3} = \frac{1}{8}$



82. (b) In two half lives, the activity becomes one fourth.

83. (a) α decay decreases the mass number by 4 and atomic number by 2, β decay increases the atomic number by 1. Here atomic number of C is same as that of A.

84. (a) Number of half lives in two days four substance 1 and 2 respectively are $n_1 = \frac{2 \times 24}{12} = 4$ and $n_2 = \frac{2 \times 24}{1.6} = 3$

$$\text{By using } N = N_0 \left(\frac{1}{2}\right)^n \Rightarrow \frac{N_1}{N_2} = \frac{(N_0)_1}{(N_0)_2} \times \frac{\left(\frac{1}{2}\right)^{n_1}}{\left(\frac{1}{2}\right)^{n_2}}$$

$$= \frac{2}{1} \times \frac{\left(\frac{1}{2}\right)^4}{\left(\frac{1}{2}\right)^3} = \frac{1}{1}$$

85. (b)

86. (c) $\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{2.3} = 0.3$

87. (d) Number of α - particles emitted $= \frac{238 - 222}{4} = 4$

This decreases atomic number to $90 - 4 \times 2 = 82$

Since atomic number of ${}_{83}\text{Y}^{222}$ is 83, this is possible if one β - particle is emitted.

88. (d) Number of half lives in 150 years $n = \frac{150}{75} = 2$

$$\text{Fraction of the atom of decayed} = 1 - \left(\frac{1}{2}\right)^n$$

$$= 1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4} = 0.75 \Rightarrow \text{Percentage decay} = 75\%$$

89. (b) $A = A_0 e^{-\lambda t} \Rightarrow 975 = 9750 e^{-\lambda \times 5} \Rightarrow e^{5\lambda} = 10$
 $\Rightarrow 5\lambda = \log_e 10 = 2.3026 \log_{10} 10 = 2.3026$
 $\Rightarrow \lambda = 0.461$

90. (a) Mass number decreases by $8 \times 4 = 32$

Atomic number decreases by $8 \times 2 - 5 = 11$

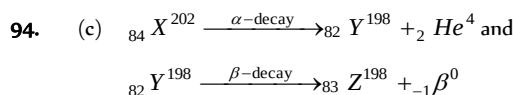
91. (b)

92. (d) $A = A_0 \left(\frac{1}{2}\right)^{t/T_{1/2}} \Rightarrow 5 \times 10^{-6} = 64 \times 10^{-5} \left(\frac{1}{2}\right)^{t/3}$
 $\Rightarrow \frac{1}{128} = \left(\frac{1}{2}\right)^{t/3} \Rightarrow t = 21 \text{ days}$

93. (c) Decayed fraction $= \frac{3}{4}$, so undecayed fraction $= \frac{1}{4}$

$$\text{Now } \frac{N}{N_0} = \left(\frac{1}{2}\right)^n \Rightarrow \frac{1}{4} = \left(\frac{1}{2}\right)^n \Rightarrow n = 2$$

$$\Rightarrow t = n \times T_{1/2} = 2 \times 3.8 = 7.6 \text{ days}$$



95. (a) $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n \Rightarrow \frac{1}{8} = \left(\frac{1}{2}\right)^n \Rightarrow n = 3$

$$\text{Now } t = n \times T_{1/2} = 3 \times 3.8 = 11.4 \text{ days}$$

96. (d) $n = \frac{72000}{24000} = 3$; Now $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n = \frac{1}{8}$

97. (d)

98. (a)

99. (b) $n_\alpha = \frac{A - A'}{4} = \frac{232 - 208}{4} = 6$

$$n_\beta = 2n_\alpha - Z + Z' = 2 \times 6 - 90 + 82 = 4$$

100. (c)

101. (a) Number of half lives $n = \frac{5}{1} = 5$

$$\text{Now } \frac{N}{N_0} = \left(\frac{1}{2}\right)^n \Rightarrow \frac{N}{N_0} = \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

102. (d)

103. (b) Number of half lives $n = \frac{10}{5} = 2$, now $\frac{N}{N_0} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

$$\text{Fraction decayed} = 1 - \frac{N}{N_0} = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\Rightarrow \text{In percentage} = \frac{3}{4} \times 100 = 75\%$$

104. (b) Number of half lives $n = \frac{19}{3.8} = 5$; Now $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n$
 $\Rightarrow \frac{N}{10.38} = \left(\frac{1}{2}\right)^5 \Rightarrow N = 10.38 \times \left(\frac{1}{2}\right)^5 = 0.32 \text{ gm}$

105. (b) $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^5$

106. (d) $T_{1/2} = \frac{\log_e 2}{\lambda} = \frac{2.303 \log_{10} 2}{\lambda}$

107. (a) ${}^A_Z X \rightarrow {}^{A-4}_{Z-2} Y + {}^4_2 \text{He} \rightarrow {}^{A-4}_{Z-3} Z + {}^0_1 B$
108. (d) $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n, n = 2 \Rightarrow \frac{N}{N_0} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$
So disintegrated part $= 1 - \frac{N}{N_0} = 1 - \frac{1}{4} = \frac{3}{4}$
109. (c) Number of half lives $n = \frac{10}{2.5} = 4$
 $\Rightarrow \frac{A}{A_0} = \frac{N}{N_0} = \left(\frac{1}{2}\right)^n \Rightarrow A = 1.6 \times \left(\frac{1}{2}\right)^4 = 0.1 \text{ curie}$
110. (b) By using $N = N_0 e^{-\lambda t}$ and $t = \tau = \frac{1}{\lambda}$
Substance remains $= N = \frac{N_0}{e} = 0.37 N_0 \simeq \frac{N_0}{3}$
 $\therefore$ Substance disintegrated $= N_0 - \frac{N_0}{3} = \frac{2N_0}{3}$
111. (c) $\frac{3}{4}$ th active decay takes place in time
 $t = 2(T_{1/2}) \Rightarrow \frac{3}{4} = 2(T_{1/2}) \Rightarrow T_{1/2} = \frac{3}{8} \text{ sec}$
112. (c) By using $N = N_0 e^{-\lambda t}$ and average life time $t = \frac{1}{\lambda}$
So $N = N_0 e^{-\lambda \times 1/\lambda} = N_0 e^{-1} \Rightarrow \frac{N}{N_0} = e^{-1} = \frac{1}{e}$
Now disintegrated fraction $= 1 - \frac{N}{N_0} = 1 - \frac{1}{e} = \frac{e-1}{e}$
113. (d) Complete reaction will be as follows
 ${}_{92}X^{235} \rightarrow {}_{82}Y^{207} + 7 {}^4_2 \text{He} + 4 {}^0_{-1}e$
i.e., seven α - particles and four β - particles will be emitted.
114. (d) $n_\alpha = \frac{A - A'}{4} = \frac{235 - 207}{4} = 7$
 $n_\beta = (2n_\alpha - Z + Z') = (2 \times 7 - 92 + 82) = 4$
115. (c) During β -decay, a neutron is transformed into a proton and an electron.
116. (b) ${}_Z X^A \xrightarrow{\alpha} {}_{Z-2} X^{A-4}$
117. (a)
118. (a)
119. (d) After emitting β -particle (e) mass of nucleus doesn't change.
120. (a) $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n \Rightarrow \frac{1}{16} = \left(\frac{1}{2}\right)^4 = \left(\frac{1}{2}\right)^n \Rightarrow n = 4$
Also $n = \frac{t}{T_{1/2}} \Rightarrow T_{1/2} = \frac{40}{4} = 10 \text{ days}$
121. (c) As the γ - particle has no charge and mass.
122. (d) With emission of an α particle (${}_2\text{He}^4$) mass number decreases by 4 unit and atomic number decrease by 2 units and with emission of $2\beta^{-1}$ particle atomic number increases by 2 units. So Z will remain same and N will become $N - 4$.
123. (a) $N = N_0 \left(\frac{1}{2}\right)^n \Rightarrow \frac{N}{N_0} = \left(\frac{1}{2}\right)^n \Rightarrow \frac{1}{100} = \left(\frac{1}{2}\right)^n \Rightarrow 2^n = 100$
 n comes out in between 6 and 7.
124. (d) $N = N_0 \left(\frac{1}{2}\right)^n \Rightarrow N = N_0 \left(\frac{1}{2}\right)^{t/T_{1/2}}$
 $\Rightarrow N = 1 \times \left(\frac{1}{2}\right)^{\frac{8.1}{2.7}} = \left(\frac{1}{2}\right)^3 = \frac{1}{8} \Rightarrow N = \frac{1}{8} \text{ mg} = 0.125 \text{ mg}$
125. (d) $N = N_0 \left(\frac{1}{2}\right)^{\frac{t}{T}} \Rightarrow \frac{N_0}{4} = N_0 \left(\frac{1}{2}\right)^{\frac{16}{T}} \Rightarrow T = 8 \text{ days}$
126. (d) ${}_{92}\text{U}^{238} \rightarrow {}_2\text{He}^4 + {}_{90}\text{X}^{234} \rightarrow {}_{-1}e^0 + {}_{91}\text{U}^{234}$
Hence, $A = 234, Z = 91$
127. (c) Mean life $= \frac{1}{\lambda} = 6.67 \times 10^8 \text{ sec}$.
128. (d) $\frac{dN}{dt} = -\lambda N \Rightarrow \left|\frac{dN}{dt}\right| = \frac{0.693}{T_{1/2}} \times N$
 $= \frac{0.693}{1.2 \times 10^7} \times 4 \times 10^{15} = 2.3 \times 10^8 \text{ atoms/sec}$
129. (c) Remaining material $N = \frac{N_0}{2^{t/T}}$
 $\Rightarrow N = \frac{10}{(2)^{20/15}} = \frac{10}{2.15} = 3.96 \text{ gm}$
So decayed material $= 10 - 3.96 = 6.04 \text{ gm}$
130. (b) Number of atoms remains undecayed $N = N_0 e^{-\lambda t}$
Number of atoms decayed $= N_0(1 - e^{-\lambda t})$
 $= N_0 \left(1 - e^{-\lambda \times \frac{1}{\lambda}}\right) = N_0 \left(1 - \frac{1}{e}\right) = 0.63 N_0 = 63\% \text{ of } N_0$
131. (d)
132. (b) $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n \Rightarrow \left(\frac{1}{16}\right) = \left(\frac{1}{2}\right)^n \Rightarrow n = 4$
also $n = \frac{t}{T_{1/2}} \Rightarrow T_{1/2} = \frac{120}{4} = 30 \text{ days}$
133. (c) $N = N_0 e^{-t/T_{1/2}} \Rightarrow \frac{1}{4} = e^{-t/10}$
 $\Rightarrow \left(\frac{1}{2}\right)^2 = \frac{1}{e^{t/10}} \Rightarrow t = 20 \text{ years}$
134. (a) $N = N_0 \left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}} = N_0 \left(\frac{1}{2}\right)^{\frac{15}{5}} = \frac{N_0}{8}$
135. (a)
136. (b) $A = A_0 \left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}} \Rightarrow \frac{1}{64} = \left(\frac{1}{2}\right)^{\frac{60}{T_{1/2}}}$

$$\Rightarrow \left(\frac{1}{2}\right)^6 = \left(\frac{1}{2}\right)^{\frac{60}{T_{1/2}}} \Rightarrow T_{1/2} = 10 \text{ sec}$$

137. (a, c)

$$138. (b) \quad (Z=92) U^{(A=238)} \xrightarrow{(8\alpha, 6\beta)} Z' X^{A'}$$

$$\text{so } A' = A - 4n_\alpha = 238 - 4 \times 8 = 206$$

$$\text{and } Z' = n_\beta - 2n_\alpha + Z = 6 - 2 \times 8 + 92 = 82.$$

$$139. (c) \quad A = A_0 e^{-\lambda t} = A_0 e^{-t/\tau}; \text{ where } \tau = \text{mean life}$$

$$\text{So } A_1 = A_0 e^{-t_1/T} \Rightarrow A_0 = \frac{A_1}{e^{-t_1/T}} = A_1 e^{t_1/T}$$

$$\therefore A_2 = A_0 e^{-t_2/T} = (A_1 e^{t_1/T}) e^{-t_2/T} \Rightarrow A_2 = A_1 e^{(t_1 - t_2)/T}$$

140. (c)

$$141. (d) \quad n_\alpha = \frac{228 - 212}{4} = 4 \text{ and } n_\beta = 2 \times 4 - 90 + 83 = 1$$

142. (c) In a gamma decay process. There is no change in either A or Z

143. (a) The radioactivity of a sample decays to $\frac{1}{16}$ th of its initial value in four half lives.

$$144. (d) \quad \frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/T} \Rightarrow \frac{1}{16} = \left(\frac{1}{2}\right)^{t/48}$$

$$\Rightarrow \left(\frac{1}{2}\right)^4 = \left(\frac{1}{2}\right)^{t/48} \Rightarrow t = 192 \text{ hour.}$$

145. (c) If λ is the decay constant of a radioactive substance than average life = $\frac{1}{\lambda}$

$$\text{Also half life} = \frac{0.693}{\lambda} = 0.693 \times (\text{Average life})$$

in single average life, more than 63% of radioactive nuclei decay

146. (b)

$$147. (d) \quad M = M_0 e^{-\lambda t}; \text{ given } t = 2\left(\frac{1}{\lambda}\right)$$

$$\Rightarrow M = 10e^{-\lambda\left(\frac{2}{\lambda}\right)} = 10\left(\frac{1}{e}\right)^2 \Rightarrow M = 1.35 \text{ gm}$$

148. (b)

$$149. (b) \quad \lambda = \frac{\log_e \frac{A_1}{A_2}}{t} = \frac{\log_e \frac{5000}{1250}}{5} = 0.4 \ln 2$$

$$150. (c) \quad Z_{\text{Resulting nucleus}} = 92 - 8 \times 2 + 4 \times 1 - 2 \times 1 = 78$$

151. (c) Radioactive nuclei that are injected into a patient collected at certain sites within its body, undergoing radioactive decay and emitting electromagnetic radiation. These radiation can then be recorded by a detector. This procedure provides an important diagnostic tool called radio tracer technique.

$$152. (a) \quad \text{By using } n_\alpha = \frac{A - A'}{4} \text{ and } n_\beta = 2n_\alpha - Z + Z'$$

$$153. (b) \quad N = N_0 \left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}}$$

$$\text{No of atoms at } t = 2hr, N_1 = 8 \times 10^{10} \left(\frac{1}{2}\right)^{\frac{2}{1}} = 2 \times 10^{10}$$

$$\text{No. of atoms at } t = 4hr, N_2 = 8 \times 10^{10} \left(\frac{1}{2}\right)^{\frac{4}{1}} = \frac{1}{2} \times 10^{10}$$

$\therefore$ No. of atoms decayed in given duration

$$= \left(2 - \frac{1}{2}\right) \times 10^{10} = 1.5 \times 10^{10}$$

154. (b)

$$155. (d) \quad A = A_0 \left(\frac{1}{2}\right)^n \Rightarrow 30 = 240 \left(\frac{1}{2}\right)^n \Rightarrow \left(\frac{1}{2}\right)^3 = \left(\frac{1}{2}\right)^n \Rightarrow n=3$$

$$\therefore \frac{t}{T_{1/2}} = 3 \Rightarrow T_{1/2} = \frac{t}{3} = \frac{1}{3} hr = 20 \text{ min.}$$

$$156. (b) \quad M = M_0 \left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}} \Rightarrow 25 = 100 \left(\frac{1}{2}\right)^{\frac{t}{1600}} \Rightarrow t = 3200 \text{ years.}$$

$$157. (c) \quad \text{Activity } R = R_0 e^{-\lambda t}$$

$$\frac{R_0}{3} = R_0 e^{-\lambda \times 9} \Rightarrow e^{-9\lambda} = \frac{1}{3} \quad \dots(i)$$

$$\text{After further 9 years } R' = R e^{-\lambda t} = \frac{R_0}{3} \times e^{-\lambda \times 9} \quad \dots(ii)$$

$$\text{From equation (i) and (ii) } R' = \frac{R_0}{9}.$$

158. (c) To reduce one fourth it takes time $t = 2(T_{1/2}) = 2 \times 40 = 80 \text{ years.}$

$$\text{Decay constant } \lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{40} = 0.0173 \text{ years}$$

$$159. (d) \quad M = M_0 \left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}} = 20 \times \left(\frac{1}{2}\right)^{\frac{3.6}{3.6}} = 20 \times \left(\frac{1}{2}\right)^1 = 0.019 \text{ mg}$$

$$160. (c) \quad N = N_0 \left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}} \Rightarrow \frac{N}{N_0} = \left(\frac{1}{2}\right)^{\frac{30}{10}} = \frac{1}{8} = 0.125.$$

$$161. (a) \quad \frac{7}{8} \text{ part decays i.e. remaining part is } \frac{1}{8}$$

$$N = N_0 \left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}} \Rightarrow \frac{1}{8} = \left(\frac{1}{2}\right)^{\frac{15}{T_{1/2}}} \Rightarrow T_{1/2} = 5 \text{ min.}$$

$$162. (d) \quad \frac{A}{A_0} = \left(\frac{1}{2}\right)^{t/T_{1/2}} \Rightarrow \frac{1}{8} = \left(\frac{1}{2}\right)^{t/8} \Rightarrow t = 24 \text{ years.}$$

163. (c) After $\beta^+ ({}_{+1}e^0)$ emission atomic number decreases by one and mass number remain unchanged. γ -emission, there will be no change on mass number and atomic number.

$$164. (c) \quad \text{New mass number } A' = A - 4n_\alpha = 232 - 4 \times 6 = 208$$

$$\text{atomic number } Z' = Z + n_{\beta} - 2n_{\alpha} = 90 + 4 - 2 \times 6 = 82$$

165. (d)

166. (d) Using conservation of momentum $P_{\text{daughter}} = P_{\alpha}$

$$\Rightarrow \frac{E_d}{E_{\alpha}} = \frac{m_{\alpha}}{m_d} \Rightarrow E_d = \frac{E_{\alpha} \times m_{\alpha}}{m_d} = \frac{6.7 \times 4}{214} = 0.125 \text{ MeV}$$

167. (c)

168. (c)

169. (b) $N = N_0 \times \left(\frac{1}{2}\right)^{11400/5700} = N_0 \left(\frac{1}{2}\right)^2 = 0.25 N_0.$

170. (d) Mean life $(T) = 1/\lambda = 100 \text{ second}$

$$\text{Half-life} = \frac{0.693}{\lambda} = \frac{0.693 \times 100}{60} = 1.155 \text{ min.}$$

171. (b) By using $n_{\alpha} = \frac{A - A'}{4}$ and $n_{\beta} = 2n_{\alpha} - Z + Z'$

172. (c)

173. (b) $\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{77} = 9 \times 10^{-3} / \text{day}.$

174. (a) By using $n_{\alpha} = \frac{A - A'}{4} = \frac{232 - 204}{4} = 7.$

175. (c) $\therefore \frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/T_{1/2}} \left(\frac{1}{2}\right)^{1/2} = \frac{1}{\sqrt{2}}.$

176. (d)

Critical Thinking Questions

1. (c) At closest distance of approach

Kinetic energy = Potential energy

$$\Rightarrow 5 \times 10^6 \times 1.6 \times 10^{-19} = \frac{1}{4\pi\epsilon_0} \times \frac{(ze)(2e)}{r}$$

For uranium $z = 92$, so $r = 5.3 \times 10^{-12} \text{ cm}$

2. (c) Speed of electron in n orbit of hydrogen atom $v = \frac{e^2}{2\epsilon_0 n h}$

$$\text{In ground state } n = 1 \Rightarrow v = \frac{e^2}{2\epsilon_0 h}$$

$$\Rightarrow \frac{v}{c} = \frac{e^2}{2\epsilon_0 c h} = \frac{(1.6 \times 10^{-19})^2}{2 \times 8.85 \times 10^{-12} \times 3 \times 10^8 \times 6.6 \times 10^{-34}} = \frac{1}{137}.$$

3. (b) Recoil momentum = momentum of photon $= \frac{h}{\lambda}$

$$= hR \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = \frac{hR \times 15}{16} = 6.8 \times 10^{-27} \text{ N} \times \text{sec}$$

4. (a) The average time that the atom spends in this excited state is equal to Δt , so by using $\Delta E \cdot \Delta t = \frac{h}{2\pi}$

$$\Rightarrow \text{Uncertainty in energy} = \frac{h/2\pi}{\Delta t}$$

$$= \frac{6.6 \times 10^{-34}}{2 \times 3.14 \times 10^{-8}} = 1.05 \times 10^{-26} \text{ J} = 6.56 \times 10^{-8} \text{ eV}$$

5. (a) After the removal of first electron remaining atom will be hydrogen like atom.

So energy required to remove second electron from the atom

$$E = 13.6 \times \frac{2^2}{1} = 54.4 \text{ eV}$$

$$\therefore \text{Total energy required} = 24.6 + 54.4 = 79 \text{ eV}$$

6. (a) Electron after absorbing 10.2 eV energy goes to its first excited state ($n=2$) from ground state ($n=1$).

$$\therefore \text{Increase in momentum} = \frac{h}{2\pi}$$

$$= \frac{6.6 \times 10^{-34}}{6.28} = 1.05 \times 10^{-34} \text{ J-s}$$

7. (a) Using $\Delta E \propto Z^2$ ($\because n_1$ and n_2 are same)

$$\Rightarrow \frac{hc}{\lambda} \propto Z^2 \Rightarrow \lambda Z^2 = \text{constant}$$

$$\Rightarrow \lambda_1 Z_1^2 = \lambda_2 Z_2^2 = \lambda_3 Z_3^2 = \lambda_4 Z_4^2$$

$$\Rightarrow \lambda_1 \times 1 = \lambda_2 \times 1^2 = \lambda_3 \times 2^2 = \lambda_4 \times 3^2$$

$$\Rightarrow \lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$$

8. (d) $mvr = \frac{h}{2\pi}$ (for first orbit)

$$\Rightarrow m\omega r^2 = \frac{h}{2\pi} \Rightarrow m \times 2\pi\nu \times r^2 = \frac{h}{2\pi} \Rightarrow \nu = \frac{h}{4\pi^2 m r^2}$$

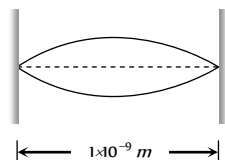
$$= \frac{6.6 \times 10^{-34}}{4(3.14)^2 \times 9.1 \times 10^{-31} \times (0.53 \times 10^{-10})^2} = 6.5 \times 10^{15} \frac{\text{rev}}{\text{sec}}$$

9. (b) It will form a stationary wave

$$\lambda = 2l = 2 \times 10^{-9} \text{ m}$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2mE}}$$

$$\Rightarrow E = \frac{h^2}{2m\lambda^2} = 6 \times 10^{-20} \text{ J}$$



10. (d) Suppose closest distance is r , according to conservation of energy.

$$400 \times 10^3 \times 1.6 \times 10^{-19} = 9 \times 10^9 \frac{(ze)(2e)}{r}$$

$$\Rightarrow 6.4 \times 10^{-14} = \frac{9 \times 10^9 \times (82 \times 1.6 \times 10^{-19}) \times (2 \times 1.6 \times 10^{-19})}{r}$$

$$\Rightarrow r = 5.9 \times 10^{-13} \text{ m} = 0.59 \text{ pm}$$

11. (a) Here radius of electron orbit $r \propto 1/m$ and energy $E \propto m$, where m is the mass of the electron.

Hence energy of hypothetical atom

$$E_0 = 2 \times (-13.6 \text{ eV}) = -27.2 \text{ eV} \text{ and radius } r_0 = \frac{a_0}{2}$$

12. (a) Electronic configuration of iodine is 2, 8, 18, 18, 7,

$$\text{Here } r_n = (0.053 \times 10^{-9} \text{ m}) \frac{n^2}{Z}$$

$$\text{Here } n = 5 \text{ and } Z = 53, \text{ hence } r_n = 2.5 \times 10^{-11} \text{ m}$$

13. (a) $N \propto \left[\frac{1}{\sin^4 \theta/2} \right] \Rightarrow N_1 = 7 \times \frac{1}{(\sin 30^\circ)^4} = 112$

$$\text{and } N_2 = 7 \times \frac{1}{(\sin 60^\circ)^4} = 12.5$$

14. (d) $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$. Required energy for said transition

$$\Delta E = E_3 - E_1 = 13.6 Z^2 \left[\frac{1}{1^2} - \frac{1}{3^2} \right]$$

$$\Rightarrow \Delta E = 13.6 \times 3^2 \left[\frac{8}{9} \right] = 108.8 \text{ eV}$$

$$\Rightarrow \Delta E = 108.8 \times 1.6 \times 10^{-19} \text{ J}$$

$$\text{Now } \Delta E = \frac{hc}{\lambda} = 108.8 \times 1.6 \times 10^{-19}$$

$$\Rightarrow \lambda = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{108.8 \times 1.6 \times 10^{-19}} = 0.11374 \times 10^{-7} \text{ m} = 113.74 \text{ \AA}$$

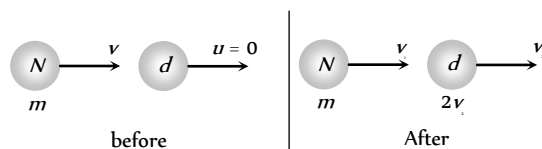
15. (c) $\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

$$\Rightarrow \frac{1}{970.6 \times 10^{-10}} = 1.097 \times 10^7 \left[\frac{1}{1^2} - \frac{1}{n_2^2} \right] \Rightarrow n_2 = 4$$

$$\therefore \text{Number of emission lines } N = \frac{n(n-1)}{2} = \frac{4 \times 3}{2} = 6$$

16. (d) Neutron velocity = v , mass = m

Deuteron contains 1 neutron and 1 proton, mass = $2m$



In elastic collision both momentum and K.E. are conserved $p_i = p_f$

$$mv = mv_f + m v_d \Rightarrow mv = mv + 2mv_d \quad \dots (i)$$

By conservation of kinetic energy

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_f^2 + \frac{1}{2}(2m)v_d^2 \quad \dots (ii)$$

By solving (i) and (ii) we get

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2} v + \frac{2m_2}{(m_1 + m_2)} v \Rightarrow v_1 = \frac{m_1 + 2m}{3m} = -\frac{v}{3}$$

$$K_i = \frac{1}{2}mv^2, \quad K_f = \frac{1}{2}mv_1^2 \Rightarrow \frac{K_i - K_f}{K_i} = 1 - \frac{v_1^2}{v^2}$$

$$= 1 - \frac{1}{9} = \frac{8}{9} \text{ (Fractional change in K.E.)}$$

17. (c) In hydrogen atom $E_n = -\frac{Rhc}{n^2}$

Also $E_n \propto m$; where m is the mass of the electron. Here the electron has been replaced by a particle whose mass is double of an electron. Therefore, for this hypothetical atom energy in

n orbit will be given by $E_n = -\frac{2Rhc}{n^2}$

The longest wavelength $\lambda_{\max}$ (or minimum energy) photon will correspond to the transition of particle from $n = 3$ to $n = 2$

$$\Rightarrow \frac{hc}{\lambda_{\max}} = E_3 - E_2 = Rhc \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$$

This gives $\lambda_{\max} = \frac{18}{5R}$.

18. (d) As the transition $n = 4$ and $n = 3$, results in *UV* radiation and infrared radiation involves smaller amounts of energy *UV*. So we require a transition involving initial values of n greater than 4 e.g. $5 \rightarrow 4$.

19. (c) $\frac{hc}{\lambda} = E = eV$

$$\Rightarrow \lambda = \frac{hc}{eV} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 4.9} = 2525 \text{ \AA}$$

20. (d) Rydberg constant $R = \frac{\epsilon_0 n^2 h^2}{\pi m Z e^2}$

Velocity $v = \frac{Ze^2}{2\epsilon_0 nh}$ and energy $E = -\frac{mZ^2 e^4}{8\epsilon_0^2 n^2 h^2}$

Now, it is clear from above expressions $R.v \propto n$

21. (b) In second excited state $n = 3$

So $l_H = l_{Li} = 3 \left(\frac{h}{2\pi} \right)$

While $E \propto Z$ and $Z_e = 1$, $Z_n = 3$

So $|E| = 9|E|$ or $|E| < |E|$

22. (c) Since the $^{133}_{55}\text{Cs}$ has larger size among the four atoms gives, thus the electrons present in the outermost orbit will be away from the nucleus and the electrostatic force experienced by electrons due to nucleus will be minimum. Therefore the energy required to liberate electron from outer will be minimum in the case of $^{133}_{55}\text{Cs}$.

23. (d)

24. (a) Potential energy $U = eV = eV_0 \ln \frac{r}{r_0}$

$\therefore$ Force $F = -\left| \frac{dU}{dr} \right| = \frac{eV_0}{r}$.

$\therefore$ The force will provide the necessary centripetal force.

Hence $\frac{mv^2}{r} = \frac{eV_0}{r} \Rightarrow v = \sqrt{\frac{eV_0}{m}}$ (i)

and $mvr = \frac{nh}{2\pi}$ (ii)

From equation (i) and (ii) $mr = \left(\frac{nh}{2\pi} \right) \sqrt{\frac{m}{eV_0}}$ or $r \propto n$

25. (d) $(r_m) = \left(\frac{m^2}{Z} \right) (0.53 \text{ \AA}) = (n \times 0.53 \text{ \AA}) \Rightarrow \frac{m^2}{Z} = n$

$m = 5$ for $^{257}_{100}\text{Fm}$ (the outermost shell)

and $z = 100 \Rightarrow n = \frac{(5)^2}{100} = \frac{1}{4}$

26. (d) Energy radiated $= 1.4 \text{ kW} / m^2$

$$= 1.4 \text{ kJ} / \text{sec } m^2 = \frac{1.4 \text{ kJ}}{\frac{1}{86400} \text{ day } m^2} = \frac{1.4 \times 86400}{\text{day } m^2}$$

Total energy radiated/day

$$= \frac{4\pi \times (1.5 \times 10^{11})^2 \times 1.4 \times 86400}{1} \frac{\text{kJ}}{\text{day}} = E$$

$\therefore E = mc^2 \Rightarrow m = \frac{E}{c^2}$

$$= \frac{4\pi (1.5 \times 10^{11})^2 \times 1.4 \times 86400}{(3 \times 10^8)^2} = 3.8 \times 10^{14} \text{ kg}$$

27. (c) The equation is $O^{17} \rightarrow O^{16} + n^1$

$\therefore$ Energy required = B.E. of O - B.E. of O

$$= 17 \times 7.75 - 16 \times 7.97 = 4.23 \text{ MeV}$$

28. (c) $\Delta = mc^2 - m_0 c^2 = \frac{m_0 c^2}{\sqrt{1 - (v^2/c^2)}} - m_0 c^2$

$$= m_0 c^2 \left(\frac{1}{\sqrt{1 - (v^2/c^2)}} - 1 \right) = 0.511 \left(\frac{1}{\sqrt{0.75}} - 1 \right)$$

$$= 0.079 \text{ MeV}$$

29. (c,d) Due to mass defect (which is finally responsible for the binding energy of the nucleus), mass of a nucleus is always less than the sum of masses of its constituent particles $^{20}_{10}\text{Ne}$ is made up of 10 protons plus 10 neutrons. Therefore, mass of $^{20}_{10}\text{Ne}$ nucleus $M_1 < 10(m_p + m_n)$

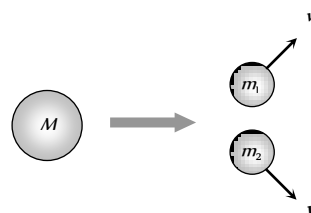
Also heavier the nucleus, more is the mass defect thus $20(m_n + m_p) - M_2 > 10(m_p + m_n) - M_1$

$$\text{or } 10(m_p + m_n) > M_2 - M_1$$

$$\Rightarrow M_2 < M_1 + 10(m_p + m_n) \Rightarrow M_2 < M_1 + M_1$$

$$\Rightarrow M_2 < 2M_1.$$

30. (a)



By conservation of momentum $m_1 v_1 = m_2 v_2$

$$\Rightarrow \frac{v_1}{v_2} = \frac{m_2}{m_1} \quad \text{..... (i)}$$

Also from $r \propto A^{1/3} \Rightarrow \frac{r_1}{r_2} = \left(\frac{A_1}{A_2} \right)^{1/3} = \left(\frac{1}{8} \right)^{1/3} = \frac{1}{2}$.

31. (a) Since nuclear density is constant hence mass $\propto$ volume.

32. (c) Mass defect $= 3 \times 2.014 - 4.001 - 1.007 - 1.008$
 $= 0.026 \text{ amu} = 0.026 \times 931 \times 10^6 \times 1.6 \times 10^{-19} \text{ J}$
 $= 3.82 \times 10^{-12} \text{ J}$

Power of star = $10^7 W$

$$\text{Number of deuterons used} = \frac{10^{16}}{\Delta M} = 0.26 \times 10^{28}$$

$$\text{Deuteron supply exhausts in } \frac{10^{40}}{0.26 \times 10^{28}} = 10^{12} s.$$

33. (a) Since electron and positron annihilate

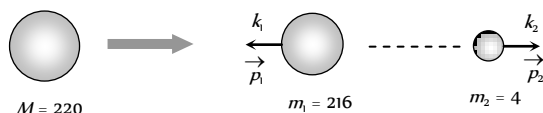
$$\lambda = \frac{hc}{E_{\text{Total}}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{(0.51 + 0.51) \times 10^6 \times 1.6 \times 10^{-19}} \\ = 1.21 \times 10^{-12} m = 0.012 \text{ \AA}.$$

34. (a) Kinetic energy of the molecules of a gas at a temp. T is $\frac{3}{2} kT$

$$\therefore \text{To initiate the reaction } \frac{3}{2} kT = 7.7 \times 10^{-14} J$$

$$\Rightarrow \frac{3}{2} \times 1.38 \times 10^{-23} T = 7.7 \times 10^{-14} \Rightarrow T = 3.7 \times 10^9 K.$$

35. (b)



Q -value of the reaction is 5.5 eV

$$\text{i.e. } k_1 + k_2 = 5.5 \text{ MeV} \quad \dots\dots(i)$$

By conservation of linear momentum

$$p_1 = p_2 \Rightarrow \sqrt{2(216)k_1} = \sqrt{2(4)k_2}$$

$$\Rightarrow k_1 = 54 k \quad \dots\dots(ii)$$

On solving equation (i) and (ii) we get $k = 5.4 \text{ MeV}$.

36. (b) By the formula $N = N_0 e^{-\lambda t}$

$$\text{Given } \frac{N}{N_0} = \frac{1}{20} \text{ and } \lambda = \frac{0.6931}{3.8} \Rightarrow 20 = e^{\frac{0.6931 \times t}{3.8}}$$

Taking log of both sides

$$\text{or } \log 20 = \frac{0.6931 \times t}{3.8} \log_{10} e$$

$$\text{or } 1.3010 = \frac{0.6931 \times t \times 0.4343}{3.8} \Rightarrow t = 16.5 \text{ days.}$$

37. (b) $N = N_0 e^{-\lambda t}$

$$\therefore 0.9 N_0 = N_0 e^{-\lambda \times 5} \Rightarrow 5\lambda = \log_e \frac{1}{0.9} \quad \dots\dots(i)$$

$$\text{and } x N_0 = N_0 e^{-\lambda \times 20} \Rightarrow 20\lambda = \log_e \left(\frac{1}{x} \right) \quad \dots\dots(ii)$$

Dividing (i) by (ii), we get

$$\frac{1}{4} = \frac{\log_e(1/0.9)}{\log_e(1/x)} = \frac{\log_{10}(1/0.9)}{\log_{10}(1/x)} = \frac{\log_{10} 0.9}{\log_{10} x}$$

$$\Rightarrow \log_{10} x = 4 \log_{10} 0.9 \Rightarrow x = 0.658 = 65.8\%$$

38. (c) If in the rock there is no Y element, then the time taken by element X to reduce to $\frac{1}{8}$ th the initial value will be equal to

$$\frac{1}{8} = \left(\frac{1}{2} \right)^n \text{ or } n = 3$$

Therefore, from the beginning three half life time is spent. Hence the age of the rock is

$$= 3 \times 1.37 \times 10^9 = 4.11 \times 10^9 \text{ years.}$$

39. (b) $\frac{N}{N_0} = \left(\frac{1}{2} \right)^n \Rightarrow \frac{1}{64} = \left(\frac{1}{2} \right)^n = \left(\frac{1}{2} \right)^n \Rightarrow n = 6.$

After 6 half lives intensity emitted will be safe.

$$\therefore \text{Total time taken} = 6 \times 2 = 12 \text{ hrs.}$$

40. (a) $\frac{dN}{dt} = \lambda N$; $\lambda = \frac{0.6931}{t_{12}} = \frac{0.6931}{1620 \times 365 \times 24 \times 60 \times 60}$,

$$N = \frac{6.023 \times 10^{23}}{226}$$

$$\therefore \frac{dN}{dt} = \frac{0.6931 \times 6.023 \times 10^{23}}{1620 \times 365 \times 24 \times 60 \times 60 \times 226} = 3.61 \times 10^{10}$$

41. (a) $\lambda = \lambda_1 + \lambda_2 \Rightarrow \frac{1}{T} = \frac{1}{T_1} + \frac{1}{T_2}$

$$\therefore T = \frac{T_1 T_2}{T_1 + T_2} = \frac{810 \times 1620}{810 + 1620} = 540 \text{ years}$$

Hence $\frac{1}{4}$ th of material remain after 1080 years.

42. (b) Similar to Q. 40.

43. (c) $(T_{1/2})_x = (t_{\text{mean}})_y$

$$\Rightarrow \frac{0.693}{\lambda_x} = \frac{1}{\lambda_y} \Rightarrow \lambda_x = 0.693 \lambda_y \text{ or } \lambda_x < \lambda_y$$

Also rate of decay = λN

Initially number of atoms (N) of both are equal but since $\lambda_y > \lambda_x$, therefore, y will decay at a faster rate than x .

44. (c) $\lambda_\alpha = \frac{1}{1620} \text{ per year}$ and $\lambda_\beta = \frac{1}{405} \text{ per year}$ and it is given

$$\text{that the fraction of the remained activity } \frac{A}{A_0} = \frac{1}{4}$$

Total decay constant

$$\lambda = \lambda_\alpha + \lambda_\beta = \frac{1}{1620} + \frac{1}{405} = \frac{1}{324} \text{ per year}$$

$$\text{We know that } A = A_0 e^{-\lambda t} \Rightarrow t = \frac{1}{\lambda} \log_e \frac{A_0}{A}$$

$$\Rightarrow t = \frac{1}{\lambda} \log_e 4 = \frac{2}{\lambda} \log_e 2 = 324 \times 2 \times 0.693 = 449 \text{ years.}$$

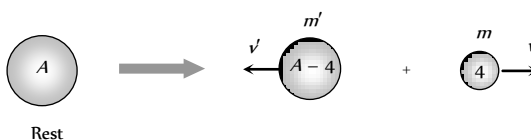
45. (d) $n = \frac{24}{24 \times 138.6} = \frac{1}{138.6}$; Now $\frac{N}{N_0} = \left(\frac{1}{2} \right)^n = \left(\frac{1}{2} \right)^{1/138.6}$

$$\Rightarrow N = 10,00000 \left(\frac{1}{2} \right)^{1/138.6} = 995011$$

So number of disintegration

$$= 1000000 - 995011 = 4989 \approx 5000.$$

46. (a)



According to conservation of momentum $4v = (A-4)v' \Rightarrow$

$$v' = \frac{4v}{A-4}.$$

47. (b) $\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{20} = 0.03465$

Now time of decay $t = \frac{2.303}{\lambda} \log \frac{N_0}{N}$

$\Rightarrow t_1 = \frac{2.303}{0.03465} \log \frac{100}{67} = 11.6 \text{ min}$

and $t_2 = \frac{2.303}{0.03465} \log \frac{100}{33} = 32 \text{ min}$

Thus time difference between points of time
 $= t_2 - t_1 = 32 - 11.6 = 20.4 \text{ min} \approx 20 \text{ min}$

48. (d) $N_1 = N_0 e^{-10\lambda t}$ and $N_2 = N_0 e^{-\lambda t}$

$\Rightarrow \frac{N_1}{N_2} = \frac{1}{e} = e^{-1} = e^{(-10\lambda + \lambda)t} = e^{-9\lambda t} \Rightarrow t = \frac{1}{9\lambda}$

49. (a) $N = N_0 \left(\frac{1}{2}\right)^{t/T_{1/2}} \Rightarrow N_A = 10 \left(\frac{1}{2}\right)^{t/1}$ and $N_B = 1 \left(\frac{1}{2}\right)^{t/2}$

Given $N_A = N_B \Rightarrow 10 \left(\frac{1}{2}\right)^t = \left(\frac{1}{2}\right)^{t/2}$

$\Rightarrow 10 = \left(\frac{1}{2}\right)^{-t/2} \Rightarrow 10 = 2^{t/2}$. Taking log both the sides.

$\log_{10} 10 = \frac{t}{2} \log_{10} 2 \Rightarrow 1 = \frac{t}{2} \times 0.3010 \Rightarrow t = 6.62 \text{ years}$

50. (b) Here $T_{1/2} = 20 \text{ minutes}$; we know $\frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/T_{1/2}}$

For 20% decay $\frac{N}{N_0} = \frac{80}{100} = \left(\frac{1}{2}\right)^{t_1/20}$ (i)

For 80% decay $\frac{N}{N_0} = \frac{20}{100} = \left(\frac{1}{2}\right)^{t_2/20}$ (ii)

Dividing (ii) by (i)

$\frac{1}{4} = \left(\frac{1}{2}\right)^{\frac{(t_2 - t_1)}{20}}$; on solving we get $t_2 - t_1 = 40 \text{ min}$

51. (d) Here the activity of the radioactive sample reduces to half in 140 days. Therefore, the half life of the sample is 140 days. 280 days is its two half lives. So before two half lives its activity was ($2^2 \times$ present activity).

$\therefore$ Initial activity = $2^2 \times 6000 = 24000 \text{ dps}$

52. (a) Excitation energy $\Delta E = E_2 - E_1 = 13.6 Z^2 \left[\frac{1}{1^2} - \frac{1}{2^2} \right]$

$\Rightarrow 40.8 = 13.6 \times \frac{3}{4} \times Z^2 \Rightarrow Z = 2$

Now required energy to remove the electron from ground state
 $= \frac{+13.6 Z^2}{(1)^2} = 13.6(Z)^2 = 54.4 \text{ eV}$

53. (b) Rate of disintegration $\frac{dN}{dt} = 10^{17} \text{ s}^{-1}$

Half life $T_{1/2} = 1445 \text{ year}$

$= 1445 \times 365 \times 24 \times 60 \times 60 = 4.55 \times 10^8 \text{ sec}$

Now decay constant

$\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{4.55 \times 10^{10}} = 1.5 \times 10^{-11} \text{ per sec}$

The rate of disintegration

$\frac{dN}{dt} = \lambda \times N_0 \Rightarrow 10^{17} = 1.5 \times 10^{-11} \times N_0$

$\Rightarrow N_0 = 6.6 \times 10^8$

54. (b) $P = \frac{nE}{t} \Rightarrow 300 \times 10^6 = \frac{n \times 170 \times 10^6 \times 1.6 \times 10^{-19}}{t}$

$\therefore$ Number of atoms per sec $\frac{n}{t} = 1.102 \times 10^{19}$

Number of atoms per hour = $1.02 \times 10^8 \times 3600$
 $= 3.97 \times 10^8$

55. (a) According to kinetic interpretation of temperature

$K.E. = \left(\frac{1}{2} m v^2\right) = \frac{3}{2} kT$

$\Rightarrow 10.2 \times 1.6 \times 10^{-19} = \frac{3}{2} \times (1.38 \times 10^{-23}) T$

$\Rightarrow T = 7.9 \times 10^4 \text{ K}$

56. (a) R_i = Initial activity = 1 micro curie = $3.7 \times 10^4 \text{ dps}$

r = Activity in 1 cm of blood at $t = 5 \text{ hrs}$

$= \frac{296}{60} \text{ dps} = 4.93 \text{ dps}$

R = Activity of whole blood at time $t = 5 \text{ hr}$,

Total volume should be $V = \frac{R}{r} = \frac{R_0 e^{-\lambda t}}{r}$

$= \frac{3.7 \times 10^4 \times 0.7927}{4.93} = 5.94 \times 10^3 \text{ cm} = 5.94 \text{ Litre}$

57. (b) Let ground state energy (in eV) be E_1

Then from the given condition

$E_{2n} - E_1 = 204 \text{ eV}$ or $\frac{E_1}{4n^2} - E_1 = 204 \text{ eV}$

$\Rightarrow E_1 \left(\frac{1}{4n^2} - 1 \right) = 204 \text{ eV}$ (i)

and $E_{2n} - E_n = 40.8 \text{ eV}$

$\Rightarrow \frac{E_1}{4n^2} - \frac{E_1}{n^2} = E_1 \left(-\frac{3}{4n^2} \right) = 40.8 \text{ eV}$ (ii)

From equation (i) and (ii), $\frac{1 - \frac{1}{4n^2}}{\frac{3}{4n^2}} = 5 \Rightarrow n = 2$

58. (b) Here $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^{1/3}$

where n = Number of half lives = $\frac{1}{3}$

$\Rightarrow \frac{N}{N_0} = \frac{1}{1.26} \Rightarrow \frac{N_U}{N_{Pb} + N_U} = \frac{1}{1.26}$

$\Rightarrow N_{Pb} = 0.26 N_U \Rightarrow \frac{N_{Pb}}{N_U} = 0.26$

59. (b) For
- K_α
- X-ray line

$$\frac{1}{\lambda_\alpha} = R(Z-1)^2 \left[\frac{1}{1^2} - \frac{1}{2^2} \right] = \frac{3R}{4} (Z-1)^2$$

On putting the given values

$$\frac{1}{0.76 \times 10^{-10}} = \frac{3}{4} \times 1.09 \times 10^7 (Z-1)^2$$

$$\Rightarrow (Z-1)^2 \approx 1600 \Rightarrow Z-1 = 40 \Rightarrow Z = 41$$

60. (a) Maximum energy is liberated for transition
- $E_n \rightarrow 1$
- and minimum energy for
- $E_n \rightarrow E_{n-1}$

$$\text{Hence } \frac{E_1}{n^2} - E_1 = 52.224 \text{ eV} \quad \dots(i)$$

$$\text{and } \frac{E_1}{n^2} - \frac{E_1}{(n-1)^2} = 1.224 \text{ eV} \quad \dots(ii)$$

Solving equations (i) and (ii) we get

$$E_1 = -54.4 \text{ eV and } n = 5$$

$$\text{Now } E_1 = -\frac{13.6 Z^2}{1^2} = -54.4 \text{ eV. Hence } Z = 2$$

61. (a) Activity of substance that has 2000 disintegration/sec

$$= \frac{2000}{3.7 \times 10^{10}} = 0.054 \times 10^{-6} \text{ ci} = 0.054 \mu\text{ci}$$

The number of radioactive nuclei having activity A

$$N = \frac{A}{\lambda} = \frac{2000 \times T_{1/2}}{\log_e 2}$$

$$= \frac{2000 \times 138.6 \times 24 \times 3600}{0.693} = 3.45 \times 10^{10}$$

62. (a) Maximum number of nuclei will be present when rate of decay = rate of formation
- $\Rightarrow \lambda N = \alpha \Rightarrow N = \frac{\alpha}{\lambda}$

63. (b) $r \propto A^{1/3} \Rightarrow \frac{r_1}{r_2} = \left(\frac{A_1}{A_2} \right)^{1/3}$

$$\Rightarrow \frac{3}{5} = \left(\frac{27}{A} \right)^{1/3} \Rightarrow \frac{27}{125} = \frac{27}{A} \Rightarrow A = 125$$

$$\text{Number of nuclei in atom } X = A - 52 = 125 - 52 = 73.$$

64. (c) 1 week
- ≈ 7
- days
- $\approx 7 \times 24 \text{ hrs} \approx 14$
- half lives

$$\text{Number of atoms left} = \frac{N_0}{(2)^{14}}, \text{ Activity} = N\lambda$$

$$\therefore \text{Activity left is } \frac{1}{(2)^{14}} \text{ times the initial}$$

$$\Rightarrow \frac{1}{(2)^{14}} \times 1 \text{ curie} = \frac{1}{16384} \times 1 \text{ curie} \approx 61 \times 10^{-6} \text{ curie}$$

$$\approx 60 \mu \text{ curie.}$$

65. (a)
- $m_0 c^2 = 0.54 \text{ MeV}$
- and
- $\text{K.E.} = mc^2 - m_0 c^2$

$$\text{Also } m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{m_0}{\sqrt{1 - (0.8)^2}} = \frac{m_0}{0.6}$$

$$\therefore E = mc^2 = \frac{m_0}{0.6} c^2 = \frac{m_0 c}{0.6} = \frac{0.54 \text{ MeV}}{0.6} = 0.9 \text{ MeV}$$

$$\therefore \text{K.E.} = (0.9 - 0.54) = 0.36 \text{ MeV.}$$

Graphical Questions

1. (a) B.E. per nucleon is maximum for
- Fe^{56}
- . For further detail refer theory.

2. (a) $\omega = 2\pi\nu = \frac{2\pi c}{\lambda} = 2\pi c \bar{\nu} \Rightarrow \omega \propto \bar{\nu}$

3. (c)

4. (d) The total number of atoms neither remains constant (as in option (a) nor can ever increase (as in option (b) and (c)). They will continuously decrease with time. Therefore option (d) is correct.

5. (c) $N = N_0 e^{-\lambda t} \Rightarrow \frac{dN}{dt} = -N_0 \lambda e^{-\lambda t}$

i.e. Rate of decay $\left(\frac{dN}{dt} \right)$ varies exponentially with time (t).

6. (d) Rate $R = -\frac{dN}{dt} = \lambda N_0 e^{-\lambda t} = \lambda N \Rightarrow \frac{R}{N} = \lambda$ (constant)

i.e. graph between $\frac{R}{N}$ and t , be a straight line parallel to the time axis.

7. (b) Read time for 50 count rate, it gives half life period of 3 hrs, one small square gives 600 counts (
- 10×60
-). The number of small squares between graph and time axis are approx 24

$$\text{Hence count rate} = 24 \times 600 = 14400$$

8. (b) Number of atoms undecayed
- $N = N_0 e^{-\lambda t}$

$$\text{Number of atoms decayed} = N_0 - N = N_0 (1 - e^{-\lambda t})$$

$$\Rightarrow \text{Decayed fraction } f = \frac{N_0 - N}{N_0} = 1 - e^{-\lambda t}$$

i.e. fraction will rise up to 1, following exponential path as shown in graph (B).

9. (c) Energy is released in a process when total Binding energy (B.E.) of the nucleus is increased or we can say when total B.E. of products is more than the reactants. By calculation we can see that only in case of option (c), this happens.

$$\text{Given } W \rightarrow 2Y$$

$$\text{B.E. of reactants} = 120 \times 75 = 900 \text{ MeV}$$

$$\text{and B.E. of products} = 2 \times (60 \times 85) = 1020 \text{ MeV}$$

i.e. B.E. of products > B.E. of reactants.

10. (d)
- $N = N_0 e^{-\lambda t}$
- and
- $A = A_0 e^{-\lambda t} = \lambda N_0 e^{-\lambda t}$

$$\therefore N_{\infty} = N_0 - N = N_0 - N_0 e^{-\lambda t} \Rightarrow N_{\infty} = N_0 - \frac{A}{\lambda}$$

This is equation of straight line with negative slope.

11. (d) Radius of n orbit $r_n \propto n^2$, graph between r_n and n is a

$$\text{parabola. Also, } \frac{r_n}{r_1} = \left(\frac{n}{1}\right)^2 \Rightarrow \log_e \left(\frac{r_n}{r_1}\right) = 2 \log_e(n)$$

Comparing this equation with $y = mx + c$,

Graph between $\log_e \left(\frac{r_n}{r_1}\right)$ and $\log_e(n)$ will be a straight line, passing from origin.

Similarly it can be proved that graph between $\log_e \left(\frac{f_n}{f_1}\right)$ and $\log_e n$ is not a straight line.

12. (d) By using $N = N_0 e^{-\lambda t}$ and $\frac{dN}{dt} = -\lambda N$.

It shows that N decreases exponentially with time.

13. (b) Activity $= -\frac{dN}{dt} = \lambda N = \lambda N_0 e^{-\lambda t}$

i.e., graph between activity and t , be exponential having negative slope.

14. (d) Activity $A = \lambda N_0 e^{-\lambda t} \Rightarrow \log_e A = \log_e \lambda N_0 + \log_e e^{-\lambda t}$

$$\Rightarrow \log_e A = \log_e C - \lambda t \quad (\text{Take } \lambda N_0 = C)$$

$$\Rightarrow \log_e A = -\lambda t + \log_e C$$

This is the equation of a straight line having negative slope ($= -\lambda$) and positive intercept on $\log A$ axis.

15. (c) Charge density is uniform inside and then falls rapidly near the surface of the nucleus.

16. (a) $R = R_0 A^{1/3}$; where $R_0 = 1.2 \times 10^{-15} \text{ m}$.

$$\Rightarrow \log R = \log R_0 + \frac{1}{3} \log_e A$$

This is the equation of a straight line with positive slope.

17. (b) $\left|\frac{dN}{dt}\right| = \lambda N \Rightarrow \left|\frac{dN}{dt}\right| \propto N$

18. (c) Number of atom decayed $N' = N_0(1 - e^{-\lambda t})$

N' will increase with time (t) exponentially.

19. (a) $A_n = \pi r_n^2 \Rightarrow \frac{A_n}{A_1} = \left(\frac{r_n}{r_1}\right)^2 = \left(\frac{n}{1}\right)^4 \quad (\because r_n \propto n^2)$

$$\text{Taking log both the side } \log_e \frac{A_n}{A_1} = 4 \log_e(n)$$

Comparing it with $y = mx + c$, graph (4) is correct.

2. (a) ${}^{90}_{38}\text{Sr}$ decays to ${}^{90}_{39}\text{Y}$ by the emission of β -rays. Sr gets absorbed in bones along with calcium.

Reason is also true. ${}^{90}_{38}\text{Sr} \xrightarrow{\beta} {}^{90}_{39}\text{Y}$ which emits β -rays of very high energy. Sr does not emit γ -rays. The damage is by the β -rays only.

3. (b) Neutron is about 0.1 more massive than proton. But the unique thing about the neutron is that while it is heavy, it has no charge (it is neutral). This lack of charge gives it the ability to penetrate matter without interacting as quickly as the beta particles or alpha particles.

4. (b) Bohr postulated that electrons in stationary orbits around the nucleus do not radiate.

This is the one of Bohr's postulate. According to this the moving electrons radiate only when they go from one orbit to the next lower orbit.

5. (c) Nuclear stability depends upon the ratio of neutron to proton. If the n/p ratio is more than the critical value, then a neutron gets converted into a proton forming a β^- particle in the process. $n \rightarrow p + e^-$

The β^- particle (e^-) is emitted from the nucleus in some radioactive transformation. So electrons do not exist in the nucleus but they result in some nuclear transformation.

6. (a) ${}_Z X^A \rightarrow 2({}_2\text{He}^4) + 2({}_{-1}\text{e}^0) + 2\gamma + {}_{Z-2} X^{A-8}$

7. (a) Experimentally, it is found that the average radius of a nucleus is given by

$$R = R_0 A^{1/3} \text{ where } R_0 = 1.1 \times 10^{-15} \text{ m} = 1.1 \text{ fm}$$

and A = mass number

8. (b)

9. (b) Rutherford confirmed the repulsive force on α -particle due to nucleus varies with distance according to inverse square law and that the positive charges are concentrated at the centre and not distributed throughout the atom.

10. (a) In α -particle scattering experiment, Rutherford found a small number of α -particles which were scattered back through an angle approaching to 180° . This is possible only if the positive charges are concentrated at the centre or nucleus of the atom.

11. (e) According to classical electromagnetic theory, an accelerated charge continuously emits radiation. As electrons revolving in circular paths are constantly experiencing centripetal acceleration, hence they will be losing their energy continuously and the orbital radius will go on decreasing and form spiral and finally the electron will fall on the nucleus.

12. (c) According to postulates of Bohr's atom model, the electron revolve round the nucleus in fixed orbit of definite radii. As long as the electron is in a certain orbits it does not radiate any energy.

13. (b) Maximum number of photon is given by all the transitions possible $= 4C_2 = 6$

Minimum number of transition = 1,

that is directly jump from 4 to 1.

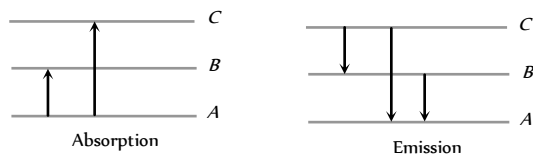
14. (b) When the atom gets appropriate energy from outside, then this electron rises to some higher energy level. Now it can return either directly to the lower energy level or come to the lowest energy level after passing through other lower energy levels,

Assertion and Reason

1. (c) In fusion, lighter nuclei are used so, fusion is not possible with ${}^{35}\text{Cl}$. Also binding energy of ${}^{35}\text{Cl}$ is not too small.

hence all possible transitions take place in the source and many lines are seen in the spectrum.

15. (d) Emission transitions can take place between any higher energy level and any energy level below it while absorption transitions start from the lowest energy level only and may end at any higher energy level. Hence number of absorptions transitions between two given energy levels is always less than the number of emission transitions between same two levels.



16. (a) We know that an electron is a very light particle as compared to an α -particle. Hence electron cannot scatter the α -particle at large angles, according to law of conservation of momentum. On the other hand, mass of nucleus is comparable with the mass of α -particle, hence only the nucleus of atom is responsible for scattering of α -particles.
17. (c) All those elements which are heavier than lead are radioactive. This is because in the nuclei of heavy atoms, besides the nuclear attractive forces, repulsive forces between the protons are also effective and these forces reduce the stability of the nucleus. Hence, the nuclei of heavier elements are being converted into lighter and lighter elements by emission of radioactive radiation. When they are converted into lead, the emission is stopped because the nucleus of lead is stable (or lead is most stable element in radioactive series).
18. (d) The penetrating power is maximum in case of gamma rays because gamma rays are an electromagnetic radiation of very small wavelength.
19. (b) β -particles, being emitted with very high speed compared to α -particles, pass very little time near the atoms of the medium. So the probability of the atoms being ionised is comparatively less. But due to this reason, their loss of energy is very slow and they can penetrate the medium through a sufficient depth.
20. (b) β -particles are emitted with very high velocity (up to $0.99c$). So, according to Einstein's theory of relativity, the mass of a β -particle is much higher compared to its rest mass (m_0). The velocity of electrons obtained by other means is very small compared to c (Velocity of light). So its mass remains nearly m_0 . But β -particle and electron both are similar particles.

21. (c) Radioactivity $= -\frac{dN}{dt} = \lambda N = \frac{0.693N}{T}$

$$= \frac{0.693 \times 10^8}{50} = \frac{0.693 \times 1.2 \times 10^8}{60} = 0.693 \times 2 \times 10^6.$$

Radioactivity is proportional to $1/T$, and not to T .

22. (c) Fragments produced in the fission of U^{235} are radioactive. When uranium undergoes fission, barium and krypton are not the only products. Over 100 different isotopes of more than 20 different elements have been detected among fission products. All of these atoms are, however, in the middle of the periodic table, with atomic numbers ranging from 34 to 58. Because the neutron-proton ratio needed for stability in this range is much smaller than that of the original uranium nucleus, the residual nuclei called fission fragments, always have too many neutrons

for stability. A few free neutrons are liberated during fission and the fission fragments undergo a series of beta decays (each of which increases Z by one and decreases N by one) until a stable nucleus is reached. During decay of the fission fragments, an average of 15 MeV of additional energy is liberated.

23. (b) Electron capture occurs more often than positron emission in heavy elements. This is because if positron emission is energetically allowed, electron capture is necessarily allowed, but the reverse is not true *i.e.* when electron capture is energetically allowed, positron emission is not necessarily allowed.
24. (e) The whole mass of the atom is concentrated at nucleus and $M_{\text{e}} < (\text{Sum of the masses of nucleons})$ because, when nucleons combine, some energy is wasted.

Atomic and Nuclear Physics

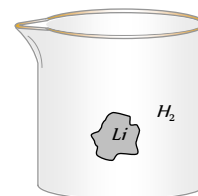
Self Evaluation Test -26

- In Bohr model of hydrogen atom, the force on the electron depends on the principal quantum number as
 - $F \propto 1/n^3$
 - $F \propto 1/n^4$
 - $F \propto 1/n^5$
 - Does not depend on n
- A nucleus X emits 9α -particles and $5p$ particle. The ratio of total protons and neutrons in the final nucleus is
 - $\frac{Z-13}{(A-Z-23)}$
 - $\frac{(Z-18)}{(A-36)}$
 - $\frac{(Z-13)}{(A-36)}$
 - $\frac{(Z-13)}{(A-Z-13)}$
- If $t_{1/2}$ is the half life of a substance then $t_{1/4}$ is the time in which substance
 - Decays $\frac{3}{4}th$
 - Remains $\frac{3}{4}th$
 - Decays $\frac{1}{2}$
 - Remains $\frac{1}{2}$
- The energy level diagram for an hydrogen like atom is shown in the figure. The radius of its first Bohr orbit is

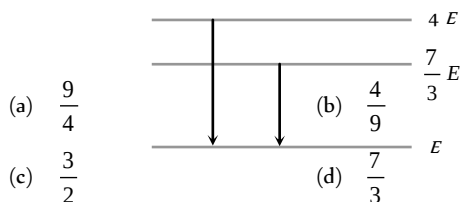
0 eV	_____	$n = \infty$
- 6.04 eV	_____	$n = 3$
- 13.6 eV	_____	$n = 2$
- 54.4 eV	_____	$n = 1$

 - 0.265 Å
 - 0.53 Å
 - 0.132 Å
 - None of these
- How much work must be done to pull apart the electron and the proton that make up the Hydrogen atom, if the atom is initially in the state with $n = 2$
 - $13.6 \times 1.6 \times 10^{-19} J$
 - $3.4 \times 1.6 \times 10^{-19} J$
 - $1.51 \times 1.6 \times 10^{-19} J$
 - 0
- The nuclide $^{131}_{53}I$ is radioactive, with a half-life of 8.04 days. At noon on January 1, the activity of a certain sample is 60089. The activity at noon on January 24 will be
 - 75 Bq
 - Less than 75 Bq
 - More than 75 Bq
 - 150 Bq
- U^{238} decays into Th^{234} by the emission of an α -particle. There follows a chain of further radioactive decays, either by α -decay or by β -decay. Eventually a stable nuclide is reached and after that, no further radioactive decay is possible. Which of the following stable nuclides is the end product of the U^{238} radioactive decay chain
 - Pb^{206}
 - Pb^{207}
 - Pb^{208}
 - Pb^{209}
- If the mass of a radioactive sample is doubled, the activity of the sample and the disintegration constant of the sample are respectively
 - Increases, remains the same
 - Decreases, increases
 - Decreases, remains same
 - Increases, decreases
- When a sample of solid lithium is placed in a flask of hydrogen gas then following reaction happened

$$^1_3H + ^6_3Li \rightarrow ^4_2He + ^4_2He$$
 This statement is
 - True
 - False
 - May be true at a particular pressure
 - None of these
- Consider an initially pure $M gm$ sample of X , an isotope that has a half life of T hour, what is its initial decay rate ($N_A =$ Avogadro No.)
 - $\frac{M N_A}{T}$
 - $\frac{0.693 M N_A}{T}$
 - $\frac{0.693 M N_A}{AT}$
 - $\frac{2.303 M N_A}{AT}$
- At a given instant there are 25% undecayed radioactive nuclei in a sample. After 10 sec the number of undecayed nuclei reduces to 6.25%, the mean life of the nuclei is
 - 14.43 sec
 - 7.21 sec
 - 5 sec
 - 10 sec
- Highly energetic electrons are bombarded on a target of an element containing 30 neutrons. The ratio of radii of nucleus to that of Helium nucleus is $14^{1/3}$. The atomic number of nucleus will be
 - 25
 - 26
 - 56
 - 30
- The ratio of ionization energy of Bohr's hydrogen atom and Bohr's hydrogen like lithium atom is
 - 1 : 1
 - 1 : 3



- (c) 1 : 9 (d) None of these
14. What is the angular momentum of an electron in Bohr's hydrogen atom whose energy is -0.544 eV .
- (a) $\frac{h}{\pi}$ (b) $\frac{2h}{\pi}$
- (c) $\frac{5h}{2\pi}$ (d) $\frac{7h}{2\pi}$
15. Consider a hypothetical annihilation of a stationary electron with a stationary positron. What is the wavelength of resulting radiation.
- (a) $\frac{h}{2m_0c}$ (b) $\frac{h}{m_0c}$
- (c) $\frac{2h}{m_0c}$ (d) $\frac{h}{m_0c^2}$
- (h = Planck's constant, c = speed of light, m_e = rest mass)
16. Nuclear reactions are given as
- (i) $\square (n, p)_{15}P^{32}$ $\square (p, \alpha)_8O^{16}$ (iii) ${}_7\square^4 (\quad p)$
 ${}_6C^{14}$
- missing particle or nuclide (in box $\square$) in these reactions are respectively
- (a) $S, F, {}_0n^1$ (b) $F, S, {}_0n^1$
- (c) $Be, F, {}_0n^1$ (d) None of these
17. In a sample of hydrogen like atoms all of which are in ground state, a photon beam containing photons of various energies is passed. In absorption spectrum, five dark lines, are observed. The number of bright lines in the emission spectrum will be (assume that all transitions takes place).
- (a) 5 (b) 10
- (c) 15 (d) None of these
18. A hydrogen atom emits a photon corresponding to an electron transition from $n = 5$ to $n = 1$. The recoil speed of hydrogen atom is almost (mass of proton $\approx 1.6 \times 10^{-27} \text{ kg}$).
- (a) 10 ms (b) $2 \times 10^{-5} \text{ ms}$
- (c) 4 ms (d) $8 \times 10^{-5} \text{ ms}$
19. Number of nuclei of a radioactive substance at time $t = 0$ are 1000 and 900 at time $t = 2 \text{ s}$. Then number of nuclei at time $t = 4 \text{ s}$ will be
- (a) 800 (b) 810
- (c) 790 (d) 700
20. The ratio between total acceleration of the electron in singly ionized helium atom and hydrogen atom (both in ground state) is
- (a) 1 (b) 8
- (c) 4 (d) 16
21. If the series limit of Lyman series for Hydrogen atom is equal to the series limit of Balmer series for a hydrogen like atom, then atomic number of this hydrogen like atom will be
- (a) 1 (b) 2
- (c) 3 (d) 4
22. Which sample contains greater number of nuclei :
- a $5.00\text{-}\mu\text{Ci}$ sample of ${}^{239}\text{Pu}$ (half-life 6560 y) or a $4.45\text{-}\mu\text{Ci}$ sample of ${}^{241}\text{Am}$ (half-life 7370 y)
- (a) ${}^{239}\text{Pu}$ (b) ${}^{241}\text{Am}$
- (c) Equal in both (d) None of these
23. The fission of ${}^{235}\text{U}$ can be triggered by the absorption of a slow neutrons by a nucleus. Similarly a slow proton can also be used. This statement is
- (a) Correct (b) Wrong
- (c) Information is insufficient (d) None of these
24. The radioactivity of a given sample of whisky due to tritium (half life 12.3 years) was found to be only 3% of that measured in a recently purchased bottle marked "7 years old". The sample must have been prepared about
- (a) 220 years back (b) 300 years back
- (c) 400 years back (d) 70 years back
25. The following diagram indicates the energy levels of a certain atom when the system moves from $4E$ level to E . A photon of wavelength λ_1 is emitted. The wavelength of photon produced during its transition from $\frac{7}{3}E$ level to E is λ_2 . The ratio $\frac{\lambda_1}{\lambda_2}$ will be



AS Answers and Solutions

(SET -26)

1. (b) $F \propto \frac{v^2}{r}$ also $v \propto \frac{1}{n}$ and $r \propto n^2 \Rightarrow F \propto \frac{1}{n^4}$
2. (a) ${}_Z X^A \xrightarrow{9\alpha} {}_{Z-18} X^{A-36} \xrightarrow{5\beta} {}_{Z-13} X^{A-36}$

Number of protons = ($Z = 13$)Number of neutrons = $(A - 36) - (Z - 13) = (A - Z - 23)$

$$\therefore \frac{P}{N} = \frac{(Z - 13)}{(A - Z - 23)}$$

3. (a) You must remember that $t_{1/2}$ is time in which substance decays

half. Hence in $t_{1/2}$ time substance decays $\frac{3}{4}$ th.

4. (a) We know that $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$ and $r_n = 0.53 \frac{n^2}{Z} (\text{\AA})$

Here for $n=1$, $E_1 = -54.4 \text{ eV}$

$$\text{Therefore } -54.4 = -13.6 \frac{Z^2}{1^2} \Rightarrow Z = 2$$

$$\text{Hence radius of first Bohr orbit } r = \frac{0.53(1)^2}{2} = 0.265 \text{ \AA}$$

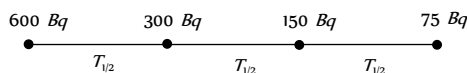
5. (b) The electrostatic P.E. is zero when the electron and proton are far apart from each other. Work done in pulling electron and proton far away from each other

$$W = E_f - E_i = 0 - E_i = -\left(-\frac{13.6}{n^2} \text{ eV}\right)$$

$$\Rightarrow W = \frac{13.6}{(2)^2} \times 1.6 \times 10^{-19} \text{ J} = 3.4 \times 1.6 \times 10^{-19} \text{ J}$$

6. (c) Number of days from January 1 to January 24 = 23 days.

$$\text{Number of half lives } n = \frac{23}{8.04} = 2.86 (< 3)$$



In three half lives activity becomes 75 Bq, but the given number of half lives are lesser than 3 so activity becomes greater than 75 Bq.

7. (a) $(4n+2)$ series starts from U^{238} and its stable end product is Pb^{206} .
8. (a) Activity depends upon mass, but λ doesn't change.
9. (b) The given reaction is a nuclear reaction, which can take place only if a proton (a hydrogen nucleus) comes into contact with a lithium nucleus. If the hydrogen is in the atomic form, the interaction between its electron cloud and the electron cloud of a lithium atom keeps the two nuclei from getting close to each other. Even if isolated protons are used, they must be fired at the Li atom with enough kinetic energy to overcome the electric repulsion between the proton and Li atom.

10. (c) $N = N_0 e^{-\lambda t} \Rightarrow \left| \frac{dN}{dt} \right| = N_0 \lambda e^{-\lambda t}$

$$\text{Initially at } t = 0, \left| \frac{dN}{dt} \right|_{t=0} = N_0 \lambda$$

where N_0 = Initial number of undecayed atoms

$$= \frac{\text{Mass of the sample}}{\text{Mass of a single atom of X}} = \frac{M}{A/N_A} = \frac{MN_A}{A}$$

$$\therefore \left| \frac{dN}{dt} \right|_{t=0} = \frac{MN_A \lambda}{A} = \frac{0.693 MN_A}{AT}$$

11. (b) In 10 sec, number of nuclei has been reduced to one fourth (25% to 6.25%).

Therefore its half life is $T_{1/2} = 5 \text{ sec}$.

$$\therefore \text{Mean life } T = \frac{T_{1/2}}{0.693} = \frac{5}{0.693} = 7.21 \text{ sec}$$

12. (b) By using $R = R_0 A^{1/3} \Rightarrow \frac{R_1}{R_2} = \left(\frac{A_1}{A_2} \right)^{1/3}$

$$\Rightarrow \frac{R}{R_{He}} = \left(\frac{A}{4} \right)^{1/3} \Rightarrow (14)^{1/3} = \left(\frac{A}{4} \right)^{1/3}$$

$$\Rightarrow A = 56 \text{ so } Z = 56 - 30 = 26.$$

13. (c) Energy of an electron in ground state of an atom (Bohr's hydrogen like atom) is given as

$$E = -13.6 Z^2 \text{ eV} \quad (Z = \text{atomic number of the atom})$$

$$\Rightarrow E_{\text{ion}} = 13.6 Z$$

$$\Rightarrow \frac{(E_{\text{ion}})_H}{(E_{\text{ion}})_{Li}} = \left(\frac{Z_H}{Z_{Li}} \right)^2 = \left(\frac{1}{3} \right)^2 = \frac{1}{9}$$

14. (c) By using $E = -\frac{13.6}{n^2} \text{ eV}$ (for H atom)

$$\Rightarrow -0.544 = -\frac{13.6}{n^2} \Rightarrow n = 25 \Rightarrow n = 5$$

$$\therefore \text{Angular momentum} = n \frac{h}{2\pi} = \frac{5h}{2\pi}$$

15. (b) From conservation of momentum, two identical photons must travel in opposite directions with equal magnitude of momentum and energy $\frac{hc}{\lambda}$

$$\text{from conservation of energy } \frac{hc}{\lambda} + \frac{hc}{\lambda} = m_0 c^2 + m_0 c^2$$

$$\Rightarrow \lambda = \frac{h}{m_0 c}$$

16. (a) (i) ${}_{16}S^{32} + {}_0n^1 \rightarrow {}_{15}P^{32} + {}_1H^1$
 (ii) ${}_9F^{19} + {}_1H^1 \rightarrow {}_2He^4 + {}_8O^{16}$
 (iii) ${}_7N^{14} + {}_0n^1 \rightarrow {}_6C^{14} + {}_1H^1$

17. (c) Number of lines in absorption spectrum = $(n-1)$

$$\Rightarrow 5 = n - 1 \Rightarrow n = 6$$

$\therefore$ Number of bright lines in the emission spectrum

$$= \frac{n(n-1)}{2} = \frac{6(6-1)}{2} = 15.$$

18. (c) The Hydrogen atom before the transition was at rest. Therefore from conservation of momentum.

$$p_{H\text{-atom}} = p_{\text{photon}} = \frac{E_{\text{radiated}}}{c} = \frac{13.6 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV}}{c}$$

$$1.6 \times 10^{-27} \times v = \frac{13.6 \left(\frac{1}{1^2} - \frac{1}{5^2} \right) \times 1.6 \times 10^{-19}}{3 \times 10^8}$$

$$\Rightarrow v = 4.352 \text{ m/s} \approx 4 \text{ m/sec}.$$

19. (b) In 2 sec only 90% nuclei are left behind. Thus in next two second 90% of 900 or 810 nuclei will be left.

20. (b) Acceleration $a \propto \frac{v^2}{r}$

$$\text{where } v \propto \frac{Z}{n} \text{ and } r \propto \frac{n^2}{Z} \Rightarrow a \propto \frac{Z^3}{n^4}$$

Since both are in ground state i.e., $n = 1$

$$\text{so } a \propto Z \Rightarrow \frac{a_{\text{He}^+}}{a_{\text{H}}} = \left(\frac{Z_{\text{He}^+}}{Z_{\text{H}}} \right)^3 = \left(\frac{2}{1} \right)^3 = \frac{8}{1}.$$

21. (b) By using $\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

$$\text{For Hydrogen atom } \frac{1}{(\lambda_{\min})_{\text{H}}} = R \left[\frac{1}{1^2} - \frac{1}{\infty} \right] = R$$

$$\Rightarrow (\lambda_{\min})_{\text{H}} = \frac{1}{R} \quad \dots(i)$$

$$\text{For hydrogen like atom } \left(\frac{1}{\lambda_{\min}} \right)_{\text{atom}} = RZ^2 \left(\frac{1}{2^2} - \frac{1}{\infty} \right)$$

$$\Rightarrow (\lambda_{\min})_{\text{atom}} = \frac{4}{RZ^2} \quad \dots(ii)$$

$$\text{From equation (i) and (ii) } \frac{1}{R} = \frac{4}{RZ^2} \Rightarrow Z = 2.$$

22. (c) The activity $\left(-\frac{dN}{dt} \right) = \lambda N \Rightarrow N = \left(-\frac{dN}{dt} \right) \left(\frac{T_{1/2}}{\log_e 2} \right)$

Taking the ratio of this expression for ^{240}Pu to this same expression for ^{243}Am ,

$$\frac{N_{\text{Pu}}}{N_{\text{Am}}} = \frac{\left(-\frac{dN_{\text{Pu}}}{dt} \right) (T_{1/2})_{\text{Pu}}}{\left(-\frac{dN_{\text{Am}}}{dt} \right) (T_{1/2})_{\text{Am}}} = \frac{(5 \mu\text{Ci}) \times (6560\text{y})}{(4.45 \mu\text{Ci}) \times (7370\text{y})} = 1$$

i.e. the two samples contains equal number of nuclei.

23. (b) Because the neutron has no electric charge, it experience no electric repulsion from a U^+ nucleus. Hence a slow moving neutron can approach and enter a U^+ nucleus, thereby providing the excitation needed to trigger fission. By contrast a slow moving proton feels a strong repulsion from a U^+ nucleus. It never get's close to the nucleus, so it cannot trigger fission.
24. (d) After one half life period, the activity of Tritium becomes 50%.
After 2 half life period 25%
After 3 half life period 12.5%
After 4 half life period 6.25%
After 5 half life period 3.12% $\approx$ 3%
It is $5 \times 12.5 \text{ years} + 7 \text{ years}$ i.e. approximately 70 years only .
25. (b) Transition from $4E$ to E

$$(4E - E) = \frac{hc}{\lambda_1} \Rightarrow \lambda_1 = \frac{hc}{3E} \quad \dots(i)$$

Transition from $\frac{7}{3}E$ to E

$$\left(\frac{7}{3}E - E \right) = \frac{hc}{\lambda_2} \Rightarrow \lambda_2 = \frac{3hc}{4E} \quad \dots(ii)$$

$$\text{From equation (i) and (ii) } \frac{\lambda_1}{\lambda_2} = \frac{4}{9}$$



Chapter 27 Electronics

Semiconductor electronics

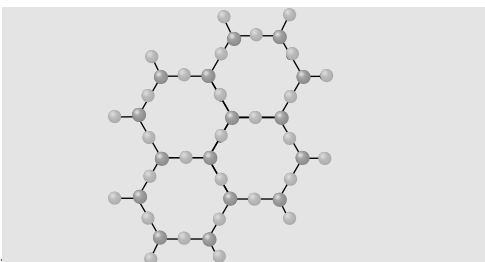


Solids

It is a state of matter which has a definite shape and a definite volume. The characteristic properties of the solid depends upon the nature of forces acting between their constituent particles (*i.e.* ions, atoms or molecules). Solids are divided into two categories.

Crystalline solids

- (1) These solids have definite external geometrical form.
- (2) Ions, atoms or molecules of these solid are arranged in a definite fashion in all it's three dimensions.



(3) Examples : Quartz, Salts, Metals, diamond etc.

(4) They have well defined facets or faces.

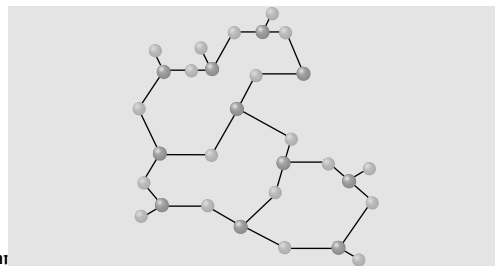
(5) They are ordered at short range as well as at long range.

(6) They are anisotropic, *i.e.* the physical properties like elastic moduli, thermal conductivity, electrical conductivity, refractive index have different values in different direction.

- (7) They have sharp melting point.
- (8) Bond strengths are identical throughout the solid.
- (9) These are considered as true solids.
- (10) An important property of crystals is their symmetry.

Amorphous or glassy solids

- (1) These solids have no definite external geometrical form.
- (2) Ions, atoms or molecules of these solids are not arranged in a definite fashion.



(3) Exa

(4) They do not possess definite facets or faces.

(5) These have short range order, and there is no long range order.

(6) They are isotropic.

(7) They do not have a sharp melting point.

(8) Bond strengths vary.

(9) These are considered as pseudo-solids or super cooled liquids.

(10) Amorphous solids do not have any symmetry.

Terms Related with Crystal Structure

(1) **Crystal lattice** : It is a geometrical arrangement of points in space where if atoms or molecules of a solid are placed, we obtain an actual crystal structure of the solid.

(2) **Basis** : The atoms or molecules attached with every lattice point in a crystal structure is called the basis of crystal structure.

